SAMSUNG TAB S-PEN STUDY EDITION · UPPCS 2026

Environment & Ecology

Complete Master Study Notes & High-Yield Syllabus Coverage

📱 **Tablet & S-Pen Reading Guide:**

• **S-Pen Margin:** Every page features a generous **26mm right margin** specifically calibrated for your S-Pen handwritten notes, keyword callouts, and active recall arrows.

• **Interactive TOC:** Tap any chapter in the Table of Contents below to jump directly to it in Samsung Notes.

• **Chapter Scratchpads:** An active recall handwriting box is provided at the conclusion of every chapter for your personal mind-maps.

• **Exam Traps & PYQs:** All examiner traps and high-yield comparison tables are expanded and highlighted.

Table of Contents (45 Chapters)

01. Environment Basics

02. Ecology & Ecosystem

03. Food Chain & Energy Flow

04. Biodiversity

05. Habitat, Flora & Fauna

06. Protected Areas & Conservation

07. Wildlife Conservation

08. Forests & Forest Management

09. Pollution & Waste Management

10. Climate Change

11. Ozone Layer

12. Acid Rain

13. Desertification & Land Degradation

14. Environmental Impact Assessment

15. Sustainable Development & Environmental Governance

16. Environmental Organizations (India)

17. Environmental Laws & Policies

18. International Environmental Agreements & Conferences

19. Climate & Environmental Institutions

20. Biodiversity Conservation Methods

21. Species & Ecology

22. Renewable Energy

23. Disaster & Environment

24. Current Environmental Issues

25. Global Environmental Geography

26. Water Resources & Water Conservation

27. Renewable & Non-Renewable Energy

28. Environmental Research & Institutions

29. Indian Environmental Movements

30. Environmental Literature & Awareness

31. Environmental Days

32. National Parks & Protected Areas (Advanced)

33. Biosphere Reserves

34. Climate Change (Advanced)

35. Atmosphere

36. Ozone Layer

37. Greenhouse Gases

38. Pollution (Advanced)

39. Acid Rain

40. Desertification

41. Environmental Monitoring

42. International Environmental Organizations

43. International Environmental Agreements

44. Current Environmental Issues

45. Environment PYQ Trend Analysis (UPPCS 2018–2025)

Topic 1 — Environment (पर्यावरण) Basics

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Environment – Meaning | Environmental Components | Classification of Environment | Human–Environment Interaction

Sources baked in: NCERT Class 11–12 (Biology Ch 13–16, Geography Ch 14–15), MoEFCC, CPCB, UNEP, Brundtland Report, UPPCS Prelims PYQs 2018–2025 **Weight:** ★★★ High — Prelims direct + Mains foundation **Last verified:** August 2026

Current Affairs

Year	Fact	Why asked
2023	**Amrit Dharohar Scheme** — wetlands conservation; launched on WED 2023	Wetlands + governance
2023	**National Green Hydrogen Mission**	SD / climate policy overlap
2019	**NCAP** (राष्ट्रीय स्वच्छ वायु कार्यक्रम) — 132 cities; 20–30% PM2.5/PM10 reduction target by 2024	Urban (नगरीय) environment governance

💡 🎯 **Consolidated — 31 Must-Score Facts (Click to Expand)**

1. Environment means the sum of **biotic and abiotic** factors that affect living organisms. Notes do not treat “only physical” or “only living” as the full definition. 2. The four (चातुर्याम) spheres are **lithosphere, hydrosphere, atmosphere, and biosphere**. Biosphere roughly spans from about **200 m below soil** to about **6 km** above sea level. 3. **Ecology** (पारिस्थितिकी) is the **study** of organism–environment links. An **ecosystem** (पारिस्थितिकी तंत्र) is the **functional unit**. A **habitat** (वास स्थान) is the **place**. A **niche** is the **role**. 4. Abiotic factors include light, temperature, water, soil, gases, pH, and salinity. **Edaphic** means soil; **climatic** means weather; **topographic** means slope and altitude. 5. Biotic structure runs **producers → consumers → decomposers**. **Rhizobium** fixes nitrogen in legumes. 6. **Liebig’s law** says growth is limited by the **scarcest** essential resource. **Shelford’s law** says a species survives only inside a **min–max tolerance** range. 7. Land is about **29%** of Earth’s surface and water about **71%**. **Lentic** means standing water; **lotic** means flowing water. Mangroves and wetlands are **transitional**. 8. Human–environment thought moves **environmental determinism (Ratzel) → possibilism (Vidal de la Blache) → neo-determinism (Griffith Taylor)**. 9. **IPAT** is Impact = Population (जनसंख्या) × Affluence × Technology. Hardin’s **Tragedy of the Commons** dates to **1968**. 10. Sustainable development is defined in the **Brundtland Report (1987)** (ब्रंटलैंड रिपोर्ट). **Rio 1992** produced **Agenda 21**, **UNFCCC** (यूएनएफसीसीसी), and **CBD**. 11. India’s **LiFE** Mission launched in **2022**; the idea was floated at **COP26 in 2021**, not COP25. 12. **Chipko** (चिपको) began in **1973** at **Reni, Uttarakhand (उत्तराखंड)**. **JFM** guidance is **1990**. **FRA** is **2006**. 13. **Article 48A** is the State DPSP (राज्य के नीति निदेशक तत्व) on environment. **Article 51A(g)** is the citizen’s Fundamental Duty. **Article 48** is about agriculture and cows — not environment. 14. **Article 21** (अनुच्छेद 21) has been read to include a **healthy environment**. The **42nd (42वां) Amendment (1976)** added 48A and 51A(g). 15. The **Environment (Protection) Act, 1986** is India’s post-**Bhopal (1984)** umbrella law. 16. **MoEF** began in **1985** and became **MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) in 2014**. Headquarters is New Delhi. 17.

Stockholm 1972 (स्टॉकहोम सम्मेलन) was the first UN human-environment conference and led to **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) at **Nairobi**. 18. World Environment Day (विश्व पर्यावरण दिवस) is **5 June**. Earth Day (पृथ्वी दिवस) is **22 April**. World Wildlife Day (विश्व वन्यजीव दिवस) is **3 March**, not 22 March. 19. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February**. International Ozone Day is **16 September**. Desertification (मरुस्थलीकरण) and Drought (सूखा) Day is **17 June**. International Day for Biological Diversity is **22 May**. 20. UNEP headquarters is **Nairobi, Kenya**. CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) and MoEFCC sit in **New Delhi**. 21. The **Brundtland Commission (WCED)** worked **1983–1987** and defined sustainable development in *Our Common Future*. 22. Uttar Pradesh (उत्तर प्रदेश) mainly falls in the **Gangetic Plain** biogeographic zone (**Zone 7**). 23. Classic UP river–city facts include Ganga (गंगा) at Varanasi (वाराणसी) and Kanpur (कानपुर), Yamuna (यमुना) at Agra, and Gomti at Lucknow (लखनऊ). 24. Pollution hotspots often tested for UP include **Kanpur tanneries**, NCR air (Ghaziabad/Noida), and Ganga ghats at **Varanasi** (वाराणसी). 25. **Ganga Action Plan** (गंगा एक्शन प्लान) began in **1985**. **Namami Gange** (नमामि गंगे) launched in **2014**. 26. A standard triad classifies environment as **physical**, **biological**, and **cultural**. **Operational environment** is **not** part of that triad. 27. Natural versus anthropogenic, terrestrial versus aquatic, and internal versus external are further common classification axes. 28. Micro-, meso-, and macro-scales describe environment size from a pond edge to a region or the globe. Competitive papers usually default to the **natural/physical** sense of environment. 29. Agenda 21 belongs to **Rio 1992**, not 1995. Mixing Brundtland 1987 with Agenda 21's year is a common trap. 30. **Carrying capacity** is the maximum load an environment can sustain indefinitely. Resource extraction must stay **below** regeneration, and waste must stay **within** absorption capacity — not the reverse. 31. **Ecological footprint** (पारिस्थितिक पदचिह्न) measures the land/water area needed to support consumption, in **global hectares (gha)**.

💡 ⚡ Confused Pairs (Click to Expand)

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Environment** | **Ecology** | Total biotic and abiotic surroundings influencing an organism vs the scientific *study* of interactions between organisms and environment | पर्यावरण / पारिस्थितिकी | | **Ecology** | **Ecosystem** | Scientific discipline/field of study (Ernst Haeckel 1866) vs functional, structural ecological *unit* of nature (A.G. Tansley 1935) | पारिस्थितिकी / पारिस्थितिकी तंत्र | | **Habitat** | **Ecological Niche** | Physical address / geographical place where an organism lives vs functional profession / ecological role of a species in an ecosystem | वास स्थान / पारिस्थितिक निकेत (Niche) | | **Biotic Components** | **Abiotic Components** | Living organisms (producers, consumers, decomposers) vs non-living physical and chemical factors (light, temperature, soil, water) | जैविक घटक / अजैविक घटक | | **Lentic Ecosystem** | **Lotic Ecosystem** | Standing or stagnant fresh water bodies (ponds, lakes, bogs, swamps) vs running or flowing fresh water bodies (rivers, streams, brooks) | स्थिर जल (लैंटिक) / प्रवाही जल (लोटिक) | | **Liebig's Law of the Minimum** | **Shelford's Law of Tolerance** | Population growth is controlled by the scarcest limiting resource vs species survival requires environmental factors to remain between minimum and maximum tolerance limits | लाइबिग का न्यूनतम नियम / शेलफोर्ड का सहनशीलता नियम | | **Internal Environment** | **External Environment** | Biochemical and physiological conditions within an organism (homeostasis) vs physical surroundings external to the organism | आंतरिक पर्यावरण / बाह्य पर्यावरण | | **Physical Environment** | **Biological Environment** | Inanimate non-living components (lithosphere, hydrosphere, atmosphere) vs living communities of flora, fauna, and microbes | भौतिक पर्यावरण / जैविक पर्यावरण |

💡 📋 Must-score facts — spheres, laws, articles, days (Click to Expand)

Core tags | Item | Lock | |---|---| | Ecology / Ecosystem / Habitat / Niche | Study / functional unit / place / role | | Four spheres | Lithosphere · hydrosphere · atmosphere · biosphere | | Biosphere span | ~200 m below soil → ~6 km above sea level | | Edaphic / Climatic / Topographic | Soil / weather / slope–altitude | | Liebig / Shelford | Scarcest resource / min–max tolerance | | Lentic / Lotic | Standing / flowing water | | Thought chain | Ratzel determinism → Vidal possibilism → Griffith Taylor neo-determinism | | IPAT / Tragedy of Commons | P×A×T / Hardin **1968** | ### Milestones & law | Item | Lock | |---|---| | Brundtland / Rio |

1987 SD definition / **1992** Agenda 21 + UNFCCC + CBD | | LiFE | Idea COP26 **2021**; launch **2022** | | Chipko / JFM / FRA | **1973** Reni / **1990** / **2006** | | Art. 48A / 51A(g) / 48 | State DPSP env / citizen duty / agri–cows (**not** env) | | 42nd Amd / EPA | **1976** adds 48A+51A(g) / EPA **1986** post-Bhopal | | MoEF → MoEFCC | **1985** → **2014**; HQ New Delhi | | Stockholm / UNEP | **1972** → UNEP **Nairobi** | | GAP / Namami Gange | **1985** / **2014** | ### Days (quick) | Item | Lock | |------|------| | WED / Earth / Wildlife | 5 Jun / 22 Apr / **3 Mar** (≠ 22 Mar) | | Wetlands / Ozone / Desertif. / Biodiversity | 2 Feb / 16 Sep / 17 Jun / 22 May | --- ![Earth's Four Interconnected Spheres & Biosphere Envelope] (images/env_ch01_01_earth_four_spheres_atmosphere_biosphere.png)

1.1 Environment – Meaning

Scope: NCERT definition | four spheres | atmosphere basics | scales & types | milestones & law

Source	Definition
NCERT (Class 12)	Sum of all **biotic + abiotic** factors influencing living organisms ← **most asked**
UNEP	Physical, chemical, and biotic factors acting on organism/community, determining form and survival
EPA 1986 (पर्यावरण संरक्षण 1986) (Legal)	Water, air, land + inter-relationship among them and with humans, other living beings, plants, and property
General	External conditions — physical, chemical, biological, social, economic, cultural — affecting life

- French *environ* means surroundings (परिवेश).
- In UPPCS, "environment" in the scientific sense means **natural/physical** factors (biotic plus abiotic). Social and economic factors are not the primary focus in most Prelims questions.
- The **lithosphere** is Earth's solid land crust. It covers about **29%** of the surface.
- The **hydrosphere** is all water on Earth in liquid, ice, and vapour form. It covers about **71%** of the surface.
- The **atmosphere** is the gaseous envelope. Roughly **78%** is nitrogen and **21%** is oxygen.
- The **biosphere** is the zone where life exists. It spans from about **200 m below soil** to about **6 km above sea level**, where lithosphere, hydrosphere, and atmosphere meet.
- **Pedosphere** (soil) is usually grouped with the lithosphere or with edaphic abiotic factors. It is not counted as a fifth sphere.
- The **troposphere** (क्षोभमंडल) extends from the surface to about **12 km**. All weather phenomena occur here.
- The **stratosphere** (समतापमंडल) lies roughly **12–50 km** above the surface. The **ozone layer** that absorbs UV radiation sits in this layer.
- **Micro-scale** environment covers immediate surroundings such as a room or a pond edge.
- **Meso-scale** covers a village or city.
- **Macro-scale** covers regional or global systems.
- **Total environment** can be natural, social, economic, or cultural. Competitive papers usually default to **natural/physical** environment.
- The **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन 1972) was the first major UN conference on the human environment.
- **UNEP** (United Nations Environment Programme) was created at Stockholm. Its headquarters is in **Nairobi, Kenya**.
- **World Environment Day (WED)** is observed every year on **5 June**. The date marks the opening day of the Stockholm Conference in 1972.
- **MoEF** was established in **1985**. It was renamed **MoEFCC in 2014** when climate change (जलवायु परिवर्तन) was added to its mandate.

- The **Environment (Protection) Act, 1986** is India's umbrella environmental law. Parliament passed it after the **Bhopal Gas Tragedy (1984)**.
- **Article 48A** is a Directive Principle. The State shall protect the environment and safeguard forests and wildlife. The **42nd Amendment (1976)** added it.
- **Article 51A(g)** is a Fundamental Duty. Every citizen shall protect and improve the natural environment. The same amendment added it.
- **Article 21** guarantees the Right to Life. The Supreme Court expanded it to include the **right to a wholesome and healthy environment** in cases such as *M.C. Mehta* and *Subhash Kumar*.

> **Logic:** UPPCS 2018 treats environment as **जैविक + अजैविक**. Trap options say "only physical" or "only living." **Article 48** deals with agriculture and cows. Environment is **Article 48A**.

PYQ — UPPCS Prelims 2025, Q62

Which of the following pairs is/are NOT correctly matched?

(Day) — (Date)

1. International Ozone Day — 16 September
2. World Environment Day — 5 June
3. World Wildlife Day — 22 March

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

💡 **Show answer**

Ans: B — Only pair 3 is wrong. World Wildlife Day = **3 March**, not 22 March.

Foundational Ecological Laws & Limiting Factor Dynamics

1.2 Environmental Components

Identity: Environment = Abiotic + Biotic | Liebig + Shelford | N-fixation

Abiotic factor category	Includes	Hindi
Edaphic	Soil texture, minerals, pH, moisture	मृदा संबंधी
Climatic	Temperature, rainfall, wind, humidity	जलवायु संबंधी
Topographic	Altitude, slope, aspect	स्थलाकृति संबंधी
Biotic group	Also called	Function
---	---	---
Producers	Autotrophs	Convert solar energy → chemical energy
Consumers	Heterotrophs	Feed on other organisms
Decomposers	Saprotrophs	Break down dead matter; recycle nutrients
Detritivores	—	Feed on detritus (earthworms, termites)

- About **99%** of biosphere energy comes from the **Sun**.

- **Liebig's Law of Minimum** states that growth is controlled by the **scarcest essential resource**. That resource is the limiting factor.
- **Shelford's Law of Tolerance** (शेलफोर्ड का सहनशीलता नियम) states that each species survives only within a **minimum–maximum range** for each factor. Beyond that range, stress or death follows even if other factors are abundant.
- **N_2** makes up about **78%** of air, but most life cannot use it directly. Organisms need **nitrogen fixation** first.
- **Rhizobium** bacteria live in legume root nodules and fix atmospheric N_2. Pulses, gram, and soybean are common legume examples.
- **Azotobacter** is a free-living soil bacterium that fixes nitrogen.
- **Cyanobacteria** (blue-green algae) also fix nitrogen in soil and water.
- **Humus** is organic matter in soil. It comes from decomposed plant and animal material, so it is **not** an abiotic component.
- Abiotic factors shape biotic life. Desert heat favours cacti and other drought-adapted plants.
- Biotic activity modifies abiotic conditions. Forests increase local rainfall. Overgrazing can trigger desertification.
- **Dissolved Oxygen (DO)** is a key aquatic abiotic indicator. Fish die in polluted rivers when DO falls too low.

> **Logic:** Liebig names **which single factor limits growth**. Shelford names the **survivable range**. Humus is organic, not abiotic.

PYQ — UPPCS Prelims 2019, Q46

Nitrogen fixing bacteria make combination with cells of the roots of

A. Pulses

B. Rice

C. Wheat

D. Sugarcane

💡 **Show answer**

Ans: A — Rhizobium associates with leguminous (pulse) root nodules.

1.3 Classification of Environment

Identity: Physical/Biological/Cultural triad | Natural/Anthropogenic | Terrestrial/Aquatic | Internal/External | India = 10 biogeographic zones

Basis	Categories
Standard triad	**Physical** / **Biological** / **Cultural** (**Operational** is **not** a category)
Origin	Natural vs **Anthropogenic** (human-modified — cities, dams, cropland)
Components	Abiotic vs Biotic
Domain	**Terrestrial** (~29% surface) vs **Aquatic** (~71%)
Organism level	**Internal** (inside body — physiology) vs **External** (outside — ecology default)
Scale	Micro / Meso / Macro
Human use	Rural / Urban / Industrial

Physical, biological, and cultural environment

- In the standard triad, environment is classified as **physical**, **biological**, and **cultural**.

- The **physical environment** (भौतिक पर्यावरण) includes landforms, water bodies, climate, soils, rocks, and minerals.
- The **biological environment** (जैविक पर्यावरण) includes plants, animals, and microorganisms.
- The **cultural environment** includes man-made features such as economic, social, and political elements.
- **Operational environment** is **not** part of this standard classification.
- **Internal environment** (आंतरिक पर्यावरण) means conditions inside an organism's body, such as blood chemistry and tissue fluids studied in physiology.
- **External environment** (बाह्य पर्यावरण) means conditions outside the organism. Ecology usually studies the external environment.

Aquatic type	Description	Indian example
Marine	Oceans/seas — high salinity	Arabian Sea, Bay of Bengal
Lentic	**Standing** freshwater	Loktak Lake (लोकटक), Dal Lake
Lotic	**Flowing** freshwater	Ganga, Yamuna
Estuarine	River meets sea — brackish	Sundarbans (सुंदरबन) delta (डेल्टा)
Wetland	Transitional — neither fully land nor water	Chilika, Keoladeo (केवलादेव)
Marine zone	Location	Key fact
---	---	---
Neritic	Shallow shelf (0–200 m)	Fisheries, coral, mangroves
Pelagic	Open water column	Plankton, tuna
Benthic	Sea floor	Crabs, decomposers
Littoral	Intertidal shore	Barnacles, molluscs

Transitional ecosystems — mangroves & wetlands

- A **wetland** is land that stays waterlogged or seasonally flooded, so it is neither fully terrestrial nor fully aquatic.
- Wetlands store floodwater, recharge groundwater, filter pollutants, and support migratory birds.
- **World Wetlands Day** is observed on **2 February** (Ramsar (रामसर) Convention anniversary).
- India's wetland rules are notified under **Wetlands (Conservation and Management) Rules, 2017**.
- Classic Indian wetlands s include **Chilika** (Odisha) and **Keoladeo** (केवलादेव) (Rajasthan).
- A **mangrove** is a salt-tolerant coastal forest growing in intertidal / brackish zones.
- Mangroves act as a storm buffer, reduce shoreline erosion, and nursery fish stocks.
- India's largest mangrove stretch is the **Sundarbans** (सुंदरबन) (West Bengal / Bangladesh).
- In classification questions, mangroves are **coastal transitional**, not "purely terrestrial."
- UPSC (संघ लोक सेवा आयोग) 2020 trap: among Desert, Mangrove, Rainforest — only Desert and Rainforest count as terrestrial.

> **Logic:** Mangrove / wetland = transitional. Do not tick mangrove as terrestrial.

Urban heat island

- An **urban heat island** means a city runs warmer than nearby rural land under similar weather.
- Typical city–rural gap s is about **2–5°C**.
- Concrete and asphalt absorb heat and re-radiate it at night.
- Dense buildings trap warm air and block wind flow.
- Less vegetation cuts evapotranspiration cooling, so city nights stay warmer than rural nights.

- Extra heat raises energy demand for cooling and worsens ground-level air pollution stress.
- Delhi NCR is a frequent Indian example of urban heat + air-quality interaction.

> **Logic:** Urban heat island = city warmer by roughly **2–5°C**, not a separate climate zone.

India's 10 biogeographic zones

Authors / year: Rodgers & Panwar | **1988** (WII delineation used s)

#	Zone	feature
1	**Trans-Himalaya**	Cold desert — Ladakh (लद्दाख), high-altitude sparse vegetation
2	**Himalaya** (हिमालय)	Alpine to subtropical belts along the main Himalayan range
3	**Desert**	Thar (थार) — Rajasthan and adjoining Gujarat
4	**Semi-Arid**	Dry scrub/savanna (सवाना) fringe of Gujarat–Maharashtra–central India
5	**Western Ghats** (पश्चिमी घाट)	Biodiversity hotspot (जैव विविधता हॉटस्पॉट); high endemism
6	**Deccan (दक्कन) Peninsula**	Plateau forests and dry deciduous tracts
7	**Gangetic Plain**	Fertile alluvial (जलोढ़) plains — covers **most of UP**
8	**Coasts**	Mangroves, estuaries, coral reefs
9	**North-East India**	Indo-Malayan biodiversity corridor
10	**Islands**	Andaman (अंडमान) & Nicobar (निकोबार) and Lakshadweep (लक्षद्वीप)

- Remember: the number **10**, not 8 or 14. Fourteen is a global biome (जीवोम) count from a different classification list.
- Most of Uttar Pradesh falls in the **Gangetic Plain** zone.
- The northern Terai (तराई) fringe of UP links to Himalayan foothill ecology.

> **Logic:** UPPCS has asked "how many biogeographic zones in India?" → **10** (Rodgers & Panwar, 1988).

CRZ — Coastal Regulation Zone

Legal basis: Notifications under the **Environment (Protection) Act, 1986** (nodal: MoEFCC)

- CRZ regulates development along India's coast to protect ecology and coastal livelihoods.
- The coastal stretch covered is generally from the High Tide Line landward and seaward as defined in the notification (focus is categories, not metre-by-metre measurement).
- **CRZ-IA** covers ecologically sensitive areas such as mangroves, coral reefs, and nesting beaches — **no new development** of the usual commercial kind.
- **CRZ-IB** is the **intertidal zone** between Low Tide Line and High Tide Line.
- **CRZ-II** covers already **developed / urbanised** coastal stretches within municipal limits.
- **CRZ-III** covers **rural** coastal areas where development is more restricted than in CRZ-II.
- The **CRZ 2019** notification revised earlier CRZ rules. Papers still test the **IA / IB / II / III** category logic.

> **Logic:** CRZ-IA = most sensitive (mangroves and corals). CRZ-II = developed urban coast. CRZ-III = rural coastal belt.

Eco-Sensitive Zones (ESZ)

Legal basis: Declared under the **Environment (Protection) Act, 1986**

- An **ESZ** is a **buffer belt** around a National Park, Wildlife Sanctuary, or other protected area.

- Eco-Sensitive Zones are notified under the **Environment (Protection) Act, 1986**, **not** under the Wildlife (Protection) Act, 1972.
- The aim is to reduce shock between a strict protected core and surrounding human use.
- The purpose is **regulation**, not a total ban on every human activity except agriculture.
- **Permitted** activities typically include ongoing local agriculture/horticulture, rainwater harvesting, and organic farming.
- **Regulated** activities include tree felling, hotels/resorts, commercial use of natural water, and road widening.
- **Prohibited** activities include polluting discharges into natural water bodies and tourism such as flying over protected areas in an aircraft or hot-air balloon.
- ESZ width is **not** a fixed all-India kilometre figure. It is notified case by case, often up to about **10 km**.
- Do not treat 10 km as a universal law for every protected area.
- An ESZ is an **EPA buffer tool**. It is **not** the same as declaring a National Park under the Wildlife (Protection) Act.

> **Logic:** ESZ = EPA 1986 buffer. "All human activity banned except agriculture" is **false**. CRZ = coastal regulation under the same Act.
>
> **Logic:** Lentic = lake or pond (standing water). Lotic = river or stream (flowing water). marks **river** as lotic.

PYQ — UPPCS Prelims 2023, Q112

Which of the following is an example of a **Lotic Ecosystem** (प्रवाही जल)?

A. Swamps

B. Ponds

C. Rivers

D. Marshy areas

💡 **Show answer**

Ans: C — Lotic = flowing water. Ponds/swamps = lentic or wetland.

1.4 Human–Environment Interaction

Identity: Reciprocal relationship | models | carrying capacity | SD governance | Indian movements

Model	Proponent	Idea	Status
Environmental Determinism	Ratzel	Environment **dictates** culture/society	Largely **rejected**
Possibilism	Vidal de la Blache	Environment offers **possibilities**; humans choose	**Accepted**
Neo-determinism	Griffith Taylor	Humans modify environment **within limits**	Balanced view
Concept	Definition / fact		
---	---		
Carrying capacity	Maximum load an environment can sustain indefinitely. Resource extraction must stay **below** regeneration. Waste must stay **within** absorption capacity		

Model	Proponent	Idea	Status
IPAT	Impact = **P**opulation × **A**ffluence × **T**echnology		
Ecological footprint (पारिस्थितिक पदचिह्न)	Land/water area needed to support consumption — measured in **global hectares (gha)**		
Tragedy of the Commons	Shared resources overexploited when individual interest > collective good — **Garrett Hardin, 1968**		
Sustainable Development	"Meet present needs without compromising future generations' ability to meet theirs" — **Brundtland Report, 1987** (*Our Common Future*)		

Brundtland Report, 1987

Commission: World Commission on Environment and Development (WCED) | Chair: Gro Harlem Brundtland

- The report's official title is ***Our Common Future*** (published **1987**).
- It popularised the classic definition of **Sustainable Development**.
- SD balances three pillars: economic growth, social equity, and environmental protection.
- The commission worked roughly **1983–1987** under a UN mandate.
- Do **not** confuse Brundtland with ***The Limits to Growth*** (Club of Rome, **1972**).
- UPPCS 2024 A/R trap: Reason saying Brundtland = *Limits to Growth* is **false**.

> **Logic:** Brundtland 1987 = *Our Common Future*. Club of Rome 1972 = *Limits to Growth*.

Rio Earth Summit, 1992

Also called: United Nations Conference on Environment and Development (UNCED) | Place: Rio de Janeiro

- Rio shifted global environment talk from awareness (Stockholm 1972) to an action framework.
- Three landmark outcomes signed/adopted (अंगीकृत) at Rio are **UNFCCC**, **CBD**, and **Agenda 21**.
- **UNFCCC** is the climate convention that later produced Kyoto (क्योटो) and Paris pathways.
- **CBD** is the Convention on Biological Diversity for conservation and benefit-sharing.
- Rio is **1992**, not Stockholm 1972 and not Agenda 21's false "1995" year.

> **Logic:** Rio 1992 package = UNFCCC + CBD + Agenda 21.

Agenda 21, 1992

- **Agenda 21** is a non-binding action blueprint for sustainable development in the 21st century.
- It was adopted at the **Rio Earth Summit, 1992**.
- It covers local, national, and global action themes (poverty, resources, governance, participation).
- The title **Agenda 21** refers to the 21st century action plan. It does **not** mean the document was signed in the year 2021.
- UPPCS 2025 A/R used the wrong year **1995** as a Reason trap — that Reason is **false**.

> **Logic:** Agenda 21 = **1992** (Rio). Never mark 1995.

LiFE Mission, 2022

Full form: Lifestyle for Environment (जीवनशैली पर्यावरण) | Nodal idea: India at climate diplomacy

- PM Modi proposed the LiFE idea at **COP26, Glasgow (2021)**.
- India formally launched **Mission LiFE** in **June 2022**.

- The mission pushes individual and community lifestyle shifts — save energy, cut waste, choose sustainable food and consumption.
- Notes often cite about **75** pro-planet actions / behavioural pledges under LiFE.
- The Mission LiFE app / campaign tracks and popularises those behaviours.
- Trap: "proposed at COP-25 Madrid" is **false**.

> **Logic:** Idea = COP26 (2021). Launch = **2022**. Not COP-25 Madrid.

Chipko Movement (चिपको आंदोलन), 1973

Place: Reni village, Chamoli (चमोली) (then Uttar Pradesh; now **Uttarakhand** (उत्तराखंड))

- Chipko means villagers hugged trees to stop commercial felling.
- The 1973 Reni episode is the classic anchor for the movement's start.
- **Sunderlal Bahuguna** was a leading Chipko activist linked to the Uttarakhand forest protests.
- **Chandi Prasad Bhatt** (चंडी प्रसाद भट्ट) was another key Chipko leader from the Garhwal (गढ़वाल) region.
- Women played a visible role in forest protection protests.
- Chipko is often cited as **passive resistance** (निष्क्रिय प्रतिरोध) / non-violent environmental action.
- Place trap: Chipko is **not** Assam, Kerala, or Karnataka.

> **Logic:** Chipko = **1973**, Reni, Uttarakhand (then UP). Appiko (अप्पिको) = Karnataka.

Appiko Movement, 1983

Place: Karnataka (Western Ghats / Uttara Kannada belt)

- Appiko is the southern Indian analogue of Chipko.
- Villagers hugged trees to resist forest felling and protect local ecology.
- UPPCS 2025 asked the state association — correct fact is **Karnataka only**.
- Do not mark Uttarakhand, UP, or Kerala for Appiko.

> **Logic:** Appiko = **Karnataka, 1983**. Chipko = Uttarakhand.

Joint Forest Management (JFM) (संयुक्त वन प्रबंधन), 1990

- JFM is a partnership between the **Forest Department** and local **village communities**.
- The national policy push is dated to the **1990** Government of India resolution / circular.
- Village communities help protect and regenerate forests under JFM.
- In return, communities share usufruct and other forest benefits.
- JFM is participatory forest governance, not a full ownership transfer law.
- Do not confuse JFM (**1990**) with FRA (**2006**) or EPA (**1986**).

> **Logic:** JFM launch year fact = **1990**.

Forest Rights Act (FRA), 2006

Full title: Scheduled Tribes and Other Traditional Forest Dwellers (Recognition (मान्यता) of Forest Rights) Act, 2006

- FRA recognises **individual** and **community** forest rights of Scheduled Tribes and **OTFDs**.
- OTFD means Other Traditional Forest Dwellers who depended on forests for livelihood.
- Rights can include habitation, cultivation, and community forest resource use, subject to the Act's process.

- FRA is framed as justice for forest dwellers, not as a blanket free-for-all to clear forests.
- Year trap: FRA = **2006**, after JFM 1990 and EPA 1986.

> **Logic:** FRA 2006 = tribal (आदिवासी)/OTFD forest rights. JFM 1990 = community–State forest partnership.

- **Polluter Pays** and **Precautionary Principle** entered Indian environmental jurisprudence through *Vellore Citizens Welfare Forum* (**1996**).
- The **Public Trust Doctrine** treats air, water, and forests as held by government in trust for the public.
- The **Bhopal Gas Tragedy (1984)** triggered the umbrella **Environment (Protection) Act, 1986**.

Landmark Environmental Classics & Pioneer Thinkers

Thinker / Landmark	Year / Publication	Core Concept & Exam Significance
Rachel Carson	**1962** (*Silent Spring*)	Exposed environmental hazards of indiscriminate synthetic pesticides (**DDT** bioaccumulation and eggshell thinning); catalytic milestone that birthed the modern global environmental movement.
John W. Bennett	**1976** (*The Ecological Transition*)	Coined and developed the concept of **'Ecological Transition'**, analyzing human society's cultural shift from balanced ecological adaptation to intensive environmental exploitation.
Garrett Hardin	**1968** (*The Tragedy of the Commons*)	Postulated in *Science* that independent rational actors deplete shared public resources (pastures, fisheries, clean air) through unconstrained self-interest.
Club of Rome	**1972** (*The Limits to Growth*)	Commissioned MIT computer-simulation model (Meadows et al.) showing ecological collapse if exponential economic and population growth outpace planetary carrying capacity. *(Trap: Brundtland Report 1987 is Our Common Future, NOT Limits to Growth).*
Ernst Haeckel	**1866** (*Generelle Morphologie*)	Coined the term **'Ecology'** (*Oekologie*) for the study of organism-environment interactions.
A.G. Tansley	**1935**	Coined the term **'Ecosystem'** as the fundamental functional unit of nature.
Arne Naess	**1973**	Coined **'Deep Ecology'** (advocating inherent biocentric rights of all living species) versus utilitarian 'Shallow Ecology'.
Joseph Grinnell	**1917**	Coined and formalized the concept of **'Ecological Niche'** as the functional role and distributional unit of a species.
Mathis Wackernagel & William Rees	**1996**	Formulated the concept of **'Ecological Footprint'**, measuring human resource appropriation in global hectares (gha).

Ecological Zonation & Organism Habitat Adaptations

- **Organism Adaptations**:
- **Arboreal**: Tree-dwelling animals adapted for life in tree canopies (e.g., squirrels, chameleons, monkeys).
- **Fossorial**: Animals adapted for digging and living underground in burrows (e.g., moles, pocket gophers, earthworms).
- **Xericole**: Animals anatomically and physiologically adapted for arid, desert environments with minimal water loss (e.g., camels, kangaroo rats, desert lizards).
- **Cursorial**: Animals adapted specifically for running fast on land (e.g., cheetah, ostrich, deer).
- **Lakes & Freshwater Zonation**:
- **Littoral Zone**: Shallow, sunlit near-shore waters where light penetrates to the bottom; supports rooted aquatic macrophytes (water lilies, reeds).

- **Limnetic Zone**: Open, well-lit surface waters away from the shoreline; prime habitat for photosynthetic phytoplankton, zooplankton, and nekton.
- **Profundal Zone**: Deep aphotic open-water layer below the limnetic zone; lacks light for photosynthesis, dominated by heterotrophic decomposers.
- **Benthic Zone**: Bottom floor and sediment layer across the entire body of water inhabited by benthos organisms.
- **Nitrogen-Fixing Symbiosis**:
- **Rhizobium** genus bacteria establish a symbiotic relationship within the root nodules of **Leguminous plants / Pulses** (Fabaceae family, e.g., gram, pea, arhar, lentil, soybean). They convert atmospheric dinitrogen (N_2) into biologically accessible ammonium (NH_4^+) and nitrates, naturally replenishing soil nitrogen fertility without chemical fertilizers.

PYQ — UPPCS Prelims 2025, Q22

With reference to 'Carrying Capacity of Environment', which of the following statements is/are correct?

1. The rate of resource extraction should remain above the rate of resource generation.
2. Generation of waste should remain within the absorption capacity of environment.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Ans: A — Only statement 2. Extraction must stay **below** regeneration, not above.

PYQ — UPPCS Prelims 2025, Q51

Assertion (A): Sustainable development should take place without harming the environment and without compromising the needs of future generations.

Reason (R): Agenda 21 was signed by world leaders in 1995.

A. Both true; R not explanation

B. A false, R true

C. A true, R false

D. Both true; R explains A

💡 **Show answer**

Ans: C — A is true (Brundtland definition). R is false — Agenda 21 = **Rio 1992**.

Complete PYQ Bank (Topic 1)

Q1. UPPCS Prelims 2025, Q22

With reference to 'Carrying Capacity of Environment', which of the following statements is/are correct?

1. The rate of resource extraction should remain above the rate of resource generation.
2. Generation of waste should remain within the absorption capacity of environment.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Ans: A — Only 2. Extraction must stay **below** regeneration.

Q2. UPPCS Prelims 2025, Q51

Assertion (A): Sustainable development should take place without harming the environment and without compromising the needs of future generations.

Reason (R): Agenda 21 was signed by world leaders in 1995.

A. Both true; R not explanation

B. A false, R true

C. A true, R false

D. Both true; R explains A

💡 **Show answer**

Ans: C — Agenda 21 = Rio **1992**.

Q3. UPPCS Prelims 2025, Q62

Which pairs is/are NOT correctly matched? (Day)—(Date): 1. Ozone Day—16 Sept 2. WED—5 June 3. Wildlife Day —22 March

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

💡 **Show answer**

Ans: B — Wildlife Day = **3 March**.

Q4. UPPCS Prelims 2025, Q139

With reference to Lifestyle for Environment (LiFE), which statements is/are correct?

1. Launched June 2022. 2. Idea proposed at COP-25 Madrid.

A. Only 2

B. Neither

C. Both

D. Only 1

💡 **Show answer**

Ans: D — Idea at COP26 Glasgow 2021.

Q5. UPPCS Prelims 2024, Q51

Assertion (A): SD was popularised by Brundtland Report. Reason (R): Brundtland Report is also known as “The Limits to Growth.”

A. Both true; R not explanation

B. A false, R true

C. Both true; R explains A

D. A true, R false

💡 **Show answer**

Ans: D — *Limits to Growth* = Club of Rome.

Q6. UPPCS Prelims 2024, Q62

Plants and animals that live on land are called:

A. Abiotic

B. Aquatic habitants

C. Terrestrial habitants

D. Biotic

💡 **Show answer**

Ans: C

Q7. UPPCS Prelims 2023, Q21

Which provision is part of both DPSP and Fundamental Duties?

A. Guardians—education to children

B. Protection of environment

C. Workers in management

D. UCC

💡 **Show answer**

Ans: B — Art 48A + Art 51A(g).

Q8. UPPCS Prelims 2023, Q112

Which is an example of a Lotic Ecosystem?

A. Swamps

B. Ponds

C. Rivers

D. Marshy areas

💡 **Show answer**

Ans: C

Q9. UPPCS Prelims 2023, Q123

Balancing resource use while conserving for future generations is called:

A. Resource Conservation

B. Sustainable Development

C. Future Resources

D. Reducing Consumption

💡 **Show answer**

Ans: B

Q10. UPPCS Prelims 2023, Q125

Assertion (A): Population control necessary for environment. Reason (R): Rapid population increase maintains environmental balance.

A. Both true; R explains A

B. Both true; R not explanation

C. A true, R false

D. A false, R true

💡 **Show answer**

Ans: C — Rapid growth **disrupts** balance.

Q11. UPPCS Prelims 2023, Q132

Agenda-21 of Rio Summit (1992) is related to:

A. Sustainable Development

B. Polluter Pays

C. Environmental Education

D. Ozone preservation

💡 **Show answer**

Ans: A

Q12. UPPCS Prelims 2022, Q11

When was the Environment (Protection) Act passed?

A. 1971

B. 1974

C. 1981

D. 1986

💡 **Show answer**

Ans: D

Q13. UPPCS Prelims 2022, Q134

Which is NOT a Fundamental Duty?

A. Protect monuments

B. Protect & improve natural environment

C. Safeguard public property

D. Respect National Anthem

💡 **Show answer**

Ans: A — Monument protection is not a listed FD.

Q14. UPPCS Prelims 2021, Q90

In which year was the Environment (Protection) Act passed?

A. 1982

B. 1986

C. 1990

D. 1992

💡 **Show answer**

Ans: B

Q15. UPPCS Prelims 2021, Q114

World Ozone Day is celebrated on:

A. 25 Dec

B. 21 Apr

C. 16 Sep

D. 30 Jan

💡 **Show answer**

Ans: C

Q16. UPPCS Prelims 2021, Q41

Assertion (A): Ecosystem components are not interdependent. Reason (R): Human activities impact environment.

A. Both true; R explains A

B. Both true; R not explanation

C. A true, R false

D. A false, R true

💡 **Show answer**

Ans: D — Components **are** interdependent.

Q17. UPPCS Prelims 2020, Q13

SD discussion started after a 1987 UN report. That report was:

A. First Climate Change Report

B. Our Common Future

C. Second Climate Report

D. Fifth Assessment Report

💡 **Show answer**

Ans: B — Brundtland Report.

Q18. UPPCS Prelims 2020, Q17

Silent Spring (1962), which set the tone for environmental movement, was written by:

A. Carolyn Merchant

B. Karl Marx

C. Rachel Carson

D. Rajgopalan

💡 **Show answer**

Ans: C

Q19. UPPCS Prelims 2020, Q142

Environment classified as Operational/Physical/Cultural/Biological. Which is NOT a standard category?

A. Operational

B. Physical

C. Cultural

D. Biological

💡 **Show answer**

Ans: A

Q20. UPPCS Prelims 2020, Q144

The concept of 'Ecological Transition' was first used by:

A. Elton

B. Bennett

C. Berkeley

D. Ratzel

💡 **Show answer**

Ans: B — **John W. Bennett** (*The Ecological Transition*). Not Ratzel.

Q21. UPPCS Prelims 2019, Q46

Nitrogen fixing bacteria make combination with cells of the roots of:

A. Pulses

B. Rice

C. Wheat

D. Sugarcane

💡 **Show answer**

Ans: A

Q22. UPPCS Prelims 2019, Q57

The environment modified by human activities is called:

A. Natural

B. Anthropogenic

C. Urban

D. Modern

💡 **Show answer**

Ans: B

Q23. UPPCS Prelims 2019, Q71

Assertion (A): SD is important for well-being. Reason (R): SD meets present needs without compromising future generations.

A. Both true; R explains A

B. Both true; R not explanation

C. A true, R false

D. A false, R true

💡 **Show answer**

Ans: A

Q24. UPPCS Prelims 2019, Q98

Provisions for protection/improvement of environment & safeguarding forests/wildlife are found in:

A. DPSP only

B. FD only

C. Both

D. Neither

💡 **Show answer**

Ans: C

Q25. UPPCS Prelims 2019, Q135

Associated with biological nitrogen fixation:

A. Red algae

B. Brown algae

C. Green algae

D. Blue-green algae

💡 **Show answer**

Ans: D

Q26. UPPCS Prelims 2019, Q146

Who propounded *Limits to Growth*?

A. Club of Rome

B. UNESCO

C. Brundtland Commission

D. Agenda 21

💡 **Show answer**

Ans: A

Q27. UPPCS Prelims 2018, Q46

Which is an example of anthropogenic biome?

A. Fresh water

B. Grassland

C. Rain forest

D. Cropland

💡 **Show answer**

Ans: D

Q28. UPPCS Prelims 2018, Q51

Which is concerned with soil?

A. Edaphic

B. Climatic

C. Biotic

D. Topography

💡 **Show answer**

Ans: A

Q29. UPPCS Prelims 2018, Q113

'Saving energy and resources for the future without sacrificing present comfort' defines:

A. Economic growth

B. Economic development

C. Sustainable development

D. Human development

💡 **Show answer**

Ans: C

Q30. UPPCS Prelims 2025, Q122

The Appiko Movement is associated with which of the following Indian States?

1. Uttarakhand 2. Uttar Pradesh 3. Kerala 4. Karnataka

A. 1 and 2

B. Only 4

C. 3 and 4

D. Only 1

💡 **Show answer**

Ans: B — Appiko = **Karnataka** (1983). Chipko = Uttarakhand.

Q31. UPPCS Prelims 2023, Q115

Desert-dwelling animals are called—

A. Arboreal

B. Terrestrial

C. Fossorial

D. Xericole

💡 **Show answer**

Ans: D — Xericole = desert-adapted animals.

Q32. UPPCS Prelims 2022, Q34

Assertion (A): Phytoplankton grow in abundance in the limnetic zone. Reason (R): The limnetic zone is the open-water zone.

A. A false, R true

B. A true, R false

C. Both true; R not explanation

D. Both true; R explains A

💡 **Show answer**

Ans: D — Limnetic = open water in lakes; phytoplankton abundant there.

Q33. UPPCS Prelims 2018, Q50

When is World Biodiversity Day observed?

A. Mar 22

B. May 22

C. Jun 23

D. Apr 16

💡 **Show answer**

Ans: B — 22 May. Trap: Mar 22 = World Water Day.

Q34. UPSC Prelims 2021 — pattern

'Tragedy of the Commons' is associated with:

A. Malthus

B. Garrett Hardin

C. Rachel Carson

D. Haeckel

💡 **Show answer**

Ans: B

Q35. UPSC Prelims 2020 — pattern

Terrestrial ecosystems among Desert, Mangrove, Rainforest:

A. Desert and Rainforest

B. Mangrove only

C. All three

D. None

💡 **Show answer**

Ans: A — Mangrove = transitional.

Q36. UPSC Prelims 2013 — pattern

Permafrost is found in:

A. Taiga and Tundra

B. Savannah

C. Desert

D. Mangrove

💡 **Show answer**

Ans: A

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D

Q1. With reference to the meaning of 'environment', which of the following statements is/are correct?

1. NCERT treats environment as the sum of biotic and abiotic factors influencing living organisms.
2. In most Prelims stems, "environment" primarily means social and economic surroundings rather than natural/physical factors.

Select the correct answer from the code given below:

A. Only 2 B. Both 1 and 2 C. Neither 1 nor 2 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Statement 1 is the NCERT fact. Statement 2 reverses the Prelims default (natural/physical sense).

Q2. Consider the following pairs:

Term	Meaning
1. Ecology	Functional unit of living + non-living interaction
2. Ecosystem	Study of organism–environment links
3. Habitat	Place where an organism lives
4. Niche	Role of a species in an ecosystem

Which of the pairs given above is/are **correctly** matched?

A. 1 and 2 only B. 3 and 4 only C. 1, 3 and 4 only D. 2, 3 and 4 only

💡 **Show answer**

Ans: B. Only habitat and niche are correctly matched. **Logic:** Ecology = study; ecosystem = functional unit. Pairs 1–2 are swapped; 3–4 are correct.

Q3. With reference to Earth's environmental spheres, which of the following statements is/are correct?

1. Lithosphere covers roughly 71% of Earth's surface and hydrosphere about 29%.
2. Biosphere extends from about 200 m below the soil surface to about 6 km above sea level.
3. Pedosphere is counted as a fifth sphere equal in status to lithosphere in standard four-sphere teaching.

Select the correct answer from the code given below:

A. Only 3 B. Only 2 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Statement 1 reverses lithosphere/hydrosphere shares (~29% land / ~71% water). Statement 2 is the usual biosphere vertical range. Statement 3 over-ranks pedosphere versus the standard four-sphere teaching.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Liebig's law states that organism growth is limited by the scarcest essential resource.

Reason (R): Shelford's law states that a species can survive only within a minimum–maximum tolerance range for environmental factors.

Select the correct answer from the code given below:

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; R does not explain A. **A/R logic:** Both laws are correctly stated, but tolerance range does not explain why the scarcest resource limits growth.

Q5. Match List-I with List-II and select the correct answer using the code given below the lists:

List-I (Factor type)	List-II (Concerned with)
A. Edaphic	1. Slope and altitude
B. Climatic	2. Soil
C. Topographic	3. Weather elements

Code:

A. A-3, B-2, C-1 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-1, B-3, C-2

💡 **Show answer**

Ans: C. A-2, B-3, C-1. **Logic:** Edaphic = soil; climatic = weather; topographic = slope/altitude.

Q6. Which of the following pairs is/are **NOT** correctly matched?

1. lotic ecosystem — flowing water such as rivers
2. Lotic ecosystem — standing water such as ponds and lakes
3. Mangrove / wetland — transitional habitat between land and water

Select the correct answer from the code given below:

A. Only 1 B. 1 and 2 C. Only 3 D. Only 2

💡 **Show answer**

Ans: D. Only pair 2 is not correctly matched. **Logic:** Lotic = flowing water; standing water is lentic — pair 2 swaps them. Pairs 1 and 3 are correctly matched.

Q7. With reference to biotic structure of an ecosystem, which of the following statements is/are correct?

1. The usual sequence is producers → consumers → decomposers.
2. Rhizobium has no role in nitrogen fixation in leguminous plants.
3. Viruses are the principal decomposers in terrestrial ecosystems.

Select the correct answer from the code given below:

A. Only 1 B. 1 and 2 C. Only 3 D. 2 and 3

💡 **Show answer**

Ans: A. Only 1 is correct. **Logic:** Statement 1 is the standard biotic sequence. Statement 2 is false — Rhizobium fixes N in legumes. Statement 3 is false — fungi/bacteria are principal terrestrial decomposers, not viruses.

Q8. Consider the following statements about human–environment schools of thought:

1. Environmental determinism is linked with Friedrich Ratzel.
2. Possibilism is linked with Vidal de la Blache.
3. Neo-determinism / stop-and-go determinism is linked with Griffith Taylor.

Which of the statements given above are correct?

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. 1, 2 and 3. **Logic:** All three associations are standard.

Q9. With reference to the IPAT identity and related ideas, which of the following statements is/are correct?

1. In IPAT, Impact = Population × Affluence × Technology.
2. Garrett Hardin's "Tragedy of the Commons" appeared in 1968.
3. If population rises while affluence and technology stay constant, impact necessarily falls.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** 1 and 2 are correct. Higher P with fixed A and T raises Impact — statement 3 is false.

Q10. Arrange the following in correct chronological order:

1. Stockholm Conference on the Human Environment
2. Brundtland Report (*Our Common Future*)
3. Rio Earth Summit / UNCED
4. Launch of India's LiFE Mission

Select the correct answer from the code given below:

A. 1-3-2-4 B. 1-2-3-4 C. 2-1-3-4 D. 1-2-4-3

💡 **Show answer**

Ans: B. 1-2-3-4. **Logic:** Stockholm 1972 → Brundtland 1987 → Rio 1992 → LiFE 2022.

Q11. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Sustainable development was popularised by the Brundtland Report of 1987.

Reason (R): Agenda 21 was adopted at the Rio Summit in 1995 as a binding climate treaty.

Select the correct answer from the code given below:

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** A is true. Agenda 21 is Rio **1992**, non-binding SD action plan — not a 1995 binding climate treaty. R is false.

Q12. With reference to India's LiFE Mission, which of the following statements is/are correct?

1. The idea was floated at COP26 (Glasgow) in 2021.
2. The Mission was launched in 2022.
3. The idea was first announced at COP25 (Madrid).

Select the correct answer from the code given below:

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** COP26 2021 + launch 2022 are correct. COP25 is the classic trap.

Q13. Match List-I with List-II and select the correct answer using the code given below the lists:

List-I (Event / law)	List-II (Year)
A. Chipko movement (Reni)	1. 1986
B. Environment (Protection) Act	2. 1973
C. Joint Forest Management guidelines	3. 2006
D. Forest Rights Act	4. 1990

Code:

A. A-2, B-4, C-1, D-3 B. A-4, B-1, C-2, D-3 C. A-2, B-1, C-3, D-4 D. A-2, B-1, C-4, D-3

💡 **Show answer**

Ans: D. A-2, B-1, C-4, D-3. **Logic:** Chipko 1973; EPA 1986; JFM 1990; FRA 2006.

Q14. Which of the following pairs is/are correctly matched?

1. Article 48A — Directive Principle directing the State to protect the environment
2. Article 51A(g) — Fundamental Duty of citizens to protect nature
3. Article 48 — Constitutional provision on environment protection

Select the correct answer from the code given below:

A. 2 and 3 only B. 1 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** 48A and 51A(g) are environment provisions. Article 48 is agriculture and cattle — not environment.

Q15. With reference to constitutional developments on environment, which of the following statements is/are correct?

1. The 42nd Amendment (1976) inserted Article 48A and Article 51A(g).
2. Article 21 has been interpreted to include the right to a healthy environment.
3. The Environment (Protection) Act, 1986 was enacted before the Bhopal gas tragedy.

Select the correct answer from the code given below:

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** 1 and 2 are correct. EPA 1986 is post-Bhopal (1984).

Q16. Consider the following statements:

1. MoEF was created in 1985 and renamed MoEFCC in 2014.
2. UNEP headquarters is in Nairobi.
3. CPCB headquarters is in Mumbai.

Which of the statements given above is/are correct?

A. 1 and 3 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** MoEF→MoEFCC and UNEP Nairobi are correct. CPCB sits in New Delhi, not Mumbai.

Q17. Which of the following pairs is/are **NOT** correctly matched?

Day	Date
1. World Environment Day	5 June
2. Earth Day	22 April
3. World Wildlife Day	22 March
4. International Day for Biological Diversity	22 May

Select the correct answer from the code given below:

A. 1 and 3 only B. 3 only C. 3 and 4 only D. 2 and 3 only

💡 **Show answer**

Ans: B. Only pair 3 is not correctly matched. **Logic:** Wildlife Day is **3 March** (CITES), not 22 March (Water Day trap). Others are correct.

Q18. Arrange the following observances in the order of their calendar dates within a year (earliest to latest):

1. World Wetlands Day
2. World Wildlife Day
3. World Environment Day

4. International Ozone Day

Select the correct answer from the code given below:

A. 2-1-3-4 B. 1-2-4-3 C. 1-2-3-4 D. 1-3-2-4

💡 **Show answer**

Ans: C. 1-2-3-4. **Logic:** 2 Feb → 3 Mar → 5 Jun → 16 Sep.

Q19. With reference to environment classification, which of the following statements is/are correct?

1. A common triad classifies environment as physical, biological, and cultural.
2. "Operational environment" is a standard fourth member of that triad in Prelims teaching.
3. Cropland is an example of an anthropogenic / human-modified environment.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: D. 1 and 3 only. **Logic:** Triad is physical–biological–cultural; operational is not part of it. Cropland is anthropogenic.

Q20. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Uttar Pradesh mainly falls in the Gangetic Plain biogeographic zone (Zone 7).

Reason (R): Uttar Pradesh has several marine national parks along its coastline.

Select the correct answer from the code given below:

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** A is true. UP is landlocked — no marine national parks — R is false.

Q21. Match List-I with List-II and select the correct answer using the code given below the lists:

List-I (River)	List-II (City association often tested)
A. Ganga	1. Agra
B. Yamuna	2. Lucknow
C. Gomti	3. Varanasi / Kanpur

Code:

A. A-1, B-3, C-2 B. A-3, B-2, C-1 C. A-3, B-1, C-2 D. A-2, B-1, C-3

💡 **Show answer**

Ans: C. A-3, B-1, C-2. **Logic:** Ganga–Varanasi/Kanpur; Yamuna–Agra; Gomti–Lucknow.

Q22. With reference to Ganga cleaning programmes, which of the following statements is/are correct?

1. Ganga Action Plan began in 1985.
2. Namami Gange was launched in 2014.
3. Namami Gange was launched in the same year as the Stockholm Conference.

Select the correct answer from the code given below:

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** GAP 1985 and Namami Gange 2014 are correct. Stockholm is 1972 — not 2014.

Q23. How many of the following are abiotic components?

1. Soil minerals
2. Rainfall
3. Green algae
4. Wind
5. Decomposers

Select the correct answer from the code given below:

A. Only two B. Only three C. Only four D. All five

💡 **Show answer**

Ans: B. Only three. **Logic:** Abiotic = minerals, rainfall, wind (three). Algae and decomposers are biotic.

Q24. Which of the following pairs is/are correctly matched?

1. Internal environment — physiological conditions inside the organism
2. External environment — surroundings outside the organism (ecology default)
3. Micro-scale environment — regional or global systems

Select the correct answer from the code given below:

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** 1 and 2 are correct. Micro-scale is immediate surroundings; macro is regional/global.

Q25. Consider the following statements about Agenda 21:

1. Agenda 21 belongs to the Rio Summit of 1992.
2. Agenda 21 is often wrongly dated to 1995 in trap options.
3. Agenda 21 is identical with the Brundtland Report of 1987.

Which of the statements given above is/are correct?

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** 1 and 2 are teaching traps. Brundtland 1987 ≠ Agenda 21 1992.

Q26. With reference to atmospheric layers in environment basics, which of the following statements is/are correct?

1. Weather phenomena occur mainly in the troposphere.
2. The protective ozone layer that absorbs UV sits mainly in the stratosphere.
3. Maximum ozone concentration is found in the troposphere.

Select the correct answer from the code given below:

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Weather = troposphere; protective ozone = stratosphere. Statement 3 swaps the layers.

Q27. Which one of the following is the correct sequence of Chipko → JFM guidance → Forest Rights Act?

A. 1973 → 2006 → 1990 B. 1990 → 1973 → 2006 C. 1986 → 1990 → 2006 D. 1973 → 1990 → 2006

💡 **Show answer**

Ans: D. 1973 → 1990 → 2006. **Logic:** Reni Chipko 1973; JFM 1990; FRA 2006. EPA 1986 is a distractor year.

Q28. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Stockholm 1972 was the first UN Conference on the Human Environment and led to the creation of UNEP.

Reason (R): World Environment Day is observed on 22 April to mark the opening of the Stockholm Conference.

Select the correct answer from the code given below:

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** A is true. WED is **5 June**; 22 April is Earth Day. R is false.

Q29. Which of the following UP-linked pairs is/are correctly matched?

1. Kanpur — tannery pollution hotspot teaching
2. Varanasi — Ganga ghat pollution hotspot teaching
3. Surajkund crafts fair — principal UP environment festival fact

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Kanpur and Varanasi are standard UP pollution facts. Surajkund is Haryana — not a UP environment festival fact.

Q30. With reference to Brundtland Commission, which of the following statements is/are correct?

1. It worked roughly during 1983–1987 as the World Commission on Environment and Development (WCED).
2. It defined sustainable development in *Our Common Future*.
3. *Limits to Growth* (Club of Rome, 1972) is another name for the Brundtland Report.

Which of the statements given above is/are correct?

A. 2 and 3 only B. 1 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** 1 and 2 are correct. *Limits to Growth* is a different 1972 Club of Rome study.

Mains Answer Framework

Human–Environment in India (250 words): Define reciprocal interaction → Chipko/JFM/FRA examples → Art 48A/51A(g)/Art 21 → challenges (Bhopal, urban pollution, dam (दाम -) conflicts) → SD + LiFE + Namami Gange → balance possibilism with ecological limits.

Constitutional environmental protection (200 words): 42nd Amendment (48A + 51A(g)) → Art 21 expansion via SC → Polluter Pays/Precautionary/Public Trust → local governance (11th/12th Schedule (12वीं)) → Concurrent List (समवर्ती सूची) forests/wildlife/pollution.

Common Traps — Don't Fall For These

Trap	Correct fact
Art **48** vs **48A**	Environment = **48A** (48 = agriculture/cows)
Humus = abiotic?	**No** — organic, biotic-origin
Mangrove = terrestrial?	**No** — coastal/transitional
MoEFCC est. 1972?	**No** — **1985** (1972 = Stockholm)
Lentic = flowing?	**No** — Lentic = **standing**
Wildlife Day = 22 March?	**No** — **3 March**
Agenda 21 = 1995?	**No** — **Rio 1992**
Brundtland = *Limits to Growth*?	**No** — Brundtland = *Our Common Future* (1987)

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 2 — Ecology (पारिस्थितिकी) & Ecosystem (पारिस्थितिकी तंत्र)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Ecology – Meaning | Ecosystem | Types of Ecosystems | Terrestrial Ecosystem | Aquatic Ecosystem | Structure and Functions of Ecosystem | Biotic and Abiotic Components (अजैविक घटक) | Ecological Niche (पारिस्थितिक निकेत) | Ecological Principles | Ecological Balance | Ecological Succession (उत्तराधिकार क्रम) | Ecosystem Services | Ecosystem Productivity | Ecological Efficiency | Genetic Level Ecology | Species Level Ecology | Community Level Ecology

Sources baked in: NCERT Class 11–12 Biology, Odum, MEA 2005, TEEB/IPBES, UPPCS Prelims PYQs 2018–2025
Weight: ★★★ Very High — core Prelims conceptual topic **Last verified:** August 2026

Current Affairs (this topic)

- Ecosystem-services and productivity stems stay conceptual — refresh MEA / IPBES wording if a CA stem cites (साइट्स) them.
- Keep Tansley / Haeckel / Odum name traps stable; they are not CA-dependent.

💡 🎯 **Consolidated — 39 Must-Score Facts (Click to Expand)**

1. **Ecology** is the **study** of organism–environment (पर्यावरण) interactions. **Haeckel (1866)** coined the term. **Ramdeo Misra** is called the father of Indian ecology. **Bionomics** (बायोनॉमिक्स) is synonymous with ecology; it does **not** mean "management of life." 2. An **ecosystem** (पारिस्थितिकी तंत्र) is a **functional unit** of biotic and abiotic interaction with energy flow and nutrient cycling. **Tansley (1935)** coined the word. 3. Ecosystems are normally **open systems**. Calling an ecosystem a closed system is wrong. 4. Four (चातुर्याम) basic components often listed together are **abiotic materials, producers, consumers, and decomposers**. 5. The seven ecological levels run **organism → population (जनसंख्या) → community → ecosystem → landscape → biome (जीवोम) → biosphere**. A frequent shorter order asked is **Population → Community → Ecosystem → Landscape**. 6. The **biosphere** is the largest ecosystem of Earth. Among ecosystem *types*, the **marine/ocean** ecosystem covers the largest area and is cited as the **most stable**. 7. Ecosystems may be natural or anthropogenic, terrestrial or aquatic, and **lentic** (standing) or **lotic** (flowing). 8. Artificial / anthropogenic examples include **farmland / rice field**, orchard, garden, aquarium, and human-made reservoir. 9. A pond's vertical/horizontal zones include **littoral → limnetic → profundal**. Marine zones include littoral, neritic, pelagic, benthic, and abyssal. 10. Structure runs **producers → primary/secondary/tertiary consumers → decomposers and detritivores**. Producers generally outnumber primary consumers. 11. Core ecosystem functions are **productivity, decomposition, nutrient cycling, and unidirectional energy flow**. 12. A **niche** is a species' **role**, not its place. A **habitat** (वास स्थान) is the place. **Joseph Grinnell (1917)** coined the ecological-niche concept. 13. A **fundamental niche** (मूल (मौलिक) निकेत) is the full potential role without competitors. A **realized niche** (वास्तविक निकेत) is the narrower role after competition. 14. **Gause's competitive exclusion** says two species with identical niches cannot stably coexist when resources are limited. 15. **Liebig** = limiting factor; **Shelford** = tolerance range; **Lindeman (1942)** ≈ **10%** energy transfer between trophic levels. 16. **NPP = GPP − plant respiration**. Plants capture only about **1%** of incident sunlight on leaves. 17. Decreasing productivity order often tested: **Mangroves > grasslands > lakes > oceans**. 18. **Primary succession** (प्राथमिक

अनुक्रमण) starts on a lifeless substrate. **Secondary succession** (द्वितीयक अनुक्रमण) starts where **soil already exists** after disturbance. 19. Full succession phases: **Nudation → Migration → Ecesis → Reaction → Stabilisation**. The full community sequence is a **sere** (क्रमक). UPPCS 2024 often starts from Migration when Nudation is omitted. 20. Pioneers such as lichen and moss lead through seral stages to a **climax community** (चरम समुदाय). **Monoclimax** theory is linked to **F.E. Clements**. **Hydrosere** is wet; **xerosere** is dry. 21. In grasslands, trees often fail to replace grasses because of **water limits and fire**. 22. An **r-strategist** produces many offspring with fast growth and short life. A **K-strategist** produces few offspring with slow growth and long life. 23. Population growth models contrast the **J-curve** (exponential) with the **S-curve** (logistic) that levels near **carrying capacity**. 24. MEA (**2005**) ecosystem services are usually **provisioning, regulating, supporting, and cultural**. Some WDR/UPSC (संघ लोक सेवा आयोग) framings add **preserving**. Supporting examples include nutrient cycling and (in that framing) crop pollination. 25. Ecosystem balance is a **dynamic equilibrium**. **Resilience** is the capacity to recover after disturbance. **Deforestation** is a main cause of ecological imbalance in India. 26. **Eutrophication** (सुपोषण) is nutrient overload that drives algal bloom and a dissolved-oxygen crash. 27. Energy flow is **one-way**. Nutrients **cycle**. Saying energy cycles is a classic wrong option. 28. A **community** is all populations of different species in an area. A **population** is one interbreeding species in one area. 29. An **ecotone** (इकोटोन) is the transition zone between two communities or ecosystems. 30. Aquatic clines: **pycnocline** = density gradient; **halocline** = salinity gradient; **thermocline** = temperature-with-depth gradient. 31. In the **phosphorus cycle**, weathering of rocks is the main source releasing the nutrient into the cycle. 32. **Upwelling** brings cold, nutrient-rich water to the surface and raises marine productivity. 33. **Eucalyptus** is often treated as an environmental hazard because it draws heavy groundwater and can lower the water table. 34. UP lotic examples include the **Ganga (गंगा), Yamuna (यमुना), and Gomti**. Terai (तराई) forest and Vindhyan scrub are terrestrial examples. 35. UP wetland examples include **Sur Sarovar (Agra)** and **Nawabganj (Unnao)**. Western UP sugarcane belts are classic agro-ecosystems. 36. Odum is linked to modern ecosystem ecology; keep him separate from Haeckel (term ecology) and Tansley (term ecosystem). **Deep ecology** was coined by **Arne Næss (1973)**. **Ecological Transition** was used by **John W. Bennett**. 37. **Allen's Rule:** colder climates → smaller appendages (less heat loss). **Bergmann's Rule:** colder climates → larger body size (heat conservation). Do not swap the pair. 38. **Gloger's Rule:** darker pigmentation is more common in warm, humid regions. 39. **Limiting similarity** means coexisting species differ enough in niche use that they do not compete for exactly the same resource at the same time.

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Ecology (Haeckel 1866)	**Ecosystem (Tansley 1935)**	Scientific academic study of organism-environment interactions vs concrete structural-functional unit of nature combining biotic and abiotic components	पारिस्थितिकी / पारिस्थितिक तंत्र
Habitat	**Ecological Niche**	Physical place or environmental zone where a species resides vs unique functional role, trophic position, and resource utilization pattern (Grinnell)	वास स्थान / पारिस्थितिक निकेत
Population	**Ecological Community**	Group of interbreeding individuals of the *same* species in an area vs assemblage of populations of *different* interacting species in an area	जनसंख्या (समष्टि) / जैविक समुदाय
Gross Primary Productivity (GPP)	**Net Primary Productivity (NPP)**	Total rate of organic matter synthesized by producers via photosynthesis vs actual biomass available to consumers after plant respiration losses (NPP = GPP - R)	सकल प्राथमिक उत्पादकता (GPP) / शुद्ध प्राथमिक उत्पादकता (NPP)
Fundamental Niche	**Realized Niche**	Full theoretical potential range of environmental conditions and resources a species can use without competition vs actual restricted niche occupied due to biotic competition	मूल (मौलिक) निकेत / वास्तविक निकेत
Primary Succession	**Secondary Succession**	Ecological succession beginning on barren, lifeless substrate with no pre-existing soil (cooled lava, bare rock) vs succession on disturbed areas where soil already exists (after forest fire, flood)	प्राथमिक अनुक्रमण / द्वितीयक अनुक्रमण
r-Selected Species	**K-Selected Species**	High reproductive rate, small body, short lifespan, little parental care (insects, rodents, weeds) vs low reproductive rate, large body, long lifespan, high parental care (elephants, whales, humans)	r-रणनीतिकारी / K-रणनीतिकारी प्रजाति

Ecotone | **Ecotype** | Sharp transition boundary zone between two distinct ecological communities (mangroves, forest-grassland edge) vs genetically distinct geographic variety or sub-population within a species | इकोटोन (संक्रमण क्षेत्र) / इकोटाइप | | **Sere** | **Climax Community** | Entire chronological transitional sequence of developmental communities during ecological succession vs final, stable, self-perpetuating terminal community in equilibrium with regional climate | क्रमक (Sere) / चरम समुदाय | | **Bionomics** | **Ecology** | Historical 19th-century synonym of ecology meaning "laws of life" (UPPCS trap: bionomics is NOT management of life) vs modern ecology | बायोनॉमिक्स / पारिस्थितिकी |

💡 📋 **Must-score facts — terms, succession, productivity (Click to Expand)**

Founders / levels | Item | Lock | |------|------| | Ecology term / Indian ecology | Haeckel **1866** / Ramdeo Misra | | Ecosystem term | Tansley **1935**; normally **open** system | | Levels (short) | Population → Community → Ecosystem → Landscape | | Niche coined | Grinnell **1917** (role ≠ place) | | Lindeman | **10%** energy transfer (**1942**) | | NPP | GPP − plant respiration; plants ~**1%** of sunlight | ### Succession / services | Item | Lock | |------|------| | Primary vs secondary | Lifeless substrate / soil already present | | Sere phases | Nudation → Migration → Ecesis → Reaction → Stabilisation | | Monoclimax | F.E. Clements | | r vs K | Many offspring–short life / few–long life | | MEA 2005 services | Provisioning · regulating · supporting · cultural | | Eutrophication | Nutrient overload → bloom → DO crash | | Energy vs nutrients | Energy **one-way**; nutrients **cycle** | | Ecotone / Deep ecology | Transition zone / Arne Næss **1973** |

2.1 Ecology – Meaning

Identity: Study of interactions | Haeckel 1866 | 7 levels | Autoecology vs Synecology | Bionomics

- **Ecology** (from Greek *Oikos* = house + *Logos* = study) is the study of relationships between living organisms and their environment.
- **Ernst Haeckel** coined the term **Ecology** in **1866**.
- **Bionomics** (Greek *bio* = life; *nomos* = law) is a comprehensive study of an organism in relation to its environment. It is **synonymous with ecology**.
- Bionomics stresses the value of natural systems that influence human systems. It does **not** mean "management of life."
- NCERT definition: study of relationships of organisms with each other and with their environment.
- Ecology is a **biological science** — distinct from environment studies (policy/law/economics).
- **Ramdeo Misra** is regarded as the father of **ecology in India** (ecology teaching at BHU).
- **Eugene P. Odum** is called the father of modern **ecosystem ecology** (*Fundamentals of Ecology*, 1953).
- **Arne Næss** coined **deep ecology** in **1973**.
- **John W. Bennett** used the concept of **Ecological Transition** in *The Ecological Transition: Cultural Anthropology and Human Adaptation*.
- Ecology is studied at **seven levels**: Organism → Population → Community → Ecosystem → Landscape → Biome → Biosphere.
- A shorter organisation order often asked from organism upward is **Population → Community → Ecosystem → Landscape**.
- **Organ** is anatomy, **not** an ecological organisation level.
- **Autoecology** studies one species in its environment.
- **Synecology** studies communities and ecosystems together.

Level	example
Organism	Camel water conservation in desert
Population	Tiger population in Dudhwa (दुधवा)

Level	example
Community	Fish, birds, and plants sharing a wetland
Ecosystem	Ganga stretch at Varanasi (वाराणसी) with its biota, water, and nutrient flows
Biome	Tropical deciduous forest of India
Biosphere	Global life-support zone

> **Logic:** Trap — Odum did **not** coin (मुद्रा) "ecosystem". **Tansley (1935)** did.

2.2 Ecosystem

Identity: Tansley 1935 | Functional unit | Open system | Natural vs anthropogenic | Four components

- **A.G. Tansley** introduced the term **ecosystem** in **1935**.
- An ecosystem is the basic **functional unit** where living organisms interact with each other and the physical environment (भौतिक पर्यावरण).
- A standard short definition is a community of organisms together with the environment in which they live.
- Every ecosystem has **biotic components** (जैविक घटक), **abiotic components**, **energy flow**, and **nutrient cycling**.
- Four components frequently listed together are **abiotic materials**, **producers**, **consumers**, and **decomposers**.
- **Energy flow** is **unidirectional** (Sun → producers → consumers → heat loss).
- **Nutrient cycling** is **cyclic** — matter is reused through decomposers.
- Most ecosystems are **open systems** (exchange energy and matter with surroundings). Calling an ecosystem a **closed system** is wrong.
- The **biosphere** itself is an open system. New substances can enter, and energy can enter and leave.
- Components of an ecosystem **are interdependent**. A rise or fall in one population can reshape the rest of the food web (खाद्य जाल).
- **Natural ecosystems** self-regulate (forest, lake, ocean).
- **Anthropogenic / artificial ecosystems** depend on human inputs (cropland, rice field, orchard, garden, aquarium, sewage-treatment wetland, spaceship life-support systems).
- Among ecosystem types, the **marine (ocean) ecosystem** covers about **71%** of Earth's surface and holds about **97%** of planetary water. It is the largest and most stable ecosystem type in common option lists.
- The **biosphere** is the largest ecosystem of Earth as a whole because it integrates atmosphere, hydrosphere, and lithosphere where life exists.
- Ecosystem size runs from **micro** (drop of pond water, garden pond) to **macro** (river basin, biosphere).

> **Logic:** Ecosystem = open, not closed. Components are interdependent. Field/rice field = artificial. Biosphere = largest overall; ocean = largest/most stable type.

2.3 Types of Ecosystems

Identity: Natural/Anthropogenic | Terrestrial/Aquatic | Lentic/Lotic

Basis	Categories
Origin	Natural vs **Anthropogenic** (human-modified)
Medium	**Terrestrial** (~29% Earth's surface) vs **Aquatic** (~71%)

Basis	Categories
Water flow (freshwater)	**Lentic** (standing) vs **Lotic** (flowing)
Salinity	Freshwater vs Brackish vs Marine

Terrestrial ecosystems

- A **terrestrial ecosystem** exists on land. Plants and animals that live on land are called **terrestrial**.
- A **forest ecosystem** is dominated by trees. The Terai sal forests and Western Ghats (पश्चिमी घाट) evergreen tracts are common Indian examples.
- A **grassland ecosystem** is dominated by grasses. Dudhwa Terai grasslands are a UP-relevant example.
- A **desert ecosystem** is water-scarce and extreme in temperature. The Thar (थार) is India's classic desert tract.
- Animals adapted to desert dryness are called **xericole**.
- A **savanna** (सवाना) mixes grassland with scattered trees. Gir (गीर) in Gujarat is India's best-known savanna-style lion (सिंह) habitat.
- **Mountain ecosystems** show vegetation change with altitude along the Himalaya (हिमालय) belts.
- **Tundra** (टुंड्रा) and **taiga** are cold-climate terrestrial types. They are not typical of the Indian mainland in Prelims options.

Aquatic ecosystems — lentic and lotic

- An **aquatic ecosystem** is water-based. **Lentic** water stands still in lakes and ponds. **Lotic** water flows in rivers and streams.
- A **lentic ecosystem** (स्थिर जल) is standing freshwater such as a lake, pond, or reservoir. Loktak and Dal Lake are lentic examples.
- A **lotic ecosystem** (प्रवाही जल) is flowing freshwater such as a river or stream. The Ganga and Yamuna are lotic.
- **Wetlands**, **estuaries**, and **mangroves** are **transitional**. They are neither purely terrestrial nor purely open aquatic.

Aquatic Ecosystem Zonation & Ecotone Edge Effect

Pond and lake zonation (lentic)

Zone	Location	Key fact
Littoral	Shallow shore with rooted plants	Warmest, highest light
Limnetic	Open water away from shore	Phytoplankton zone
Profundal	Deep water below light penetration	Low light, decomposers dominate

- The **littoral zone** is the shallow margin where rooted plants receive strong sunlight.
- The **limnetic zone** is the open-water body of a lake or pond away from the shore.
- **Phytoplankton** grow abundantly in the limnetic zone because open water receives enough light for photosynthesis.
- The **profundal zone** lies below the lighted layer. Decomposers are important here because dead organic matter sinks down.

Marine zones

- The **littoral (intertidal) zone** is the shore area alternately submerged and exposed by tides.
- The **neritic zone** covers the shallow continental shelf down to about 200 m. Fisheries and coral reefs are concentrated here.
- The **pelagic zone** is the open ocean water column away from the bottom.
- The **benthic zone** is the sea floor and its sediments.

- The **abyssal zone** is the deep ocean beyond the continental slope.
- A human-made **cropland** is an **anthropogenic biome**. It depends on irrigation, fertiliser, and harvest removal unlike a self-regulating natural forest.

> **Logic:** River = **lotic**. Lake = **lentic**. Cropland = **anthropogenic**, not natural grassland.

2.4 Structure, Functions & Components

Identity: Biotic structure | Four ecosystem functions | Nutrient cycling

Biotic components

- **Producers (autotrophs)** convert solar energy into food through photosynthesis or chemosynthesis. Green plants, algae, and cyanobacteria are the main producers.
- **Consumers (heterotrophs)** feed on other organisms. They include herbivores, carnivores, and omnivores arranged in trophic levels.
- Consumers do **not** make food by photosynthesis. That statement is false.
- **Decomposers** break down dead organic matter and return minerals to the environment. Bacteria and fungi are the classic decomposers. Viruses are **not** treated as decomposer organisms in standard food-chain questions.
- Decomposers convert **organic matter into inorganic** nutrients usable by producers. The reverse statement is false.
- **Producers and decomposers** act as intermediaries between biotic and abiotic components.
- **Detritivores** feed directly on detritus. Earthworms are a common example.
- In a typical stable ecosystem, **producers outnumber primary consumers**.
- Air is an **abiotic** component. Plants, animals, and bacteria are biotic.

Abiotic components

- **Abiotic components** are the non-living physical and chemical factors of an ecosystem.
- **Climatic factors** include temperature, rainfall, sunlight, wind, and humidity.
- **Edaphic factors** relate to soil texture, minerals, pH, and moisture.
- **Topographic factors** include altitude, slope, and aspect of the land.
- **Aquatic abiotic factors** include salinity, dissolved oxygen, and water current.
- Abiotic factors decide which species can survive in a habitat. The scarcest limiting factor (Liebig) and tolerance range (Shelford) govern which organisms persist.

Four functions of an ecosystem

Function	What it means
Productivity	Rate at which biomass or energy is produced
Decomposition	Breakdown of dead organic matter
Nutrient cycling	Movement of chemical elements through biotic and abiotic pools
Energy flow	One-way transfer of energy through trophic levels

- The movement of nutrient elements through living and non-living parts of an ecosystem is called a **biogeochemical cycle**.
- In the **phosphorus cycle**, weathering of rocks is the main source that releases the nutrient into the cycle. Carbon and nitrogen cycles are not rock-weathering dominated in the same way.

- **Food chains**, **food webs**, and **ecological pyramids** are covered in the next syllabus topic. Here the fact is that energy flow is unidirectional through trophic levels.

Ecotone and aquatic clines

- An **ecotone** is the transition zone or boundary between two distinct communities or ecosystems.
- **Pycnocline** shows a **density** gradient in a water column.
- **Halocline** shows a **salinity** gradient.
- **Thermocline** shows temperature change with depth.
- **Marine upwelling** replaces surface water with colder, nutrient-rich deep water and raises biological productivity and fisheries.

> **Logic:** A **sere** is the community sequence in succession. It is **not** the name for nutrient cycling. Ecotone ≠ ecosphere.

2.5 Ecological Niche

Identity: Role not place | Grinnell 1917 | Fundamental vs realized | Competitive exclusion

- A **habitat** is the place where a species lives. It answers the "where" question.
- A **niche** is the functional role of a species in an ecosystem. It answers the "what job" question — resources used, conditions tolerated, and interaction with other species.
- **Joseph Grinnell (1917)** coined the ecological-niche idea and treated it as a micro-habitat / distributional unit of species.
- Each species has a **distinct niche**. The niche covers the range of conditions a species can tolerate, the resources it uses, and its functional role.
- The **fundamental niche** is the full role a species could play if no competition or predation limited it.
- The **realized niche** is the narrower role a species actually occupies after competitors, predators, or resource limits restrict it.
- **Gause's competitive exclusion principle** states that two species competing for exactly the same niche cannot coexist forever. One species is excluded, or the niche gets divided between them.
- When resources are abundant and spread fairly, **more species** can coexist in a community. UPPCS 2024 tests the false claim that species number becomes **small** when resources are sufficient.
- **Dominant species** often occupy a **broader niche** because they control more resources in the habitat.

> **Logic:** Habitat = place. Niche = role (Grinnell). More resources → more species, not fewer.

PYQ — UPPCS Prelims 2024, Q9

Which statement is **not true** about "ecological niche"?

Options: A. Dominant species occupy extensive and broader ecological niche. B. Number of species becomes large in ecological niche of natural ecosystem, if resources are sufficient. C. Number of species becomes small in ecological niche, if resources are sufficient. D. Dominance of single species becomes minimum if the resource distribution is equitable in a habitat.

💡 **Show answer**

Ans: C — More resources support **more** species, not fewer.

2.6 Ecological Principles

Identity: Allen/Bergmann/Gloger | Limiting similarity | Liebig + Shelford

- **Liebig's Law of Minimum** states that growth is limited by the scarcest essential resource. In deserts, water is often that limiting factor even when sunlight is abundant.
- **Shelford's Law of Tolerance** (शेलफोर्ड का सहनशीलता नियम) states that each species survives only within a minimum–maximum range for each factor. Beyond that range, stress or death follows even if other factors are plentiful.
- **Limiting similarity** means coexisting species differ enough in niche use that they do not compete for exactly the same resource at the same time.
- **Allen's Rule** states that animals in colder climates tend to have smaller appendages such as ears and limbs. Smaller surface area reduces heat loss.
- **Bergmann's Rule** states that animals in colder climates tend to have larger body size. Bulk helps conserve body heat.
- **Gloger's Rule** states that darker pigmentation is more common in warm and humid regions.

> **Logic:** Allen = smaller appendages in cold. Bergmann = larger body in cold. Do not swap the pair.

2.7 Ecological Balance

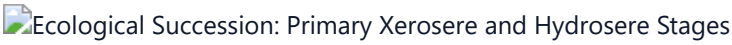
Identity: Dynamic equilibrium | Resilience | Deforestation | Eutrophication

- Ecological balance is **dynamic equilibrium**. An ecosystem looks stable but species populations constantly adjust through births, deaths, and interactions.
- **Feedback loops** such as predator–prey cycles and nutrient recycling help ecosystems absorb small disturbances without collapsing.
- **Resilience** is the ability of an ecosystem to recover after a disturbance such as fire, flood, or pollution spike.
- A **keystone species** (कीस्टोन प्रजाति) has an influence on community structure far greater than its numbers suggest. Removing it can reshape the whole ecosystem.
- Rapid human population growth **disrupts** environmental balance.
- **Deforestation** is one of the main causes of ecological imbalance in India. Desertification (मरुस्थलीकरण), floods, famines, and rainfall variation are related stresses, but forest loss is the primary driver in many stems.
- Acts that disturb balance include **lumbering** / commercial felling. Afforestation, rainwater management, biosphere reserves (जैवमंडल आरक्षित क्षेत्र), and wildlife protection (वन्यजीव सुरक्षा) help restore balance.
- Industrial management alone is **not** treated as an ecological-balance tool in standard option sets that contrast it with water management, afforestation, and wildlife protection.
- **Eucalyptus** plantations are often called an environmental hazard because the tree draws heavy groundwater and can lower the water table.

Eutrophication — imbalance in aquatic ecosystems

- **Eutrophication** begins when excess nutrients (often nitrates and phosphates from fertiliser runoff) enter a water body.
- The nutrient surge triggers rapid **algal blooms** on the surface.
- When algae die, decomposers multiply and consume **dissolved oxygen** faster than it can be replenished.
- Fish and other aerobic organisms die from oxygen starvation. marks **both** nutrient enrichment and oxygen depletion as correct.

> **Logic:** Eutrophication statement 1 (nutrient enrichment) and statement 2 (DO depletion) are **both true**.


Ecological Succession: Primary Xerosere and Hydrosere Stages

2.8 Ecological Succession

Identity: Primary vs secondary | Nudation → M-E-R-S | Sere | Pioneer → climax | Monoclimax

- **Ecological succession** is the orderly, predictable replacement of one community by another over time in the same area.
- The full set of communities that appear one after another is called a **sere**. Each intermediate community is a **seral stage**.
- The final relatively stable community is the **climax community**.
- **F.E. Clements** described succession toward climax and is linked to the **monoclimax** theory — one climatic climax for a region.

Primary vs secondary succession

- **Primary succession** starts on bare lifeless substrate where no soil yet exists. Examples include cooled lava, bare rock, and new sand dunes.
- **Secondary succession** starts where soil and some organisms already remain after a disturbance. Examples include abandoned farmland, forest after fire, and flooded land once water recedes.
- **Pioneer species** are the first colonisers. Lichens on rock and moss on thin soil are classic terrestrial pioneers. Phytoplankton may pioneer a new pond.

Lichens as pioneer and pollution indicator

- **Lichens** often pioneer bare rock in primary succession because they tolerate harsh exposed surfaces.
- Lichens are also sensitive **bio-indicators of air pollution**. They absorb pollutants from the air and fail to grow in dirty air.
- Trap: “lichens grow well in polluted areas” is **false**.

Phases of biotic succession (full sequence)

Phase	What happens
Nudation	Creation of a bare site free of organisms (erosion, fire, glaciation, human clearance)
Migration	Seeds, spores, or other propagules reach the bare or disturbed area
Ecesis	The newcomers successfully establish, germinate, and grow (colonisation)
Reaction	Established species modify the habitat (soil, shade, moisture) for the next community
Stabilisation	Change slows and the climax community dominates

- Full Odum/Clements-style order: **Nudation → Migration → Ecesis → Reaction → Stabilisation**.
- UPPCS 2024 starts from Migration when Nudation is omitted: **Migration → Ecesis → Reaction → Stabilisation**.
- Trap: putting **Ecesis before Migration** is always wrong.
- In many grasslands, trees do not replace grasses in succession because of **water limits and fire**.

Hydrosere and xerosere

- A **hydrosere** is succession in an aquatic habitat that gradually becomes terrestrial. A pond may pass through floating plants, reeds, marsh, meadow, and finally forest if sediment fills it in.

- A **xerosere** is succession on dry land starting from bare rock or sand. The usual sequence runs lichen → moss → grass → shrub → forest as soil builds up.

> **Logic: Sere** = the community **sequence**. Full five phases begin with **Nudation**.

PYQ — UPPCS Prelims 2024, Q26

Correct sequence of phases of Biotic succession?

Options: A. Ecesis – Migration – Reaction – Stabilisation B. Ecesis – Migration – Stabilisation – Reaction C. Migration – Reaction – Stabilisation – Ecesis D. Migration – Ecesis – Reaction – Stabilisation

💡 **Show answer**

Ans: D

2.9 Ecosystem Services

Identity: MEA 2005 | Four (sometimes five) categories | Supporting traps

Source: Millennium Ecosystem Assessment (MEA), **2005**

Category	What humans get	Example
Provisioning	Direct material products	Food, water, timber
Regulating	Processes that stabilise the environment	Climate/disease control
Supporting	Services that underpin all other services	Nutrient cycling, soil formation
Cultural	Non-material benefits	Recreation, spiritual value
Preserving (WDR/UPSC five-fold framing)	Long-term diversity maintenance	Maintenance of diversity

- **Provisioning services** supply goods humans take directly from nature. Timber from a forest is a provisioning service. Production of food and water sits here.
- **Regulating services** control natural processes that affect human welfare. Climate and disease control sit here. Bee pollination of crops is often treated as regulating, though some stems place crop pollination with supporting.
- **Supporting services** maintain the basic conditions for life. Nutrient cycling and soil formation are supporting services. In the UPSC MEA framing that lists five categories, **nutrient cycling and crop pollination** are the supporting pair tested.
- **Cultural services** are non-material benefits such as recreation at a national park or spiritual value of a sacred grove.
- **Preserving services**, when listed separately, cover maintenance of diversity over long time scales.
- **Supporting services** underlie provisioning and regulating services. Without soil formation and primary production, food and timber cannot exist.
- **TEEB** and **IPBES** later pushed economic valuation of ecosystem services for policy.

> **Logic:** Timber / food-water = **Provisioning**. Climate-disease control = **Regulating**. Nutrient cycling = **Supporting**. Maintenance of diversity = **Preserving** in five-fold lists.

2.10 Ecosystem Productivity

Identity: GPP | NPP | 1% sunlight rule | Standing crop

Term	Meaning
GPP (Gross Primary Productivity)	Total energy or biomass fixed by producers through photosynthesis
R	Energy lost by producers through respiration
NPP (Net Primary Productivity)	**GPP − R** = biomass actually available to consumers
NSP	NPP remaining after heterotroph consumption at producer level
Standing crop	Biomass present in an ecosystem at a given instant

- **Productivity** measures how fast an ecosystem produces organic matter. Standard units are **g/m²/yr** or **kcal/m²/yr**.
- Terrestrial green plants capture only about **1%** of the solar energy that falls on their leaves (NCERT standard; statement 1).
- Most incoming sunlight is reflected, transmitted through leaves, or used in ways that do not become stored biomass.
- **GPP** counts all energy fixed by photosynthesis. Plants immediately use part of it in **respiration (R)**.
- **NPP** is what remains for herbivores and other consumers after plant respiration is subtracted.
- **Tropical rainforests** and **coral reefs** rank among the highest-NPP ecosystems because they combine strong sunlight with ample moisture.
- **Deserts** show low NPP despite bright sunlight because water availability is the limiting factor.
- A decreasing productivity sequence often tested is **Mangroves → grasslands → lakes → oceans**.
- Most of the world's coral reefs lie in **tropical waters**. More than one-third are in the territories of **Australia, Indonesia, and the Philippines**.
- Coral reefs host far more animal **phyla** than tropical rainforests, despite covering a tiny share of ocean floor.

> **Logic:** The **1% rule** applies to energy **captured by producers**. The **10% rule** applies to energy **transferred between trophic levels**. Both can be true in the same question. Productivity order ≠ stability order.

2.11 Ecological Efficiency

Identity: Lindeman 10% rule | Trophic levels | Energy loss as heat

- **Raymond Lindeman (1942)** showed that only about **10%** of energy or organic matter passes from one trophic level to the next higher level.
- About **90%** is lost as heat through respiration, movement, excretion, and energy in uneaten parts of organisms.
- Because of this loss, energy **decreases** at each higher trophic level.
- **Trophic levels** are the feeding steps in a food chain (खाद्य श्रृंखला). They are formed by organisms linked in a food chain.
- **Ecological efficiency** compares energy at two trophic levels: (energy at higher level ÷ energy at lower level) × 100.
- A **pyramid (पिरामिड) of energy** (ऊर्जा पिरामिड) is always upright because energy cannot increase as you move up the chain.

PYQ — UPPCS Prelims 2025, Q5

With reference to 'Ecosystem', which statements is/are correct?

1. Green plants in terrestrial ecosystem capture about 1% of energy of the sunlight that falls on their leaves.
2. Around 10% of organic matter reaches to the next higher level of consumers.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Ans: C — Statement 1 = 1% capture by producers. Statement 2 = 10% transfer to next trophic level. Both true together.

> **Logic:** Trap — "energy increases up the food chain" is **false**. Trap — rejecting statement 1 because statement 2 exists is **false**.

2.12 Genetic, Species & Community Ecology

Identity: Three study levels below ecosystem | Population curves | Species interactions

Genetic level ecology

- **Genetic-level ecology** studies gene pools, allele frequencies, and genetic variation within populations.
- High **genetic diversity** (आनुवंशिक विविधता) helps a population survive disease outbreaks and environmental change.
- The **Hardy–Weinberg equilibrium** is an ideal model where allele frequencies stay constant when no evolutionary forces act.

Species level ecology (population ecology)

- A **population** is a group of individuals of the same species living in one area and interbreeding.
- **Carrying capacity (K)** is the maximum population size an environment can support indefinitely without degradation, given food, habitat, water, and other necessities.
- **Biotic potential (r)** is the maximum reproductive capacity of an organism under optimum conditions. Carrying capacity is the environmental limit beyond which that growth cannot be sustained.
- Exceeding carrying capacity stresses the ecosystem and can cause collapse.
- A **J-curve** shows exponential population growth when resources seem unlimited at first.
- An **S-curve** shows logistic growth slowing as the population approaches carrying capacity K.
- An **r-strategist** species produces many offspring quickly and invests little in each individual. Mosquitoes and weeds are examples.
- A **K-strategist** species produces few offspring and invests heavily in parental care. Elephants, whales, and tigers are examples.

Community level ecology

- A **community** includes all populations of different species living together in one area.
- **Species richness** counts how many species are present. **Evenness** measures how evenly individuals are spread among those species.
- Among common options, **coral reefs** show relatively **highest species diversity (प्रजाति विविधता)**.
- The **marine (ocean) ecosystem** is often cited as the **most stable** ecosystem type. Do not swap diversity with stability.
- **Habitat destruction** is the leading direct cause of biodiversity loss.

Interaction	What happens	Example
Competition	Two species vie for the same limited resource	Two grazers on one pasture

Interaction	What happens	Example
Predation	One organism kills and eats another	Tiger hunting deer
Parasitism	One benefits while the host is harmed	Tick on cattle
Mutualism	Both species benefit	**Azolla** with *Anabaena* in rice fields
Commensalism	One benefits and the other is unaffected	Cattle egret feeding near grazing cattle

> **Logic:** Coral reefs = high **diversity**. Open ocean = high **stability**. These are different 2018 questions.

2.13 Ecological Succession, Seral Stages & Biogeochemical Dynamics

- **Ecological Succession**: The universal and predictable process by which an ecological community undergoes orderly and gradual changes in species composition over time until a stable state is established.
- **Sere (Seral Community)**: The entire sequential progression of transitional communities that successively replace one another in a given habitat: \text{Pioneer Community} \longrightarrow \text{Transitional Seral Stages} \longrightarrow \text{Climax Community}
- **Pioneer Species**: The first hardy organisms to colonize a previously uninhabited or barren area (e.g., **Crustose Lichens** and mosses on bare rock; **Phytoplankton** in barren water).
- **Climax Community**: The final, stable, mature, and self-perpetuating terminal community in dynamic equilibrium with the regional climate (e.g., a mature forest).
- **Primary Succession vs Secondary Succession**:
  - *Primary Succession*: Initiates on newly exposed barren surfaces where no previous soil or life existed (bare volcanic rock, cooled lava, newly formed sand dune).
  - *Secondary Succession*: Occurs on previously vegetated lands where pre-existing communities were destroyed by disturbance (forest fires, abandoned agricultural fields, floods), having pre-existing soil; proceeds substantially faster than primary succession.
- **Hydrosere vs Xerosere**:
  - *Hydrosere*: Succession sequence starting in an open body of fresh water (Phytoplankton \to Submerged macrophyte \to Floating \to Reed-swamp \to Marsh-meadow \to Woodland \to Forest).
  - *Xerosere (Lithosere)*: Succession sequence starting on dry, bare rock (Crustose lichen \to Foliose lichen \to Moss \to Herbaceous \to Shrub \to Forest).
- **Biogeochemical Cycles (Nutrient Cycling)**:
- Circulation of chemical elements (carbon, nitrogen, phosphorus, sulphur, water) through biotic and abiotic components.
- *Gaseous Cycles*: Reservoir in atmosphere or hydrosphere (e.g., Nitrogen, Carbon, Oxygen).
- *Sedimentary Cycles*: Reservoir in Earth's crust / lithosphere rocks (e.g., Phosphorus, Sulphur).

2.14 Ecotones, Edge Effect, Bioindicators & Biofertilizers

- **Ecotone & Edge Dynamics**:
- **Ecotone**: The transitional boundary or junction zone between two distinct, adjoining ecosystems or biomes (e.g., **Mangroves** between marine and terrestrial ecosystems; **Estuaries** between river and sea; **Marshes** between aquatic and dry land).
- **Edge Effect**: The ecological phenomenon where the boundary ecotone zone exhibits a significantly higher species richness, population density, and diversity than either of the adjacent communities alone.

- **Edge Species**: Organisms primarily or most abundantly found in the ecotone boundary zone (e.g., certain warblers, edge-nesting birds).
- **Ecocline**: A physiochemical or environmental gradient (such as elevation, moisture, salinity, or temperature) along which species varieties and community structures gradually shift.
- **Ecological Bioindicators**:
- **Lichens**: Dual organisms consisting of symbiotic fungus and alga/cyanobacterium. Highly sensitive bioindicators of atmospheric **Sulphur Dioxide (SO_2)** pollution and heavy metals. They completely disappear ("lichen desert") in polluted urban and industrial atmospheres.
- **Mosses & Algae**: Sensitive to acid precipitation and heavy metals.
- **Aquatic Macrophytes & Biofertilizers in Agriculture**:
- **Azolla**: A free-floating aquatic freshwater fern that harbors the symbiotic cyanobacterium ***Anabaena azollae*** in its dorsal leaf cavities. Widely inoculated in flooded paddy (rice) fields as a potent green biofertilizer, fixing 30-50\text{ kg N/ha}.
- **Other Aquatic Macrophytes**:
  - *Lemna* (Duckweed): Fast-growing floating macrophyte used in wastewater phytoremediation.
  - *Wolffia*: Rootless, microscopic aquatic plant (smallest flowering plant on Earth).
  - *Trapa* (Water Chestnut / Singhara): Rooted floating aquatic macrophyte cultivated in freshwater ponds for edible nutritious kernels.
- **Pioneer Concept Milestones (Direct Exam Locks)**:
- **Raymond Lindeman (1942)**: Formulated the **10% Law of Trophic Efficiency** (only \approx 10\% of energy stored as biomass in one trophic level is transferred to the next higher level; 90\% is lost via respiration and metabolic heat).
- **John W. Bennett (1976)**: Introduced the concept of **'Ecological Transition'** (*The Ecological Transition: Cultural Anthropology and Human Adaptation*).

Complete PYQ Bank (Topic 2)

Q1. UPPCS Prelims 2025, Q5

With reference to 'Ecosystem', which of the following statements is/are correct?

1. Green plants in terrestrial ecosystem capture about 1% of energy of the sunlight that falls on their leaves.
2. Around 10% of organic matter reaches to the next higher level of consumers.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Ans: C — Both NCERT standard facts.

Q2. UPPCS Prelims 2025, Q61

With reference to lichens, which of the following statements is/are correct?

1. Lichens are very good indicators of pollution.
2. Lichens grow well in polluted areas.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Ans: D — Sensitive bio-indicators; fail in polluted air.

Q3. UPPCS Prelims 2024, Q9

Which of the following statements is not true about “ecological niche”?

Options: A. Dominant species occupy extensive and broader ecological niche. B. Number of species becomes large in ecological niche of natural ecosystem, if resources are sufficient. C. Number of species becomes small in ecological niche, if resources are sufficient. D. Dominance of single species becomes minimum if the resource distribution is equitable in a habitat.

💡 **Show answer**

Ans: C — More resources → more species.

Q4. UPPCS Prelims 2024, Q26

Which of the following is the correct sequence of phases of Biotic succession?

Options: A. Ecesis – Migration – Reaction – Stabilisation B. Ecesis – Migration – Stabilisation – Reaction C. Migration – Reaction – Stabilisation – Ecesis D. Migration – Ecesis – Reaction – Stabilisation

💡 **Show answer**

Ans: D

Q5. UPPCS Prelims 2024, Q62

Plants and animals that live on land are called:

A. Abiotic

B. Aquatic habitants

C. Terrestrial habitants

D. Biotic

💡 **Show answer**

Ans: C

Q6. UPPCS Prelims 2023, Q100

Most important factor for loss of biodiversity?

A. Genetic assimilation

B. Control of predators

C. Destruction of natural habitat

D. Pest control

💡 **Show answer**

Ans: C

Q7. UPPCS Prelims 2023, Q104

Aquatic plant used as biofertilizer in water-logged rice fields:

A. Lemna

B. Azolla

C. Wolffia

D. Trapa

💡 **Show answer**

Ans: B — Azolla–Anabaena mutualism.

Q8. UPPCS Prelims 2023, Q112

Which of the following is an example of a Lotic Ecosystem?

A. Swamps

B. Ponds

C. Rivers

D. Marshy areas

💡 **Show answer**

Ans: C

Q9. UPPCS Prelims 2023, Q115

Desert-dwelling animals are called:

A. Arboreal

B. Terrestrial

C. Fossorial

D. Xericole

💡 **Show answer**

Ans: D

Q10. UPPCS Prelims 2023, Q117

Trophic levels are formed by—

A. Organisms linked in a food chain

B. Plants only

C. Animals only

D. Carnivorous animals only

💡 **Show answer**

Ans: A

Q11. UPPCS Prelims 2023, Q125

Assertion (A): Population control is necessary to maintain the environment of the country.

Reason (R): Due to the rapid increase in population, environmental balance is maintained.

A. Both true; R explains A

B. Both true; R not explanation

C. A true, R false

D. A false, R true

💡 **Show answer**

Ans: C — Rapid growth disrupts balance.

Q12. UPPCS Prelims 2022, Q34

Assertion (A): Phytoplankton grow in abundance in the limnetic zone.

Reason (R): The limnetic zone is the open-water zone.

A. A false, R true

B. A true, R false

C. Both true; R not explanation

D. Both true; R explains A

💡 **Show answer**

Ans: D

Q13. UPPCS Prelims 2022, Q94

With reference to 'ecological niche', which statements is/are correct?

1. It represents the range of conditions that a species can tolerate, the resources it utilizes, and its functional role in ecological systems.
2. Each species has a distinct niche.

A. Neither 1 nor 2

B. Both 1 and 2

C. Only 2

D. Only 1

💡 **Show answer**

Ans: B

Q14. UPPCS Prelims 2021, Q41

Assertion (A): Various components of an ecosystem are not interdependent.

Reason (R): Human activities have impacts on the environment.

A. Both true; R explains A

B. Both true; R not explanation

C. A true, R false

D. A false, R true

💡 **Show answer**

Ans: D

Q15. UPPCS Prelims 2021, Q111

Indicator of air pollution:

A. Puffballs

B. Algae

C. Lichen

D. Moss

💡 **Show answer**

Ans: C

Q16. UPPCS Prelims 2020, Q79

Movement of nutrient elements through the various components of an ecosystem is called:

A. Bio-geo-chemical cycle

B. Bio-geological cycle

C. Ecological succession

D. Biological cycle

💡 **Show answer**

Ans: A

Q17. UPPCS Prelims 2020, Q85

Entire sequence of communities that successively change in a given area are called:

A. Ecological succession

B. Sere

C. Community dynamics

D. Pyramid of biomass

💡 **Show answer**

Ans: B

Q18. UPPCS Prelims 2020, Q144

The concept of 'Ecological Transition' was first used by:

A. Elton

B. Bennett

C. Berkeley

D. Ratzel

💡 **Show answer**

Ans: B — **John W. Bennett** (*The Ecological Transition*, 1976).

Q19. UPPCS Prelims 2019, Q46

Nitrogen fixing bacteria make combination with cells of the roots of:

A. Pulses

B. Rice

C. Wheat

D. Sugarcane

💡 **Show answer**

Ans: A

Q20. UPPCS Prelims 2019, Q55

What is true about ecosystem?

A. Primary consumers are least dependent upon producer

B. Primary consumers are out-number producers

C. Producers are more than primary consumers

D. Secondary consumers are the largest and most powerful

💡 **Show answer**

Ans: C

Q21. UPPCS Prelims 2019, Q56

With reference to Eutrophication, which statements is/are correct?

1. It is a phenomenon of nutrient enrichment of a water body.
2. It depletes dissolved oxygen of the water.

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

💡 **Show answer**

Ans: C

Q22. UPPCS Prelims 2019, Q57

The environment modified by human activities is called:

A. Natural environment

B. Anthropogenic environment

C. Urban environment

D. Modern environment

💡 **Show answer**

Ans: B

Q23. UPPCS Prelims 2019, Q59

Well-known bioindicator of air pollution:

A. Lichens

B. Methyl mercury

C. Rose

D. Sunflower

💡 **Show answer**

Ans: A

Q24. UPPCS Prelims 2019, Q134

Energy during transfer from one trophic level to another in an ecosystem:

A. Increases

B. Decreases

C. Remains constant

D. May increase or decrease

💡 **Show answer**

Ans: B

Q25. UPPCS Prelims 2019, Q135

Associated with biological nitrogen fixation:

A. Red algae

B. Brown algae

C. Green algae

D. Blue-green algae

💡 **Show answer**

Ans: D

Q26. UPPCS Prelims 2018, Q46

Which of the following is an example of anthropogenic biome?

A. Fresh water

B. Grassland

C. Rain forest

D. Cropland

💡 **Show answer**

Ans: D

Q27. UPPCS Prelims 2018, Q51

Which of the following is concerned with soil?

A. Edaphic

B. Climatic

C. Biotic

D. Topography

💡 **Show answer**

Ans: A

Q28. UPPCS Prelims 2018, Q122

In which ecosystem is species diversity relatively higher?

A. Deep sea

B. Tropical rain forest

C. Coral reefs

D. Desert

> 💡 **Show answer**
>
> **Ans: C**

Q29. UPPCS Prelims 2018, Q127

Which is the most stable ecosystem?

A. Marine (ocean)

B. Forest

C. Mountain

D. Desert

> 💡 **Show answer**
>
> **Ans: A**

Q30. UPSC Prelims 2016 — pattern

In food chains, energy at successive trophic levels:

A. Increases

B. Decreases; ~10% transfers

C. Stays same

D. Becomes cyclic

> 💡 **Show answer**
>
> **Ans: B** — Lindeman rule. Energy decreases at each higher trophic level.

Extra drill — Ghatnachakra ecology stems

Q31. The term ecosystem was proposed by:

A. G. Taylor

B. E. Hartigan

C. D.R. Stoddart

D. A.G. Tansley

> 💡 **Show answer**
>
> **Logic:** These stems recycle Tansley / open system / hierarchy / artificial ecosystem / marine stability / succession / MEA / productivity order traps. **Ans: D.** Tansley coined *ecosystem* in **1935**.

Q32. Which is NOT correct about the ecosystem?

A. It comprises abiotic and biotic components

B. It has its productivity

C. It is a closed system

D. It is a natural resource system

💡 **Show answer**

Ans: C. An ecosystem is normally an **open** system.

Q33. Correct order from organism toward biosphere:

A. Population–Ecosystem–Community–Landscape

B. Landscape–Community–Ecosystem–Population

C. Population–Community–Ecosystem–Landscape

D. Population–Landscape–Community–Ecosystem

💡 **Show answer**

Ans: C. Population → Community → Ecosystem → Landscape.

Q34. Which is an artificial ecosystem?

A. Pond (natural)

B. Rice field / farmland

C. Forest

D. Lake (natural)

💡 **Show answer**

Ans: B. Farmland / rice field is human-made and controlled.

Q35. Most stable / largest-area ecosystem type among common options:

A. Desert

B. Grassland

C. Forest

D. Marine (ocean)

💡 **Show answer**

Ans: D. Marine ecosystem covers ~71% of Earth and is treated as most stable.

Q36. Full biotic succession sequence:

A. Migration–Nudation–Ecesis–Reaction–Stabilisation

B. Nudation–Migration–Ecesis–Reaction–Stabilisation

C. Ecesis–Migration–Nudation–Reaction–Stabilisation

D. Stabilisation–Reaction–Nudation–Migration–Ecesis

💡 **Show answer**

Ans: B. Starts with **Nudation**.

Q37. Decreasing productivity order:

A. Oceans–lakes–grasslands–mangroves

B. Mangroves–oceans–grasslands–lakes

C. Mangroves–grasslands–lakes–oceans

D. Oceans–mangroves–lakes–grasslands

💡 **Show answer**

Ans: C. Mangroves > grasslands > lakes > oceans.

Q38. Transition zone between two communities is:

A. Ecosphere

B. Ecoboundary

C. Ecotone

D. Ecotype

💡 **Show answer**

Ans: C. **Ecotone**.

Q39. Density gradient in a water system is shown by:

A. Ecocline

B. Halocline

C. Pycnocline

D. Thermocline

💡 **Show answer**

Ans: C. **Pycnocline** = density; Halocline = salt; Thermocline = temperature with depth.

Q40. Which tree is treated as an environmental hazard for groundwater drawdown?

A. Babul

B. Eucalyptus

C. Neem

D. Peepal

💡 **Show answer**

Ans: B. **Eucalyptus** draws heavy groundwater.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D

Q1. With reference to ecology and ecosystem, which of the following statements is/are correct?

1. Ernst Haeckel coined the term ecology in 1866.
2. A.G. Tansley coined the term ecosystem in 1945.
3. Ecosystems are normally closed systems with no energy exchange.

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

Show answer

Ans: D. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q2. Consider the following pairs:

Term	fact
1. Habitat	Role of a species
2. Niche	Place where an organism lives
3. Community	Populations of different species in an area

Which of the pairs given above is/are correctly matched?

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1, 2 and 3

Show answer

Ans: B. **Logic:** Habitat = place; niche = role — pairs 1–2 are swapped. Community is correctly matched.

Q3. Arrange the following ecological levels in the correct ascending order of organisation:

1. Community 2. Population 3. Ecosystem 4. Landscape

A. 1-2-3-4 B. 2-1-3-4 C. 2-1-4-3 D. 1-2-4-3

Show answer

Ans: B. **Logic:** Population → Community → Ecosystem → Landscape.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Energy flow in an ecosystem is unidirectional.

Reason (R): Nutrients also flow in only one direction and do not cycle.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: C. **A/R logic:** A is true. Nutrients cycle — R is false.

Q5. Match List-I with List-II and select the correct answer using the code given below the lists:

List-I	List-II
A. Liebig	1. 10% energy transfer
B. Shelford	2. Limiting factor / scarcest resource
C. Lindeman	3. Tolerance range

Code:

A. A-3, B-2, C-1 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-1, B-3, C-2

Show answer

Ans: C. **Logic:** Liebig = limiting; Shelford = tolerance; Lindeman = 10%.

Q6. Which of the following pairs is/are NOT correctly matched?

1. Primary succession — starts where soil already exists
2. Secondary succession — starts on a lifeless substrate
3. Climax community — relatively stable end community of a sere

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. **Logic:** Primary = lifeless; secondary = soil present. Pair 3 is correct.

Q7. With reference to ecological succession, which of the following statements is/are correct?

1. Full succession phases exclude Nudation and Ecesis and jump straight from Migration to Stabilisation.
2. Monoclimax theory is linked with F.E. Clements.
3. Hydrosere develops in dry habitats and xerosere in wet habitats.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: A. Only 2 is correct. **Logic:** Statement 2 is true (Clements–monoclimax). Statement 1 is false — the full phase sequence includes Nudation → Migration → Ecesis → Reaction → Stabilisation. Statement 3 swaps hydrosere (wet) and xerosere (dry).

Q8. With reference to productivity, which of the following statements is/are correct?

1. NPP = GPP + plant respiration.
2. Plants capture only about 1% of incident sunlight on leaves.
3. Among common options, oceans are more productive per unit area than mangroves.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: A. Only 2 is correct. **Logic:** Statement 2 is the usual ~1% capture teaching. Statement 1 is false — NPP = GPP − plant respiration. Statement 3 is false — oceans are less productive per unit area than mangroves among common options.

Q9. Consider the following statements:

1. A fundamental niche is the full potential role without competitors.
2. A realized niche is narrower after competition.
3. Gause's competitive exclusion says identical niches can stably coexist when resources are limited.

A. 1 and 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** Identical niches cannot stably coexist when resources are limited.

Q10. Match List-I with List-II:

List-I (Cline)	List-II
A. Pycnocline	1. Temperature-with-depth gradient

List-I (Cline)	List-II
B. Halocline	2. Density gradient
C. Thermocline	3. Salinity gradient

Code:

A. A-1, B-3, C-2 B. A-2, B-1, C-3 C. A-3, B-2, C-1 D. A-2, B-3, C-1

💡 **Show answer**

Ans: D. **Logic:** Pycnocline = density; Halocline = salinity; Thermocline = temperature.

Q11. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): An ecotone is the transition zone between two communities or ecosystems.

Reason (R): Edge effect often raises species richness in ecotones compared with adjoining communities.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. **A/R logic:** Both true. Edge effect is associated, but does not define what an ecotone is.

Q12. With reference to r- and K-strategists, which of the following statements is/are correct?

1. r-strategists produce many offspring with fast growth and short life.
2. K-strategists produce few offspring with slow growth and long life.
3. Humans are classic pure r-strategists in all ecological classifications.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** Humans are generally K-leaning — not pure r.

Q13. Which of the following is/are correctly matched?

1. J-curve — exponential growth
2. S-curve — logistic growth leveling near carrying capacity
3. Carrying capacity — unlimited maximum population with no resource limit

A. 2 and 3 only B. 1 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. **Logic:** Carrying capacity is the sustainable max under resource limits.

Q14. With reference to MEA (2005) ecosystem services, which of the following statements is/are correct?

1. Provisioning, regulating, supporting, and cultural are the usual four categories.
2. Nutrient cycling is a supporting service.
3. Roads and bridges are counted as natural capital provisioning services.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** Roads are not natural capital.

Q15. Consider the following statements about eutrophication:

1. It is nutrient overload that drives algal bloom and a dissolved-oxygen crash.
2. It is primarily an air-pollution phenomenon linked to ozone holes.
3. Nitrogen and phosphorus enrichment are common drivers in water bodies.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** Eutrophication is aquatic — not an ozone-hole story.

Q16. Which of the following pairs is/are NOT correctly matched?

1. Deep ecology — Arne Næss (1973)
2. Ecological niche concept — Joseph Grinnell (1917)
3. Father of Indian ecology — A.G. Tansley

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** Ramdeo Misra is father of Indian ecology; Tansley coined ecosystem.

Q17. With reference to pond zones, which of the following is the correct sequence from shore toward deeper water?

A. Profundal → limnetic → littoral B. Limnetic → littoral → profundal C. Littoral → limnetic → profundal D. Littoral → profundal → limnetic

💡 **Show answer**

Ans: C. **Logic:** Littoral → limnetic → profundal.

Q18. Arrange the following in correct chronological order of coining/publication association:

1. Haeckel — ecology term
2. Grinnell — niche concept
3. Tansley — ecosystem term
4. Lindeman — trophic-dynamic / 10% idea era

A. 1-2-3-4 B. 1-3-2-4 C. 2-1-3-4 D. 1-2-4-3

💡 **Show answer**

Ans: A. **Logic:** 1866 → 1917 → 1935 → 1942.

Q19. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Among ecosystem types, the marine/ocean ecosystem covers the largest area of Earth.

Reason (R): The biosphere is not considered an ecosystem at any scale.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. **A/R logic:** A is true. Biosphere is the largest ecosystem of Earth — R is false.

Q20. With reference to UP examples, which of the following statements is/are correct?

1. Ganga, Yamuna and Gomti are lotic examples.
2. Sur Sarovar (Agra) and Nawabganj (Unnao) are wetland examples.
3. Western UP sugarcane belts are classic natural climax forests.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** Sugarcane belts are agro-ecosystems — not climax forests.

Q21. Match List-I with List-II:

List-I	List-II
A. Artificial ecosystem	1. Ganga river
B. Lotic ecosystem	2. Aquarium / rice field
C. Lentic ecosystem	3. Pond / lake

Code:

A. A-1, B-2, C-3 B. A-2, B-3, C-1 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. **Logic:** Artificial = aquarium/rice field; lotic = river; lentic = pond/lake.

Q22. With reference to phosphorus cycle, which of the following statements is/are correct?

1. Weathering of rocks is a main source releasing phosphorus into the cycle.
2. Upwelling brings cold, nutrient-rich water to the surface and raises marine productivity.
3. Phosphorus has a large gaseous reservoir like nitrogen in the atmosphere.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. **Logic:** Phosphorus lacks a major atmospheric gaseous reservoir like N_2.

Q23. Which one of the following correctly describes resilience of an ecosystem?

A. Complete absence of any disturbance forever B. Capacity to recover after disturbance C. Permanent conversion to a desert biome D. Energy becoming cyclic like nutrients

💡 **Show answer**

Ans: B. **Logic:** Resilience = recovery capacity after disturbance.

Q24. Consider the following statements:

1. Eucalyptus is often treated as an environmental concern because it can draw heavy groundwater.
2. Deforestation is a main cause of ecological imbalance in India.
3. Ecosystem balance is a static equilibrium that never changes.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** Balance is dynamic equilibrium — statement 3 is false.

Q25. Which of the following pairs is/are correctly matched?

1. Bionomics — synonymous with ecology in older usage
2. Bionomics — management of life / wildlife budgets in Prelims fact
3. Ramdeo Misra — father of Indian ecology

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** Bionomics ≠ management of life; that is a trap.

Q26. With reference to producers and consumers, which of the following statements is/are correct?

1. Producers generally outnumber primary consumers.
2. Core functions include productivity, decomposition, nutrient cycling, and unidirectional energy flow.
3. Calling energy cyclic like nutrients is a correct statement.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** Energy is one-way — statement 3 is false.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Secondary succession begins where soil already exists after disturbance.

Reason (R): Primary succession always begins on farmland abandoned for one season.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. **A/R logic:** A is true. Farmland abandonment is typically secondary — R is false.

Q28. How many of the following are anthropogenic ecosystems?

1. Rice field 2. Aquarium 3. Natural coral reef 4. Orchard 5. Human-made reservoir

A. Only two B. Only three C. Only four D. All five

💡 **Show answer**

Ans: C. **Logic:** Rice field, aquarium, orchard, reservoir = four.

Q29. Which of the following statements about Odum / Haeckel / Tansley is/are correct?

1. Haeckel is linked to the term ecology.
2. Tansley is linked to the term ecosystem.
3. Odum is linked to modern ecosystem ecology teaching.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. **Logic:** All three associations are standard.

Q30. With reference to grassland succession limits, which of the following statements is/are correct?

1. In many grasslands, trees often fail to replace grasses because of water limits and fire.
2. Every grassland globally must succeed to tropical rainforest climax within a decade.
3. A sere is the full community sequence in succession.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** Statement 2 is false.

Mains Answer Framework

Ecosystem structure & function (200 words): Define ecosystem (Tansley) → biotic + abiotic → four functions (productivity, decomposition, nutrient cycling, energy flow) → 1% capture + 10% transfer → Indian example (Ganga basin (गंगा) / Dudhwa).

Ecological succession (200 words): Define → primary vs secondary → pioneer (lichen) → phases M-E-R-S → hydrosere/xerosere → climax → conservation link (restoration ecology).

Common Traps — Don't Fall For These

Trap	Correct fact
Odum coined "ecosystem"	**Tansley 1935** coined ecosystem
Ecosystem is a closed system	Ecosystem is normally **open**
Ecesis before Migration	Full: **Nudation** → **Migration** → **Ecesis** → **Reaction** → **Stabilisation**
Sere = nutrient cycle	**Sere** = succession community **sequence**
Lichens thrive in polluted air	Lichens are **sensitive** indicators
1% and 10% cannot both be true	**Both true** in different stages
Energy increases up food chain	Energy **decreases** (~10% transfer)
Coral reefs = most stable ecosystem	Coral reefs = high **diversity**; marine = high **stability**
Primary succession has soil already	Primary = **no soil**; secondary = soil present
Bionomics = management of life	Bionomics = **ecology**
Niche coined by Odum	Niche concept linked to **Grinnell (1917)**
Ecological Transition = Ratzel	**John W. Bennett**
ESZ under WPA 1972 / total human ban	ESZ under **EPA 1986**; activities regulated, not all banned
Eucalyptus is eco-friendly	Often treated as **groundwater hazard**
Productivity: oceans first	Decreasing order **Mangroves > grasslands > lakes > oceans**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 3 — Food Chain (खाद्य श्रृंखला) & Energy Flow

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Food Chain | Food Web (खाद्य जाल) | Trophic Levels | Energy Flow in Ecosystem (पारिस्थितिकी तंत्र) | Ecological Pyramid (पिरामिड) (Ecological Pyramids)

Sources baked in: NCERT Class 12 Biology Ch 14, Odum, Elton (1927), Lindeman (1942), UPPCS/UPSC PYQs 2018–2025 **Weight:** ★★★ Very High — direct Prelims; links to pollution (biomagnification) **Last verified:** August 2026

Current Affairs (this topic)

- Energy-flow and trophic-level traps recycle yearly — 10% rule and pyramid types stay core.
- Bioaccumulation (जैव संचय) / biomagnification (जैव आवर्धन) CA usually rides on pesticide or heavy-metal news; keep the mechanism pair clear.

💡 🎯 **Consolidated — 30 Must-Score Facts (Click to Expand)**

1. A **food chain** is a single linear who-eats-whom sequence. A **food web** is a network of interconnected chains and is more stable than one chain alone. 2. A **grazing food chain** starts from **living green plants**. A **detritus food chain** starts from **dead organic matter**. 3. Classic terrestrial chain: **Grass → Grasshopper → Frog → Snake**. Classic human-linked chain: **Grass → Goat → Human**. 4. Classic marine chain: **Diatoms → Crustaceans → Herrings**. 5. **Trophic levels** are feeding ranks counted from producers: **T1 producers, T2 herbivores, T3–T5 carnivores**. 6. The **Sun is the energy source**, not a trophic level. Never count sunlight as T1. **Solar energy** is the driving force of the ecosystem. 7. Energy flow is **unidirectional**: Sun → producers → consumers → heat. Nutrients cycle; **energy does not**. 8. About **10%** of energy passes to the next trophic level (**Lindeman, 1942**). About **90%** is lost as heat, respiration, and movement. Energy **decreases** upward. 9. Food chains stay short, usually about **3–5 levels**, because usable energy runs out. Food chains are **not** found within a single species. 10. **Humans** are **primary as well as secondary** consumers (plants and meat). 11. Main marine primary producers are **phytoplankton**. 12. Primary consumers are mainly herbivores such as **deer**; **ants** may also act as primary consumers (and scavengers/decomposers in broader roles). Fox and tiger are not primary consumers. 13. Decomposers are **fungi and bacteria**, not viruses. 14. The **energy pyramid** (kcal/m^2/yr) is **always upright**. An inverted energy pyramid is never correct. 15. A **biomass pyramid** can invert in **pond / marine** systems. A **numbers pyramid** can invert in a parasitic chain (one tree, many insects). 16. **Biomagnification** raises toxin concentration up the chain. For DDT, the **snake** (tertiary consumer) shows higher concentration than grasshopper or toad. 17. Butterfly decline harms **pollination** and can cut populations of predators such as wasps, spiders, and birds. It does not by itself force a drastic rise in fungal infections of crops. 18. Grazing does **not** start from dead matter. Detritus does **not** start from living green plants. 19. More links in a food web raise stability, but energy still flows **one way**. 20. Decomposers and detritivores close the detritus path and recycle nutrients back to producers. 21. Apex predators sit at the top trophic levels and are often scarce because little energy remains. 22. Dudhwa (दुधवा)'s grass → deer → tiger chain is a classic UP **grazing** example with tiger as apex predator. 23. A Ganga (गंगा) algae → fish → dolphin web is a standard UP **lotic aquatic** food-web example. 24. Terai (तराई) sal leaf litter → earthworm is a UP **detritus** start. Do not call it grazing. 25. Western UP sugarcane → pest → predator is a short

cropland chain of about three levels. 26. Vulture decline after veterinary **diclofenac** (डाइक्लोफेनाक) disrupts the **detritus / scavenging** pathway. 27. Ecological pyramids may show energy, biomass, or numbers — always note which quantity is plotted. 28. Saying "energy is cyclic like nutrients" is a frequent false statement. 29. Match pyramid type to exception: energy never inverts; biomass may invert at sea/pond; numbers may invert with parasites. 30. Keep Lindeman's **10% rule** as an approximate key fact, not an exact laboratory constant for every ecosystem.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Food Chain** | **Food Web** | Single linear sequence of organisms where nutrients and energy transfer through feeding vs complex network of interconnected, interlocking food chains providing alternative feeding pathways | खाद्य श्रृंखला / खाद्य जाल | | **Grazing Food Chain (GFC)** | **Detritus Food Chain (DFC)** | Begins with living green photosynthetic plants (producers) and drives terrestrial energy vs begins with dead organic matter (detritus) decomposed by fungi and bacteria | चराई खाद्य श्रृंखला / अपरद (अवशेष) खाद्य श्रृंखला | | **Upright Pyramid of Biomass** | **Inverted Pyramid of Biomass** | Characteristic of terrestrial grasslands and forests where producer biomass far exceeds herbivores vs characteristic of aquatic/pond ecosystems where standing crop of phytoplankton is smaller than zooplankton/fish | सीधा जैवभार पिरामिड (स्थलीय) / उल्टा जैवभार पिरामिड (जलीय/तालाब) | | **10% Energy Transfer Law** | **Bioaccumulation** | Raymond Lindeman (1942) principle that only ~10% of energy transfers from one trophic level to the next vs progressive buildup of non-biodegradable chemicals within an individual organism over time | 10% ऊर्जा नियम (लिंडमैन) / जैव संचय (बायोएक्यूमुलेशन) | | **Bioaccumulation** | **Biomagnification** | Increase in toxicant concentration within a single organism's tissues through direct absorption from water/food vs increase in toxic chemical concentration (e.g. DDT, mercury) across successive trophic levels of a food chain | जैव संचय (एक जीव में) / जैव आवर्धन (खाद्य श्रृंखला में) | | **Primary Consumer** | **Secondary Consumer** | Herbivores feeding directly on autotrophic green plants (cow, deer, zooplankton) vs primary carnivores that hunt and feed on herbivores (frog, fox, small fish) | प्राथमिक उपभोक्ता (शाकाहारी) / द्वितीयक उपभोक्ता (मांसाहारी) | | **Pyramid of Energy** | **Pyramid of Numbers** | Always universally upright in all ecosystems because energy is irretrievably lost as metabolic heat at each level vs can be upright (grassland) or inverted (single large tree supporting thousands of birds/parasites) | ऊर्जा पिरामिड (सदैव सीधा) / संख्या पिरामिड |

💡 📋 **Must-score facts — chains, pyramids, biomagnification (Click to Expand)**

Chain types | Item | Lock | |---|---| | Food chain vs web | Linear sequence / interconnected network (more stable) | | Grazing vs detritus | Living green plants / dead organic matter | | Trophic ranks | T1 producers → T2 herbivores → T3–T5 carnivores | | Sun | Energy **source** — **not** a trophic level | | Lindeman 10% | ~10% up; ~90% lost; energy ↓ upward | | Humans | Primary **and** secondary consumers | | Decomposers | Fungi + bacteria (≠ viruses) | ### Pyramids / toxins | Item | Lock | |---|---| | Energy pyramid | **Always upright** | | Biomass / numbers invert | Pond–marine / parasitic chain | | Biomagnification | Toxin ↑ up chain (DDT highest in apex) | | Diclofenac | Vulture crash → scavenging pathway break | | UP grazing example | Dudhwa grass → deer → tiger |

3.1 Food Chain

Identity: Linear feeding sequence | Grazing vs detritus | 3–5 levels max

- A **food chain** is a linear sequence in which one organism is eaten by the next. Energy and nutrients pass step by step through **trophic levels**.
- The **Sun** is the ultimate energy source for almost all ecosystems. The **immediate** starting point of a chain depends on chain type.

- A **grazing (pastoral (पशुपालक)) food chain** starts from **living green plants** (producers). Energy then passes to herbivore, carnivore, and top predator.
- A **detritus food chain** starts from **dead organic matter** such as leaf litter, dead plankton, or animal carcasses. Detritivores and decomposers dominate this pathway.
- Both chain types run **in parallel** in the same ecosystem. A tiger may hunt living deer (grazing chain) and also feed on carrion (detritus chain).
- Food chains are usually **short (3–5 levels)** because only about **10%** of energy passes to the next level. Too little energy remains to support a sixth or seventh level.
- Removing one link can **break the chain** and trigger a **trophic cascade** in the levels above.
- A **terrestrial grazing chain** in UP runs Terai grass to barasingha or chital to tiger in Dudhwa.
- A classic textbook terrestrial chain is **Grass → Grasshopper → Frog → Snake** (producer → primary → secondary → tertiary consumer).
- A human-linked chain is **Grass → Goat → Human** (autotroph → herbivore → omnivore).
- An **aquatic grazing chain** runs phytoplankton to zooplankton to small fish to large fish.
- A common marine chain is **Diatoms → Crustaceans → Herrings**.
- A **detritus chain** on the Terai forest floor runs fallen sal leaves to earthworm to bacteria and fungi.
- Food chains illustrate the **order** in which organisms feed upon each other. They are **not** found within the population (जनसंख्या) of a single species.
- Humans act as **primary as well as secondary consumers** because they eat both plant products and meat.
- In the marine environment (पर्यावरण), the main primary producers are **phytoplankton** (micro-algae), not seaweeds alone.
- Main primary consumers are herbivores such as **deer**. **Ants** may also feed as primary consumers. Fox and tiger are not primary consumers.
- A large fall in butterfly populations can harm **pollination** of some plants and reduce prey for wasps, spiders, and birds. It does not by itself force a drastic rise in fungal infections of cultivated plants.

Type	Starts from	UP / textbook example
Grazing	Living producers	Grass → deer → tiger (Dudhwa); Grass → grasshopper → frog → snake
Detritus	Dead organic matter	Leaf litter → earthworm → fungi
Marine	Phytoplankton/diatoms	Diatoms → crustaceans → herrings
Parasitic (special)	Living host tissue	Tree → insects → parasites

> **Logic:** Grazing chain starts from a **living producer**. Detritus chain starts from **dead matter**. Trap: "all chains begin with green plants" is **false**.

3.2 Food Web

Identity: Network of chains | Stability through alternatives | Omnivores as connectors

- A **food web** is a network formed when **multiple food chains interconnect** in one ecosystem.
- Food webs represent nature more accurately than a single chain because most animals eat more than one prey species.
- Webs are **more stable** than isolated chains. If one prey species declines, predators can switch to alternatives.
- **Omnivores** such as bears, crows, and humans link grazing and detritus pathways. They can occupy **more than one trophic level** at once.
- **Keystone predators** such as tigers regulate entire webs. Removing them causes trophic cascade across several chains.

- Simpler webs in agricultural monoculture are **less stable**. Removing one pest predator can cause a pest outbreak.
- Energy still flows **unidirectionally** through every pathway in the web. A web shows many one-way paths, not cyclic energy.

> **Logic:** Trap — "a food web makes energy flow cyclic" is **false**. Only **nutrients** cycle through decomposers.

3.3 Trophic Levels

Identity: T1 = producers | T2–T5 consumers | Decomposers beside the ladder

- A **trophic level** is the feeding rank of an organism in a food chain. Count from **producer = T1**. The Sun is an energy **source**, not a numbered trophic level.
- **T1 (Producers)** are autotrophs such as green plants, algae, and phytoplankton. They fix solar energy into chemical energy.
- **T2 (Primary consumers)** are herbivores that eat producers directly. Examples include deer, zooplankton, and caterpillars.
- **T3 (Secondary consumers)** are carnivores that eat primary consumers. Examples include frogs and small fish.
- **T4 (Tertiary consumers)** eat secondary consumers. Examples include snakes and large fish.
- **T5 (Quaternary consumers)** are top predators with few or no natural enemies. Tigers and eagles appear here when the chain is long enough.
- **Omnivores** eat both plants and animals. They span **multiple trophic levels** simultaneously.
- **Decomposers** break down dead matter from **all levels**. They recycle nutrients but are usually drawn **beside** the main chain, not as the top of an energy pyramid.
- Decomposer organisms in standard stems are **fungi and bacteria**. **Viruses** are not counted as decomposers here.
- **Detritivores** such as earthworms fragment dead material and bridge the detritus chain toward decomposers.

Bioaccumulation vs Biomagnification and DDT Trophic Cascade

Bioaccumulation and Biomagnification

- **Biomagnification** means a non-degradable toxin (classic example **DDT**, also mercury) rises in concentration at successive trophic levels and lodges in fatty tissues. The highest concentration appears in top consumers such as a **snake** in a grasshopper–toad–snake chain.

Level	Role	UP / India example
T1	Producer	Terai grass, wheat, Ganga algae
T2	Primary consumer (प्राथमिक उपभोक्ता)	Barasingha, chital, rohu (juvenile)
T3	Secondary consumer (द्वितीयक उपभोक्ता)	Frog, catfish
T4	Tertiary consumer	Python, fishing eagle
T5	Top carnivore	Tiger (Dudhwa), Gangetic dolphin (river apex)

PYQ — UPPCS Prelims 2023, Q117

Trophic levels are formed by—

A. Organisms linked in a food chain

B. Plants only

C. Animals only

D. Carnivorous animals only

> 💡 **Show answer**
>
> **Ans: A**

3.4 Energy Flow in Ecosystem

Identity: Unidirectional flow | Lindeman 10% | Nutrients cycle, energy does not

- **Energy flow** is the movement of solar energy through living systems via food chains and food webs.
- Flow is **unidirectional**: Sun → producers (chemical energy in biomass) → consumers → **heat** lost to the environment.
- **Nutrients** such as carbon and nitrogen **cycle** through decomposers and return to producers. **Energy does not cycle.**
- **Raymond Lindeman (1942)** showed that roughly **10%** of energy or organic matter passes to the next higher trophic level.
- About **90%** is lost at each transfer through respiration, movement, egestion, uneaten body parts, and heat radiation.
- This loss explains why **top predators are few**, why **food chains are short**, and why eating lower on the chain is more energy-efficient for humans.
- Terrestrial plants capture about **1%** of incident sunlight on leaves. That capture stage is separate from the **10% transfer** between consumer levels. Both facts can be true in one question.
- **Ecological efficiency** equals energy at the higher level divided by energy at the lower level, multiplied by 100. The NCERT standard is about **10%**.

Where energy goes at each level	Key fact
Respiration and heat	Largest share — why pyramid of energy (ऊर्जा पिरामिड) is upright
Movement and metabolism	Not passed to next consumer
Egestion and uneaten parts	Lost from food chain pathway
Passed to next trophic level	**~10%** (Lindeman 1942)

PYQ — UPPCS Prelims 2019, Q134

The amount of energy during transfer from one trophic level to another in an ecosystem:

A. Increases

B. Decreases

C. Remains constant

D. May increase or decrease

> 💡 **Show answer**
>
> **Ans: B** — Lindeman 10% rule. Only about 10% moves up; the rest is lost.

Trophic Energy Flow, Lindeman's 10% Law & Ecological Pyramids

3.5 Ecological Pyramid (Ecological Pyramids)

Identity: Elton 1927 | Energy / biomass / numbers | Only energy pyramid always upright

- An **ecological pyramid** is a diagram of trophic structure. It shows energy, biomass, or numbers at each level.

- **Charles Elton (1927)** developed the early pyramid concept in trophic studies.
- A **pyramid of energy** plots energy flow in **kcal/m²/yr**. It is **always upright** because energy is lost at every transfer.
- A **pyramid of biomass** plots standing dry weight in **g/m²**. It is usually upright on land but can be **inverted in open ocean and pond ecosystems**.
- In a **pond**, producers are tiny organisms with low standing biomass, so biomass increases toward higher consumers and the pyramid looks inverted.
- In the sea, phytoplankton reproduce so fast that at any instant their **standing biomass** may be less than fish biomass, even though energy flow still moves upward correctly.
- A **pyramid of numbers** (संख्या पिरामिड) counts **individual organisms** at each level. It is usually upright but can be **inverted in parasitic chains**.
- One large tree (one individual at T1) can support millions of insects and parasites at higher levels. That produces an inverted number pyramid.
- The **pyramid of energy** is the best measure of ecosystem function because it shows actual energy availability and is never inverted.
- **Decomposers** are usually omitted from standard pyramid diagrams. They act on all levels and return nutrients to producers.
- **Biomagnification** of toxins such as DDT increases concentration up the food chain. Top predators suffer most even as energy decreases upward.

Pyramid	Unit	Always upright?	Inverted example
Energy	kcal/m²/yr	**Yes — always**	Never inverted
Biomass	g/m²	Usually	Open ocean (phytoplankton turnover)
Numbers	Individual count	Usually	One tree → many insects/parasites

> **Logic:** Trap — "all ecological pyramids are always upright" is **false**. Only the **energy** pyramid is always upright.

Complete PYQ Bank (Topic 3)

Q1. UPPCS Prelims 2025, Q5

With reference to 'Ecosystem', which of the following statements is/are correct?

1. Green plants in terrestrial ecosystem capture about 1% of energy of the sunlight that falls on their leaves.
2. Around 10% of organic matter reaches to the next higher level of consumers.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Ans: C — 1% capture at producer stage. 10% transfer between trophic levels.

Q2. UPPCS Prelims 2023, Q117

Trophic levels are formed by—

A. Organisms linked in a food chain

B. Plants only

C. Animals only

D. Carnivorous animals only

💡 **Show answer**

Ans: A

Q3. UPPCS Prelims 2019, Q134

The amount of energy during transfer from one trophic level to another in an ecosystem:

A. Increases

B. Decreases

C. Remains constant

D. May increase or decrease

💡 **Show answer**

Ans: B

Q4. UPSC Prelims 2016 — pattern

With reference to food chains in ecosystems, which of the following is correct?

A. Energy increases at higher trophic levels

B. Energy decreases; ~10% transfers

C. Energy is cyclic

D. 50% transfers between levels

💡 **Show answer**

Ans: B

Q5. UPSC Prelims 2011 — pattern

Why are food chains generally short?

A. Predators are lazy

B. Progressive energy loss at each trophic transfer

C. Too many species

D. Nutrients are scarce

💡 **Show answer**

Ans: B

Q6. UPSC Prelims 2017 — pattern

Inverted pyramid of biomass is seen in:

A. Grassland

B. Open ocean

C. Desert

D. Tropical rainforest

💡 **Show answer**

Ans: B

Q7. UPSC Prelims 2014 — pattern

Inverted pyramid of numbers is seen in:

A. Grassland

B. Parasitic food chain on a tree

C. Open ocean

D. Pond

💡 **Show answer**

Ans: B

Q8. UPSC Prelims 2018 — pattern

Which occupies the lowest trophic level?

A. Tiger

B. Phytoplankton

C. Snake

D. Eagle

💡 **Show answer**

Ans: B

Q9. UPSC Prelims 2014 — pattern

In grassland chain Grass → Grasshopper → Frog → Snake → Hawk, trophic levels are:

A. T1–T5 in order

B. All T2

C. Hawk = T1

D. Grass = T2

💡 **Show answer**

Ans: A

Q10. UPPCS Prelims 2019 — pattern

Which pyramid is always upright?

A. Pyramid of biomass

B. Pyramid of numbers

C. Pyramid of energy

D. All three

💡 **Show answer**

Ans: C

Extra drill — food-chain / energy / biomagnification stems

Q31. Correct grassland food-chain order:

A. Grass–Grasshopper–Frog–Snake

B. Grass–Grasshopper–Snake–Frog

C. Grasshopper–Frog–Grass–Snake

D. Grasshopper–Snake–Frog–Grass

💡 **Show answer**

Logic: Classic chain order, human trophic role, marine producers, 10% law, inverted biomass, DDT biomagnification. **Ans: A.** Producer → primary → secondary → tertiary consumer.

Q32. Which constitutes a food chain?

A. Grass, wheat and mango

B. Grass, goat and human

C. Goat, cow and elephant

D. Grass, fish and goat

💡 **Show answer**

Ans: B. Grass → goat → human.

Q33. In a food chain, man is:

A. Producer only

B. Only primary consumer

C. Only secondary consumer

D. Primary as well as secondary consumer

💡 **Show answer**

Ans: D. Humans eat plants and meat.

Q34. Main primary producers in the marine environment:

A. Phytoplankton

B. Seaweeds only

C. Marine angiosperms only

D. Aquatic bryophytes only

💡 **Show answer**

Ans: A. **Phytoplankton**.

Q35. Decomposer organisms include:

A. Virus only

B. Fungi and bacteria

C. Virus and bacteria only

D. Virus, fungi and bacteria

💡 **Show answer**

Ans: B. Fungi and bacteria — not virus in this framing.

Q36. Correct marine food-chain sequence:

A. Diatoms–Crustaceans–Herrings

B. Crustaceans–Diatoms–Herrings

C. Diatoms–Herrings–Crustaceans

D. Crustaceans–Herrings–Diatoms

💡 **Show answer**

Ans: A.

Q37. 10 percent law relates to:

A. Heat transfer between metals

B. Energy transfer as food between trophic levels

C. Bird migration between zones

D. Water transfer between aquifers

💡 **Show answer**

Ans: B. **Lindeman** ~10% rule.

Q38. Inverted pyramid of biomass is typical of:

A. Forest

B. Pond / marine

C. Grassland

D. Drylands

💡 **Show answer**

Ans: B. Pond and open ocean standing biomass can invert.

Q39. Highest DDT concentration after biomagnification is likely in:

A. Grasshopper

B. Toad

C. Snake

D. Cattle

💡 **Show answer**

Ans: C. Tertiary consumer holds the highest load.

Q40. Primary source / driving force of energy in an ecosystem:

A. Fermentation heat

B. Stored plant sugar alone

C. Solar energy

D. None of the above

💡 **Show answer**

Ans: C. **Solar energy**.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D

Q1. With reference to food chain and food web, which of the following statements is/are correct?

1. A food chain is a single linear who-eats-whom sequence.
2. A food web is a single linear chain and is generally less stable than one isolated food chain.
3. Food chains commonly occur entirely within a single species.

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Statement 1 is true. Statement 2 reverses the food-web definition (network of chains; usually more stable). Statement 3 is false — chains link different species/trophic levels.

Q2. Which of the following pairs is/are NOT correctly matched?

1. Grazing food chain — starts from dead organic matter
2. Detritus food chain — starts from living green plants
3. Marine primary producers — mainly phytoplankton

A. 2 and 3 only B. 1 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. **Logic:** Grazing from living plants; detritus from dead matter. Pair 3 correct.

Q3. With reference to trophic levels, which of the following statements is/are correct?

1. T1 denotes herbivores and T2 denotes producers.
2. The Sun is counted as trophic level T1 in standard pyramids.
3. Energy decreases as one moves to higher trophic levels.

A. Only 1 B. 1 and 2 C. Only 3 D. 2 and 3

💡 **Show answer**

Ans: C. Only 3 is correct. **Logic:** Statement 3 is true (energy declines up the chain). Statement 1 swaps T1/T2. Statement 2 is false — the Sun is the energy source, not trophic level T1; T1 is producers.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): About 10% of energy passes to the next trophic level (Lindeman, 1942).

Reason (R): Energy flow is cyclic in ecosystems just like nutrient cycling.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. **A/R logic:** A true; energy is unidirectional — R false.

Q5. Match List-I with List-II:

List-I	List-II
A. Energy pyramid	1. May invert in pond/marine systems
B. Biomass pyramid	2. Always upright
C. Numbers pyramid	3. May invert in a parasitic chain

Code:

A. A-1, B-2, C-3 B. A-2, B-3, C-1 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. **Logic:** Energy always upright; biomass may invert at sea/pond; numbers with parasites.

Q6. Consider the following statements:

1. Humans are primary as well as secondary consumers.
2. Decomposers are mainly green plants and herbivores.
3. Viruses are the principal decomposers in terrestrial systems.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: C. Only 1 is correct. **Logic:** Statement 1 is true (humans as primary/secondary consumers). Statement 2 is false — decomposers are mainly fungi and bacteria. Statement 3 is false — viruses are not principal terrestrial decomposers.

Q7. Arrange a classic terrestrial grazing chain in correct order:

1. Frog 2. Grass 3. Snake 4. Grasshopper

A. 2-4-1-3 B. 2-1-4-3 C. 4-2-1-3 D. 2-4-3-1

💡 **Show answer**

Ans: A. **Logic:** Grass → Grasshopper → Frog → Snake.

Q8. With reference to biomagnification, which of the following statements is/are correct?

1. Toxin concentration generally rises up the food chain.
2. For DDT-type stems, secondary consumers usually show higher toxin concentration than apex consumers.
3. Biomagnification means energy increases at each trophic level.

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Statement 1 is true (concentration rises up the chain). Statement 2 reverses biomagnification (apex > secondary for DDT-type toxins). Statement 3 confuses toxins with energy flow.

Q9. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Food chains usually remain short (about 3–5 levels).

Reason (R): Usable energy runs out as transfers continue upward.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. **A/R logic:** Both true and R explains why chains stay short.

Q10. Which of the following UP-linked examples is/are correctly matched?

1. Dudhwa grass → deer → tiger — grazing chain with tiger as apex
2. Terai sal leaf litter → earthworm — detritus start
3. Terai leaf litter → earthworm — grazing food chain start

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** Leaf litter path is detritus — not grazing.

Q11. Match List-I with List-II:

List-I	List-II
A. Classic marine chain start	1. Phytoplankton / diatoms
B. Primary consumer example	2. Deer
C. Apex predator example	3. Tiger

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-1, B-3, C-2 D. A-3, B-2, C-1

💡 **Show answer**

Ans: B. **Logic:** Diatoms/phytoplankton → deer → tiger associations.

Q12. With reference to ecological pyramids, which of the following statements is/are correct?

1. An inverted energy pyramid is a normal feature of terrestrial forests.
2. A biomass pyramid can invert in aquatic systems.
3. A numbers pyramid can invert in a parasitic chain.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** Energy pyramid never inverts.

Q13. Consider the following statements about diclofenac and vultures:

1. Veterinary diclofenac is linked to India's vulture crash.
2. Vulture decline disrupts scavenging / detritus pathways.
3. Diclofenac primarily boosted vulture populations through better carcass quality.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** Statement 3 is false.

Q14. Which one of the following correctly states Lindeman's approximate rule?

A. 90% of energy passes to the next trophic level B. 50% of energy passes to the next trophic level C. About 10% of energy passes to the next trophic level D. Energy doubles at each trophic level

💡 **Show answer**

Ans: C. **Logic:** ~10% transfer; ~90% lost.

Q15. With reference to butterfly decline impacts, which of the following statements is/are correct?

1. Decline can harm pollination services.
2. It can cut populations of predators such as wasps, spiders, and birds that feed on them.
3. It necessarily forces a drastic rise in fungal infections of all crops as the only outcome.

A. 2 and 3 only B. 1 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. **Logic:** Statement 3 is an overclaim / false trap.

Q16. Arrange the following in the direction of energy flow:

A. Consumers → producers → Sun → heat B. Sun → producers → consumers → heat C. Heat → consumers → producers → Sun D. Producers → Sun → consumers → heat

💡 **Show answer**

Ans: B. **Logic:** Sun → producers → consumers → heat.

Q17. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): More links in a food web generally raise stability compared with a single chain.

Reason (R): Energy begins to cycle in a closed loop once a food web forms.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. **A/R logic:** A true; energy still one-way — R false.

Q18. Which of the following pairs is/are correctly matched?

1. Detritivores — help close the detritus pathway
2. Apex predators — often scarce because little energy remains at the top
3. Primary consumers — typically tigers in all terrestrial systems

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. **Logic:** Tigers are apex — not primary consumers.

Q19. With reference to cropland chains, which of the following statements is/are correct?

1. Western UP sugarcane → pest → predator is a short cropland chain of about three levels.
2. Cropland chains are always longer than marine chains because of higher energy.

3. Food chains stay short mainly because usable energy runs out.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** Statement 2 is false.

Q20. Match List-I with List-II:

List-I	List-II
A. Grazing start	1. Dead organic matter
B. Detritus start	2. Living green plants
C. Driving force	3. Solar energy

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. **Logic:** Grazing = living plants; detritus = dead matter; driver = solar.

Q21. How many of the following statements are correct?

1. Energy pyramid is always upright.
2. Nutrients cycle while energy flows one way.
3. About 90% energy is typically lost between trophic levels.

A. Only one B. Only two C. All three D. None

💡 **Show answer**

Ans: C. **Logic:** All three are standard facts.

Q22. Which of the following is/are correct about ants in consumer roles?

A. Ants can never be primary consumers B. Ants may act as primary consumers (and in broader roles as scavengers) C. Ants are always apex predators only D. Ants replace phytoplankton as marine producers

💡 **Show answer**

Ans: B. **Logic:** Teaching allows ants as primary consumers / broader roles.

Q23. With reference to Ganga aquatic webs, which of the following statements is/are correct?

1. Algae → fish → dolphin is a standard UP lotic aquatic food-web teaching example.
2. Gangetic dolphin is a freshwater river dolphin.
3. Gangetic dolphin is primarily a marine open-ocean species.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. **Logic:** Dolphin is freshwater — not marine open-ocean.

Q24. Consider the following statements:

1. Saying “energy is cyclic like nutrients” is a frequent false statement.

2. Ecological pyramids may show energy, biomass, or numbers.
3. Lindeman's 10% rule is an exact laboratory constant for every ecosystem on Earth.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** **Logic:** 10% is approximate — not an exact universal constant.

Q25. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): A biomass pyramid can be inverted in some aquatic systems.

Reason (R): Phytoplankton biomass can be lower than consumer biomass at a given time while productivity remains high.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** **A/R logic:** Both true; R explains the aquatic invert case.

Q26. Which of the following pairs is/are NOT correctly matched?

1. Fox — typical primary consumer in all stems
2. Tiger — typical primary consumer in all stems
3. Deer — typical primary consumer / herbivore

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** **Logic:** Fox/tiger are not primary consumers; deer is.

Q27. Arrange trophic ranks from producers upward for a simple chain ending at tertiary consumer:

A. Herbivore → producer → tertiary → secondary B. Producer → herbivore → secondary carnivore → tertiary carnivore C. Tertiary → secondary → herbivore → producer D. Producer → tertiary → herbivore → secondary

> 💡 **Show answer**
>
> **Ans: B.** **Logic:** T1 producer → T2 herbivore → T3 → T4.

Q28. With reference to energy loss, which of the following statements is/are correct?

1. About 90% is lost as heat, respiration, and movement between levels.
2. Energy increases steadily toward apex predators.
3. Food webs raise stability but do not make energy cyclic.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** **Logic:** Energy decreases toward apex — statement 2 false.

Q29. Which one of the following is a correct marine chain teaching sequence?

A. Herrings → Crustaceans → Diatoms B. Diatoms → Crustaceans → Herrings C. Crustaceans → Diatoms → Herrings D. Herrings → Diatoms → Crustaceans

💡 **Show answer**

Ans: B. **Logic:** Diatoms → Crustaceans → Herrings.

Q30. With reference to pyramid exceptions, which of the following statements is/are correct?

1. Match pyramid type before claiming an invert is possible.
2. Energy pyramid invert is never correct.
3. Numbers pyramid invert is impossible even in parasitic chains.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. **Logic:** Numbers can invert with parasites — statement 3 false.

Mains Answer Framework

Energy flow and pyramids (200 words): Define unidirectional energy flow. State Lindeman 10% rule and why chains are short. Contrast with nutrient cycling. Describe three pyramid types and inversion cases (sea biomass, tree-parasite numbers). Indian example: Dudhwa grazing chain or Ganga aquatic web.

Common Traps — Don't Fall For These

Trap	Correct fact
Energy flow is cyclic	**Unidirectional** — nutrients cycle, energy does not
All pyramids always upright	Only **energy** pyramid is always upright
Grazing chain starts from dead matter	Starts from **living green plants**
Sun is Trophic Level 1	Sun = energy **source**; **Producer = T1**
10% rule by Darwin	**Lindeman 1942**
Food web makes energy cyclic	Each pathway is **one-way**
Biomass pyramid measures energy flow	Biomass = **g/m²**; energy = **kcal/m²/yr**
Open ocean = inverted number pyramid	Open ocean / pond = inverted **biomass** pyramid
Man is only secondary consumer	Man is **primary + secondary** consumer
Virus is a decomposer	Decomposers = **fungi + bacteria**
Food chain within one species	Food chains are **across** feeding links, not within one species
Highest DDT in grasshopper	Highest in top consumer (e.g. **snake**)

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 4 — Biodiversity

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Biodiversity | Levels of Biodiversity | Genetic Diversity (आनुवंशिक विविधता) | Species Diversity (प्रजाति विविधता) | Ecosystem (पारिस्थितिकी तंत्र) Diversity (पारिस्थितिक तंत्र विविधता) | Global Biodiversity | Indian Biodiversity | Importance (Values) of Biodiversity | Biodiversity Assessment | Biodiversity Registers | Biodiversity and Environment (पर्यावरण) | Biodiversity Loss | Biodiversity Conservation | Biodiversity Hotspots | Endemic Species (स्थानिक प्रजाति) | Endangered Species | Threatened Species (संकटग्रस्त प्रजातियां) | Extinct Species | Invasive Alien Species (आक्रामक विदेशी प्रजाति) | Keystone Species (कीस्टोन प्रजाति) | Flagship Species (प्रतीक (फ्लैगशिप) प्रजाति) | Umbrella Species (छत्र (अंब्रेला) प्रजाति) | Indicator Species (जैव संकेतक प्रजाति)

Sources baked in: NCERT Class 12 Biology Ch 15–16, CBD/Nagoya/Cartagena, Biological Diversity Act 2002, IUCN Red List, Conservation International hotspots, Ramsar (~100 sites; UP 12), UPPCS/UPSC PYQs 2018–2025 **Weight:** ★★★ Very High — definitions, hotspots, IUCN, loss drivers, conservation, Ramsar **Last verified:** September 2026

Current Affairs (this topic)

- Track new Ramsar (रामसर) and IUCN (आईयूसीएन) Red List updates against the Must-Score baselines (India Ramsar **~100**; UP **12**).
- Invasive-species and hotspot stems recycle yearly — keep Western Ghats (पश्चिमी घाट) / Indo-Burma trap pairs stable.

💡 🎯 **Consolidated — 36 Must-Score Facts (Click to Expand)**

1. Biodiversity is variability among living organisms at **genetic, species, and ecosystem** levels — the range of plants and animals in an area. 2. **Walter G. Rosen (1985)** coined the word *biodiversity*. **E.O. Wilson** popularised it. **Whittaker (1960)** advanced **alpha, beta, and gamma** diversity. 3. **Genetic diversity** is variation **within** a species. **Species diversity** is variety **among** species. **Ecosystem diversity** is variety of habitats and ecosystems. 4. **Alpha** = mean species diversity at a local habitat (वास स्थान). **Beta** = change between ecosystems. **Gamma** = overall regional diversity. 5. The most significant aspect of biodiversity is **maintenance of the ecosystem**, not food or drugs alone. 6. Species richness is generally highest in the **tropics / tropical rainforest** (optimum biome (जीवोम)) and **increases toward the equator**; it decreases toward the poles and with altitude. 7. On land, tropical rainforest is richest; in the sea, **coral reefs** are among the richest. UPPCS has accepted both rainforest and coral reefs for "relatively higher diversity." 8. India holds about **2.4%** of world land but about **7–8%** of global species (rough shares often cited: mammals ~7.6%, birds ~12.6%, reptiles ~6.2%, amphibians ~4.4%). 9. India's four (चातुर्याम) biodiversity hotspots are **Himalaya (हिमालय), Indo-Burma, Western Ghats–Sri Lanka, and Sundaland**. **Andamans** fall under **Indo-Burma**; **Nicobar** (निकोबार) under **Sundaland**. Eastern Ghats (पूर्वी घाट) / Vindhyas / Naro Hills are **not** hotspots. 10. A hotspot needs about **≥1500 endemic vascular plants** and **≥70%** original habitat lost. Criteria also stress **species richness, endemism, and threat**. There are **36** global hotspots (~2.5% of land; >50% endemic plants). 11. **Silent Valley (साइलेंट वैली) (Kerala)** is among India's richest rainforest biodiversity sites (NP **1984**; Silent Valley Project = Kerala). **Valley of Flowers** (फूलों की घाटी) is in **Chamoli (चमोली), Uttarakhand (उत्तराखंड)**. Himalayan richness reflects confluence of biogeographic zones. 12. The top cause of biodiversity loss is **habitat destruction**. HIPPO =

Habitat, Invasive species (आक्रामक विदेशी प्रजाति), Pollution, Population (जनसंख्या) pressure, Over-exploitation. Vegetarianism is **not** a threat. Fewer trophic levels reduce diversity. 13. India aims for **Land Degradation (भू-क्षरण) Neutrality by 2030**. 14. **In-situ** (स्थान पर) = natural habitat (NP, sanctuary, BR, sacred grove). **Ex-situ** (स्थानांतरित) = outside habitat (zoo, botanical garden (वनस्पति उद्यान), seed bank (बीज बैंक)). Cryo-banks use **liquid nitrogen**. Biosphere reserves (जैवमंडल आरक्षित क्षेत्र) best join biodiversity with traditional human life / cultural diversity. 15. The **CBD (1992, Rio)** rests on three pillars: conserve, sustainable use, fair benefit-sharing. **Cartagena Protocol** = biosafety / LMOs. **Nagoya (2010)** = ABS. 16. India's **Biological Diversity Act, 2002** creates **NBA (Chennai, 2003)**, SBBs, BMCs, and **PBRs**. BMCs are central to Nagoya ABS, including collection fees. 17. On the IUCN Red List, **Threatened** = **VU + EN + CR**. **Endangered** = **EN** only. Red Data Book (रेड डेटा बुक) lists threatened species. A species not seen for **50 years** may be treated as extinct in older definitional stems. 18. An **endemic** species occurs only in one area. An **invasive alien** is non-native and harmful (**Prosopis juliflora** reduces native biodiversity). 19. Keystone / flagship / umbrella / indicator roles stay distinct. 20. International Day for Biological Diversity is **22 May** (earlier linked to **29 December**). UN **Decade on Biodiversity = 2011–2020**. 21. India's first **National Centre for Marine Biodiversity (NCMB)** is at **Jamnagar (Gujarat)**. 22. The **Kunming–Montreal GBF (2022)** includes the **30×30** target. CBD COP-10 = **Nagoya 2010**; COP-11 = **Hyderabad (हैदराबाद) 2012**. 23. Hotspot concept = **Norman Myers** (not BirdLife). **BirdLife International** identifies Important Bird and Biodiversity Areas. 24. **TRAFFIC** (WWF (डब्ल्यूडब्ल्यूएफ)–IUCN alliance, 1976) monitors wildlife trade (पण्याध्यक्ष) — not a UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) bureau. 25. India coral reefs: **Lakshadweep (लक्षद्वीप) (atolls), Gulf of Mannar, Gulf of Kachchh, Andaman (अंडमान) & Nicobar** — **not** Sundarbans (सुंदरबन) (mangroves). Coral bleaching (प्रवाल विरंजन) mainly from **warming seawater / global warming (भूमंडलीय तापन)**. 26. **Diclofenac** (डाइक्लोफेनाक) veterinary painkiller caused India's vulture crash. **Dugong** = herbivorous marine mammal (Schedule I); Gulf of Mannar / Palk. **Indian wild ass** = Rann of Kachchh. **Lion-tailed macaque** = Western Ghats (Kerala, Karnataka, Tamil Nadu (तमिलनाडु)). 27. India has **80 Ramsar sites**. **Tamil Nadu 16** (highest), **Uttar Pradesh (उत्तर प्रदेश) 10**, **Punjab 6**. **Montreux Record** lists Ramsar sites with adverse ecological change from human interference. 28. UP Terai (Dudhwa (दुधवा), Pilibhit) sits on the **Himalaya hotspot** fringe. Key UP species: tiger, rhino, barasingha, Gangetic dolphin; state bird **Sarus crane**. 29. Common UP invasives: water hyacinth, Parthenium, Lantana. 30. NBA HQ = **Chennai**. CBD ≠ Nagoya year. Threatened ≠ Endangered. 31. **Sea buckthorn** (Ladakh (लद्दाख) / cold Himalaya) checks erosion and is nutritious — not primarily a biodiesel (बायो-डीजल) timber crop. 32. **TEEB** makes nature's economic values visible for decision-makers. **Wetlands International** is an independent NGO (HQ Netherlands), not a Ramsar intergovernmental organ. 33. **CEBPOL** (Centre for Biodiversity Policy and Law) sits with NBA, **Chennai** (India–Norway). 34. Maximum butterflies in tropics because they **cannot tolerate low temperatures**. 35. **Gulf of Mannar Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र)** is often cited as India's richest marine biodiversity region (21 islands mosaic). 36. Afforestation does **not** cause biodiversity loss; road expansion, urbanisation, and agriculture expansion do (UKPCS-style stems).

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Genetic Diversity	**Species Diversity**	Variation of genes and alleles *within* a single species across its geographic range vs the variety and abundance of *different species* within a given ecological region	आनुवंशिक विविधता / प्रजाति विविधता
Species Diversity	**Ecosystem Diversity**	Number of species and their relative abundance (species richness + evenness) vs variety of distinct habitats, biotic communities, and ecological processes in a landscape	प्रजाति विविधता / पारिस्थितिक तंत्र विविधता
Alpha Diversity	**Beta Diversity**	Species richness measured *within* a particular local habitat or community vs rate of species turnover / change in species composition *between* two different habitats across an environmental gradient	अल्फा विविधता (स्थानीय) / बीटा विविधता (आवासीय परिवर्तन)
In-situ Conservation	**Ex-situ Conservation**	Protecting endangered species within their natural evolutionary habitats (National Parks, Sanctuaries, Biosphere Reserves) vs conserving species outside their natural habitats in artificial environments (Zoological Parks, Botanical Gardens, Gene Banks)	स्व-स्थाने संरक्षण (In-situ) / बाह्य-स्थाने संरक्षण

(Ex-situ) | | **Endangered (EN)** | **Critically Endangered (CR)** | Very high risk of extinction in the wild (e.g. Asiatic Lion, Tiger) vs extremely high imminent risk of extinction in the wild (e.g. Great Indian Bustard, Gharial, Hangul) | लुप्तप्राय (EN) / अति संकटग्रस्त (CR) | | **Endemic Species** | **Invasive Alien Species** | Native species strictly restricted to a specific unique geographic territory and found nowhere else (e.g. Lion-tailed Macaque in Western Ghats) vs non-native species introduced from outside that outcompete and displace indigenous wildlife (e.g. Lantana camara, Water Hyacinth) | स्थानिक प्रजाति (Endemic) / आक्रामक विदेशी प्रजाति (Invasive) | | **Biodiversity Hotspot** | **Hope Spot** | Terrestrial regions with exceptionally high endemic species richness under severe threat (≥1,500 endemic vascular plants and ≥70% primary vegetation lost; Norman Myers) vs marine protected areas critical to the health of global oceans (Sylvia Earle / Mission Blue) | जैव विविधता हॉटस्पॉट (स्थलीय) / होप स्पॉट (समुद्री) | | **People's Biodiversity Register (PBR)** | **Red Data Book** | Mandated under Biological Diversity Act 2002 prepared by local Biodiversity Management Committees (BMCs) documenting traditional ecological knowledge vs official compendium published by IUCN documenting global extinction risks | जन जैव विविधता रजिस्टर (PBR) / रेड डेटा बुक (IUCN) |

💡 📋 **Must-score facts — levels, hotspots, CBD, IUCN (Click to Expand)**

Concepts | Item | Lock | |------|------| | Term / populariser | Rosen **1985** / E.O. Wilson | | Alpha / Beta / Gamma | Local / between ecosystems / regional | | Genetic / Species / Ecosystem | Within species / among species / habitats | | Richest land / sea | Tropical rainforest / coral reefs | | India land vs species | ~**2.4%** land; ~**7–8%** species | | 4 Indian hotspots | Himalaya · Indo-Burma · WG–SL · Sundaland | | Andaman / Nicobar | Indo-Burma / Sundaland | | Hotspot criteria | ≥1500 endemic plants + ≥70% habitat lost; **36** global | ### Law / lists | Item | Lock | |------|------| | In-situ / Ex-situ | Natural habitat / outside (zoo, seed bank) | | CBD / Cartagena / Nagoya | 1992 Rio / biosafety LMOs / ABS **2010** | | BDA / NBA | **2002** / NBA **Chennai 2003** | | Threatened (IUCN) | **VU + EN + CR** (Endangered ≠ all Threatened) | | Biodiversity Day | **22 May** | | Kunming–Montreal | **2022**; 30×30 | | India Ramsar | **80** sites; TN highest; UP **10** class fact | | Hotspot concept | Norman Myers (≠ BirdLife) |

4.1 Biodiversity – Meaning

Identity: CBD Art 2 | Three nested levels | Rosen coined · Wilson popularised

- **Biodiversity** is the variety and variability of life at **genetic**, **species**, and **ecosystem** (पारिस्थितिकी तंत्र) levels in a given area — the diverse range of plants and animals in an ecosystem.
- The **Convention on Biological Diversity (CBD, 1992)** defines it as variability among living organisms from all sources, including terrestrial, marine, and aquatic ecosystems.
- **Walter G. Rosen (1985)** coined the word *biodiversity* (from "biological diversity") for a National Research Council seminar title. **E.O. Wilson** popularised the term thereafter.
- Biodiversity is **not** just a species count. Genes within populations and the variety of habitats also matter.
- The most significant aspect of biodiversity is **maintenance of the ecosystem**, ahead of food, drugs, or industrial use alone.
- **International Day for Biological Diversity** is **22 May** (UN moved it from the older **29 December** CBD entry-into-force date). The UN declared **2011–2020** the **Decade on Biodiversity**.
- India's first **National Centre for Marine Biodiversity (NCMB)** is at **Jamnagar (Gujarat)**.

> **Logic:** "Biodiversity = only number of species" is false — all three CBD/NCERT levels are required. Rosen coined; Wilson popularised — do not reverse.

4.2 Levels of Biodiversity

Identity: Genetic | Species | Ecosystem | Whittaker α–β–γ

Genetic diversity

- **Genetic diversity** is variation in genes and alleles **within** a species.
- It gives populations resistance to disease and ability to adapt to environmental change.
- Crop **landraces** and wild relatives of rice, mango, and pulses are India's genetic wealth.

Species diversity

- **Species diversity** depends on **species richness** (how many species) and **evenness** (how evenly individuals are spread).
- **R.H. Whittaker (1960)** advanced the **alpha–beta–gamma** scale for species diversity.
- **Alpha diversity** (अल्फा विविधता) is the **mean species diversity in habitats at a local scale**.
- **Beta diversity** (बीटा विविधता) is the comparison / change of species **between** ecosystems.
- **Gamma diversity** is overall diversity across different ecosystems in a **region**.
- Species diversity is highest in the **tropics** and generally **decreases toward the poles** (latitudinal diversity gradient). Along mountains, diversity is normally greater at **lower altitudes** than at higher ones.

Ecosystem diversity

- **Ecosystem diversity** is the variety of habitats and ecosystems in a region such as forests, wetlands, grasslands, and deserts.
- India spans **10 biogeographic zones**, so ecosystem diversity is high despite moderate (नरम दल) land area.
- On land, **tropical rainforest** (about **23.5°N–23.5°S**) is the richest biome — often called the **optimum biome**. In the sea, **coral reefs** rank among the richest systems; both can be accepted when a stem asks for relatively higher diversity.

> **Logic:** "Species diversity increases from equator toward poles" is **false**. Alpha = local mean; beta = between ecosystems; gamma = regional total.

4.3 Global & Indian Biodiversity

Identity: ~8.7M species estimated | 17 megadiverse nations | India 2.4% land, 7–8% species

- About **8.7 million** species may exist on Earth (Mora et al., 2011). Only about **1.8 million** are described so far.
- The **tropics** hold the greatest species diversity because of long evolutionary history (इतिहास), stable climate, and high productivity.
- **17 megadiverse countries** together hold most of the world's species. **India** is one of them.
- **Russia is not** on the standard megadiverse list of 17.

Region	Megadiverse countries (17)
South America	Brazil, Colombia, Ecuador, Peru, Venezuela
North / Central America	Mexico, United States
Asia	China, India, Malaysia, Indonesia, Philippines
Africa	Democratic Republic of Congo, South Africa, Madagascar
Oceania	Australia, Papua New Guinea

- India covers about **2.4%** of the world's land but holds roughly **7–8%** of recorded global species.

- India has about **45,000** plant species and **91,000+** animal species (NCERT order of magnitude).

Group	India's rough share of global species
Mammals	~**7.6%**
Birds	~**12.6%**
Reptiles	~**6.2%**
Amphibians	~**4.4%**
Indian endemism centre	Note
----------	------
Western Ghats (पश्चिमी घाट)	Richest among common Indian options
Northeast India	High endemism belt
Himalaya (हिमालय)	Richness from confluence of biogeographic zones
Andaman & Nicobar Islands	Island endemism

- Among common Indian options, the **Western Ghats** is the most biodiversity-rich region.
- **Silent Valley (Kerala, Palakkad)** is remnant moist evergreen rainforest of the Western Ghats.
- Silent Valley was declared a **National Park in 1984** (inaugurated **1985**).
- The **Silent Valley Project** was the Kerala hydel controversy linked to this forest.
- Among named Indian valleys in older stems, **Silent Valley** (साइलेंट वैली) is the richest.
- The **Valley of Flowers** lies in **Chamoli, Uttarakhand**.
- Himalayan species richness is best explained as a **confluence of different biogeographic zones**, not rainfall alone.

> **Logic:** Western Ghats beats Gangetic Plain, Trans-Himalaya, and Central India for species richness. Silent Valley = Kerala, not Uttarakhand.

PYQ — UPPCS Prelims 2019, Q139

The most biodiversity rich area in India is

A. Gangetic plain

B. Trans Himalayas

C. Western Ghats

D. Central India

💡 **Show answer**

Logic: Stem tests India's richest mainland biodiversity region among standard distractors — Gangetic Plain, Trans-Himalaya, and Central India are species-poorer than the Western Ghats hotspot. **Ans: C.** The **Western Ghats** hold the highest species richness among these options. Gangetic Plain, Trans-Himalaya, and Central India are classic weaker distractors. Silent Valley (Kerala) sits inside this Western Ghats richness belt — do not confuse it with Valley of Flowers (Chamoli, Uttarakhand).

4.4 Importance & Biodiversity–Environment Link

Identity: Direct + indirect + option values | Ecosystem maintenance first

- The most significant aspect of biodiversity is **maintenance of the ecosystem**.

- **Direct values** include food, fuel, fibre, medicine, and timber taken from nature.
- **Indirect values** include soil formation, prevention of soil erosion, recycling of waste, pollination of crops, climate regulation, and flood control.
- **Ethical and aesthetic values** cover cultural, spiritual, and recreational benefits.
- **Option value** is the potential future use of species and genes not yet exploited.
- Higher biodiversity often means **greater ecosystem stability** and resilience after disturbance.
- Biodiverse systems recover faster from drought (सूखा), pests, and pollution shocks.
- Maximum butterfly species occur in **tropical** countries because butterflies **cannot tolerate low temperatures**.

> **Logic:** Food/drugs matter, but stems asking the "most significant aspect" key **ecosystem maintenance**. Soil formation + pollination + waste recycling are classic multi-statement true sets.

4.5 Biodiversity Assessment & Registers

Identity: IUCN Red List | BDA 2002 | PBR + BMC

IUCN Red List categories

Category	Meaning
LC	Least Concern
NT	Near Threatened
VU	Vulnerable
EN	Endangered
CR	Critically Endangered
EW	Extinct in the Wild
EX	Extinct

- **Threatened species** = **VU + EN + CR**.
- **Endangered** means **EN only**, not all threatened categories.
- **DD** (Data Deficient) and **NE** (Not Evaluated) mean insufficient data for a full assessment.
- **IUCN** (आईयूसीएन) is headquartered in **Gland, Switzerland**. It assesses species status but does **not** regulate international trade (**CITES** (साइट्स) does that).
- India's **Red Data Book** lists threatened flora (वनस्पति) (BSI) and fauna (प्राणीजात) (ZSI) separately.
- The **Living Planet Index (LPI)** tracks global vertebrate population trends. **WWF** publishes it.
- **IPBES** (Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services) is headquartered in **Bonn, Germany**. It gives biodiversity assessment reports similar to IPCC (आईपीसीसी) for climate.

People's Biodiversity Register (India)

- The **Biological Diversity Act, 2002** implements the CBD in India.
- The **National Biodiversity Authority (NBA)** (राष्ट्रीय जैव विविधता प्राधिकरण) is in **Chennai**.
- **State Biodiversity Boards (SBB)** and **Biodiversity Management Committees (BMC)** operate at state and local levels.
- A **People's Biodiversity Register (PBR)** (जन जैव विविधता रजिस्टर) documents local biodiversity and traditional knowledge at the panchayat level.
- The **Nagoya Protocol (2010)** sets rules for **Access and Benefit Sharing (ABS)** of genetic resources under the CBD.

- **Biodiversity Management Committees (BMCs)** are central to Nagoya-style ABS; they may levy **collection fees** on access to biological resources in their jurisdiction.
- **CEBPOL** (Centre for Biodiversity Policy and Law), with Norway, sits with the NBA in **Chennai**.

> **Logic:** NBA HQ is **Chennai**, not New Delhi. **Threatened ≠ Endangered**. NBA does **not** directly run GEAC genetic-modification supervision.

4.6 Biodiversity Loss

Identity: Habitat destruction #1 | HIPPO | Jhum · invasives · trophic levels

- **Habitat loss and fragmentation** is the **leading direct cause** of biodiversity loss worldwide.
- **Habitat fragmentation** splits large habitats into small patches. Species face edge effects, inbreeding, and local extinction.
- **Over-exploitation** through poaching, overfishing, and unsustainable harvesting removes species faster than they reproduce.
- **Invasive alien species** outcompete or prey on natives. **Water hyacinth** choking UP ponds and **Prosopis juliflora** wiping native flora are classic examples.
- **Pollution** kills pollinators, poisons aquatic life, and degrades breeding habitats.
- **Climate change (जलवायु परिवर्तन) / global warming** shifts ranges faster than many species can move or adapt.
- **HIPPO** summarises major drivers: **H**abitat loss, **I**nvasive species, **P**ollution, **P**opulation pressure, **O**ver-exploitation.
- **Shifting cultivation (jhum)** in eastern and northeastern India degrades habitats when practised intensively.
- India's bird populations shrink mainly from **habitat loss** and **pesticides / chemical fertilisers / mosquito repellents** (e.g. **DDT** eggshell thinning) — not from large-scale use of bio-pesticides.
- **Promotion of vegetarianism** is **not** a biodiversity threat. A **fewer number of trophic levels** indicates dominance of few organisms and **lowers** diversity — so it does **not** increase biodiversity.
- Afforestation of barren land is **not** a cause of biodiversity loss; road expansion, urbanisation, and agricultural extension are.
- Current extinction rates may be **100–1,000 times** the natural background rate.
- Scientists describe the present era as a **sixth mass extinction** driven mainly by human activity.
- The **species–area relationship** states that larger habitats generally support more species.
- **IPBES (2019)** reported that about **1 million** animal and plant species face extinction risk if trends continue.
- India aims for **Land Degradation Neutrality (LDN) (भूमि क्षरण तटस्थता) by 2030**.

> **Logic:** Destruction of **natural habitat** is the #1 loss factor. Pollution or climate change is not #1 in standard Prelims keys. Vegetarianism is a classic false threat option.

PYQ — UPPCS Prelims 2023, Q100

Which of the following is the **most important factor** responsible for the loss of biodiversity?

A. Genetic assimilation

B. Control of predators

C. Destruction of natural habitat

D. Pest control

💡 **Show answer**

Ans: C

4.7 Biodiversity Conservation

Identity: In-situ preferred | Ex-situ backup | Biosphere + culture | CBD 3 objectives | 30×30

- **In-situ conservation** (स्व-स्थाने संरक्षण) protects species in their **natural habitat**. National parks, wildlife sanctuaries, biosphere reserves, reserved/protected forests, nature reserves, and **sacred groves** (पवित्र उपवन) are examples.
- **Ex-situ conservation** (बाह्य-स्थाने संरक्षण) protects species **outside** natural habitat in zoos, aquaria, botanical gardens, seed banks, and captive breeding centres.
- Cryo-banks store seeds, pollen, tissue, or embryos in **liquid nitrogen** for long-term ex-situ storage.
- **In-situ is preferred** because whole ecosystems and natural selection processes stay intact.
- **Ex-situ** is a **supplementary** insurance for critically endangered species.
- **Ex-situ** examples include the **National Bureau of Plant Genetic Resources (NBPGR)** seed bank in New Delhi, zoos, and botanical gardens.
- The **most important strategy for biodiversity together with traditional human life / cultural diversity** is the establishment of **biosphere reserves** (UNESCO MAB logic).
- India has **18** biosphere reserves; a majority are on the UNESCO World Network of Biosphere Reserves. **Gulf of Mannar** is often cited as India's richest **marine** biodiversity biosphere mosaic (21 islands with reefs, seagrass, mangroves).
- India's coral reefs (~**5,790 km²**) lie mainly in **Lakshadweep (atolls)**, **Gulf of Mannar**, **Gulf of Kachchh**, and **Andaman & Nicobar**. **Sundarbans** (सुंदरबन) is mangrove-dominated, **not** a coral-reef region. **Coral bleaching** is driven mainly by **rise in sea-water temperature / global warming**.
- The **Wildlife (Protection) Act, 1972** gives maximum protection to **Schedule I** species.
- The **CBD (1992)** has three objectives: **conserve** biodiversity, **use it sustainably**, and share benefits **fairly and equitably**.
- The **Cartagena Protocol on Biosafety** covers safe handling of **living modified organisms (LMOs)**; India implements it through **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय).
- The **Kunming–Montreal Global Biodiversity Framework (2022)** targets protecting **30%** of land and oceans by **2030** (30×30).
- CBD COP chronology: **COP-10 Nagoya (2010)**, **COP-11 Hyderabad (2012)**, COP-12 Korea, COP-13 Cancun, COP-14 Egypt (मिस्र), COP-15 Kunming–Montreal.
- **Project Tiger (प्रोजेक्ट टाइगर) (1973)** (प्रोजेक्ट टाइगर) uses the **umbrella/flagship** approach to protect forest ecosystems.
- **BirdLife International** is a global partnership of conservation organisations that identifies **Important Bird and Biodiversity Areas** — it did **not** invent biodiversity hotspots (**Norman Myers** did).
- **TRAFFIC** (Trade Records Analysis of Flora and Fauna in Commerce, **1976**) is a **WWF–IUCN** alliance monitoring wildlife trade — **not** a UNEP bureau.
- **TEEB** (The Economics of Ecosystems and Biodiversity) makes nature's economic values visible for decision-makers.
- **Wetlands International** is an independent global NGO (HQ **Netherlands**) working on wetlands — not an intergovernmental Ramsar organ.
- **UN-REDD+** (FAO–UNDP (यूएनडीपी)–UNEP) can protect biodiversity, forest resilience, and support poverty reduction when well designed.

> **Logic:** Botanical garden = ex-situ, not in-situ. Biosphere reserve wins stems that join biodiversity with traditional human life. Cryo gas = **nitrogen**.

PYQ — UPPCS Prelims 2023, Q118

In National Parks, genetic diversity is maintained through—

A. In-situ conservation

B. Ex-situ conservation

C. Gene pool

D. None of the above

> 💡 **Show answer**
>
> **Ans: A**

India's 4 Biodiversity Hotspots — Geographic Distribution & Norman Myers Criteria

4.8 Biodiversity Hotspots

Identity: Myers 1988 | ≥1500 endemic plants + ≥70% habitat lost | India has 4 of 36

- A **biodiversity hotspot** (जैव विविधता हॉटस्पॉट) is a region with **high endemism** and **severe habitat loss**.
- **Norman Myers (1988)** set the hotspot concept. Conservation International later updated the global list to **36 hotspots** (about **2.5%** of Earth's land; >**50%** of endemic plant species). The **36th** hotspot added via CEPF (2016) is the **North American Coastal Plain**.
- **Two formal criteria:** at least **1,500 endemic vascular plant species** and at least **70% loss** of original primary vegetation. Recognition (मान्यता) also stresses **species richness, endemism, and threat perception** — not vegetation density or ethnobotany alone.
- Hotspots are **not** confined to the tropics; temperate examples include the California Floristic Province and Japan.
- **India has four hotspots:**

Hotspot	Indian region	UP link
Himalaya (often called Eastern Himalaya in older stems)	Himalaya + Terai belt; Sikkim is a botanist's paradise here	Dudhwa, Pilibhit Terai
Indo-Burma / Indo-Myanmar	Northeast India + **Andaman Islands**	—
Western Ghats & Sri Lanka	West coast mountains	—
Sundaland	**Nicobar Islands** (not Andaman)	Not mainland

- **Eastern Ghats, Vindhyas, Aravallis, Gangetic Plain, Naro Hills, and Silent Valley alone** are **not** global hotspots (Silent Valley sits inside the Western Ghats hotspot).
- The **Gangetic Plain** is fertile and species-rich in places but is **not** a global biodiversity hotspot.

> **Logic:** Hotspot needs **endemism + threat**. Andaman ≠ Nicobar for hotspot mapping. Western Himalayas / Eastern Ghats are classic wrong options.

PYQ — UPPCS Prelims 2024, Q33

Consider the following statements regarding biodiversity hotspots:

1. They have high species endemism.

2. They have experienced significant habitat loss.

A. Only 1

B. Only 2

C. Both 1 and 2

D. Neither 1 nor 2

💡 **Show answer**

Ans: C

4.9 Species Categories

Identity: Nine roles | IUCN status vs ecological role

Category	One-line fact	India example
Endemic	Found **nowhere else** in the world	Asiatic lion (सिंह) (Gir (गीर)), lion-tailed macaque (Western Ghats)
Endangered	IUCN **EN** — very high extinction risk	Gangetic dolphin, Indian vulture
Threatened	**VU + EN + CR** together	Broader than endangered alone
Extinct	No surviving individuals (**EX**) or wild population (**EW**)	Cheetah in India (1952 wild extinction)
Invasive alien	Non-native + ecological or economic harm	Water hyacinth, lantana, Parthenium
Keystone	Disproportionately large ecosystem impact if removed	Sea otter (kelp forest), tiger (apex regulation)
Flagship	Charismatic species that mobilises public support	Tiger, elephant, one-horned rhino
Umbrella	Protecting its large range saves many other species	Tiger reserve network
Indicator	Reflects environmental quality or change	**Lichens** (air pollution), frog (wetland health)

- An **endemic species** is found only in one defined geographic area. The Asiatic lion occurs only in Gir, Gujarat.
- An **endangered species** faces a very high risk of extinction in the wild under IUCN category **EN**.
- A **threatened species** is any species in **VU**, **EN**, or **CR** combined. It is a wider group than endangered alone.
- An **extinct species (EX)** has no surviving individuals. **Extinct in the Wild (EW)** survives only in captivity.
- An **invasive alien species** is non-native and harms native ecosystems or the economy. Water hyacinth chokes UP wetlands.
- A **keystone species** has ecosystem impact far greater than its numbers suggest. Robert **Paine (1969)** showed this in starfish experiments.
- A **flagship species** is a charismatic animal used to raise public support and funds for conservation, such as the tiger on campaign logos.
- An **umbrella species** needs a very large home range, so protecting it also protects many other species in the same habitat.
- An **indicator species** shows habitat or pollution conditions. Lichens fail in polluted air and signal air quality decline.
- Older definitional stems treat a species as **extinct** if not seen in its natural habitat for **50 years**.
- **Diclofenac sodium** (veterinary NSAID) caused India's vulture crash via contaminated carcasses.
- The **dodo** of Mauritius helped disperse **Tambalacoque** seeds; plant failure followed bird extinction.

- **Indian flying fox** is a **bat** (fruit bat), not a bird.
- **Dugong** is a **herbivorous marine mammal** ("sea cow"), Schedule I, mainly Gulf of Mannar / Palk / parts of Gujarat and A&N — **not** the entire Indian coast.
- **Indian wild ass** lives in the sandy saline (लवणीय) **Rann of Kachchh**; threat is habitat destruction, not predators.
- **Lion-tailed macaque** is endemic to the **Western Ghats** (Kerala, Karnataka, Tamil Nadu).
- **Cheetah** was declared extinct in India in **1952** (reintroduction is a later policy story).
- **Red panda** occurs in the eastern Himalayan belt (Sikkim, West Bengal, Meghalaya (मेघालय), Arunachal Pradesh, etc.) — **not** western Himalayas only. **Slow loris** lives in dense forests of the Northeast.
- **Shrew and tapir** are Himalayan-range animals, **not** Malabar Western Ghats.
- **Salt-water crocodile** occurs in the Andaman and Nicobar Islands.
- **Sea buckthorn** grows in cold deserts of **Ladakh** (लद्दाख) / Lahaul–Spiti and related Himalayan tracts; it checks erosion and is nutritious — not mainly a biodiesel/timber crop.
- **Prosopis juliflora** is an invasive exotic that reduces native biodiversity where it spreads.
- **Sequoia (Redwood)** of California is the tallest known living tree. Deep sea has **minimum** annual biomass production among common biome options.
- Hibernation occurs in **bats, bears, and many rodents**. Author of *Tomorrow's Biodiversity* is **Vandana Shiva**.
- **Bhitarkanika** (Odisha) sits where **Brahmani** (South Koel in Jharkhand) and **Vaitarni** meet near the Bay of Bengal.

Ramsar wetlands (biodiversity overlap)

- **Ramsar Convention (1971, Iran)** conserves wetlands of international importance under **Wise Use**. **World Wetlands Day (विश्व आर्द्रभूमि दिवस) = 2 February**.
- India has about **100** Ramsar sites. State counts often tested: **Tamil Nadu 16** (highest), **Uttar Pradesh 10**, **Punjab 6**. Bihar's first is **Kabartal / Kanwar Lake (Begusarai)**.
- **Montreux Record** lists Ramsar sites where ecological character has changed or is likely to change from human interference.
- Wetlands act as "kidneys" mainly by aquatic plants absorbing **heavy metals and excess nutrients**; they also cycle nutrients, hold sediment, and buffer floods.
- **Sambhar Lake (Rajasthan)** is India's largest inland saline wetland. Wetland Rules **2010/2017** and catchment coverage are Indian law — not every wetland worldwide is a mandatory Ramsar obligation.

> **Logic:** Lichens are pollution indicators because they **fail** in polluted air. Flying fox = bat. Dugong = mammal. Sundarbans ≠ coral reef. Ramsar count and TN/UP ranks move with notifications — verify live MoEFCC list before the paper year.

Complete PYQ Bank (Topic 4)

Q1. UPPCS Prelims 2024, Q33

Consider the following statements regarding biodiversity hotspots:

1. They have high species endemism.
2. They have experienced significant habitat loss.

A. Only 1

B. Only 2

C. Both 1 and 2

D. Neither 1 nor 2

💡 **Show answer**

Ans: C

Q2. UPPCS Prelims 2023, Q100

Which of the following is the **most important factor** responsible for the loss of biodiversity?

A. Genetic assimilation

B. Control of predators

C. Destruction of natural habitat

D. Pest control

💡 **Show answer**

Ans: C

Q3. UPPCS Prelims 2023, Q118

In National Parks, genetic diversity is maintained through—

A. In-situ conservation

B. Ex-situ conservation

C. Gene pool

D. None of the above

💡 **Show answer**

Ans: A

Q4. UPPCS Prelims 2019, Q139

The most biodiversity rich area in India is

A. Gangetic plain

B. Trans Himalayas

C. Western Ghats

D. Central India

💡 **Show answer**

Ans: C

Q5. UPPCS Prelims 2019, Q58

Which of the following statement is NOT true?

A. Species diversity increases from equator towards poles

B. Tropics harbour more species than temperate areas

C. Greatest biodiversity occurs in Amazon rainforest

D. Species diversity decreases from temperate areas towards poles

💡 **Show answer**

Ans: A

Q6. UPPCS Prelims 2018, Q50

When the world biodiversity Day is observed?

A. March 22

B. May 22

C. June 23

D. April 16

💡 **Show answer**

Ans: B

Q7. UPPCS Prelims 2025, Q61

With reference to lichens, which of the following statements is/are correct?

1. Lichens are very good indicators of pollution.
2. Lichens grow well in polluted areas.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Ans: D — Indicator species fact; succession indicators also sit in the Ecology & Ecosystem chapter.

Q8. UPSC Prelims 2012 — pattern

How many biodiversity hotspots are located in India?

A. 2

B. 4

C. 6

D. 8

💡 **Show answer**

Ans: B

Q9. UPSC Prelims 2014 — pattern

Which is the most effective strategy for biodiversity conservation?

A. Ex-situ only

B. In-situ protected areas

C. Captive breeding only

D. Commercial farming

💡 **Show answer**

Ans: B

Q10. UPSC Prelims 2015 — pattern

Lion-tailed macaque is endemic to:

A. Himalaya

B. Western Ghats

C. Sundarbans

D. Thar Desert

💡 **Show answer**

Ans: B

Extra Drill — Ghatnachakra-pattern (Q31–Q40)

Logic (set): These stems recycle definition, hotspot geography, conservation method, species identity, and Ramsar traps from the Biodiversity chapter map.

Q31. Who first used the term 'biodiversity'?

A. E.O. Wilson B. Norman Myers C. Walter G. Rosen D. A.G. Tansley

💡 **Show answer**

Ans: C. Walter G. Rosen coined it in **1985**; Wilson popularised it later.

Q32. Among mathematical indices, which shows mean species diversity in a habitat at a local scale?

A. Alpha B. Beta C. Gamma D. None

💡 **Show answer**

Ans: A. Alpha = local; beta = between ecosystems; gamma = regional.

Q33. India's first National Centre for Marine Biodiversity is at:

A. Mumbai B. Jamnagar C. Bhavnagar D. Puducherry

💡 **Show answer**

Ans: B. NCMB is at **Jamnagar, Gujarat**.

Q34. Which statement is **NOT** true?

A. Tropics harbour more species than temperate areas. B. Species diversity increases from equator toward poles. C. Greatest biodiversity occurs in Amazon rainforest. D. Species diversity decreases from temperate areas toward poles.

💡 **Show answer**

Ans: B. Diversity **increases toward the equator**, not toward the poles.

Q35. Which three criteria recognise Western Ghats–Sri Lanka and Indo-Burma as hotspots?

A. Species richness, endemism, threat perception B. Vegetation density, ethnobotany, warm-humid adaptation C. Only rainfall and altitude D. Only invasive-species count

💡 **Show answer**

Ans: A. Richness + endemism + threat — not density or ethnobotany alone.

Q36. Cryo-bank for ex-situ conservation commonly uses which gas?

A. Oxygen B. Carbon dioxide C. Nitrogen D. Methane

> 💡 **Show answer**
>
> **Ans: C.** Storage uses **liquid nitrogen**.

Q37. Which regions have coral reefs?

1. Andaman and Nicobar
2. Gulf of Kachchh
3. Gulf of Mannar
4. Sundarbans

A. 1, 2 and 3 only B. 2 and 4 only C. 1 and 3 only D. All four

> 💡 **Show answer**
>
> **Ans: A.** Sundarbans is mangrove country, not a coral-reef region.

Q38. With reference to TRAFFIC, which is correct?

A. It is a UNEP bureau. B. Its mission is to ensure wildlife trade is not a threat to nature. C. Both A and B D. Neither

> 💡 **Show answer**
>
> **Ans: B.** TRAFFIC is a **WWF–IUCN** alliance (1976), not UNEP.

Q39. India currently has about how many Ramsar sites, and which state has the most?

A. 26; Rajasthan B. 80; Tamil Nadu (16) C. 80; Uttar Pradesh (16) D. 50; Punjab

> 💡 **Show answer**
>
> **Ans: B.** About **80** sites; **Tamil Nadu 16**, UP **10**, Punjab **6**.

Q40. Assertion (A): Maximum butterfly species are found in tropical countries. Reason (R): Butterflies cannot tolerate low temperatures.

A. Both true; R explains A B. Both true; R does not explain A C. A true; R false D. A false; R true

> 💡 **Show answer**
>
> **A/R logic:** A tests tropical richness; R tests temperature intolerance as the causal link. **Ans: A.** Both true and R correctly explains A.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D

Q1. With reference to biodiversity, which of the following statements is/are correct?

1. Biodiversity spans only the species level, not genetic or ecosystem levels.
2. Walter G. Rosen (1985) coined the word biodiversity; E.O. Wilson popularised it.
3. The most significant aspect is only pharmaceutical drug discovery.

A. Only 3 B. Only 2 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Statement 2 is the Rosen/Wilson naming fact. Statement 1 is false — biodiversity includes genetic, species, and ecosystem levels. Statement 3 is too narrow (not only pharma).

Q2. Match List-I with List-II:

List-I	List-II
A. Alpha diversity	1. Overall regional diversity
B. Beta diversity	2. Local habitat mean diversity
C. Gamma diversity	3. Change between ecosystems

Code:

A. A-2, B-3, C-1 B. A-3, B-2, C-1 C. A-2, B-1, C-3 D. A-1, B-3, C-2

💡 **Show answer**

Ans: A. **Logic:** Alpha local; beta between; gamma regional.

Q3. Which of the following pairs is/are NOT correctly matched?

1. Genetic diversity — variation among different ecosystems only
2. Species diversity — variety among species
3. Ecosystem diversity — variety of habitats/ecosystems

A. 1 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. **Logic:** Genetic diversity is within a species — pair 1 is wrong.

Q4. With reference to India's biodiversity hotspots, which of the following statements is/are correct?

1. The four are Himalaya, Indo-Burma, Western Ghats–Sri Lanka, and Sundaland.
2. Andamans fall under Indo-Burma and Nicobar under Sundaland.
3. Eastern Ghats are one of India's four hotspots.

A. Only 1 B. 1 and 2 C. Only 3 D. 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2 are correct. **Logic:** Statements 1–2 match the four-hotspot map (Andaman → Indo-Burma; Nicobar → Sundaland). Statement 3 is false — Eastern Ghats are not one of India's four hotspots.

Q5. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): A biodiversity hotspot needs high endemism and severe habitat loss.

Reason (R): Hotspot criteria often cite about ≥1500 endemic vascular plants and ≥70% original habitat lost.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. **A/R logic:** Both true; R states the quantitative criteria that define A.

Q6. Arrange richness ideas correctly:

1. Species richness generally increases toward the equator.
2. On land, tropical rainforest is among the richest biomes.
3. In the sea, coral reefs are among the richest systems.

A. Only 1 B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. **Logic:** All three are standard.

Q7. With reference to in-situ and ex-situ conservation, which of the following statements is/are correct?

1. In-situ protects biodiversity in the natural habitat.
2. Ex-situ includes zoos, seed banks, and botanic gardens.
3. A National Park is an ex-situ method.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. **Logic:** National Park is in-situ.

Q8. Match List-I with List-II:

List-I	List-II
A. CBD	1. Biosafety / LMOs
B. Cartagena Protocol	2. ABS
C. Nagoya Protocol	3. 1992 Rio biodiversity convention

Code:

A. A-3, B-1, C-2 B. A-1, B-3, C-2 C. A-3, B-2, C-1 D. A-2, B-1, C-3

💡 **Show answer**

Ans: A. **Logic:** CBD 1992; Cartagena biosafety; Nagoya ABS.

Q9. Consider the following statements:

1. On IUCN Red List, Threatened = VU + EN + CR.
2. Endangered alone equals the full Threatened set.
3. India's Biological Diversity Act is 2002 with NBA at Chennai.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** Endangered ≠ all Threatened.

Q10. Which of the following pairs is/are correctly matched?

1. Silent Valley — Kerala
2. Valley of Flowers — Chamoli, Uttarakhand
3. Hotspot concept — BirdLife International

A. 2 and 3 only B. 1 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. **Logic:** Hotspot concept = Norman Myers — not BirdLife.

Q11. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Habitat destruction is the top cause of biodiversity loss.

Reason (R): Vegetarianism is classified as a primary HIPPO threat driver.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. **A/R logic:** A true; vegetarianism is not a HIPPO threat — R false.

Q12. With reference to India land/species share, which of the following statements is/are correct?

1. India holds about 2.4% of world land but about 7–8% of global species.
2. India aims for Land Degradation Neutrality by 2030.
3. India has only one global biodiversity hotspot.

A. 2 and 3 only B. 1 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. **Logic:** India has four hotspots — statement 3 false.

Q13. Arrange the following in correct chronological order:

1. CBD opened at Rio
2. Nagoya Protocol on ABS
3. Kunming–Montreal Global Biodiversity Framework

A. 2-1-3 B. 1-3-2 C. 1-2-3 D. 3-1-2

💡 **Show answer**

Ans: C. **Logic:** 1992 → 2010 → 2022.

Q14. Which of the following coral areas is/are correctly listed for India?

1. Gulf of Mannar
2. Lakshadweep
3. Sundarbans mangrove belt as a coral-reef hotspot
4. Gulf of Kutch

A. 1, 2 and 3 only B. 1, 2 and 4 only C. 2, 3 and 4 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: B. **Logic:** Sundarbans = mangroves — not coral reef list.

Q15. With reference to Ramsar, which of the following statements is/are correct?

1. India has about 80 Ramsar sites in recent teaching counts.
2. Tamil Nadu often leads state counts; Uttar Pradesh is also among high-count states.
3. Montreux Record lists sites with adverse ecological change from human interference.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. **Logic:** All three are teaching facts.

Q16. Match List-I with List-II:

List-I	List-II
A. Dugong	1. Western Ghats
B. Lion-tailed macaque	2. Gulf of Mannar / Palk teaching
C. Indian wild ass	3. Rann of Kachchh

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. **Logic:** Dugong marine herbivore; LTM Western Ghats; wild ass Kachchh.

Q17. Which of the following pairs is/are NOT correctly matched?

1. NBA headquarters — Chennai
2. International Day for Biological Diversity — 22 May
3. TRAFFIC — UNEP bureau for climate finance

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** TRAFFIC is WWF–IUCN wildlife trade monitor — not UNEP climate finance.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Endemic species occur only in a restricted area.

Reason (R): Prosopis juliflora is an invasive alien that can reduce native biodiversity.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. **A/R logic:** Both true but R does not explain endemism.

Q19. With reference to UP biodiversity fringe facts, which of the following statements is/are correct?

1. UP Terai (Dudhwa, Pilibhit) sits on the Himalaya hotspot fringe.
2. Sarus crane is the state bird of Uttar Pradesh.
3. Water hyacinth, Parthenium and Lantana are common UP invasives teaching.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. **Logic:** All three correct.

Q20. How many of the following are biodiversity threats under HIPPO framing?

Habitat destruction, Invasive species, Pollution, Population pressure, Over-exploitation

A. All five B. Only two C. Only three D. Only four

💡 **Show answer**

Ans: A. **Logic:** All five letters of HIPPO.

Q21. With reference to cryo-banks, which of the following statements is/are correct?

1. Cryo-banks use liquid nitrogen.
2. Ex-situ methods can include cryopreservation.
3. Cryo-banks replace the need for any in-situ protected areas.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** Ex-situ is backup — does not replace in-situ.

Q22. Which one of the following is India's first National Centre for Marine Biodiversity (NCMB) location teaching?

A. Kochi B. Jamnagar (Gujarat) C. Port Blair D. Chennai

💡 **Show answer**

Ans: B. **Logic:** Jamnagar, Gujarat.

Q23. Consider the following statements:

1. UN Decade on Biodiversity was 2011–2020.
2. Kunming–Montreal GBF includes the 30×30 target.
3. CBD COP-11 was held at Hyderabad in 2012.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. **Logic:** All three correct.

Q24. Which of the following pairs is/are correctly matched?

1. Sea buckthorn — Ladakh / cold Himalaya erosion check + nutrition
2. Sea buckthorn — primarily a coastal mangrove tree of Sundarbans
3. Maximum butterflies in tropics — linked to intolerance of low temperatures

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. **Logic:** Pair 2 is wrong.

Q25. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Afforestation does not cause biodiversity loss in standard threat lists.

Reason (R): Road expansion, urbanisation, and agriculture expansion can cause biodiversity loss.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. **A/R logic:** Both true; R does not explain A (different point).

Q26. With reference to Biological Diversity Act institutions, which of the following statements is/are correct?

1. NBA sits at national level; SBBs at state; BMCs at local level.
2. BMCs prepare People's Biodiversity Registers (PBRs).
3. NBA headquarters is New Delhi.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** NBA HQ is Chennai — not Delhi.

Q27. Arrange hotspot / protocol chronology:

1. Myers hotspot concept era (1980s teaching)
2. CBD 1992
3. Nagoya 2010

A. 2-1-3 B. 1-2-3 C. 1-3-2 D. 3-2-1

💡 **Show answer**

Ans: B. **Logic:** Hotspot concept → CBD → Nagoya.

Q28. Which of the following statements about coral bleaching is/are correct?

1. Bleaching is mainly linked to warming seawater / global warming stress.
2. Zooxanthellae expulsion is part of the bleaching story.
3. Bleaching is caused only by freshwater flooding of the Sahara.

A. 2 and 3 only B. 1 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. **Logic:** Statement 3 is false.

Q29. With reference to keystone / flagship / umbrella / indicator roles, which statement is correct?

A. All four terms mean exactly the same thing B. The roles stay conceptually distinct in teaching C. Flagship means only invasive aliens D. Indicator species never reflect ecosystem health

💡 **Show answer**

Ans: B. **Logic:** Roles remain distinct.

Q30. Which of the following UP / India traps is/are correctly stated?

1. Threatened ≠ Endangered alone
2. NBA HQ = Chennai
3. CBD year = Nagoya year

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** CBD 1992 ≠ Nagoya 2010.

Mains Answer Framework

Biodiversity loss and conservation in India (250 words): Define three-level biodiversity. State habitat loss as primary driver (HIPPO). India as megadiverse nation with four hotspots. In-situ vs ex-situ with Indian examples (Dudhwa, NBPGR seed bank). CBD, BDA 2002, Nagoya ABS, 30×30 target. End with UP Terai/Himalaya hotspot and Project Tiger umbrella model.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q33

Which of the following statements about causes of biodiversity loss is/are correct?

1. Habitat loss and fragmentation are the most important cause driving animals and plants to extinction.
2. Human activities lead to over-exploitation of natural resources.
3. When a species becomes extinct, the plant and animal species associated with it do not become extinct.

A. Only 3 B. Only 1 C. Both 1 and 2 D. Only 2

💡 **Show answer**

Logic: Statement 3 denies co-extinction — that is the planted error. **Ans: C (Series B provisional key).** Statements 1 and 2 are correct. Statement 3 is wrong because associated species can disappear with their host (co-extinction).

Q. UKPCS Prelims 2025, Q97

The word bio-diversity was coined by

A. W. G. Rosen B. Tansley C. Darwin D. David

💡 **Show answer**

Logic: Tansley and Darwin belong to other concepts. **Ans: A (Series B provisional key).** Walter G. Rosen coined biodiversity in 1985. Tansley = ecosystem; Darwin = natural selection.

Common Traps — Don't Fall For These

Trap	Correct fact
Biodiversity = species count only	**Genetic + species + ecosystem**
Wilson coined "biodiversity"	**Walter G. Rosen (1985)** coined; Wilson popularised
Alpha = regional diversity	Alpha = **local** mean; gamma = regional
Russia is megadiverse	**Not** in the standard 17 list
India has 6 hotspots	**4** — Himalaya, Indo-Burma, WG–Sri Lanka, Sundaland
Sundaland = Andaman + Nicobar	**Nicobar** under Sundaland; **Andaman** under Indo-Burma
Eastern Ghats / Vindhyas = hotspot	**Not** hotspots
Hotspots only in tropics	Temperate hotspots also exist
Threatened = Endangered	Threatened = **VU + EN + CR**
Botanical garden = in-situ	**Ex-situ**

Trap	Correct fact
Cryo-bank gas = oxygen / CO_2	**Liquid nitrogen**
Lichens thrive in pollution	**Sensitive** indicators — fail in pollution
Ex-situ preferred over in-situ	**In-situ preferred**
NBA headquarters Delhi	**Chennai**
BirdLife invented hotspots	**Norman Myers**; BirdLife = IBAs
TRAFFIC = UNEP bureau	**WWF–IUCN** NGO alliance
Coral reefs in Sundarbans	Sundarbans = **mangroves**; reefs = GoM, GoK, A&N, Lakshadweep
Vegetarianism threatens biodiversity	**Not** a threat
NCMB = Mumbai / Bhavnagar	**Jamnagar**
Silent Valley = Uttarakhand	**Kerala**; Valley of Flowers = Chamoli UK
India Ramsar ≈ 26 forever	About **80**; TN **16**, UP **10**, Punjab **6**
Gangetic Plain is a hotspot	**Not** a global hotspot

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 5 — Habitat (वास स्थान), Flora (वनस्पति) & Fauna (प्राणीजात)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Habitat | Habitat Conservation | Flora and Fauna

Sources baked in: NCERT Class 12 Biology, BSI/ZSI, MoEFCC, NUHHP 2007, MISHTI (Budget 2023–24), UPPCS PYQs 2018–2025 **Weight:** ★★ Medium–High — habitat definitions, MISHTI/NUHHP, BSI/ZSI, UP fauna **Last verified:** August 2026

Current Affairs (this topic)

- Habitat-loss and corridor news often refresh Terai (तराई) / Himalaya (हिमालय) colour for UP–UK.
- Do not swap endemic vs endangered wording when CA names a new Red List update.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. A **habitat** (वास स्थान) is the physical **place** where an organism lives. A **niche** is the **role**. An **ecosystem** (पारिस्थितिकी तंत्र) is the functional unit. **Environment** (पर्यावरण) is the total surroundings. 2. Habitats are commonly classed as **terrestrial, aquatic, or transitional** (mangrove/wetland). Transitional habitats are not purely terrestrial. 3. Adaptation labels include **arboreal** (trees), **fossorial** (burrows), **cursorial** (open plains), and **xericole** (desert). 4. **Habitat loss** is the number-one biodiversity threat. **Fragmentation** splits patches; **corridors** reconnect them. 5. **MISHTI** is the Mangrove Initiative for Shoreline Habitats & Tangible Income from Budget **2023–24** under MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय). It is **not** a 2019 scheme. 6. **NUHHP** is the National Urban (नगरीय) Housing and Habitat Policy of **2007** (MoHUA). 7. Habitat–scheme chronology: **JNNURM 2005 → NUHHP 2007 → AMRUT 2015 → Jal Jeevan Mission 2019**. 8. **Flora** means plant life of a region. **Fauna** means animal life. Do not swap the two. 9. India has roughly **45,000** plant species and **91,000+** animal species on about **2.4%** of world land. 10. **BSI (1890, Kolkata)** surveys plants. **ZSI (1916, Kolkata)** surveys animals. Both sit in Kolkata; kingdoms differ. 11. Engler is linked to world floral regions. **C.O. Sauer** is linked to planting hearths in cultural geography. 12. A **microhabitat** (सूक्ष्म आवास) is a small subdivision inside a larger habitat. 13. In-situ (स्थान पर) conservation (स्व-स्थाने संरक्षण) protects species in natural habitat; ex-situ (स्थानांतरित) is outside habitat — same fact as biodiversity sheets. 14. Uttar Pradesh (उत्तर प्रदेश)'s richest wildlife habitat belt is the **Terai** (तराई) (Dudhwa (दुधवा), Pilibhit, Kishanpur). 15. UP state animal is **barasingha**. UP state bird is **Sarus crane**. 16. The **Terai Arc Landscape** corridor links UP Terai protected areas toward Nepal. 17. Aquatic fauna facts for UP include the **Gangetic dolphin** in the Ganga (गंगा)–Yamuna (यमुना) system. 18. **Water hyacinth** is a common invasive threat in UP village ponds. 19. Uttar Pradesh has **no mangrove habitat** purposes — it is landlocked for that category. 20. Habitat is an address; niche is a job description. Mixing the two loses marks on definition stems. 21. Wetlands and mangroves are **transitional**, not "purely terrestrial forest" options. 22. MISHTI is mangrove-focused coastal income and protection — not a Terai tiger scheme. 23. Jal Jeevan Mission (**2019**) is the latest fact in the urban/habitat chronology set above; do not date NUHHP as 2019. 24. Corridor questions for UP should start from Terai Arc links, not from Western Ghat mangrove corridors.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Habitat** | **Ecological Niche** | Physical geographic environment and specific spatial location where an organism resides vs functional status, behavioral role, and trophic interactions of that species in the community | आवास / पारिस्थितिक निकेत | | **Flora** | **Fauna** | Total native plant species assemblages inhabiting a particular geographic region or geological era vs total native animal species inhabiting a region or era | वनस्पति (फ्लोरा) / प्राणीजात (फॉना) | | **Microhabitat** | **Ecotone** | Small, localized physical niche within a broader habitat (e.g., under a rotting log, a tree cavity) vs macro-scale transitional boundary ecosystem between two adjacent distinct biomes (e.g., marsh between dry land and water) | सूक्ष्म आवास (माइक्रोहैबिटैट) / इकोटोन (संक्रमणिका) | | **Keystone Species** | **Foundation Species** | Species having a disproportionately massive impact on ecosystem structure relative to its biomass (loss triggers ecological collapse; e.g. Fig tree, Sea Otter) vs dominant primary producer that physically creates and structures the habitat (e.g. Corals, Kelp, Mangroves) | कीस्टोन प्रजाति / आधारभूत (फाउंडेशन) प्रजाति | | **Umbrella Species** | **Flagship Species** | Wide-ranging species whose habitat protection automatically shields hundreds of co-occurring species (e.g. Tiger, Asian Elephant) vs charismatic, culturally iconic species chosen as marketing symbols to rally conservation campaigns (e.g. Giant Panda, Polar Bear) | छत्र (अंब्रेला) प्रजाति / प्रतीक (फ्लैगशिप) प्रजाति | | **Indicator Species** | **Invasive Species** | Species whose presence, absence, or abundance directly signals environmental stress or health (e.g. Lichens for SO_2 air pollution, Mayfly larvae for clean water) vs non-native species proliferating aggressively and threatening native ecology | जैव संकेतक प्रजाति / आक्रामक विदेशी प्रजाति | | **In-situ Conservation** | **Ex-situ Conservation** | On-site conservation in natural ecosystems (Sacred Groves, Wildlife Sanctuaries) vs off-site conservation under artificial human custody (Seed Banks, Tissue Culture Labs) | स्व-स्थाने संरक्षण / बाह्य-स्थाने संरक्षण |

💡 📋 **Must-score facts — habitat, flora–fauna, schemes (Click to Expand)**

Tags | Item | Lock | |---|---| | Habitat / Niche / Ecosystem | Place / role / functional unit | | Habitat classes | Terrestrial · aquatic · transitional | | Arboreal / Fossorial / Cursorial / Xericole | Trees / burrows / open plains / desert | | #1 threat | Habitat loss (+ fragmentation; corridors reconnect) | | Flora / Fauna | Plants / animals — do not swap | | India species share | ~45k plants; ~91k+ animals on ~2.4% land | ### Schemes | Item | Lock | |---|---| | MISHTI | Mangrove initiative; Budget **2023–24** (≠ 2019) | | NUHHP | Urban Housing & Habitat Policy **2007** | | Chronology | JNNURM 2005 → NUHHP 2007 → AMRUT 2015 → JJM 2019 |

5.1 Habitat

Identity: Place where species lives | Terrestrial / aquatic / transitional | Adaptation-based types

- A **habitat** is the natural place where a plant or animal normally lives and finds food, shelter, and breeding conditions.
- NCERT treats habitat as the **address** of a species. It answers the question “where does it live?”
- A **habitat** is not the same as a **niche**. The niche is the species **role** in the ecosystem.
- A **habitat** is also not the same as an **ecosystem**. An ecosystem includes all interactions between living and non-living parts of an area.
- **Terrestrial habitats** exist on land. Organisms that live on land are called **terrestrial habitants**.
- **Aquatic habitats** exist in freshwater or marine water. Rivers are lotic and lakes are lentic aquatic habitats.
- **Transitional habitats** such as wetlands, mangroves, and estuaries are neither purely terrestrial nor purely aquatic.
- **Abiotic factors** in a habitat include temperature, moisture, light, soil, pH, and salinity.
- **Biotic factors** include food availability, predators, competitors, and symbiotic partners.

- A **microhabitat** is a small unit within a larger habitat, such as bark crevices or the underside of a leaf.
- A **macrohabitat** is a landscape-scale habitat unit such as an entire forest, lake, or desert tract.
- A **biotope** is a European ecology (पारिस्थितिकी) term for an area with relatively uniform environmental conditions. Notes often treat it as similar to habitat.

Whittaker's Global Biome Classification Matrix

Major habitat types (classification)

Type	Meaning	India example
Terrestrial	Land-based	Terai sal forest, Thar (थार) desert
Aquatic freshwater	Ponds, lakes, rivers	Ganga, Loktak
Marine	Oceans and seas	Arabian Sea coral/ fisheries belt
Wetland	Seasonally or permanently waterlogged	Keoladeo (केवलादेव), UP Ramsar (रामसर) sites
Mangrove / coastal	Tidal saline (लवणीय) coast	Sundarbans (सुंदरबन)
Grassland / savanna (सवाना)	Grass-dominated open tract	Dudhwa grasslands
Alpine / montane	High-altitude cold zone	Himalayan snow leopard belt
Desert / arid	Low rainfall extreme temperature	Thar
Arboreal	Tree-dwelling	Lion-tailed macaque
Fossorial	Burrowing underground	Indian pangolin
Cursorial	Open plain runners	Blackbuck
Xericole	Desert-adapted	Camel, desert fox
Amphibious	Both land and water phases	Crocodile, bullfrog

- India's richest natural wildlife habitats in UP lie in the **Terai belt** at Dudhwa, Kishanpur, and Pilibhit.
- The **Bushmen** live in the Kalahari Desert.
- The **Eskimo** live in Arctic Canada.
- The **Pygmies** live in the Congo Basin.
- The **Masai** pastoralists live in **East Africa** in Kenya and Tanzania.

PYQ — UPPCS Prelims 2024, Q62

Plants and animals that live on land are called:

A. Abiotic

B. Aquatic habitants

C. Terrestrial habitants

D. Biotic

💡 **Show answer**

Ans: C

PYQ — UPPCS Prelims 2025, Q23

Which of the following pairs is/are NOT correctly matched?

(Tribe) — (Habitat)

1. Bushman — Kalahari Desert
2. Eskimo — Northern Canada

3. Pygmy — Congo Basin
4. Masai — West Africa

A. Only 1, 2 and 3

B. Only 4

C. Only 3 and 4

D. Only 1

💡 **Show answer**

Ans: B — Masai = **East Africa**, not West Africa.

5.2 Habitat Conservation

Identity: Protect the place first | Loss / fragmentation / degradation | MISHTI + NUHHP

- **Habitat conservation** protects the natural dwelling place of species so ecological processes can continue.
- Without habitat protection, species-level measures such as captive breeding alone cannot succeed long term.
- **Habitat loss** completely removes an area's ability to support native species.
- **Habitat fragmentation** breaks a large habitat into small isolated patches and blocks movement and gene flow.
- **Habitat degradation** reduces habitat quality without removing it entirely, such as river pollution affecting dolphin habitat.
- **In-situ conservation** protects species inside their natural habitat. National parks and wildlife sanctuaries are examples.
- **Wildlife corridors** link fragmented habitats. The **Terai Arc Landscape** connects Dudhwa with Katarniaghat across the Nepal border.
- **Habitat restoration** rebuilds damaged ecosystems through afforestation, wetland revival, and mangrove planting.

Key schemes and policies

Scheme / policy	Year	Ministry	Fact
MISHTI	Budget **2023–24**	MoEFCC	Mangrove shoreline habitat restoration + community income
NUHHP	**2007**	MoHUA	Urban housing and sustainable urban habitat
JNNURM	**2005**	MoHUA	Urban infrastructure renewal
AMRUT	**2015**	MoHUA	Urban water, sewerage, green spaces
Jal Jeevan Mission	**2019**	Jal Shakti (शक्ति)	Rural tap water connectivity
CAMPA	2009 / Act 2016	MoEFCC	Compensatory afforestation for diverted forest land
Green India Mission	2014	MoEFCC	Forest cover (वन आवरण) and habitat quality improvement under NAPCC

- **MISHTI** stands for **Mangrove Initiative for Shoreline Habitats and Tangible Income**.
- MISHTI targets coastal mangrove habitats for shoreline protection and blue carbon (नीला कार्बन) restoration.
- **NUHHP** addresses urban habitat quality through housing and green-space planning in cities.

PYQ — UPPCS Prelims 2025, Q93

With reference to Mangrove Initiative for Shoreline Habitats and Tangible Income (MISHTI), which of the following statements is/are correct?

1. It was initiated in 2019.
2. It aims to restore mangrove forests.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

> 💡 **Show answer**
>
> **Ans: A** — Only statement 2. Launch = **2023–24** Budget.

PYQ — UPPCS Prelims 2025, Q72

Consider the following and arrange them in the correct chronological order:

1. Atal Mission for Rejuvenation and Urban Transformation (AMRUT)
2. Jawaharlal Nehru (जवाहरलाल नेहरू) National Urban Renewal Mission (JNNURM)
3. Jal Jeevan Mission
4. National Urban Housing and Habitat Policy

A. 2, 4, 1, 3

B. 4, 2, 1, 3

C. 2, 4, 3, 1

D. 4, 2, 3, 1

> 💡 **Show answer**
>
> **Ans: A** — JNNURM 2005 → NUHHP 2007 → AMRUT 2015 → JJM 2019.

Ecological Species Roles, Functional Guilds & Invasive Alien Species

5.3 Flora and Fauna

Identity: Biotic inhabitants of habitats | BSI + ZSI | India megadiversity stats

- **Flora** is the collective **plant** life of a particular region, habitat, or geological period.
- **Fauna** is the collective **animal** life of a particular region, habitat, or geological period.
- Flora includes trees, shrubs, herbs, grasses, ferns, and algae. It is **not** limited to flowering plants.
- Fauna includes invertebrates and vertebrates across insects, fish, birds, and mammals.
- Together, flora and fauna form the **biotic content** of a habitat.
- India has about **45,000** plant species and **91,000+** animal species on roughly **2.4%** of Earth's land area.
- The **Botanical Survey of India (BSI)** (भारतीय वनस्पति सर्वेक्षण) was established in **1890**. Its headquarters is in **Kolkata**. It surveys Indian **plants**.
- The **Zoological Survey of India (ZSI)** (भारतीय प्राणी सर्वेक्षण) was established in **1916**. Its headquarters is in **Kolkata**. It surveys Indian **animals** and publishes the *Fauna of India* series.
- **Adolf Engler** mapped the world's major **floral regions**.
- **C.O. Sauer** introduced the concept of **planting hearth**, the region where domesticated plants first originated.
- The **Puszta (पुस्ज़टा) grassland** is associated with **Hungary**.

- The **cryophyte steppe (स्टेपी)** is associated with **Arctic tundra (टुंड्रा)** vegetation.
- **Red Sanders (*Pterocarpus santalinus*)** is a South India dry-deciduous endemic (Palkonda–Seshachalam / southern Eastern Ghats (पूर्वी घाट)) — **not** a tropical rainforest tree.
- **Taxus** occurs naturally in the Himalaya, is Red Data–listed, and yields **taxol** (anti-cancer; older stems also cite Parkinson's).
- Himalayan associates include **oak** and **rhododendron**; **sandalwood** is South Indian deciduous — not a Himalayan natural associate.
- Desert flora often shows hard/waxy leaves, tiny leaves, or thorns to cut water loss.

UP flora and fauna facts

Category	UP example	Note
State animal	**Barasingha** (swamp deer)	Terai grassland specialist
State bird	**Sarus crane**	Wetland and farmland mosaic
Dominant Terai flora	**Sal** (*Shorea robusta*)	Dudhwa forest tree
Aquatic fauna	**Gangetic dolphin**	Ganga–Yamuna stretch
Flagship habitat fauna	Tiger, rhino, elephant	Dudhwa and Pilibhit
Invasive flora	Parthenium, Lantana, water hyacinth	Degrade native habitats

- UP has **no coastline**, so mangrove flora is **not** found in the state.

> **Logic:** BSI = **plants**. ZSI = **animals**. Both HQ **Kolkata**. Trap — BSI documents animal diversity is **false**.

PYQ — UPPCS Prelims 2024, Q29

Match List-I with List-II:

List-I: A. Planting Hearth B. Cryophyte Steppe C. World Floral Regions D. Puszta Grassland List-II: 1. Hungary 2. Adolf Engler 3. C.O. Sauer 4. Arctic Tundra

A. 3 4 2 1

B. 2 1 4 3

C. 3 4 1 2

D. 2 3 4 1

💡 **Show answer**

Ans: A — Sauer–3, Arctic–4, Engler–2, Hungary–1.

Complete PYQ Bank (Topic 5)

Q1. UPPCS Prelims 2024, Q62

Plants and animals that live on land are called:

A. Abiotic

B. Aquatic habitants

C. Terrestrial habitants

D. Biotic

💡 **Show answer**

Ans: C

Q2. UPPCS Prelims 2024, Q29

Match List-I with List-II:

List-I: A. Planting Hearth B. Cryophyte Steppe C. World Floral Regions D. Puszta Grassland List-II: 1. Hungary 2. Adolf Engler 3. C.O. Sauer 4. Arctic Tundra

A. 3 4 2 1

B. 2 1 4 3

C. 3 4 1 2

D. 2 3 4 1

💡 **Show answer**

Ans: A

Q3. UPPCS Prelims 2025, Q93

With reference to MISHTI, which statements is/are correct?

1. It was initiated in 2019.
2. It aims to restore mangrove forests.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Ans: A

Q4. UPPCS Prelims 2025, Q72

Arrange chronologically: AMRUT, JNNURM, Jal Jeevan Mission, NUHHP.

A. 2, 4, 1, 3

B. 4, 2, 1, 3

C. 2, 4, 3, 1

D. 4, 2, 3, 1

💡 **Show answer**

Ans: A

Q5. UPPCS Prelims 2025, Q23

Which pairs is/are NOT correctly matched? (Tribe — Habitat) — Masai — West Africa

A. Only 1, 2 and 3

B. Only 4

C. Only 3 and 4

D. Only 1

💡 **Show answer**

Ans: B

Q6. UPPCS Prelims 2023, Q115

Desert-dwelling animals are called:

A. Arboreal

B. Terrestrial

C. Fossorial

D. Xericole

💡 **Show answer**

Ans: D

Q7. UPSC Prelims 2020 — pattern

Among Desert, Mangrove, and Rainforest, which are terrestrial ecosystems?

💡 **Show answer**

Ans: Desert and Rainforest — Mangrove is transitional.

Q8. UPPCS Prelims 2019 — pattern

Which is an example of in-situ conservation?

A. Botanical garden

B. Seed bank

C. National park

D. Zoo

💡 **Show answer**

Ans: C

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D

Q1. With reference to habitat and niche, which of the following statements is/are correct?

1. Habitat is the functional role of a species in the community.
2. Niche is the role of a species.
3. Habitat and niche are interchangeable terms in all Prelims keys.

A. Only 1 B. 1 and 2 C. Only 3 D. Only 2

💡 **Show answer**

Ans: D. Only 2 is correct. **Logic:** Statement 2 is true (niche = role). Statement 1 swaps habitat for niche. Statement 3 is false — habitat and niche are not interchangeable.

Q2. Which of the following habitat classes is/are correctly matched?

1. Terrestrial habitats are water-only systems such as oceans and lakes.
2. Aquatic — water
3. Mangrove/wetland — purely terrestrial only

A. Only 1 B. 1 and 2 C. Only 3 D. Only 2

> 💡 **Show answer**
>
> **Ans: D.** Only 2 is correct. **Logic:** Statement 2 is true (aquatic = water). Statement 1 misdefines terrestrial. Statement 3 is false — mangrove/wetland is transitional, not purely terrestrial.

Q3. Match List-I with List-II:

List-I (Adaptation)	List-II
A. Arboreal	1. Burrows
B. Fossorial	2. Trees
C. Cursorial	3. Open plains
D. Xericole	4. Desert

Code:

A. A-1, B-2, C-3, D-4 B. A-2, B-1, C-3, D-4 C. A-2, B-3, C-1, D-4 D. A-3, B-1, C-2, D-4

> 💡 **Show answer**
>
> **Ans: B.** **Logic:** Arboreal trees; fossorial burrows; cursorial plains; xericole desert.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Habitat loss is the number-one biodiversity threat.

Reason (R): Habitat corridors are used to reconnect fragmented patches.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** **A/R logic:** Both true; corridors respond to fragmentation but do not define why loss is #1.

Q5. With reference to MISHTI, which of the following statements is/are correct?

1. MISHTI stands for Mangrove Initiative for Shipping, Highways and Trade Infrastructure.
2. It comes from Union Budget 2023–24 under MoEFCC.
3. It was launched as a 2019 mangrove mission.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 2 is correct. **Logic:** Statement 2 is true (Union Budget 2023–24 / MoEFCC). Statement 1 invents a wrong expansion — MISHTI is Mangrove Initiative for Shoreline Habitats & Tangible Income. Statement 3 is false — it is not a 2019 mission launch.

Q6. Arrange the following in correct chronological order:

1. JNNURM 2. NUHHP 3. AMRUT 4. Jal Jeevan Mission

A. 2-1-3-4 B. 1-2-3-4 C. 1-3-2-4 D. 1-2-4-3

💡 **Show answer**

Ans: B. **Logic:** 2005 → 2007 → 2015 → 2019.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Flora — animal life of a region
2. Fauna — plant life of a region
3. BSI — plant survey (Kolkata)
4. ZSI — animal survey (Kolkata)

A. 3 and 4 only B. 1 and 2 only C. 1, 2 and 3 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: B. **Logic:** Flora/fauna swapped in 1–2; BSI/ZSI correct.

Q8. With reference to India species counts, which of the following statements is/are correct?

1. Roughly 45,000 plant species and 91,000+ animal species are cited on about 2.4% of world land.
2. BSI (1890) surveys plants; ZSI (1916) surveys animals.
3. Both BSI and ZSI have headquarters in Mumbai.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. **Logic:** Both sit in Kolkata — not Mumbai.

Q9. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NUHHP is the National Urban Housing and Habitat Policy of 2007.

Reason (R): NUHHP is administered as a MoEFCC forest-conservation statute.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. **A/R logic:** A true; NUHHP is MoHUA housing/habitat policy — R false.

Q10. Match List-I with List-II:

List-I	List-II
A. Engler	1. Planting hearths
B. C.O. Sauer	2. World floral regions
C. Microhabitat	3. Small subdivision inside a larger habitat

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. **Logic:** Engler floral regions; Sauer planting hearths; microhabitat subdivision.

Q11. With reference to fragmentation, which of the following statements is/are correct?

1. Fragmentation splits habitat into patches.
2. Corridors help reconnect patches.
3. Fragmentation always increases genetic diversity of every species.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** **Logic:** Fragmentation typically harms connectivity/genetics — statement 3 false.

Q12. Which one of the following correctly distinguishes flora and fauna?

A. Flora = animals; fauna = plants B. Flora = plants; fauna = animals C. Both mean only microorganisms D. Both mean only domesticated species

> 💡 **Show answer**
>
> **Ans: B.** **Logic:** Flora plants; fauna animals.

Q13. Consider the following statements:

1. Transitional habitats include mangroves and wetlands.
2. Transitional habitats are identical to purely terrestrial forests in all classifications.
3. Aquatic habitats include both lentic and lotic systems.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** **Logic:** Statement 2 is false.

Q14. With reference to Budget-linked mangrove policy, which of the following statements is/are correct?

1. MISHTI is from Budget 2023–24.
2. Dating MISHTI to 2019 is a common trap.
3. MISHTI is a Ministry of Coal statute for underground mining.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** **Logic:** Statement 3 is false.

Q15. Arrange adaptation labels to habitats:

1. Desert specialist label among options — xericole
2. Tree-dwelling — arboreal
3. Burrowing — fossorial

A. Only 1 B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: D.** **Logic:** All three correct.

Q16. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): BSI and ZSI both sit in Kolkata.

Reason (R): BSI surveys animals and ZSI surveys plants.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. **A/R logic:** A true; kingdoms are swapped in R — R false.

Q17. Which of the following pairs is/are correctly matched?

1. JNNURM — 2005
2. NUHHP — 2007
3. AMRUT — 2019
4. Jal Jeevan Mission — 2015

A. 2 and 3 only B. 1 and 2 only C. 1 and 4 only D. 3 and 4 only

💡 **Show answer**

Ans: B. **Logic:** AMRUT 2015; JJM 2019 — 1 and 2 correct.

Q18. With reference to environment vs ecosystem vs habitat, which statements is/are correct?

1. Environment is total surroundings.
2. Ecosystem is the functional unit.
3. Habitat is the place; niche is the role.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. **Logic:** All three correct.

Q19. How many of the following are correctly associated with MoEFCC?

1. MISHTI mangrove initiative teaching
2. NUHHP as a MoEFCC Act
3. Wildlife protected-area governance broadly under environment/forest ministry world

A. Only two B. Only one C. All three D. None

💡 **Show answer**

Ans: A. **Logic:** MISHTI MoEFCC yes; NUHHP MoHUA no; PA governance broadly yes → two.

Q20. Which of the following is/are correct about microhabitat?

A. It is always larger than a biome B. It is a small subdivision inside a larger habitat C. It replaces the biosphere concept D. It means only deep ocean trenches

💡 **Show answer**

Ans: B. **Logic:** Small subdivision inside larger habitat.

Q21. With reference to habitat threat pathways, which of the following statements is/are correct?

1. Loss removes habitat area.
2. Fragmentation splits remaining patches.

3. Corridors are irrelevant once fragmentation begins.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** Corridors matter — statement 3 false.

Q22. Match scheme to year:

List-I	List-II
A. AMRUT	1. 2007
B. NUHHP	2. 2015
C. JJM	3. 2019

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. **Logic:** AMRUT 2015; NUHHP 2007; JJM 2019.

Q23. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Cursorial adaptations suit open plains.

Reason (R): Fossorial organisms are specialised for tree canopies.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. **A/R logic:** A true; fossorial = burrows — R false.

Q24. Which of the following statements is/are correct?

1. India'a species richness on small land share is a recurring Prelims fact.
2. Flora/fauna swap is a recurring trap.
3. MISHTI year trap is 2019 vs 2023–24.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. **Logic:** All three correct.

Q25. With reference to urban habitat policy, which of the following statements is/are correct?

1. NUHHP 2007 is a housing/habitat policy.
2. JNNURM preceded NUHHP.
3. Jal Jeevan Mission is primarily a 2005 urban renewal mission identical to JNNURM.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

💡 **Show answer**

Ans: D. **Logic:** JJM is 2019 drinking-water mission — not JNNURM.

Q26. Which one of the following is transitional habitat teaching?

A. Pure continental desert only B. Mangrove / wetland C. Inner core of a Himalayan glacier only D. Stratospheric ozone layer

> 💡 **Show answer**
>
> **Ans: B.** **Logic:** Mangrove/wetland transitional.

Q27. Consider the following statements about BSI/ZSI:

1. BSI year fact is 1890.
2. ZSI year fact is 1916.
3. Both publish Red Data Book components for their kingdoms.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: D.** **Logic:** All three teaching facts.

Q28. Which of the following pairs is/are NOT correctly matched?

1. Arboreal — desert only
2. Xericole — trees only
3. Fossorial — burrows

A. 2 and 3 only B. 1 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** **Logic:** Pair 3 correct; 1 and 2 wrong.

Q29. With reference to MoHUA vs MoEFCC, which of the following statements is/are correct?

1. NUHHP sits with MoHUA teaching.
2. MISHTI sits with MoEFCC teaching.
3. Both NUHHP and MISHTI are identical CAMPA statutes.

A. 2 and 3 only B. 1 and 3 only C. 1, 2 and 3 D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** **Logic:** Statement 3 false.

Q30. Arrange from broadest to narrowest conceptual scale among the options:

A. Microhabitat → habitat → biome-level surroundings B. Biome-level surroundings → habitat → microhabitat C. Habitat → biome → microhabitat D. Microhabitat → biome → habitat

> 💡 **Show answer**
>
> **Ans: B.** **Logic:** Broad surroundings/biome → habitat → microhabitat.

Mains Answer Framework

Habitat conservation in India (250 words): Define habitat and link to flora/fauna. India's megadiversity on 2.4% land. Threats — loss, fragmentation, degradation with UP Terai and Ganga examples. Strategies — in-situ

protected areas, Terai Arc corridor, MISHTI for coastal mangrove habitat, NUHHP for urban habitat. Legal brief — WPA 1972 and FCA 1980. Conclude with habitat-first conservation logic.

Common Traps — Don't Fall For These

Trap	Correct fact
Habitat = niche	Habitat = **place**; niche = **role**
Mangrove = terrestrial	Mangrove = **transitional**
MISHTI launched 2019	**Budget 2023–24**
BSI surveys animals	BSI = **plants**; ZSI = **animals**
Masai in West Africa	Masai = **East Africa**
Zoos = in-situ	Zoos = **ex-situ**
NUHHP before JNNURM	JNNURM **2005** → NUHHP **2007**
Mangrove flora in UP Terai	UP has **no coastline**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 6 — Protected Areas & Conservation

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 Covers syllabus (click to expand)

Wildlife Sanctuary | National Park | Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र) | Conservation Reserve (संरक्षण आरक्षित क्षेत्र) | Community Reserve (सामुदायिक आरक्षित क्षेत्र) | Tiger Reserve | Elephant Reserve (हाथी अभयारण्य) | Marine Protected Areas | Ramsar (रामसर) Sites | UNESCO World Heritage (विश्व धरोहर) Sites | Eco-Sensitive Zones

Sources baked in: WPA 1972 (amended 2002, 2006), EPA 1986, Ramsar Convention, UNESCO MAB/WH, NTCA, MoEFCC, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — PA comparison, location matching, UP Ramsar traps **Last verified:** August 2026

Current Affairs (this topic)

- Track new Ramsar notifications and tiger-reserve counts against the Must-Score totals.
- Keep UP Ramsar **12** and India **~100** as the teaching baseline unless a stem cites (साइट्स) a later official total.

💡 🎯 Consolidated — 32 Must-Score Facts (Click to Expand)

1. The **Wildlife (Protection) Act, 1972** is India's main protected-area statute. 2. Terrestrial WPA strictness runs **National Park > Wildlife Sanctuary > Conservation/Community Reserve**. 3. A **Wildlife Sanctuary** (Section **18**) allows limited human use with **Chief Wildlife Warden** permission. 4. A **National Park** (Section **35**) is the strictest WPA protected area. Grazing and forestry are largely banned. Boundary change needs a resolution of the **State Legislature** (not Parliament). 5. A **Biosphere Reserve** (जैवमंडल आरक्षित क्षेत्र) follows the UNESCO MAB three-zone model: **core** (no exploitative use) → **buffer** (research, education, regulated tourism/grazing) → **transition** (settlements, cultivation, livelihoods). It has **no WPA statutory basis**. 6. India has **18** biosphere reserves (जैवमंडल आरक्षित क्षेत्र); about **12–13** are in the UNESCO network (count drifts with new WNBR listings). The first was **Nilgiri (नीलगिरि) (1986)**. **Nokrek** is in **Meghalaya** (मेघालय), not Manipur. 7. A **Conservation Reserve** (Section **36A**, 2002) is a government/community forest buffer near a protected area. 8. A **Community Reserve** (Section **36B**, 2002) sits on private or community land with voluntary consent. 9. A **Tiger Reserve** rests on WPA Section **38V** (2006 path) with **core + buffer** under **NTCA**. India has **58** tiger reserves. **Corbett** was first in **1973**. 10. An **Elephant Reserve** comes from **Project Elephant (प्रोजेक्ट एलीफेंट) (1992)** (प्रोजेक्ट एलिफेंट) by executive notification. It has **no WPA statutory basis**. India has about **33** elephant reserves. 11. India's first Marine National Park is **Gulf of Kutch, Gujarat (1980)**. **Gulf of Mannar (TN, 1980)** is also a Marine NP. Uttar Pradesh (उत्तर प्रदेश) has **no marine PA**. 12. **Ramsar (1971)** marks wetlands of international importance. India joined in **1982**. Uttar Pradesh has **12** Ramsar sites. 13. **Rudrasagar** is in **Tripura**. **Sultanpur** is in **Haryana**. Neither is an Uttar Pradesh Ramsar site. 14. An **Eco-Sensitive Zone (ESZ)** (पर्यावरण-संवेदनशील क्षेत्र) is an **EPA 1986** (पर्यावरण संरक्षण 1986) buffer around protected areas that restricts mining and polluting industry. ESZ is **not** a WPA category. 15. Protected-area stock (ENVIS-style ~2023): about **106** National Parks, **573** Wildlife Sanctuaries, **123** Conservation Reserves, **220** Community Reserves. **Madhya Pradesh** leads NP count; **Andaman (अंडमान) & Nicobar (निकोबार)** leads sanctuary count. 16. In a National Park, local people are **not** allowed to collect/use biomass the way limited use is permitted in a sanctuary. BR is for landscape conservation of **all forms of

life**, not “a few species only.” 17. WPA basis yes: National Park, Sanctuary, Tiger Reserve, Conservation Reserve, Community Reserve. WPA basis no: Biosphere Reserve, Elephant Reserve, Ramsar, ESZ. 18. Uttar Pradesh’s only classic National Park fact is **Dudhwa** (दुधवा) (Lakhimpur Kheri; rhino rehabilitation from **1984**, later phases including ~**2018**). 19. UP tiger reserves are **Dudhwa, Pilibhit, and Ranipur**. 20. Important UP sanctuaries include **Katarniaghat (1975), Kishanpur, Chandra Prabha (Chandauli), Nawabganj, and Sur Sarovar**. 21. UP elephant reserves include **UP ER (2009)** and **Terai (तराई) ER (2022)**. 22. UP Ramsar sites include Upper Ganga (गंगा) (Brijghat–Narora), Nawabganj, Parvati Arga, Saman, Samaspur, Sandi, Sarsai Nawar, Sur Sarovar, Haiderpur, Bakhira, Patna (पटना), and Shekha Jheel. 23. River–park traps: **Corbett = Ramganga (रामगंगा) + Kosi (कोसी)** (not Ganga); **Kaziranga (काजीरंगा) = Brahmaputra (ब्रह्मपुत्र) / Diphlu system** (not Manas); **Silent Valley (साइलेंट वैली) = Kunthipuzha** (not Kaveri (कावेरी)). 24. The **Terai Arc** corridor links Dudhwa–Kishanpur–Katarniaghat toward Nepal. 25. Tiger Reserves are statutory under WPA; Elephant Reserves are not — that swap is a high-yield trap. 26. Biosphere Reserve zoning is landscape management under MAB; it is not the same as a National Park notification under Section 35. 27. Community Reserve is private/community land; Conservation Reserve is mainly government forest buffer — do not reverse Sections **36A/36B**. 28. ESZ regulates harmful activity around a PA; it does not itself create a new WPA Schedule animal list. 29. Sanctuary permits regulated use; National Park does not allow the same degree of use — strictness order must stay NP > WLS. 30. Match “firsts” carefully: first BR = **Nilgiri 1986**; first TR = **Corbett 1973**; first Marine NP (प्रथम समुद्री उद्यान) = **Gulf of Kutch 1980**; world first NP = **Yellowstone 1872**. 31. **International Tiger Day = 29 July** (St Petersburg 2010). **M-STrIPES** monitors tiger reserves (patrol + ecology (पारिस्थितिकी) GIS tool). 32. **Sathyamangalam TR (TN)** is a classic ecological link between Eastern and Western Ghats (पश्चिमी घाट).

💡 ⚡ Confused Pairs (Click to Expand)

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **National Park (NP)** | **Wildlife Sanctuary (WLS)** | Declared under WPA 1972 Sec 35; highest protection, zero human settlement, grazing banned, boundaries altered only by State Legislature on NBWL advice vs WPA Sec 18; regulated grazing and traditional rights may be permitted by Chief Wildlife Warden | राष्ट्रीय उद्यान (NP) / वन्यजीव अभयारण्य (WLS) | | **Conservation Reserve** | **Community Reserve** | Established on government-owned lands acting as buffer corridors between PAs under WPA Sec 36A vs established on private or community-owned lands with voluntary community participation under WPA Sec 36C | संरक्षण आरक्षित क्षेत्र / सामुदायिक आरक्षित क्षेत्र | | **Tiger Reserve** | **Elephant Reserve** | Statutory protected area notified under Wildlife Protection Act 1972 (amended 2006) with legally delineated Core and Buffer zones supervised by NTCA vs administrative/executive designation under Project Elephant without statutory backing under WPA | बाघ अभयारण्य (सांविधिक) / हाथी अभयारण्य (प्रशासनिक) | | **Biosphere Reserve (MAB)** | **National Park** | Landscape-level conservation framework under UNESCO MAB with 3 distinct concentric zones (Core, Buffer, Transition) integrating human development vs strict statutory preservation excluding human commercial activity | जैवमंडल आरक्षित क्षेत्र / राष्ट्रीय उद्यान | | **Eco-Sensitive Zone (ESZ)** | **Protected Area (PA)** | Buffer zone up to 10 km notified around National Parks and Sanctuaries under Section 3 of Environment (Protection) Act 1986 to regulate commercial activities vs statutory core area notified under Wildlife Protection Act 1972 | पर्यावरण-संवेदनशील क्षेत्र (ESZ) / संरक्षित क्षेत्र (PA) | | **Ramsar Wetland Site** | **Natural World Heritage Site** | International designation under 1971 Ramsar Convention specifically targeting ecological wise use of waterfowl/wetland habitats vs UNESCO World Heritage designation recognizing universal natural value under 1972 Convention | रामसर आर्द्रभूमि स्थल / प्राकृतिक विश्व धरोहर स्थल | | **Core Zone (Biosphere)** | **Transition Zone (Biosphere)** | Legally protected untouched sanctum sanctorum where all human activity except non-destructive research is barred vs outermost open area where local sustainable agriculture, settlements, and economic activities are actively fostered | कोर क्षेत्र (पूर्ण सुरक्षित) / संक्रमण क्षेत्र (सहअस्तित्व) | ### Protected-area hierarchy — one card | Category | Legal home | Strictness / use | Classic trap | |---|---|---|---| | **National Park** | WPA Sec **35** | Strictest WPA PA | Boundary change needs **State Legislature** | | **Wildlife Sanctuary** | WPA Sec **18** | Limited use with **CWLW** permission | Not as strict as NP | | **Tiger Reserve** | WPA Sec **38V** + NTCA |

Core + buffer mandatory | Statutory — unlike Elephant Reserve | | **Conservation Reserve** | WPA Sec **36A** | Forest buffer near PA | Not private-land Community Reserve | | **Community Reserve** | WPA Sec **36B** | Private/community land + consent | Needs voluntary consent | | **Biosphere Reserve** | UNESCO MAB (executive in India) | Core–buffer–transition | **No** WPA statutory basis | | **Elephant Reserve** | Project Elephant 1992 | Corridor / conflict focus | **No** WPA statutory basis | | **Ramsar site** | Ramsar Convention 1971 | Wise-use wetland | Not a forest or ozone treaty | | **Natural World Heritage** | UNESCO WH Convention | Outstanding universal value | ≠ Ramsar ≠ BR WNBR automatically | | **ESZ** | EPA **1986** | Buffer restrictions around PA | **Not** a WPA category | - **UP teaching row:** Dudhwa NP / TR, Pilibhit TR, Ranipur TR, **12 Ramsar**, Terai corridors, **no marine PA**, **no biosphere reserve** inside present UP. - **Three UNESCO instruments:** World Heritage · Biosphere (MAB / WNBR) · Intangible Cultural Heritage — do not force one site into all three without checking the list.

💡 📋 **Must-score facts — WPA categories, tiger, Ramsar (Click to Expand)**

Strictness / categories | Item | Lock | |------|------| | WPA year | **1972** | | Strictness (terrestrial) | NP > Sanctuary > Conservation/Community Reserve | | NP / Sanctuary | Sec. **35** strictest / Sec. **18** limited use | | Biosphere Reserve | UNESCO MAB; core → buffer → transition (**not** WPA) | | India BRs | **18** national; first **Nilgiri 1986** | | Conservation / Community Res. | Sec. 36A / 36B (**2002**) | | Tiger Reserve | Sec. **38V**; NTCA; **58** TRs; Corbett first **1973** | | Elephant Reserve | Project Elephant **1992**; ~**33**; **no** WPA statute | ### Marine / wetland / ESZ | Item | Lock | |------|------| | First Marine NP | Gulf of Kutch, Gujarat **1980** | | Ramsar | **1971**; India joined **1982** | | Rudrasagar / Sultanpur | Tripura / Haryana (**not** UP) | | ESZ | EPA **1986** buffer (≠ WPA category) |

6.1 Wildlife Sanctuary

Identity: WPA Sec 18 | State declares | Human use with CWLW permission

- A **Wildlife Sanctuary (WLS)** (वन्यजीव अभयारण्य) is a protected area where wildlife is conserved in natural habitat (वास स्थान).
- The **State Government** declares a sanctuary under **WPA Section 18**.
- Hunting, poaching, and capturing wildlife are prohibited inside a sanctuary.
- Other activities such as grazing or collection of minor forest produce may continue if the **Chief Wildlife Warden (CWLW)** permits them.
- India has about **573 Wildlife Sanctuaries**.
- Sanctuaries are **in-situ** (स्थान पर) conservation tools that protect whole habitats, not just one species.
- A sanctuary can later be upgraded to a **National Park** with stricter rules.

Feature	Wildlife Sanctuary	National Park
WPA section	Sec 18–27	Sec 35
Human activity	Allowed with CWLW permission	Largely prohibited
Grazing / forestry	May be permitted	Prohibited
Biomass collection	Limited use may be allowed	**Not** allowed as in sanctuary
Boundary change	State process	Needs **State Legislature** resolution
Strictness	Moderate (नरम दल)	**Highest** among WPA terrestrial PAs

> **Logic:** Trap — "no human activity is allowed in any wildlife sanctuary" is **false**. NP boundary change = **State Legislature**, not Parliament.

6.2 National Park

Identity: WPA Sec 35 | Strictest terrestrial PA | First = Corbett 1936

- A **National Park (NP)** (राष्ट्रीय उद्यान) is the **strictest** category of terrestrial protected area under WPA.
- The **State Government** notifies a national park under **WPA Section 35**.
- Grazing, hunting, forestry, and private exploitation are **prohibited** inside a national park.
- Altering a national park boundary requires a resolution of the **Legislature of the State** (WPA Sec 35) — **not Parliament**.
- Local people are **not** allowed to collect and use biomass in a National Park the way limited biotic interference may be permitted in a sanctuary.
- India's first national park was **Hailey National Park**, now **Corbett National Park (1936)** in Uttarakhand (उत्तराखंड).
- Uttar Pradesh's major national park is **Dudhwa National Park** (दुधवा राष्ट्रीय उद्यान) in the Terai (rhino rehabilitation from **1984**).

> **Logic:** tests NP location matching. Keoladeo (केवलादेव) is in **Rajasthan** (थार). Dachigam is in **Jammu & Kashmir**.

PYQ — UPPCS Prelims 2025, Q90

Match List-I with List-II:

List-I: A. Dachigam B. Keoladeo C. Kanha (कान्हा) D. Periyar (पेरियार) List-II: 1. Madhya Pradesh 2. Rajasthan 3. Kerala 4. Jammu and Kashmir

A. 4 2 1 3

B. 2 4 3 1

C. 4 2 3 1

D. 4 2 1 3

💡 **Show answer**

Ans: D — Dachigam=J&K; Keoladeo=Rajasthan; Kanha=MP; Periyar=Kerala.

In-Situ vs Ex-Situ Conservation & UNESCO Biosphere Reserve Zonation Architecture

6.3 Biosphere Reserve

Identity: UNESCO MAB 1971 | Core–Buffer–Transition | 18 notified | No WPA basis

- A **Biosphere Reserve (BR)** (जैवमंडल आरक्षित क्षेत्र) is a large landscape for conservation, research, and sustainable community use under UNESCO's **Man and the Biosphere (MAB) Programme**.
- India notified its first BR, **Nilgiri** (नीलगिरि), in **1986**.
- India has **18 notified Biosphere Reserves**. **13** are in UNESCO's World Network of Biosphere Reserves.
- Every BR has three zones. The **core** zone bans exploitative human activity (strict protection / genetic diversity (आनुवंशिक विविधता)). The **buffer** allows limited research, education, regulated eco-tourism and grazing. The **transition** zone permits settlements, cultivation, and regulated sustainable livelihoods.
- BRs conserve **in situ all forms of life** with support systems — they are **not** declared merely to conserve "a few specific species."
- BRs are **executive notifications** by MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय). They do **not** have a separate statutory category under WPA, though core areas often overlap with national parks or sanctuaries.
- **Nokrek** is in **Meghalaya** (Garo (गारो) Hills; Nokrek peak), not Manipur. **Simlipal** is in **Odisha** (ओडिशा). **Agasthyamalai** spans **Kerala and Tamil Nadu (तमिलनाडु)** (Neyyar, Peppara, Shendurney WLS + Kalakad–

Mundanthurai TR).

- The **largest** national BR by area is **Great Rann of Kutch** (Gujarat). Among UNESCO-certified Indian BRs by area, **Gulf of Mannar** is often cited as the largest. The **smallest** is **Nokrek** (Meghalaya).
- Area order among common options: **Gulf of Mannar > Sundarbans (सुंदरबन) > Pachmarhi > Khangchendzonga**.

India's 18 Biosphere Reserves & UNESCO WNBR Global Recognition Network

#	Biosphere Reserve	State(s)
1	Nilgiri	Tamil Nadu, Karnataka, Kerala
2	Nanda Devi (नंदा देवी)	Uttarakhand
3	Nokrek	**Meghalaya**
4	Gulf of Mannar	Tamil Nadu
5	Sundarbans	West Bengal
6	Manas	Assam
7	Great Nicobar	Andaman & Nicobar
8	Simlipal	Odisha
9	Dibru-Saikhowa	Assam
10	Dihang-Dibang	Arunachal Pradesh
11	Pachmarhi	Madhya Pradesh
12	Khangchendzonga	Sikkim
13	Agasthyamalai	Kerala, Tamil Nadu
14	Achanakmar-Amarkantak	MP, Chhattisgarh
15	Great Rann of Kutch	Gujarat
16	Cold Desert	Himachal (हिमाचल) Pradesh
17	Seshachalam Hills	Andhra Pradesh
18	Panna	Madhya Pradesh

PYQ — UPPCS Prelims 2025, Q144

Which of the following pairs is/are NOT correctly matched?

(Biosphere Reserve) — (State)

1. Nokrek — Manipur
2. Manas — Assam
3. Agasthyamalai — Kerala

A. Only 1

B. Only 2

C. Only 3

D. Only 1 and 3

💡 **Show answer**

Ans: A — Nokrek = **Meghalaya**.

6.4 Conservation Reserve

Identity: WPA Sec 36A (2002) | Buffer forest near PA

- A **Conservation Reserve** protects forest areas that act as **buffers or corridors** next to national parks or sanctuaries.
- It was added by the **WPA Amendment, 2002** under **Section 36A**.
- The **State Government** declares it on **government-owned or community-owned forest land**.
- Human activity is more flexible than in a national park. Traditional uses may continue under regulation.
- **Asan (आसन) Conservation Reserve** in Uttarakhand is a known example and is also a Ramsar site.

> **Logic:** Conservation Reserve = **forest buffer**. Community Reserve = **private/community land**. Do not swap them.

6.5 Community Reserve

Identity: WPA Sec 36B (2002) | Voluntary community consent

- A **Community Reserve** is declared on **private or community land** when local people agree to conserve wildlife.
- It was added by the **WPA Amendment, 2002** under **Section 36B**.
- The **State Government** notifies it after receiving **voluntary community consent**.
- It is the most community-led WPA protected-area category.
- Community Reserves supplement strict PAs where full national park notification is socially difficult.

> **Logic:** Community Reserve requires **consent**. Conservation Reserve does **not** require private individual land.

6.6 Tiger Reserve

Identity: WPA Sec 38V (2006) | NTCA 2005 | Core + Buffer mandatory

- A **Tiger Reserve (TR)** is a protected area notified specifically for tiger conservation under **WPA Section 38V**.
- The **National Tiger Conservation Authority (NTCA)** was created in **2005** and given statutory status by the **2006 WPA amendment**.
- Every tiger reserve must have a **core** area kept inviolate for tigers and a **buffer** area for regulated human use.
- India's first tiger reserve was **Corbett (1973)** under **Project Tiger (प्रोजेक्ट टाइगर) (1973)** (प्रोजेक्ट टाइगर).
- India had **58 tiger reserves** after the notification of **Madhav Tiger Reserve** (2025).
- UP tiger reserves include **Dudhwa (1987)**, **Pilibhit (2014)**, and **Ranipur (2023)**.

> **Logic:** Tiger Reserves have **WPA statutory backing**. Elephant Reserves do **not**.

6.7 Elephant Reserve

Identity: Project Elephant 1992 | Executive notification | No WPA basis

- An **Elephant Reserve (ER)** is declared under **Project Elephant** (प्रोजेक्ट एलीफेंट), launched in **1992**.
- It is an **executive/administrative** category. It is **not** a separate statutory category under WPA.
- ERs aim to protect elephant corridors and reduce human–elephant conflict across **14 elephant-range states**.
- India has **33 Elephant Reserves**.
- UP has the **Uttar Pradesh Elephant Reserve (2009)** and the **Terai Elephant Reserve (2022)**.

> **Logic:** Trap — "Elephant Reserves are declared under WPA like Tiger Reserves" is **false**.

6.8 Marine Protected Areas

Identity: Marine NP/WLS under WPA | CRZ under EPA | UP has none

- **Marine Protected Areas (MPAs)** include **Marine National Parks** and **Marine Wildlife Sanctuaries** notified under WPA.
- India's first marine national park is **Gulf of Kutch Marine National Park, Gujarat (1980)**.
- **Gahirmatha Marine Sanctuary** in Odisha is famous for olive ridley turtle nesting.
- Coastal regulation also uses **CRZ** rules under the **EPA 1986**.
- **Uttar Pradesh has no marine protected area** because it is landlocked.

> **Logic:** Negative questions may use UP as the state with **zero** marine PAs.

6.9 Ramsar Sites

Identity: Ramsar Convention 1971 | India joined 1982 | Wise use of wetlands

- A **Ramsar site** is a wetland of **international importance** designated under the **Ramsar Convention (1971)** (रामसर कन्वेंशन).
- India became a contracting party in **February 1982**.
- **World Wetlands Day** (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर).
- India has about **100 Ramsar sites** (among the highest globally).
- Ramsar designation follows the **wise use** principle. Wetlands may support fishing, tourism, and local livelihoods if managed sustainably.
- Ramsar status does **not** change land ownership. WPA protections on the site continue.
- **Uttar Pradesh has 12 Ramsar sites**.
- **Rudrasagar Lake** is in **Tripura**, not UP.
- **Sultanpur National Park** is in **Haryana**, not UP.

PYQ — UPPCS Prelims 2025, Q58

Which of the following Ramsar sites are NOT located in Uttar Pradesh?

1. Sarsai Nawar Jheel
2. Samaspur Bird Sanctuary
3. Rudrasagar Lake
4. Sultanpur National Park

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Ans: B — Rudrasagar = Tripura. Sultanpur = Haryana. Samaspur and Sarsai Nawar are in UP.

6.10 UNESCO World Heritage Sites (Natural)

Identity: WH Convention 1972 | Natural vs cultural | India natural WH sites

- **UNESCO World Heritage Sites** are places of outstanding universal value under the **World Heritage Convention (1972)**.
- **Natural** World Heritage Sites are recognised for exceptional natural beauty or biodiversity.
- India's major **natural** World Heritage Sites include **Kaziranga**, **Manas**, **Keoladeo**, **Sundarbans** (सुंदरबन), **Nanda Devi and Valley of Flowers (फूलों की घाटी)**, **Western Ghats**, **Great Himalayan National Park** (हिमाचल), and **Khangchendzonga**.
- **Keoladeo** is both a Ramsar wetland and a UNESCO natural World Heritage Site (प्राकृतिक विश्व धरोहर स्थल).
- **Uttar Pradesh has no natural UNESCO World Heritage Site**. Nanda Devi is in **Uttarakhand** (उत्तराखंड).

> **Logic:** Do not confuse **natural WH sites** with **cultural WH sites** such as Jaipur city.

6.11 Eco-Sensitive Zones (ESZ)

Identity: EPA 1986 | Buffer around PA | Restricts mining and polluting industry

- An **Eco-Sensitive Zone (ESZ)** is a regulated **buffer belt** around a national park, wildlife sanctuary, or other protected area.
- ESZs are notified under the **Environment (पर्यावरण) (Protection) Act, 1986**, not under WPA.
- The Supreme Court has guided ESZ notification to reduce mining, polluting industries, and major construction near protected areas.
- ESZ width is notified case by case. It is **not** a fixed 10 km rule for every protected area in all states.
- ESZs aim to reduce shock between strict protected cores and surrounding development.

> **Logic:** ESZ = **EPA 1986**. National Park = **WPA 1972**. Do not swap the two laws.

Complete PYQ Bank (Topic 6)

Q1. UPPCS Prelims 2025, Q58

Which of the following Ramsar sites are NOT located in Uttar Pradesh?

1. Sarsai Nawar Jheel 2. Samaspur Bird Sanctuary 3. Rudrasagar Lake 4. Sultanpur National Park

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Ans: B

Q2. UPPCS Prelims 2025, Q90

Match List-I with List-II: A. Dachigam B. Keoladeo C. Kanha D. Periyar | 1. MP 2. Rajasthan 3. Kerala 4. J&K

A. 4 2 1 3

B. 2 4 3 1

C. 4 2 3 1

D. 4 2 1 3

💡 **Show answer**

Ans: D

Q3. UPPCS Prelims 2025, Q144

Which pairs is/are NOT correctly matched? 1. Nokrek — Manipur 2. Manas — Assam 3. Agasthyamalai — Kerala

A. Only 1

B. Only 2

C. Only 3

D. Only 1 and 3

💡 **Show answer**

Ans: A

Q4. UPPCS Prelims 2024, Q10

Match: A. Simlipal B. Nokrek C. Agasthyamalai D. Manas | 1. Kerala 2. Odisha 3. Assam 4. Meghalaya

A. 1 2 4 3

B. 3 1 4 2

C. 2 4 3 1

D. 2 4 1 3

💡 **Show answer**

Ans: D — Simlipal=Odisha(2); Nokrek=Meghalaya(4); Agasthyamalai=Kerala(1); Manas=Assam(3).

Q5. UPPCS Prelims 2023, Q21

Chandraprabha Wildlife Sanctuary is situated in which district of Uttar Pradesh?

A. Chandauli

B. Mirzapur

C. Sonbhadra

D. Varanasi

💡 **Show answer**

Ans: A

Q6. UPPCS Prelims 2023, Q118

In National Parks, genetic diversity is maintained through—

A. In-situ conservation

B. Ex-situ conservation

C. Gene pool

D. None of the above

💡 **Show answer**

Ans: A

Q7. UPPCS Prelims 2022, Q11

Dudhwa National Park is situated in which district of Uttar Pradesh?

A. Lakhimpur Kheri

B. Pilibhit

C. Bahraich

D. Bijnor

💡 **Show answer**

Ans: A

Q8. UPPCS Prelims 2022, Q134

Keibul Lamjao National Park is located in which lake?

A. Loktak

B. Wular

C. Dal

D. Chilika

💡 **Show answer**

Ans: A

Q9. UPPCS Prelims 2018, Q46

When was Project Tiger launched in India?

A. 1972

B. 1973

C. 1992

D. 2006

💡 **Show answer**

Ans: B

Q10. UPPCS Prelims 2018, Q54

Which of the following is NOT an example of in-situ conservation strategy?

A. Biosphere reserve

B. Botanical garden

C. National Park

D. Sacred groves

💡 **Show answer**

Ans: B — Botanical garden is **ex-situ**.

Q11. UPPCS Prelims 2018, Q58

Which pair is NOT correctly matched?

A. Bandipur NP — Karnataka

B. Manas WLS — Assam

C. Periyar WLS — Kerala

D. Simlipal NP — Madhya Pradesh

💡 **Show answer**

Ans: D — Simlipal is in **Odisha**.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D | **Ans first, then Logic**

Q1. With reference to the Wildlife (Protection) Act, 1972, which of the following statements is/are correct?

1. It is India's main protected-area statute.
2. Terrestrial WPA strictness runs Community Reserve > Wildlife Sanctuary > National Park.
3. A Community Reserve is stricter than a National Park under WPA.

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Statement 1 is true (WPA 1972). Statement 2 reverses the strictness order (NP > WLS > Conservation/Community Reserve). Statement 3 is false — Community Reserve is the least strict of the three.

Q2. Consider the following pairs:

Category	WPA section teaching
1. Wildlife Sanctuary	Section 18
2. National Park	Section 35
3. Tiger Reserve	Section 38V (2006 path)

Which of the pairs given above is/are correctly matched?

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: C. All three pairs are correct. **Logic:** Sanctuary Sec. 18; NP Sec. 35; Tiger Reserve Sec. 38V.

Q3. Which of the following pairs is/are NOT correctly matched?

1. Biosphere Reserve — Wildlife (Protection) Act statutory category like National Park
2. Biosphere Reserve — UNESCO MAB three-zone model (core → buffer → transition)
3. India's first Biosphere Reserve — Nilgiri (1986)

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 2 only D. 1 only

💡 **Show answer**

Ans: D. Only pair 1 is not correctly matched. **Logic:** BR is MAB/UNESCO model — not a WPA category. Pairs 2–3 are correct.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): A National Park is the strictest terrestrial WPA protected area.

Reason (R): Grazing and forestry are largely banned in National Parks, and boundary change needs the State Legislature.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the strictness of National Parks. **A/R logic:** R states the hard restrictions that make NP the strictest WPA category.

Q5. Match List-I with List-II and select the correct answer using the code given below the lists:

List-I	List-II
A. Conservation Reserve	1. Private/community land with voluntary consent (Sec. 36B)
B. Community Reserve	2. Govt/community forest buffer near PA (Sec. 36A)
C. Elephant Reserve	3. Project Elephant 1992; no WPA statutory basis

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** 36A conservation; 36B community; Elephant Reserve executive (1992).

Q6. With reference to Tiger Reserves, which of the following statements is/are correct?

1. They rest on WPA Section 38V with core + buffer under NTCA.
2. India has 58 tiger reserves; Corbett was the first (1973).
3. Elephant Reserves are also notified under the same WPA Section 38V.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Elephant Reserves lack WPA statutory basis — not Sec. 38V.

Q7. Arrange the following in correct chronological order:

1. Wildlife (Protection) Act
2. First Tiger Reserve (Corbett / Project Tiger start)
3. First Biosphere Reserve (Nilgiri)
4. Project Elephant

A. 2-1-3-4 B. 1-2-4-3 C. 1-2-3-4 D. 1-3-2-4

💡 **Show answer**

Ans: C. 1972 → 1973 → 1986 → 1992. **Logic:** WPA 1972; Project Tiger/Corbett 1973; Nilgiri BR 1986; Project Elephant 1992.

Q8. With reference to marine protected areas, which of the following statements is/are correct?

1. India's first Marine National Park is Gulf of Kutch, Gujarat (1980).
2. Gulf of Mannar (Tamil Nadu) is also a Marine National Park.
3. Uttar Pradesh has two coastal Marine National Parks.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** UP is landlocked — zero marine NPs.

Q9. Which of the following pairs is/are correctly matched?

1. Rudrasagar — Tripura Ramsar site
2. Sultanpur — Haryana
3. Both Rudrasagar and Sultanpur — Uttar Pradesh Ramsar sites

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Neither Rudrasagar nor Sultanpur is a UP Ramsar site.

Q10. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): An Eco-Sensitive Zone (ESZ) is notified as a buffer around protected areas.

Reason (R): ESZ is itself a WPA Schedule category equal to National Park.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** ESZ flows from EPA 1986 buffering — not a WPA category.

Q11. Match List-I with List-II:

List-I	List-II
A. Ramsar Convention	1. 1982
B. India joins Ramsar	2. 1971
C. WPA	3. 1972

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** Ramsar 1971; India 1982; WPA 1972.

Q12. With reference to Sanctuary vs National Park, which of the following statements is/are correct?

1. Sanctuary allows limited human use with CWLW permission.
2. National Park is stricter; grazing/forestry largely banned.
3. Sanctuary boundary change always needs a Constitutional amendment.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** NP boundary change needs State Legislature — not a Constitutional amendment for sanctuary as stated.

Q13. How many of the following are WPA-linked protected area categories?

National Park, Wildlife Sanctuary, Conservation Reserve, Community Reserve, Biosphere Reserve

A. Only two B. Only three C. All five D. Only four

💡 **Show answer**

Ans: D. Only four. **Logic:** Biosphere Reserve is MAB — not a WPA category.

Q14. With reference to India Biosphere Reserves, which of the following statements is/are correct?

1. India has 18 notified national Biosphere Reserves.
2. About 12–13 are on the UNESCO World Network (count drifts).
3. Uttar Pradesh has the highest number of Biosphere Reserves in India.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** UP has zero Biosphere Reserves.

Q15. Which of the following is the correct MAB zonation sequence?

A. Transition → buffer → core B. Core → buffer → transition C. Buffer → core → transition D. Core → transition → buffer

💡 **Show answer**

Ans: B. Core → buffer → transition. **Logic:** Core has no exploitative use; buffer regulated; transition livelihoods.

Q16. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Elephant Reserves are created under Project Elephant (1992).

Reason (R): Elephant Reserves have the same WPA statutory basis as Tiger Reserves under Section 38V.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Elephant Reserves are executive notifications — no WPA statute like TR.

Q17. Arrange PA strictness (strictest to least strict among WPA terrestrial set):

A. Sanctuary > National Park > Community Reserve B. Community Reserve > Sanctuary > National Park C. National Park > Sanctuary > Conservation/Community Reserve D. Conservation Reserve > National Park > Sanctuary

💡 **Show answer**

Ans: C. NP > Sanctuary > Conservation/Community Reserve. **Logic:** Standard WPA terrestrial strictness ladder.

Q18. Which of the following pairs is/are correctly matched?

1. NTCA — governs Tiger Reserves
2. Project Elephant — about 33 elephant reserves class figure
3. ESZ — EPA 1986 buffer restricting mining/polluting industry near PAs

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three are correct. **Logic:** NTCA–TR; ~33 elephant reserves; ESZ from EPA.

Q19. With reference to Ramsar and UP, which of the following statements is/are correct?

1. Ramsar marks wetlands of international importance.
2. India joined Ramsar in 1982.
3. Calling Sultanpur a UP Ramsar site is correct.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Sultanpur is Haryana.

Q20. Match List-I with List-II:

List-I	List-II
A. Core (BR)	1. Livelihoods / sustainable use
B. Buffer (BR)	2. No exploitative use
C. Transition (BR)	3. Research, education, regulated use

Code:

A. A-3, B-2, C-1 B. A-2, B-3, C-1 C. A-2, B-1, C-3 D. A-1, B-3, C-2

💡 **Show answer**

Ans: B. A-2, B-3, C-1. **Logic:** Core no exploitation; buffer research/education; transition livelihoods.

Q21. Which of the following statements about Community Reserve is/are correct?

1. It sits on private or community land with voluntary consent.
2. It was inserted via 2002 WPA path (Sec. 36B).
3. It is identical to a Tiger Reserve core.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Not identical to TR core.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Conservation Reserve (Sec. 36A) is typically a buffer near a protected area.

Reason (R): Conservation Reserves are notified only under the Forest Rights Act, 2006.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Conservation Reserve is WPA 36A (2002) — not FRA.

Q23. With reference to firsts, which of the following statements is/are correct?

1. First Tiger Reserve associated teaching — Corbett, 1973.
2. First Biosphere Reserve — Nilgiri, 1986.
3. First Marine NP — Gulf of Kutch, 1980.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three are correct. **Logic:** Corbett TR; Nilgiri BR; Gulf of Kutch Marine NP.

Q24. Which one of the following correctly states the legal base of Eco-Sensitive Zones?

A. Environment (Protection) Act, 1986 buffer framework B. WPA Schedule I only C. Forest Rights Act, 2006 alone D. Biological Diversity Act, 2002 alone

💡 **Show answer**

Ans: A. EPA 1986 buffer framework. **Logic:** ESZ is EPA-linked, not a WPA schedule category.

Q25. How many of the following UP negative geography facts are correct?

1. Rudrasagar ≠ UP
2. Sultanpur ≠ UP
3. UP has marine national parks

A. Only one B. Only two C. All three D. None

💡 **Show answer**

Ans: B. Only two. **Logic:** 1 and 2 correct; 3 is false (UP has no marine NPs).

Q26. With reference to Tiger Reserve structure, which of the following statements is/are correct?

1. Core + buffer model under NTCA.
2. NTCA became statutory through 2005/2006 WPA path.
3. NTCA also notifies all Elephant Reserves under the same section.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Elephant Reserves are Project Elephant executive path.

Q27. Arrange: Conservation Reserve insertion → Community Reserve insertion teaching year path:

A. Both via EPA 1986 only B. Both via 1972 original WPA only C. Both via FRA 2006 only D. Both via 2002 WPA amendments (36A / 36B)

💡 **Show answer**

Ans: D. Both via 2002 WPA path (36A/36B). **Logic:** 2002 inserted Conservation and Community Reserves.

Q28. Which of the following pairs is/are NOT correctly matched?

1. Wildlife Sanctuary — limited regulated use possible
2. National Park — generally no grazing/forestry
3. Biosphere Reserve core — intensive mining as the default use

A. 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only pair 3 is not correctly matched. **Logic:** Core = no exploitative use.

Q29. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): India has about 58 tiger reserves in recent teaching counts.

Reason (R): Project Tiger began in 1992 along with Project Elephant.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Project Tiger began in 1973; Project Elephant in 1992.

Q30. With reference to protected-area governance, which of the following statements is/are correct?

1. Chief Wildlife Warden is the State implementing officer under WPA.
2. NBWL is the PM-chaired national advisory body.
3. ESZ notification makes a site a UNESCO Biosphere Reserve automatically.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** ESZ ≠ automatic UNESCO BR listing.

Mains Answer Framework

Protected area network in India (250 words): Define protected areas under WPA. Compare NP, sanctuary, TR, BR, Ramsar, ESZ. India counts and firsts (Corbett, Gulf of Kutch, Nilgiri BR). UP case — Dudhwa NP/TR, Pilibhit TR, 12 Ramsar sites, Terai corridor, no marine PA. Conclude with in-situ hierarchy and ESZ buffer role under EPA.

Common Traps — Don't Fall For These

Trap	Correct fact
No human activity in sanctuary	Activity **possible with CWLW permission**
Nokrek in Manipur	**Meghalaya**
Sultanpur in UP	**Haryana**
Rudrasagar in UP	**Tripura**
ESZ under WPA	**EPA 1986**

Trap	Correct fact
Elephant Reserve under WPA	**No** — Project Elephant only
Biosphere Reserve under WPA	**No** — executive/MAB
UP has marine PA	**No** — landlocked
Grazing allowed in NP	**Prohibited** in NP

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 7 — Wildlife Conservation (वन्यजीव संरक्षण)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Wildlife Conservation | Wildlife Protection (वन्यजीव सुरक्षा) | Project Tiger (प्रोजेक्ट टाइगर) | Project Elephant (प्रोजेक्ट एलीफेंट) | Crocodile Conservation Project | Asiatic Lion (सिंह) Conservation | Rhino Conservation | Snow Leopard Conservation | Vulture Conservation

Sources baked in: WPA 1972 (amended 2006), MoEFCC, NTCA, BNHS, IUCN, NCERT Biology Ch 15–16, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — launch years, statutory vs executive traps, species ↔ habitat **Last verified:** August 2026

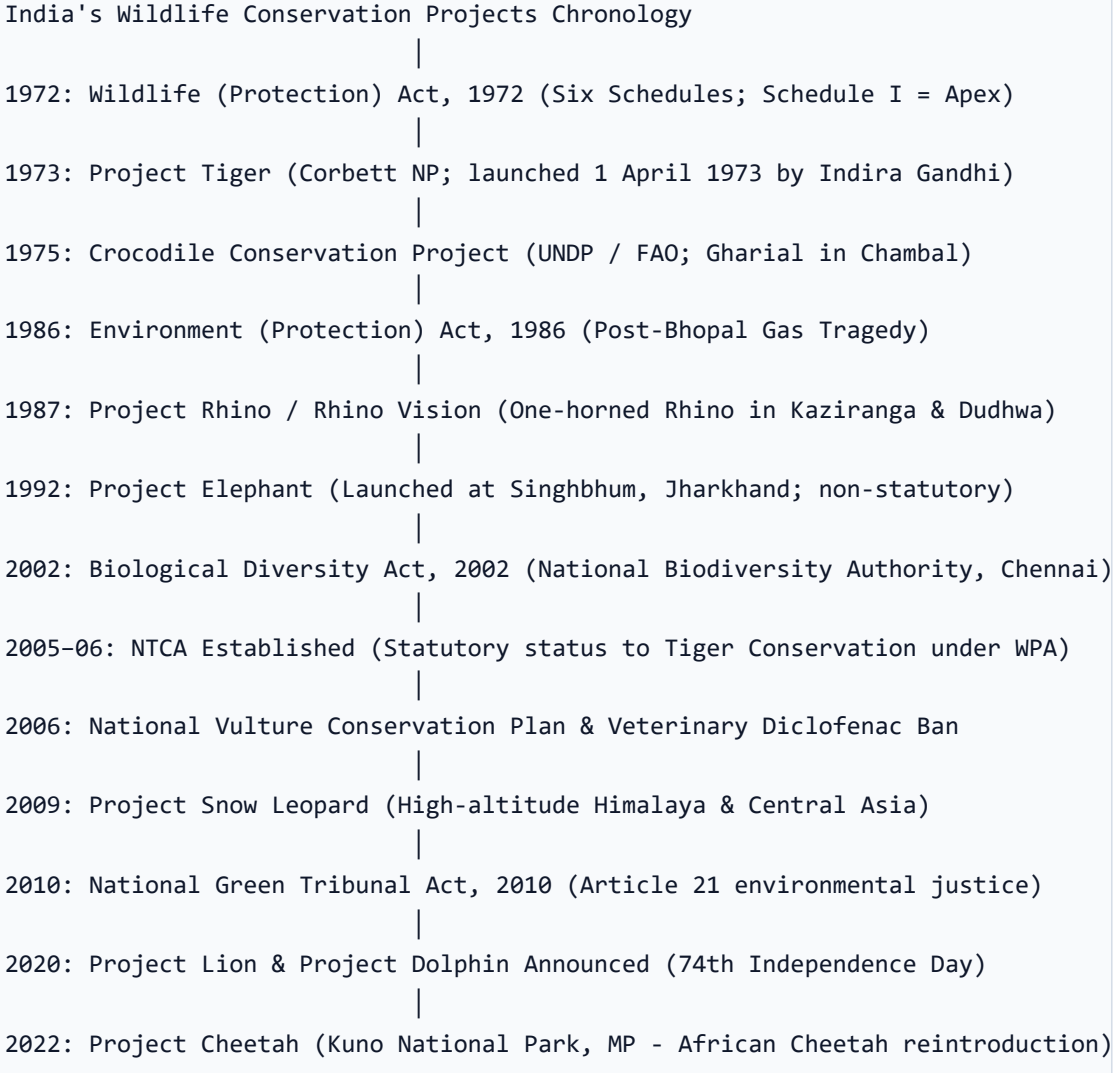
Current Affairs (this topic)

- Track Project Tiger / Project Elephant / species project headlines against the statutory vs scheme trap.
- NTCA tiger-reserve count drifts — teach the concept; verify the live total before a count stem.

💡 🎯 **Consolidated — 36 Must-Score Facts (Click to Expand)**

1. Wildlife conservation protects species and habitats, with **in-situ** (स्थान पर) primary and **ex-situ** (स्थानांतरित) as backup. 2. Major threats include habitat (वास स्थान) loss, poaching, human–wildlife conflict, invasive species (आक्रामक विदेशी प्रजाति), and veterinary **diclofenac** (डाइक्लोफेनाक) for vultures. 3. The **Wildlife (Protection) Act, 1972** followed the Stockholm moment and uses **six Schedules**. **Schedule I** is maximum protection. 4. Schedule I facts include tiger, elephant, rhino, gharial, and vultures among frequently asked species. A **Schedule I tortoise** enjoys the **same protection level as the tiger**. 5. **Schedule VI** plants cannot be cultivated without a **licence** from the Chief Wildlife Warden (or authorised officer) — not “banned under all circumstances.” 6. The **Chief Wildlife Warden** is the State implementing officer. **NBWL** is the PM-chaired national advisory body. 7. **NTCA** became statutory through the **2005/2006** WPA amendments and governs **Tiger Reserves only**. 8. **Project Tiger** (प्रोजेक्ट टाइगर) began in **1973**. Corbett was the first tiger reserve. India has **58** tiger reserves. The **2022** census figure often cited is **3,682** tigers. 9. **Project Elephant** (प्रोजेक्ट एलीफेंट) began in **1992** with about **33** elephant reserves and **no WPA statutory basis** — it covers **wild and captive** elephants. 10. The **Crocodile Project (1975)** covers gharial, mugger, and saltwater crocodile. Best natural gharial view: **Chambal** (चंबल) (also Girwa); IUCN (आईयूसीएन) status **Critically Endangered**. 11. **Project Lion (2020)** focuses on Asiatic lions at **Gir (गीर), Gujarat**. Uttar Pradesh (उत्तर प्रदेश) has **no** wild Asiatic lion population (जनसंख्या). 12. **IRV 2020** planned rhino recovery across seven PAs, including **Dudhwa** (दुधवा) in Uttar Pradesh. 13. **Project Snow Leopard (2009)** and **GSLEP (2013)** cover high Himalaya (हिमालय). UP has **no** snow leopard habitat. 14. The **Vulture Action Plan (2006)** and veterinary **diclofenac ban (2006)** address renal failure in Indian vultures. **SAVE** (Nepal–India) targets **vulture** recovery. 15. An **umbrella** species (tiger, elephant) protects wide ecosystems. A **flagship** species mobilises public support. A **keystone** species’ loss collapses function (vulture example). 16. World Wildlife Day (विश्व वन्यजीव दिवस) is **3 March**. Wildlife Week in India is **2–8 October**. WWF (डब्ल्यूडब्ल्यूएफ) logo animal is the **Giant Panda**. 17. **Operation Olivia** is an Indian Coast Guard (तटरक्षक बल) nesting patrol on the **Odisha** (ओडिशा) coast. 18. UP tiger reserves are **Dudhwa, Pilibhit, and Ranipur**. 19. UP rhino fact is **Dudhwa National Park** (दुधवा राष्ट्रीय उद्यान) (reintroduced population). 20. UP gharial sites include **Katarniaghat

Wildlife Sanctuary** and the **Chambal** sanctuary landscape. 21. UP elephant reserves are **UP ER (2009)** and **Terai (तराई) ER (2022)**. 22. Programme-year chain: WPA **1972** → Project Tiger **1973** → Crocodile Project **1975** → Project Elephant **1992** → Biodiversity Act **2002** → NTCA statutory **2005/2006** → Vulture ban **2006** → Project Snow Leopard **2009** → Project Lion **2020**. 23. **Whale shark** is India's largest fish; seasonal aggregations off **Gujarat** (Mar–May); protection programme from **2004**. 24. **Kharai camel** (Kutch, Gujarat) can swim in seawater and graze mangroves; can be domesticated — **not** "wild only." 25. **Pugmark** technique estimates (प्राकलन समिति) wild animal populations from footprints. 26. **Spider monkey** is Central/South American — **not** naturally found in India (Star tortoise, Monitor lizard, Pygmy hog are). 27. Scientific names: Asiatic wild ass (**Equus hemionus**), Barasingha (**Rucervus/Cervus duvaucelii**), Chinkara (**Gazella bennettii**), Nilgai (**Boselaphus tragocamelus**). 28. **Community Reserve** (सामुदायिक आरक्षित क्षेत्र) (WPA): CWLW authority applies; hunting banned; NTFP collection allowed; **traditional agriculture / jhum not** allowed after notification. 29. **Animal Welfare Board of India** (1962) is under the **Prevention of Cruelty to Animals Act, 1960** — **not** EPA 1986 (पर्यावरण संरक्षण 1986). NTCA is statutory; NGRBA was PM-chaired. 30. Gharial, Indian wild ass, and wild buffalo cannot be hunted except under limited WPA provisions. 31. India holds about **2.4%** of world land area but about **8%** of known biodiversity (~**7.28%** of known animal species in Alfred-type figures). 32. First International Tiger Forum / World Tiger Summit **2010** was at **Saint Petersburg**, Russia. 33. **New York Declaration on Forests (2014)** is voluntary/non-binding; India was **not** an original signatory. 34. **UNCCD (1994)** (मरुस्थलीकरण रोकथाम संधि) is the sole legally binding desertification (मरुस्थलीकरण) convention; bottom-up local participation; no exclusive South Asia finance monopoly claim. 35. Conservation is ecological protection; **Wildlife Protection** language often means the **WPA enforcement** track. 36. NTCA does not run Elephant Reserves; confusing NTCA with Project Elephant administration is a trap.



💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Wildlife Conservation** | **Wildlife Protection** | Holistic scientific management and preservation of species, populations, and habitats to prevent extinction vs legal punitive enforcement and regulatory prohibition of hunting/poaching under Wildlife Protection Act 1972 | वन्यजीव संरक्षण (वैज्ञानिक प्रबंधन) / वन्यजीव सुरक्षा (कानूनी प्रवर्तन) | | **Umbrella Species** | **Flagship Species** | Ecologically wide-ranging species whose expansive habitat protection indirectly shelters the broader biological community (e.g. Tiger, Rhino) vs popular charismatic animal used as an emotional figurehead in conservation marketing (e.g. Giant Panda, Snow Leopard) | छत्र प्रजाति / प्रतीक प्रजाति | | **Keystone Species** | **Indicator Species** | Species whose functional role exerts disproportionate control on community structure (loss causes trophic cascade; e.g. Fig tree, Sea Otter) vs species whose physiological response or presence directly indicates environmental conditions (e.g. Lichen, Frogs) | मुख्य आधार (कीस्टोन) प्रजाति / जैव संकेतक प्रजाति | | **Project Tiger (1973)** | **Project Elephant (1992)** | Centrally Sponsored Scheme given statutory backing in 2006 under WPA creating the National Tiger Conservation Authority (NTCA) vs Centrally Sponsored Scheme operating through administrative guidelines without statutory authority | प्रोजेक्ट टाइगर (1973, सांविधिक) / प्रोजेक्ट एलिफेंट (1992, गैर-सांविधिक) | | **Diclofenac** | **Meloxicam** | Non-steroidal anti-inflammatory drug (NSAID) administered to cattle causing lethal visceral gout and acute renal failure in Gyps vultures (banned in 2006) vs safe, vulture-tested veterinary NSAID recommended as its non-toxic replacement | डाइक्लोफेनाक (गिद्ध नाशक) / मेलोक्सिकैम (सुरक्षित विकल्प) | | **Wildlife Protection Act 1972** | **Biological Diversity Act 2002** | Focuses on species protection, hunting bans, protected area network, and trade schedules (MoEFCC/WCCB) vs focuses on sovereign rights over biological resources, fair & equitable benefit sharing (ABS), and traditional knowledge protection (NBA/SBB/BMC) | वन्यजीव संरक्षण अधिनियम 1972 / जैव विविधता अधिनियम 2002 | | **Central Zoo Authority (CZA)** | **National Board for Wildlife (NBWL)** | Statutory body under WPA Sec 38A overseeing standards, recognition, and animal exchanges in Indian zoos vs apex advisory body chaired by the Prime Minister framing national wildlife policies and approving PA boundary alterations | केंद्रीय चिड़ियाघर प्राधिकरण / राष्ट्रीय वन्यजीव बोर्ड (NBWL) |

💡 📋 **Must-score facts — projects, schedules, NTCA (Click to Expand)**

Law / schedules | Item | Lock | |------|------| | WPA Schedules | Six; **Schedule I** = maximum protection | | Schedule VI | Plants; licence from CWLW (not absolute (निरपेक्ष) ban) | | NBWL / CWLW | PM-chaired advisory / State implementing officer | | NTCA | Statutory via 2005–06 path; Tiger Reserves only | ### Projects | Item | Lock | |------|------| | Project Tiger | **1973**; Corbett first; **58** TRs; ~3682 (2022) | | Project Elephant | **1992**; ~33 reserves; wild + captive | | Crocodile Project | **1975**; gharial–mugger–saltwater; Chambal | | Project Lion | **2020**; Gir only (no wild UP lions) | | IRV 2020 | Rhino recovery; includes **Dudhwa** | | Project Snow Leopard / GSLEP | **2009** / **2013** (no UP habitat) | | Vulture plan / diclofenac ban | **2006** / **2006** |

7.1 Wildlife Conservation

Identity: In-situ primary | Species programmes + community participation | MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) nodal

- **Wildlife conservation** means protecting wild plants and animals and the habitats they need to survive.
- The main strategy is **in-situ conservation** (स्व-स्थाने संरक्षण), which protects species where they naturally live.
- **Ex-situ conservation** (बाह्य-स्थाने संरक्षण) uses zoos, captive breeding, and gene banks as backup when wild populations are critically low.
- A **flagship species** (प्रतीक (फ्लैगशिप) प्रजाति) such as the tiger attracts funding and public attention for conservation.

- An **umbrella species** (छत्र (अंब्रेला) प्रजाति) such as the tiger or elephant needs large habitat, so protecting it also protects many other species.
- A **keystone species** (कीस्टोन प्रजाति) such as the vulture plays an irreplaceable ecological role. Its loss harms the whole system.
- Major threats include habitat fragmentation, poaching, human–wildlife conflict, veterinary drug poisoning, and invasive species.
- India runs **species-specific programmes** under MoEFCC on top of the protected area network.
- **World Wildlife Day** is observed on **3 March**, marking the adoption of CITES (साइट्स) in 1973.
- **Wildlife Week** in India is observed from **2 to 8 October**.
- **Operation Olivia** is an **Indian Coast Guard** initiative to protect **Olive Ridley turtles** along the **Odisha coast**.
- Olive Ridley **nesting season** runs from **November to February**, when mass **arribada** occurs at Gahirmatha and Rushikulya.

PYQ — UPPCS Prelims 2025, Q62

Which pairs is/are NOT correctly matched? (Day) — (Date): 1. Ozone Day — 16 Sept 2. WED — 5 June 3. World Wildlife Day — 22 March

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

> 💡 **Show answer**
>
> **Ans: B** — World Wildlife Day = **3 March**.

PYQ — UPPCS Prelims 2025, Q143

With reference to Operation Olivia, which is/are correct?

1. ICG initiative to protect Olive Ridley turtles along Odisha coast.
2. Carried out in August during nesting.

A. Only 2

B. Neither

C. Both

D. Only 1

> 💡 **Show answer**
>
> **Ans: D** — Nesting is **Nov–Feb**, not August.

Statutory Wildlife Governance & Legal Reforms — NBWL, NTCA, WCCB & WPA 2022

7.2 Wildlife Protection

Identity: WPA 1972 | 6 Schedules | CWLW + NBWL | Art 48A, 51A(g)

- **Wildlife (Protection) Act, 1972** is India's main law for protecting wild animals, birds, and plants.
- The Act was enacted in the same year as the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन).
- The Act divides species into **six schedules** based on protection level.
- **Schedule I** gives maximum protection. Hunting and trade (पण्याध्यक्ष) are banned for tiger, elephant, rhino, lion, gharial, snow leopard, and Indian vultures.

- If a tortoise (or any species) is placed in **Schedule I**, it enjoys the **same protection level as the tiger**.
- **Schedule V** lists vermin such as the common crow and rat, which may be hunted.
- **Schedule VI** lists specified plants whose cultivation is **prohibited except with a licence** from the Chief Wildlife Warden or authorised officer (classic pitcher-plant / specified-plant pattern).
- Gharial, Indian wild ass, and wild buffalo cannot be hunted except under limited WPA provisions.
- The **Chief Wildlife Warden (CWLW)** is the state-level authority for permits and sanctuary management.
- The **National Board for Wildlife (NBWL)** (राष्ट्रीय वन्यजीव बोर्ड) is chaired by the **Prime Minister** and advises on protected area policy.
- Offences against Schedule I species can lead to imprisonment up to **7 years**.
- **Article 48A** directs the State to protect wildlife. **Article 51A(g)** makes it a citizen's duty to protect wildlife.
- Once an area is notified as a **Community Reserve**, hunting is banned, NTFP collection may continue, but **traditional agriculture / jhum is not** allowed; CWLW authority under WPA applies.
- **Animal Welfare Board of India** was established in **1962** under the **Prevention of Cruelty to Animals Act, 1960** — not under EPA 1986.

Schedule	Protection	Examples
I	Maximum — hunting banned	Tiger, elephant, rhino, gharial, vulture
II	High	Himalayan black bear, cobra
III–IV	Moderate (नरम दल)/lower fauna (प्राणीजात)	Chital, sambar, hare
V	Vermin — may be hunted	Crow, rat
VI	Specified plants — licence needed	Pitcher plant and other listed taxa

> **Logic:** WPA = **1972**. Project Tiger = **1973**. Do not swap the two years. Schedule VI ≠ absolute cultivation ban.

7.3 Project Tiger

Identity: CSS 1973 | NTCA 2005/2006 | 58 TRs | Umbrella species model

- **Project Tiger** was launched in **April 1973** by PM Indira Gandhi (गांधी) at Jim Corbett (जिम कॉर्बेट) National Park.
- The project began with **9 tiger reserves**. **Corbett** was the first reserve.
- India now has **58 tiger reserves**. **Madhav Tiger Reserve** (MP) is the 58th.
- The **National Tiger Conservation Authority (NTCA)** became a statutory body under the **WPA 2006 amendment**.
- Every tiger reserve must have a **core zone** (कोर क्षेत्र) kept inviolate and a **buffer zone** (बफर क्षेत्र) for regulated use.
- The **All India Tiger Estimation** runs every **4 years** using camera traps and the **M-STrIPES** app (Monitoring System for Tigers — Intensive Protection and Ecological Status: field patrol protocols + GIS software for management).
- **International Tiger Day** is **29 July** (decided at the **2010 St Petersburg** Tiger Summit).
- **Nagarjunsagar–Srisailam** holds the largest critical tiger habitat among common options. **Bandipur** (बांदीपुर) pioneered drone monitoring among common stems.
- **Sathyamangalam TR (Tamil Nadu (तमिलनाडु))** sits at the Eastern–Western Ghats (पश्चिमी घाट) junction.
- The **2022 census** recorded **3,682 tigers** in India.
- **Madhya Pradesh** has the most tiger reserves among states.
- India hosts about **70% of the world's wild tigers**.

- The tiger is an **umbrella species**. Protecting it also protects prey, co-predators, and forest habitat.
- UP tiger reserves are **Dudhwa (1987)**, **Pilibhit (2014)**, and **Ranipur (2023)**.

> **Logic:** Tiger Reserve has **WPA statutory basis (Sec 38V)**. Elephant Reserve (हाथी अभयारण्य) does **not**.

7.4 Project Elephant

Identity: CSS 1992 | 33 ERs | 150 corridors | No WPA statutory basis

- **Project Elephant** was launched in **1992** as a Centrally Sponsored Scheme.
- It aims to protect elephants, restore corridors, and reduce **human–elephant conflict**.
- It covers welfare of **wild Asian elephants and captive elephants** — "wild only" is a false stem.
- **Elephant Reserves** are notified by state governments on MoEFCC recommendation.
- India has **33 Elephant Reserves** across **14 states**.
- About **150 elephant corridors** have been identified for genetic exchange between herds.
- Elephant Reserves have **no separate statutory category under WPA 1972**.
- Legal protection inside an ER comes from overlapping sanctuaries, national parks, and forest laws.
- India's elephant population is estimated at about **30,000** (2020 census trend).
- UP has **Uttar Pradesh Elephant Reserve (2009)** and **Terai Elephant Reserve (2022)**.
- The Asian elephant is listed in **WPA Schedule I**.

> **Logic:** Trap — "Elephant Reserve is declared under WPA like Tiger Reserve" is **false**.

7.5 Crocodile Conservation Project

Identity: 1975 | Crocodile Breeding and Management Project | 3 Indian species

- The **Crocodile Conservation Project** was launched in **1975** with UNDP (यूएनडीपी)/FAO assistance.
- Its official name is the **Crocodile Breeding and Management Project**.
- It covers **three species**: **gharial**, **mugger (marsh crocodile)**, and **saltwater crocodile**.
- The **gharial** has a long narrow snout and lives in clean deep rivers such as the **Chambal** and **Girwa**.
- To see gharials in a natural habitat, **Chambal River / National Chambal Sanctuary** is the standard answer among common options.
- The **mugger** is the most widespread Indian crocodilian and occurs in rivers, lakes, and tanks.
- The **saltwater crocodile** is the largest reptile and lives in coastal mangroves, notably at **Bhitar Kanika** (Odisha).
- The gharial is **Critically Endangered** on the IUCN Red List.
- **National Chambal Sanctuary** (UP–MP–Rajasthan) is India's main gharial stronghold.
- **Katarniaghat Wildlife Sanctuary** in Bahraich, UP, protects gharial on the Girwa river.
- **Whale shark** is India's (and among the world's) largest fish; seasonal aggregations off the **Gujarat** coast (about March–May); a dedicated protection programme began around **2004**.
- **Kharai camel** of Kutch can swim several kilometres in seawater and graze mangroves; it can be domesticated (statement "lives only wild / cannot be domesticated" is false).
- **Pugmark** technique estimates wild animal numbers from species-specific footprints.
- **Spider monkeys** occur in Central and South American tropical forests — they are **not** naturally found in India.
- Scientific name locks: Asiatic wild ass (**Equus hemionus**), Barasingha (**Rucervus/Cervus duvaucelii**), Chinkara (**Gazella bennettii**), Nilgai (**Boselaphus tragocamelus**).

- WWF has used the **Giant Panda (*Ailuropoda melanoleuca*)** as its logo since **1961**.
- The first International Tiger Forum / World Tiger Summit (**2010**) was held at **Saint Petersburg**, Russia.
- Captive breeding centres operate at Nandankanan, Hyderabad (हैदराबाद), Chennai, Lucknow (लखनऊ), and Trivandrum.
- **Ganges River Dolphin** (*Platanista gangetica*) is India's **National Aquatic Animal (notified 2009)** — one of four (चातुर्याम) obligate freshwater dolphins; Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)–Meghna systems. Decline drivers: dams/barrages, fishing nets, agrochemicals — not mainly rising crocodile numbers. UP also uses a state aquatic-animal / Meri Ganga Meri Dolphin awareness framing.
- IUCN teaching: gharial often **Critically Endangered**; mugger/saltwater crocodile commonly **Vulnerable** in bank notes.

> **Logic:** The crocodile project covers **three species**, not gharial alone. National aquatic animal = **dolphin**, not gharial.

7.6 Asiatic Lion Conservation

Identity: Gir, Gujarat only | Project Lion 2020 | Kuno second home planned

- The **Asiatic lion** (*Panthera leo persica*) survives in the wild only in the **Gir landscape of Gujarat**.
- India's wild Asiatic lion population is about **750+** (2025 trend).
- Historically lions ranged from Greece to Bihar. Hunting reduced them to Gir by the early 20th century.
- **Gir Conservation Programme** focused on habitat, prey base, and disease control after a **canine distemper** outbreak in 2018.
- **Project Lion** was launched in **August 2020** to strengthen Gir management and create a second population.
- **Kuno-Palpur Wildlife Sanctuary** in Madhya Pradesh is the planned second home for translocated lions.
- The Asiatic lion is in **WPA Schedule I** and is the **state animal of Gujarat**.
- **Uttar Pradesh has no wild Asiatic lion**. Dudhwa holds tiger and rhino, not lion.

> **Logic:** Trap — "Asiatic lions in Dudhwa/UP" is **false**. Gir = Gujarat only.

7.7 Rhino Conservation

Identity: Greater one-horned rhino | Kaziranga (काजीरंगा) largest | IRV 2020 | Dudhwa UP site

- The **Greater One-Horned Rhinoceros** (*Rhinoceros unicornis*) lives in Terai and Brahmaputra floodplain grasslands.
- **Kaziranga National Park** (Assam) holds the **largest rhino population** (~2,600+).
- Poaching for horn remains the main threat despite strict anti-poaching in Kaziranga.
- **Indian Rhino Vision 2020 (IRV 2020)** aimed to spread rhinos across **7 protected areas**.
- Target sites include Kaziranga, Pobitora, Orang, and Manas in Assam, Jaldapara and Gorumara in West Bengal, and **Dudhwa** in UP.
- **Dudhwa National Park** in Lakhimpur Kheri received **reintroduced rhinos** under IRV 2020.
- India hosts over **80% of the global** greater one-horned rhino population.
- The rhino is **WPA Schedule I** and **IUCN Vulnerable**.

> **Logic:** UP rhinos are in **Dudhwa**, not Kaziranga. Kaziranga is in Assam.

7.8 Snow Leopard Conservation

Identity: Project Snow Leopard 2009 | GSLEP 2013 | Himalaya only | 500–700 in India

- The **snow leopard** (*Panthera uncia*) lives at **3,000–5,400 m** in the Himalayas and trans-Himalayas.
- India has an estimated **500–700 snow leopards** across Ladakh (लद्दाख), HP, Uttarakhand (उत्तराखंड), Sikkim, Arunachal Pradesh, and J&K.
- **Project Snow Leopard** was launched in **2009** to strengthen high-altitude protected area management.
- **GSLEP (Global Snow Leopard & Ecosystem (पारिस्थितिकी तंत्र) Protection Programme)** was launched in **2013** by **12 range countries**.
- **Hemis National Park** (हेमिस) in Ladakh is a key snow leopard habitat.
- **SECURE Himalaya** is a UNDP-GEF project for high-altitude biodiversity in Himachal (हिमाचल), Uttarakhand, Sikkim, and Ladakh.
- The snow leopard is an **umbrella species** for alpine ecosystems.
- **International Snow Leopard Day** is **23 October**.
- The snow leopard is **WPA Schedule I**.

> **Logic:** Snow leopard is **not** found in Gir, Western Ghats, or Terai plains.

7.9 Vulture Conservation

Identity: Diclofenac crisis | Vet ban 2006 | Action Plan 2006 | Pinjore breeding

- **Vultures** are keystone scavengers that quickly remove carcasses and limit disease spread.
- India lost about **99% of its vultures** between the 1980s and 2007 — one of the fastest avian declines on record.
- **BNHS research (2003)** identified **diclofenac**, a veterinary NSAID given to cattle, as the main cause of vulture renal failure.
- **Veterinary diclofenac was banned in 2006**. **Meloxicam** (मेलोक्सिकैम) is promoted as a safe alternative.
- The **Action Plan for Vulture Conservation 2006** has four pillars: ban toxic drugs, breeding centres, Vulture Safe Zones, and monitoring.
- **Vulture Safe Zones** are roughly **100 km radius** areas where diclofenac use is eliminated.
- Captive breeding centres operate at **Pinjore** (Haryana), **Rajabhatkhawa** (West Bengal), **Rani** (Assam), and **Bhopal** (MP).
- The **SAVE programme** (Saving Asia's Vultures from Extinction) coordinates international vulture recovery.
- Key Indian vulture species — white-rumped, long-billed, and slender-billed — are **Critically Endangered**.

> **Logic:** UPPCS 2018 pattern — cattle painkiller linked to vulture decline is **diclofenac**.

PYQ — UPPCS Prelims 2018, Q131

Which painkiller given to cattle is responsible for near extinction of vultures in India?

A. Ibuprofen

B. Acetaminophen

C. Aspirin

D. Diclofenac

💡 **Show answer**

Ans: D

Complete PYQ Bank (Topic 7)

Q1. UPPCS Prelims 2025, Q62

Which of the following pairs is/are **NOT** correctly matched? (Day — Date)

1. Ozone Day — 16 September
2. World Environment Day — 5 June
3. World Wildlife Day — 22 March

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1 only

💡 **Show answer**

Logic: Tests international environment/wildlife observance dates; only one listed date is wrong. **Ans: B.** World Wildlife Day is observed on **3 March**, not 22 March. **Trap:** Ozone Day (16 September) and World Environment Day (5 June) are correct — do not pick options mixing them with the WWD error.

Q2. UPPCS Prelims 2025, Q143

With reference to **Operation Olivia**, which of the following statements is/are correct?

1. It is an Indian Coast Guard initiative to protect Olive Ridley turtles along the Odisha coast.
2. It is carried out mainly in August during the nesting season.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

💡 **Show answer**

Logic: Statement 1 tests agency, species, and coast; statement 2 tests nesting season timing. **Ans: D.** Operation Olivia is an ICG Odisha-coast turtle protection drive, but nesting runs **November–February**, not August. **Trap:** August sounds like a monsoon-adjacent season — Olive Ridley mass nesting on Odisha beaches peaks in winter.

Q3. UPPCS Prelims 2018, Q46

Project Tiger was launched in India in the year—

A. 1972 B. 1973 C. 1992 D. 2006

💡 **Show answer**

Logic: Separates WPA enactment year from flagship species programme launch year. **Ans: B.** Project Tiger began in **April 1973** under PM Indira Gandhi at Jim Corbett National Park. **Trap:** **1972** is the Wildlife (Protection) Act year — not Project Tiger.

Q4. UPPCS Prelims 2018, Q131

Which painkiller given to cattle is responsible for the near extinction of vultures in India?

A. Ibuprofen B. Acetaminophen C. Aspirin D. Diclofenac

💡 **Show answer**

Logic: Links veterinary NSAID use to gyps vulture population collapse through kidney failure after feeding on carcasses. **Ans: D.** **Diclofenac** in cattle carcasses caused catastrophic vulture mortality; India banned

veterinary diclofenac in **2006**. **Trap:** Human painkillers in options are distractors — the crisis was from **veterinary** use on livestock.

Q5. UPPCS Prelims 2019, Q67

Arrange the following events in chronological order of their commencement:

I. Wild Life (Protection) Act II. Biological Diversity Act III. Project Tiger IV. Project Elephant

A. I, II, III, IV B. I, III, IV, II C. II, III, IV, I D. II, III, I, IV

💡 **Show answer**

Logic: Chronology of wildlife statutes and flagship species programmes from 1972 to 2002. **Ans: B.** Order: WLPA **1972** → Project Tiger **1973** → Project Elephant **1992** → Biological Diversity Act **2002**. **Trap:** Do not place BDA before Project Tiger — biodiversity legislation came **three decades** after the tiger programme.

Q6. UPPCS Prelims 2023, Q90

Assertion (A): Twelve cheetahs arrived from South Africa to India in February 2023.

Reason (R): Restoring the cheetah population is considered to be a priority for India.

A. Both (A) and (R) are true and (R) is the correct explanation of (A). B. Both (A) and (R) are true but (R) is not the correct explanation of (A). C. (A) is true but (R) is false. D. (A) is false but (R) is true.

💡 **Show answer**

A/R logic: A tests the February 2023 reintroduction fact under Project Cheetah; R tests whether cheetah restoration is a stated national priority that explains that translocation. **Ans: A.** Twelve cheetahs were airlifted from South Africa in February 2023 to Kuno National Park, and restoring cheetah is an explicit Project Cheetah priority — R explains A. **Trap:** Namibia cheetahs arrived later — do not confuse the **South Africa batch (Feb 2023)** with subsequent translocations.

Q7. UPPCS Prelims 2022, Q150

The Union Ministry of Environment, Forest and Climate Change has announced **'Dolphin Day'**, which will be celebrated on—

A. April 15 B. October 5 C. July 15 D. October 28

💡 **Show answer**

Logic: Tests MoEFCC observance for the Ganges river dolphin — India's National Aquatic Animal. **Ans: B.** **National Dolphin Day** is observed on **5 October** to raise awareness for freshwater dolphin conservation. **Trap:** Do not swap with Wildlife Week (**2–8 October**) or World Environment Day (**5 June**).

Extra Drill — Species, Schedules & Institutions (Q31–Q40)

Q31. If a tortoise is listed under Schedule I of WPA 1972, it implies:

A. It is extinct in the wild

B. It enjoys the same protection level as the tiger

C. It is endemic to one state only

D. It may be hunted as vermin

💡 **Show answer**

Logic: Schedule levels, Indian vs exotic fauna, and institution years are high-yield traps. **Ans: B.** Schedule I = maximum protection, same band as tiger.

Q32. Schedule VI plant cultivation under WPA:

A. Is allowed freely everywhere

B. Is banned under all circumstances

C. Needs a licence from CWLW / authorised officer

D. Applies only to genetically modified crops

💡 **Show answer**

Ans: C. Cultivation of Schedule VI plants needs a licence — not an absolute ban.

Q33. Best place among options to see gharial in natural habitat:

A. Pulicat Lake

B. Deepor Beel

C. Chambal River

D. Bhitarkanika only

💡 **Show answer**

Ans: C. Chambal / National Chambal Sanctuary; also Girwa–Katarniaghat. IUCN: Critically Endangered.

Q34. Kharai camel — which statements are correct? 1. Can swim in seawater 2. Grazes mangroves 3. Cannot be domesticated

A. 1 and 2 only

B. 3 only

C. 1 and 3 only

D. 1, 2 and 3

💡 **Show answer**

Ans: A. Kutch swimming camel; domesticated by graziers. Statement 3 is false.

Q35. Which is **not** naturally found in India?

A. Star tortoise

B. Monitor lizard

C. Pygmy hog

D. Spider monkey

💡 **Show answer**

Ans: D. Spider monkeys are Central/South American. Pugmark technique estimates wild populations from footprints.

Q36. Animal Welfare Board of India was established under:

A. Environment (Protection) Act, 1986

B. Prevention of Cruelty to Animals Act, 1960

C. Wildlife (Protection) Act, 1972

D. Biological Diversity Act, 2002

💡 **Show answer**

Ans: B. AWBI **1962** under PCA **1960**. NTCA is statutory; NGRBA was PM-chaired.

Q37. Symbol of WWF since inception:

A. Tiger

B. Giant Panda

C. Hornbill

D. White Bear

💡 **Show answer**

Ans: B. *Ailuropoda melanoleuca*; WWF founded **1961**. World Tiger Summit **2010** = Saint Petersburg.

Q38. India's largest fish among common options:

A. Hilsa

B. Marlin

C. Whale shark

D. Stonefish

💡 **Show answer**

Ans: C. Whale shark; seasonal aggregations off Gujarat; protection push from ~2004.

Q39. Match correctly: Asiatic wild ass — Barasingha — Chinkara — Nilgai

A. Equus hemionus; Cervus/Rucervus duvaucelii; Gazella bennettii; Boselaphus tragocamelus

B. Boselaphus; Equus; Gazella; Cervus

C. Gazella; Equus; Cervus; Boselaphus

D. Cervus; Gazella; Equus; Boselaphus

💡 **Show answer**

Ans: A. Keep one name–one sentence locks; do not swap Nilgai with wild ass.

Q40. Community Reserve notification — which is incorrect?

A. Hunting is not allowed

B. People may collect non-timber forest produce

C. Traditional agriculture / jhum continues freely after notification

D. WPA provisions / CWLW authority apply

💡 **Show answer**

Ans: C. After notification, hunting stops and agriculture/jhum is not allowed; NTFP collection may continue.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D | **Ans first, then Logic**

Q1. With reference to wildlife conservation strategy, which of the following statements is/are correct?

1. ex-situ conservation is the preferred primary strategy.
2. Ex-situ methods are backup tools.
3. Ex-situ alone is enough; in-situ parks are optional in all cases.

A. Only 1 B. 1 and 2 C. Only 3 D. Only 2

💡 **Show answer**

Ans: D. Only 2 is correct. **Logic:** Only statement 2 remains true among the numbered claims. Statement 1 is false. Statement 2 is true. Statement 3 is false.

Q2. Which of the following is/are major wildlife threats in standard teaching?

1. Habitat loss
2. Poaching
3. Veterinary diclofenac for vultures
4. Human–wildlife conflict

A. 1 and 2 only B. 1, 2 and 4 only C. 1, 2, 3 and 4 D. 2, 3 and 4 only

💡 **Show answer**

Ans: C. All four. **Logic:** All are standard threat facts including diclofenac.

Q3. With reference to WPA Schedules, which of the following statements is/are correct?

1. Schedule I gives maximum protection.
2. Schedule VI covers specified plants needing licence for cultivation.
3. Schedule I animals may be hunted freely with a district collector's oral order.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Schedule I is maximum protection — not free hunting.

Q4. Match List-I with List-II:

List-I	List-II
A. Project Tiger	1. 1992
B. Project Elephant	2. 1973
C. Crocodile Project	3. 1975

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Tiger 1973; Elephant 1992; Crocodile 1975.

Q5. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NTCA governs Tiger Reserves.

Reason (R): NTCA became statutory through the 2005/2006 WPA amendments path.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains NTCA's statutory tiger role. **A/R logic:** Statutory NTCA is tied to the TR governance path.

Q6. With reference to Project Tiger, which of the following statements is/are correct?

1. It began in 1973; Corbett was the first tiger reserve.
2. India has 58 tiger reserves in recent counts.
3. The 2022 census figure often cited is about 3,682 tigers.

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

💡 **Show answer**

Ans: B. All three. **Logic:** 1973/Corbett; 58 TRs; ~3682 (2022).

Q7. Which of the following pairs is/are NOT correctly matched?

1. Project Lion (2020) — Asiatic lions at Gir, Gujarat
2. Project Snow Leopard (2009) — high Himalaya
3. Project Lion — wild Asiatic lions across Uttar Pradesh plains

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** UP has no wild Asiatic lion population.

Q8. Arrange the following in chronological order:

1. Project Tiger
2. Crocodile Project
3. Project Elephant
4. Project Snow Leopard

A. 1-2-4-3 B. 2-1-3-4 C. 1-3-2-4 D. 1-2-3-4

💡 **Show answer**

Ans: D. 1973 → 1975 → 1992 → 2009. **Logic:** Tiger 1973; Crocodile 1975; Elephant 1992; Snow Leopard 2009.

Q9. With reference to gharial, which of the following statements is/are correct?

1. Crocodile Project covers gharial, mugger, and saltwater crocodile.
2. Chambal is a classic natural gharial viewing landscape.
3. Gharial is absent from Schedule I teaching lists.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: A. Only 1 and 2. **Logic:** Gharial is a frequent Schedule I fact.

Q10. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Veterinary diclofenac ban (2006) addresses vulture decline.

Reason (R): Diclofenac caused renal failure in Indian vultures feeding on treated carcasses.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: D. Both true and R explains the ban. **A/R logic:** Causal link is renal failure from diclofenac residues.

Q11. Match List-I with List-II:

List-I	List-II
A. NBWL	1. State implementing officer
B. CWLW	2. PM-chaired national advisory body
C. IRV 2020	3. Rhino recovery plan including Dudhwa

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

Show answer

Ans: B. A-2, B-1, C-3. **Logic:** NBWL PM-chaired; CWLW state; IRV includes Dudhwa.

Q12. With reference to Project Elephant, which of the following statements is/are correct?

1. It began in 1992 with about 33 elephant reserves.
2. It covers wild and captive elephants.
3. It has the same WPA Section 38V statutory basis as Tiger Reserves.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

Show answer

Ans: C. Only 1 and 2. **Logic:** No WPA statutory basis like TR.

Q13. Which of the following Schedule I species facts is/are correctly stated?

1. Tiger, elephant, rhino, gharial, and vultures appear in frequent Schedule I teaching.
2. A Schedule I tortoise enjoys the same maximum protection principle.
3. Schedule VI plants need CWLW (or authorised officer) licence for cultivation.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: D. All three. **Logic:** Schedule I max protection; Schedule VI licence regime.

Q14. With reference to GSLEP, which of the following statements is/are correct?

1. GSLEP (2013) is linked to snow leopard landscapes.
2. Uttar Pradesh is the core snow leopard state of India.
3. Project Snow Leopard began in 2009.

A. 1 and 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 3. **Logic:** UP has no snow leopard habitat.

Q15. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Schedule VI plants cannot be cultivated without a licence from the CWLW (or authorised officer).

Reason (R): Schedule VI plants are banned from cultivation under all circumstances with no licence route.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Licence route exists — not an absolute ban slogan.

Q16. Arrange: Vulture Action Plan / diclofenac ban teaching year:

A. 1992 B. 2006 C. 2013 D. 2020

💡 **Show answer**

Ans: B. 2006. **Logic:** Vulture Action Plan and diclofenac ban teaching = 2006.

Q17. Which of the following pairs is/are correctly matched?

1. IRV 2020 — rhino recovery across seven PAs including Dudhwa
2. Project Lion — Gir focus
3. SAVE — Nepal–India vulture conservation cooperation teaching

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: C. All three. **Logic:** IRV/Dudhwa; Gir lions; SAVE vulture cooperation.

Q18. With reference to in-situ vs ex-situ in this chapter, which statements is/are correct?

1. Protected areas are in-situ.
2. Captive breeding centres are ex-situ.
3. Kukrail-type crocodile centres are in-situ National Parks.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Captive centres are ex-situ — not NP in-situ.

Q19. How many of the following projects began before 1990?

Project Tiger, Crocodile Project, Project Elephant, Project Snow Leopard

A. Only two B. Only one C. Only three D. All four

💡 **Show answer**

Ans: A. Only two. **Logic:** Tiger 1973 and Crocodile 1975; Elephant 1992; Snow Leopard 2009.

Q20. Match List-I with List-II:

List-I	List-II
A. Schedule I	1. Vermin teaching
B. Schedule V	2. Maximum protection
C. Schedule VI	3. Specified plants

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** I max; V vermin; VI plants.

Q21. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Corbett was the first tiger reserve under Project Tiger.

Reason (R): Project Tiger began in 1992.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Project Tiger began in 1973 — not 1992.

Q22. With reference to human–wildlife conflict, which of the following statements is/are correct?

1. It is listed among major wildlife threats.
2. Habitat loss can intensify conflict.
3. Conflict means wildlife laws are repealed automatically.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Statement 3 is false.

Q23. Which one of the following correctly identifies NBWL?

A. State pollution board B. Only a municipal forest cell C. UNEP Nairobi office D. PM-chaired national wildlife advisory body

💡 **Show answer**

Ans: D. PM-chaired national advisory body. **Logic:** NBWL ≠ SPCB / UNEP.

Q24. With reference to Dudhwa and rhinos, which of the following statements is/are correct?

1. IRV 2020 includes Dudhwa among rhino recovery PAs.
2. Dudhwa is in Uttar Pradesh.
3. Project Lion's wild lion focus is also Dudhwa.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Project Lion = Gir, not Dudhwa.

Q25. Arrange Schedule protection idea from strongest animal protection teaching:

A. Schedule V > Schedule I B. Schedule I > lower animal schedules C. Schedule VI > Schedule I for animals D. Schedule V = Schedule I

💡 **Show answer**

Ans: B. Schedule I is maximum for animals. **Logic:** Schedule VI is plants; V is vermin teaching.

Q26. Which of the following pairs is/are NOT correctly matched?

1. Project Elephant — 1973
2. Project Tiger — 1973
3. Project Lion — 2020

A. 1 and 2 only B. 2 and 3 only C. 1 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 1 is not correctly matched. **Logic:** Project Elephant = 1992.

Q27. With reference to SAVE, which of the following statements is/are correct?

1. It targets vulture conservation cooperation (Nepal–India teaching).
2. It replaces the Wildlife (Protection) Act.
3. It sits in the diclofenac / vulture recovery neighbourhood.

A. 1 and 2 only B. 1, 2 and 3 C. 2 and 3 only D. 1 and 3 only

💡 **Show answer**

Ans: D. Only 1 and 3. **Logic:** Does not replace WPA.

Q28. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Invasive species are a major wildlife/biodiversity threat.

Reason (R): Poaching is never considered a wildlife threat in Indian teaching.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Poaching is a core threat.

Q29. How many of the following are correctly paired with Gujarat/Assam classics?

1. Asiatic lion — Gir
2. One-horned rhino — Kaziranga teaching stronghold
3. Snow leopard — Gir

A. Only two B. Only one C. All three D. None

💡 **Show answer**

Ans: A. Only two. **Logic:** Snow leopard is high Himalaya — not Gir.

Q30. With reference to WPA after Stockholm, which of the following statements is/are correct?

1. WPA 1972 followed the Stockholm environmental moment.
2. WPA uses six Schedules.
3. Stockholm 1972 created Project Tiger in the same week as a UN statute.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Project Tiger is India's 1973 programme — not a Stockholm UN statute.

Mains Answer Framework

Wildlife conservation in India (250 words): Define conservation vs protection. WPA 1972 framework and schedules. Flagship/umbrella/keystone species logic. Major programmes with years (Tiger 1973, Elephant 1992, Crocodile 1975, Snow Leopard 2009, Lion 2020, Vulture 2006). Statutory vs executive distinction (TR vs ER). Threats and community role. UP examples — Dudhwa tiger/rhino, Terai elephant, Katarniaghat gharial. Conclude with NTCA and species-programme layering over PA network.

Common Traps — Don't Fall For These

Trap	Correct fact
WPA enacted in 1973	**1972** — Project Tiger = 1973
World Wildlife Day = 22 March	**3 March**; Wildlife Week = **2–8 October**
Operation Olivia nesting in August	**Nov–Feb**
Elephant Reserve under WPA	**No** — executive only; covers wild **and captive**
All rhinos in Assam	Also **WB and Dudhwa UP**
Asiatic lion in UP	**Gir, Gujarat only**
Schedule VI = absolute ban	Cultivation needs a **licence**
Schedule I tortoise ≠ tiger level	**Same** maximum protection band
Spider monkey in India	**Central/South America** only
Kharai cannot be domesticated	**Can** be domesticated; swims + mangrove graze
AWBI under EPA 1986	Under **PCA 1960** (est. 1962)
WWF logo = tiger	**Giant Panda**
Best gharial site = Pulicat	**Chambal**
Tiger Summit 2010 = Delhi	**Saint Petersburg**
Snow leopard in Terai/Gir	**High Himalaya only**
Crocodile project = gharial only	**3 species**
Diclofenac ban = human medicine ban	**Veterinary** use caused crisis

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 8 — Forests & Forest Management

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Forests and their Types | Forest Types in India | Forest Conservation | Forest Management | Afforestation | Social Forestry (सामाजिक वानिकी) | Agroforestry (कृषि वानिकी) | Joint Forest Management (JFM) (संयुक्त वन प्रबंधन) | Forest Rights Act

Sources baked in: Champion & Seth 1968, IFA 1927, FCA 1980, NFP 1988, FRA 2006, ISFR 2023 (FSI), MoEFCC, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — ISFR data, act years, FCA vs FRA, UP district traps **Last verified:** August 2026

Current Affairs (this topic)

- Keep ISFR / FSI function traps stable; new ISFR cycle is the usual forest CA refresh.
- FCA / forest-conservation wording may appear in CA — separate from WPA PA categories.

💡 🎯 **Consolidated — 36 Must-Score Facts (Click to Expand)**

1. Forest cover (वन आवरण) in ISFR means a patch of at least **1 ha** with canopy **≥10%**, on any land ownership. 2. Canopy classes: **VDF ≥70%**, **MDF 40–70%**, **OF 10–40%**. 3. ISFR **2023** India figures: forest cover about **21.76%**, tree cover (वृक्ष आवरण) about **3.41%**, total green cover about **25.17%**. ISFR **2021** held forest cover at about **21.71%** and forest+tree cover about **24.62%** (still used in older stems). 4. **Madhya Pradesh** has the largest forest **area**. **Mizoram** has the highest **percentage** cover among States. **Arunachal Pradesh** often leads **very dense forest** area. 5. **Forest cover ≠ Recorded Forest Area ≠ tree cover**. Recorded forest area is a legal-notification figure near about **23%**. 6. Under the **Indian Forest Act, 1927**, **Reserved Forest** is strictest (acts banned unless permitted). **Protected Forest** is State-regulated with some customary rights possible. 7. **Champion and Seth (1968)** classify Indian forests into **16 types in 6 groups**. The largest share is **Tropical Dry Deciduous** (about **38%**). 8. **Tropical wet evergreen** (>**200 cm** rain) holds maximum plant diversity / biomass among common options — Western Ghats (पश्चिमी घाट), NE hills, A&N (not Himachal (हिमाचल) as wet evergreen). 9. **Teak** dominates **tropical moist deciduous** forests; **Madhya Pradesh** has the maximum teak area among common options. 10. **Red Sanders (*Pterocarpus santalinus*)** is a dry-deciduous South India endemic (Palkonda–Seshachalam / southern Eastern Ghats (पूर्वी घाट)) — **not** a tropical rainforest tree. 11. **Taxus** (Himalaya (हिमालय); Red Data) yields **taxol** (anti-cancer; also cited vs Parkinson's in older stems). 12. **Amazon rainforest** is often called the **"lungs of the Earth"** (~**20%** of terrestrial photosynthetic oxygen share in coaching figures). 13. UP type facts: Terai (तराई) = moist deciduous; Vindhyan = dry deciduous; Bundelkhand = thorn. 14. Law chronology: **IFA 1927 → FCA 1980 → NFP 1988 → JFM 1990 → FRA 2006 → CAMPA Act 2016**. 15. The **Forest (Conservation) Act, 1980** requires **Central** approval before diverting forest land. Compensatory afforestation is mandatory. 16. **National Forest Policy 1988** targets about **33%** national forest/tree cover and **66%** in hills. **NFP 1952** classes: Protected/Conserved, National, Village Forests, Tree Lands — **not** "National Park" as a forest class. 17. NFP 1988 includes afforestation/wasteland, reforestation, and wood substitutes — **not** wide promotion of insecticides/pesticides. 18. **Chipko (1973, Reni, Uttarakhand (उत्तराखंड))** — Bahuguna, Bhatt, **Gaura Devi**. The **Godavarman (1996)** line broadened "forest." 19. Afforestation tools include **NAP (2000)**,

Green India Mission (2014), **Miyawaki** mini-forests, and **CAMPA**. 20. **Social forestry** grows trees on **non-forest** land. **Agroforestry** integrates trees with crops/livestock (**CAFRI Jhansi (झांसी)**). 21. **JFM (1990)** is State–village co-management of **degraded forest**. **FRA 2006** recognises dweller rights (**IFR ~4 ha**). 22. **FCA** controls **diversion**. **FRA** recognises **rights**. **JFM** is executive co-management — three different tools. 23. UP ISFR **2023** forest cover is about **6.24%** of State area. 24. **Sonbhadra** (सोनभद्र) has UP's highest district forest cover. Districts above **20%** include **Chandauli** and **Shravasti**. 25. **Urbanisation is a cause of deforestation**, not an "impact/effect" of deforestation (classic trap). 26. Forests are a **renewable** resource; they also enhance environment (पर्यावरण) quality — but the second fact does **not** alone explain renewability. 27. **Aluminium** is called a **"green metal"** for recyclability/eco profile — not primarily "to save forests by replacing wood." 28. **Living root bridges** are characteristic of **Meghalaya** (मेघालय). 29. ISFR 2021 top absolute (निरपेक्ष) gains: **Andhra Pradesh, Telangana, Odisha**. NE states showed large losses. 30. Mangrove area rank (ISFR 2021 style): **WB > Gujarat > A&N (~616–617 km^2) > AP**. **Coringa** = Godavari (गोदावरी) delta (डेल्टा) (AP). 31. **Andaman (अंडमान) & Nicobar (निकोबार)** uniquely combine mangrove + evergreen + deciduous in one option set. 32. International Day of Forests = **21 March** (UN **2012**). Wildlife Week = **2–8 October**. WWF (डब्ल्यूडब्ल्यूएफ) logo = **Giant Panda**. 33. Bhutan constitutionally aims for **≥60%** forest cover (actual cover much higher). 34. Bamboo on **non-forest** land was omitted from "tree" under the **Indian Forest (Amendment) Act, 2017**. 35. **FRI Dehradun (देहरादून) (1906)**; **ICFRE HQ Dehradun**; **IIFM Bhopal**; Survey of India ≠ environment agency. 36. CAMPA manages compensatory afforestation funds after diversion — it is not FRA Gram Sabha (ग्राम सभा) rights recognition (मान्यता).

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Forest Cover	**Tree Cover**	All land patches with tree canopy density ≥10% and area ≥1 hectare irrespective of ownership or legal status (FSI remote sensing) vs scattered tree patches, linear trees along canals/roads, and isolated trees <1 hectare outside recorded forest area	वन आवरण (≥1 हे., ≥10% छतरी) / वृक्ष आवरण (<1 हेक्टेयर)
Reserved Forest (RF)	**Protected Forest (PF)**	Highest degree of protection under Indian Forest Act 1927; all activities (grazing, timber felling) are strictly prohibited unless expressly permitted vs forest where all activities are permitted unless specifically prohibited by notification	आरक्षित वन (RF, पूर्णतः प्रतिबंधित) / संरक्षित वन (PF, सशर्त अनुमत)
Recorded Forest Area (RFA)	**Forest Cover**	Land legally recorded or notified as forest in government records (RF + PF + Unclassed) regardless of actual tree presence vs physical presence of tree canopy detected by satellite imagery (even on private orchards/tea estates)	अभिलिखित वन क्षेत्र (कानूनी अभिलेख) / वन आवरण (भौतिक वृक्ष आच्छादन)
Forest Conservation Act (FCA 1980)	**Forest Rights Act (FRA 2006)**	Central regulatory statute preventing de-reservation or diversion of forest land for non-forest purposes without prior central approval vs restorative rights law vesting individual and community forest rights in traditional forest-dwelling tribes and OTFDs	वन संरक्षण अधिनियम 1980 / वन अधिकार अधिनियम 2006
Social Forestry	**Agroforestry**	Growing trees on non-forest communal lands, roadside strips, and wastelands to meet rural fodder and fuel needs (National Commission on Agriculture 1976) vs deliberate integration of woody perennials (trees/shrubs) with agricultural crops and/or livestock on the same land unit	सामाजिक वानिकी / कृषि वानिकी
Joint Forest Management (JFM)	**Community Forest Resource (CFR)**	Collaborative institutional partnership between state forest department and village committees to protect and regenerate degraded forests vs statutory community ownership and governance right recognized under Section 3(1)(i) of FRA 2006	संयुक्त वन प्रबंधन (JFM) / सामुदायिक वन संसाधन अधिकार (CFR)
Tropical Wet Evergreen Forest	**Tropical Moist Deciduous Forest**	Dense multi-layered forest with rainfall >200 cm, no distinct leafless season (Ebony, Mahogany, Rosewood) vs most widespread forest in India with rainfall 100–200 cm, shedding leaves for 6–8 weeks in dry spring (Teak, Sal, Shisham)	उष्णकटिबंधीय आर्द्र सदाबहार / उष्णकटिबंधीय नम पर्णपाती वन

💡 📋 Must-score facts — ISFR, types, law chain (Click to Expand)

Cover definitions | Item | Lock | |------|------| | Forest cover (ISFR) | ≥1 ha; canopy ≥**10%**; any ownership | | VDF / MDF / OF | ≥70% / 40–70% / 10–40% | | ISFR 2023 class | Forest ~**21.76%**; tree ~3.41%; total green ~25.17% | | Largest area / highest % | MP / Mizoram | | Cover ≠ RFA ≠ tree cover | Legal RFA ~23% class figure | ### Types / law | Item | Lock | |------|------| | Champion & Seth | **1968**; 16 types / 6 groups; Tropical Dry Deciduous largest | | Wet evergreen rain | >**200 cm**; max diversity/biomass | | Teak / Red Sanders | Moist deciduous / dry-deciduous S. India endemic | | UP belts | Terai moist deciduous · Vindhyan dry · Bundelkhand thorn | | Law chain | IFA 1927 → FCA 1980 → NFP 1988 → JFM 1990 → FRA 2006 → CAMPA 2016 | | Reserved vs Protected Forest | Strictest IFA / State-regulated lighter | --- ![India Forest Canopy Architecture & ISFR Forest Types](images/env_ch08_01_india_forest_canopy_classification.png)

8.1 Forests and their Types

Identity: Tree-dominated ecosystem (पारिस्थितिकी तंत्र) | Global + legal + ISFR classes

- A **forest** is a large area where trees form the dominant life form and support layered ecosystems.
- Forests regulate climate, conserve soil and water, store carbon, and provide timber and non-timber forest produce.
- **Global forest types** include tropical rainforest, tropical deciduous, temperate, boreal, Mediterranean scrub, and mangrove.
- India has no boreal forest. Its forests range from Western Ghats evergreen to Himalayan alpine scrub.
- Under the **Indian Forest Act, 1927**, forests are classed as **Reserved Forest**, **Protected Forest**, or **Unclassed Forest**.
- **Reserved Forest** has the strictest protection. Most acts are prohibited unless permitted.
- **Protected Forest** is state-regulated forest where some grazing, collection, or customary use may continue under rules.
- **Unclassed Forest** is forest land not yet formally placed in the Reserved or Protected category.
- **ISFR forest cover** means any patch of **1 hectare or more** with **10% or more** tree canopy, regardless of ownership.
- **Very Dense Forest** has canopy density **70% or more**. **Moderately Dense Forest** has **40–70%**. **Open Forest** has **10–40%**.
- **Forest cover** is not the same as **Recorded Forest Area**. Much forest cover exists outside legally notified forest land.
- **Tree cover** counts scattered trees in orchards and farms that fall below the 1-hectare forest patch rule.
- **Mangrove and littoral swamp forests** are distinct coastal types with salt-tolerant species such as sundari in the Sundarbans (सुंदरबन).

IFA class	Protection level
Reserved Forest	Strictest — hunting, grazing, felling banned unless permitted
Protected Forest	Moderate (नरम दल) — state controls but some rights allowed
Unclassed Forest	Not yet formally classified

> **Logic:** Trap — "forest cover counts only government forest land" is **false**.

8.2 Forest Types in India

Identity: Champion & Seth 1968 | 16 types | UP Terai–Vindhyan–Bundelkhand

- **Champion and Seth (1968)** is India's standard forest type classification with **16 major types** in **6 groups**.

- The **six groups** are Moist Tropical, Dry Tropical, Montane Subtropical, Montane Temperate, Sub-Alpine, and Alpine.
- **Tropical Wet Evergreen** forest occurs where rainfall exceeds about **200 cm**, mainly on the Western Ghats, Northeast hills (e.g. Arunachal, Mizoram/Khasi (खासी)–Jaintia (जयंतिया) belt), and Andaman & Nicobar — **not** Himachal (pine/temperate belts).
- These forests hold **maximum plant diversity / biomass** among common biome (जीवोम) options; equatorial rainforest is the densest high-biomass biome because of year-round high rainfall and temperature.
- **Tropical Semi-Evergreen** forest is a transition type with rainfall around **1500–2500 mm**.
- **Tropical Moist Deciduous** forest (about **100–200 cm** rain) drops leaves in the dry season; **teak** is the dominant commercial species. **Madhya Pradesh** has the maximum teak area among common state options. Other species: bamboo, sal, shisham, sandalwood, khair, arjun.
- **Red Sanders (*Pterocarpus santalinus*)** is endemic to dry deciduous hills of South India (Palkonda–Seshachalam / southern Eastern Ghats) — valued red wood; **not** a tropical rainforest species.
- **Taxus** grows naturally in the Himalaya, is Red Data–listed, and yields **taxol** (anti-cancer; older stems also cite Parkinson's).
- **Littoral and Swamp** forest includes mangroves on the Sundarbans and Odisha coast (**Sundari** is the famous Sundarbans tree).
- **Andaman & Nicobar Islands** uniquely show a combination of **mangrove + evergreen + deciduous** forests (mangrove cover ~**616–617 km²** in ISFR 2021-style figures).
- **Amazon rainforest** is popularly called the **"lungs of the Earth."**
- Oak and rhododendron are Himalayan/subtropical–temperate associates; **sandalwood** is South Indian deciduous — not a Himalayan natural associate.
- **Tropical Dry Deciduous** is India's **largest forest type** by area, covering about **38%** of forest area.
- **Tropical Thorn** forest occurs in low-rainfall zones such as **Bundelkhand** with acacia and babool at **200–800 mm**.
- **Subtropical Pine** forest is dominated by **chir pine** in the outer Himalaya between **1000–1800 m**.
- **Himalayan Moist Temperate** forest holds oak and deodar between **1500–3300 m**.
- **Sub-Alpine and Alpine scrub** occur above **3000 m** near the treeline.
- **ISFR 2023** recorded India forest cover at **21.76%** (**7,15,343 sq km**) and total green cover at **25.17%**. ISFR **2021** forest cover was about **21.71%**; forest+tree cover about **24.62%**.
- **Madhya Pradesh** has the **largest forest area** among states. **Mizoram** has the **highest percentage** forest cover. Top percentage cluster: Mizoram, Arunachal, Meghalaya, Manipur, Nagaland.
- ISFR 2021 absolute gains were led by **Andhra Pradesh, Telangana, Odisha**; the Northeast showed large losses.
- About **17** States/UTs have **>33%** forest cover; five exceed **75%** (Lakshadweep (लक्षद्वीप), Mizoram, A&N, Arunachal, Meghalaya).
- **Uttar Pradesh** (उत्तर प्रदेश) forest cover is **6.24%** of state area (ISFR 2023). **Sonbhadra** has the highest district share at **34.87%**.
- Only **Chandauli (21.63%)** and **Shravasti (20.95%)** exceed **20%** forest cover in UP.
- **Bahraich** is Terai moist deciduous by type but its forest cover is only about **10%**, below the 20% threshold.

#	Champion & Seth type	Key region
1	Tropical Wet Evergreen	Western Ghats, NE India
2	Tropical Semi-Evergreen	Transition to evergreen belt
3	Tropical Moist Deciduous	Terai, central India
4	Littoral & Swamp	Sundarbans, coastal Odisha

#	Champion & Seth type	Key region
5	Tropical Dry Deciduous	**Largest area (~38%)**
6	Tropical Thorn	Rajasthan, Bundelkhand
7	Tropical Dry Evergreen	East coast Karnataka/AP
8	Subtropical Broadleaved Hill	Nilgiri (नीलगिरि), NE hills
9	Subtropical Pine	Chir pine — outer Himalaya
10	Subtropical Dry Evergreen	Siwalik, Jammu
11	Montane Wet Temperate	Nilgiri, eastern Himalaya
12	Himalayan Moist Temperate	Oak, deodar — Uttarakhand, HP
13	Himalayan Dry Temperate	Inner Himalaya conifers
14	Sub-Alpine	Rhododendron, fir — >3000 m
15	Moist Alpine Scrub	High Himalaya
16	Dry Alpine Scrub	Cold desert Himalaya

PYQ — UPPCS Prelims 2025, Q8

As per ISFR 2023, which UP districts had forest over more than 20% of total area?

1. Bahraich 2. Chandauli 3. Shravasti

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

💡 **Show answer**

Ans: C — Bahraich is below 20%.

UP region	Forest type	Key districts
Terai-Duar	Tropical Moist Deciduous	Pilibhit, Bahraich, Shravasti
Vindhyan	Tropical Dry Deciduous	Sonbhadra, Chandauli, Mirzapur (मिर्ज़ापुर)
Bundelkhand	Thorn / Dry Deciduous	Jhansi, Banda, Lalitpur

PYQ — UPPCS Prelims 2023, Q59

Consider the following statements about Uttar Pradesh:

1. Forests constitute about 12.8% of the total geographical area of the State.
2. The districts of Jaunpur (जौनपुर), Ghazipur and Ballia have no forest land.

A. Only 1

B. Both 1 and 2

C. Only 2

D. Neither

💡 **Show answer**

Ans: C — ISFR 2023 gives **~6.24%** forest cover, not 12.8%. Statement 2 is correct.

8.3 Forest Conservation

Identity: FCA 1980 | Chipko 1973 | NFP 1988 | CAMPA | Godavarman 1996

- **Forest conservation** protects existing forests and restores degraded ones, not just new planting.
- Major threats include deforestation, mining, forest fires, overgrazing, shifting cultivation (**jhum**), and invasive species (आक्रामक विदेशी प्रजाति).
- The **Chipko Movement (चिपको आंदोलन) (1973)** (चिपको आंदोलन) began at **Reni village** in Uttarakhand when villagers hugged trees to stop commercial felling.
- **Sunderlal Bahuguna**, **Chandi Prasad Bhatt** (चंडी प्रसाद भट्ट) (Dasholi Gram Swarajya (स्वराज्य) Mandal, 1964), and **Gaura Devi** (26 March 1974 Reni episode) were key Chipko leaders.
- Major **causes** of deforestation include agriculture, housing, urbanisation, industry, and jhum. **Nagaland** hills becoming barren is classically linked to **shifting cultivation**. **Urbanisation is a cause**, not an "impact/effect," of deforestation — drying springs, biodiversity loss, and soil erosion are impacts.
- Forests are a **renewable** resource under sustainable use. They also absorb CO_2 and release O_2, but that environmental role does **not** alone define renewability (classic A/R trap).
- **Aluminium** is called a **"green metal"** for recyclability/eco-profile — not primarily because it "saves forests by replacing wood."
- **Living root bridges** of trained Ficus roots are characteristic of **Meghalaya**.
- The **Tiwari Committee (1980)** recommended strict control on forest diversion and led to **FCA 1980**.
- **Forest (Conservation) Act, 1980** requires **prior Central Government approval** before forest land is used for non-forest purposes.
- **Compensatory afforestation** must replace diverted forest area, with payment of **net present value (NPV)** into the **CAMPA fund**.
- **CAMPA** was set up in **2009**. The **CAMPA Act, 2016** created national and state fund authorities.
- **National Forest Policy, 1988** replaced the **1952 revenue-oriented policy** and targets **33% forest and tree cover** nationally and **66% in hills and mountains**.
- **NFP 1952** forest classes: **Protected/Conserved Forests, National Forests, Village Forests, Tree Lands** — **National Park is not** a 1952 forest class.
- NFP 1988 includes afforestation of wastelands, reforestation, and wood substitutes / alternate fuels — **not** promoting wide insecticide/pesticide use.
- In **Godavarman v. Union of India (1996)**, the Supreme Court expanded **forest** to include areas meeting the dictionary meaning, not only notified land.
- **Green India Mission (2014)** aims to improve forest quality and carbon sequestration (कार्बन पृथक्करण) under India's climate action plan.
- **Miyawaki** planting creates dense mini-forests quickly on degraded/urban (नगरीय) plots using native species.
- **Article 48A** (DPSP (राज्य के नीति निदेशक तत्व)) directs the State to protect forests. **Article 51A(g)** makes environmental protection a fundamental duty of citizens.
- International Day of Forests = **21 March** (UNGA **2012**). Bhutan constitutionally aims for **≥60%** forest cover.

> **Logic:** FCA = **1980**. WPA = **1972**. NFP current version = **1988**, not 1952. "Urbanisation as deforestation impact" is the classic trap.

8.4 Forest Management

Identity: IFA 1927 | Working plans | FSI/ISFR | Concurrent List (समवर्ती सूची)

- **Forest management** uses scientific planning to harvest forest produce without destroying regeneration capacity.

- The **Indian Forest Act, 1927** empowers states to declare Reserved and Protected Forests and regulate felling.
- A **Working Plan** is the **10-year** scientific management document for each forest division covering inventory, regeneration, and felling schedules.
- **Sustainable yield** means harvest must not exceed the forest's regeneration capacity.
- **Forest Survey of India (FSI) (भारतीय वन सर्वेक्षण), Dehradun** publishes the biennial **India State of Forest Report (ISFR)**.
- **Forest Research Institute (FRI) (वन अनुसंधान संस्थान), Dehradun** (roots in **1906** Imperial FRI) is a premier forestry research/training institute and deemed university.
- **Indian Council of Forestry Research and Education (ICFRE), Dehradun** coordinates forestry research and revised type classification work.
- **Indian Institute of Forest Management (IIFM)** is at **Bhopal**. **Survey of India** is a mapping agency under DST — **not** an environment body.
- **Recorded Forest Area** is land legally notified as forest, about **23%** of India's geographical area.
- About **27%** of India's forest cover lies **outside** Recorded Forest Area according to ISFR 2023.
- **Forests** are on the **Concurrent List** (समवर्ती सूची), so both Centre and states can legislate.
- **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) sets national policy. State Forest Departments implement protection, working plans, and JFM on the ground.
- **Van Panchayats** in Uttarakhand manage local forest patches through customary community institutions.

> **Logic:** ISFR is published by **FSI**, not CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड), ICFRE, or MoEFCC directly.

8.5 Afforestation

Identity: New forest on degraded/non-forest land | NAP | GIM | compensatory planting

- **Afforestation** creates forest cover on land that had no forest or lost canopy below the 10% threshold.
- **Reforestation** replants harvested forest. **Restoration** rebuilds full ecological function.
- **Compensatory afforestation** under FCA is **mandatory** when notified forest land is diverted.
- The developer pays **net present value (NPV)** of forest services into the **CAMPA fund**.
- **National Afforestation Programme (NAP)** was launched in **2000** through **Forest Development Agencies (FDAs)**.
- **Green India Mission (2014)** targets improvement of **5 million hectares** of forest and tree cover.
- **Nagar Van Scheme (2020)** promotes urban forests in cities.
- The **Miyawaki method** creates dense urban mini-forests with native species planted close together on degraded or urban land.
- **National Afforestation and Eco-Development Board (NAEB)** was set up in **1992**.
- Bamboo grown on **non-forest** land was omitted from the definition of "tree" under the **Indian Forest (Amendment) Act, 2017** — freer harvest/transport outside forests; FRA still treats bamboo as **minor forest produce**.
- **Native species** planting is preferred under NFP 1988 and Green India Mission over exotic monocultures such as eucalyptus.
- Farm and agroforestry plantations add to **tree cover** and **Trees Outside Forests (TOF)** in ISFR, which helps plain states like UP.

> **Logic:** CAMPA money is used mainly for afforestation and forest protection, not wildlife alone.

8.6 Social Forestry

Identity: NCA 1976 | Non-forest land | Fuelwood–fodder focus

- **Social forestry** grows trees on **non-forest land** to meet local fuelwood, fodder, and timber needs.
- The **National Commission on Agriculture (1976)** promoted it to address India's **fuelwood famine**.
- **Farm forestry** is trees planted by farmers on private land, such as poplar in western UP for plywood markets.
- **Community forestry** uses village commons and panchayat land for fuelwood and fodder.
- **Extension forestry** lines roadsides, canals, and institutional land with trees.
- **NFP 1988** endorsed social forestry as part of the **33% cover target** and stressed women's participation.
- Social forestry **reduces pressure** on natural forests by supplying needs locally.
- **Kerala** was an early successful social forestry state in the 1970s–80s.
- **Social forestry is not JFM**. JFM co-manages existing degraded forest, not non-forest land.
- **Social forestry is not agroforestry**. Agroforestry integrates trees with active crop or livestock systems on farmland.

> **Logic:** Social forestry roots lie in **NCA 1976**, not JFM 1990.

8.7 Agroforestry

Identity: National Policy 2014 | ICAR-CAFRI Jhansi | Trees + crops/livestock

- **Agroforestry** deliberately combines trees with crops and/or livestock on the same land unit.
- The **National Agroforestry Policy, 2014** was India's first comprehensive agroforestry policy.
- **ICAR-CAFRI (Central Agroforestry Research Institute)** is located in **Jhansi, Uttar Pradesh**.
- **ICRAF (World Agroforestry Centre)** is headquartered in **Nairobi**, not India.
- Common systems include **silvopasture**, **alley cropping**, **home gardens**, and **windbreaks**.
- **Multi-Purpose Tree Species (MPTS)** such as poplar, eucalyptus, teak, and bamboo are chosen for market value and fast growth.
- Agroforestry improves soil fertility, diversifies farmer income, and adds to **tree cover** in ISFR.
- **Taungya** allows temporary farming between rows of planted trees in forest department plantations.
- FAO treats agroforestry as a **climate-smart agriculture** practice.

> **Logic:** Agroforestry Policy = **2014**, not 2006. CAFRI = **Jhansi** (झांसी), not Banda.

PYQ — UPPCS Prelims 2021, Q15

Which is NOT correctly matched? A. ICRAF — Nairobi B. IIFM — Bhopal C. CAFRI — Banda D. TERI — New Delhi

💡 **Show answer**

Ans: C — Central Agroforestry Research Institute is in **Jhansi, UP**.

8.8 Joint Forest Management (JFM)

Identity: MoEF resolution 1990 | VSS/FPC | 50% benefit share

- **Joint Forest Management (JFM)** is a partnership between the State Forest Department and village communities.
- JFM was launched through a **MoEF resolution in June 1990**.

- It aims to **regenerate degraded forest** with community protection and shared benefits.
- Village bodies called **Van Samrakshan Samiti (समिति) (VSS)** or **Forest Protection Committee (FPC)** implement JFM.
- Communities receive about **50% of net income** from regenerated forest produce after final felling.
- **Intermediate produce** such as grass, fodder, and fallen wood is often shared fully with the village during protection.
- **Micro-plans** guide silviculture, protection, and livelihood activities for each JFM unit.
- **NFP 1988** provided the policy foundation for community participation that JFM operationalized.
- India has over **100,000 JFM committees**, one of the world's largest participatory forestry programmes.
- **JFM is an executive scheme**, not a statute like FRA 2006.
- **JFM does not grant permanent land title**. FRA can grant recognized forest rights.

> **Logic:** UPPCS 2021 asked JFM launch year — answer **1990**.

8.9 Forest Rights Act

Identity: FRA 2006 | ST + OTFD | Gram Sabha | IFR 4 ha cap

- The **Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006** corrects historical injustice from colonial forest laws.
- **Rules were notified in 2008**.
- It covers **Scheduled Tribes** and **Other Traditional Forest Dwellers (OTFDs)** who depend on forests.
- An **OTFD** must have depended on forest land for livelihood for at least **three generations** prior to **13 December 2005**.
- **Individual Forest Rights (IFR)** allow habitation and cultivation on forest land up to **4 hectares** per household.
- Claimants must have occupied the land as of **13 December 2005**.
- **Community Forest Rights (CFR)** cover grazing, NTFP collection, nistar, and community forest management.
- **Community Forest Resource (CFR) (सामुदायिक वन संसाधन अधिकार) management rights** allow a village to protect and manage its traditional forest boundary.
- The **Gram Sabha** (ग्राम सभा) receives, verifies, and passes resolutions on forest rights claims.
- Claims then go to **SDLC (Sub-Divisional Level Committee)** and **DLC (District Level Committee)** for final approval.
- Rights are **heritable** but **not transferable or alienable**. Titles are registered in both spouses' names.
- **No eviction** of eligible claimants is allowed until the recognition process is complete.
- **Critical Wildlife Habitat (वास स्थान)** provisions allow relocation from protected areas only after strict scientific and consent procedures.
- In **Niyamgiri (2013)**, Gram Sabhas rejected Vedanta (उत्तर मीमांसा) bauxite mining under FRA community consent powers.
- **FRA grants rights**. **FCA controls diversion**. Both can apply to the same forest landscape.

> **Logic:** IFR cap = **4 ha**, not 5 or 10. Cutoff date = **13 December 2005**.

Complete PYQ Bank (Topic 8)

Q1. UPPCS Prelims 2025, Q8

As per **ISFR 2023**, which districts of Uttar Pradesh had forest cover on **more than 20%** of their total geographical area?

1. Bahraich
2. Chandauli
3. Shravasti

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1 only

> 💡 **Show answer**
>
> **Logic:** ISFR district-level UP trap — only two districts cross the 20% forest-cover threshold. **Ans: C.** **Chandauli** and **Shravasti** exceed 20%; Bahraich is below that mark despite Terai forest presence. **Trap:** Bahraich is a Terai forest district but **does not** cross the 20% ISFR threshold — a frequent wrong pick.

Q2. UPPCS Prelims 2024, Q86

With reference to the **"Piral Lao – Paise Pao"** campaign, consider the following statements:

1. It helps reduce forest fires and provides income to villagers by collecting pine needles.
2. It was launched by the Chief Minister of Uttarakhand.

A. 2 only B. Neither 1 nor 2 C. 1 only D. Both 1 and 2

> 💡 **Show answer**
>
> **Logic:** Tests Uttarakhand chir-pine fire-fuel management campaign — purpose and launch authority. **Ans: D.** Villagers collect pine needles (**piral**) for cash while reducing summer fuel load; the campaign was launched under the **Uttarakhand CM**. **Trap:** This is a **Uttarakhand** scheme — not a default Uttar Pradesh flagship programme.

Q3. UPPCS Prelims 2021

Joint Forest Management (JFM) in India was launched through a MoEF resolution in—

A. 1982 B. 1986 C. 1990 D. 1992

> 💡 **Show answer**
>
> **Logic:** Year fact for community–State co-management of degraded forest — distinct from FRA 2006. **Ans: C.** JFM began with a **June 1990** MoEF resolution creating Van Samrakshan Samitis / Forest Protection Committees. **Trap:** **1992** is Project Elephant — JFM is **1990**, eight years earlier.

Q4. UPPCS Prelims 2018, Q44

According to the **India State of Forest Report, 2017**, what percentage of India's total geographical area was under forest cover?

A. 20.34% B. 22.34% C. 21.54% D. 23.54%

> 💡 **Show answer**
>
> **Logic:** National ISFR 2017 headline forest-cover percentage from FSI Dehradun. **Ans: C.** ISFR 2017 reported forest cover at about **21.54%** of India's geographical area. **Trap:** Do not confuse with the **33% national policy target** — that is aspirational, not the 2017 measured cover.

Q5. UPPCS Prelims 2023, Q59

Consider the following statements about **Uttar Pradesh**:

1. Forests constitute about **12.8%** of the total geographical area of the State.
2. The districts of **Jaunpur, Ghazipur and Ballia** have no forest land.

A. 1 only B. Both 1 and 2 C. 2 only D. Neither 1 nor 2

> 💡 **Show answer**
>
> **Logic:** ISFR 2023 UP figure trap — outdated high percentage vs eastern UP zero-forest districts. **Ans: C.** ISFR 2023 gives UP forest cover about **6.24%**, not 12.8%; Jaunpur–Ghazipur–Ballia indeed have negligible forest land. **Trap:** Statement 1 uses an **outdated/higher** figure — do not accept both statements as true.

Q6. UPPCS Prelims 2022, Q61

As per **India State of Forest Report–2021**, which State registered the **maximum increase in forest cover** during the last two years?

A. Kerala B. Assam C. Arunachal Pradesh D. Andhra Pradesh

> 💡 **Show answer**
>
> **Logic:** ISFR 2021 state-wise absolute gain ranking — distinct from highest percentage cover (Mizoram) or largest area (MP). **Ans: D.** **Andhra Pradesh** led absolute forest-cover gains (~647 km^2), followed by Telangana and Odisha in ISFR 2021. **Trap:** Mizoram leads **percentage** cover; Madhya Pradesh leads **total area** — neither is the ISFR 2021 **maximum increase** answer.

Q7. UPSC pattern

Under the **Forest Rights Act, 2006**, the maximum area recognised under an **Individual Forest Right (IFR)** for cultivation/habitation is—

A. 2 hectares B. 4 hectares C. 10 hectares D. No statutory cap

> 💡 **Show answer**
>
> **Logic:** FRA IFR ceiling per household — a standard coaching fact distinct from JFM benefit-sharing. **Ans: B.** IFR is capped at **4 hectares** per eligible household under FRA 2006. **Trap:** **10 hectares** is a common wrong guess — the statutory cap is **4 ha**.

Q8. UPSC pattern

The **Forest (Conservation) Act, 1980** was enacted primarily to—

A. Grant permanent land titles to forest dwellers B. Regulate diversion of forest land for non-forest use with prior central approval C. Establish Joint Forest Management committees nationwide D. Fix a 33% national forest-cover target by 1988

> 💡 **Show answer**
>
> **Logic:** Distinguishes FCA (diversion control) from FRA (rights), JFM (co-management), and NFP targets. **Ans: B.** FCA 1980 requires **prior approval** from the Centre/State for using forest land for non-forest purposes — the core anti-deforestation check. **Trap:** Forest dweller rights belong to **FRA 2006**, not FCA 1980.

Q9. UPPCS Prelims 2021, Q15

Which of the following is **NOT** correctly matched?

A. ICRAF — Nairobi B. IIFM — Bhopal C. CAFRI — Banda D. TERI — New Delhi

> 💡 **Show answer**
>
> **Logic:** Institution–headquarters matching for forestry/agroforestry research bodies. **Ans: C.** **Central Agroforestry Research Institute (CAFRI)** is at **Jhansi, Uttar Pradesh**, not Banda. **Trap:** CAFRI sounds like it could be in Bundelkhand's Banda — the correct UP city is **Jhansi**.

Q10. UPPCS Prelims 2019, Q98

In which of the following are provisions for the **protection and improvement of the environment** and **safeguarding of forests and wildlife** found in the Constitution of India?

A. Only in the Directive Principles of State Policy B. Only in the Fundamental Duties C. Both (a) and (b) D. None of the above

> 💡 **Show answer**
>
> **Logic:** Constitutional environment mandate spans DPSP **Article 48A** and Fundamental Duty **Article 51A(g)**. **Ans: C.** **Art 48A** (DPSP) directs the State to protect environment, forests, and wildlife; **Art 51A(g)** makes it every citizen's duty. **Trap:** Neither provision sits alone — both pillars carry forest/environment language.

Extra Drill — Ghatnachakra Forest & Wildlife (Q31–Q40)

Q31. Tropical evergreen forests in India are typically found in:

A. Malwa Plateau

B. Chhotanagpur Plateau

C. Western Ghats

D. Only Himachal Pradesh

> 💡 **Show answer**
>
> **Logic:** These stems recycle evergreen/teak/mangrove/NFP/ISFR traps from standard banks. **Ans: C.** Evergreen belts need >**200 cm** rain — Western Ghats, NE hills, A&N. Himachal is mainly pine/temperate, not tropical wet evergreen.

Q32. Teak is the dominant tree species of which forest type?

A. Tropical rainforest

B. Tropical moist deciduous forest

C. Tropical thorn scrub

D. Temperate grassland forest

> 💡 **Show answer**
>
> **Ans: B.** Teak dominates tropical moist deciduous (100–200 cm). **Madhya Pradesh** leads teak area among common options.

Q33. With reference to Red Sanders, which is correct?

A. It is a tropical rainforest tree of South India

B. It is endemic to dry deciduous South India / southern Eastern Ghats

C. It is a Himalayan conifer

D. It is found only in Sundarbans

💡 **Show answer**

Ans: B. *Pterocarpus santalinus* — Palkonda–Seshachalam / southern Eastern Ghats dry deciduous. Statement pairing it with tropical rainforest is false.

Q34. Mangroves function as a safety hedge against coastal calamities mainly because:

A. People never live near mangrove coasts

B. Trees provide post-disaster food and medicine only

C. Dense canopies alone stop every cyclone

D. Extensive roots keep trees from being uprooted by storms and tides

💡 **Show answer**

Ans: D. Classic root-system explanation. Odisha is most cyclone-prone; Mahanadi mangrove loss raises risk.

Q35. Which region has mangrove + evergreen + deciduous forests together?

A. Southern Saurashtra

B. North Coastal Andhra Pradesh

C. Andaman and Nicobar Islands

D. Himalayan Terai

💡 **Show answer**

Ans: C. A&N mangrove cover ~**616–617 km^2** (ISFR 2021-style). Coringa = Godavari (AP); Bhitarkanika = Brahmani–Baitarni (Odisha).

Q36. National Forest Policy 1952 classified forests as Protected, National, Village Forests and:

A. National Parks

B. Tree Lands

C. Biosphere Reserves

D. Community Reserves

💡 **Show answer**

Ans: B. Four classes include **Tree Lands**. National Park is not a 1952 forest class. Current cover target is from **NFP 1988** (~**33%**).

Q37. Which is **not** an impact of deforestation?

A. Soil erosion

B. Loss of biodiversity

C. Urbanisation

D. Drying of water sources

💡 **Show answer**

Ans: C. Urbanisation is a **cause**, not an effect. Nagaland barren hills → jhum; Jharkhand climate stress often linked to forest fires.

Q38. Assertion (A): Forests are a renewable resource. Reason (R): They enhance environmental quality by absorbing CO_2 and releasing O_2.

A. Both true; R explains A

B. Both true; R does not explain A

C. A true; R false

D. A false; R true

💡 **Show answer**

Ans: B. Both true, but renewability comes from regenerative capacity / sustainable reuse — not from the CO_2–O_2 services alone.

Q39. Living root bridges are found in:

A. Meghalaya

B. Himachal Pradesh

C. Jharkhand

D. Tamil Nadu

💡 **Show answer**

Ans: A. Meghalaya Ficus-trained bridges. Valparai (TN) is a plantation town fact, not root bridges.

Q40. As per ISFR 2021-style ranking, maximum increase in forest cover among common options was in:

A. Assam

B. Andhra Pradesh

C. Arunachal Pradesh

D. Kerala

💡 **Show answer**

Ans: B. Top gains: Andhra Pradesh (~647 km^2), Telangana, Odisha. Mizoram leads **percentage** cover; MP leads **area**.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D | **Ans first, then Logic**

Q1. With reference to ISFR forest cover definition, which of the following statements is/are correct?

1. Forest cover means a patch of at least 1 ha with canopy ≥10%.
2. Forest cover is counted only on Recorded Forest Area land.
3. Forest cover is identical to Recorded Forest Area in all cases.

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

💡 **Show answer**

Ans: B. Only 1 is correct. **Logic:** Statement 1 is the FSI canopy definition (≥1 ha, ≥10%). Statement 2 is false — forest cover is irrespective of ownership and is not limited to RFA. Statement 3 is false — forest cover ≠ Recorded Forest Area.

Q2. Match canopy classes:

List-I	List-II
A. VDF	1. 10–40%
B. MDF	2. ≥70%
C. OF	3. 40–70%

Code:

A. A-2, B-1, C-3 B. A-3, B-2, C-1 C. A-2, B-3, C-1 D. A-1, B-3, C-2

💡 **Show answer**

Ans: C. A-2, B-3, C-1. **Logic:** VDF ≥70; MDF 40–70; OF 10–40.

Q3. With reference to ISFR 2023 class figures, which of the following statements is/are correct?

1. Forest cover about 21.76%.
2. Tree cover about 3.41%; total green cover about 25.17%.
3. Mizoram has the largest forest area among States.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Largest area = Madhya Pradesh; Mizoram = highest percentage.

Q4. Which of the following pairs is/are correctly matched?

1. Largest forest area — Madhya Pradesh
2. Highest % cover among States — Mizoram
3. Forest cover = Recorded Forest Area always

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Cover ≠ RFA.

Q5. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Under the Indian Forest Act, 1927, Reserved Forest is the strictest category.

Reason (R): In Reserved Forests, many acts are banned unless permitted.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains Reserved Forest strictness. **A/R logic:** IFA reserved = strictest.

Q6. Arrange the law chain in correct order:

1. Indian Forest Act
2. Forest (Conservation) Act
3. National Forest Policy
4. JFM guidelines
5. FRA
6. CAMPA Act

A. 2-1-3-4-5-6 B. 1-2-3-4-5-6 C. 1-3-2-4-5-6 D. 1-2-4-3-5-6

💡 **Show answer**

Ans: B. 1927 → 1980 → 1988 → 1990 → 2006 → 2016. **Logic:** IFA–FCA–NFP–JFM–FRA–CAMPA.

Q7. With reference to Champion and Seth (1968), which of the following statements is/are correct?

1. They classify Indian forests into 16 types in 6 groups.
2. Tropical Dry Deciduous has the largest share (~38%).
3. Tropical wet evergreen occurs mainly below 50 cm rainfall.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Wet evergreen needs >200 cm rain.

Q8. Match List-I with List-II:

List-I (UP belt)	List-II
A. Terai	1. Thorn
B. Vindhyan	2. Moist deciduous
C. Bundelkhand	3. Dry deciduous

Code:

A. A-1, B-3, C-2 B. A-3, B-2, C-1 C. A-2, B-1, C-3 D. A-2, B-3, C-1

💡 **Show answer**

Ans: D. A-2, B-3, C-1. **Logic:** Terai moist; Vindhyan dry; Bundelkhand thorn.

Q9. Which of the following pairs is/are NOT correctly matched?

1. Teak — tropical moist deciduous dominance teaching
2. Red Sanders — dry-deciduous South India endemic
3. Red Sanders — Himalayan wet evergreen endemic of Himachal only

A. 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only pair 3 is not correctly matched. **Logic:** Red Sanders = Palkonda–Seshachalam / southern Eastern Ghats.

Q10. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Tropical wet evergreen forests hold maximum plant diversity/biomass among common options.

Reason (R): They typically receive more than 200 cm rainfall and occur in Western Ghats, NE hills, and A&N.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the richness conditions. **A/R logic:** High rainfall + WG/NE/A&N distribution.

Q11. With reference to Taxus, which of the following statements is/are correct?

1. It is linked with Himalaya and Red Data teaching.
2. It yields taxol (anti-cancer association).
3. It is the dominant tree of tropical dry deciduous plains of Tamil Nadu only.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Himalayan Taxus — not TN dry deciduous dominant.

Q12. Which one of the following correctly distinguishes forest cover and tree cover?

A. They are always identical percentages B. Tree cover replaces Recorded Forest Area legally C. Forest cover and tree cover are different ISFR concepts; total green combines both D. Forest cover counts only reserved forests

💡 **Show answer**

Ans: C. Different ISFR concepts; green cover combines both. **Logic:** Cover ≠ RFA; tree cover is separate.

Q13. Arrange canopy density from densest to open:

A. OF → MDF → VDF B. VDF → OF → MDF C. MDF → VDF → OF D. VDF → MDF → OF

💡 **Show answer**

Ans: D. VDF → MDF → OF. **Logic:** ≥70 → 40–70 → 10–40.

Q14. With reference to Amazon teaching, which of the following statements is/are correct?

1. Amazon rainforest is often called the “lungs of the Earth” in coaching figures.
2. This replaces India’s ISFR definition of forest cover.
3. It is a global biogeography teaching hook, not an ISFR India % fact.

A. 1 and 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 3. **Logic:** Does not replace ISFR definitions.

Q15. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Protected Forest under IFA is State-regulated with some acts allowed unless prohibited.

Reason (R): Protected Forest is stricter than Reserved Forest under IFA.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Reserved is stricter than Protected.

Q16. Match List-I with List-II:

List-I	List-II
A. FCA	1. 2006
B. FRA	2. 1980
C. CAMPA Act	3. 2016

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** FCA 1980; FRA 2006; CAMPA 2016.

Q17. With reference to Madhya Pradesh forest facts, which of the following statements is/are correct?

1. MP has the largest forest area among common State options.
2. MP has the maximum teak area among common options.
3. MP has the highest forest percentage among all States, above Mizoram.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Highest % = Mizoram.

Q18. Which of the following pairs is/are correctly matched?

1. JFM guidelines — 1990
2. National Forest Policy — 1988
3. IFA — 1980

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** IFA is 1927; FCA is 1980.

Q19. How many of the following receive >200 cm rain in wet evergreen teaching?

Western Ghats, NE hills, Andaman & Nicobar, Thar Desert interior

A. Only three B. Only two C. Only one D. All four

💡 **Show answer**

Ans: A. Only three. **Logic:** Thar is not wet evergreen.

Q20. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Recorded Forest Area is a legal-notification figure.

Reason (R): ISFR forest cover always equals Recorded Forest Area percentage.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Cover and RFA differ.

Q21. With reference to Red Sanders, which of the following statements is/are correct?

1. Botanical name teaching: *Pterocarpus santalinus*.
2. Dry-deciduous South India endemic.
3. It is the state tree of Himachal Pradesh wet alpine zone.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not HP wet alpine.

Q22. Arrange densest canopy class letter to open forest:

A. OF, MDF, VDF B. MDF, OF, VDF C. VDF, MDF, OF D. VDF, OF, MDF

💡 **Show answer**

Ans: C. VDF, MDF, OF. **Logic:** Same density ladder.

Q23. Which of the following UP type facts is/are correct?

1. Terai — moist deciduous
2. Vindhyan — dry deciduous
3. Bundelkhand — thorn

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Standard UP belt facts.

Q24. With reference to FSI/ISFR, which of the following statements is/are correct?

1. FSI publishes ISFR every two years.
2. ISFR is an annual CPCB product.
3. Forest cover mapping is FSI's job.

A. 1 and 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 3. **Logic:** ISFR is biennial FSI — not CPCB annual.

Q25. Match List-I with List-II:

List-I	List-II
A. NFP	1. 1990
B. JFM	2. 1988
C. FRA	3. 2006

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** NFP 1988; JFM 1990; FRA 2006.

Q26. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Arunachal Pradesh often leads very dense forest area among States in ISFR teaching.

Reason (R): Mizoram has the largest absolute forest area in India.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Largest area = MP; Mizoram = highest %.

Q27. Which of the following pairs is/are NOT correctly matched?

1. Wet evergreen — >200 cm
2. Tropical Dry Deciduous — largest share among Champion & Seth types
3. Open Forest — canopy ≥70%

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** OF is 10–40%; ≥70% is VDF.

Q28. With reference to CAMPA, which of the following statements is/are correct?

1. CAMPA Act teaching year is 2016.
2. It sits in the forest clearance / compensatory afforestation world with FCA.
3. CAMPA repealed the Wildlife (Protection) Act.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Does not repeal WPA.

Q29. How many of the following are correctly true?

1. Forest cover ≥1 ha and ≥10% canopy
2. RFA ≈ legal notification figure near ~23% class teaching
3. Tree cover is the same as VDF

A. Only two B. Only one C. All three D. None

💡 **Show answer**

Ans: A. Only two. **Logic:** Tree cover ≠ VDF.

Q30. Which one of the following is the correct FCA year?

A. 1927 B. 1980 C. 1988 D. 2006

💡 **Show answer**

Ans: B. 1980. **Logic:** IFA 1927; FCA 1980; NFP 1988; FRA 2006.

Mains Answer Framework

Forest governance in India (250 words): Define forest cover vs RFA. Champion & Seth types and regional distribution. Conservation chain — Chipko, FCA, Godavarman, NFP, CAMPA. Management — IFA, working plans, FSI/ISFR. Community instruments — social forestry, agroforestry, JFM, FRA (compare purpose and legal status). UP case — 6.24% cover, Sonbhadra, Terai types, CAFRI Jhansi, Chandauli-Shravasti ISFR trap. Conclude with 33% target and participatory forestry.

Common Traps — Don't Fall For These

Trap	Correct fact
Bahraich >20% forest	**Chandauli + Shravasti** only
Forest cover = RFA only	Cover exists **outside** notified forest
FCA enacted 1972	**1980**
JFM = FRA same year	JFM **1990**, FRA **2006**
Social forestry = JFM	Social = **non-forest land**; JFM = degraded forest
FRA IFR cap 10 ha	**4 ha**
Agroforestry Policy 2006	**2014**
ISFR by CPCB	**FSI Dehradun**
NFP 1952 vs 1988	Current policy = **1988**; 1952 classes ≠ National Park
CAFRI location	**Jhansi**, not Banda
MP vs Mizoram	MP = **largest area**; Mizoram = **highest %**
RF vs Protected Forest	RF = strictest IFA class
Red Sanders = rainforest	**Dry deciduous** South India / Eastern Ghats
Teak = rainforest	**Tropical moist deciduous**; MP max teak
HP = tropical wet evergreen	**Pine/temperate** belts
Urbanisation = deforestation impact	Urbanisation is a **cause**
Mangrove safety = canopy shelter only	**Extensive roots**
Bhitarkanika in West Bengal	**Odisha** — Brahmani–Baitarni
Gujarat < A&N mangrove	**Gujarat 2nd** after WB (ISFR 2021 style)
Aluminium green = saves forests	Recyclability/eco-profile; wood-substitute R often false

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 9 — Pollution & Waste Management

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Environmental Pollution (पर्यावरणीय प्रदूषण) | Environmental Degradation (पर्यावरणीय क्षरण) | Air Pollution | Water Pollution | Soil Pollution | Noise Pollution | Radioactive Pollution | Thermal Pollution | Marine Pollution | Plastic Pollution | Solid Waste Management | Biomedical Waste | Hazardous Waste | E-waste

Sources baked in: Water Act 1974, Air Act 1981, EPA 1986, SWM/BMW/HW/E-waste Rules, CPCB/CGWB, NCAP 2019, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — standards, org↔report traps, UP rivers/cities **Last verified:** August 2026

Current Affairs (this topic)

- NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) / CAQM / solid-waste and e-waste rule updates are the living pollution CA layer.
- Keep primary pollutant vs secondary pollutant and BOD vs COD pairs stable.

💡 🎯 **Consolidated — 39 Must-Score Facts (Click to Expand)**

1. **Pollution** is harmful addition beyond the environment (पर्यावरण)'s absorption capacity. **Degradation** is the broader fall in environmental quality and productivity. 2. Human-caused pollution / human-modified environment = **anthropogenic**. Anthropogenic pollution often raises persistent **non-biodegradable** loads. 3. **Biodegradable** pollutants (sewage, faecal matter, urine, domestic waste, rubber in some stems) break down by microbes. **Non-biodegradable** include pesticides, plastics, mercury, asbestos. 4. **Point sources** are chimneys and drains. **Non-point sources** include farm runoff and roads. 5. **Primary pollutants** are emitted directly (SO_2, PM, CO, NOx, lead). **Secondary pollutants** form in air (O_3, PAN, photochemical smog, acid rain (अम्ल वर्षा)). 6. Law years: **Water Act 1974 → Air Act 1981 → EPA 1986 (पर्यावरण संरक्षण 1986)**. Noise Rules date to **2000**. Air Act amended **1987** to include noise. 7. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) sets ambient air and surface-water standards. **SPCB** enforces in States. **CGWB** handles **groundwater** assessment — not CPCB. 8. India's AQI uses **eight** pollutants: **PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb** — **not CO_2 or CH_4** (those are climate gases). Bands: **0–50 Good** … **401–500 Severe**. 9. London smog is **SO_2 + smoke**. Los Angeles / photochemical smog is **NOx + VOCs + sunlight → O_3 / PAN**. **PAN** is a powerful eye irritant. 10. **CO** binds haemoglobin **~200–300×** more strongly than O_2 (carboxyhaemoglobin). Lead antiknock in petrol harms IQ/brain development. 11. **Fly ash** from coal TPPs: usable in bricks / Portland cement replacement; contains SiO_2, Al_2O_3, CaO **plus toxic metals** — not "oxides only." 12. **Lichens** are SO_2 bio-indicators and die in polluted air. **Radon** is the classic major **indoor** air pollutant (lung cancer risk). 13. **NCAP (2019)** covers non-attainment cities with a **20–30%** PM reduction target. **BS-VI** fuel arrived in **2020**. **WAYU** air-purifying units were installed in **Delhi**. 14. **CCAC** (2012, with UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम)) focuses on **methane, black carbon, HFCs** — not a G20-only club. 15. Water facts: high **BOD** means worse organic pollution; **COD ≥ BOD**; clean DO is often **>6 mg/L**, while **<5 mg/L** / **<4 mg/L** stresses aquatic life. 16. **Eutrophication** (सुपोषण) is nutrient overload (N/P) → algal bloom → dissolved-oxygen crash — a **water** phenomenon, not air pollution. 17. Disease locks: **Minamata–Hg**, **Itai-itai–Cd**, **Knock-knee / fluorosis–F**, **Black Foot–As**, blue-baby–nitrate. 18. **Damodar** (दामोदर) is classically called a **biological desert** from mining/industry pollution. Ganga (गंगा) BOD often highest **Kanpur (कानपुर)–Allahabad (इलाहाबाद)**

stretch. 19. Arsenic groundwater hotspots: Ganga–Brahmaputra (ब्रह्मपुत्र) / Bangladesh Padma (पद्मा)–Meghna; Indian states include WB, Bihar, UP, Jharkhand, Assam, Manipur, Chhattisgarh. 20. About **80%** of marine pollution comes from **land**. Ocean acidification (महासागरीय अम्लीकरण) from CO_2 harms calcareous plankton and corals. **IMO** HQ = **London**. 21. Noise is measured in **decibels**. Under **Noise Pollution (Regulation and Control) Rules, 2000**, a person may complain when noise exceeds the ambient standard by **10 dB(A) or more**. A **10 dB** rise also means **10×** intensity on the logarithmic scale. **Green muffler** = roadside tree belts for noise (and dust). 22. Plastic: **PWM Rules 2016**, SUP ban push **2022**, microplastics **<5 mm**; **microbeads** harm marine ecosystems. Polythene = ethylene polymer, non-biodegradable. 23. **Bioremediation** uses microbes to clean pollution; **cannot** readily/completely treat **Cd/Pb** heavy metals. **Oilzapper** (TERI) is for oily sludge/spills. Genetic engineering can create specialised strains (e.g. *Pseudomonas putida*). 24. Waste rules cluster in **2016**: SWM, BMW, Hazardous Waste, and Plastic Waste. **E-waste Rules** updated in **2022**. **EPR** first featured strongly in plastic/e-waste rules **2011+**. 25. **3R** priority is **Reduce > Reuse > Recycle**. **Swachh Bharat** began in **2014**. NAQI launched **17 Oct 2014**. 26. BMW colour codes: **Yellow** incineration; **Red** contaminated recyclables (IV tubes/sets, catheters, urine bags, syringes **without** needles) for disinfection then recycle; **White** sharps (needles); **Blue** glass vials / metal. 27. **Euro / BS** emission norms limit CO, HC, NOx, PM. Euro-II ultra-low sulphur diesel ≈ **≤0.05% S**. India skipped BS-V to **BS-VI (1 Apr 2020)**. 28. Catalytic converters use **Pt / Pd / Rh** to convert CO → CO_2, NOx → N_2, HC → CO_2 + H_2O. 29. UP air NCAP cities include Lucknow (लखनऊ), Kanpur, Ghaziabad, Agra, Varanasi (वाराणसी), Meerut (मेरठ), Noida, Prayagraj (प्रयागराज), Moradabad, and Bareilly. 30. UP issue facts: Kanpur Ganga tanneries; Ghaziabad/Noida air; **Moradabad** noise; **Taj Trapezium** at Agra; Varanasi ghat/plastic pressure. 31. FSI publishes forest cover (वन आवरण), not pollution standards. CWC (केंद्रीय जल आयोग) is surface-water resources, not the groundwater quality report. 32. EPA **1986** is the umbrella for many later rules; it does not replace Water and Air Acts. 33. Secondary ozone at ground level is a pollutant; do not confuse it with stratospheric protective ozone. 34. Hazardous waste follows HW Rules **2016** plus Basel controls — not SWM municipal rules alone. 35. **Asian Brown Cloud** / atmospheric brown cloud is especially noted over **South Asia** (winter aerosols, black carbon, sulphate). 36. Bag filters trap finer particulates (**<~50 µm**); cyclone separators/collectors handle coarser (**>~50 µm**). **Cyclone divider** is not a standard control device name. 37. **Radioactive pollution** is ionizing radiation beyond safe dose. **Becquerel (Bq)** counts disintegrations per second; **Sievert (Sv)** is biological dose; **half-life** is time for **50%** decay — not complete disappearance. 38. **AERB** (Atomic Energy Regulatory Board) regulates nuclear safety in India. **Chernobyl (26 April 1986)** is the classic nuclear-accident teaching case; **Bhopal (1984)** is MIC industrial gas, not radioactive pollution. 39. **Thermal pollution** raises water temperature (mainly power-plant cooling water). Warm water holds **less DO**; even a **2–3°C** rise can stress aquatic life. **Cooling towers/ponds** mitigate discharge under Water Act consent conditions.

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Pollution	**Environmental Degradation**	Direct or indirect release of harmful substances, radiation, or thermal energy into air, water, or soil causing toxic injury to living organisms vs broader long-term deterioration of environmental quality, natural capital, and ecosystem productivity	पर्यावरणीय प्रदूषण / पर्यावरणीय क्षरण
Primary Air Pollutant	**Secondary Air Pollutant**	Emitted directly from identifiable domestic or industrial exhaust sources into the atmosphere (CO, SO_2, NO, PM, Lead) vs formed in the atmosphere through chemical or photochemical reactions among primary precursors (Tropospheric Ozone, PAN, Smog, Nitric acid)	प्राथमिक प्रदूषक (सीधे उत्सर्जित) / द्वितीयक प्रदूषक (रासायनिक क्रिया)
Biochemical Oxygen Demand (BOD)	**Chemical Oxygen Demand (COD)**	Amount of dissolved oxygen needed by aerobic microorganisms to decompose biodegradable organic matter in water at 20°C over 5 days vs total amount of oxygen required to chemically oxidize all organic matter (both biodegradable and non-biodegradable) using a strong chemical oxidant	जैव रासायनिक ऑक्सीजन मांग (BOD) / रासायनिक ऑक्सीजन मांग (COD)
Oligotrophic Lake	**Eutrophic Lake**	Deep, crystal-clear water body with low nutrient levels,

low biological productivity, and high dissolved oxygen vs shallow, murky water body rich in nitrates/phosphates, choked with algal blooms, and experiencing severe hypolimnetic oxygen crashes | अल्पपोषी झील (स्वच्छ/कम पोषक) / सुपोषित झील (अत्यधिक ब्लूम/ऑक्सीजन ह्रास) | | **Extended Producer Responsibility (EPR)** | **Polluter Pays Principle** | Policy mandate making manufacturers legally and financially responsible for post-consumer collection, recycling, and safe disposal of their products (E-waste, plastic packaging) vs international environmental law doctrine requiring polluters to bear the financial costs of pollution mitigation and cleanup | विस्तारित उत्पादक दायित्व (EPR) / प्रदूषक भुगतान सिद्धांत | | **Central Pollution Control Board (CPCB)** | **Central Ground Water Board (CGWB)** | Statutory apex pollution regulator established under Water Act 1974 for framing air/water standards and national monitoring vs subordinate scientific office under Ministry of Jal Shakti assessing groundwater resources and aquifer dynamics | केंद्रीय प्रदूषण नियंत्रण बोर्ड (CPCB) / केंद्रीय भूजल बोर्ड (CGWB) | | **Classical (London) Smog** | **Photochemical (Los Angeles) Smog** | Occurs in cool, humid winter mornings; high concentrations of SO_2 and coal soot; chemically reducing in nature vs occurs in hot, dry, sunny afternoons; vehicle exhaust NOx + volatile organic compounds (VOCs); chemically oxidizing in nature with high ozone and PAN | लंदन स्मॉग (सल्फ्यूरस/अपचायक) / प्रकाश रासायनिक स्मॉग (ऑक्सीकारक) |

💡 📋 **Must-score facts — pollutants, AQI, smog, laws (Click to Expand)**

Basics | Item | Lock | |------|------| | Primary / Secondary | Emitted direct / formed in air (O_3, PAN, acid rain) | | Point / Non-point | Chimney–drain / farm runoff–roads | | Law years | Water **1974** → Air **1981** → EPA **1986**; Noise Rules **2000** | | AQI 8 pollutants | PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb (**not** CO_2/CH_4) | | London / LA smog | SO_2+smoke / NOx+VOCs+sun → O_3+PAN | | CO / Lead | Hb affinity ~200–300× / IQ–brain harm | | Lichens / Radon | SO_2 bio-indicator / top indoor pollutant | | NCAP / BS-VI | **2019** / **2020** |

9.1 Environmental Pollution

Identity: Beyond absorption capacity | Point/non-point | Primary/secondary | Water–Air–EPA laws

- **Environmental pollution** is the harmful addition of substances or energy when the rate exceeds the environment's **absorption or assimilative capacity**.
- Pollutants may be **physical** (heat, noise, radiation), **chemical** (metals, pesticides, gases), or **biological** (pathogens).
- Pollution caused or strongly influenced by humans is called **anthropogenic** pollution; the human-modified environment is the **anthropogenic environment**.
- **Biodegradable** pollutants (sewage, faecal matter, urine, domestic organic waste) can be broken down by microbes. **Non-biodegradable** pollutants (pesticides, plastics, many metals, asbestos) persist or degrade very slowly.
- **Point sources** have identifiable outlets such as factory (फैक्टरी) chimneys and sewage pipes.
- **Non-point sources** are diffuse, such as fertilizer runoff from farms and urban (नगरीय) stormwater.
- **Primary pollutants** are emitted directly, including **SO_2**, **CO**, **NOx**, and **particulate matter**.
- **Secondary pollutants** form in the atmosphere, including **ground-level ozone**, **PAN**, photochemical smog, and **acid rain**.
- The **Water (Prevention and Control of Pollution) Act, 1974** controls water pollution.
- The **Air (Prevention and Control of Pollution) Act, 1981** controls air pollution (noise included by **1987** amendment).
- The **Environment (Protection) Act, 1986** is the umbrella law for hazardous substances, standards, and EIA.
- **CPCB** sets national standards. **SPCBs** enforce pollution control at the state level.

> **Logic:** Pollution requires exceeding tolerance — not every human addition counts. Pesticide ≠ biodegradable.

PYQ — UPPCS Prelims 2025, Q22

Carrying capacity: (1) Resource extraction should remain above regeneration. (2) Waste should stay within absorption capacity.

A. Only 2

B. Neither

C. Both

D. Only 1

💡 **Show answer**

Ans: A — Extraction must be **at or below** regeneration.

9.2 Environmental Degradation

Identity: Broader than pollution | Carrying capacity | Deforestation–mining–urbanization chains

- **Environmental degradation** is the decline in environmental quality and productivity from pollution, resource depletion, and ecosystem (पारिस्थितिकी तंत्र) damage.
- Degradation is **broader than pollution**. Deforestation without toxic chemicals still causes degradation.
- A common chain runs from **deforestation → soil erosion → river siltation → flood risk → productivity loss**.
- **Urbanization** degrades the environment through sewage overload, heat islands, and waste accumulation.
- **Mining** removes vegetation and can cause acid drainage and toxic tailings. **Sonbhadra (सोनभद्र) (UP)** faces mining-related degradation.
- **Intensive agriculture** degrades soil through pesticide buildup, salinization, and organic matter loss.
- Degradation accelerates when **resource extraction exceeds regeneration** or **waste exceeds absorption capacity**.
- About **30% of India's land area** is degraded according to ISRO desertification (मरुस्थलीकरण) assessments.
- **EIA under EPA 1986** aims to prevent degradation before major projects start.

> **Logic:** Afforestation **reverses** degradation; it is not a cause of it.

9.3 Air Pollution

Identity: PM2.5/PM10 | Smog types | AQI | NCAP | Lichens as bio-indicators

- **Air pollution** is the presence of harmful gases, dust, and smoke at levels that injure health or visibility.
- **Natural sources** include volcanoes and forest fires. **Man-made sources** include fuel combustion, vehicles, and industry.
- **PM2.5** particles (≤2.5 µm) penetrate deep into the lungs and bloodstream; rain cleans coarser dust better than ultrafines.
- **PM10** particles (≤10 µm) mainly affect the upper respiratory tract.
- **SO_2** from coal combustion causes respiratory illness, acid rain, and building damage.
- **NOx** from vehicles and power plants helps form **photochemical smog** and **ground-level ozone**.
- **CO** from incomplete combustion binds haemoglobin **~200–300×** more strongly than oxygen (carboxyhaemoglobin) and can kill at high saturation.
- **Lead** from petrol antiknock additives harms children's brain development and adult heart/respiratory health.
- **London/classical smog** forms from **SO_2 and smoke** in cool, humid conditions.

- **Los Angeles/photochemical smog** forms when **NOx + hydrocarbons** react in sunlight to yield **O_3, formaldehyde, and PAN**; **PAN** is a strong eye irritant.
- India's **AQI** is based on **eight** pollutants: **PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb** — **not CO_2 or CH_4**. Bands: **0–50 Good** … **401–500 Severe**. **NAQI** launched **17 Oct 2014**.
- **NCAP (National Clean Air Programme, 2019)** targets non-attainment cities for about **20–30% PM reduction** (2017 baseline framing).
- **Fly ash** from coal TPPs is used in bricks and as Portland cement replacement; main oxides are SiO_2, Al_2O_3, CaO, plus toxic metals (Pb, As, Cu).
- Control devices: **bag filters** for finer particulates (**<~50 µm**); **cyclone separator/collector** and wet scrubbers for coarser loads; **cyclone divider** is not a standard device name.
- **Catalytic converters** (Pt–Pd–Rh) convert CO → CO_2, NOx → N_2, and unburnt HC → CO_2 + H_2O.
- **Lichens** are sensitive **bio-indicators of air pollution**, especially **SO_2**. They **do not grow well in polluted areas**.
- **Radon** (from radium/soil) is the classic major **indoor** air pollutant and raises lung-cancer risk in poorly ventilated homes.
- **Asian Brown Cloud** aerosols are especially noted over **South Asia** (winter; black carbon, sulphate).
- **WAYU** (Wind Augmentation Purifying Unit) was installed at Delhi traffic junctions. **CCAC** (2012, UNEP partners) targets short-lived climate pollutants: **methane, black carbon, HFCs** — not a G20-only club.
- **Ringelmann scale** (रिंगेलमैन पैमाना) grades smoke density/opacity.

> **Logic:** — lichens indicate pollution but **decline** in polluted air. AQI ≠ greenhouse-gas index.

PYQ — UPPCS Prelims 2025, Q61

1. Lichens are very good indicators of pollution. 2. Lichens grow well in polluted areas.

A. Only 2

B. Neither

C. Both

D. Only 1

💡 **Show answer**

Ans: D

Pollutant	CPCB 24-h standard (µg/m^3)
PM2.5	60
PM10	100
SO_2	80
NO_2	80

Aquatic Degradation Dynamics: Phelps-Streeter Dissolved Oxygen Sag Curve & Cultural Eutrophication

9.4 Water Pollution

Identity: BOD/COD/DO | Eutrophication | Namami Gange (नमामि गंगे) | CGWB trap

- **Water pollution** alters water quality so it becomes unfit for drinking, aquatic life, or irrigation.

- **BOD (Biochemical Oxygen Demand)** measures oxygen consumed by microbes decomposing **organic** matter. **Higher BOD = more organic pollution** (and usually lower DO).
- **COD (Chemical Oxygen Demand)** measures total oxidizable matter. **COD is always ≥ BOD**.
- **DO (Dissolved Oxygen)** above **6 mg/L** suits healthy rivers. Below about **5 mg/L** stresses fish; teaching banks often call **<4 mg/L** highly polluted.
- **Eutrophication** begins when excess **nitrogen and phosphorus** cause algal blooms, bacterial decomposition, and **DO collapse** — a **water** phenomenon, not air pollution.
- **Faecal coliform** bacteria indicate sewage contamination. Safe drinking water should have **zero faecal coliform**.
- Highest **BOD** on the Ganga is classically taught between **Kanpur and Allahabad (Prayagraj)** from heavy pollutant load.
- **Damodar** is called a **biological desert** along the Giridih–Durgapur industrial/mining stretch.
- **Leather industry** effluents rank among the worst chemical polluters of water and soil.
- Disease locks: **arsenic** → Black Foot / cancers (WHO limit **0.01 mg/L**); **fluoride** → fluorosis / **knock-knee**; **mercury** → Minamata; **cadmium** → Itai-itai.
- **Occupational Lung Diseases / Pneumoconiosis (High-Yield UPPCS Match):**

Disease	Causative Pollutant / Dust	Industry / Occupational Source	Key feature / Nickname
Silicosis	Free silica (SiO_2) dust	Stone crushing, quarrying, mining, glass factories	Progressive lung fibrosis; Rajasthan first state to frame Silicosis Policy (2019)
Asbestosis	Asbestos fibers	Roofing sheets, pipe insulation, shipbuilding	Leads to pulmonary fibrosis and **Mesothelioma** (cancer of pleural lining)
Anthracosis	Coal dust	Coal mining (Jharia, Raniganj, Singrauli)	**Black Lung Disease** / Coal Workers' Pneumoconiosis
Byssinosis	Cotton fiber dust	Textile mills, ginning factories	**Brown Lung Disease**; called **"Monday Fever"** (symptoms peak after weekend return)
Bagassosis	Sugarcane bagasse dust	Sugar mills, paper & cardboard manufacturing	Fungal spores (*Thermoactinomyces*) on decomposing crushed sugarcane stalk
Siderosis	Iron / iron oxide dust	Iron foundries, arc welding, iron ore mining	Iron dust accumulation in lung parenchyma
Farmer's Lung	Moldy hay / straw dust	Agriculture, silos, threshing	Hypersensitivity reaction to thermophilic actinomycetes
- Arsenic groundwater hotspots: Ganga–Brahmaputra plains and Bangladesh's Padma–Meghna; Indian states include **West Bengal** (first reported **1978**), Bihar, UP, Jharkhand, Assam, Manipur, and Chhattisgarh.			
- **Kanpur's tanneries** discharge **chromium** into the Ganga as a major UP point source.			
- The **Gomti** through **Lucknow** (लखनऊ) has been flagged among India's most polluted stretches. The **Yamuna** (यमुना) is heavily polluted in the Mathura (मथुरा)–Agra–Delhi stretch.			

Disease	Causative Pollutant / Dust	Industry / Occupational Source	Key feature / Nickname
- **Namami Gange (2014)** is India's flagship Ganga cleaning mission. **NMCG** is the nodal agency.			
- The **Annual Groundwater Quality Report** is prepared by **CGWB**, not CPCB.			
- Rail **bio-toilets** use **anaerobic bacterial** inoculum (DRDO–Railways), producing water and gases after chlorination — not fungal-only, and not ammonia-only end products.			

> **Logic:** — groundwater report = **CGWB only**. Eutrophication ≠ air pollution.

PYQ — UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 prepared by: 1. CPCB 2. CWC 3. CGWB

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

Show answer

Ans: B

Parameter	Clean river	Polluted river
BOD (mg/L)	<3	>10
DO (mg/L)	>6	<4
pH	6.5–8.5	<6 or >9

9.5 Soil Pollution

Identity: Pesticides | Heavy metals | Acid rain | Kanpur chromium

- **Soil pollution** is the buildup of toxic chemicals, salts, or waste that reduces fertility and enters the food chain (खाद्य श्रृंखला).
- **Organochlorine pesticides** such as **DDT** persist for years and **biomagnify** in food chains.
- **Heavy metals** including **chromium, cadmium, lead, and arsenic** accumulate in topsoil from industry and sludge.
- **Kanpur tanneries** release **hexavalent chromium**, contaminating soil and river sediments in UP.
- **Acid rain** from **SO_2 and NOx** acidifies soil and releases toxic aluminium.
- **Plastic mulch and microplastics** alter soil structure and microbial activity.
- **Phytoremediation** uses plants to absorb contaminants. **Bioremediation** uses microbes to detoxify soil/water by enhancing natural biodegradation.
- Bioremediation **cannot** readily and completely treat many **heavy metals** such as **cadmium and lead**. Genetic engineering can create specialised strains (e.g. *Pseudomonas putida* for oil spills).
- **Oilzapper** (TERI) is a bacterial consortium for oily sludge and oil-spill remediation.

- Indiscriminate fertilisers drive soil, water, and air pollution; residues can enter the food chain. Soil pollution often **leaches into groundwater** over time.

> **Logic:** Soil **erosion** is physical degradation. Chemical contamination = soil **pollution**. Heavy metals ≠ easy bioremediation.

9.6 Noise Pollution

Identity: dB scale | CPCB zone limits | Noise Rules 2000 | Health effects

- **Noise pollution** is unwanted sound that harms hearing, sleep, and health.
- **Decibel (dB)** (डेसिबल) uses a **logarithmic scale**. A **10 dB** rise means **10 times** the sound intensity.
- **Day** is **6 AM–10 PM**. **Night** is **10 PM–6 AM**. Night limits are stricter.
- **CPCB residential limits** are **55 dB (day)** and **45 dB (night)**.
- **Silence zones** near hospitals and schools allow **50 dB (day)** and **40 dB (night)**.
- **Industrial zones** allow **75 dB (day)** and **70 dB (night)**.
- Under **Noise Pollution (Regulation and Control) Rules, 2000**, a complaint may be made when noise exceeds the prescribed ambient standard by **10 dB(A) or more** (Rule 7). Do not key 5 / 8 / 9 dB.
- Prolonged exposure above **85 dB** can cause permanent **Noise-Induced Hearing Loss (NIHL)**.
- Rough dB anchors: rustling leaves **~20 dB**, normal talk **~60 dB**, truck **~80–85 dB**, jet engine **~120 dB**, jet landing **~150 dB**, rocket **~180 dB**.
- Noise from aircraft faster than sound is a **sonic boom** (Mach scale: Mach-1 = speed of sound).
- **Green muffler** means rows of trees/shrubs planted to absorb noise (and dust) near roads and industry.
- **Moradabad (UP)** ranked among the world's most noise-polluted cities in UNEP's Frontier Report 2022.
- **Noise Pollution (Regulation and Control) Rules, 2000** operate under **EPA 1986**. Jet flights are among the most intense common noise sources.

> **Logic:** Complaint threshold = **+10 dB(A)** above ambient, not 5/8/9. **45 dB** is residential **night**, not silence zone night (**40 dB**). Green muffler = noise control, not soil.

9.7 Radioactive Pollution

Identity: Ionizing radiation | Bq/Sv | Half-life | AERB

- **Radioactive pollution** comes from ionizing radiation exceeding safe dose limits.
- **Becquerel (Bq)** measures radioactivity as disintegrations per second.
- **Sievert (Sv)** measures biological radiation dose.
- **Half-life** is the time for **50%** of a radioactive isotope to decay.
- **Natural sources** include cosmic rays and **radon gas** in buildings.
- **Anthropogenic sources** include nuclear power plants, medical isotopes, and uranium mining.
- **Chernobyl (26 April 1986, Ukraine / then USSR)** is the classic nuclear accident case for radioactive pollution teaching.
- Radioactive pollution can cause **hereditary damage**, blood-circulation harm, and **carcinogenesis** — it is not mainly a soil-mineral imbalance story.
- **Bhopal Gas Tragedy (2–3 December 1984)** at Union Carbide released **methyl isocyanate (MIC, CH_3NCO)** — India's worst industrial gas disaster.
- **Alpha particles** are dangerous if ingested. **Gamma rays** penetrate the body from outside.

- **AERB (Atomic Energy Regulatory Board)** regulates nuclear safety in India.
- Radioactive biomedical waste needs **shielded containers** under **BMW Rules 2016**.

> **Logic:** Half-life means **50% decay**, not complete safety. Bhopal = MIC 1984, not CO/SO_2.

9.8 Thermal Pollution

Identity: Heated effluent | ↓DO | Power plants | Cooling towers

- **Thermal pollution** raises water temperature above the natural range, mainly from industrial cooling water.
- **Warm water holds less dissolved oxygen**. Fish need more oxygen as metabolism rises with heat.
- Even a **2–3°C** rise can stress aquatic ecosystems.
- **Coal and nuclear power plants** are major sources of thermal discharge.
- **Cooling towers** and **cooling ponds** reduce thermal pollution before release.
- Thermal pollution is regulated through **consent conditions under the Water Act 1974**.

> **Logic:** Fish kills mainly follow **DO reduction**, not direct heat toxicity alone.

9.9 Marine Pollution

Identity: 80% land-based | MARPOL | Oil/plastic/nutrients | Coral bleaching (प्रवाल विरंजन)

- **Marine pollution** contaminates oceans through sewage, oil, plastics, and nutrients.
- About **80% of marine pollution originates on land** via rivers and coastal discharge.
- **Oil spills** form slicks that block oxygen exchange and coat wildlife.
- **Plastic waste** and **microplastics** enter marine food chains globally.
- **Nutrient runoff**, estuary (ज्वारनदमुख) discharges, monsoon run-off, and coastal **upwelling** can all drive harmful **algal blooms**.
- **Ocean acidification** is falling ocean pH from atmospheric **CO_2** uptake; it harms **calcareous phytoplankton**, **coral reefs**, and some animals with phytoplankton larvae (cloud seeding is **not** the usual acidification concern).
- Oceans absorb about **one-third** of anthropogenic CO_2.
- **MARPOL 73/78** is the international convention to prevent pollution from ships (नावाध्यक्ष). The **International Maritime Organization (IMO)** HQ is in **London** (IMCO renamed IMO in **1982**).
- **SACEP** = **South Asia Cooperative Environment Programme** (oil/chemical pollution MoU framing in papers).
- **MARPOL Annex V** prohibits plastic discharge from vessels.
- **Coral bleaching** in the Andaman (अंडमान) and Nicobar (निकोबार) Islands was linked to **Indian Ocean warming of about 2°C** (2024 PYQ pattern).
- **Oilzapper**-type bioremediation has been used for oily sludge/spills (e.g. Mumbai port (बंदरगाह) spill teaching note).

> **Logic:** — bleaching from **temperature rise**, not temperature decrease. Acidification concern ≠ cloud seeding.

PYQ — UPPCS Prelims 2024, Q19

Coral bleaching in Andaman & Nicobar caused by:

A. Increased coastal industry

B. **+2°C Indian Ocean warming**

C. Heavy marine pollution

D. −4°C temperature drop

> 💡 **Show answer**
>
> **Ans: B** — Warming is the primary trigger; pollution is a secondary stressor.

9.10 Plastic Pollution

Identity: PWM Rules 2016 | SUP ban 2022 | Microplastics | EPR

- **Plastic pollution** persists because polymers take **100–1000 years** to break down.
- **Polythene** is a polymer of **ethylene (C_2H_4)** — carbon–hydrogen chains that do not biodegrade readily.
- **Microplastics** are fragments **smaller than 5 mm**. **Microbeads** (often polyethylene in cosmetics) harm **marine ecosystems**.
- **Plastic Waste Management Rules, 2016** regulate manufacture, recycling, and **EPR**.
- The **SUP (single-use plastic) ban notification of July 2022** banned identified items such as plastic cutlery, straws, and certain films.
- **EPR** makes producers and brand owners collect equivalent plastic waste they place in the market; EPR featured early in **Plastic/E-waste Rules 2011**.
- **India is the 3rd largest plastic waste generator** globally after the USA and China.
- World Environment Day (विश्व पर्यावरण दिवस) **2023** theme focused on solutions to plastic pollution (**Beat Plastic Pollution**).
- The **Ganga** (गंगा) carries large amounts of plastic waste from ghats and urban dumping.
- **MARPOL Annex V** bans plastic discharge from ships into the sea.
- **Incinerators** burn waste. **Pyrolysis / plasma gasification** are waste-to-energy thermal routes.
- **R2 Code of Practices** promotes environmentally responsible **electronics recycling**.

> **Logic:** PWM Rules = **2016**. **SUP ban = 2022**. Microbeads = marine harm, not mainly skin cancer.

9.11 Solid Waste Management

Identity: SWM Rules 2016 | 3R hierarchy | Segregation | Swachh Bharat

- **Solid waste management** covers collection, segregation, treatment, and disposal of municipal and commercial waste.
- The **3R hierarchy** is **Reduce > Reuse > Recycle > Recovery > Disposal**.
- **Solid Waste Management Rules, 2016** make **segregation at source** mandatory into wet, dry, and domestic hazardous fractions.
- The Rules provide **exact and elaborate criteria** for identifying landfill and waste-processing sites.
- **Wet waste** is treated by **composting** or **biomethanation**.
- **Dry waste** goes to **Material Recovery Facilities (MRF)** for recycling.
- **Landfill** is the **last resort** for inert rejects only, in lined sites.
- **Swachh Bharat Mission (2014)** targeted open defecation free status and waste management. **SBM 2.0 (2021–2026)** targets garbage-free cities.
- Biomedical, hazardous, and e-waste follow **separate rules**, not SWM Rules alone.

> **Logic:** SWM Rules = **2016**. Segregation is **mandatory**, not optional.

9.12 Biomedical Waste

Identity: BMW Rules 2016 | Colour coding | CBWTF | 48-hour limit

- **Biomedical waste (BMW)** comes from hospitals, clinics, labs, and veterinary institutions.
- **BMW Rules 2016** replaced the 1998 rules and require separate collection from municipal waste.
- **Yellow bags** hold anatomical waste, soiled dressings, and expired medicines for **incineration or autoclaving**.
- **Red bags** hold contaminated recyclable disposables for **disinfection then recycling**. Classic Red list items include **IV tubes and sets**, **catheters**, **urine bags**, and **syringes without needles**.
- **White translucent containers** hold **sharps** such as **needles** from used syringes and scalpels.
- **Blue bags** hold **glassware** (including contaminated glass vials) and metallic implants after disinfection.
- **CBWTF (Common Biomedical Waste Treatment Facility)** serves clusters of healthcare units.
- Untreated BMW must not be stored beyond **48 hours**.

> **Logic:** Sharps = **white**, not yellow. Needles are **not** Red. Contaminated glass vials are **Blue**, not Red. Yellow = anatomical/infectious soiled waste.

Colour	Waste type	Treatment
Yellow	Anatomical, soiled, medicines	Incineration/autoclave
Red	Contaminated recyclables (IV tubes, catheters, urine bags, syringes without needles)	Disinfection → recycle
White	Sharps (needles)	Autoclave → disposal
Blue	Glass vials, metal	Disinfection → recycle

9.13 Hazardous Waste

Identity: HW Rules 2016 | Basel Convention | 90-day storage | Kanpur chromium sludge

- **Hazardous waste** threatens health or the environment because it is toxic, reactive, corrosive, or flammable.
- **Hazardous and Other Wastes Rules, 2016** list generating processes in **Schedule I** and hazardous constituents in **Schedule II**.
- Hazard characteristics include **ignitability, corrosivity, reactivity, and toxicity**.
- Storage is allowed up to **90 days** unless an extension is granted.
- The **manifest system** tracks waste from generator to transporter to treatment facility.
- The **Basel Convention (1989)** (बासेल कन्वेंशन) controls **transboundary movement** of hazardous waste. India ratified it in **1992**.
- **Kanpur tannery chromium sludge** is a classic UP hazardous waste example.

> **Logic:** Basel = **hazardous waste movement**, not climate or biodiversity.

9.14 E-waste

Identity: E-waste Rules 2022 | 3rd largest generator | EPR targets | Lead/mercury/cadmium

- **E-waste** is discarded electrical and electronic equipment and its components.
- India is the **3rd largest e-waste generator** globally after China and the USA.
- **E-waste (Management) Rules, 2022** replaced the 2016 rules and strengthened **EPR**.
- **Schedule I** covers **67 categories** of EEE from phones and computers to refrigerators and lamps.
- Hazardous components include **lead** (CRT screens), **mercury** (CFLs), and **cadmium** (batteries).

- Valuable metals such as **gold, copper, and palladium** can be recovered from circuit boards.
- **EPR collection targets** for producers rise from about **60% toward 80%** of market placement.
- Consumers must channel e-waste only to **authorized collectors and recyclers**.
- **Moradabad and Noida** region has significant e-waste generation and informal processing in UP.

> **Logic:** Current rules = **2022**. India rank = **3rd**, not 1st.

Complete PYQ Bank (Topic 9)

Q1. UPPCS Prelims 2025, Q22

With reference to **carrying capacity**, consider the following statements:

1. Resource extraction should remain **above** the rate of regeneration.
2. Waste generation should stay within the environment's **absorption capacity**.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

💡 **Show answer**

Logic: Sustainable use requires extraction at or below regeneration and waste within absorption limits. **Ans: A.** Statement 2 is correct; statement 1 is wrong — extraction must be **at or below** regeneration, not above it. **Trap:** Statement 2 alone sounds incomplete — but statement 1 reverses the regeneration rule and makes "Both" wrong.

Q2. UPPCS Prelims 2025, Q61

Consider the following statements about **lichens**:

1. Lichens are very good indicators of air pollution.
2. Lichens grow well in polluted areas.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

💡 **Show answer**

Logic: Bio-indicator sensitivity — lichens signal SO_2 pollution by declining, not thriving. **Ans: D.** Lichens are excellent pollution indicators but **die or disappear** in polluted air — they do not grow well there. **Trap:** Statement 1 is true and tempts "Both" — statement 2 is the classic reverse trap.

Q3. UPPCS Prelims 2025, Q70

The **Annual Groundwater Quality Report 2024** was prepared by which of the following?

1. Central Pollution Control Board (CPCB)
2. Central Water Commission (CWC)
3. Central Ground Water Board (CGWB)

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1 only

💡 **Show answer**

Logic: Org–report matching — groundwater quality is CGWB's domain, not CPCB's air/water pollution mandate. **Ans: B.** The groundwater quality report is published by **CGWB only**. **Trap:** CPCB sets pollution standards but **does not** prepare the national groundwater quality report.

Q4. UPPCS Prelims 2024, Q19

Coral bleaching in the **Andaman & Nicobar** region was primarily caused by—

A. Increased coastal industrial effluents alone B. About **+2°C warming** of the Indian Ocean C. Heavy marine plastic pollution alone D. A **−4°C** drop in sea-surface temperature

💡 **Show answer**

Logic: Bleaching trigger is thermal stress on symbiotic zooxanthellae — warming, not cooling or pollution alone. **Ans: B.** Indian Ocean warming of about **+2°C** drove mass bleaching; pollution is a secondary stressor. **Trap:** **−4°C cooling** is physically opposite to bleaching mechanism — temperature **rise** expels algae.

Q5. UPPCS Prelims 2023, Q111

The **Water (Prevention and Control of Pollution) Act** was enacted in—

A. 1974 B. 1981 C. 1986 D. 2000

💡 **Show answer**

Logic: Pollution law chronology — Water Act precedes Air Act and EPA in India's environmental statute stack. **Ans: A.** The Water Act was passed in **1974**; Air Act followed in **1981** and EPA in **1986**. **Trap:** **1981** is the Air Act — do not swap the two earliest pollution statutes.

Q6. UPPCS Prelims 2022, Q46

Photochemical smog always contains—

A. Phosphorus B. Ozone C. Methane D. Aluminium ion

💡 **Show answer**

Logic: Los Angeles-type smog is a secondary-pollution product — O_3 and PAN from NOx + VOCs in sunlight. **Ans: B.** Photochemical smog characteristically includes **ground-level ozone** (plus PAN and other oxidants). **Trap:** London smog is **SO_2 + smoke** — ozone chemistry defines photochemical, not primary stack gases alone.

Q7. UPPCS Prelims 2022, Q81

According to UNEP's **Frontier Report–2022**, which Indian city ranked as the **world's second most noise-polluted city**?

A. Kota B. Moradabad C. Indore D. Patna

💡 **Show answer**

Logic: UP noise hotspot from UNEP Frontier 2022 — brassware/industrial hub in western UP. **Ans: B.** **Moradabad (UP)** was flagged as the world's **second noisiest city** in Frontier Report 2022. **Trap:** Ghaziabad/Kanpur are air-pollution NCAP cities — **Moradabad** is the noise-specific PYQ city.

Q8. UPPCS Prelims 2021, Q15

Which of the following is the best **bio-indicator of air pollution**, especially sulphur dioxide?

A. Moss on rocks in clean hills B. Lichens C. Mangrove pneumatophores D. Deep-sea sponges

💡 **Show answer**

Logic: Standard bio-indicator fact — lichens lack protective cuticle and die when SO_2 rises. **Ans: B.** **Lichens** are sensitive SO_2 bio-indicators and decline sharply in polluted air. **Trap:** Moss grows in clean environments too — lichens are the textbook **air-pollution indicator** pair.

Q9. UPSC pattern

The **Basel Convention (1989)** primarily deals with—

A. Transboundary movement of hazardous wastes B. Climate change mitigation commitments C. Ozone-depleting substances phase-out D. Wetland conservation (Ramsar sites)

💡 **Show answer**

Logic: Treaty-purpose matching — Basel is the hazardous-waste shipment control regime. **Ans: A.** Basel controls **international movement and disposal of hazardous wastes**; India ratified in 1992. **Trap:** Montreal = ozone; UNFCCC/Kyoto/Paris = climate — Basel is **waste movement**.

Q10. UPSC pattern

India ranks **how high globally** in **e-waste generation**?

A. 1st B. 2nd C. 3rd D. 5th

💡 **Show answer**

Logic: India e-waste volume rank after China and the USA under E-waste Rules 2022 framing. **Ans: C.** India is the **3rd largest e-waste generator** globally after China and the USA. **Trap:** **1st** is a frequent overestimate — India is top-tier but **third**, not first.

Extra Drill (Ghatnachakra Pollution chapter)

Q31. Anthropogenic pollution mainly raises—

A. Only always instantly biodegradable gases

B. Often persistent non-biodegradable pollutant loads

C. Only pure rainwater

D. Only stratospheric ozone repair

💡 **Show answer**

Logic: Bank stems on types, diseases, disasters, standards, and remediation so the chapter dump is absorbed. **Ans: B.** Human-driven pollution often adds slow/non-biodegradable loads.

Q32. Secondary pollutants include—

A. Only stack CO emitted directly

B. Ozone, PAN, and photochemical smog

C. Only elemental nitrogen in air always

D. Only pure argon

💡 **Show answer**

Ans: B. Formed in the atmosphere from primary reactants + sunlight/chemistry.

Q33. CO binds haemoglobin relative to oxygen—

A. Weaker always

B. About 200–300 times more strongly

C. Never binds

D. Only binds calcium

💡 **Show answer**

Ans: B. Forms carboxyhaemoglobin and cuts O_2 delivery.

Q34. Bhopal Gas Tragedy gas / year—

A. SO_2 in 1974

B. Methyl isocyanate (MIC), 2–3 December 1984

C. Only methane in 1991

D. Only CO_2 in 2000

💡 **Show answer**

Ans: B. Union Carbide pesticide plant, Bhopal.

Q35. Chernobyl is related to—

A. Only acid rain in Manchester

B. Nuclear accident (1986, Ukraine / then USSR)

C. Only noise Rules 2000

D. Only SUP ban 2022

💡 **Show answer**

Ans: B. Classic radioactive pollution disaster teaching case.

Q36. Green muffler is for—

A. Only soil fertility

B. Noise pollution control by planting tree belts

C. Only nuclear shielding

D. Only coral bleaching

💡 **Show answer**

Ans: B. Multi-row greenery absorbs sound (and dust).

Q37. Ocean acidification mainly harms—

A. Only cloud seeding machines

B. Calcareous phytoplankton, coral reefs, and some planktonic larvae stages

C. Only Himalayan snowmelt volume always

D. Only desert sand dunes

💡 **Show answer**

Ans: B. Falling ocean pH from CO_2 uptake — not the cloud-seeding option.

Q38. Microbeads concern—

A. Mainly skin cancer in children as the bank key

B. Harmful to marine ecosystems

C. Only tiger census

D. Only Ramsar birds

> 💡 **Show answer**
>
> **Ans: B.** Tiny plastic beads enter marine food webs.

Q39. Euro-II ultra-low sulphur diesel sulphur content—

A. About 20% sulphur mandatory

B. ≤0.05% (or less)

C. Exactly 50% always

D. Zero relevance to Euro norms

> 💡 **Show answer**
>
> **Ans: B.** Sulphur limit for Euro-II ultra-low sulphur diesel teaching.

Q40. Correct disease match—

A. Minamata–Hg; Itai-itai–Cd; Knock-knee–fluoride; Black Foot–arsenic

B. Minamata–fluoride; Knock-knee–ozone only

C. All diseases = only noise

D. All diseases = only SO_2 smell

> 💡 **Show answer**
>
> **Ans: A.** Standard pollution–disease pairs for prelims.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D | **Ans first, then Logic**

Q1. With reference to pollution vs degradation, which of the following statements is/are correct?

1. Pollution is harmful addition beyond absorption capacity.
2. Degradation means a rise in environmental quality and productivity.
3. The two terms are identical in all Prelims keys.

A. Only 3 B. Only 1 C. 1 and 2 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 is correct. **Logic:** Statement 1 is true. Statement 2 is false — degradation is decline in quality/productivity, not a rise. Statement 3 is false — pollution and degradation are related but not identical.

Q2. Which of the following pairs is/are correctly matched?

1. Secondary pollutant — emitted directly (SO_2, PM, CO, NOx, lead)

2. Secondary pollutant — formed in air (O_3, PAN, acid rain)
3. CO_2 — one of India's AQI eight pollutants

A. Only 1 B. 1 and 2 C. 2 and 3 D. Only 2

> 💡 **Show answer**
>
> **Ans: D.** Only 2 is correct. **Logic:** Statement 2 correctly defines secondary pollutants (formed in air). Statement 1 mislabels primary pollutants as secondary. Statement 3 is false — CO_2 is not among India's AQI eight.

Q3. Arrange the law years in correct order:

1. Water Act
2. Air Act
3. EPA
4. Noise Rules

A. 1-2-4-3 B. 2-1-3-4 C. 1-3-2-4 D. 1-2-3-4

> 💡 **Show answer**
>
> **Ans: D.** 1974 → 1981 → 1986 → 2000. **Logic:** Water–Air–EPA–Noise Rules.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): London smog is associated with SO_2 + smoke.

Reason (R): Los Angeles / photochemical smog needs NOx + VOCs + sunlight forming O_3/PAN.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; R describes a different smog type — does not explain London smog. **A/R logic:** Two different smog chemistries; R not explanation of A.

Q5. Match List-I with List-II:

List-I	List-II
A. Point source	1. Farm runoff / roads
B. Non-point source	2. Chimneys / drains
C. PAN	3. Strong eye irritant in photochemical smog

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: A.** A-2, B-1, C-3. **Logic:** Point chimney; non-point runoff; PAN eye irritant.

Q6. With reference to India's AQI, which of the following statements is/are correct?

1. Eight pollutants: PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb.
2. CO_2 and CH_4 are included in the eight.
3. AQI bands run from Severe (0–50) toward Good (401–500).

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

💡 **Show answer**

Ans: B. Only 1 is correct. **Logic:** Statement 1 lists the eight AQI pollutants correctly. Statement 2 is false — CO_2 and CH_4 are not among the eight. Statement 3 reverses Good/Severe bands.

Q7. Which of the following pairs is/are NOT correctly matched?

1. CPCB — sets NAAQS / ambient air and surface-water standards coordination
2. SPCB — State enforcement / consent regime
3. CGWB — same as CPCB for ambient air standards

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** CGWB = groundwater — not CPCB air.

Q8. With reference to CO and lead, which of the following statements is/are correct?

1. CO binds haemoglobin ~200–300× more strongly than O_2.
2. Lead antiknock in petrol harms IQ/brain development.
3. CO forms carboxyhaemoglobin.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Classic CO/lead facts.

Q9. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Lichens are SO_2 bio-indicators and die in polluted air.

Reason (R): Radon is a classic major indoor air pollutant linked to lung cancer risk.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; radon indoor fact does not explain lichen SO_2 sensitivity. **A/R logic:** Two separate air-pollution facts.

Q10. Match List-I with List-II:

List-I	List-II
A. NCAP	1. 2020
B. BS-VI	2. 2019
C. Air Act noise inclusion amendment	3. 1987

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** NCAP 2019; BS-VI 2020; noise via Air Act 1987.

Q11. With reference to fly ash, which of the following statements is/are correct?

1. It comes from coal thermal power plants.
2. It is usable in bricks / Portland cement replacement.
3. It contains only harmless sugar molecules and no metals.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** Contains SiO_2/Al_2O_3/CaO plus toxic metals — not "oxides only / harmless".

Q12. Which one of the following correctly describes biodegradable pollutants?

A. They never break down B. They are only radioactive isotopes C. They break down by microbes (sewage, domestic waste, etc.) D. They are identical to PAN always

> 💡 **Show answer**
>
> **Ans: C.** Microbial breakdown. **Logic:** Vs persistent non-biodegradable loads.

Q13. Arrange smog types to chemistry:

1. London — SO_2 + smoke
2. Photochemical — NOx + VOCs + sunlight → O_3/PAN

A. Only 1 B. Only 2 C. Neither D. Both 1 and 2

> 💡 **Show answer**
>
> **Ans: D.** Both. **Logic:** Classic London vs LA smog.

Q14. With reference to NCAP, which of the following statements is/are correct?

1. NCAP (2019) covers non-attainment cities.
2. It targets about 20–30% particulate reduction.
3. NCAP repealed the Water Act, 1974.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1 and 2. **Logic:** Does not repeal Water Act.

Q15. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): India's AQI does not treat CO_2 as one of its eight pollutants.

Reason (R): CO_2 is irrelevant to climate change.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** CO_2 is a climate GHG — just not an AQI eight pollutant.

Q16. Which of the following pairs is/are correctly matched?

1. WAYU — air-purifying units installed in Delhi teaching
2. CCAC (2012) — methane, black carbon, HFCs focus with UNEP

3. CCAC — G20-only club with no UNEP link

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** CCAC is not a G20-only club slogan.

Q17. How many of the following are secondary pollutants?

O_3, PAN, photochemical smog components, SO_2 emitted from a stack

A. Only one B. Only two C. Only three D. All four

> 💡 **Show answer**
>
> **Ans: C.** Only three. **Logic:** SO_2 from stack is primary; O_3/PAN/photochemical products secondary.

Q18. With reference to Noise Rules, which of the following statements is/are correct?

1. Noise Rules date to 2000.
2. Air Act was amended in 1987 to include noise.
3. Noise is measured in Dobson Units.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Noise uses decibels — Dobson is ozone.

Q19. Match List-I with List-II:

List-I	List-II
A. Water Act	1. 1986
B. Air Act	2. 1974
C. EPA	3. 1981

Code:

A. A-2, B-3, C-1 B. A-3, B-2, C-1 C. A-2, B-1, C-3 D. A-1, B-3, C-2

> 💡 **Show answer**
>
> **Ans: A.** A-2, B-3, C-1. **Logic:** 1974 / 1981 / 1986.

Q20. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Anthropogenic pollution often raises persistent non-biodegradable loads.

Reason (R): Biodegradable wastes never exist in cities.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** Sewage/domestic wastes are biodegradable examples.

Q21. Which of the following is/are correct about photochemical smog?

A. Needs only pure nitrogen with no sunlight B. Needs NOx + VOCs + sunlight; yields ozone and PAN C. Is identical to London sulphurous smog always D. Is caused only by argon

> 💡 **Show answer**
>
> **Ans: B.** NOx + VOCs + sunlight → O_3/PAN. **Logic:** Not London SO_2 smog.

Q22. With reference to CPCB vs CGWB, which of the following statements is/are correct?

1. CPCB coordinates air/water pollution standards and SPCBs.
2. CGWB handles groundwater assessment/quality reports.
3. CPCB publishes the Annual Groundwater Quality Report as its main statutory product.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** GW report is CGWB.

Q23. Arrange AQI idea from good toward severe numerically:

A. 401–500 is Good B. AQI has no numeric bands C. 0–50 is Severe D. 0–50 is Good; 401–500 is Severe

> 💡 **Show answer**
>
> **Ans: D.** 0–50 Good … 401–500 Severe. **Logic:** Standard CPCB AQI bands.

Q24. Which of the following pairs is/are NOT correctly matched?

1. Primary pollutant — O_3 formed in atmosphere from precursors
2. Secondary pollutant — SO_2 emitted from a chimney
3. Lead — primary pollutant when emitted

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** 1 and 2 are not correctly matched. **Logic:** O_3 secondary; SO_2 from chimney primary; lead primary.

Q25. With reference to BS-VI, which of the following statements is/are correct?

1. BS-VI fuel norms arrived in 2020.
2. Cleaner fuel helps cut sulphur-related pollution.
3. BS-VI repealed the Environment (Protection) Act.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** Does not repeal EPA.

Q26. How many of the following are AQI pollutants?

PM2.5, NO_2, CO, O_3, CH_4

A. Only two B. Only three C. Only four D. All five

> 💡 **Show answer**

Ans: C. Only four. **Logic:** CH_4 is not in AQI eight.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): PAN is a powerful eye irritant in photochemical smog.

Reason (R): Photochemical smog forms without sunlight.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: C. A true, R false. **A/R logic:** Sunlight is required.

Q28. Which of the following statements about non-biodegradable pollution is/are correct?

1. Plastics and many heavy metals persist.
2. Anthropogenic loads often raise persistent fractions.
3. All pollutants are biodegradable within one hour.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

Show answer

Ans: D. Only 1 and 2. **Logic:** Statement 3 false.

Q29. With reference to Air Act 1981, which of the following statements is/are correct?

1. It covers air pollution and NAAQS framework with CPCB/SPCB.
2. Noise was brought in via 1987 amendment path.
3. It was enacted after EPA 1986.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: A. Only 1 and 2. **Logic:** Air 1981 before EPA 1986.

Q30. Which one of the following is a point source example?

A. Diffuse farm runoff across a watershed B. Industrial chimney C. City-wide road dust with no single outlet D. Open agricultural field volatilisation only

Show answer

Ans: B. Industrial chimney. **Logic:** Point = chimney/drain; non-point = runoff/roads.

Mains Answer Framework

Pollution and waste governance in India (250 words): Define pollution vs degradation and carrying capacity. Media-wise controls — air (NCAP, AQI, CPCB standards), water (BOD/DO, Namami Gange, CGWB trap), soil, noise, thermal, marine, plastic. Legal stack Water 1974 → Air 1981 → EPA 1986. Waste streams — SWM 3R, BMW colour coding, hazardous waste + Basel, e-waste EPR. UP case — Kanpur Ganga, Gomti Lucknow, Ghaziabad air, Moradabad noise, Varanasi plastic. Conclude with EPR and source segregation as policy direction.

Common Traps — Don’t Fall For These

Trap	Correct fact
Lichens grow well in polluted air	They **die** in pollution
CPCB publishes groundwater report	**CGWB**
O_3 at ground level is beneficial	**Harmful** — unlike stratospheric ozone
Yellow BMW bag for sharps	**White** container
PWM Rules year = SUP ban year	**2016 vs 2022**
All marine pollution from ships	**~80% from land**
BOD measures heavy metals	**Organic pollution only**
Gomti flows through Kanpur	**Lucknow**
E-waste Rules still 2016	**2022** rules current
Thermal kills fish by heat toxicity alone	Mainly **DO reduction**
E-waste rank 1st globally	**3rd**
Complete decay in one half-life	**50% decay**
Pesticide = biodegradable	**Not**
AQI includes CO_2/CH_4	**Eight** pollutants — no
PAN = primary pollutant	**Secondary**; eye irritant
Knock-knee = mercury	**Fluoride**
Eutrophication = air pollution	**Water**
Cyclone divider = control device	**Not**
Bhopal = CO/methane	**MIC, Dec 1984**
Bioremediation cleans all Cd/Pb	**Not** readily
Green muffler = soil only	**Noise** tree belts
Most important indoor = outdoor SO_2	Often **radon**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 10 — Climate Change (जलवायु परिवर्तन)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Global Warming (भूमंडलीय तापन) | Greenhouse Effect | Greenhouse Gases (GHGs) | Carbon Footprint | Carbon Sink (कार्बन सिंक) | Carbon Sequestration (कार्बन पृथक्करण) | Climate Change Impacts | Climate Adaptation (जलवायु अनुकूलन) | Climate Mitigation (जलवायु शमन)

Sources baked in: IPCC AR6, NCERT Geography/Biology, India NDC 2022, NAPCC 2008, MoEFCC, IMD, WMO GHG Bulletin shares, UPPCS/UPSC PYQs 2018–2025 **Weight:** ★★★ High — GHG traps, adaptation vs mitigation, treaty years, India targets **Last verified:** September 2026

Current Affairs (this topic)

- Keep India NDC / Net Zero 2070 / LiFE as living climate CA; treaty years stay in the ladder below.
- Do not put **Montreal** into a climate-mitigation answer slot when Kyoto (क्योटो) / Paris are present.

💡 🎯 **Consolidated — 36 Must-Score Facts (Click to Expand)**

1. **Global warming** is the long-term rise in average surface temperature. It is a subset of broader **climate change**, which also includes rainfall, extremes, and sea-level shifts. 2. IPCC (आईपीसीसी) AR6 facts about **1.1°C** above 1850–1900 globally. India has warmed about **0.7°C** since 1901 (IMD class figure). Past-century surface rise is about **1°C ≈ 1.8°F**. 3. The natural greenhouse effect (प्राकृतिक ग्रीनहाउस प्रभाव) keeps Earth near about **+15°C**. Without it, Earth would be near about **−18°C**. 4. **Joseph Fourier (1820s)** postulated the greenhouse-effect idea. Sun sends **shortwave** (UV + visible + IR); Earth re-emits **longwave IR**. Ozone absorbs harmful UV; GHGs trap outgoing IR. 5. Enhanced greenhouse effect (मानवजनित संवर्धित प्रभाव) means excess anthropogenic GHGs trap outgoing infrared radiation. Main cause of recent warming is **CO_2** (highest concentration among long-lived GHGs). 6. Kyoto-basket majors: **CO_2** (GWP **1**), **CH_4** (GWP about **28–36**), **N_2O** (GWP about **265**), plus HFCs, PFCs, **SF_6** (and **NF_3** in later inventories). 7. Rough **WMO-style relative (सापेक्ष) shares** often used in Indian papers: **CO_2 ~64%, CH_4 ~19%, N_2O ~6%, CFCs + others ~11%**. Decreasing order: **CO_2 > CH_4 > CFCs > N_2O**. 8. **Water vapour** is Earth's most abundant natural GHG (NASA: ~half of the natural greenhouse effect) but mainly a **feedback**, not the primary policy driver. 9. **Not GHGs:** **argon, nitrogen (N_2), oxygen, hydrogen, propane**. **SO_2 / NOx** do **not** contribute to warming **directly** (indirect / other roles). 10. **Direct** GHGs include CO_2, CH_4, N_2O, CFCs, SF_6, NF_3. **Indirect** agents include **NOx, CO, NMVOCs, SO_2**. 11. India is about the **third**-largest total emitter (**China** first); **Bhutan** is often cited as **carbon-negative**. 12. A **carbon footprint** totals GHGs in tCO_2e across Scope **1** (direct), **2** (energy), and **3** (supply chain). 13. A **carbon sink** absorbs more CO_2 than it releases (forests, oceans, soils, mangroves / **blue carbon** (नीला कार्बन)). **Sequestration** is the storage process. 14. **Carbon fertilization** = more plant growth from higher atmospheric CO_2 (not the same as ocean acidification (महासागरीय अम्लीकरण) or warming). 15. Pre-industrial CO_2 ~**278–280 ppm (~0.03%)**; mid-20th century ~**315 ppm**; now **415–420+ ppm**. 16. **Methane ("marsh gas")** sources: paddies, cattle, coal mines, landfills, termites; natural wetlands dominate natural CH_4 (~**75–76%** of natural). **Rice** is a key anthropogenic source of **both CH_4 and N_2O**. 17. **Methane hydrates** under Arctic tundra (टुंड्रा) / seafloor can release CH_4 if warmed; atmospheric CH_4 oxidises to CO_2 in roughly a decade or two. 18. Impacts include heat waves, glacier melt,

sea-level rise, coral bleaching (प्रवाल विरंजन), disease spread, stronger storms. **Arctic / Greenland ice** is among the most fragile first-hit systems. 19. Rough IPCC-style risk: beyond about **+2°C** pre-industrial → widespread coral mortality; beyond about **+3°C** → terrestrial biosphere tends toward a **net carbon source**. 20. **Adaptation** = coping with impacts. **Mitigation** = cutting emissions and enhancing sinks. 21. **NAPCC (2008)** has **eight** national missions — **not** nuclear power as a listed mission. 22. India's updated **NDC** path includes about **45%** emission-intensity cut and about **50%** non-fossil electricity capacity by **2030** (not "60% energy from renewables" as a false paraphrase). 23. India's **Net Zero** pledge is **2070**. Net zero means emissions balanced by removals — **not** literally zero every emission. 24. **LiFE Mission** launched **June 2022**; idea at **COP26 (2021)**, not COP25. 25. Climate treaty chain: **UNFCCC (यूएनएफसीसीसी) 1992 → Kyoto 1997 (force 2005) → Paris 2015**. **Montreal 1987** = ozone, not climate. 26. **CDM** (Kyoto Art. 12) lets Annex-I parties fund projects in developing countries for **CERs** (1 CER ≈ 1 tCO_2). **Carbon credit** (कार्बन क्रेडिट) concept originates from Kyoto. 27. **Green Climate Fund** was set up at **Cancun COP-16 (2010)**, not Durban. **Earth Hour** (अर्थ आवर) = **WWF** (डब्ल्यूडब्ल्यूएफ), last Saturday of March (lights off ~8:30–9:30 pm). 28. **GHG Protocol** = **WRI + WBCSD** accounting tool. **Blue carbon** = ocean/coastal carbon (mangroves, seagrass, salt (लवणाध्यक्ष) marsh). 29. UP heat vulnerability often cites (साइट्स) Lucknow (लखनऊ), Agra, Prayagraj (प्रयागराज), and NCR. 30. UP drought (सूखा) fact is **Bundelkhand's seven districts**. Flood facts include Ballia, Ghazipur, and Varanasi (वाराणसी) belts. 31. Emissions hotspot language for UP–MP often points to the **Singrauli** coal belt. 32. Terai (तराई) forests at Dudhwa (दुधवा) and Pilibhit are UP **carbon-sink** examples. 33. **Milankovitch** orbital theory (eccentricity, obliquity, precession) — **solar irradiance** is **not** part of that astronomical set. 34. **Ice cores** are the classic **cryogenic** climate archive (Greenland / Antarctica). 35. **Solar flares** are **not** a primary cause of recent climate change in standard NASA/IPCC framing. 36. Farm practices that help soil carbon storage include **contour bunding, relay cropping, and zero tillage**.

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Global Warming	**Climate Change**	Long-term monotonic increase in Earth's average surface and tropospheric temperature driven by greenhouse gas buildup vs broad spectrum of shifting weather patterns, rising sea levels, altering precipitation, and intensifying extreme climate anomalies	भूमंडलीय तापन (ग्लोबल वार्मिंग) / जलवायु परिवर्तन
Climate Mitigation	**Climate Adaptation**	Interventions designed to reduce greenhouse gas emissions at the source or enhance carbon sinks (renewable energy, afforestation, carbon capture) vs adjustments in natural or human systems to moderate harm, exploit beneficial opportunities, and cope with actual/expected climate impacts (flood barriers, drought-tolerant seeds)	जलवायु शमन (उत्सर्जन रोकथाम) / जलवायु अनुकूलन (सहनशीलता वृद्धि)
Natural Greenhouse Effect	**Enhanced Greenhouse Effect**	Essential natural phenomenon keeping Earth's surface habitable at an average ~15°C (without it Earth would freeze at -18°C) vs excessive anthropogenic heat-trapping caused by fossil fuel combustion and industrial emissions driving global warming	प्राकृतिक ग्रीनहाउस प्रभाव / मानवजनित संवर्धित प्रभाव
Global Warming Potential (GWP)	**Radiative Forcing**	Metric measuring the cumulative heat-trapping ability of 1 kg of a specific GHG over 100 years relative to CO_2 (=1.0) vs the net change in Earth's radiative energy balance (measured in Watts/m^2) caused by changes in atmospheric gas concentrations	वैश्विक तापन क्षमता (GWP) / विकिरण दबाव
Carbon Sink	**Carbon Sequestration**	Natural or artificial reservoir that absorbs and stores more carbon than it releases into the atmosphere (oceans, boreal forests, soil) vs the actual physical, biological, or chemical process of capturing and securing carbon dioxide in long-term storage	कार्बन सिंक (अवशोषक भंडार) / कार्बन पृथक्करण (भंडारण प्रक्रिया)
Carbon Credit	**Carbon Offset**	Formal tradable permit or certificate representing the reduction or removal of 1 metric tonne of CO_2 equivalent under a compliance cap-and-trade system vs voluntary purchase of emission reduction units to neutralize an individual's or company's carbon footprint	कार्बन क्रेडिट (1 टन CO_2e प्रमाण-पत्र) / कार्बन ऑफसेट
Kyoto Protocol (1997)	**Paris Agreement (2015)**	Top-down legally binding emission reduction targets imposed exclusively on developed nations (Annex-I) with Common But Differentiated Responsibilities vs bottom-up universal framework requiring all countries to submit voluntary Nationally Determined

Contributions (NDCs) aiming to cap warming well below 2.0°C/1.5°C | क्योटो प्रोटोकॉल (बाध्यकारी/विकसित देश) / पेरिस समझौता (सार्वभौमिक NDCs) |

💡 📋 **Must-score facts — warming, GHGs, sinks (Click to Expand)**

Frame | Item | Lock | |------|------| | Warming vs climate change | Temp rise subset / broader incl. rainfall & extremes | | Natural GHG effect | ~+15°C with; ~−18°C without | | Fourier | 1820s greenhouse-effect idea | | Main recent driver | Anthropogenic **CO_2** | | Kyoto basket | CO_2, CH_4, N_2O, HFCs, PFCs, SF_6 (+NF_3 later) | | Relative shares (WMO-style) | CO_2~64% > CH_4~19% > CFCs+~11% > N_2O~6% | | Not GHGs | Ar, N_2, O_2, H_2, propane | | India rank / Bhutan | ~3rd absolute (निरपेक्ष) / often carbon-negative | | Sink / sequestration | Net CO_2 absorb / storage process | | Blue carbon | Ocean–coastal sinks (mangroves etc.) |

10.1 Global Warming

Identity: Temperature rise since industrial era | IPCC 1.1°C | Anthropogenic dominant

- **Global warming** is the long-term increase in Earth's average near-surface temperature.
- **Climate change** is broader and includes shifts in rainfall, monsoon timing, cyclones, and extremes.
- Since the industrial era, atmospheric **CO_2** has risen from about **280 ppm to over 420 ppm**.
- **IPCC AR6** estimates (प्राकलन समिति) warming of about **1.1°C** above the **1850–1900** pre-industrial baseline.
- **Human activities** — fossil fuels, deforestation, industry, agriculture — are the **dominant cause** of recent warming.
- **India** has warmed about **0.7°C** since 1901 according to **IMD** data.
- **Oceans absorb about 90%** of excess heat, causing thermal expansion and marine stress.
- **Arctic warming** is about **2–3 times faster** than the global average.
- A single cold day is **weather**. Multi-decade warming trend is **climate**.

> **Logic:** "Warming is entirely natural" is **false**. IPCC attributes recent trend mainly to human activity.

Earth Radiation Budget & Greenhouse Gas Radiative Forcing

10.2 Greenhouse Effect

Identity: Fourier 1820s | Shortwave in · longwave out | Natural vs enhanced | ≠ ozone depletion (ओजोन क्षरण)

- The **greenhouse effect** is the process by which atmospheric gases absorb and re-emit **outgoing infrared radiation**, warming the surface and lower atmosphere.
- **Joseph Fourier (1820s)** postulated the greenhouse-effect concept.
- The Sun emits **high-energy shortwave** radiation (UV, visible, near-IR). Harmful **UV** is largely absorbed by the **ozone layer**; visible and IR reach the surface as heat.
- Earth, being cooler, re-emits energy as **longwave IR**. Greenhouse gases absorb much of that longwave and re-radiate it in all directions.
- The **natural greenhouse effect** keeps Earth's average temperature near **+15°C**.
- Without it, Earth's average would be about **−18°C** and largely uninhabitable.
- The **enhanced greenhouse effect** comes from human-caused GHG buildup faster than natural sinks can absorb — this drives **global warming**.
- **Water vapour** is the most abundant natural GHG (about **half** of the natural greenhouse effect in NASA-style framing) but acts mainly as a **feedback**, not a primary emitted driver like fossil CO_2.
- The greenhouse effect is **not ozone depletion**. Ozone blocks UV; GHGs trap IR.

Logic: Natural greenhouse effect is **essential**. The problem is the **enhanced** anthropogenic effect. Fourier coined the concept — do not invent modern IPCC authors for that stem.

10.3 Greenhouse Gases (GHGs)

Identity: Kyoto basket | WMO share order | Argon/H_2/propane not GHG | Direct vs indirect

- **Greenhouse gases** absorb and emit infrared radiation in the atmosphere.
- **CO_2** is the main anthropogenic driver by concentration and total forcing. IPCC-style forcing share is often ~**76%**; older WMO-bulletin style relative shares used in many Indian papers are **CO_2 ~64%, CH_4 ~19%, N_2O ~6%, CFCs + others ~11%**.
- Decreasing contribution order commonly tested: **CO_2 > CH_4 > CFCs > N_2O**.
- **CO_2** has **GWP = 1**. Pre-industrial ~**278–280 ppm (~0.03%)**; ~**315 ppm** around 1950s; now **415–420+ ppm**.
- **Methane (CH_4)** — also called **marsh gas** — has **GWP around 28–36** (100-yr). Sources: paddies, cattle, coal mines, landfills, termites; natural **wetlands** dominate natural CH_4 (~**75–76%**).
- **Rice** is a major anthropogenic source of **both methane and nitrous oxide** (anaerobic soils + nitrogen fertiliser).
- **Nitrous oxide (N_2O)** has **GWP around 265** and links strongly to synthetic fertilisers.
- The **Kyoto Protocol** covers **CO_2, CH_4, N_2O, HFCs, PFCs, and SF_6** (later inventories also track **NF_3**).
- **Not greenhouse gases: argon, N_2, O_2, hydrogen, propane**. **Nitrogen oxide (NO_x)** is **not** listed among the classic naturally occurring GHGs in UPPCS-style keys.
- **SO_2 and oxides of nitrogen** do **not** contribute to global warming **directly** (acid rain (अम्ल वर्षा) / air pollution / indirect chemistry roles).
- **IPCC direct GHGs** include CO_2, CH_4, N_2O, halocarbons, SF_6, NF_3. **Indirect** agents include **NO_x, CO, NMVOCs, SO_2**.
- **Carbon fertilization** means increased plant growth from higher atmospheric CO_2 — distinct from warming or ocean acidification.
- Oceans remove much atmospheric CO_2 via **phytoplankton photosynthesis** (biological pump) — destroying phytoplankton harms the ocean carbon sink and marine food webs, not seawater density.
- **Black carbon (soot)** is a climate forcer but **not a GHG**.
- **India is the 3rd largest total GHG emitter** (**China** #1); **low per capita**. **Bhutan** is often described as **carbon-negative**.

GHG / group	Rough relative share (WMO-style stems)	GWP (100-yr)
CO_2	~64%	1
CH_4	~19%	28–36
N_2O	~6%	265
CFCs + others	~11%	Very high (F-gases)

Logic: Highest **GWP per molecule** ≠ largest total warming. **CO_2 dominates by volume**. Water vapour can win "most abundant natural GHG / largest share of natural effect" stems; CO_2 wins "main component / highest contribution to anthropogenic warming" stems.

10.4 Carbon Footprint

Identity: tCO_2e | Scope 1/2/3 | India low per capita, high total

- A **carbon footprint** is the total GHG emissions caused by a person, product, organisation, or country.
- It is measured in **tonnes of CO_2 equivalent (tCO_2e)** using GWP conversion.
- **Scope 1** covers **direct** emissions from owned sources such as factory (फैक्टरी) boilers.
- **Scope 2** covers **indirect** emissions from purchased electricity or heat.
- **Scope 3** covers other indirect emissions such as supply chains, travel, and product disposal.
- **India's per capita footprint** is about **2 tCO_2e**, below the global average of about **4.7**.
- The **energy/electricity sector** is India's largest emissions source.
- **LiFE Mission (2022)** promotes lifestyle changes to reduce individual footprints.

> **Logic:** India has low **per capita** footprint but high **total** emissions due to population (जनसंख्या) size.

10.5 Carbon Sink

Identity: Net CO_2 absorption | Forests, oceans, soil | Opposite = source

- A **carbon sink** absorbs more CO_2 than it releases over time.
- **Forests** store carbon in biomass and soil through photosynthesis.
- **Oceans** absorb about **25% of anthropogenic CO_2** through dissolution (विघटन) and the biological pump.
- **Soils and wetlands** store organic carbon. Degraded land can become a **carbon source**.
- **Mangroves** are **blue carbon** sinks with very high sequestration rates per hectare.
- **Deforestation** converts a sink into a **source** by releasing stored carbon and removing uptake capacity.
- India's NDC targets an additional **2.5–3 GtCO_2e forest carbon sink** by 2030.

> **Logic:** A coal plant is a **source**. A growing forest is a **sink**.

10.6 Carbon Sequestration

Identity: Long-term carbon storage | Biological + geological | Farm practices | CCS (सुरक्षा कैबिनेट समिति) sites

- **Carbon sequestration** is the process of capturing and storing atmospheric carbon for long periods.
- **Biological sequestration** uses afforestation, soil carbon build-up, and mangrove restoration.
- Farm practices that help soil carbon storage include **contour bunding**, **relay cropping**, and **zero tillage**.
- **Geological sequestration / CCS** can use abandoned coal seams, depleted oil and gas reservoirs, and deep saline (लवणीय) formations; other discussed methods include crushed basalt on farmland and ocean alkalinity enhancement.
- **Afforestation and reforestation** are India's main practical sequestration tools under **Green India Mission**.
- **Blue carbon** is carbon captured by oceans and coastal ecosystems (mangroves, seagrasses, salt marshes).
- **CAMPA** and the **Green Credit Programme (2023)** finance ecosystem (पारिस्थितिकी तंत्र) restoration in India.
- Stored carbon can be **re-released** by fire or deforestation, creating permanence risk.

> **Logic:** Sequestration is the **process**. Sink is the **reservoir**. Blue carbon ≠ forest biomass carbon label.

10.7 Climate Change Impacts

Identity: Heat, glaciers, monsoon extremes, agriculture, coral bleaching | Arctic first

- **Heat waves** are increasing in frequency across India, especially in **northwest India and UP**.

- **Himalayan glaciers** such as Gangotri (गंगोत्री) are retreating, affecting lean-season Ganga (गंगा) flow.
- **Monsoon** shows more **extreme rainfall bursts** and longer dry spells rather than uniform change.
- **Bundelkhand** faces recurring **drought** and water scarcity.
- **Eastern UP** along the Ganga faces **flood risk** from intense monsoon events.
- **Wheat** in UP plains is sensitive to **March heat spikes** during the rabi (रबी) season.
- **Sea-level rise** threatens coastal India and increases delta (डेल्टा) salinity pressures.
- **Coral bleaching** occurs when sea surface temperature rises about **1–2°C above normal** for sustained periods.
- Impacts of warming commonly listed together: rise in sea level, melting glaciers, spread of diseases, coral bleaching, stronger cyclones/storms/hurricanes.
- The **Arctic and Greenland ice sheet** are among the most fragile systems hit first by warming.
- Rough IPCC risk framing: above about **+2°C** pre-industrial → widespread coral mortality; above about **+3°C** → terrestrial biosphere tends toward a **net carbon source** (wetlands do not all permanently disappear; cereal farming does not become impossible everywhere).
- Climate stress on agriculture and food security can raise **social tension** through resource pressure and food insecurity.
- **Milankovitch** astronomical climate theory covers **eccentricity, obliquity, and precession** — **not** solar irradiance as a listed element of that set.
- **Ice cores** provide the classic **cryogenic** record of past climate and atmospheric composition.
- Primary causes of recent climate change: fossil-fuel burning, oil-driven automobile growth, deforestation — **not** increasing solar flares as a major driver.

> **Logic:** — Andaman (अंडमान) coral bleaching linked to **+2°C Indian Ocean warming**. Do not mark "all wetlands vanish" or "no cereals anywhere" as automatic +3°C outcomes.

PYQ — UPPCS Prelims 2024, Q19

Coral bleaching in Andaman & Nicobar (निकोबार) caused by:

A. Coastal industry

B. +2°C Indian Ocean warming

C. Heavy pollution

D. −4°C cooling

💡 **Show answer**

Ans: B

10.8 Climate Adaptation

Identity: Build resilience | NAPCC 2008 | 8 missions | ≠ mitigation

- **Climate adaptation** means adjusting to actual or expected climate impacts to reduce harm.
- **Mitigation** reduces emissions. **Adaptation** copes with impacts already occurring or already committed in the climate system.
- **NAPCC (National Action Plan on Climate Change, 2008)** is India's master climate framework with **8 national missions**.
- Adaptation-focused missions include **National Water Mission**, **Sustainable Agriculture Mission**, and **Himalayan Ecosystem Mission**.
- **National Water Mission** goals include a public water database and climate-impact assessment, citizen/state conservation action, focus on over-exploited areas, **~20% water-use efficiency gain**, and basin-level IWRM.

Urban (नगरीय) wastewater recycling fits; nationwide Himalaya (हिमालय)–peninsula river linking and free borewell reimbursement do **not**.

- **Early warning systems** from IMD and CWC (केंद्रीय जल आयोग) for heat, floods, and cyclones are adaptation tools.
- **Climate-resilient crops**, micro-irrigation, and **PM Fasal Bima Yojana** support farm adaptation.
- **Bundelkhand water harvesting** and **Lucknow heat action plans** are UP adaptation examples.

#	NAPCC Mission
1	National Solar Mission
2	Enhanced Energy Efficiency
3	Sustainable Habitat (वास स्थान)
4	National Water Mission
5	Sustaining Himalayan Ecosystem
6	Green India
7	Sustainable Agriculture
8	Strategic Knowledge for Climate Change

- **Atomic / nuclear power is not** one of the eight NAPCC missions.
- Climate-smart village approaches linked to **CCAFS / CGIAR** (HQ France; **ICRISAT** is a CGIAR centre) sit in the adaptation–agriculture research lane.

> **Logic:** Building sea walls = **adaptation**. Closing coal plants = **mitigation**. Nuclear ≠ listed NAPCC mission.

10.9 Climate Mitigation

Identity: Cut emissions + sinks | UNFCCC/Kyoto/Paris | CDM · GCF | India NDC | LiFE

Treaty ladder — climate vs ozone (read the stem)

Ladder	Instruments	What it controls	Classic trap
Climate	UNFCCC (1992) (जलवायु परिवर्तन संधि) → **Kyoto** (क्योटो) (1997) → **Paris** (2015)	GHG mitigation / NDCs	Montreal is **not** climate mitigation
Ozone	Vienna (1985) → **Montreal** (1987) → **Kigali** (HFCs, 2016)	ODS / HFC phase-down	Kyoto is **not** ozone

- **UNFCCC** (यूएनएफसीसीसी) is the parent climate convention. **Kyoto** gave binding Annex-I cuts and CDM. **Paris** uses voluntary **NDCs** for all parties.
- **CBDR** (common but differentiated responsibilities) is the climate equity line — do not paste it onto Montreal automatically.
- **Montreal** protects the ozone layer; **Kigali** amends Montreal for HFCs (also climate-relevant, but still ozone-family architecture).
- **Climate mitigation** means reducing GHG sources or enhancing sinks to limit future warming.
- **UNFCCC (1992, Rio)** entered into force **21 March 1994**; it frames greenhouse-gas mitigation.
- **Kyoto Protocol (1997; force 16 February 2005)** was the first binding emission-cut treaty for **developed (Annex I) countries**.
- Kyoto's **Clean Development Mechanism (CDM)** lets Annex-I parties fund projects in developing countries and earn **CERs** (1 CER ≈ **1 tCO_2**). Carbon credits are **market-traded**, not fixed by UNEP (संयुक्त राष्ट्र पर्यावरण

कार्यक्रम) price.

- **Paris Agreement (पेरिस समझौता) (2015)** (पेरिस समझौता) uses voluntary **NDCs** to limit warming well below **2°C** and pursue **1.5°C**. The "1000 billion/year" paraphrase is wrong — finance language was about mobilising **100 billion** class climate finance.
- **Green Climate Fund (GCF)** (जीसीएफ) was established at **Cancun COP-16 (2010)** under UNFCCC to help developing countries with adaptation and mitigation — **not** founded at Durban and **not** under UNEP/OECD/ADB/World Bank as its founding aegis.
- **India's updated NDC (2022)** targets **45% lower emissions intensity of GDP by 2030** (vs 2005).
- India targets **50% cumulative electric installed capacity from non-fossil sources by 2030** (not a "60% of total energy from renewables" paraphrase).
- India pledged **Net Zero by 2070** at **COP-26 Glasgow (2021)**. **Net zero** means emissions are **balanced by absorption/removal** — not that every emission falls to absolute zero.
- **LiFE (Lifestyle for Environment (जीवनशैली पर्यावरण))** was launched in **June 2022**. The idea was proposed at **COP-26**, not COP-25.
- **Earth Hour** is a **WWF** awareness event (usually last Saturday of March, lights off ~1 hour) — not UNEP/UNESCO.
- **Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) (1987)** (मॉन्ट्रियल प्रोटोकॉल) controls **ozone-depleting substances**, not climate change.
- **GHG Protocol** (WRI + WBCSD) is an accounting tool for governments and business — not a UN treaty.

> **Logic:** Climate mitigation treaty stem → **Kyoto**. Montreal = ozone. GCF = Cancun 2010. Net zero ≠ absolute zero emissions.

PYQ — UPPCS Prelims 2024, Q35

Which agreement relates to climate change mitigation?

A. Montreal Protocol

B. Ramsar (रामसर)

C. Kyoto Protocol

D. CITES

💡 **Show answer**

Ans: C

PYQ — UPPCS Prelims 2025, Q139

LiFE: (1) Launched June 2022. (2) Idea proposed at COP-25 Madrid.

A. Only 2

B. Neither

C. Both

D. Only 1

💡 **Show answer**

Ans: D — Idea at **COP-26 Glasgow**.

Treaty	Year	Focus
UNFCCC	1992	Framework

Treaty	Year	Focus
Kyoto Protocol	1997	Binding cuts (developed countries)
Paris Agreement	2015	NDC pledges
Montreal Protocol	1987	**Ozone** (not climate)

Complete PYQ Bank (Topic 10)

Q1. UPPCS Prelims 2025, Q139

With reference to the **LiFE (Lifestyle for Environment)** Mission, consider the following statements:

1. LiFE was launched in **June 2022**.
2. The idea was first proposed at **COP-25 in Madrid**.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

💡 **Show answer**

Logic: LiFE launch date vs COP venue where PM Modi first floated the concept. **Ans: D.** LiFE launched **June 2022**; the idea was announced at **COP-26 Glasgow (2021)**, not COP-25 Madrid. **Trap:** COP-25 Madrid is a plausible-sounding UN climate venue — the correct pitch was **Glasgow 2021**.

Q2. UPPCS Prelims 2024, Q35

Which of the following international agreements relates to **climate change mitigation**?

A. Montreal Protocol B. Ramsar Convention C. Kyoto Protocol D. CITES

💡 **Show answer**

Logic: Treaty-family trap — Montreal controls ozone; Kyoto/Paris address greenhouse-gas mitigation. **Ans: C.** The **Kyoto Protocol (1997)** imposed binding emission cuts on developed (Annex I) parties. **Trap:** **Montreal Protocol** is the most picked wrong answer — it phases out **ODS**, not CO_2 mitigation.

Q3. UPPCS Prelims 2024, Q19

Coral bleaching in the **Andaman & Nicobar Islands** was primarily caused by—

A. Increased coastal industrial effluents alone B. About **+2°C warming** of the Indian Ocean C. Heavy marine pollution alone D. A **−4°C** drop in sea-surface temperature

💡 **Show answer**

Logic: Climate-impact stem — thermal stress on reef symbionts drives bleaching events. **Ans: B.** **+2°C Indian Ocean warming** triggered bleaching; pollution adds stress but warming is primary. **Trap:** Cooling (−4°C) would not expel zooxanthellae — bleaching follows **heat stress**.

Q4. UPPCS Prelims 2023, Q109

What was India's rank in the world in **greenhouse gas emissions** in 2022 (as framed in the paper)?

A. 9th B. 10th C. 11th D. 12th

💡 **Show answer**

Logic: Rank framing trap — paper used a 9th–12th band; absolute total-emission rank is ~3rd. **Ans: B.** The keyed answer is **10th** (performance-index style framing in the paper). **Trap:** India is ~**3rd in absolute total emissions** — do not confuse with the paper's **10th** option key.

Q5. UPPCS Prelims 2022, Q103

Which of the following is **NOT** a greenhouse gas?

A. Argon B. Methane C. Carbon dioxide D. Nitrous oxide

💡 **Show answer**

Logic: Non-GHG noble gas trap among Kyoto-basket gases. **Ans: A.** **Argon** is an inert atmospheric component and **not** a greenhouse gas. **Trap:** **Water vapour** is a natural GHG but absent here — **argon** is the keyed non-GHG distractor set.

Q6. UPPCS Prelims 2018 pattern

Global warming is most directly linked to an increased frequency and intensity of—

A. Total solar eclipses B. **Extreme weather events** such as heat waves and intense rainfall C. Milankovitch ice-age cycles alone D. Stratospheric ozone recovery

💡 **Show answer**

Logic: Climate-impact consequence — enhanced greenhouse effect raises extreme-event probability. **Ans: B.** Warming intensifies **heat waves, droughts, floods, and cyclone rainfall** — core IPCC impact framing. **Trap:** Milankovitch cycles operate on **millennial** scales — they do not explain recent decadal extreme-weather rise.

Q7. UPSC pattern

Under the **Kyoto Protocol**, quantified emission-reduction commitments applied primarily to—

A. All UN member states equally B. **Annex I developed countries** C. Only Least Developed Countries D. Only island nations

💡 **Show answer**

Logic: CBDR architecture — Kyoto bound industrialised Annex I parties, not India or other developing countries. **Ans: B.** Kyoto's binding cuts targeted **Annex I (developed) countries**; developing nations had no binding targets. **Trap:** India is a major emitter today but was **not** under Kyoto binding cuts — Paris uses voluntary NDCs instead.

Extra Drill — Ghatnachakra-pattern (Q31–Q40)

Logic (set): Mechanism, non-GHG traps, methane/rice, impacts thresholds, treaties/finance, and net-zero wording.

Q31. The greenhouse-effect concept was postulated by:

A. C.C. Park B. Joseph Fourier C. L. Zobler D. Arrhenius only in 1990s

💡 **Show answer**

Ans: B. Joseph Fourier in the **1820s**.

Q32. Which is **not** responsible for global warming?

A. Methane B. Water vapour C. Argon D. Carbon dioxide

> 💡 **Show answer**
>
> **Ans: C.** Argon is **not** a greenhouse gas.

Q33. Decreasing order of contribution commonly tested:

A. CO_2 > CH_4 > CFCs > N_2O B. CH_4 > CO_2 > CFCs > N_2O C. CO_2 > N_2O > CH_4 > CFCs D. CFCs > CO_2 > CH_4 > N_2O

> 💡 **Show answer**
>
> **Ans: A.** WMO-style share order used in many Indian papers.

Q34. Among the following crops, the most important anthropogenic source of **both** methane and nitrous oxide is:

A. Cotton B. Rice C. Sugarcane D. Wheat

> 💡 **Show answer**
>
> **Ans: B.** Paddy anaerobic CH_4 + fertiliser N_2O.

Q35. Carbon fertilization best describes:

A. Ocean acidification from CO_2 B. Increased plant growth from higher atmospheric CO_2 C. Global warming from CO_2 alone D. Adaptation of all species to climate change

> 💡 **Show answer**
>
> **Ans: B.** More photosynthesis / greening from elevated CO_2.

Q36. Without the greenhouse effect, Earth's average surface temperature would be about:

A. 0°C B. −18°C C. 5°C D. −20°C

> 💡 **Show answer**
>
> **Ans: B.** Roughly **−18°C**.

Q37. If warming exceeds about **3°C** above pre-industrial levels, which impacts are plausible?

1. Terrestrial biosphere tends toward a net carbon source
2. Widespread coral mortality
3. All global wetlands permanently disappear
4. Cereals cannot be cultivated anywhere

A. 1 only B. 1 and 2 only C. 2, 3 and 4 only D. All four

> 💡 **Show answer**
>
> **Logic:** +2°C already implies widespread coral mortality; +3°C adds net carbon-source risk. Absolute "all wetlands" / "no cereals anywhere" are overstated. **Ans: B**

Q38. Green Climate Fund was established at:

A. Durban COP B. Cancun COP-16 (2010) C. Paris COP-21 only as first idea D. Stockholm 1972

> 💡 **Show answer**
>
> **Ans: B.** Cancun **2010** under UNFCCC.

Q39. With reference to net-zero for 2050/2070 pathways, which is correct?

A. A country must bring every emission to absolute zero. B. Residual emissions are compensated by absorption and removal. C. Both A and B D. Neither

> 💡 **Show answer**
>
> **Ans: B.** Balance of emissions and removals — Bhutan-style negative emissions possible if removals exceed emissions.

Q40. Assertion (A): Climate change can increase social tension in India. Reason (R): Extreme weather frequency/intensity has serious repercussions on food security.

A. Both true; R explains A B. Both true; R does not explain A C. A true; R false D. A false; R true

> 💡 **Show answer**
>
> **A/R logic:** A tests socio-political stress; R tests the food-security pathway linking climate extremes to social tension. **Ans: A**

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill (not a PYQ dump) | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced A–D | **Ans first, then Logic**

Q1. With reference to global warming and climate change, which of the following statements is/are correct?

1. Global warming is the long-term fall in average surface temperature.
2. Climate change is broader and also includes rainfall and extremes.
3. The two terms are identical in every Prelims key.

A. Only 2 B. Only 1 C. 2 and 3 D. 1 and 2

> 💡 **Show answer**
>
> **Ans: A.** Only 2 is correct. **Logic:** Only statement 2 remains true among the numbered claims. Statement 1 is false. Statement 2 is true. Statement 3 is false.

Q2. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Without the natural greenhouse effect Earth would average about −18°C.

Reason (R): With the natural greenhouse effect the surface averages about +15°C.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; R states the with-GHG temperature, not why without-GHG is −18°C — related pair, not explanation of A's counterfactual. **A/R logic:** Both are standard temperature facts; R does not explain the −18°C claim.

Q3. Match List-I with List-II:

List-I	List-II
A. CO_2 GWP	1. ~28–36

List-I	List-II
B. CH_4 GWP	2. 1
C. N_2O GWP	3. ~265

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** CO_2=1; CH_4~28–36; N_2O~265.

Q4. With reference to relative GHG shares (WMO-style teaching), which of the following statements is/are correct?

1. Order often tested: CO_2 > CH_4 > CFCs+others > N_2O.
2. Rough shares reverse CO_2 and CH_4 (~19% CO_2 and ~64% CH_4).
3. Argon is the largest Kyoto-basket GHG by share.

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

💡 **Show answer**

Ans: B. Only 1 is correct. **Logic:** Statement 1 is the usual relative-share order. Statement 2 invents reversed CO_2/CH_4 shares. Statement 3 is false — argon is not a Kyoto-basket GHG.

Q5. Which of the following is/are NOT greenhouse gases?

1. Argon
2. Nitrogen (N_2)
3. Oxygen
4. Carbon dioxide

A. Only 3 B. 1 and 2 C. 1, 2 and 3 D. Only 4

💡 **Show answer**

Ans: C. Argon, nitrogen and oxygen are not greenhouse gases; CO_2 is. **Logic:** Major GHGs include CO_2, CH_4, N_2O, ozone and water vapour. N_2, O_2 and argon dominate dry air but are not greenhouse gases.

Q6. Arrange IPCC AR6 warming idea:

A. About 5°C cooling since 1900 B. About 1.1°C above 1850–1900 globally C. No human influence in AR6 D. Exact 0.00°C change

💡 **Show answer**

Ans: B. About 1.1°C above 1850–1900. **Logic:** AR6 global warming fact.

Q7. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Water vapour is Earth's most abundant natural GHG.

Reason (R): Water vapour is mainly treated as a feedback, not the primary policy lever like CO_2.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; R does not explain why vapour is most abundant — it states policy/feedback role. **A/R logic:** Abundance vs feedback/policy distinction.

Q8. With reference to enhanced greenhouse effect, which of the following statements is/are correct?

1. Excess anthropogenic GHGs trap outgoing infrared.
2. Main cause of recent warming is CO_2.
3. Joseph Fourier (1820s) framed the greenhouse-effect idea.

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: C. All three. **Logic:** Enhanced GHG + CO_2 driver + Fourier.

Q9. Match List-I with List-II:

List-I	List-II
A. Carbon sink	1. Storage process
B. Sequestration	2. Net CO_2 absorption
C. Blue carbon	3. Ocean/coastal sinks

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

💡 **Show answer**

Ans: D. A-2, B-1, C-3. **Logic:** Sink absorbs; sequestration stores; blue = coastal/ocean.

Q10. Which of the following pairs is/are correctly matched?

1. India — about third-largest absolute emitter
2. China — largest absolute emitter
3. Bhutan — often cited as carbon-negative

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** China #1; India ~3rd; Bhutan carbon-negative teaching.

Q11. With reference to carbon footprint, which of the following statements is/are correct?

1. It totals GHGs in tCO_2e.
2. Scope 1 is direct; Scope 2 purchased energy; Scope 3 value chain.
3. Ecological footprint uses the same unit tCO_2e only.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Ecological footprint uses gha — not only tCO_2e.

Q12. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Montreal Protocol addresses ozone-depleting substances.

Reason (R): Kyoto/Paris instruments are climate/GHG instruments — not ozone ODS phase-out treaties.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: A. Both true; R contrasts climate treaties but does not explain Montreal's ODS purpose by itself as causal explanation — still both true, R not explanation of A. **A/R logic:** Issue-tag separation; R is true contrast, not the definitional explanation of Montreal.

Q13. Which of the following are direct GHGs in teaching lists?

1. CO_2
2. CH_4
3. N_2O
4. SF_6

A. 1 and 2 only B. 1, 2 and 3 only C. 1, 2, 3 and 4 D. 2, 3 and 4 only

Show answer

Ans: C. All four. **Logic:** Direct GHGs include F-gases too.

Q14. With reference to India warming, which of the following statements is/are correct?

1. IMD-class figure often cites about 0.7°C warming since 1901.
2. AR6 global figure is about 1.1°C above 1850–1900.
3. India has cooled by 3°C since 1901 in all datasets.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

Show answer

Ans: D. Only 1 and 2. **Logic:** Statement 3 false.

Q15. Arrange radiation idea:

1. Sun sends shortwave insolation
2. Earth re-emits longwave IR
3. GHGs trap outgoing IR

A. 1, 2 and 3 B. 1 and 2 only C. 2 and 3 only D. Only 1

Show answer

Ans: A. All three. **Logic:** Shortwave in → longwave out → GHG trap.

Q16. Which of the following pairs is/are NOT correctly matched?

1. Carbon fertilization — more plant growth from higher CO_2
2. Ocean acidification — identical to carbon fertilization
3. Blue carbon — mangroves/seagrass/coastal sinks

A. 1 and 3 only B. 2 only C. 2 and 3 only D. 1, 2 and 3

Show answer

Ans: B. Only pair 2 is not correctly matched. **Logic:** Acidification ≠ fertilization.

Q17. Match Kyoto-basket teaching:

List-I	List-II
A. HFCs	1. Later inventories also
B. PFCs	2. Kyoto F-gas
C. NF_3	3. Kyoto F-gas

Code:

A. A-3, B-2, C-1 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

💡 **Show answer**

Ans: C. A-2, B-3, C-1. **Logic:** HFCs/PFCs Kyoto basket; NF_3 later inventories.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): SO_2 and NOx do not contribute to warming as primary direct GHGs in the same way as CO_2.

Reason (R): They may have indirect/other atmospheric roles but are not classic Kyoto-basket GHGs.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the classification. **A/R logic:** Indirect agents vs direct Kyoto GHGs.

Q19. With reference to carbon sink examples, which of the following statements is/are correct?

1. Forests, oceans, soils, mangroves can act as sinks.
2. A sink absorbs more CO_2 than it releases.
3. Every coal plant is a carbon sink by definition.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Coal plants are sources.

Q20. Which one of the following correctly states GWP?

A. A measure comparing heat-trapping to CO_2 over a stated horizon B. A measure of ocean salinity only C. A unit of forest canopy density D. Identical to Dobson Unit

💡 **Show answer**

Ans: A. Heat-trapping vs CO_2 over a horizon (often 100 years). **Logic:** GWP ≠ DU.

Q21. How many of the following are Kyoto-basket gases (classic six + note NF_3 later)?

CO_2, CH_4, N_2O, HFCs, PFCs, SF_6

A. Only three B. All six C. Only five D. Only four

💡 **Show answer**

Ans: B. All six. **Logic:** Classic Kyoto basket six.

Q22. With reference to propane/hydrogen/argon, which of the following statements is/are correct?

1. They are listed among gases that are not GHGs in standard traps.
2. CO_2 is also not a GHG.
3. N_2O is a GHG.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 3. **Logic:** CO_2 is a GHG.

Q23. Arrange decreasing relative share teaching:

A. $N_2O > CO_2 > CH_4$ B. CFCs > $CO_2 > CH_4 > N_2O$ C. $CH_4 > CO_2 > N_2O$ D. $CO_2 > CH_4$ > CFCs+others > N_2O

💡 **Show answer**

Ans: D. $CO_2 > CH_4$ > CFCs+ > N_2O. **Logic:** WMO-style order.

Q24. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Carbon fertilization can increase plant growth from higher atmospheric CO_2.

Reason (R): Carbon fertilization is the same process as stratospheric ozone depletion.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Different processes.

Q25. With reference to India's emitter rank, which of the following statements is/are correct?

1. Absolute emissions place India near the top (about 3rd).
2. Per capita emissions remain relatively low versus many developed countries.
3. Absolute rank and per capita rank are always the same number.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Absolute ≠ per capita.

Q26. Which of the following pairs is/are correctly matched?

1. Direct GHG — CO_2
2. Indirect agent teaching — NOx, CO, NMVOCs, SO_2
3. Direct GHG — argon

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Argon not GHG.

Q27. With reference to Scope emissions, which of the following statements is/are correct?

1. Scope 1 — direct
2. Scope 2 — purchased energy
3. Scope 3 — wider value chain

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: C. All three. **Logic:** Standard Scope 1/2/3.

Q28. Which one of the following best states the main recent warming driver in teaching?

A. Argon increase B. Montreal Protocol alone C. Purely orbital cooling only D. Anthropogenic CO_2

💡 **Show answer**

Ans: D. Anthropogenic CO_2. **Logic:** Main recent driver fact.

Q29. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Blue carbon refers to ocean and coastal carbon sinks such as mangroves.

Reason (R): Blue carbon is identical to black carbon soot from incomplete combustion.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Blue ≠ black carbon.

Q30. With reference to HFCs, which of the following statements is/are correct?

1. HFCs are high-GWP greenhouse gases.
2. Kigali Amendment phases down HFCs.
3. HFCs are classic ozone-depleting substances under Montreal's original ODS list.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** HFCs are not classic ODS; Kigali is climate/GWP.

Mains Answer Framework

Climate change and India (250 words): Define warming, greenhouse effect, major GHGs. India profile — 3rd total emitter, low per capita, sector split. Impacts on agriculture, water, heat, Himalaya, coast. Adaptation — NAPCC 8 missions, UP Bundelkhand/flood/heat examples. Mitigation — NDC 2030 targets, net zero 2070, renewables, forest sink, LiFE. Treaty chain UNFCCC → Kyoto → Paris; distinguish Montreal. Conclude with both adaptation and mitigation needed under CBDR.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q115

The 'Panchamrit' targets announced by Prime Minister Modi in Glasgow are related with

A. Science and technology B. Foreign policy C. Climate change D. Investment plans in other countries

💡 **Show answer**

Logic: Glasgow venue points straight to the climate package. **Ans: C (Series B provisional key).** Panchamrit is the five climate commitments at COP26 Glasgow (2021), including 500 GW non-fossil by 2030 and net zero by 2070.

Common Traps — Don't Fall For These

Trap	Correct fact
Montreal = climate treaty	**Ozone** — Kyoto/Paris = climate
LiFE proposed at COP-25	**COP-26 Glasgow**
Eliminate greenhouse effect	**Natural effect essential** — problem is enhanced
Wilson/IPCC invented greenhouse effect	**Joseph Fourier (1820s)**
CH_4 highest total contribution	**CO_2** (~64% WMO-style / ~76% IPCC forcing)
Water vapour = main Kyoto inventory driver	Most abundant **natural** GHG / feedback; policy driver = **CO_2**
Argon / H_2 / propane / N_2 are GHGs	**Not** GHGs
SO_2 / NOx warm Earth directly	**Not direct** GHGs
India highest per capita emitter	**Low per capita**, ~3rd total; **China** #1
Net zero = absolute zero emissions	Emissions **balanced** by removals
Nuclear power is an NAPCC mission	**Not** among the eight
GCF founded at Durban	**Cancun COP-16 (2010)**
Earth Hour = UNEP/UNESCO	**WWF**
Carbon credit price fixed by UNEP	**Market-traded**
Adaptation = cut emissions	**Mitigation** cuts emissions
Kyoto bound India to cuts	Bound **Annex I developed** countries
Solar flares = primary recent climate cause	**Not** major (NASA/IPCC framing)
Milankovitch includes solar irradiance	Eccentricity, obliquity, precession only
Blue carbon = petroleum carbon	**Ocean/coastal** ecosystems

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 11 — Ozone Layer

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Ozone Layer | Ozone Depletion (ओजोन क्षरण) | Ozone Hole (ओजोन छिद्र) | Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल)

Sources baked in: NCERT Geography/Biology, UNEP Ozone Secretariat, Vienna Convention 1985, Montreal Protocol 1987, WMO assessments, UPPCS PYQs 2018–2025 **Weight:** ★★ Medium–High — CFC mechanism, treaty traps, ozone hole geography **Last verified:** August 2026

Current Affairs (this topic)

- Montreal / Kigali compliance headlines stay ozone-family CA — do not park them under Paris NDC.
- ODS vs GHG trap is permanent; HFC phase-down is Kigali, not Kyoto (क्योटो).

💡 🎯 **Consolidated — 30 Must-Score Facts (Click to Expand)**

1. Ozone (O_3) is triatomic oxygen. **Stratospheric ozone** (समतापमंडलीय ओजोन) (about **15–30/35 km**) is protective "good" ozone. **Tropospheric ozone** (क्षोभमंडलीय ओजोन) is a ground-level pollutant. 2. About **90%** of ozone is stratospheric; about **10%** is tropospheric. Stratospheric ozone is regulated in part by **NO_2**. 3. The **Chapman cycle** describes natural formation and destruction balance of stratospheric ozone (UV acting on O_2). 4. **UV-A (315–400 nm)** reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is fully blocked. Ozone blocks ~**93–99%** of the most damaging UV. 5. Ozone-depleting substances release chlorine and bromine radicals that catalytically destroy ozone. One chlorine atom can destroy on the order of **100,000** ozone molecules. 6. Main ODS include **CFCs, Halon-1211/1301, carbon tetrachloride, methyl chloroform, methyl bromide, and transitional HCFCs**. Fridge coolants often sold as **Mafron**; large plants may use **ammonia**. 7. An **ozone hole** is not a physical hole. It means column ozone below **220 Dobson Units**. **1 DU ≈ 0.01 mm** pure ozone at 0°C, 1 atm. 8. The Antarctic ozone hole peaks in Southern Hemisphere spring (**September–October**) under a polar vortex with polar stratospheric clouds + CFCs. 9. The Antarctic ozone hole was discovered in **1985** by **Farman (फ़रमान), Gardiner, Shanklin** (British Antarctic Survey). **Kent Moore** reported Tibetan Plateau ozone feature in **2005**. 10. Treaty chain: **Vienna Convention 1985 → Montreal Protocol 1987 (force 1 Jan 1989) → Kigali Amendment 2016 (force 1 Jan 2019)**. 11. Montreal is the binding ODS phase-out treaty. International Ozone Day is **16 September** (UN observance from **1994**). 12. **Kigali (2016)** phases down **HFCs**. HFCs are not classic ODS, but they have high global-warming potential. 13. Montreal achieved near-universal ratification and cut ODS emissions by about **99%** from peak levels. 14. Treaty trap: **Montreal = ozone**; **Kyoto/Paris = climate**. Never swap the issue tags. Carbon trading ≠ Montreal. 15. Global average ozone column is often near about **300 DU**; the hole threshold is **<220 DU**. 16. India ratified Montreal in **1992**. The nodal unit is the **Ozone Cell, MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय)**. 17. India CFC phase-out fact is **2010**. HCFC freeze is **2020**; complete HCFC phase-out aims at **2030**. 18. India's Kigali HFC phase-down commitment begins from **2028** in standard coaching facts. 19. Stratospheric ozone protects life; tropospheric ozone harms lungs and crops — same molecule, opposite verdict by layer. 20. Vienna is the framework convention; Montreal is the binding control protocol — parallel to UNFCCC (यूएनएफसीसीसी)/Kyoto structure, but for ozone. 21. Dobson Unit measures total ozone column, not ground-level ppm of pollution ozone. 22. Polar stratospheric clouds enable chlorine activation — that is why

the deep hole is polar-spring specific. 23. HCFCs were transitional substitutes on the way out of CFCs; they are still ODS, just weaker/longer phase-out. 24. Kigali sits under the Montreal family, not under UNFCCC as a climate protocol. 25. Ozone Day is **16 September**; do not confuse it with World Environment (पर्यावरण) Day (विश्व पर्यावरण दिवस) (**5 June**) or Wildlife Day (**3 March**). 26. CFCs are both ODS and GHGs; CFC source is not primarily automobiles; CFC use is not tubeless tyres. 27. Extra UV mainly raises **skin cancer** risk (classic stem). 28. Homosphere (सममंडल) ≈ troposphere (क्षोभमंडल) + stratosphere (समतापमंडल) + mesosphere (मध्यमंडल); **ionosphere not** in that set. 29. Column thickness often highest in **spring**, lowest in **rainy season** (general seasonal pattern). 30. Full teaching depth for this chapter also lives in Topic **36**.

💡 ⚡ **Confused Pairs (Click to Expand)**

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Stratospheric Ozone	**Tropospheric Ozone**	Essential atmospheric shield absorbing harmful solar UV-B radiation (20–30 km altitude) vs toxic secondary pollutant and greenhouse gas causing smog and respiratory ailments	समतापमंडलीय ओजोन (सुरक्षा कवच) / क्षोभमंडलीय ओजोन (हानिकारक प्रदूषक)
Ozone Depletion	**Ozone Hole**	Widespread gradual thinning of the stratospheric ozone layer globally caused by halogenated gases vs severe, seasonal Antarctic springtime drop where column ozone drops below 220 Dobson Units (DU)	ओजोन क्षरण (वैश्विक स्तर पर) / ओजोन छिद्र (अंटार्कटिक <220 DU)
Vienna Convention (1985)	**Montreal Protocol (1987)**	Non-binding multilateral framework agreement establishing international cooperation on ozone research and monitoring vs legally binding international treaty mandating precise, phase-out timetables for ozone-depleting substances (ODS)	वियना कन्वेंशन (ढांचागत समझौता) / मॉन्ट्रियल प्रोटोकॉल (बाध्यकारी प्रतिबंध)
Chlorofluorocarbons (CFCs)	**Hydrofluorocarbons (HFCs)**	Powerful ozone-depleting substances with high chlorine and high GWP (phased out under Montreal) vs zero-ODP fluorinated gases that do not harm ozone but are super greenhouse gases (phased down under 2016 Kigali Amendment)	सीएफसी (ओजोन विनाशक + GHG) / एचएफसी (केवल GHG, किगाली द्वारा नियंत्रित)
Dobson Unit (DU)	**Parts Per Million (ppm)**	Standard unit measuring the total vertical column abundance of atmospheric ozone (1 DU = 0.01 mm thickness at STP; normal baseline ~300 DU) vs standard volumetric concentration ratio of trace gases in ambient air	डॉबसन यूनिट (स्तंभ ओजोन मोटाई) / पीपीएम (सान्द्रता अनुपात)
Polar Stratospheric Clouds (PSCs)	**Cirrus Clouds**	High-altitude clouds forming in extreme Antarctic winter cold (-78°C / Type I and II) that catalyze chlorine release from reservoirs vs normal high-altitude wispy ice-crystal weather clouds in the upper troposphere	ध्रुवीय समतापमंडलीय बादल (PSCs) / पक्षाभ मेघ (ट्रोपोस्फेरिक)
Kigali Amendment (2016)	**Paris Agreement (2015)**	Legally binding amendment to Montreal Protocol committing countries to progressively phase down consumption of hydrofluorocarbons (HFCs) by 80–85% vs UNFCCC climate framework agreement targeting global warming mitigation	किगाली संशोधन (HFC कटौती) / पेरिस समझौता (जलवायु लक्ष्य)

💡 📋 **Must-score facts — good/bad ozone (बुरा ओजोन), Montreal, Kigali (Click to Expand)**

Science

Item	Lock
Good / bad ozone	Stratosphere (~15–35 km) / tropospheric pollutant
Share	~90% stratospheric; ~10% tropospheric
UV-A / B / C	Reaches surface / mostly blocked / fully blocked
Ozone hole	Column <**220 DU** (not a physical hole)
Antarctic peak	SH spring Sep–Oct; PSCs + vortex + CFCs
Discovery	**1985** Farman–Gardiner–Shanklin (BAS)

Treaties

Item	Lock
Chain	Vienna **1985** → Montreal **1987** → Kigali **2016**
Montreal vs Kyoto/Paris	Ozone ODS / climate GHGs — never swap
Kigali	Phases down **HFCs** (high GWP; not classic ODS)
Ozone Day	**16 September**

(images/env_ch11_01_ozone_layer_formation_depletion_mechanism.png)

11.1 Ozone Layer

Identity: O_3 in stratosphere | UV shield | Chapman cycle | Dobson Unit

- **Ozone (O_3)** is a molecule of three oxygen atoms that absorbs harmful ultraviolet radiation.

- The **ozone layer** lies mainly in the **stratosphere** (समतापमंडल) at about **15–35 km**, with peak concentration near **25 km**.
- The **Chapman cycle** describes natural ozone formation when UV splits O_2 and O combines with O_2 to form O_3.
- Ozone absorbs UV energy and releases heat, which is why **stratosphere temperature rises with height**.
- **UV-A** largely reaches the surface. **UV-B** is mostly blocked by ozone and causes sunburn and skin cancer if exposure rises.
- **UV-C** is fully absorbed by ozone and oxygen under normal conditions.
- **Column ozone** is measured in **Dobson Units (DU)**. Global average is about **300 DU**.
- **Stratospheric ozone** is protective. **Tropospheric ozone** is a harmful air pollutant and greenhouse gas.
- Without the ozone layer, lethal UV levels would reach Earth's surface and damage DNA, crops, and marine plankton.

> **Logic:** UPPCS 2023 pattern — ozone layer is in the **stratosphere**, not troposphere.

PYQ — UPPCS Prelims 2025, Q45

1. The ozone layer protects Earth from ultraviolet radiation.
2. Ozone depletion has been linked to CFCs.

A. Only 2

B. Neither

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Ans: C — Both statements are correct.

UV type	Wavelength	Effect if unfiltered
UVA	320–400 nm	Ageing, tanning
UVB	280–320 nm	Sunburn, skin cancer, cataracts
UVC	100–280 nm	Lethal — normally blocked

11.2 Ozone Depletion

Identity: CFCs/ODS | Catalytic chlorine | Rowland–Molina 1974

- **Ozone depletion** is the sustained reduction of stratospheric ozone below natural levels.
- **Ozone-Depleting Substances (ODS)** contain chlorine and/or bromine that destroy ozone in the stratosphere.
- **CFCs** were used in refrigerants, aerosols, foam blowing, and solvents.
- CFCs are stable in the troposphere and take decades to reach the stratosphere.
- UV breaks CFC bonds and releases **chlorine atoms (Cl·)** that destroy ozone in a **catalytic cycle**.
- **One chlorine atom can destroy up to about 100,000 ozone molecules** before removal.
- **Halons** used in fire suppressants release **bromine**, which is even more destructive per atom.
- Other ODS include **carbon tetrachloride**, **methyl bromide**, and transitional **HCFCs**.
- The **Rowland–Molina hypothesis (1974)** linked CFCs to ozone loss and won the Nobel Prize in 1995.
- Increased UV-B raises **skin cancer, cataracts, crop damage, and marine food chain (खाद्य श्रृंखला) stress**.
- After the Montreal Protocol, ODS emissions fell about **99%**. Mid-latitude recovery is expected around **2040**.

> **Logic:** CFCs destroy ozone by **UV chemistry**, not by greenhouse trapping — that is a separate effect.

ODS	Main use
CFCs	Refrigerants, aerosols, foam
Halons	Fire extinguishers
Carbon tetrachloride	Solvent
Methyl bromide	Pesticide/fumigant
HCFCs	Transitional refrigerants

11.3 Ozone Hole

Identity: < 220 DU | Antarctic spring | PSCs | Not a physical hole

- The **ozone hole** is **not a hole in the atmosphere**. It is a region where ozone falls below **220 DU**.
- The most severe and recurring depletion occurs over **Antarctica** in **Southern Hemisphere spring (September–October)**.
- The **polar vortex** isolates cold Antarctic stratospheric air during winter darkness.
- **Polar Stratospheric Clouds (PSCs)** (ध्रुवीय समतापमंडलीय बादल) form at very low temperatures and enable chlorine activation.
- Spring sunlight triggers rapid ozone destruction when the polar vortex breaks down.
- The ozone hole was discovered in **1985** by **Farman and colleagues** at the British Antarctic Survey.
- **Arctic** ozone loss can occur in cold winters but is **less severe and less regular** than Antarctic depletion.
- India is not under the ozone hole, but **Himalayan high altitude** increases UV-B exposure risk.
- WMO projects **Antarctic recovery to 1980 levels around 2066** if Montreal controls continue.

> **Logic:** Trap — "ozone hole over Arctic every winter" is **false**. Classic hole = **Antarctic spring**.

Feature	Antarctic	Arctic
Severity	Severe, recurring	Occasional, milder
Season	Sep–Oct (austral spring)	Mar–Apr if event occurs
Polar vortex	Strong, very cold	Weaker, warmer

11.4 Montreal Protocol

Identity: Vienna 1985 + Montreal 1987 | ODS phase-out | Kigali HFCs | India compliance

- The **Vienna Convention (1985)** (वियना कन्वेंशन) created the framework to protect the ozone layer through cooperation and monitoring.
- The **Montreal Protocol (1987)** (मॉन्ट्रियल प्रोटोकॉल) set **binding schedules** to phase out production and consumption of ODS.
- **International Ozone Day** is **16 September**, the date the Montreal Protocol was signed.
- The **Multilateral Fund (1991)** helps developing countries switch to ODS-free technologies.
- The Montreal Protocol is the **only UN treaty ratified by every country**.
- ODS emissions have declined about **99%** since their peak in the 1980s–1990s.
- **India ratified in 1992**. The **Ozone Cell under MoEFCC** coordinates compliance.

- India ended **CFC production in 2010** and targets **HCFC phase-out by 2030**.
- The **Kigali Amendment (2016)** (किगाली संशोधन) phases down **HFCs**, which do not deplete ozone but have **high global warming (भूमंडलीय तापन) potential**.
- **Montreal controls ozone**. **Kyoto controls climate**. Do not swap them.

PYQ — UPPCS Prelims 2024, Q35

Which agreement relates to climate change (जलवायु परिवर्तन) mitigation?

A. Montreal Protocol

B. Ramsar (रामसर)

C. Kyoto Protocol

D. CITES (साइट्स)

> 💡 **Show answer**
>
> **Ans: C** — Montreal = **ozone/ODS**.

Treaty / step	Year	Role
Vienna Convention	1985	Framework
Montreal Protocol	1987	Binding ODS phase-out
London Amendment	1990	Multilateral Fund
Kigali Amendment	2016	HFC phase-down

Complete PYQ Bank (Topic 11)

Q1. UPPCS Prelims 2025, Q45

Consider the following statements about the **ozone layer**:

1. The ozone layer protects Earth from harmful **ultraviolet radiation**.
2. Ozone depletion has been linked to **chlorofluorocarbons (CFCs)**.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

> 💡 **Show answer**
>
> **Logic:** Basic ozone-layer function and anthropogenic ODS link — both are standard textbook facts. **Ans: C.** Stratospheric O_3 absorbs UV-B/C; CFCs release catalytic chlorine that destroys ozone. **Trap:** Tropospheric ozone is a **pollutant** — the protective layer is **stratospheric**.

Q2. UPPCS Prelims 2025, Q62

Which of the following pairs is/are **NOT** correctly matched? (Day — Date)

1. Ozone Day — 16 September
2. World Environment Day — 5 June
3. World Wildlife Day — 22 March

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1 only

> 💡 **Show answer**
>
> **Logic:** International observance dates — Ozone Day and WED are correct; one wildlife date is wrong. **Ans: B.** **World Wildlife Day is 3 March**, not 22 March; Ozone Day (16 Sept) and WED (5 June) are

correct. **Trap:** Do not confuse **Ozone Day (16 September)** with **World Environment Day (5 June)**.

Q3. UPPCS Prelims 2024, Q35

Which international agreement relates to **climate change mitigation**?

A. Montreal Protocol B. Ramsar Convention C. Kyoto Protocol D. CITES

💡 **Show answer**

Logic: Cross-topic treaty trap — Montreal is ozone/ODS; Kyoto is climate mitigation. **Ans: C.** **Kyoto Protocol** targets greenhouse-gas cuts; Montreal controls ozone-depleting substances. **Trap:** Montreal is the most common wrong pick in "climate treaty" stems — it is **not** a climate convention.

Q4. UPPCS Prelims 2023, Q52

The **ozone layer**, which absorbs ultraviolet radiation, exists in which layer of the atmosphere?

A. Troposphere B. Mesosphere C. Stratosphere D. Thermosphere

💡 **Show answer**

Logic: Atmospheric layer fact — peak ozone concentration and UV shield sit in the stratosphere. **Ans: C.** The ozone layer lies mainly in the **stratosphere** (about 15–35 km), not the troposphere. **Trap:** Ground-level ozone in the **troposphere** is harmful smog — opposite of the protective stratospheric layer.

Q5. UPPCS Prelims 2022

International Ozone Day is observed every year on—

A. 5 June B. 16 September C. 22 March D. 2 February

💡 **Show answer**

Logic: Montreal Protocol anniversary observance date. **Ans: B.** World Ozone Day / International Ozone Day is **16 September** — the date the Montreal Protocol was signed (1987). **Trap:** **5 June** is World Environment Day — not Ozone Day.

Q6. UPPCS Prelims 2021, Q96 pattern

World Ozone Day is celebrated on—

A. 3 March B. 5 June C. 16 September D. 22 April

💡 **Show answer**

Logic: Repeat observance-date fact for ozone protection awareness. **Ans: C.** World Ozone Day falls on **16 September** each year. **Trap:** **3 March** is World Wildlife Day — another frequent date-swap distractor.

Q7. UPPCS Prelims 2018, Q44 pattern

Maximum concentration of **atmospheric ozone** is found in the—

A. Troposphere B. Stratosphere C. Mesosphere D. Ionosphere

💡 **Show answer**

Logic: Vertical ozone profile — Chapman production peaks in stratosphere near ~25 km. **Ans: B.** Ozone concentration peaks in the **stratosphere**, where UV photolysis builds the ozone layer. **Trap:**

Troposphere has some ozone as **pollution**, but not the maximum column density.

Q8. UPSC pattern

The **Montreal Protocol** primarily controls—

A. Transboundary hazardous waste trade B. Phase-out of **ozone-depleting substances (ODS)** C. Wetland encroachment in Ramsar sites D. Carbon credit trading under Kyoto

💡 **Show answer**

Logic: Montreal = ODS phase-out under Vienna Convention framework — universal ratification success story. **Ans: B.** Montreal mandates phased elimination of **CFCs, halons, carbon tetrachloride**, and later HCFCs/HFCs (Kigali). **Trap:** Kyoto/Paris handle **GHGs** — Montreal is exclusively **ozone/ODS**.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to ozone, which of the following statements is/are correct?

1. tropospheric ozone is protective “good” ozone.
2. Tropospheric ozone is a ground-level pollutant.
3. About 90% of ozone is tropospheric.

A. Only 2 B. Only 1 C. 2 and 3 D. 1 and 2

💡 **Show answer**

Ans: A. Only 2 is correct. **Logic:** Only statement 2 remains true among the numbered claims. Statement 1 is false. Statement 2 is true. Statement 3 is false.

Q2. Match List-I with List-II:

List-I	List-II
A. UV-A	1. Fully blocked
B. UV-B	2. Reaches surface
C. UV-C	3. Mostly blocked by ozone

Code:

A. A-2, B-1, C-3 B. A-3, B-2, C-1 C. A-2, B-3, C-1 D. A-1, B-3, C-2

💡 **Show answer**

Ans: C. A-2, B-3, C-1. **Logic:** UV-A reaches; UV-B mostly blocked; UV-C fully blocked.

Q3. Which of the following pairs is/are NOT correctly matched?

1. Ozone hole — physical cavity in the sky
2. Ozone hole — column ozone often below 220 DU
3. 1 DU ≈ 0.01 mm pure ozone at 0°C, 1 atm

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 2 only D. 1 only

💡 **Show answer**

Ans: D. Only pair 1 is not correctly matched. **Logic:** Hole is a column-ozone definition — not a physical hole.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): The Antarctic ozone hole peaks in Southern Hemisphere spring (September–October).

Reason (R): Polar stratospheric clouds and the polar vortex enable chlorine/bromine catalytic destruction.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the spring peak chemistry. **A/R logic:** PSC + vortex + ODS chemistry.

Q5. Arrange the treaty chain:

1. Vienna Convention
2. Montreal Protocol
3. Kigali Amendment

A. 1-2-3 B. 2-1-3 C. 1-3-2 D. 3-2-1

💡 **Show answer**

Ans: A. 1985 → 1987 → 2016. **Logic:** Vienna–Montreal–Kigali.

Q6. With reference to ODS, which of the following statements is/are correct?

1. Main ODS include CFCs, Halons, CCl_4, methyl chloroform, methyl bromide, HCFCs.
2. One chlorine atom destroys only about ten ozone molecules before becoming inert.
3. HFCs are classic ODS under the original Montreal ODS list.

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Statement 1 correctly lists main ODS. Statement 2 understates catalytic destruction (~10^5 ozone molecules per Cl). Statement 3 is false — HFCs are not classic Montreal ODS (Kigali/climate, not original ODS list).

Q7. Which of the following pairs is/are correctly matched?

1. Discovery of Antarctic ozone hole — 1985 Farman–Gardiner–Shanklin (BAS)
2. International Ozone Day — 16 September
3. Montreal Protocol — climate Kyoto GHG targets

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Montreal = ozone ODS — not Kyoto climate.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Kigali Amendment phases down HFCs.

Reason (R): HFCs are classic ozone-depleting substances identical to CFCs.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** HFCs are high GWP — not classic ODS.

Q9. With reference to the Chapman cycle, which of the following statements is/are correct?

1. It describes natural formation/destruction balance of stratospheric ozone.
2. UV acting on O_2 is central to formation.
3. It means tropospheric smog is caused only by argon.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Statement 3 false.

Q10. Match List-I with List-II:

List-I	List-II
A. Montreal	1. 2016
B. Vienna	2. 1987
C. Kigali	3. 1985

Code:

A. A-2, B-3, C-1 B. A-3, B-2, C-1 C. A-2, B-1, C-3 D. A-1, B-3, C-2

> 💡 **Show answer**
>
> **Ans: A.** A-2, B-3, C-1. **Logic:** Montreal 1987; Vienna 1985; Kigali 2016.

Q11. Which one of the following correctly states stratospheric ozone height teaching?

A. Only below 2 km B. About 15–30/35 km C. Only in the exosphere D. Only inside Earth's core

> 💡 **Show answer**
>
> **Ans: B.** About 15–30/35 km. **Logic:** Good ozone layer height.

Q12. With reference to Montreal vs Kyoto/Paris, which of the following statements is/are correct?

1. Montreal addresses ODS / ozone.
2. Kyoto/Paris address climate GHGs.
3. The two issue tags may be freely swapped in answers.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** Never swap ozone vs climate tags.

Q13. Arrange UV types from longest to shortest wavelength teaching:

A. UV-C → UV-B → UV-A B. UV-A → UV-C → UV-B C. UV-B → UV-A → UV-C D. UV-A → UV-B → UV-C

💡 **Show answer**

Ans: D. UV-A → UV-B → UV-C. **Logic:** 315–400; 280–315; 100–280 nm.

Q14. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Montreal achieved near-universal ratification and large ODS cuts from peak levels.

Reason (R): International Ozone Day marks the Montreal Protocol (16 September).

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; R commemorates Montreal but does not explain ratification success. **A/R logic:** Related commemorative fact ≠ explanation of success.

Q15. Which of the following statements about HCFCs is/are correct?

1. They are transitional ODS in teaching lists.
2. They contain chlorine/bromine ozone risk pathways.
3. They are identical to nitrogen gas (N_2).

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not N_2.

Q16. With reference to NO_2 and ozone, which of the following statements is/are correct?

1. Stratospheric ozone is regulated in part by NO_2.
2. About 10% of ozone is tropospheric.
3. Tropospheric ozone is always beneficial UV shield only.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Tropospheric ozone is a pollutant.

Q17. How many of the following are ODS teaching items?

CFCs, Halon-1211/1301, carbon tetrachloride, methyl bromide, argon

A. Only two B. Only three C. Only four D. All five

💡 **Show answer**

Ans: C. Only four. **Logic:** Argon is not ODS.

Q18. Match List-I with List-II:

List-I	List-II
A. Ozone Day	1. Climate GHGs
B. Kyoto/Paris	2. 16 September
C. Good ozone layer	3. Stratosphere

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

💡 **Show answer**

Ans: D. A-2, B-1, C-3. **Logic:** Ozone Day 16 Sep; Kyoto/Paris climate; good ozone stratosphere.

Q19. Which of the following pairs is/are correctly matched?

1. Fridge coolants — often ODS/halocarbon history teaching
2. Ammonia — used in many large plants
3. CFCs — tubeless-tyre production as the main use

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** CFCs not mainly tubeless-tyre production.

Q20. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): An ozone hole means column ozone below about 220 DU.

Reason (R): Average column is often cited near about 300 DU.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; average 300 DU does not by itself define the 220 DU hole threshold explanation. **A/R logic:** Related DU facts; R not the definitional explanation.

Q21. With reference to Montreal force date teaching, which of the following statements is/are correct?

1. Montreal Protocol 16 September 1987.
2. Entered into force 1 January 1989.
3. Kigali entered into force 1 January 2019 teaching.

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

💡 **Show answer**

Ans: B. All three. **Logic:** 1987/1989 and Kigali force 2019 teaching.

Q22. Which one of the following is correct about carbon trading vs Montreal?

A. Carbon trading = Montreal ODS phase-out B. Montreal sets India's NDCs C. Carbon trading ≠ Montreal; climate markets ≠ ozone treaty D. Montreal repealed UNFCCC

💡 **Show answer**

Ans: C. Carbon trading ≠ Montreal. **Logic:** Issue separation.

Q23. Arrange discovery → Vienna → Montreal:

A. 1987 → 1985 → 1985 discovery B. 1972 → 1987 → 1985 C. 2016 → 1985 → 1987 D. 1985 discovery → 1985 Vienna → 1987 Montreal

💡 **Show answer**

Ans: D. 1985 discovery & Vienna → 1987 Montreal. **Logic:** Same-year Vienna + discovery teaching; Montreal 1987.

Q24. With reference to chlorine catalysis, which of the following statements is/are correct?

1. ODS release Cl/Br radicals.
2. Catalytic cycles destroy ozone.
3. Ozone absorbs infrared as its main protective role against UV.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Main protective role is UV absorption — not IR.

Q25. Which of the following statements is/are correct?

1. Kent Moore reported Tibetan Plateau ozone thinning teaching in some stems.
2. Antarctic hole is the classic global teaching case.
3. Ozone hole peaks in Antarctic midsummer January only always.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Peak is SH spring Sep–Oct.

Q26. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Tropospheric ozone is a pollutant.

Reason (R): About 10% of atmospheric ozone is tropospheric.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; share does not explain why it is a pollutant. **A/R logic:** Definition vs share.

Q27. How many of the following treaty issue tags are correct?

1. Montreal — ozone
2. Kyoto — climate
3. Paris — climate

A. Only one B. Only two C. All three D. None

💡 **Show answer**

Ans: C. All three. **Logic:** Standard tags.

Q28. With reference to Halons, which of the following statements is/are correct?

1. Halon-1211/1301 appear in ODS lists.
2. They are bromine/chlorine carriers in fire-fighting history teaching.
3. Halons are identical to nitrogen fertilizer N_2O only.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not N_2O.

Q29. Which of the following pairs is/are NOT correctly matched?

1. Good ozone — stratosphere
2. Bad ozone — troposphere pollutant
3. Ozone Day — 22 April

A. 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only pair 3 is not correctly matched. **Logic:** Ozone Day 16 Sep; 22 Apr is Earth Day.

Q30. With reference to Kigali, which of the following statements is/are correct?

1. It is an amendment under the Montreal Protocol family.
2. It phases down HFCs for climate reasons.
3. It replaces UNFCCC entirely.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Does not replace UNFCCC.

Mains Answer Framework

Ozone layer protection (250 words): Define ozone layer and UV shield function. Explain depletion mechanism — CFCs, catalytic chlorine, health/ecosystem (पारिस्थितिकी तंत्र) impacts. Describe ozone hole — Antarctic spring, PSCs, 220 DU threshold. Treaty response — Vienna framework, Montreal Protocol, Multilateral Fund, universal ratification, recovery timeline. India compliance — Ozone Cell, 2010 CFC phase-out, HCFC 2030, Kigali HFCs. Conclude with Montreal as model environmental treaty; distinguish from Kyoto/Paris.

Common Traps — Don't Fall For These

Trap	Correct fact
Montreal = climate treaty	**Ozone / ODS**
Ozone layer in troposphere	**Stratosphere**
Ozone hole = physical hole	**Low ozone column < 220 DU**
Ozone hole over Arctic always	**Antarctic spring**
Tropospheric ozone is good	**Pollutant**
CFCs controlled under Kyoto	**Montreal Protocol**
Ozone Day = 5 June	**16 September**
Kigali targets CFCs	**HFC phase-down**
UV-C reaches surface normally	**Blocked by O_3/O_2**
Depletion permanent with no recovery	Recovery expected **~2040/2066**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 12 — Acid Rain (अम्ल वर्षा)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Acid Rain | Causes | Effects | Prevention

Sources baked in: NCERT Geography/Biology, CPCB norms, MoEFCC, CLRTAP/Gothenburg Protocol, Taj Trapezium case, UPPCS PYQs 2018–2025 **Weight:** ★★ Medium — pH traps, CO_2 vs SO_2/NOx, Taj Mahal, treaty confusion **Last verified:** August 2026

Current Affairs (this topic)

- Acid-rain CA is rare; keep SO_2 / NO_x chemistry and dry vs wet deposition stable.
- Do not confuse acid rain with ozone hole (ओजोन छिद्र) or greenhouse trapping.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. **Acid rain** means wet or dry acidic deposition with pH below **5.6**. 2. Normal unpolluted rain is already about **pH 5.6** because dissolved CO_2 forms weak carbonic acid — it is **not** pH 7. 3. **Wet deposition** is acids in rain, snow, or fog. **Dry deposition** is acidic gases and particles settling without rain. 4. **SO_2** oxidises to **H_2SO_4** (sulphuric acid), mainly from coal TPPs, smelters, and refineries. 5. **NOx** oxidises to **HNO_3** (nitric acid), mainly from vehicles and high-temperature combustion. 6. CO_2 forms only weak H_2CO_3 and sets the **5.6 baseline**; it is **not** the main culprit of severe acid rain. 7. On the pH scale, acidic is **<7**, basic is **>7**, neutral is **7**. Each step down multiplies H^+ by **10**. 8. Freshwater below about **pH 5** stresses fish and can release toxic **Al^{3+}**. 9. Acid deposition leaches soil **Ca^{2+}** and **Mg^{2+}** and can free toxic aluminium. 10. Conifers are especially sensitive forest receptors. Marble and limestone monuments form a **gypsum crust** ($CaSO_4$) under sulphuric attack — the Taj story. 11. **Lichens** die in high SO_2 and act as pollution bio-indicators. 12. Prevention tools include **FGD scrubbers** (lime/limestone), catalytic converters, low-sulphur coal, CNG/LPG switches, and **BS-VI** fuel (about **10 ppm** sulphur). 13. The **Gothenburg Protocol (1999)** (गोथेनबर्ग प्रोटोकॉल) under **CLRTAP (1979)** cuts transboundary acidifying emissions. 14. Treaty trap: Gothenburg/CLRTAP = acidifying air pollution; Montreal = ozone; Kyoto (क्योटो) = climate GHGs. 15. The **Taj Trapezium Zone** is about **10,400 km^2** around the Taj Mahal (ताज महल), strengthened by Supreme Court orders from **1996**. 16. **Mathura (मथुरा) refinery** is a classic SO_2 source in the Taj Trapezium narrative. 17. The **Singrauli–Sonbhadra (सोनभद्र)** coal TPP belt is a major UP–MP SO_2 source region. 18. UP NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) cities linked to acidifying/urban (नगरीय) air pressure include Kanpur (कानपुर), Ghaziabad, Lucknow (लखनऊ), and Noida. 19. SO_2 is denser than air — a physical property sometimes used in distractors. 20. Acid rain is **not** "rain only"; dry deposition counts too. 21. Secondary pollutants H_2SO_4 and HNO_3 form in the atmosphere from primary SO_2 and NOx. 22. FGD is flue-gas desulphurisation at power-plant chimneys — a mitigation hardware fact, not a treaty name. 23. Calling CO_2 the chief severe acid-rain driver is wrong; SO_2 and NOx are the pair. 24. Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) does **not** control acid rain. Keep ozone chemistry out of acid-rain treaty options.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Normal Rain** | **Acid Rain** | Slightly acidic precipitation with pH ~5.6 caused by dissolved atmospheric carbon dioxide forming weak carbonic acid (H_2CO_3) vs precipitation with pH strictly below 5.6 caused by anthropogenic emissions of sulfur dioxide (SO_2) and nitrogen oxides (NOx) | सामान्य वर्षा (pH ~5.6) / अम्ल वर्षा (pH < 5.6) | | **Sulfuric Acid (H_2SO_4)** | **Nitric Acid (HNO_3)** | Accounts for ~60–70% of total acid deposition acidity (primarily emitted by coal-fired thermal power plants and metal smelters) vs accounts for ~30–40% of acidity (primarily emitted by automobile exhausts and nitrogenous fertilizer use) | सल्फ्यूरिक अम्ल (60–70% भारांक) / नाइट्रिक अम्ल (30–40% भारांक) | | **Wet Acid Deposition** | **Dry Acid Deposition** | Direct transfer of acidic pollutants to Earth's surface incorporated into falling rain, fog, sleet, or snow vs settling of acidic gases, sulfate aerosols, and nitrate dust directly onto soil, vegetation, and buildings without moisture | आर्द्र निक्षेपण (वर्षा/कोहरा) / शुष्क निक्षेपण (गैस व धूलिकण) | | **Marble Cancer** | **Eutrophication** | Chemical weathering and yellowing/spalling of calcium carbonate structures (e.g. Taj Mahal) reacting with sulfuric acid to form soluble gypsum ($CaSO_4$) vs excessive nutrient enrichment of water bodies leading to dense algal blooms and dissolved oxygen crashes | संगमरमर कैंसर (ताजमहल क्षरण) / सुपोषण (जलीय पोषक संवर्धन) | | **Gothenburg Protocol (1999)** | **Montreal Protocol (1987)** | UNECE agreement under CLRTAP targeting multi-pollutant abatement of acidification, eutrophication, and ground-level ozone (SO_2, NOx, VOCs, NH_3) vs UNEP treaty targeting the global phase-out of ozone-depleting substances | गोथेनबर्ग प्रोटोकॉल (अम्लीकरण नियंत्रण) / मॉन्ट्रियल प्रोटोकॉल (ओजोन) | | **Taj Trapezium Zone (TTZ)** | **Eco-Sensitive Zone (ESZ)** | Specifically demarcated ~10,400 km^2 protective geographical zone covering Agra, Firozabad, Mathura, and Bharatpur to safeguard the Taj Mahal from industrial air pollution vs buffer belt up to 10 km declared around National Parks under EPA 1986 | ताज ट्रेपेज़ियम ज़ोन (10,400 वर्ग किमी) / पारिस्थितिक संवेदनशील क्षेत्र |

💡 📋 **Must-score facts — pH, acids, Taj, Gothenburg (Click to Expand)**

Chemistry | Item | Lock | |------|------| | Acid rain pH | Below **5.6** | | Normal rain (सामान्य वर्षा) | ~pH **5.6** from $CO_2 \rightarrow H_2CO_3$ ($\neq 7$) | | Wet / Dry | Rain–snow–fog / gases–particles without rain | | SO_2 / NOx paths | $\rightarrow H_2SO_4$ (TPPs) / $\rightarrow HNO_3$ (vehicles) | | CO_2 role | Sets 5.6 baseline — **not** main severe-acid culprit | | Receptors | Fish <~pH 5; Al^{3+} release; conifers; marble → gypsum | | Lichens | SO_2 bio-indicators | | Gothenburg / CLRTAP | **1999** / **1979** acidifying emissions | --- ![Acid Rain Pathways Landscape Profile & Taj Mahal Stone Cancer](images/env_ch12_01_acid_rain_mechanisms_impacts.png)

12.1 Acid Rain

Identity: pH < 5.6 | H_2SO_4 + HNO_3 | Wet + dry deposition | Secondary pollutant

- **Acid rain** is precipitation or atmospheric deposition with **pH below 5.6**, mainly from **sulphuric acid (H_2SO_4)** and **nitric acid (HNO_3)** (नाइट्रिक अम्ल).
- **Normal unpolluted rain** has pH about **5.6**, not 7, because atmospheric **CO_2** forms weak **carbonic acid (H_2CO_3)**.
- Anthropogenic acid rain often reaches **pH 4.0–4.5** in heavily industrial downwind areas.
- The **pH scale** runs 0–14. **pH 7 is neutral**. **Below 7 is acidic**. **Above 7 is basic/alkaline**.
- Each one-unit drop in pH means **10 times more hydrogen ions**, so pH 4 is 100× more acidic than pH 6.
- **Wet deposition** carries acids in rain, snow, fog, or dew.
- **Dry deposition** means **SO_2**, **NO_2**, and sulphate/nitrate particles settle on soil, leaves, and buildings **without rainfall**.
- **SO_2** and **NOx** are **primary pollutants**. Strong acids form in the atmosphere, so acid rain is a **secondary pollutant** phenomenon.
- Acidifying pollutants can travel **hundreds to thousands of km**, causing **transboundary damage**.
- Soils and lakes with **limestone or calcium carbonate** have **buffering capacity** and resist acidification better than granite or thin acidic soils.

PYQ — UPPCS Prelims 2025, Q103

1. Acidic solutions have pH more than 7.
2. Basic solutions have pH less than 7.

A. Only 2

B. Neither

C. Both

D. Only 1

💡 **Show answer**

Ans: B — Acidic = **< 7**; Basic = **> 7**.

PYQ — UPPCS Prelims 2024, Q108

(A) CO_2, SO_2, and NO_2 dissolve in rainwater and produce acid rain. (R) Air is polluted due to high concentration of CO_2, SO_2, and oxides of nitrogen.

A. Both true; R explains A

B. A false, R true

C. Both true; R not explanation

D. A true, R false

💡 **Show answer**

Ans: C — All three dissolve (A true). Air pollution statement (R) is true but does **not explain** why **CO_2 gives only weak acid** while **SO_2/NOx give strong acids**.

pH	Meaning
< 5.6	Anthropogenic acid rain threshold
~5.6	Normal rain (CO_2 equilibrium)
7	Neutral distilled water — not natural rain
> 7	Basic/alkaline

12.2 Causes

Identity: $SO_2 \rightarrow H_2SO_4$ | NOx → HNO_3 | Coal TPPs + vehicles | CO_2 weak only

- **Sulphur dioxide (SO_2)** is the main source of **sulphuric acid rain**.
- SO_2 comes from burning **sulphur-containing coal and oil** in **thermal power plants, refineries, smelters, brick kilns, and diesel generators**.
- In the atmosphere, SO_2 oxidises and reacts with water to form **H_2SO_4**.
- **Nitrogen oxides (NOx)** form at **high combustion temperatures** when nitrogen in air reacts with oxygen.
- NOx mainly comes from **vehicles, power plants, industrial boilers, and lightning**.
- NOx oxidises to form **HNO_3 (nitric acid)** in the atmosphere.
- **CO_2** dissolves as weak carbonic acid and sets the **pH 5.6 baseline**. It is **not** the main driver of severe acid rain.
- **Coal thermal power plants** are India's largest **SO_2** point sources.
- **Vehicles** are a major **NOx** source in cities.
- **SO_2 is heavier than air** and tends to accumulate near ground-level emission sources.

- Natural sources include **volcanic SO_2** and **lightning NOx**, but anthropogenic sources dominate in industrial regions.
- The **Singrauli–Sonbhadra belt** on the UP–MP border is a major Indian **SO_2 hotspot** from super thermal power plants.

PYQ — UPPCS Prelims 2025, Q43

Which gas/gases is/are lighter than air?

1. Hydrogen chloride
2. Sulfur dioxide
3. Helium

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

💡 **Show answer**

Ans: B — HCl and SO_2 are **heavier** than air.

Pollutant	Main acid	Main sources
SO_2	H_2SO_4	Coal TPPs, smelters, refineries
NOx	HNO_3	Vehicles, high-temp combustion
CO_2	H_2CO_3 (weak)	Natural baseline only

12.3 Effects

Identity: Aquatic kill | Soil Al^{3+} | Conifers | Taj gypsum | Lichens

- **Aquatic ecosystems** fail when lake/river pH drops **below about 5.0** — fish eggs do not hatch and adult fish die.
- **Aluminium (Al^{3+})** leached from acidified soil is **toxic to fish gills** and plant roots.
- **Soil acidification** leaches essential **calcium (Ca^{2+})** and **magnesium (Mg^{2+})** and releases toxic **Al^{3+}**.
- **Coniferous forests** (pine, spruce, fir) are **most sensitive** to acid rain and acid fog.
- **Crops** suffer from direct leaf damage and nutrient loss in soil.
- **Marble and limestone ($CaCO_3$)** react with sulphuric acid to form **gypsum ($CaSO_4$)**, causing yellow crust and pitting.
- The **Taj Mahal (Agra, UP)** is the classic Indian example — **SO_2 from Mathura refinery and Firozabad industries** damaged white marble.
- **Metal structures** rust faster under acid deposition.
- **SO_2 and NOx** cause **respiratory illness** — asthma, bronchitis, and lung irritation from acid aerosols and $PM_{2.5}$.
- **Lichens** are **bio-indicators of clean air**. They **die in high SO_2** and do **not** thrive in polluted areas.
- Transboundary teaching: **Norway** (and Canada/Scandinavia) are classic heavily affected countries; acidified lakes are sometimes called **"lake killers."**
- The **Acid Rain Information Centre (ARIC)** is associated with **Manchester, England**.
- The **1985 Helsinki Protocol** targeted about a **30%** cut in sulphur emissions / transboundary fluxes.

> **Logic:** — lichens as pollution indicators = true; grow well in polluted areas = **false**.

PYQ — UPPCS Prelims 2025, Q61

1. Lichens are very good indicators of pollution.
2. Lichens grow well in polluted areas.

A. Only 2

B. Neither

C. Both

D. Only 1

> 💡 **Show answer**
>
> **Ans: D** — Statement 1 only.

Receptor	Key impact
Lakes/rivers	Fish kills below pH ~5
Soil	Nutrient leaching, Al^{3+} toxicity
Forests	Conifer needle damage
Marble monuments	Gypsum crust (Taj Mahal)
Lichens	Die in SO_2 pollution

12.4 Prevention

Identity: FGD | Catalytic converters | BS-VI | Gothenburg | Taj Trapezium

- **Flue Gas Desulphurisation (FGD) scrubbers** pass chimney gas through **lime or limestone slurry** to absorb **SO_2** and can remove **90%+** of it.
- **Coal washing** and **low-sulphur coal** cut sulphur before combustion.
- **Catalytic converters** in vehicles convert **NO to N_2** and reduce **NOx** that forms nitric acid.
- **BS-VI norms (2020)** brought **10 ppm sulphur** in petrol and diesel, cutting both SO_2 and NOx precursors.
- **CNG, LPG, and natural gas** produce **negligible SO_2** and lower NOx than coal.
- **CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) emission standards** under the Air Act cap SO_2 and NOx from TPPs, cement, steel, and refineries.
- **Tall chimneys alone are not prevention** — they only disperse pollution farther.
- The **Gothenburg Protocol (1999)** under **CLRTAP (1979)** sets binding cuts in SO_2 and NOx for transboundary acid rain control in Europe.
- **Montreal Protocol controls ozone**, not acid rain — do not confuse treaties.
- The **Taj Trapezium Zone (~10,400 km^2)** was created to protect the Taj Mahal. The **1996 Supreme Court order** directed polluting industries near Agra to switch to gas or relocate.
- **NCAP (2019)** targets PM reduction in 122 cities with co-benefits for SO_2 and NOx cuts.

Measure	Target
FGD scrubbers	SO_2 from TPP chimneys
Catalytic converters	NOx from vehicles
BS-VI fuel	SO_2 + NOx precursors

Measure	Target
CNG/LPG switch	Near-zero SO_2
Gothenburg Protocol	Transboundary SO_2/NOx
Taj Trapezium	Heritage protection near Agra

Complete PYQ Bank (Topic 12)

Q1. UPPCS Prelims 2024, Q108

Assertion (A): CO_2, SO_2 and NO_2 dissolve in rainwater and produce acid rain.

Reason (R): Air is polluted due to high concentration of CO_2, SO_2 and oxides of nitrogen.

A. Both (A) and (R) are true and (R) is the correct explanation of (A). B. (A) is false but (R) is true. C. Both (A) and (R) are true but (R) is **not** the correct explanation of (A). D. (A) is true but (R) is false.

> 💡 **Show answer**
>
> **A/R logic:** A tests whether all three gases dissolve in rain; R tests a generic air-pollution statement and whether it explains **strong acid rain** from SO_2/NOx vs weak carbonic acid from CO_2. **Ans: C.** All three dissolve (A true) and polluted air (R true), but R does **not explain** why **SO_2/NOx** drive severe acid rain while **CO_2** gives only weakly acidic normal rain (~pH 5.6). **Trap:** "Both true; R explains A" ignores that **CO_2 alone** cannot explain coal-driven **H_2SO_4/HNO_3** acid rain.

Q2. UPPCS Prelims 2025, Q103

Consider the following statements about **pH**:

1. Acidic solutions have pH **more than 7**.
2. Basic solutions have pH **less than 7**.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

> 💡 **Show answer**
>
> **Logic:** Fundamental pH scale reversal trap — acidic < 7, basic > 7. **Ans: B.** Both statements reverse the scale — **acidic < 7**, **basic > 7**. **Trap:** Statement 1 alone sounds plausible if you confuse "more acid" with "higher number" — pH numbers **decrease** as acidity rises.

Q3. UPPCS Prelims 2025, Q43

Which of the following gas/gases is/are **lighter than air**?

1. Hydrogen chloride
2. Sulfur dioxide
3. Helium

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1 only

> 💡 **Show answer**
>
> **Logic:** Molecular weight vs air (≈29 g/mol) — HCl and SO_2 are heavier; helium is lighter. **Ans: B.** Only **helium** is lighter than air; **HCl** and **SO_2** are denser and accumulate near ground. **Trap:** SO_2 is linked to acid rain but is **heavier than air** — do not assume all acid-rain gases rise quickly.

Q4. UPPCS Prelims 2025, Q61

Consider the following statements about **lichens**:

1. Lichens are very good indicators of air pollution.
2. Lichens grow well in polluted areas.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

💡 **Show answer**

Logic: Bio-indicator sensitivity to SO_2 — presence signals clean air; absence signals pollution. **Ans: D.** Lichens indicate pollution levels but **decline or die** in high SO_2 — they do not thrive there. **Trap:** "Good indicator" does not mean "grows well in pollution" — sensitivity means **loss**, not growth.

Q5. UPSC pattern

Normal unpolluted rain is slightly acidic (pH ≈ **5.6**) primarily because of dissolved—

A. Sulphur dioxide from coal plants B. **Carbon dioxide** forming weak carbonic acid C. Nitrogen oxides from vehicles D. Hydrochloric acid from volcanoes

💡 **Show answer**

Logic: Natural rain acidity vs anthropogenic acid rain — CO_2 equilibrium vs SO_2/NOx strong acids. **Ans: B.** Atmospheric **CO_2** dissolves to form weak **H_2CO_3**, giving natural rain pH about **5.6**. **Trap:** **SO_2/NOx** cause severe acid rain (pH < 5.6) — CO_2 alone cannot drop rain to ~4.

Q6. UPSC pattern

Acid rain is primarily caused by atmospheric emissions of—

A. CO_2 and methane B. **SO_2 and NOx** C. CFCs and halons D. Ozone and PAN only

💡 **Show answer**

Logic: Primary acid-rain precursors form H_2SO_4 and HNO_3 in the atmosphere. **Ans: B.** **SO_2** (coal, smelters) and **NOx** (vehicles, high-temperature combustion) are the main acid-rain precursors. **Trap:** CO_2 is emitted with coal but produces only **weak** acidity — not the main anthropogenic acid-rain driver.

Q7. UPSC pattern

Flue Gas Desulphurisation (FGD) scrubbers are installed mainly at—

A. Vehicle exhaust pipes B. **Thermal power plant chimneys** C. Municipal landfill vents D. Agricultural field boundaries

💡 **Show answer**

Logic: FGD hardware removes SO_2 from stack gas before release — TPP acid-rain prevention. **Ans: B.** FGD passes flue gas through lime/limestone slurry at **coal thermal power plants**, removing **90%+ SO_2**. **Trap:** Vehicle NOx uses **catalytic converters**, not FGD scrubbers.

Q8. UPSC pattern

Yellowing and pitting of the **Taj Mahal** marble is primarily attributed to—

A. Carbon dioxide from tourist breathing B. **SO_2 pollution** and acid deposition forming gypsum crust C. Ozone hole ultraviolet alone D. Plastic waste micro-abrasion only

💡 **Show answer**

Logic: Taj Trapezium classic case — sulphuric acid attack on calcium carbonate marble. **Ans: B.** **SO_2** from Mathura refinery and Firozabad industries caused **gypsum ($CaSO_4$) crust** on marble — acid deposition damage. **Trap:** CO_2 is present everywhere but **SO_2 + acid rain** is the documented Taj damage pathway.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to acid rain, which of the following statements is/are correct?

1. Acid rain means wet or dry acidic deposition with pH below 5.6.
2. Normal unpolluted rain is exactly pH 7 because CO_2 never dissolves in rain.
3. Normal rain is exactly pH 7.

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

Show answer

Ans: B. Only 1 is correct. **Logic:** Statement 1 is the acid-rain <5.6 definition. Statement 2 is false — normal rain is already ~pH 5.6 from $CO_2 \rightarrow H_2CO_3$. Statement 3 repeats the pH-7 trap.

Q2. Match List-I with List-II:

List-I	List-II
A. Wet deposition	1. Gases/particles without rain
B. Dry deposition	2. Acids in rain/snow/fog
C. SO_2 path	3. → H_2SO_4

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

Show answer

Ans: C. A-2, B-1, C-3. **Logic:** Wet rain; dry particles; $SO_2 \rightarrow H_2SO_4$.

Q3. Which of the following pairs is/are correctly matched?

1. NOx → HNO_3 comes mainly from quiet soil respiration, never from vehicles or high-temperature combustion.
2. $SO_2 \rightarrow H_2SO_4$ — coal TPPs, smelters, refineries
3. CO_2 → severe acid rain main culprit

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

Show answer

Ans: A. Only 2 is correct. **Logic:** Statement 2 is true ($SO_2 \rightarrow H_2SO_4$ from coal TPPs/smelters). Statement 1 denies the vehicle/combustion NOx pathway. Statement 3 is false — CO_2 sets weak baseline rain, not severe acid rain.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Freshwater below about pH 5 stresses fish and can release toxic Al^{3+}.

Reason (R): Each pH step down multiplies H^+ by 10.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; logarithmic scale fact does not specifically explain fish/Al^{3+} stress. **A/R logic:** Related chemistry; R not the biological explanation.

Q5. With reference to receptors, which of the following statements is/are correct?

1. Conifers are among the least sensitive forest receptors to acid deposition.
2. Marble/limestone can form gypsum crust under sulphuric attack.
3. Lichens thrive best in high SO_2 air.

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Statement 2 is true (marble/limestone → gypsum crust). Statement 1 is false — conifers are especially sensitive. Statement 3 is false — lichens decline in high SO_2 air.

Q6. Arrange prevention tools teaching:

A. Only ban all rain B. FGD scrubbers, catalytic converters, low-sulphur coal, cleaner fuels/BS-VI C. Increase SO_2 emissions only D. Remove all oxygen from air

💡 **Show answer**

Ans: B. FGD, converters, cleaner fuels. **Logic:** Standard prevention set.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Gothenburg Protocol — acidifying emissions under CLRTAP family
2. Montreal — ozone ODS
3. Gothenburg — identical to Montreal ozone treaty

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Different problems.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Taj marble → gypsum is a classic Indian acid-deposition teaching fact.

Reason (R): The Taj Trapezium is a pollution-control belt designed to protect the Taj.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; Trapezium policy does not chemically explain gypsum formation. **A/R logic:** Chemistry vs policy belt.

Q9. Match List-I with List-II:

List-I	List-II
A. Acidic	1. =7
B. Neutral	2. <7
C. Basic	3. >7

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

💡 **Show answer**

Ans: D. A-2, B-1, C-3. **Logic:** Acidic <7; neutral 7; basic >7.

Q10. With reference to soil effects, which of the following statements is/are correct?

1. Acid deposition can leach Ca^{2+} and Mg^{2+}.
2. It can free toxic aluminium.
3. Acid rain makes all soils alkaline forever.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Statement 3 false.

Q11. Arrange CLRTAP → Gothenburg teaching years:

A. 1999 → 1979 B. 1979 → 1999 C. 1987 → 2016 D. 1972 → 1992

💡 **Show answer**

Ans: B. CLRTAP 1979 → Gothenburg 1999. **Logic:** Acidifying protocol family.

Q12. Which one of the following correctly states CO_2's role in rain chemistry?

A. Main culprit of severe acid rain always B. Makes rain strongly alkaline pH 11 C. Sets ~5.6 baseline via weak carbonic acid D. Identical to H_2SO_4 pathway

💡 **Show answer**

Ans: C. Baseline ~5.6 via H_2CO_3. **Logic:** Not severe-acid main path.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Acid rain is a secondary-pollutant story built from primary SO_2 and NOx.

Reason (R): SO_2 and NOx are emitted as primary pollutants then form strong acids.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Primary → secondary acid pathway.

Q14. With reference to SO_2 travel, which of the following statements is/are correct?

1. SO_2 is denser than air and can travel long distances before deposition.
2. Transboundary acidification is why CLRTAP/Gothenburg matter.

3. Acid rain never crosses borders.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Transboundary is real.

Q15. How many of the following are true?

1. Acid rain pH <5.6
2. Normal rain ~5.6
3. Neutral =7

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Core pH facts.

Q16. Which of the following pairs is/are correctly matched?

1. FGD — flue-gas desulphurisation with lime/limestone teaching
2. Catalytic converters — cut vehicle NOx/others
3. Increasing coal sulphur — best acid-rain prevention

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Higher sulphur worsens problem.

Q17. With reference to wet vs dry deposition, which of the following statements is/are correct?

1. Wet = rain/snow/fog acids.
2. Dry = acidic gases/particles without rain.
3. Dry deposition cannot occur.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Dry deposition is real.

Q18. Match List-I with List-II:

List-I	List-II
A. H_2SO_4 path	1. NOx
B. HNO_3 path	2. SO_2
C. Baseline weak acid	3. $CO_2 \rightarrow H_2CO_3$

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

💡 **Show answer**

Ans: D. A-2, B-1, C-3. **Logic:** SO_2 sulphuric; NOx nitric; CO_2 baseline.

Q19. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Calling normal rain alkaline is false.

Reason (R): Normal rain is already slightly acidic near pH 5.6.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Baseline acidity.

Q20. Which of the following statements about monuments is/are correct?

1. Sulphuric attack on marble can yield gypsum ($CaSO_4$).
2. Taj Trapezium aims to protect the Taj from pollution.
3. Acid rain only affects oceans, never stone.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Stone damage is classic.

Q21. Arrange sources to acids:

1. Coal TPP $SO_2 \rightarrow H_2SO_4$
2. Vehicles NOx $\rightarrow HNO_3$

A. Only 1 B. Both 1 and 2 C. Only 2 D. Neither

💡 **Show answer**

Ans: B. Both. **Logic:** Two main strong-acid paths.

Q22. With reference to lichens, which of the following statements is/are correct?

1. They are SO_2 bio-indicators.
2. They die in high SO_2.
3. They prove rain is always pH 7.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Statement 3 false.

Q23. Which one of the following treaty tags is correct?

A. Kyoto = Taj Trapezium statute B. Gothenburg = ozone hole only C. Montreal = acid rain only D. Gothenburg/CLRTAP = acidifying air pollution

💡 **Show answer**

Ans: D. Gothenburg/CLRTAP = acidifying air. **Logic:** Keep treaty tags straight.

Q24. How many of the following prevention tools are valid?

FGD, catalytic converters, low-sulphur coal, BS-VI cleaner fuel norms

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** Standard toolkit.

Q25. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Acid rain includes dry deposition.

Reason (R): Dry deposition requires rainfall by definition.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Dry = without rain.

Q26. With reference to pH mathematics teaching, which of the following statements is/are correct?

1. Each step down multiplies H^+ by 10.
2. pH 4 has 10× H^+ of pH 5.
3. Higher pH always means more acidic.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Lower pH = more acidic.

Q27. Which of the following pairs is/are correctly matched?

1. Smelters/refineries — SO_2 sources
2. High-temperature combustion — NOx sources
3. Pure argon air — main acid-rain precursor

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Argon not acid precursor.

Q28. With reference to fish stress, which of the following statements is/are correct?

1. Waters below ~pH 5 can kill/stress fish.
2. Toxic Al^{3+} release is linked to acidification.
3. Acid lakes are always safer for fish than neutral lakes.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Statement 3 false.

Q29. Arrange false statements that exams love:

1. “Normal rain is alkaline”
2. “Acid rain is alkaline”

A. Both are false traps B. Only 2 is a false trap C. Only 1 is a false trap D. Both are true

💡 **Show answer**

Ans: A. Both are false traps. **Logic:** Normal ~5.6 acidic; acid rain more acidic.

Q30. With reference to secondary pollutant framing, which of the following statements is/are correct?

1. Strong acids form from primary SO_2/NOx.
2. Acid rain/deposition is secondary in nature.
3. Calling SO_2 from a chimney a secondary pollutant is always correct.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** SO_2 from chimney is primary.

Mains Answer Framework

Acid rain (250 words): Define acid rain (pH < 5.6) and distinguish from normal rain (~5.6 from CO_2). Explain wet vs dry deposition and secondary acid formation from SO_2 and NOx. Causes — coal TPPs, vehicles, smelters; India/UP hotspots (Singrauli, Taj Trapezium). Effects — aquatic (pH < 5, Al^{3+}), soil nutrient loss, conifer damage, Taj marble gypsum crust, health, lichen bio-indicators. Prevention — FGD, catalytic converters, BS-VI, CNG, emission standards, Gothenburg/CLRTAP, Taj Trapezium SC order. Conclude with transboundary nature and treaty distinction (Montreal ≠ acid rain).

Common Traps — Don't Fall For These

Trap	Correct fact
Normal rain pH = 7	**~5.6** from CO_2
CO_2 = main acid rain cause	**Weak acid only**; SO_2/NOx = strong acids
Coal acid rain explained by CO_2 alone	A true (coal → acid rain via **SO_2**); R true (CO_2 emitted) but R **does not explain** A
Acid rain = rain only	**Wet + dry deposition**
Acidic pH > 7	**Acidic < 7**
Montreal = acid rain	**Gothenburg / CLRTAP**
SO_2 lighter than air	**Heavier**
Lichens thrive in pollution	**Die in SO_2**
Tall chimney = prevention	**Dispersal only**
Taj damage from CO_2	**SO_2 + acid deposition**
Acid rain stays in source country	**Transboundary**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 13 — Desertification (मरुस्थलीकरण) & Land Degradation (भू-क्षरण)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Desertification | Land Degradation | Soil Erosion

Sources baked in: NCERT Geography/Biology, ISRO Desertification Atlas 2016, UNCCD, MoEFCC/NRSA, UPPCS PYQs 2018–2025 **Weight:** ★★ Medium — concept traps, erosion sequence, Chambal ravines, UNCCD **Last verified:** August 2026

Current Affairs (this topic)

- UNCCD / LDN and Indian desertification atlas updates are the CA refresh.
- Bundelkhand / Rajasthan land-degradation colour stays high-yield for UP.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** areas from climate variation plus human pressure. 2. Desertification is **not** only "desert expanding." Any dryland losing productivity can count. 3. **Land degradation** is broader: decline in productive capacity on any land through erosion, salinization, waterlogging, mining, or pollution. 4. India has about **96.4 million ha** degraded land — roughly **29–30%** (ISRO Atlas **2016** class figure). 5. **Normal geological erosion** is slow and roughly balanced by soil formation. **Accelerated erosion** is human-driven and faster than formation — the focus. 6. Water-erosion sequence: **sheet → rill → gully → ravine**. Chambal (चंबल) badlands are the ravine end-member. 7. Wind erosion works by **deflation** (lifting) and **abrasion** (sandblasting). 8. Leading causes include overgrazing in drylands, deforestation, unsustainable farming, excessive irrigation (salinization/waterlogging), and mining. 9. India hotspots: **Thar** (थार) wind erosion; **Chambal** (चंबल) ravines (MP–UP–Rajasthan); **Bundelkhand** drought (सूखा)–degradation cycle; Punjab–Haryana salinization belts. 10. Conservation tools include contour bunding, terraces, shelter belts, afforestation, gully plugging, check dams, and watershed management (वाटरशेड (जलग्रहण) प्रबंधन). 11. Earthworms are soil-fertility allies in restoration narratives. 12. **UNCCD** was adopted (अंगीकृत) in **1994**. Desertification and Drought Day is **17 June**. 13. **LDN** (Land Degradation Neutrality) under **SDG 15.3** targets no net loss of healthy land by **2030**. 14. **DPAP** launched in **1973**. **DDP** is **1977–78**. **IWMP** is **2009**. **MGNREGA (2005)** funds bunds and check dams. 15. Treaty trap: **UNCCD ≠ UNFCCC (यूएनएफसीसीसी)**. Desertification is not the climate convention. 16. Afforestation helps **prevent** degradation; it is not a cause of desertification. 17. Gully erosion (अवनलिका अपरदन) is not the same as sheet erosion (परत (चादर) अपरदन) — sheet is the first thin uniform stage. 18. Chambal ravine area is about **3 million ha** in standard coaching numbers. 19. UP Chambal ravine districts include **Etawah (इटावा)** and Agra** (with Morena–Bhind on the MP side). 20. Bundelkhand's **seven UP districts** are the drought-plus-degradation fact. 21. Thar-margin UP questions emphasise wind erosion and dune encroachment. 22. **Sonbhadra** (सोनभद्र) is a UP mining-related land-degradation hotspot. 23. Wasteland means degraded land poorly suited to cultivation — related to, but not identical with, desertification's ASDM definition. 24. LDN by 2030 is an SDG land target; do not date UNCCD itself as 2030.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Desertification** | **Land Degradation** | Persistent reduction or loss of biological and economic productivity occurring specifically in Arid, Semi-Arid, and Dry Sub-Humid areas (drylands) vs general decline in the productive capacity and ecosystem health of any land anywhere | मरुस्थलीकरण (शुष्क भूमियों में) / भू-क्षरण (सामान्य भूमि अवनति) | | **Drought** | **Desertification** | Temporary natural meteorological period of deficient precipitation over months or a few years from which ecosystems can recover vs persistent, semi-permanent ecological degradation and productivity loss driven by unsustainable human use and climate stress | सूखा (अस्थायी मौसमी संकट) / मरुस्थलीकरण (स्थायी भूमि ह्रास) | | **Sheet Erosion** | **Gully Erosion** | Uniform, imperceptible removal of a thin layer of topsoil over large flat cultivated fields by sheetwash runoff vs cutting of deep, severe channels and ravines by concentrated runoff carving badland topography (e.g. Chambal ravines) | परत (चादर) अपरदन / अवनलिका अपरदन (बीहड़ निर्माण) | | **Splash Erosion** | **Rill Erosion** | Initial detachment and airborne displacement of soil particles caused directly by the physical impact of falling raindrops vs intermediate erosion stage where tiny, shallow, visible finger-like channels are formed on sloped ground | आघात (स्पलैश) अपरदन / क्षुद्र सरिता (रिल) अपरदन | | **UNCCD (1994)** | **UNFCCC (1992)** | Sole legally binding international treaty linking environment and development to sustainable land management, desertification, and drought in drylands vs framework convention dedicated to addressing anthropogenic climate change and greenhouse gases | मरुस्थलीकरण रोकथाम संधि (UNCCD) / जलवायु परिवर्तन संधि (UNFCCC) | | **Land Degradation Neutrality (LDN)** | **Net Zero Emissions** | Target under UN SDG 15.3 ensuring that the quantity and quality of land resources necessary to support ecosystem functions remain stable or increase by 2030 vs balancing greenhouse gas emissions with equivalent removals | भूमि क्षरण तटस्थता (LDN, SDG 15.3) / नेट जीरो उत्सर्जन |

💡 📋 **Must-score facts — drylands, erosion, UNCCD, schemes (Click to Expand)**

Concepts | Item | Lock | |------|------| | Desertification | Degradation in arid–semi-arid–dry sub-humid (ASDM) | | ≠ expanding desert only | Any dryland productivity loss can count | | India degraded land | ~**96.4 mha**; ~**29–30%** (ISRO Atlas 2016) | | Water erosion sequence | Sheet → rill → gully → ravine (Chambal) | | Wind tools | Deflation + abrasion | | UNCCD / Day | **1994** / **17 June** | | LDN | SDG **15.3**; no net loss by **2030** | | DPAP / DDP / IWMP | **1973** / **1977–78** / **2009** |

13.1 Desertification

Identity: ASDM drylands | Productivity loss | Overgrazing | UNCCD 1994

- **Desertification** is **land degradation in arid, semi-arid, and dry sub-humid (ASDM) areas** caused by climate variation and human activities.
- It is **not** simply existing deserts like the Thar expanding. Any qualifying dryland that loses fertility and vegetation can desertify.
- Drylands cover about **41% of Earth's land** and support roughly **2 billion people**.
- **Overgrazing** is the **main human cause** in drylands. Livestock remove vegetation, compact soil, and expose topsoil to wind and water erosion.
- **Deforestation**, **fuelwood collection**, and **unsustainable cropping** remove root binding and reduce local moisture retention.
- **Improper irrigation** can cause **salinization** and **waterlogging**, turning fertile irrigated land barren — common in parts of Rajasthan canal commands and the Indo-Gangetic fringe.
- **Drought and climate change (जलवायु परिवर्तन)** kill vegetation and trigger flash erosion on bare soil, creating a feedback loop of further degradation.
- **Wind erosion** dominates true arid zones like the **Thar Desert**, where dunes migrate and bury fields.

- Early desertification can be **reversed** through afforestation, controlled grazing, and water harvesting. Advanced stages may take decades to restore.
- India's **ISRO Desertification Atlas (2016)** mapped **96.4 million hectares (~29.3% of geographical area)** under desertification/land degradation. **Rajasthan** (थार) is the worst-affected state.

> **Logic:** Desertification applies to **dry sub-humid Bundelkhand**, not only sand deserts.

- The **UNCCD (United Nations Convention to Combat Desertification)** was adopted on **17 June 1994** in Paris and entered force in **1996**.
- **17 June** is observed as **World Desertification and Drought Day**.
- **SDG 15.3** targets **Land Degradation Neutrality (LDN) (भूमि क्षरण तटस्थता) by 2030** — avoid new degradation and restore degraded land.

PYQ — UPPCS Prelims 2019, Q42

India aims to achieve land degradation neutrality by:

A. 2025

B. 2030

C. 2035

D. 2040

💡 **Show answer**

Ans: B — SDG 15.3 target year **2030**.

Concept	Scope
Desertification	ASDM drylands only
Drought	Rainfall deficit event
Land degradation	All land types, all processes

13.2 Land Degradation

Identity: Productive capacity decline | Erosion + salinity + mining | Wasteland end-stage

- **Land degradation** is the **broader decline in a land's productive capacity** from natural or human factors.
- **Desertification** is one **subset** limited to ASDM drylands. **Soil erosion** is one **major process** within degradation.
- **Human causes dominate** India's degradation statistics: overgrazing, deforestation, unsustainable agriculture, excessive fertilizers, mining, industrial pollution, and urban (नगरीय) sprawl.
- Major degradation **processes** include **soil erosion**, **waterlogging**, **salinization**, **alkalinization**, **mining spoil**, and **deforestation**.
- **Salinization** from excessive canal irrigation affects **Punjab, Haryana, and western UP**.
- **Mining degradation** is severe in **Sonbhadra (UP), Jharkhand, and Odisha**.
- Continued degradation creates **wasteland** — land unsuitable for cultivation. All wasteland is degraded, but not all degraded land is yet wasteland.
- Consequences include **food insecurity**, **rural poverty**, **migration** (Bundelkhand to cities), **reservoir silting**, and **higher flood risk** on silted riverbeds.
- **Salinization and waterlogging** can often be reversed through drainage, gypsum, and drip irrigation. **Deep ravines** are hardest to restore.

- **Afforestation prevents degradation** — it is **not** a cause. Pick it when asked "NOT a cause."
- Policy tools include **Integrated Watershed Management Programme (IWMP)**, **MGNREGA** (contour bunds, check dams), **DPAP**, **Desert Development Programme (DDP)**, and **Soil Health Card Scheme**.

> **Logic:** DPAP (1973)** targets drought-prone arid/semi-arid areas for soil moisture conservation.

Process	Example region
Salinization	Punjab-Haryana canal belt
Mining degradation	Sonbhadra, Jharkhand
Waterlogging	Over-irrigated plains
Ravine formation	Chambal basin

Soil Erosion 4-Stage Water Progression & Wind Transport Dynamics

13.3 Soil Erosion

Identity: A-horizon loss | Sheet→ravine | Wind deflation | Conservation measures

- **Topsoil (A-horizon)** is the fertile organic-rich layer, usually **15–30 cm deep**. It can take **centuries to form** but may be lost in **one heavy monsoon** on bare slopes.
- **Normal (geological) erosion** is slow and balanced by soil formation. **Accelerated erosion** from human activity is **10–100× faster** than formation — this is the usual default meaning of "soil erosion."
- **Water erosion** dominates humid and monsoon India. Raindrop impact dislodges soil and runoff carries it downslope.
- **Water erosion sequence:Sheet erosion** (thin uniform layer) → **rill erosion** (क्षुद्र सरिता (रिल) अपरदन) (small channels) → **gully erosion** (deep cuts) → **ravines** (extensive badland networks).
- **Chambal ravines** (चंबल) cover about **3 million hectares** across **MP (Morena, Bhind), UP (Etawah, Agra), and Rajasthan (Dholpur, Karauli)**. They are India's classic **gully/ravine erosion** landscape.
- **Wind erosion** dominates arid zones through **deflation** (lifting fine particles) and **abrasion** (sandblasting surfaces). It affects the **Thar**, Rann of Kutch, and western Rajasthan-UP margins.
- **Gravity erosion** (landslides, slumping) matters in **Himalayan and NE hill states**, especially after deforestation and road cutting.
- **Vegetation cover** is the best natural erosion shield. Deforestation and overgrazing are the top acceleration factors.
- **Conservation measures:contour bunding/ploughing**, **terrace farming**, **shelter belts/wind breaks**, **afforestation**, **gully plugging**, **check dams**, **strip cropping**, and **cover crops**.
- **Earthworms** aerate soil, improve structure, and boost fertility — healthy soil resists erosion.

PYQ — UPPCS Prelims 2025, Q73

1. Earthworms increase soil fertility and are known as friends of farmers.
2. Earthworms have two pairs of hearts.

A. Only 2

B. Neither

C. Both

D. Only 1

💡 **Show answer**

Ans: D — Statement 1 only. Earthworms have **5 pairs** of hearts, not 2.

Erosion stage	Feature
Sheet	Thin uniform topsoil loss
Rill	Small grooves after rain
Gully	Deep channels, uncrossable by plough
Ravine	Chambal-type badland maze
Agent	Main region
-------	----------
Water	Monsoon slopes, Chambal
Wind	Thar, arid west
Gravity	Himalaya (हिमालय), NE hills

Complete PYQ Bank (Topic 13)

Q1. UPPCS Prelims 2019, Q42

India aims to achieve **Land Degradation Neutrality (LDN)** by—

A. 2025 B. 2030 C. 2035 D. 2040

💡 **Show answer**

Logic: SDG 15.3 target year for no net loss of healthy land. **Ans: B.** LDN target year is **2030** under **SDG 15.3**. **Trap:** **2025** is a near-term distractor — LDN is pegged to the **2030 Agenda** timeline.

Q2. UPPCS Prelims 2025, Q73

Consider the following statements about **earthworms**:

1. Earthworms increase soil fertility and are known as friends of farmers.
2. Earthworms have **two pairs** of hearts.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

💡 **Show answer**

Logic: Soil-biology fact plus anatomy trap — earthworms aid fertility but have five aortic arches, not two hearts. **Ans: D.** Statement 1 is correct; earthworms have **5 pairs of hearts** (aortic arches), not 2. **Trap:** "Friends of farmers" makes statement 1 obviously true — statement 2's wrong heart count is the eliminator.

Q3. UPPCS Prelims 2025, Q134

Arrange the following programmes **chronologically**:

1. Food for Work Programme
2. Community Development Programme (CDP)
3. Training of Rural Youth for Self-Employment (TRYSEM)
4. Drought Prone Areas Programme (DPAP)

A. 4, 2, 3, 1 B. 2, 4, 1, 3 C. 2, 4, 3, 1 D. 4, 2, 1, 3

💡 **Show answer**

Logic: Dryland/rural development scheme chronology — CDP earliest, TRYSEM latest in the set. **Ans: B.** Order: CDP **1952** → DPAP **1973** → Food for Work **1977** → TRYSEM **1979**. **Trap:** **1952** belongs to **CDP**, not DPAP — DPAP is **1973**.

Q4. UPSC pattern

Which of the following is **NOT** a cause of land degradation?

A. Overgrazing B. Unsustainable mining C. **Afforestation** D. Improper irrigation leading to salinization

💡 **Show answer**

Logic: Cause vs remedy trap — tree planting prevents degradation; other options accelerate it. **Ans: C.** **Afforestation** prevents degradation; overgrazing, mining, and salinization **cause** it. **Trap:** Deforestation causes degradation — **afforestation** is the opposite intervention.

Q5. UPSC pattern

The international treaty to combat **desertification** is—

A. UNFCCC B. **UNCCD** C. Montreal Protocol D. MARPOL

💡 **Show answer**

Logic: Treaty-purpose matching — UNCCD = desertification; UNFCCC = climate. **Ans: B.** **UNCCD (1994, Paris)** is the desertification convention; World Desertification Day is **17 June**. **Trap:** **UNFCCC** is climate — do not swap the two "UN" environmental treaties.

Q6. UPSC pattern

A major consequence of **soil erosion** in river basins is—

A. Increased stratospheric ozone B. **Silting of rivers and reservoirs** C. Expansion of coral reefs D. Permanent ocean acidification reversal

💡 **Show answer**

Logic: Erosion sediment pathway — topsoil enters streams and reduces dam storage capacity. **Ans: B.** Eroded soil **silts rivers and reservoirs**, raising flood risk and cutting irrigation/hydropower storage. **Trap:** Erosion removes topsoil on land first — **downstream silting** is the classic basin-scale consequence.

Q7. UPSC pattern

The **Chambal ravines** are a classic example of—

A. Wind deflation in the Thar Desert B. **Gully/ravine erosion** across MP–UP–Rajasthan C. Glacier-scoured valleys in the Himalaya D. Coastal retreat from sea-level rise

💡 **Show answer**

Logic: Badland ravine geography — water erosion end-stage in soft alluvial banks of the Chambal. **Ans: B.** Chambal badlands (~3 million ha) span **MP, UP, and Rajasthan** — advanced **gully/ravine erosion**. **Trap:** Thar has **wind erosion** — Chambal is **water/gully** erosion, not aeolian deflation.

Q8. UPSC pattern

Which pair is **correctly** matched?

A. Thar Desert — wind erosion hotspot B. Chambal — stratospheric ozone hole C. **MARPOL — marine pollution**; **UNCCD — desertification** D. Afforestation — primary cause of land degradation

> 💡 **Show answer**
>
> **Logic:** Treaty and process matching — MARPOL vs UNCCD is a standard confused-pair trap. **Ans: C.** **MARPOL** regulates **ship-source marine pollution**; **UNCCD** addresses **land degradation/desertification**. **Trap:** UNCCD is **not** marine pollution — that is **MARPOL's** domain.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to desertification, which of the following statements is/are correct?

1. It is land degradation in arid, semi-arid, and dry sub-humid (ASDM) areas.
2. It is only "the desert expanding physically."
3. Desertification is driven only by pure geology, never by climate variation or human pressure.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 is correct. **Logic:** Statement 1 is the UNCCD-style ASDM definition. Statement 2 is the classic trap (not merely desert expansion). Statement 3 denies climate + human drivers.

Q2. Which of the following pairs is/are correctly matched?

1. Land degradation applies only inside hot deserts and never on other lands.
2. Desertification — ASDM dryland subset
3. Desertification — identical to all waterlogging worldwide only

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 2 is correct. **Logic:** Statement 2 is true (desertification = ASDM subset). Statement 1 is false — land degradation is the broader productivity decline on any land. Statement 3 is false — desertification ≠ all waterlogging worldwide.

Q3. With reference to India degraded land, which of the following statements is/are correct?

1. About 96.4 million ha / ~29–30% (ISRO Desertification Atlas 2008 class).
2. Rajasthan is often the worst state hotspot teaching.
3. India has zero degraded land.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 2 is correct. **Logic:** Statement 2 is true (Rajasthan hotspot teaching). Statement 1 uses the right magnitude but the wrong atlas year — ISRO Desertification Atlas class figure is **2016**, not 2008. Statement 3 is false.

Q4. Arrange water-erosion sequence:

A. Sheet → rill → gully → ravine B. Ravine → gully → rill → sheet C. Gully → sheet → ravine → rill D. Rill → sheet → ravine → gully

💡 **Show answer**

Ans: A. Sheet → rill → gully → ravine. **Logic:** Chambal = ravine end-member.

Q5. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Chambal badlands are the ravine end-member of water erosion.

Reason (R): Wind erosion works by deflation and abrasion.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; wind processes do not explain Chambal ravines. **A/R logic:** Water vs wind tools.

Q6. Match List-I with List-II:

List-I	List-II
A. Thar	1. Salinization belt teaching
B. Chambal	2. Wind erosion
C. Punjab–Haryana	3. Ravines

Code:

A. A-3, B-2, C-1 B. A-2, B-3, C-1 C. A-2, B-1, C-3 D. A-1, B-3, C-2

💡 **Show answer**

Ans: B. A-2, B-3, C-1. **Logic:** Thar wind; Chambal ravines; Punjab–Haryana salinity.

Q7. With reference to UNCCD, which of the following statements is/are correct?

1. Adopted in 1994.
2. Desertification and Drought Day is 17 June.
3. UNCCD is identical to UNFCCC.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Different conventions.

Q8. Which of the following pairs is/are NOT correctly matched?

1. LDN — no net loss of healthy land by 2030 (SDG 15.3)
2. DPAP — 1973
3. DPAP — 1952

A. 1 and 2 only B. 1, 2 and 3 C. 2 and 3 only D. 3 only

💡 **Show answer**

Ans: D. Only pair 3 is not correctly matched. **Logic:** DPAP = 1973 not 1952.

Q9. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Accelerated erosion is human-driven and faster than soil formation.

Reason (R): Normal geological erosion is slow and roughly balanced by formation.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; R contrasts normal erosion but does not define accelerated erosion's human driver as explanation. **A/R logic:** Contrast pair.

Q10. Arrange schemes: DPAP → DDP → IWMP teaching:

A. 1973 → 1977-78 → 2009 B. 2009 → 1973 → 1977-78 C. 1977-78 → 1973 → 2009 D. 1973 → 2009 → 1977-78

💡 **Show answer**

Ans: A. 1973 → 1977–78 → 2009. **Logic:** DPAP–DDP–IWMP.

Q11. With reference to Bundelkhand, which of the following statements is/are correct?

1. It is a classic UP drought–degradation hotspot (seven districts teaching).
2. Overgrazing and water stress reinforce degradation.
3. Bundelkhand is a coastal mangrove biosphere in Kerala.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not Kerala mangroves.

Q12. Match List-I with List-II:

List-I	List-II
A. Deflation	1. Sandblasting
B. Abrasion	2. Lifting of particles
C. Shelter belts	3. Wind-erosion control

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Deflation lift; abrasion sandblast; shelter belts control.

Q13. Which one of the following is a leading dryland cause in coaching facts?

A. Montreal Protocol B. Building more check dams always worsens deserts only C. Earthworms causing deserts D. Overgrazing

💡 **Show answer**

Ans: D. Overgrazing. **Logic:** Top dryland driver teaching.

Q14. With reference to conservation tools, which of the following statements is/are correct?

1. Contour bunding, terraces, check dams, watershed management help.
2. Afforestation and gully plugging help.
3. Earthworms are soil-fertility allies in restoration narratives.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: A.** All three. **Logic:** Standard restoration toolkit.

Q15. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Excessive irrigation can cause salinization and waterlogging.

Reason (R): Salinity from poor irrigation is a human-made degradation path.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains A. **A/R logic:** Irrigation mismanagement path.

Q16. How many of the following India hotspots are correctly paired?

1. Thar — wind
2. Chambal — ravines
3. Bundelkhand — drought–degradation

A. Only one B. All three C. Only two D. None

> 💡 **Show answer**
>
> **Ans: B.** All three. **Logic:** Classic hotspot map.

Q17. With reference to MGNREGA link, which of the following statements is/are correct?

1. MGNREGA (2005) can fund bunds and check dams.
2. It can support watershed-type works.
3. MGNREGA repealed UNCCD.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** Does not repeal UNCCD.

Q18. Which of the following pairs is/are correctly matched?

1. SDG 15.3 — LDN
2. UNCCD — desertification convention
3. UNFCCC — desertification-only treaty

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** UNFCCC is climate.

Q19. Arrange meteorological → agricultural → hydrological drought idea:

A. Meteorological → agricultural → hydrological B. Hydrological → meteorological → agricultural C. Agricultural → hydrological → meteorological only always reverse D. No sequence exists

> 💡 **Show answer**
>
> **Ans: A.** Met → agri → hydro. **Logic:** Standard drought ladder (variants exist; this is teaching order used here).

Q20. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Desertification and drought reinforce each other in a feedback loop.

Reason (R): DPAP began in 1973 to address drought-prone areas.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; DPAP year does not explain the feedback loop. **A/R logic:** Related but not explanation.

Q21. With reference to sheet erosion, which of the following statements is/are correct?

1. It is the earliest stage in the water-erosion sequence teaching.
2. Ravine is deeper than gully in the sequence.
3. Sheet erosion comes after ravine.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** Sheet is first — not after ravine.

Q22. Which of the following statements about wind erosion is/are correct?

1. Deflation lifts particles.
2. Abrasion sandblasts surfaces.
3. Wind erosion is impossible in the Thar.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** Thar is wind hotspot.

Q23. Match List-I with List-II:

List-I	List-II
A. DPAP	1. 2009
B. DDP	2. 1973
C. IWMP	3. 1977–78

Code:

A. A-1, B-3, C-2 B. A-3, B-2, C-1 C. A-2, B-1, C-3 D. A-2, B-3, C-1

💡 **Show answer**

Ans: D. A-2, B-3, C-1. **Logic:** 1973 / 1977–78 / 2009.

Q24. With reference to LDN, which of the following statements is/are correct?

1. Target year 2030.
2. Linked to SDG 15.3.
3. Means unlimited land degradation is a goal.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Neutrality = no net loss.

Q25. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Afforestation helps prevent further dryland degradation.

Reason (R): Deforestation is a leading degradation driver.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains why afforestation helps. **A/R logic:** Driver ↔ remedy.

Q26. How many of the following causes are standard?

Overgrazing, deforestation, unsustainable farming, excessive irrigation, mining/pollution pathways

A. Only two B. All five C. Only four D. Only three

💡 **Show answer**

Ans: B. All five. **Logic:** Leading cause set.

Q27. Which of the following UP-linked facts is/are correct?

1. Chambal ravines touch MP–UP–Rajasthan teaching.
2. Bundelkhand drought–degradation cycle.
3. UP has the densest tropical rainforest hotspot identical to Amazon.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Statement 3 false.

Q28. With reference to Day facts, which of the following statements is/are correct?

1. Desertification and Drought Day — 17 June
2. World Environment Day — 5 June
3. Desertification Day — 22 April

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** 22 April is Earth Day.

Q29. Which one of the following correctly states desertification geography?

A. ASDM drylands B. Only polar ice caps C. Only deep ocean trenches D. Only stratospheric ozone

💡 **Show answer**

Ans: A. ASDM drylands. **Logic:** Arid–semi-arid–dry sub-humid.

Q30. With reference to check dams/watershed, which of the following statements is/are correct?

1. They are conservation tools against erosion/degradation.
2. Watershed is land draining to a common outlet.
3. Check dams increase sheet erosion by design.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Check dams reduce erosion.

Mains Answer Framework

Desertification and land degradation (250 words): Define desertification (ASDM drylands, UNCCD) vs broader land degradation. Causes — overgrazing, deforestation, unsustainable agriculture, improper irrigation (salinization), mining, drought/climate. India data (~29% degraded, Rajasthan worst). Effects — food insecurity, migration, reservoir silting, biodiversity loss. Soil erosion types — sheet to ravine; Chambal case; wind erosion in Thar. Prevention — contour bunding, terraces, shelter belts, watershed management, afforestation, DPAP/IWMP/MGNREGA. International — UNCCD 1994, 17 June, SDG 15.3 LDN 2030. UP angle — Bundelkhand, Chambal ravines, Thar margin.

Common Traps — Don't Fall For These

Trap	Correct fact
Desertification = desert expansion only	**Any ASDM dryland losing productivity**
Land degradation = desertification	**Degradation is broader**
Afforestation causes degradation	**It prevents degradation**
Sheet erosion = deep ravines	**Sheet is first, mildest stage**
Chambal ravines in one state	**MP + UP + Rajasthan**
UNCCD = climate treaty	**UNCCD = desertification; UNFCCC = climate**
Normal rain pH 7 stops erosion	**Unrelated — erosion is topsoil removal**
Wind erosion in Chambal ravines	**Water/gully erosion**
LDN target 2025	**2030 (SDG 15.3)**
UNCCD signed at Rio 1992	**Adopted 1994, Paris**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 14 — Environmental Impact Assessment

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Environmental Impact Assessment (EIA) (पर्यावरण प्रभाव आकलन) | Strategic Environmental Assessment (SEA) (रणनीतिक पर्यावरण आकलन)

Sources baked in: NCERT Geography/Biology, EPA 1986, EIA Notification 2006, NGT Act 2010, UNECE SEA Protocol 2003, PARIVESH, UPPCS/UPSC PYQs **Weight:** ★★ Medium–High — Category A/B traps, four stages, MoEFCC vs SEIAA vs CPCB, EIA vs SEA **Last verified:** August 2026

Current Affairs (this topic)

- EIA notification amendments and Category A/B traps are the living CA layer.
- Keep screening–scoping–appraisal–monitoring order stable.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. **EIA** studies a project's likely environmental effects **before** approval. The aim is anticipatory prevention, not reactive cleanup. 2. Legal base is the **Environment (पर्यावरण) (Protection) Act, 1986** (Section **3**). First EIA Notification was **1994**; the current framework is **EIA Notification 2006 (14 September)**. 3. **Environmental Clearance** is mandatory prior clearance for listed projects. 4. **Category A** is appraised centrally by **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) through the **EAC**. 5. **Category B** is appraised by **SEIAA** through **SEAC**. **B1** needs full EIA (usually with public hearing); **B2** is simplified. 6. Under a **General Condition**, a Category B project in a sensitive area is treated as Category A. 7. The four (चातुर्याम) classic stages are **screening → scoping → public consultation → appraisal**. Screening applies to Category B only. 8. **Scoping** fixes the **Terms of Reference (ToR)** (संदर्भ की शर्तें) for the EIA study. 9. An EIA report covers baseline, impact prediction, mitigation, and an **Environmental Management Plan (EMP)** (पर्यावरण प्रबंधन योजना). 10. **Rapid EIA** uses one season; **comprehensive EIA** covers all seasons. 11. MoEFCC/SEIAA **grant** EC. EAC/SEAC only **recommend** (सिफारिश). CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड)/SPCB set standards and monitor — they do **not** grant EC. 12. **NGT (Act 2010)** hears EC and pollution appeals. **PARIVESH** is MoEFCC's online EC portal. 13. **SEA** assesses policies, plans, and programmes upstream of projects. India has project-level EIA law but **no standalone SEA statute**. 14. SEA references include the Kiev SEA Protocol **2003** (Espoo family) and the EU SEA Directive. 15. Rio Principle **17** supports EIA for projects with significant impacts. 16. EC must come **before** construction. Ex-post facto clearance is illegal in the standard line. 17. Timeline: Bhopal **1984** → EPA **1986** → EIA Notification **1994** → EIA Notification **2006** → NGT Act **2010** → draft EIA amendment debates **2020** (not a full replacement of 2006). 18. UP examples: Ganga (गंगा) Expressway as Category A; Singrauli TPP expansions; Upper Ganga ESZ plan-level restrictions before project EC. 19. EIA is project-level; SEA is plan/policy-level — do not merge the acronyms. 20. Water Act **1974**/Air Act **1981** create consent regimes; they are **not** the EIA enabling Act (**EPA 1986** (पर्यावरण संरक्षण 1986) is). 21. DEIAA appears in the minor-minerals / B2 district track — not as the Category A central authority. 22. Public consultation includes hearing plus written comments; skipping it on B1 where required is a process defect. 23. EMP is mitigation and monitoring inside the EIA package — not a substitute for the EC grant itself. 24. PARIVESH tracks applications; it does not replace NGT's appellate role.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Environmental Impact Assessment (EIA)** | **Strategic Environmental Assessment (SEA)** | Formal analytical process evaluating the potential ecological impacts of a specific *project* (e.g. thermal plant, highway) before sanction vs higher-level systemic evaluation applied to government *policies, plans, and programmes* | पर्यावरण प्रभाव आकलन (EIA, परियोजना स्तर) / रणनीतिक पर्यावरण आकलन (SEA, नीति स्तर) | | **Category A Projects** | **Category B Projects** | Large-scale developmental projects requiring mandatory full-scale EIA appraised centrally at Union level by MoEFCC on recommendations of EAC vs smaller projects appraised at State level by SEIAA on recommendations of SEAC | श्रेणी 'ए' (केंद्रीय मूल्यांकन) / श्रेणी 'ब' (राज्य मूल्यांकन) | | **Category B1 Projects** | **Category B2 Projects** | State-level projects that undergo complete EIA process and mandatory public hearing due to significant ecological footprint vs smaller Category B projects exempted from EIA study and public consultation, appraised solely on application summary | श्रेणी 'ब1' (EIA अनिवार्य) / श्रेणी 'ब2' (EIA से छूट) | | **Terms of Reference (ToR)** | **Environmental Management Plan (EMP)** | Comprehensive boundary document outlining all environmental baseline issues and studies that the project proponent must investigate vs actionable mitigation, monitoring, and institutional plan submitted in the final EIA report | संदर्भ की शर्तें (ToR, आरंभिक ढांचा) / पर्यावरण प्रबंधन योजना (EMP, शमन कार्ययोजना) | | **Public Hearing (Consultation)** | **Appraisal** | Statutory public stage where local affected communities and citizens voice concerns and objections to the draft EIA vs detailed scientific scrutiny of final EIA and public hearing proceedings by EAC or SEAC | जन सुनवाई (स्थानीय सहभागिता) / वैज्ञानिक मूल्यांकन (EAC/SEAC) | | **National Green Tribunal (NGT)** | **Appellate Authority under Air/Water Acts** | Specialized judicial tribunal established under NGT Act 2010 hearing original environmental disputes and appeals against environmental clearances (EC) vs administrative appellate authorities designated under Water Act 1974 / Air Act 1981 | राष्ट्रीय हरित अधिकरण (न्यायिक अधिकरण) / राज्य अपीलीय प्राधिकरण |

💡 📋 **Must-score facts — EIA stages, categories, SEA (Click to Expand)**

Framework | Item | Lock | |---|---| | Legal base | EPA **1986**; Notification **1994** → **2006** frame | | Category A / B | MoEFCC–EAC central / SEIAA–SEAC state | | B1 / B2 | Full EIA (+ hearing usual) / simplified | | General Condition | Sensitive-area B treated as A | | Four stages | Screening → scoping → public consultation → appraisal | | Screening | Category **B** only | | Rapid vs comprehensive | One season / all seasons | | EC granter | MoEFCC/SEIAA (EAC/SEAC recommend only) | | NGT / PARIVESH | Act **2010** appeals / MoEFCC online portal | | SEA | Policies–plans–programmes; India has no standalone SEA statute | --- ![EIA Notification 2006 Statutory Stages & Categorization Architecture] (images/env_ch14_01_eia_notification_2006_workflow.png)

14.1 Environmental Impact Assessment (EIA)

Identity: EPA 1986 | EIA Notification 2006 | Category A/B | Four stages | EC before construction

- **Environmental Impact Assessment (EIA)** is a systematic study of a proposed project's likely environmental, social, and health effects **before** major decisions are taken.
- The goal is **anticipatory prevention** — reject, redesign, or condition projects rather than repair damage after construction.
- The **Environment (Protection) Act, 1986** is the legal base. **Section 3** empowers the Centre to notify rules requiring prior clearance.
- The **first EIA Notification** came in **1994**. The current framework is the **EIA Notification 2006** dated **14 September 2006**.
- **Environmental Clearance (EC)** is mandatory for **listed projects** in the Schedule before construction or land preparation.

- **Category A** projects are large, high-impact projects appraised centrally by **MoEFCC** through the **Expert Appraisal Committee (EAC)**.
- **Category B** projects are appraised by the **State Environment Impact Assessment Authority (SEIAA)** through the **State Expert Appraisal Committee (SEAC)**.
- **Screening** applies to **Category B only** and splits projects into **B1** (full EIA required) or **B2** (no full EIA report).
- Under the **General Condition**, a Category B project in a **sensitive area** — national park, sanctuary, ESZ, CRZ, inter-state boundary, or critically polluted zone — is treated as **Category A**.
- The **four-stage process** runs: **(1) Screening**, **(2) Scoping** (ToR fixed), **(3) Public Consultation** (public hearing + written submissions), **(4) Appraisal** (EAC/SEAC recommendation).
- An **EIA report** covers baseline data, impact prediction, mitigation measures, and an **Environmental Management Plan (EMP)**.
- **Rapid EIA** uses one season of baseline data. **Comprehensive EIA** covers all seasons and is more reliable for monsoon-dry variation.
- **CPCB and SPCBs** set pollution standards and monitor compliance. They **do not grant EC**.
- **PARIVESH** is MoEFCC's online single-window portal for EC applications and tracking.
- **NGT (National Green Tribunal)**, under the **NGT Act 2010**, hears appeals against EC decisions.
- EC must be obtained **before** construction. **Ex-post facto clearance** after work has begun is illegal and controversial.

> **Logic:** Carrying capacity — waste generation must stay within the environment's absorption limit; that is what EIA evaluates.
>
> **Logic:** Org trap — **CGWB** reports groundwater quality; **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) does not grant EC. EC = **MoEFCC/SEIAA**.

Category	Authority	Screening?	Full EIA?
A	MoEFCC + EAC	No	Usually yes
B1	SEIAA + SEAC	Yes	Yes
B2	SEIAA/DEIAA	Yes	No full report
B + General Condition	MoEFCC (elevated)	—	As Category A
Stage	What happens		
---	---		
Screening	B → B1 or B2		
Scoping	ToR for EIA study		
Public Consultation	Public hearing + comments		
Appraisal	Recommend grant/reject EC		

14.2 Strategic Environmental Assessment (SEA)

Identity: PPP level | Upstream of EIA | No standalone Indian law | SEA Protocol 2003

- **Strategic Environmental Assessment (SEA)** evaluates the environmental consequences of **policies, plans, and programmes (PPP)** at the earliest planning stage.
- **SEA sits upstream of EIA**. It asks whether a development path is wise **before** individual projects are designed.
- **EIA** can only accept, modify, or reject **one pre-selected project**. It cannot question whether an entire sector plan — such as many dams in one basin — is environmentally sound.

- **SEA assesses cumulative impacts** when multiple projects together may overwhelm an ecosystem (पारिस्थितिकी तंत्र) even if each passes individual EIA.
- **Public participation** in SEA happens when policy choices are still flexible — earlier and cheaper than late-stage project hearings.
- **Rio Declaration (1992) Principle 17** calls for **EIA** for proposed activities with significant adverse impact. It does **not** mandate SEA by name.
- **Espoo Convention (1991)** covers **transboundary EIA**. The **SEA Protocol (2003)** signed at **Kiev** extends assessment to plans and programmes.
- The **EU SEA Directive (2001/42/EC)** makes SEA mandatory for listed public plans in member states.
- **India has robust project-level EIA** under the 2006 Notification but **no standalone SEA legislation** as of current law.
- Partial strategic tools in India include **CRZ notifications**, **Eco-Sensitive Zone (ESZ) (पर्यावरण-संवेदनशील क्षेत्र) notifications**, and basin-level planning — but these are not a full SEA law.

> **Logic:** Statement "India has mandatory SEA law equivalent to EIA Notification" = **false**.

Feature	EIA	SEA
Level	Individual project	Policy/plan/programme
Timing	After project concept	Before projects chosen
India law	Mandatory (2006)	No standalone law
Cumulative impacts	Limited	Core focus
Example	EC for one thermal plant	National sector plan before siting
Instrument	Year	Scope
----------	------	-------
Rio Principle 17	1992	Project EIA norm
Espoo Convention	1991	Transboundary EIA
SEA Protocol (Kiev)	2003	Plans/programmes
EU SEA Directive	2001/42/EC	EU mandatory SEA
India EIA Notification	2006	Project EC only

Complete PYQ Bank (Topic 14)

Q1. UPPCS Prelims 2025, Q22 (crossover)

With reference to **carrying capacity**, consider the following statements:

1. Resource extraction should remain **above** the rate of regeneration.
2. Waste generation should stay within the environment's **absorption capacity**.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

💡 **Show answer**

Logic: EIA/sustainable-development crossover — anticipatory assessment assumes waste and resource use within environmental limits. **Ans: A.** Sustainable carrying capacity requires waste within absorption limits; extraction must be **at or below** regeneration — not above. **Trap:** EIA stems often embed carrying-capacity logic — statement 2 alone is the sustainable-use rule.

Q2. UPPCS Prelims 2025, Q70 (org trap crossover)

Environmental Clearance (EC) for large Category A projects is granted by—

A. Central Pollution Control Board (CPCB) B. **Ministry of Environment, Forest and Climate Change (MoEFCC)** C. Central Ground Water Board (CGWB) D. Botanical Survey of India (BSI)

💡 **Show answer**

Logic: Parallel to CGWB≠CPCB trap — EC authority is MoEFCC/SEIAA, not pollution boards. **Ans: B.** **MoEFCC** (through EAC recommendation) grants **Category A EC**; CPCB sets standards but does **not** grant clearance. **Trap:** CPCB is the default wrong pick when stems mention "central environment body."

Q3. UPPCS Prelims 2025 / Topic 1, Q33

The **Environment (Protection) Act, 1986** — which enables EIA notification — was preceded by which chronological pair of pollution statutes?

A. Air Act 1974; Water Act 1981 B. Water Act 1974; Air Act 1981 C. Water Act 1981; Air Act 1974 D. EPA 1986; Water Act 1974

💡 **Show answer**

Logic: Environmental law stack order before EPA 1986 as EIA enabling Act. **Ans: B.** **Water Act 1974** came first, then **Air Act 1981**, then **EPA 1986** (EIA Notification 2006 derives from EPA). **Trap:** Swapping Water and Air years (**1981/1974**) is the standard chronology trap.

Q4. UPSC pattern

Environmental Clearance for Category A projects is finally granted by—

A. Expert Appraisal Committee (EAC) directly B. **MoEFCC on EAC recommendation** C. Central Pollution Control Board D. National Green Tribunal

💡 **Show answer**

Logic: EAC recommends; MoEFCC/SEIAA grants — tribunal hears appeals only. **Ans: B.** **EAC/SEAC recommend**; **MoEFCC** (Category A) or **SEIAA** (Category B) **grant** EC. **Trap:** EAC **does not** issue final clearance — it only appraises and recommends.

Q5. UPSC pattern

In India, **EIA is mandatory** for—

A. Every economic activity in the country B. **Projects listed in the EIA Notification Schedule** C. Only foreign-investment projects D. Only mining projects below 1 ha

💡 **Show answer**

Logic: Project EIA applies to scheduled categories, not universal mandatory assessment. **Ans: B.** EIA applies to **listed Schedule projects** under EIA Notification 2006 — not all activities. **Trap:** "All projects" sounds environmentally strict but is **legally false** — only scheduled categories.

Q6. UPSC pattern

India's **EIA Notification** is issued under powers of the—

A. Water (Prevention and Control of Pollution) Act, 1974 B. Air (Prevention and Control of Pollution) Act, 1981 C. **Environment (Protection) Act, 1986** D. Wildlife (Protection) Act, 1972

💡 **Show answer**

Logic: EPA 1986 is the umbrella enabling Act for EIA, ESZ, CRZ notifications — not Water/Air Acts alone. **Ans: C.** EIA Notification is promulgated under **EPA 1986** Section 3 powers. **Trap:** Water/Air Acts create **consent regimes** (SPCB) — they do **not** authorise project EC notification.

Q7. UPSC pattern

The **National Green Tribunal** was established under an Act passed in—

A. 1986 B. 2005 C. **2010** D. 2015

💡 **Show answer**

Logic: NGT Act year distinct from EPA 1986 — tribunal operational from 2011. **Ans: C.** **NGT Act, 2010** established the tribunal; it became operational in **2011**. **Trap:** EPA **1986** followed Bhopal — NGT came **24 years later**, not the same statute.

Q8. UPSC pattern

Strategic Environmental Assessment (SEA) is conducted at the level of—

A. Individual factory chimneys only B. **Policies, plans and programmes** C. Single-tree plantation plots D. Household kitchen waste only

💡 **Show answer**

Logic: SEA vs EIA level — upstream PPP assessment before project-level EIA. **Ans: B.** SEA evaluates **policies, plans, and programmes (PPP)**; EIA is **project-level**. **Trap:** India has mandatory **project EIA** but **no standalone SEA statute** — yet SEA concept applies at PPP level internationally.

Q9. UPSC pattern

The **Espoo Convention** mainly concerns—

A. Transboundary **Environmental Impact Assessment** B. Whaling quotas in the Southern Ocean C. Acid rain only in Europe D. CFC phase-out schedules

💡 **Show answer**

Logic: Espoo = transboundary EIA; SEA Protocol (Kiev 2003) extends to plans/programmes. **Ans: A.** **Espoo Convention (1991)** requires transboundary EIA for projects likely to cause significant cross-border impact. **Trap:** Gothenburg/CLRTAP = acid rain; Montreal = ODS — Espoo is **EIA-specific**.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to EIA, which of the following statements is/are correct?

1. EIA studies likely environmental effects before approval.
2. EIA's aim is only reactive cleanup after damage is complete.
3. EIA is only done after project completion always.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: C. Only 1 is correct. **Logic:** Statement 1 is true. Statement 2 is false — EIA is anticipatory prevention, not post-damage cleanup. Statement 3 is false — EIA is before approval, not only after completion.

Q2. Which of the following pairs is/are correctly matched?

1. Legal base — EPA 1986 (Section 3)
2. First EIA Notification — 2004
3. Current framework — EIA Notification 2016 teaching

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: C. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q3. Match List-I with List-II:

List-I	List-II
A. Category A	1. SEIAA / SEAC
B. Category B	2. MoEFCC / EAC
C. B1	3. Full EIA (usually with public hearing)

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

💡 **Show answer**

Ans: D. A-2, B-1, C-3. **Logic:** A central; B state; B1 full EIA.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Under a General Condition, a Category B project in a sensitive area is treated as Category A.

Reason (R): Category A is appraised centrally by MoEFCC through the EAC.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; central appraisal fact does not by itself define the General Condition rule. **A/R logic:** Related but GC has its own trigger.

Q5. Arrange EIA stages:

A. Screening → scoping → public consultation → appraisal B. Appraisal → screening → scoping → public consultation C. Public consultation → screening → appraisal → scoping D. Scoping → appraisal → screening → public consultation

💡 **Show answer**

Ans: A. Screening → scoping → public consultation → appraisal. **Logic:** Four classic stages.

Q6. With reference to screening, which of the following statements is/are correct?

1. Screening applies to Category B only.
2. Scoping fixes Terms of Reference (ToR).
3. Screening applies to every Category A project as the first statutory filter identical to B.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** Screening is B-side teaching.

Q7. Which of the following pairs is/are NOT correctly matched?

1. EAC/SEAC — recommend
2. MoEFCC/SEIAA — grant EC
3. CPCB — grants Environmental Clearance

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only pair 3 is not correctly matched. **Logic:** CPCB does not grant EC.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Rapid EIA uses one season; comprehensive EIA covers all seasons.

Reason (R): EMP is part of the EIA report package.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; EMP presence does not explain rapid vs comprehensive seasons. **A/R logic:** Related report facts.

Q9. With reference to B1 vs B2, which of the following statements is/are correct?

1. B1 needs full EIA (usually with public hearing).
2. B2 is simplified.
3. B2 always requires a full multi-season EIA identical to B1.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** B2 simplified.

Q10. Match List-I with List-II:

List-I	List-II
A. NGT	1. MoEFCC online EC portal
B. PARIVESH	2. Act 2010; EC/pollution appeals
C. SEA	3. Policies/plans/programmes upstream

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** NGT 2010; PARIVESH portal; SEA upstream.

Q11. With reference to SEA, which of the following statements is/are correct?

1. SEA assesses policies, plans, and programmes.
2. India has project-level EIA law but no standalone SEA statute.
3. SEA is identical to Consent to Operate under Air Act.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** SEA ≠ CTO.

Q12. Arrange Notification chronology teaching:

A. 2006 → 1994 → EPA 1986 B. 1994 → EPA 1986 → 2006 C. EPA 1986 → EIA Notification 1994 → 2006 framework D. 2010 NGT → EPA 1986 only

💡 **Show answer**

Ans: C. 1986 → 1994 → 2006. **Logic:** Legal base then notifications.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Environmental Clearance is mandatory prior clearance for listed projects.

Reason (R): EIA reports include baseline, impact prediction, mitigation, and EMP.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; report contents do not define why EC is mandatory. **A/R logic:** Related EIA package facts.

Q14. Which one of the following correctly states who grants EC?

A. Only CPCB always B. UNEP Nairobi only C. Only village panchayat for Category A D. MoEFCC/SEIAA (EAC/SEAC recommend)

💡 **Show answer**

Ans: D. MoEFCC/SEIAA grant; committees recommend. **Logic:** Grant vs recommend.

Q15. With reference to NGT and EC, which of the following statements is/are correct?

1. NGT hears EC and pollution appeals.
2. NGT Act is 2010.
3. NGT replaces MoEFCC for granting all Category A ECs.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** NGT is appellate — not the EC granter.

Q16. How many of the following EIA report parts are standard?

Baseline, impact prediction, mitigation, EMP

A. Only one B. All four C. Only three D. Only two

💡 **Show answer**

Ans: B. All four. **Logic:** Standard EIA report spine.

Q17. Which of the following pairs is/are correctly matched?

1. Kiev SEA Protocol 2003 — Espoo family teaching
2. EU SEA Directive — SEA reference
3. SEA — identical to Forest Rights Act titles

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** SEA ≠ FRA.

Q18. With reference to public consultation, which of the following statements is/are correct?

1. It is one of the four classic EIA stages.
2. B1 usually involves public hearing.
3. Public consultation means EC is granted by the public by vote only.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** EC still via MoEFCC/SEIAA.

Q19. Match List-I with List-II:

List-I	List-II
A. Rapid EIA	1. All seasons
B. Comprehensive EIA	2. One season
C. ToR	3. Fixed at scoping

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Rapid one season; comprehensive all; ToR at scoping.

Q20. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): SEAC recommends at state level for Category B.

Reason (R): SEIAA grants Category B Environmental Clearance.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R pairs with SEAC's recommend role in the state track. **A/R logic:** Recommend vs grant at state.

Q21. Which of the following statements about Category A is/are correct?

1. Appraised centrally by MoEFCC through EAC.
2. EAC grants EC by itself without MoEFCC.
3. Sensitive-area General Condition can pull B into A track.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 3. **Logic:** EAC recommends — MoEFCC grants.

Q22. With reference to PARIVESH, which of the following statements is/are correct?

1. It is MoEFCC's online EC portal.
2. It replaces the need for any EIA study.
3. It is part of the clearance workflow digitisation.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 3. **Logic:** Portal ≠ skip EIA.

Q23. Arrange who does what:

1. EAC/SEAC recommend
2. MoEFCC/SEIAA grant
3. CPCB/SPCB set standards/monitor — not grant EC

A. Only 1 B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Grant/recommend/monitor separation.

Q24. Which one of the following is correct about screening?

A. Applies to Category B only in classic teaching B. Applies only to UNEP projects abroad C. Means NGT grants EC D. Means FRA title distribution

💡 **Show answer**

Ans: A. Category B only. **Logic:** Classic stage rule.

Q25. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): India lacks a standalone SEA statute though project EIA exists.

Reason (R): SEA sits upstream of projects at policy/plan/programme level.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; upstream nature explains SEA's role but does not by itself prove absence of statute — still often keyed as both true R not explanation of the legal gap. **A/R logic:** Legal gap vs conceptual level — R does not explain the missing statute.

Q26. With reference to EMP, which of the following statements is/are correct?

1. Environmental Management Plan is part of EIA reporting.
2. It addresses mitigation/management.
3. EMP is issued by the Election Commission.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not Election Commission.

Q27. How many of the following are true?

1. EC is prior clearance for listed projects
2. Category A — central
3. Category B — state

A. Only one B. Only two C. All three D. None

💡 **Show answer**

Ans: C. All three. **Logic:** Core clearance map.

Q28. Which of the following pairs is/are NOT correctly matched?

1. Scoping — fixes ToR
2. Appraisal — final stage among the four
3. Screening — Category A only

A. 1 and 2 only B. 1, 2 and 3 C. 2 and 3 only D. 3 only

💡 **Show answer**

Ans: D. Only pair 3 is not correctly matched. **Logic:** Screening = Category B.

Q29. With reference to sensitive areas and General Condition, which of the following statements is/are correct?

1. B can be treated as A.
2. This shifts appraisal toward the central track.
3. General Condition abolishes all State Pollution Control Boards.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Does not abolish SPCBs.

Q30. Which one of the following best states EIA's purpose?

A. Only post-disaster cleanup manuals B. Anticipatory prevention before approval C. Replacing the Constitution D. Setting GST rates

💡 **Show answer**

Ans: B. Anticipatory prevention. **Logic:** Before approval.

Mains Answer Framework

EIA and SEA (250 words): Define EIA as anticipatory project-level tool under **EPA 1986** and **EIA Notification 2006**. Explain Category A/B, four stages, public consultation, EC before construction, MoEFCC/SEIAA vs CPCB, NGT appeals, PARIVESH. Define SEA at PPP level upstream of EIA; cumulative impacts; international frameworks (Rio Principle 17, Espoo, SEA Protocol 2003, EU Directive). India gap — no standalone SEA law; partial tools (CRZ, ESZ). Controversies — draft EIA 2020, ex-post facto clearance, Lafarge/Great Nicobar (निकोबार) examples. Conclude: strengthen compliance and consider strategic assessment for sector plans.

Common Traps — Don't Fall For These

Trap	Correct fact
CPCB grants EC	**MoEFCC/SEIAA grant EC**
Screening for Category A	**Screening = Category B only**
EIA after construction	**EC before construction**
EIA = SEA	**EIA = project; SEA = plan/policy**
India has mandatory SEA law	**Project EIA only**
EIA for all projects	**Listed Schedule projects only**
Water Act 1981 base for EIA	**EPA 1986**
NGT established 1986	**NGT Act 2010**
EAC grants final EC	**EAC recommends; MoEFCC/SEIAA grant**
Rio Principle 17 = SEA	**Principle 17 = project EIA**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 15 — Sustainable Development & Environmental Governance

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Sustainable Development: SD | SDGs | SDGs 2030 | Objectives | Indicators | Dimensions | Brundtland Report | Natural capital (प्राकृतिक पूंजी) | Sustainable agriculture | NMSA **Environmental Governance:** Governance | Environmental Justice | Environmental Ethics | Green Economy (हरित अर्थव्यवस्था) | Circular Economy (चक्रीय अर्थव्यवस्था)

Sources baked in: Brundtland Report 1987, UN SDGs 2030, Rio 1992/Agenda 21, UNEP Green Economy 2011, NCERT, NITI Aayog SDG Index, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — SD A/R traps, 17 SDGs, Brundtland vs Limits to Growth, LiFE, UP Planning Dept **Last verified:** August 2026

Current Affairs (this topic)

- SDG progress reports and LiFE / Mission LiFE headlines refresh this chapter's CA.
- Brundtland / Agenda 21 / Rio years stay fixed — do not relocate them into Paris-only answers.

💡 🎯 **Consolidated — 32 Must-Score Facts (Click to Expand)**

1. **Sustainable development** (Brundtland **1987**, *Our Common Future*) means meeting present needs without compromising future generations' ability to meet theirs. 2. Brundtland is **not** *Limits to Growth* (Club of Rome, **1972**). 3. SD rests on **economic, social, and environmental** pillars (plus institutional capacity). It is not environment-only. 4. Core objectives include basic needs, intra- and inter-generational equity, ecological limits, participation, and integrated planning. 5. Priority objectives also include **poverty reduction**, **biodiversity conservation**, and **pollution control**. 6. Indicators include **HDI (UNDP (यूएनडीपी))**, ecological footprint (पारिस्थितिक पदचिह्न), NITI (नीति) Aayog (नीति आयोग) SDG Index, global SDG indicator sets, and EPI. 7. **Natural capital** means renewable and non-renewable environmental resources (air, water, land, minerals, forests). **Roads** are not natural capital. 8. **SDGs** were adopted (अंगीकृत) in **September 2015**, run to **2030**, and include **17 goals** and **169 targets**. 9. SDGs replaced **MDGs** (8 goals, **2000–2015**) and apply universally to all countries. 10. Goal facts worth ratta include SDG **6** water, **7** energy, **13** climate, **14** life below water, and **15** life on land. 11. Global SDG Index ranks often place **Finland** first. India was about **116th (2017)** and about **112th (2023)**. 12. NITI Aayog **SDG India Index** has repeatedly placed **Kerala** first; Himachal (हिमाचल) Pradesh and Tamil Nadu (तमिलनाडु) sit near the top. **Gujarat** was **not** in the 2019–20 top-five set often tested. 13. Conference chain: **Stockholm 1972 (स्टॉकहोम सम्मेलन) → Brundtland 1987 → Rio and Agenda 21 in 1992 → SDGs 2015**. 14. **Agenda 21** is Rio **1992**, not 1995. 15. **LiFE Mission** launched in **June 2022**; the idea was floated at **COP26 Glasgow 2021**, not COP25. 16. MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is the nodal environment (पर्यावरण) ministry. **NGT Act 2010** creates the green tribunal. 17. Rio principles frequently tested are **Precautionary Principle, Polluter Pays, and Public Trust**. 18. Constitutional hooks are **Article 48A**, **Article 51A(g)**, and **Article 21** (अनुच्छेद 21) read with a healthy environment. 19. UP SDG reporting typically sits with the **Planning Department**. 20. Environmental justice means fair distribution of environmental benefits and burdens plus fair participation in decisions. 21. Bhopal **1984** led to EPA **1986**. UP justice/pollution examples include Kanpur (कानपुर) tanneries and the Singrauli coal belt. 22.

Ethics labels: **anthropocentrism** (human-centred), **biocentrism** (all life), **ecocentrism** (ecosystems). Leopold's Land Ethic is **1949**; deep ecology (पारिस्थितिकी) links to **Arne Næss**. 23. Gandhi (गांधी)'s **trusteeship** is a frequent Indian environmental-ethics fact. 24. **Green economy** (UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) **2011** line) means low-carbon, resource-efficient, socially inclusive growth. 25. **Circular economy** redesigns take-make-dispose into reduce-reuse-recycle loops, often with **EPR** rules. Circular sits inside, but is not identical to, green economy. 26. **Sustainable agriculture** means using land so its **quality remains intact**, with mixed cropping, organic manures, nitrogen-fixing plants, and pest-resistant varieties as the environment-friendly strategy set. 27. **NMSA** (National Mission for Sustainable Agriculture) is a **NAPCC** mission on productivity, water-use efficiency, nutrient management, and livelihood diversification. 28. Example scheme–SDG matches: Jal Jeevan / Swachh Bharat → SDG **6**; SAUBHAGYA / UJALA → SDG **7**; NAPCC / Net Zero 2070 → SDG **13**. 29. MDGs had 8 goals; SDGs have 17 — do not reverse the counts. 30. HDI is an UNDP development indicator; it is not itself an MoEFCC clearance tool. 31. Polluter Pays allocates cleanup cost to the polluter; Precautionary Principle acts under scientific uncertainty — different tools. 32. Public Trust treats certain natural resources as held for public benefit, not absolute (निरपेक्ष) private exhaustion.

💡 ⚡ Confused Pairs (Click to Expand)

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Brundtland Report (1987)** | **Limits to Growth (1972)** | UN World Commission on Environment & Development (*Our Common Future*) coining the definition of Sustainable Development vs Club of Rome computer-simulation study modeling impending resource depletion and economic collapse | ब्रंटलैंड रिपोर्ट (1987, सतत विकास) / लिमिट्स टू ग्रोथ (1972) | | **Millennium Development Goals (MDGs)** | **Sustainable Development Goals (SDGs)** | 8 anti-poverty targets for 2000–2015 focused primarily on developing nations funded by rich countries vs 17 universal, integrated economic-social-environmental goals (Agenda 2030) adopted in 2015 applying to all nations equally | सहस्राब्दी विकास लक्ष्य (MDGs) / सतत विकास लक्ष्य (SDGs 2030) | | **Green Economy** | **Circular Economy** | Macro-economic development paradigm promoting low-carbon growth, resource efficiency, and social equity (UNEP) vs regenerative industrial design philosophy eliminating waste and keeping materials in closed continuous cycles | हरित अर्थव्यवस्था (व्यापक मॉडल) / चक्रीय अर्थव्यवस्था (पुनर्चक्रण लूप) | | **Linear Economy** | **Circular Economy** | Traditional unsustainable industrial model based on "take, make, use, and dispose" vs restorative circular model based on "reduce, reuse, repair, refurbish, and recycle" | रैखिक अर्थव्यवस्था (कचरा उत्पादक) / चक्रीय अर्थव्यवस्था (शून्य अपशिष्ट) | | **Mission LiFE (2022)** | **Agenda 21 (1992)** | Mass global citizen movement conceptualized by India to replace mindless consumption with mindful individual sustainable lifestyle actions vs Rio 1992 comprehensive non-binding intergovernmental action plan for sustainable governance | मिशन लाइफ (LiFE, व्यक्तिगत जीवनशैली) / एजेंडा 21 (सरकारी कार्ययोजना) | | **Natural Capital** | **Manufactured Capital** | World's natural ecological assets including geology, soil, air, water, biodiversity, and ecosystem services vs human-created physical infrastructure, machinery, factories, and financial assets | प्राकृतिक पूंजी (पारिस्थितिक संपदा) / निर्मित पूंजी (मानव-निर्मित संपत्ति) |

💡 📋 Must-score facts — Brundtland, SDGs, NITI Index (Click to Expand)

Spine | Item | Lock | |---|---| | Brundtland | **1987** *Our Common Future* (≠ Limits to Growth 1972) | | Three pillars | Economic · social · environmental | | SDGs | **Sep 2015** → **2030**; **17** goals / **169** targets | | MDGs → SDGs | 8 goals 2000–15 / universal to all countries | | Key goals | 6 water · 7 energy · 13 climate · 14 ocean · 15 land | | Conference chain | Stockholm 72 → Brundtland 87 → Rio 92 → SDGs 15 | | Agenda 21 | Rio **1992** (≠ 1995) | | NITI SDG Index top | Kerala repeatedly first; HP/TN near top |

15.1 Sustainable Development

Identity: Brundtland 1987 | Three pillars | Inter/intra-generational equity | Indicators | Natural capital

- **Sustainable development** means meeting the needs of the present **without compromising** the ability of future generations to meet their own needs.
- This definition comes from the **Brundtland Report (1987)** (ब्रंटलैंड रिपोर्ट), formally titled ***Our Common Future***, by the **World Commission on Environment and Development (WCED)** chaired by **Gro Harlem Brundtland**.
- The Brundtland Report is **not** *The Limits to Growth*. That 1972 study was by the **Club of Rome**.
- **Inter-generational equity** protects future generations. **Intra-generational equity** ensures fair sharing among people alive today.
- Sustainable development is a matter of **inter-generational sensibility** in the use of **natural resources**.
- The **three dimensions/pillars** are **economic** (viable growth), **social** (equity, health, education), and **environmental** (ecological integrity). Some frameworks add **institutional governance**.
- **Objectives** include fulfilling **basic needs**, respecting **ecological limits** and carrying capacity, integrating environment into economic planning, and ensuring **public participation**.
- Core practical objectives also include **poverty reduction**, **biodiversity conservation**, and **prevention and control of pollution**.
- Development is unsustainable if resource extraction exceeds regeneration or waste exceeds environmental absorption capacity.
- **Natural capital** includes all renewable and non-renewable environmental resources such as air, water, land, minerals, and forests. **Roads** and other purely human-made assets are **not** natural capital.
- SD road-mapping often uses information from **remote sensing**, **GIS**, statistical methods, and environmental information systems.
- **Indicators** include **HDI** (UNDP), **ecological footprint** (पारिस्थितिक पदचिह्न), **SDG Index**, **Environmental Performance Index (EPI)**, and **231 SDG indicators**.
- **NITI Aayog** (नीति आयोग) publishes the **SDG India Index** ranking states on SDG progress.
- The timeline runs **Stockholm 1972** (स्टॉकहोम सम्मेलन) → **Brundtland 1987** → **Rio Earth Summit and Agenda 21 (1992) (एजेंडा 21)** → **SDGs adopted 2015**.

PYQ — UPPCS Prelims 2025, Q51

(A) SD should occur without harming the environment or compromising future generations' needs. (R) Agenda 21 was signed by world leaders in 1995.

A. Both true; R not explanation

B. A false, R true

C. A true, R false

D. Both true; R explains A

💡 **Show answer**

Ans: C — Agenda 21 = **Rio 1992**, not 1995.

PYQ — UPPCS Prelims 2024, Q51

(A) SD was popularised by the Brundtland Report. (R) The Brundtland Report is also known as *The Limits to Growth*.

A. Both true; R not explanation

B. A false, R true

C. Both true; R explains A

D. A true, R false

> 💡 **Show answer**
>
> **Ans: D** — Brundtland = *Our Common Future* (1987).

Pillar	Focus
Economic	Employment, innovation, infrastructure
Social	Poverty, health, education, equity
Environmental	Pollution, biodiversity, climate
Institutional	Governance, rule of law, participation

15.2 SDGs & SDGs 2030

Identity: 17 goals | 2030 deadline | 2015 adoption | MDGs replaced | NITI coordination

- The **2030 Agenda for Sustainable Development** was adopted by all UN member states on **25 September 2015**.
- It sets **17 Sustainable Development Goals (SDGs) (सतत विकास लक्ष्य)** with **169 targets** and **231 indicators** to be achieved by **2030**.
- **SDGs replaced MDGs**, which had **8 goals (2000–2015)** and focused mainly on developing countries.
- **SDGs are universal** — they apply to all countries, including developed ones.
- **Leave No One Behind** requires progress to reach the poorest and most vulnerable groups.
- **NITI Aayog** coordinates SDG localisation in India. **MoSPI** maintains the National Indicator Framework.
- India's large population (जनसंख्या) share makes its SDG performance critical for global success.
- A global **SDG Index / Dashboard** ranks countries on progress toward the 17 goals. **Finland**, **Sweden**, and **Denmark** typically lead.
- India ranked about **116th** in the **2017** global SDG Index and about **112th** among **166** countries in the **2023** reading (रीडिंग).
- **NITI Aayog's SDG India Index** ranks states and UTs. **Kerala** has repeatedly held the **first** position (including the 2019–20 and 2020–21 editions).
- In the 2019–20 / 2020-style top-five set often tested, leading states include **Kerala**, **Himachal Pradesh**, **Tamil Nadu**, and **Andhra Pradesh**. **Gujarat** was **not** in that top-five list.

PYQ — UPPCS Prelims 2025, Q142

Goal 1 / Goal 3 / Goal 4 / Goal 6 matched to: (1) Clean Water (2) Quality Education (3) No Poverty (4) Good Health

> 💡 **Show answer**
>
> **Ans: B (3 4 2 1)** — Goal 1 = No Poverty; 3 = Health; 4 = Education; 6 = Water.

PYQ — UPPCS Prelims 2025, Q120

(A) India's success is critical for global SDG success. (R) India accounts for nearly one-sixth of world population.

> 💡 **Show answer**
>
> **Ans: D** — Both true; R explains A.

PYQ — UPPCS Prelims 2024, Q98

Essential for SDG 4 (Quality Education):

1. Free and compulsory education
2. School infrastructure and digital transformation
3. Expansion of agricultural programmes
4. Increasing investment in technology

💡 **Show answer**

Ans: A (1 and 2) — Agricultural expansion is not core to SDG 4.

Goal	Official name
1	No Poverty
2	Zero Hunger
3	Good Health and Well-being
4	Quality Education
5	Gender Equality
6	Clean Water and Sanitation
7	Affordable and Clean Energy
8	Decent Work and Economic Growth
9	Industry, Innovation and Infrastructure
10	Reduced Inequalities
11	Sustainable Cities and Communities
12	Responsible Consumption and Production
13	Climate Action
14	Life Below Water
15	Life on Land
16	Peace, Justice and Strong Institutions
17	Partnerships for the Goals
Feature	MDGs
---	---
Goals	8
Period	2000–2015
Scope	Mainly developing countries

15.3 Environmental Governance, Justice & Ethics

Identity: Multi-level governance | Rio principles | Bhopal-EJ | Anthropocentrism vs ecocentrism | LiFE

- **Environmental governance** is the system of rules, institutions, and processes through which environmental decisions are made and enforced.
- It operates at **global** (UNEP, treaties), **national** (MoEFCC, Parliament), **state** (SPCBs), and **local** (panchayats, municipalities) levels.
- **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is India's nodal ministry for environment, forest, and climate policy.

- **Rio principles** include **Precautionary principle**, **Polluter Pays**, **Public Trust doctrine**, and **inter-generational equity** — adopted in Indian judgments like *Vellore Citizens' Welfare Forum*.
- Constitutional anchors include **Article 48A** (state shall protect environment), **Article 51A(g)** (citizens' duty to protect environment), and **Article 21** (right to healthy environment).
- **NGT (2010)** is the specialised tribunal for environmental cases and EC appeals.
- **Agenda 21 (1992)** is a Rio blueprint for sustainable development planning, including local-level action.
- **Environmental justice** means fair treatment of all in environmental benefits and burdens — both **distributive** (who gets pollution) and **procedural** (who participates in decisions).
- The **Bhopal Gas Tragedy (1984)** exposed environmental injustice and led to the **EPA 1986** (पर्यावरण संरक्षण 1986).
- UP hotspots include **Kanpur-Unnao tanneries** and the **Singrauli–Sonbhadra (सोनभद्र) coal belt**.
- **Environmental ethics** asks what moral duties humans owe to nature.
- **Anthropocentrism** values nature mainly for human use.
- **Biocentrism** gives intrinsic worth to all living beings.
- **Ecocentrism** values whole ecosystems.
- **Aldo Leopold's Land Ethic (1949)** stresses preserving the integrity of the biotic community.
- **Gandhi's trusteeship** holds that resources are held in trust for society and future generations.
- The **LiFE Mission** promotes individual pro-planet behaviour. The idea was announced at **COP26 Glasgow (2021)** and launched in **June 2022**.

PYQ — UPPCS Prelims 2025, Q111

UP department preparing SDG National Indicator Framework Progress Report:

1. Finance 2. Education 3. Planning

> 💡 **Show answer**
>
> **Ans: B (Only 3)** — **Planning Department**.

PYQ — UPPCS Prelims 2025, Q139

1. LiFE was launched in June 2022.
2. The idea was proposed at COP-25 in Madrid.

> 💡 **Show answer**
>
> **Ans: D (Only 1)** — Idea at **COP26 Glasgow**, not COP-25.

Concept	Fact
Environmental justice	Fair distribution + participation
Anthropocentrism	Human-centred value
Biocentrism	All life has worth
Ecocentrism	Ecosystem (पारिस्थितिकी तंत्र) wholes have worth
Leopold	Land Ethic, 1949
Gandhi	Trusteeship, limit wants

15.4 Green Economy & Circular Economy

Identity: UNEP 2011 green growth | 3R/EPR | Linear vs circular | SDG 12

- The **green economy** (UNEP, 2011) aims for improved well-being and social equity while **reducing environmental risk and ecological scarcity**.
- It seeks **low-carbon, resource-efficient, socially inclusive** growth — decoupling GDP from environmental harm.
- India examples include **solar expansion**, **National Green Hydrogen Mission (2023)**, **Net Zero 2070**, and **LiFE Mission**.
- The **circular economy** replaces the linear **take-make-dispose** model with loops of **reduce, reuse, recycle, repair, and recovery**.
- In a circular system, **waste is treated as a design flaw**, not an afterthought.
- **Extended Producer Responsibility (EPR)** (विस्तारित उत्पादक दायित्व) under Plastic, E-Waste, Battery, and Tyre rules makes producers responsible for end-of-life collection.
- **Circular economy** focuses on **material flows and waste elimination**.
- **Green economy** is **broader** — it also covers renewable energy (नवीकरणीय ऊर्जा), biodiversity, jobs, and equity. Circular economy is a **component** of green economy.
- Both link strongly to **SDG 12** (Responsible Consumption and Production).

Feature	Green economy	Circular economy
Focus	Overall low-carbon inclusive growth	Material/waste loop design
Scope	Broader	Narrower — subset of green
Example	Solar parks, green hydrogen	EPR, plastic recycling
SDG link	7, 8, 13 and more	Mainly SDG 12

15.5 Sustainable Agriculture & NMSA

Identity: Land quality intact | Mixed cropping strategy | NMSA under NAPCC

- **Sustainable agriculture** means producing food, fibre, and related products with farming techniques that protect the environment, public health, communities, and animal welfare.
- In short form, it means utilising land so that its **quality remains intact**.
- Heavy chemical fertilisers and pesticides have damaged soil health, which is why sustainable pathways emphasise nutrient management and ecological farming.
- An environment-friendly strategy set for Indian agriculture is **mixed cropping**, **organic manures**, **nitrogen-fixing plants**, and **pest-resistant crop varieties**.
- **Sikkim** became India's (and often cited as the world's) first **100% organic state** (process from **2003**; declared around **2016**) — NPOP is under Commerce/APEDA, not Rural Development.
- Expansion of cultivable land with heavy urea/superphosphate use, monoculture, and intensive aerial pesticide spray is **not** the sustainable strategy option.
- The **National Mission for Sustainable Agriculture (NMSA)** is a mission under the **National Action Plan on Climate Change (जलवायु परिवर्तन) (NAPCC)**.
- NMSA focuses on agricultural productivity, **water-use efficiency**, nutrient management, and **livelihood diversification** through sustainable pathways.
- **Conservation Agriculture** (FAO framing) rests on **minimum tillage**, **permanent soil cover with crop residues**, and **crop rotation / sequencing**. Avoiding plantation crops is **not** a Conservation Agriculture principle.

Idea	Correct fact
Sustainable agriculture	Land quality remains intact

Idea	Correct fact
Best strategy set	Mixed cropping + organic + N-fixing + pest-resistant varieties
NMSA	NAPCC mission
Conservation Agriculture	Min tillage + residue cover + rotation

---

Complete PYQ Bank (Topic 15)

Q1. UPPCS Prelims 2025, Q51

Assertion (A): Sustainable development should occur without harming the environment or compromising the needs of future generations.

Reason (R): Agenda 21 was signed by world leaders in **1995**.

A. Both (A) and (R) are true but (R) is not the correct explanation of (A). B. (A) is false but (R) is true. C. (A) is true but (R) is false. D. Both (A) and (R) are true and (R) is the correct explanation of (A).

💡 **Show answer**

A/R logic: A tests Brundtland-style SD definition; R tests Agenda 21 signing year at Rio. **Ans: C.** A states the core SD principle correctly; R is false — **Agenda 21 was adopted at Rio 1992**, not 1995. **Trap:** 1995 sounds like a plausible post-Rio follow-up year — the Earth Summit was **1992**.

Q2. UPPCS Prelims 2024, Q51

Assertion (A): Sustainable development was popularised by the Brundtland Report.

Reason (R): The Brundtland Report is also known as *The Limits to Growth*.

A. Both (A) and (R) are true but (R) is not the correct explanation of (A). B. (A) is false but (R) is true. C. Both (A) and (R) are true and (R) is the correct explanation of (A). D. (A) is true but (R) is false.

💡 **Show answer**

A/R logic: A tests Brundtland authorship of SD concept; R tests alternate book-title trap. **Ans: D.** Brundtland did popularise SD in ***Our Common Future* (1987)**; *The Limits to Growth* is a **different 1972 Club of Rome report**. **Trap:** Both are famous environment reports — titles are **not interchangeable**.

Q3. UPPCS Prelims 2025, Q142

Match **SDG number** with **official goal name**:

List I (Goal)	List II (Name)
1. Goal 1	A. Clean Water and Sanitation
2. Goal 3	B. Quality Education
3. Goal 4	C. No Poverty
4. Goal 6	D. Good Health and Well-being

Row order is not the answer code.

A. 1–C, 2–D, 3–B, 4–A B. 1–D, 2–C, 3–A, 4–B C. 1–A, 2–B, 3–C, 4–D D. 1–B, 2–A, 3–D, 4–C

💡 **Show answer**

Logic: SDG number–name matching for high-frequency goals 1, 3, 4, 6. **Ans: B.** Goal **1 = No Poverty**; **3 = Good Health**; **4 = Quality Education**; **6 = Clean Water and Sanitation** (code **3 4 2 1** in paper notation). **Trap:** Swapping Goals **3 and 4** (health vs education) is the most common match-list error.

Q4. UPPCS Prelims 2025, Q120

Assertion (A): India's success is critical for global SDG success.

Reason (R): India accounts for nearly **one-sixth** of the world's population.

A. Both (A) and (R) are true and (R) is the correct explanation of (A). B. Both (A) and (R) are true but (R) is not the correct explanation of (A). C. (A) is true but (R) is false. D. (A) is false but (R) is true.

💡 **Show answer**

A/R logic: A tests India's weight in global SDG outcomes; R tests demographic scale as the explaining factor. **Ans: A.** With ~**1/6 of world population**, India's SDG performance heavily shapes global totals — R correctly explains A. **Trap:** "Critical for global success" follows from **population scale + development footprint**, not geography alone.

Q5. UPPCS Prelims 2025, Q111

The **SDG National Indicator Framework Progress Report** for Uttar Pradesh is prepared by the State's—

A. Finance Department B. **Planning Department** C. Education Department D. Forest Department

💡 **Show answer**

Logic: UP governance fact — SDG indicator reporting sits with state planning machinery. **Ans: B.** UP's **Planning Department** prepares the SDG National Indicator Framework Progress Report. **Trap:** Finance Department handles budgets — **Planning Department** owns SDG indicator reporting.

Q6. UPPCS Prelims 2025, Q139

With reference to **LiFE (Lifestyle for Environment)**, consider the following:

1. LiFE was launched in **June 2022**.
2. The idea was proposed at **COP-25 in Madrid**.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

💡 **Show answer**

Logic: Governance/mission fact — LiFE launch vs COP venue of original announcement. **Ans: D.** LiFE launched **June 2022**; the concept was announced at **COP-26 Glasgow (2021)**, not COP-25. **Trap:** COP-25 Madrid is geographically plausible — Glasgow **2021** is correct.

Q7. UPPCS Prelims 2024, Q98

Which of the following are essential for achieving **SDG 4 (Quality Education)**?

1. Free and compulsory education
2. School infrastructure and digital transformation
3. Expansion of agricultural programmes
4. Increasing investment in technology

A. 1 and 2 only B. 1, 2 and 4 only C. 2 and 3 only D. All four

💡 **Show answer**

Logic: SDG 4 core targets vs unrelated agricultural expansion distractor. **Ans: A.** SDG 4 centres on **access, quality, and infrastructure** — agricultural programme expansion is not a core SDG 4 pillar. **Trap:** Statement 4 (technology investment) sounds education-relevant but the keyed combo is **1 and 2** — statement 3 is clearly off-topic.

Q8. UPPCS Prelims 2025, Q22 (crossover)

With reference to **carrying capacity**, consider the following:

1. Resource extraction should remain **above** regeneration.
2. Waste should stay within environmental **absorption capacity**.

A. 2 only B. Neither 1 nor 2 C. Both 1 and 2 D. 1 only

💡 **Show answer**

Logic: SD operational test — sustainable development requires living within regeneration and absorption limits. **Ans: A.** Statement 2 defines sustainable waste limits; statement 1 reverses the regeneration rule. **Trap:** SD governance stems often embed this carrying-capacity pair — only statement 2 survives scrutiny.

Q9. UPSC pattern

The **Bhopal Gas Tragedy (1984)** directly catalysed enactment of the—

A. Stockholm Declaration enforcement Act B. **Environment (Protection) Act, 1986** C. Biological Diversity Act, 2002 D. NGT Act, 2010

💡 **Show answer**

Logic: Environmental governance timeline — Bhopal exposed regulatory gaps leading to EPA. **Ans: B.** Bhopal **1984** led to India's comprehensive **EPA 1986** and later EIA notification. **Trap:** NGT (**2010**) is much later — the **immediate legislative response** was EPA **1986**.

Q10. UPSC pattern

The **green economy** (UNEP, 2011) emphasises—

A. High-coal growth with waste maximisation B. **Low-carbon, resource-efficient, socially inclusive growth** C. Ozone-layer depletion as a development goal D. Elimination of all industrial activity

💡 **Show answer**

Logic: UNEP green economy definition — well-being + equity + lower environmental risk. **Ans: B.** Green economy pursues improved well-being and equity while **cutting environmental risk and ecological scarcity**. **Trap:** Green economy is **not** anti-industry — it redesigns growth to be **resource-efficient and inclusive**.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to sustainable development, which of the following statements is/are correct?

1. Brundtland 1997 (*Our Common Future*) defines meeting present needs without compromising future generations.
2. Brundtland is identical to *Limits to Growth* (1972).
3. SD rests on economic, social, and environmental pillars.

A. Only 3 B. Only 1 C. 1 and 2 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 3 is correct. **Logic:** Only statement 3 remains true among the numbered claims. Statement 1 is false. Statement 2 is false. Statement 3 is true.

Q2. Arrange conference chain:

A. SDGs 2015 → Rio 1992 → Stockholm 1972 B. Rio 1992 → Stockholm 1972 → Brundtland 1987 C. Stockholm 1972 → Brundtland 1987 → Rio 1992 → SDGs 2015 D. Brundtland 1987 → Stockholm 1972 → SDGs 2015

> 💡 **Show answer**
>
> **Ans: C.** 1972 → 1987 → 1992 → 2015. **Logic:** Stockholm–Brundtland–Rio–SDGs.

Q3. With reference to SDGs, which of the following statements is/are correct?

1. Adopted September 2007; run to 2030.
2. 17 goals and 169 targets.
3. SDGs apply only to developing countries and did not replace the MDGs.

A. 1 and 2 B. Only 3 C. Only 2 D. Only 1

> 💡 **Show answer**
>
> **Ans: C.** Only 2 is correct. **Logic:** Statement 2 is correct (17/169). Statement 1 is false — SDGs were adopted in 2015, not 2007. Statement 3 is false — SDGs replaced MDGs and apply universally.

Q4. Match List-I with List-II:

List-I	List-II
A. SDG 6	1. Climate
B. SDG 7	2. Water
C. SDG 13	3. Energy
D. SDG 15	4. Life on land

Code:

A. A-2, B-3, C-1, D-4 B. A-3, B-2, C-1, D-4 C. A-2, B-1, C-3, D-4 D. A-1, B-3, C-2, D-4

> 💡 **Show answer**
>
> **Ans: A.** A-2, B-3, C-1, D-4. **Logic:** 6 water; 7 energy; 13 climate; 15 land.

Q5. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Agenda 21 belongs to Rio 1992.

Reason (R): Agenda 21 was signed in 1995 as a binding climate treaty.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** 1992 non-binding SD action plan — not 1995 climate treaty.

Q6. Which of the following pairs is/are correctly matched?

1. Natural capital — air, water, land, minerals, forests
2. Roads — natural capital in standard teaching
3. HDI — UNEP

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

💡 **Show answer**

Ans: B. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q7. With reference to NITI SDG Index, which of the following statements is/are correct?

1. Kerala has repeatedly placed first.
2. Himachal Pradesh and Tamil Nadu sit near the top in many editions.
3. The Index is published by UNEP Nairobi as India's only SDG tool.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** NITI Aayog product — not UNEP-only.

Q8. Arrange MDGs → SDGs:

A. SDGs 2000–2015 → MDGs 2015–2030 B. MDGs started in 2030 C. Both identical 8 goals forever D. MDGs 8 goals 2000–2015 → SDGs 17 goals 2015–2030

💡 **Show answer**

Ans: D. MDGs 2000–15 (8) → SDGs 2015–30 (17). **Logic:** Replacement fact.

Q9. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): SD is not environment-only.

Reason (R): Economic and social pillars sit with the environmental pillar.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Three pillars.

Q10. Which of the following indicators is/are used in SD teaching?

1. HDI
2. Ecological footprint
3. NITI SDG Index
4. EPI

A. 1, 2, 3 and 4 B. 1, 2 and 3 only C. 2, 3 and 4 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All four. **Logic:** Indicator set.

Q11. With reference to SDG 14, which of the following statements is/are correct?

1. Life below water.
2. Links to oceans/blue economy neighbourhood.
3. SDG 14 is identical to Article 370.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not Art. 370.

Q12. Match List-I with List-II:

List-I	List-II
A. Brundtland Report	1. 1972 Club of Rome
B. Limits to Growth	2. 1987
C. Rio Agenda 21	3. 1992

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** 1987 / 1972 / 1992.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Global SDG Index ranks often place Nordic countries like Finland near the top.

Reason (R): India's rank has been far from first in recent global SDG Index teaching figures.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; India's rank does not explain Finland's top place. **A/R logic:** Two separate rank facts.

Q14. With reference to equity in SD, which of the following statements is/are correct?

1. Intra-generational equity matters.
2. Inter-generational equity matters.
3. Ecological limits and participation matter.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Core SD objectives.

Q15. Which of the following pairs is/are NOT correctly matched?

1. SDG 13 — climate

2. SDG 7 — energy
3. SDG 6 — life on land

A. 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only pair 3 is not correctly matched. **Logic:** SDG 6 = water; 15 = life on land.

Q16. Arrange poverty reduction / biodiversity / pollution control as SD priority objectives teaching:

A. They are excluded from SD B. They are among priority objectives alongside equity and ecological limits C. Only pollution control exists D. Only GDP growth is SD

> 💡 **Show answer**
>
> **Ans: B.** Among priority objectives. **Logic:** Not GDP-only.

Q17. With reference to institutional capacity, which of the following statements is/are correct?

1. Governance capacity supports SD delivery.
2. Integrated planning is a core idea.
3. SD means ignoring future generations.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** Future generations are central.

Q18. How many of the following conference facts are correct?

1. Stockholm 1972 first UN human environment conference
2. Rio 1992 produced Agenda 21
3. SDGs 2015

A. Only one B. Only two C. None D. All three

> 💡 **Show answer**
>
> **Ans: D.** All three. **Logic:** Spine facts.

Q19. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): MDGs had 8 goals.

Reason (R): SDGs have 17 goals and apply to all countries.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; SDG design does not explain why MDGs had 8. **A/R logic:** Sequential frameworks; R not explanation of MDG count.

Q20. Which one of the following correctly states ecological footprint?

A. Resource demand in global hectares (gha) B. Identical to tCO_2e only always C. A forest canopy % only D. A UN Security Council seat

💡 **Show answer**

Ans: A. gha demand. **Logic:** ≠ carbon footprint unit only.

Q21. With reference to EPI, which of the following statements is/are correct?

1. Environmental Performance Index is used among SD-related indicators teaching.
2. It replaces the Constitution of India.
3. It can appear beside HDI/footprint in indicator lists.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 3. **Logic:** Does not replace Constitution.

Q22. Match List-I with List-II:

List-I	List-II
A. SDG 14	1. Energy
B. SDG 7	2. Life below water
C. SDG 6	3. Water

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** 14 ocean; 7 energy; 6 water.

Q23. With reference to Gujarat in older NITI Index traps, which of the following statements is/are correct?

1. Kerala has been repeatedly first.
2. Assuming Gujarat was first in every 2019-style stem is unsafe.
3. NITI Index is irrelevant to Indian SD teaching.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Index matters; Gujarat-first is a trap in some years.

Q24. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Rio 1992 is a pivot for Agenda 21, UNFCCC, and CBD openings.

Reason (R): Stockholm 1972 created UNEP.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; UNEP origin does not explain Rio's 1992 products. **A/R logic:** Two conference facts.

Q25. Which of the following statements about basic needs in SD is/are correct?

1. Meeting basic needs is a core objective.

2. Participation and integrated planning matter.
3. SD ignores poverty reduction.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Poverty reduction is a priority.

Q26. Arrange 169 targets association:

A. MDGs only B. SDGs C. Only Montreal Protocol D. Only IFA 1927

💡 **Show answer**

Ans: B. SDGs. **Logic:** 17 goals / 169 targets.

Q27. With reference to WCED, which of the following statements is/are correct?

1. Brundtland Commission worked ~1983–1987.
2. It produced *Our Common Future*.
3. It is also called the Club of Rome Limits report.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Not Limits to Growth.

Q28. Which of the following pairs is/are correctly matched?

1. Intra-generational equity — within present generation
2. Inter-generational equity — across generations
3. Inter-generational equity — only within one city ward forever

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Pair 3 wrong.

Q29. How many of the following are SDG environment-facing goals in common ratta?

6, 7, 13, 14, 15

A. All five B. Only three C. Only four D. Only two

💡 **Show answer**

Ans: A. All five. **Logic:** Water, energy, climate, ocean, land.

Q30. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Sustainable development requires respecting ecological limits.

Reason (R): Unlimited resource extraction above regeneration always supports SD.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Extraction above regeneration breaks SD.

Mains Answer Framework

Sustainable development and governance (250 words): Define SD (Brundtland 1987) and three pillars. Explain objectives, indicators (HDI, SDG Index), and timeline (Stockholm, Rio, SDGs 2030). Cover all 17 SDGs briefly with India scheme mapping. Governance — MoEFCC, NGT, Rio principles, constitutional duties, Agenda 21, LiFE. Environmental justice — distributive/procedural, Bhopal, UP hotspots. Ethics — anthropocentrism/biocentrism/ecocentrism, Gandhi, Leopold. Green vs circular economy with EPR and SDG 12. Conclude with India's global SDG role and integrated policy need.

Common Traps — Don't Fall For These

Trap	Correct fact
Agenda 21 = 1995	**Rio 1992**
Brundtland = Limits to Growth	***Our Common Future* 1987**
SD = environment only	**Three pillars balanced**
SD = no development	**Sustainable, not zero, development**
LiFE at COP-25	**COP26 2021; launched June 2022**
UP SDG report = Finance Dept	**Planning Department**
Circular = identical to green	**Circular ⊂ Green**
MDGs = 17 goals	**MDGs = 8; SDGs = 17**
SDGs only for developing countries	**Universal**
Bhopal → Stockholm 1972	**Bhopal 1984 → EPA 1986**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 16 — Environmental Organizations (India)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) | CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) | SPCB | NGT | NEERI | WII | BSI | ZSI | FSI

Sources baked in: EPA 1986, Water Act 1974, Air Act 1981, NGT Act 2010, MoEFCC, ISFR 2023, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — org ↔ function ↔ HQ ↔ report matching **Last verified:** August 2026

Current Affairs (this topic)

- MoEFCC / CPCB / SPCB / NBA HQ and function traps stay stable; scheme rebrands are the CA layer.
- NEERI = Nagpur remains a classic organisation location trap.

💡 🎯 **Consolidated — 28 Must-Score Facts (Click to Expand)**

1. **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is India's nodal ministry for environment (पर्यावरण), forests, wildlife, and climate change (जलवायु परिवर्तन). It began as MoEF in **1985** and was renamed MoEFCC in **2014**. Headquarters is **New Delhi**. 2. MoEFCC grants **Category A Environmental Clearance**, runs the **PARIVESH** portal, and administers EPA, Wildlife Act, Forest Conservation Act, and Biodiversity Act. It does **not** set NAAQS pollution standards. 3. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) was created under the **Water Act, 1974**, sits in **New Delhi**, sets **NAAQS**, and coordinates State Pollution Control Boards. The Air Act **1981** widened its air-pollution role. 4. CPCB does **not** grant Environmental Clearance, does **not** publish ISFR, and does **not** prepare the Annual Groundwater Quality Report — that report belongs to **CGWB**. 5. Every state has an **SPCB**. SPCBs issue **Consent to Establish** and **Consent to Operate** under the Water and Air Acts. They do **not** grant Category B EC — that is **SEIAA**. 6. **UPPCB** is Uttar Pradesh (उत्तर प्रदेश)'s board with headquarters at **Lucknow** (लखनऊ). It enforces norms on Kanpur (कानपुर) tanneries, Noida–Ghaziabad industry, and Ganga (गंगा)–Yamuna (यमुना) discharges. 7. **NGT** was created by the **NGT Act, 2010** (operational **2011**). Headquarters is New Delhi, with regional benches at Bhopal, Pune (पुणे), Kolkata, and Chennai. 8. NGT hears environmental appeals, including EC appeals, and can award compensation under Polluter Pays. The Act aims for disposal within **six months**. NGT does **not** itself grant EC. 9. The NGT Chairperson must be a retired **Supreme Court** judge or a High Court Chief Justice. 10. **WII** (**1982**, Dehradun (देहरादून)) does wildlife research and trains forest officers. It helped frame India's **ten** biogeographic zones and tiger-census methods. It is **not** the plant or animal survey body. 11. **BSI** (**1890**, Kolkata) surveys **plants** and publishes *Flora (वनस्पति) of India* and the plant Red Data Book (रेड डेटा बुक). **ZSI** (**1916**, Kolkata) surveys **animals** and publishes *Fauna (प्राणीजात) of India* and the animal Red Data Book. 12. **FSI** (**1981**, Dehradun) maps forest cover (वन आवरण) and publishes the **India State of Forest Report** every **two years**. ISFR is **not** an annual CPCB product. 13. **NEERI** (**1958**, **Nagpur**) began as **CPHERI** and is a CSIR laboratory for environmental science and engineering under the Ministry of Science and Technology. 14. Headquarters fact: **Delhi** — MoEFCC, CPCB, NGT; **Dehradun** (देहरादून) — WII, FSI; **Kolkata** — BSI, ZSI; **Nagpur** — NEERI; **Lucknow** — UPPCB. 15. Category A EC → **MoEFCC**. Category B EC → **SEIAA**. Industry consent → **SPCB**. EC appeal → **NGT**. 16. Pollution standards → **CPCB**. Forest cover report → **FSI**. Plant taxonomy → **BSI**. Animal taxonomy → **ZSI**. Wildlife training → **WII**. Environmental engineering research → **NEERI**. 17.

MoEFCC was **not** created in **1972**. That year marks the Stockholm Conference and the Wildlife (Protection) Act. 18. Consent under Water/Air Acts is **not** the same as Environmental Clearance under EPA/EIA rules. Confusing CTE/CTO with EC is a classic trap. 19. Forest clearance for diversion of forest land sits with **MoEFCC** under the Forest (Conservation) Act — not with CPCB or NGT. 20. BSI and ZSI both sit in Kolkata, but BSI is plants only and ZSI is animals only. Swapping kingdoms is a frequent wrong match. 21. WII and FSI both sit in Dehradun, but WII is wildlife science/training and FSI is forest-cover assessment. Do not give ISFR to WII. 22. NGT adjudicates; it does **not** replace CPCB on standards or MoEFCC on clearances. The older **NEAA (1997)** was dissolved after NGT was set up. 23. SPCBs work under CPCB technical guidance and state administrative control. They enforce standards; they do not write national NAAQS. 24. PARIVESH is MoEFCC's online EC application system — not a CPCB monitoring portal. 25. Match report to publisher: ISFR → **FSI**; Groundwater Quality Report → **CGWB**; NAAQS bulletins → **CPCB**; *Flora* → **BSI**; *Fauna* → **ZSI**. 26. NEERI is **Nagpur 1958**, not New Delhi 1970 and not a pollution-standards board. 27. **CAZRI = Jodhpur**; **IIFM = Bhopal**; operational Antarctica (भारती) stations = **Maitri + Bharati** (मैत्री). 28. **INCOIS = Hyderabad (हैदराबाद)**; ENVIS population (जनसंख्या)–environment centre = **IIPS Mumbai**.

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Botanical Survey of India (BSI)	**Zoological Survey of India (ZSI)**	Apex national taxonomic survey of plant resources (founded 1890, HQ: Kolkata) publishing the Red Data Book of Indian Plants vs apex national survey of faunal resources (founded 1916, HQ: Kolkata) documenting animal biodiversity	भारतीय वनस्पति सर्वेक्षण (BSI, 1890) / भारतीय प्राणी सर्वेक्षण (ZSI, 1916)
Forest Survey of India (FSI)	**Forest Research Institute (FRI)**	National organization (HQ: Dehradun) responsible for periodic remote-sensing assessment of forest cover and publishing the biennial India State of Forest Report (ISFR) vs premier academic research and deemed university under ICFRE (HQ: Dehradun)	भारतीय वन सर्वेक्षण (FSI, ISFR प्रकाशक) / वन अनुसंधान संस्थान (FRI)
Wildlife Institute of India (WII)	**Wildlife Crime Control Bureau (WCCB)**	Autonomous research and training institute under MoEFCC (Dehradun) conducting wildlife ecology research and census vs statutory enforcement bureau created under WPA 1972 Sec 38Y (New Delhi) combating organized wildlife poaching and illegal trade	भारतीय वन्यजीव संस्थान (WII, अनुसंधान) / वन्यजीव अपराध नियंत्रण ब्यूरो (WCCB, प्रवर्तन)
Central Pollution Control Board (CPCB)	**State Pollution Control Board (SPCB)**	Statutory apex body established under Water Act 1974 advising the Union Government, setting national ambient standards, and coordinating inter-state river/air monitoring vs state-level statutory body granting Consent to Establish/Operate to industrial units	केंद्रीय प्रदूषण नियंत्रण बोर्ड (मानक निर्माता) / राज्य प्रदूषण नियंत्रण बोर्ड (अनुमति प्रदाता)
National Green Tribunal (NGT)	**Central Empowered Committee (CEC)**	Specialized statutory environmental court established under NGT Act 2010 with judicial and expert members deciding civil environmental disputes vs Supreme Court monitoring committee overseeing forest conservation and wildlife sanctuary clearances	राष्ट्रीय हरित अधिकरण (NGT, सांविधिक पीठ) / केंद्रीय अधिकार प्राप्त समिति (CEC)
National Environmental Engineering Research Institute (NEERI)	**National Institute of Oceanography (NIO)**	CSIR premier lab specializing in water/air treatment, environmental engineering, and monitoring (founded 1958, Nagpur) vs CSIR oceanographic lab researching marine resources and ecology (Dona Paula, Goa)	नीरी (NEERI, नागपुर) / राष्ट्रीय समुद्र विज्ञान संस्थान (NIO, गोवा)

💡 📋 Must-score facts — MoEFCC, CPCB, NGT, surveys (Click to Expand)

HQ map

Item	Lock
MoEF → MoEFCC	**1985** → **2014**; Delhi
CPCB	Water Act **1974**; Delhi; NAAQS; coordinates SPCBs
UPPCB	Lucknow
NGT	Act **2010** (ops 2011); Delhi + Bhopal/Pune/Kolkata/Chennai
NGT Chair	Retired SC judge **or** HC Chief Justice
WII / FSI	**1982** / **1981**; both **Dehradun**
BSI / ZSI	**1890** / **1916**; both **Kolkata**
NEERI	**1958** Nagpur; CSIR (≠ MoEFCC)

16.1 Ministry of Environment, Forest and Climate Change (MoEFCC)

Identity: Nodal ministry | 1985/2014 | Delhi | Category A EC | PARIVESH

- **MoEFCC** is India's **nodal ministry** for environment, forests, wildlife, and climate change policy.
- It was established as **Ministry of Environment & Forests (MoEF) in 1985**.
- It was renamed **MoEFCC in 2014** when climate change was formally added to the mandate.
- **Headquarters** is **New Delhi**.
- MoEFCC administers **EPA 1986** (पर्यावरण संरक्षण 1986), **Wildlife (Protection) Act 1972**, **Forest (Conservation) Act 1980**, and **Biological Diversity Act 2002** (जैव विविधता अधिनियम 2002).
- It grants **Category A Environmental Clearance** through the **Expert Appraisal Committee (EAC)**.
- The **PARIVESH portal** is MoEFCC's online system for EC applications.
- MoEFCC approves **forest land diversion** under FCA 1980.
- It coordinates India's role in **UNFCCC (यूएनएफसीसीसी), CBD, and Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल)**.
- MoEFCC **does not** set **NAAQS** pollution standards — that is **CPCB's** role.
- MoEFCC **does not** publish **ISFR** — that is **FSI's** role.

> **Logic:** MoEFCC was **not** created in 1972. **1972** = Stockholm Conference + WPA.

Function	Body
Category A EC	**MoEFCC**
Forest clearance	**MoEFCC**
NAAQS standards	**CPCB**
Category B EC	**SEIAA**
EC appeals	**NGT**

16.2 Central Pollution Control Board (CPCB) (केंद्रीय प्रदूषण नियंत्रण बोर्ड) & State Pollution Control Board (SPCB) (राज्य प्रदूषण नियंत्रण बोर्ड)

Identity: Water Act 1974 | CPCB Delhi | SPCB consent | NOT EC authority

CPCB

- **CPCB** was established in **1974** under the **Water (Prevention and Control of Pollution) Act, 1974**.
- The **Air Act 1981** extended its role to air pollution control.
- **Headquarters** is **New Delhi**.
- CPCB advises the Centre, sets **National Ambient Air Quality Standards (NAAQS)**, and coordinates all **SPCBs**.
- It runs national **air and water monitoring** programmes.
- CPCB administers **EPR portals** for plastic, e-waste, battery, and tyre waste.
- CPCB **does not grant Environmental Clearance** — MoEFCC/SEIAA do.
- CPCB **does not publish ISFR** — FSI does.
- CPCB **does not prepare the Annual Groundwater Quality Report** — **CGWB** does.

PYQ — UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 prepared by:

1. CPCB 2. Central Water Commission 3. CGWB

💡 **Show answer**

Ans: B (Only 3) — **CGWB**, not CPCB.

SPCB

- Every state has a **State Pollution Control Board** under the **Water Act 1974** and **Air Act 1981**.
- SPCBs issue **Consent to Establish (CTE)** and **Consent to Operate (CTO)** to industries.
- They enforce **CPCB-prescribed standards** through inspection, sampling, and penalties.
- SPCBs work under **CPCB technical guidance** and **state government** administrative control.
- **UPPCB** is Uttar Pradesh's board with HQ at **Lucknow**.
- UPPCB enforces norms on **Kanpur tanneries**, **Ghaziabad/Noida industries**, and **Ganga/Yamuna** discharges.
- SPCB **does not grant Environmental Clearance** — **SEIAA/MoEFCC** do.

Term	Body	Purpose
Consent (CTE/CTO)	**SPCB**	Pollution control for industry operation
Environmental Clearance	**MoEFCC/SEIAA**	Project impact assessment
Forest Clearance	**MoEFCC**	Forest land diversion

16.3 National Green Tribunal (NGT) (राष्ट्रीय हरित अधिकरण)

Identity: NGT Act 2010 | Delhi HQ | Appeals | 6-month disposal target

- **NGT** was established under the **National Green Tribunal Act, 2010** and became operational in **2011**.
- **Headquarters** is **New Delhi** with regional benches at **Bhopal, Pune, Kolkata, and Chennai**.
- The **Chairperson** must be a retired **Supreme Court judge** or Chief Justice of a High Court.
- NGT hears **civil cases** with substantial environmental questions under EPA, WPA, FCA, Water Act, and Air Act.
- It hears **appeals against Environmental Clearance** decisions by MoEFCC/SEIAA.
- NGT can award **compensation** for environmental damage under the **Polluter Pays** principle.
- The Act targets disposal within **6 months**.
- NGT **does not grant EC** — it **adjudicates appeals**.
- NGT **does not set pollution standards** — CPCB does.
- NGT was established by an **Act of Parliament**. CPCB was constituted under the **Water Act, 1974** — it is **not** a mere executive order body.
- The older **National Environment Appellate Authority (NEAA / NEA)** was created under the **NEAA Act, 1997** to hear appeals on restricted-area industrial siting under EPA.
- After NGT began functioning (**18 October 2010**), the **NEAA stood dissolved**.

> **Logic:** NGT Act = **2010**, not 1986. NEAA 1997 → dissolved after NGT.

16.4 Wildlife Institute of India (WII) (भारतीय वन्यजीव संस्थान)

Identity: 1982 Dehradun | Wildlife research + training | NOT taxonomy

- **WII** was established in **1982** at **Dehradun** under **MoEFCC**.
- It is an autonomous institute for **wildlife research, science, and training**.
- WII trains **Indian Forest Service** and state forest officers.
- It helped delineate India's **10 biogeographic zones** (Rodgers & Panwar, 1988).

- WII develops **tiger census methodology** with NTCA and states.
- WII studies **protected areas**, **human-wildlife conflict**, and habitat (वास स्थान) management.
- WII **does not** maintain national plant/animal species catalogues — **BSI/ZSI** do.
- WII **does not** publish **ISFR** — **FSI** does.

16.5 Botanical Survey of India (BSI) (भारतीय वनस्पति सर्वेक्षण)

Identity: 1890 Kolkata | PLANTS | *Flora of India* | Red Data Book (plants)

- **BSI** was established in **1890** and is headquartered at **Kolkata**.
- It surveys and documents **Indian plant diversity** — flowering plants, ferns, algae, and fungi.
- BSI publishes ***Flora of India*** and state floras.
- It maintains the **Central National Herbarium (पादप संग्रहालय)** at Shibpur, Howrah.
- BSI publishes the **Red Data Book of Indian Plants**.
- BSI surveys **plants only** — **not animals**.

16.6 Zoological Survey of India (ZSI) (भारतीय प्राणी सर्वेक्षण)

Identity: 1916 Kolkata | ANIMALS | *Fauna of India* | Red Data Book (animals)

- **ZSI** was established in **1916** and is headquartered at **Kolkata**.
- It surveys and documents **Indian animal diversity** from invertebrates to mammals.
- ZSI publishes the ***Fauna of India*** series.
- It maintains **National Zoological Collections** for species identification.
- ZSI publishes the **Red Data Book of Indian Animals**.
- ZSI supports **CITES** (साइट्स) and **WPA Schedule** listing with scientific data.
- ZSI surveys **animals only** — **not plants**.

Body	Est.	HQ	Kingdom
BSI	1890	Kolkata	**Plants**
ZSI	1916	Kolkata	**Animals**
WII	1982	Dehradun	Wildlife research
FSI	1981	Dehradun	Forest cover

16.7 Forest Survey of India (FSI) (भारतीय वन सर्वेक्षण)

Identity: 1981 Dehradun | Biennial ISFR | Satellite forest mapping

- **FSI** was established in **1981** at **Dehradun** under **MoEFCC**.
- FSI assesses **forest cover** using satellite imagery and ground verification.
- It publishes the **India State of Forest Report (ISFR)** every **two years** — biennial, not annual.
- ISFR reports **forest cover classes**, **tree cover** (वृक्ष आवरण), and **carbon stock** estimates (प्राकलन समिति).
- FSI **does not** monitor industrial air/water pollution — **CPCB/SPCB** do.
- FSI **does not** survey plant/animal species taxonomy — **BSI/ZSI** do.

PYQ — UPPCS Prelims 2025, Q8 (ISFR link)

ISFR 2023 is published by **FSI**. UP districts with >20% forest cover included **Chandauli and Shravasti** — **Ans: C (2 and 3)**.

Report	Publisher
ISFR	**FSI**
Groundwater Quality Report	**CGWB**
NAAQS bulletin	**CPCB**
Flora of India	**BSI**
Fauna of India	**ZSI**

16.8 National Environmental Engineering Research Institute (NEERI)

Identity: 1958 Nagpur | Ex-CPHERI | CSIR | Environmental science & engineering

- **NEERI** is India's pioneering CSIR laboratory for **environmental science and engineering**.
- It was established in **1958** at **Nagpur** as the **Central Public Health Engineering Research Institute (CPHERI)**.
- Early focus was water supply, sewage disposal, related diseases, and industrial pollution control.
- NEERI works under the **Council of Scientific and Industrial Research (CSIR)** and falls under the **Ministry of Science and Technology**.
- It has zonal laboratories at **Chennai, Delhi, Hyderabad, Kolkata, and Mumbai**.
- NEERI is **not** a pollution-control board and **does not** set NAAQS — that is **CPCB**.
- HQ city: **Nagpur**, not Bengaluru, Hyderabad, or New Delhi.

Body	HQ	Job
NEERI	**Nagpur**	Env. engineering research (CSIR)
CPCB	New Delhi	Pollution standards / monitoring
NGT	New Delhi	Environmental adjudication

16.9 Institutes, Stations & Match Facts

Identity: Antarctica bases | CAZRI Jodhpur | IIFM Bhopal | FRI / IGFRI | ENVIS | INCOIS

- India's Antarctic stations: **Dakshin Gangotri (गंगोत्री) (1983–84)** first base (now supply/transit); **Maitri (1989)**; **Bharati (March 2013)**. Permanent **operational** pair today = **Maitri + Bharati** (not Dakshin Gangotri as active science base).
- **Wildlife Institute of India** — **Dehradun**. **Forest Research Institute** — **Dehradun**. **Indian Institute of Forest Management** — **Bhopal** (not Kolkata).
- **Central Arid Zone Research Institute (CAZRI)** — **Jodhpur** (ICAR; arid-zone research).
- **Indian Grassland and Fodder Research Institute** — **Jhansi** (झांसी). **IARI** — **New Delhi**.
- **National Institute of Nutrition** — **Hyderabad**. **National Institute of Ayurveda** — **Jaipur**. **National Institute of Naturopathy** — **Pune**.
- **Indian Institute of Ecology (पारिस्थितिकी) and Environment** — **New Delhi** (founded **5 June 1980** teaching).
- **ENVIS Centre on Population and Environment** — **IIPS Mumbai**.
- **Indian National Centre for Ocean Information Services (INCOIS)** — **Hyderabad** (1999; Ministry of Earth Sciences).
- **G.B. Pant Institute of Himalayan Environment and Development** — **Almora** (अल्मोड़ा). **Centre for Ecological Sciences** — **Bengaluru**.
- **Kailash Sacred Landscape Conservation and Development Initiative (KSLCDI)** partners: **India, China, Nepal**.

> **Logic:** IIFM = **Bhopal**. CAZRI = **Jodhpur**. Operational Antarctica = **Maitri + Bharati**.

Complete PYQ Bank (Topic 16)

Q1. UPPCS Prelims 2025, Q70

The **Annual Groundwater Quality Report 2024** was prepared by—

1. Central Pollution Control Board (CPCB)
2. Central Water Commission (CWC)
3. Central Ground Water Board (CGWB)

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1 only

💡 **Show answer**

Logic: Org–report matching trap — CPCB is pollution standards, not groundwater quality publisher. **Ans: B.** **CGWB** publishes the national groundwater quality report — not CPCB or CWC. **Trap:** CPCB is the reflex "central environment data" answer — groundwater quality is **CGWB's** mandate.

Q2. UPPCS Prelims 2025, Q8

India State of Forest Report (ISFR) 2023 is published by **FSI**. Which UP districts had **>20% forest cover**?

1. Bahraich
2. Chandauli
3. Shravasti

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1 only

💡 **Show answer**

Logic: FSI publishes ISFR — district-level UP forest-cover threshold fact. **Ans: C.** **Chandauli** and **Shravasti** exceed 20% forest cover; Bahraich does not cross the threshold. **Trap:** Bahraich is Terai forest country but **below 20%** in ISFR 2023 — not Chandauli-Shravasti pair.

Q3. UPSC pattern

Environmental Clearance for Category A projects is granted by—

A. Central Pollution Control Board B. **Ministry of Environment, Forest and Climate Change** C. National Green Tribunal D. Forest Survey of India

💡 **Show answer**

Logic: MoEFCC vs CPCB authority split — clearance ≠ pollution consent. **Ans: B.** **MoEFCC** (on EAC recommendation) grants **Category A EC**; CPCB does **not**. **Trap:** CPCB coordinates pollution control — **EC is MoEFCC/SEIAA**, not CPCB.

Q4. UPSC pattern

The **Central Pollution Control Board (CPCB)** was established under the—

A. Environment (Protection) Act, 1986 B. **Water (Prevention and Control of Pollution) Act, 1974** C. NGT Act, 2010 D. Wildlife (Protection) Act, 1972

💡 **Show answer**

Logic: CPCB statutory birth under Water Act — Air Act 1981 widened its air role. **Ans: B.** CPCB was constituted under the **Water Act, 1974**; Air Act **1981** extended its air-pollution functions. **Trap:** EPA **1986** is umbrella environmental law — CPCB predates it by **12 years**.

Q5. UPSC pattern

The **National Green Tribunal** was established under an Act of Parliament passed in—

A. 1986 B. 2005 C. **2010** D. 2015

💡 **Show answer**

Logic: NGT Act year — tribunal operational 2011, distinct from EPA 1986. **Ans: C.** **NGT Act, 2010** created the tribunal for environmental civil appeals. **Trap:** EPA **1986** and NGT **2010** are often confused — tribunal came **after** EIA notification 2006.

Q6. UPSC pattern

Consent to Operate (CTO) for industrial units is issued by the—

A. MoEFCC B. **State Pollution Control Board (SPCB)** C. Forest Survey of India D. Wildlife Institute of India

💡 **Show answer**

Logic: Consent (CTE/CTO) under Water/Air Acts vs EC under EPA — SPCB enforces operational pollution consent. **Ans: B.** **SPCB** issues **Consent to Establish** and **Consent to Operate** under Water Act 1974 and Air Act 1981. **Trap:** **SEIAA** grants Category B **EC** — SPCB grants **pollution consent**, a different clearance track.

Q7. UPSC pattern

The **Botanical Survey of India (BSI)** is headquartered at **Kolkata** and surveys—

A. Animals and birds only B. **Plants only** C. Forest cover area nationwide D. Groundwater aquifers

💡 **Show answer**

Logic: BSI vs ZSI kingdom split — Kolkata HQ for plant taxonomy since 1890. **Ans: B.** **BSI (1890, Kolkata)** surveys **plants** and publishes *Flora of India*; animals are **ZSI's** domain. **Trap:** Both BSI and ZSI are in **Kolkata** — swapping plant/animal mandates is the classic trap.

Q8. UPSC pattern

The **Red Data Book of Indian Plants** is published by—

A. Zoological Survey of India B. **Botanical Survey of India** C. Forest Survey of India D. Central Pollution Control Board

💡 **Show answer**

Logic: Report–publisher matching — plant Red Data Book = BSI; animal version = ZSI. **Ans: B.** **BSI** publishes the plant Red Data Book; **ZSI** publishes the fauna Red Data Book. **Trap:** FSI publishes **ISFR** forest-cover reports — not species Red Data Books.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to MoEFCC, which of the following statements is/are correct?

1. It began as MoEF in 1995 and was renamed MoEFCC in 2014.
2. It is India's nodal ministry for environment, forests, wildlife, and climate change.
3. It sits in Mumbai as its only headquarters fact.

A. Only 3 B. Only 2 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Statement 2 is true. Statement 1 is false — MoEF was created in **1985** (renamed MoEFCC in 2014), not 1995. Statement 3 is false — MoEFCC is New Delhi–centred, not Mumbai-only.

Q2. Which of the following pairs is/are correctly matched?

1. CPCB — Water Act 1974; Mumbai; NAAQS
2. SPCB — Consent to Establish / Consent to Operate
3. CPCB — grants Category A Environmental Clearance

A. Only 3 B. Only 2 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Only statement 2 remains true among the numbered claims. Statement 1 is false. Statement 2 is true. Statement 3 is false.

Q3. Match List-I with List-II:

List-I	List-II
A. WII	1. Nagpur
B. FSI	2. Dehradun
C. NEERI	3. Dehradun

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-3, B-2, C-1 D. A-2, B-3, C-1

💡 **Show answer**

Ans: D. A-2, B-3, C-1. **Logic:** WII & FSI Dehradun; NEERI Nagpur.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NGT was created by the NGT Act, 2010 (operational 2011).

Reason (R): The NGT Chairperson must be a retired Supreme Court judge or a High Court Chief Justice.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; chair qualification does not explain why the Act was created in 2010. **A/R logic:** Related NGT facts.

Q5. Arrange HQ map teaching:

1. Delhi — MoEFCC, CPCB, NGT
2. Dehradun — WII, FSI
3. Kolkata — BSI, ZSI
4. Lucknow — UPPCB

A. 1, 2, 3 and 4 B. Only 2 and 3 C. 1, 2 and 3 only D. Only 1 and 2

💡 **Show answer**

Ans: A. All four. **Logic:** Standard HQ facts.

Q6. With reference to NGT benches, which of the following statements is/are correct?

1. HQ New Delhi.
2. Regional benches include Bhopal, Pune, Kolkata, and Chennai.
3. NGT grants all Category A ECs instead of MoEFCC.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** NGT is appellate — not EC granter.

Q7. Which of the following pairs is/are NOT correctly matched?

1. BSI — 1890 Kolkata plants
2. ZSI — 1916 Kolkata animals
3. FSI — annual CPCB forest cover report

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** ISFR is biennial FSI — not CPCB.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): UPPCB headquarters is Lucknow.

Reason (R): UPPCB enforces norms on Kanpur tanneries and Noida–Ghaziabad industry teaching.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R illustrates UPPCB's state enforcement role. **A/R logic:** HQ + UP enforcement examples.

Q9. With reference to WII, which of the following statements is/are correct?

1. Established 1982, Dehradun.
2. Does wildlife research and trains forest officers.
3. Publishes ISFR every year from Delhi.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** ISFR is FSI — not WII.

Q10. Match List-I with List-II:

List-I	List-II
A. PARIVESH	1. CSIR lab Nagpur
B. NEERI	2. MoEFCC online EC portal
C. NBA	3. Chennai (BDA)

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: A.** A-2, B-1, C-3. **Logic:** PARIVESH portal; NEERI Nagpur; NBA Chennai.

Q11. Arrange MoEF → MoEFCC rename:

A. 2014 → 1985 B. 1985 → 2014 C. 1972 → 1986 D. 2006 → 2010

> 💡 **Show answer**
>
> **Ans: B.** MoEF 1985 → MoEFCC 2014. **Logic:** Rename fact.

Q12. With reference to CPCB limits, which of the following statements is/are correct?

1. Does not grant EC.
2. Does not publish ISFR.
3. Does not prepare Annual Groundwater Quality Report (CGWB does).

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

> 💡 **Show answer**
>
> **Ans: C.** All three. **Logic:** CPCB negative facts.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): SPCBs do not grant Category A Environmental Clearance.

Reason (R): Category A is appraised centrally by MoEFCC through EAC.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains why SPCB is not the A-track granter. **A/R logic:** Central A track.

Q14. Which one of the following correctly states NGT's compensation idea?

A. NGT replaces Supreme Court permanently B. NGT sets GST rates C. NGT issues passports D. Polluter Pays awards possible

💡 **Show answer**

Ans: D. Polluter Pays compensation. **Logic:** Environmental appeals body.

Q15. With reference to BSI/ZSI, which of the following statements is/are correct?

1. Both sit in Kolkata.
2. BSI surveys plants; ZSI surveys animals.
3. Both are identical to CPCB.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Survey bodies ≠ CPCB.

Q16. How many of the following Delhi HQ facts are correct?

MoEFCC, CPCB, NGT

A. Only one B. All three C. Only two D. None

💡 **Show answer**

Ans: B. All three. **Logic:** Delhi trio.

Q17. Which of the following pairs is/are correctly matched?

1. FSI 1981 — Dehradun — ISFR biennial
2. FRI — Dehradun forestry research
3. NEERI — Kolkata plant Red Data Book only

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** NEERI is Nagpur CSIR env lab.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NGT aims for disposal within six months under the Act's design.

Reason (R): NGT regional benches include Guwahati and Shimla as the only two benches.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Benches: Bhopal, Pune, Kolkata, Chennai (+ Delhi HQ).

Q19. With reference to MoEFCC functions, which of the following statements is/are correct?

1. Grants Category A EC.
2. Runs PARIVESH.
3. Administers EPA, WPA, FCA neighbourhood statutes.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Nodal ministry package.

Q20. Match List-I with List-II:

List-I	List-II
A. UPPCB	1. 1974 creation path
B. CPCB	2. Lucknow
C. NGT Act	3. 2010

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** UPPCB Lucknow; CPCB Water Act 1974; NGT 2010.

Q21. Arrange BSI / ZSI years:

A. ZSI 1890 → BSI 1916 B. BSI 1890 → ZSI 1916 C. Both 2010 D. Both 1974

💡 **Show answer**

Ans: B. BSI 1890 → ZSI 1916. **Logic:** Plant survey older than ZSI.

Q22. Which of the following statements about CTE/CTO is/are correct?

1. Issued by SPCBs under Water/Air Acts.
2. Not the same as Environmental Clearance under EPA/EIA.
3. CTE/CTO is granted by UNEP Nairobi only.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Consent ≠ EC; not UNEP.

Q23. With reference to WII biogeography link, which of the following statements is/are correct?

1. Helped frame India's ten biogeographic zones teaching.
2. Supports tiger-census methodology teaching.
3. WII is identical to NBA Chennai.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** WII ≠ NBA.

Q24. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NEERI began as CPHERI and is a CSIR laboratory.

Reason (R): NEERI is under the Ministry of Science and Technology world — not MoEFCC.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; ministry home does not explain the CPHERI origin by itself. **A/R logic:** Origin vs administrative home.

Q25. How many of the following are Dehradun facts?

WII, FSI, FRI

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Dehradun cluster.

Q26. Which of the following pairs is/are correctly matched?

1. NGT — environmental appeals including EC
2. CPCB — NAAQS
3. FSI — People's Biodiversity Register local body

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** PBR is BMC/BDA ladder — not FSI.

Q27. With reference to Air Act and CPCB, which of the following statements is/are correct?

1. Air Act 1981 strengthened the air pollution mandate alongside Water Act origins of CPCB.
2. CPCB coordinates SPCBs.
3. CPCB sits in Dehradun.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** CPCB Delhi.

Q28. Arrange creation years: CPCB path → NGT Act:

A. 2010 → 1974 B. Both 2006 C. Both 1985 D. 1974 → 2010

💡 **Show answer**

Ans: D. CPCB Water Act 1974 → NGT 2010. **Logic:** Institution chronology.

Q29. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): BSI publishes plant Red Data Book teaching; ZSI animal Red Data Book teaching.

Reason (R): Both organisations are headquartered in Chennai.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Both Kolkata — not Chennai.

Q30. Which one of the following correctly separates EC and consent?

A. EC = MoEFCC/SEIAA EPA-EIA; consent = SPCB Water/Air B. Both are identical UNEP products C. EC is only issued by UPPCB for Category A D. Consent replaces NGT appeals

💡 **Show answer**

Ans: A. EC vs CTE/CTO tracks differ. **Logic:** Classic separation.

Mains Answer Framework

Environmental organizations (250 words): Map India's governance chain — MoEFCC (policy, Category A EC, forest/wildlife laws), CPCB (national standards/monitoring), SPCB (state consent/enforcement), NGT (appeals). Scientific bodies — WII (wildlife research), BSI (plants), ZSI (animals), FSI (ISFR). HQ and report matching. UP angle — UPPCB Lucknow, Kanpur enforcement, ISFR UP districts. Distinguish consent vs clearance vs appeal. Conclude with coordination need and common traps (CPCB ≠ EC ≠ CGWB).

Common Traps — Don't Fall For These

Trap	Correct fact
CPCB grants EC	**MoEFCC/SEIAA**
CPCB publishes ISFR	**FSI**
CPCB prepares groundwater report	**CGWB**
MoEFCC created 1972	**1985**
BSI surveys animals	**Plants only**
ZSI surveys plants	**Animals only**
NGT grants EC	**Hears appeals**
ISFR is annual	**Biennial**
BSI/ZSI in Dehradun	**Kolkata**
SPCB grants Category B EC	**SEIAA**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 17 — Environmental Laws & Policies

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Acts: EPA 1986 (पर्यावरण संरक्षण 1986) | WPA 1972 | FCA 1980 | BDA 2002 | Water Act 1974 | Air Act 1981 | NGT Act 2010 | FRA 2006 **Policies:** National Forest Policy 1952 | Environmental Law | Wildlife Protection (वन्यजीव सुरक्षा) Laws

Sources baked in: NCERT, MoEFCC, Constitution Arts 48A/51A(g)/21, SC environmental jurisprudence, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — Act ↔ year, Bhopal→EPA, WPA schedules, FCA vs FRA **Last verified:** August 2026

Current Affairs (this topic)

- New Acts / Rules / NGT orders are living CA; WPA / EPA / FCA year spine stays fixed.
- Do not confuse NGT with NBA (biodiversity) on acronym stems.

💡 🎯 **Consolidated — 28 Must-Score Facts (Click to Expand)**

1. Memorise the act-year chain: **WPA 1972 → Water 1974 → FCA 1980 → Air 1981 → EPA 1986 → BDA 2002 → FRA 2006 → NGT Act 2010**. 2. **Article 48A** (DPSP (राज्य के नीति निदेशक तत्व)) directs the State to protect the environment (पर्यावरण). **Article 51A(g)** is the citizen's Fundamental Duty to protect nature. Do not swap them. 3. **Article 21** (अनुच्छेद 21) has been read to include a healthy environment. **Article 48** is about agriculture and cattle — it is **not** the environment article (**48A** is). 4. The **Environment (Protection) Act, 1986** is the post-Bhopal umbrella law under **Article 253**. Section **3** powers support EIA, CRZ, ESZ, **GEAC**, and **CGWA**. It **supplements** Water and Air Acts; it does not repeal them. 5. EIA Notification and related clearances flow from **EPA**, not from the Water Act. 6. The **Wildlife (Protection) Act, 1972** uses **six** Schedules. Schedules **I–IV** protect animals; Schedule **V** is vermin; Schedule **VI** covers plants. **Schedule I** is the highest protection; Schedule **V** animals may be hunted as vermin. 7. Under WPA, a **National Park** (Section **35**) is stricter than a **Sanctuary** (Section **18**). Tiger Reserves rest on Section **38V** (2006 amendment path). 8. The **Forest (Conservation) Act, 1980** requires prior **Central** approval before diverting forest land. CAMPA and NPV sit in this clearance world. 9. The **Forest Rights Act, 2006** recognises rights of Scheduled Tribes and Other Traditional Forest Dwellers. Individual forest rights are capped around **4 hectares**, with a cutoff of **13 December 2005**, and recognition (मान्यता) begins at the **Gram Sabha** (ग्राम सभा). 10. **FCA** controls **diversion** of forest land. **FRA** recognises **rights** of forest dwellers. Do not treat them as one Act. 11. The **Biological Diversity Act, 2002** implements the CBD through **NBA** (national), **SBB** (state), and **BMC** (local) with People's Biodiversity Registers. 12. The **Water Act, 1974** created **CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड)/SPCB** and the consent regime. The **Air Act, 1981** covers air pollution and NAAQS. Water is **1974**; Air is **1981** — do not reverse the years. 13. Industry **consent** (CTE/CTO) is under Water/Air Acts via **SPCB**. **Environmental Clearance** is under EPA/EIA via **MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय)/SEIAA**. Forest clearance is under **FCA** via MoEFCC. 14. The **NGT Act, 2010** creates the green tribunal for environmental appeals and compensation, with a six-month disposal target. 15. National Forest Policy **1952** was revenue-leaning. Later practice follows National Forest Policy **1988** targets of about **33%** forest cover (वन आवरण) nationally and **66%** in hills. 16. EPA is **1986**, not 1972. WPA is **1972**, not 1986. Mixing those years is a standard trap. 17. Schedule **V** under WPA is **not** the highest protection

tier — Schedule **I** is. 18. Consent is not EC, and EC is not forest clearance. Keep the three tools on separate legal tracks. 19. FRA's nodal line is tribal (आदिवासी) forest rights, not MoEFCC forest diversion under FCA. 20. BDA is about biodiversity access and benefit-sharing institutions — not the same as WPA schedules or NGT appeals. 21. Article **48A** binds the State as a Directive Principle; **51A(g)** binds the citizen as a duty. Both environment hooks sit beside Article **21** case law. 22. Water Act creates the pollution-board architecture; EPA is the later umbrella for broader environmental regulation including EIA. 23. Air Act year fact is **1981**. Pairing Air with **1974** or Water with **1981** is wrong. 24. NGT hears appeals; it does not rewrite the year of EPA or replace CPCB's standards role. 25. CAMPA/NPV issues arise after forest diversion under FCA — they are not FRA Gram Sabha rights tools. 26. Match clearance labels carefully: **SPCB consent**, **MoEFCC/SEIAA EC**, **MoEFCC forest clearance**. 27. Wildlife and forests sit on the **Concurrent List** (समवर्ती सूची) — Centre and states both legislate. 28. Indian courts have adopted Rio principles — **Precautionary**, **Polluter Pays**, and **Public Trust** (*Vellore Citizens' Welfare Forum*).

💡 ⚡ Confused Pairs (Click to Expand)

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Environment (Protection) Act (EPA 1986)** | **Wildlife (Protection) Act (WPA 1972)** | Umbrella legislation enacted under Article 253 post-Bhopal disaster enabling the Union to frame rules for EIA, CRZ, ESZ, and hazardous waste vs dedicated legislation protecting wild animal/plant species, regulating hunting, and establishing National Parks/Sanctuaries | पर्यावरण संरक्षण अधिनियम 1986 (छाता कानून) / वन्यजीव संरक्षण अधिनियम 1972 | | **Forest (Conservation) Act (FCA 1980)** | **Forest Rights Act (FRA 2006)** | Central law barring de-reservation or diversion of forest land for non-forest purposes without prior Union MoEFCC approval vs rights-based law administered by Ministry of Tribal Affairs restoring land tenure and customary rights to forest-dwelling STs and OTFDs | वन संरक्षण अधिनियम 1980 (विकास नियंत्रण) / वन अधिकार अधिनियम 2006 (जनजातीय अधिकार) | | **Water (Prevention and Control of Pollution) Act 1974** | **Air (Prevention and Control of Pollution) Act 1981** | India's first dedicated environmental pollution statute that established CPCB and SPCBs under Article 252 vs legislation enacted under Article 253 to implement 1972 Stockholm decisions that empowered existing Water Boards to regulate air pollution and noise | जल अधिनियम 1974 (प्रदूषण बोर्डों का गठन) / वायु अधिनियम 1981 (स्टॉकहोम अनुपालन) | | **Biological Diversity Act (BDA 2002)** | **Protection of Plant Varieties and Farmers' Rights Act (PPVFRA 2001)** | Implements 1992 CBD via three-tier structure (NBA, SBB, BMC) controlling commercial bio-resource access and benefit sharing (ABS) vs sui generis IPR regime granting legal rights to commercial plant breeders while safeguarding traditional farmers' seed-saving rights | जैव विविधता अधिनियम 2002 / पौधा किस्म व कृषक अधिकार संरक्षण 2001 | | **Genetic Engineering Appraisal Committee (GEAC)** | **Central Ground Water Authority (CGWA)** | Statutory apex regulatory body constituted under Rules 1989 of EPA 1986 approving environmental release of GMOs (e.g. Bt Cotton) vs regulatory authority constituted under Section 3(3) of EPA 1986 regulating commercial groundwater extraction NOCs | जीईएसी (GEAC, जीएमओ नियामक) / केंद्रीय भूजल प्राधिकरण (CGWA) | | **Coastal Regulation Zone (CRZ)** | **Eco-Sensitive Zone (ESZ)** | Coastal belt up to 500 m from High Tide Line classified into CRZ-I to IV under EPA 1986 to regulate beach and marine development vs buffer belt around protected areas notified under Section 3 of EPA 1986 to curb polluting industrial operations | तटीय विनियमन क्षेत्र (CRZ) / पर्यावरण-संवेदनशील क्षेत्र (ESZ) |

💡 📋 Must-score facts — act years, articles, clearances (Click to Expand)

Act chain | Item | Lock | |---|---| | Spine | WPA 72 → Water 74 → FCA 80 → Air 81 → EPA 86 → BDA 02 → FRA 06 → NGT 10 | | 48A / 51A(g) / 48 | State DPSP env / duty / agri–cattle (**not** env) | | EPA umbrella | Post-Bhopal; Art. 253; EIA/CRZ/ESZ/GEAC power | | WPA Schedules | I–IV animals; V vermin; VI plants | | NP vs Sanctuary | Sec. 35 stricter / Sec. 18 | | FCA vs FRA | Central diversion approval / forest-dweller rights | | BDA ladder | NBA → SBB → BMC + PBR | | Consent vs EC vs forest | SPCB Water/Air / MoEFCC–SEIAA EPA / FCA Central | --- ![India Environmental Legal Architecture & Statutory Chronology] (images/env_ch17_01_india_environmental_legislation_timeline.png)

17.1 Environmental Law (Constitutional Framework)

Identity: Art 48A | Art 51A(g) | Art 21 expansion | Concurrent List (समवर्ती सूची) | Rio principles in SC

- **Environmental law** in India combines **constitutional duties**, **statutory Acts**, and **Supreme Court/NGT jurisprudence**.
- **Article 48A** (DPSP) directs the **State** to protect and improve the environment and safeguard forests and wildlife.
- **Article 51A(g)** (Fundamental Duty) requires every **citizen** to protect and improve the natural environment.
- **Article 21** (Right to Life) was expanded by the Supreme Court to include the **right to a wholesome environment** (*M.C. Mehta*, *Subhash Kumar*).
- **Article 48** deals with **agriculture and animal husbandry** — it is **not** the environment article. Do not confuse with **48A**.
- Wildlife and forests fall under the **Concurrent List** (समवर्ती सूची) — Centre and states both legislate.
- The **pollution law chain** runs Water Act 1974 → Air Act 1981 → EPA 1986 as umbrella supplement.
- Indian courts have adopted (अंगीकृत) **Rio principles** — **Precautionary**, **Polluter Pays**, and **Public Trust** (*Vellore Citizens' Welfare Forum*).

Article	Type	Content
48A	DPSP	State protects environment
51A(g)	Fundamental Duty	Citizen protects environment
21	FR (expanded)	Healthy environment as part of life
48	DPSP	Agriculture — **not** environment

17.2 Environment (Protection) Act, 1986

Identity: Post-Bhopal 1984 | Umbrella law | Article 253 | Section 3 | EIA/CRZ/ESZ | GEAC | CGWA

- The **EPA was enacted in 1986** after the **Bhopal Gas Tragedy (1984)** exposed gaps in environmental regulation.
- Parliament passed it under **Article 253** to implement decisions of the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन) on the human environment.
- It is India's **umbrella environmental legislation** giving the Centre broad powers under **Section 3** to protect and improve the environment.
- **Section 6** empowers the Centre to make **rules** for environmental protection.
- EPA legally defines **"environment"** as water, air, land, and their inter-relationships with living beings.
- The **EIA Notification**, **CRZ rules**, and **ESZ notifications** are issued under **EPA Section 3** — not under the Water Act.
- EPA **supplements** the Water Act 1974 and Air Act 1981. It does **not replace** them.
- **Environmental Clearance** under EIA is distinct from SPCB **consent** under Water/Air Acts. Large projects may need **both**.
- The **Genetic Engineering Appraisal Committee (GEAC)** (जीईएसी) is constituted under the **Environment (Protection) Act, 1986**.
- It was earlier called the Genetic Engineering **Approval** Committee and was renamed **Appraisal** Committee on **22 July 2010**.
- GEAC is the apex body in MoEFCC for manufacture, use, import, export, and storage of hazardous microorganisms and genetically engineered organisms under the 1989 Rules.

- The **Central Ground Water Authority (CGWA)** (केंद्रीय भूजल प्राधिकरण) was constituted under **Section 3(3)** of the EPA, 1986 to regulate and control development and management of groundwater.
- **CGWA** (regulatory authority under EPA) is **not** the same as **CGWB** (assessment board under Jal Shakti (शक्ति)).
- Violations can attract imprisonment up to **5 years** and fines, extendable on continued violation.
- **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is the nodal ministry for EPA implementation.

PYQ — UPPCS Prelims 2022, Q11

When was the Environment (Protection) Act passed?

A. 1971

B. 1974

C. 1981

D. 1986

💡 **Show answer**

Ans: D

> **Logic:** Pair "EPA — 1974" or "Water Act — 1981" = **false**.

17.3 Wildlife (Protection) Act, 1972 & Wildlife Protection Laws

Identity: 6 Schedules | Hunting ban I–IV | NP Sec 35 | Tiger Reserve Sec 38V

- The **WPA was enacted in 1972** as India's unified **wildlife protection law**.
- It protects wild animals, birds, and plants and provides the legal basis for **National Parks**, **Sanctuaries**, and **Tiger Reserves**.
- **Six Schedules** classify species by protection level.

Schedule	Content
I	Highest protection — severe penalties
II–IV	Protected species — hunting restricted
V	**Vermin** — may be hunted
VI	Protected **plants**

- **Hunting** is banned for Schedule **I–IV** species with narrow exceptions for research or public safety with permission.
- **National Park (Section 35)** has the **strictest** protection — no grazing or exploitation.
- **Wildlife Sanctuary (Section 18)** allows more regulated human use under the Chief Wildlife Warden.
- **Tiger Reserves** are notified under **Section 38V** (2006 amendment) with **NTCA** oversight.
- The **National Board for Wildlife (NBWL)** (राष्ट्रीय वन्यजीव बोर्ड) is chaired by the **Prime Minister**.
- States implement WPA through the **Chief Wildlife Warden (CWLW)**.
- WPA does **not** cover forest land diversion (**FCA**) or community forest rights (**FRA**).

> **Logic:** Schedule **V = vermin**, not highest protection. Schedule **I** = highest.

17.4 Water (Prevention and Control of Pollution) Act, 1974 & Air Act, 1981

Identity: Water 1974 → CPCB/SPCB | Air 1981 → NAAQS | Consent mechanism

Water Act 1974

- The **Water Act was enacted in 1974** — India's first major post-Stockholm pollution law.
- It created the **Central Pollution Control Board (CPCB)** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) and **State Pollution Control Boards (SPCBs)**.
- SPCBs issue **Consent to Establish (CTE)** and **Consent to Operate (CTO)** for discharging effluents.
- CPCB sets **water quality standards** and coordinates state boards.

PYQ — UPPCS Prelims 2023, Q109 pattern

Water (Prevention and Control of Pollution) Act enacted in:

A. 1974

B. 1975

C. 1976

D. 1977

💡 **Show answer**

Ans: A

Air Act 1981

- The **Air Act was enacted in 1981** to prevent and control **air pollution**.
- It extended CPCB's role to **air quality** and **National Ambient Air Quality Standards (NAAQS)**.
- SPCBs enforce air emission norms alongside water consent under the same board structure.

Act	Year	Key output
Water Act	**1974**	CPCB, SPCB, water consent
Air Act	**1981**	NAAQS, air pollution control

17.5 Forest (Conservation) Act, 1980 | Forest Rights Act, 2006 | National Forest Policy, 1952

Identity: FCA = diversion control | FRA = dweller rights | NFP 1952 revenue → NFP 1988 ecology (पारिस्थितिकी)

FCA 1980

- The **FCA was enacted in 1980** on **Tiwari Committee** recommendations to stop unchecked deforestation.
- **Prior Central Government approval** is mandatory before using **forest land** for **non-forest purposes** like mining, dams, or industry.
- Diversion requires **compensatory afforestation** and payment of **Net Present Value (NPV)** into **CAMPA** funds.
- The **Godavarman case (1996)** expanded "forest" to include areas meeting the dictionary meaning of forest, not only notified forests.

FRA 2006

- The **Forest Rights Act 2006** recognizes **Scheduled Tribes** and **Other Traditional Forest Dwellers (OTFDs)** who depend on forests.

- It grants **Individual Forest Rights (IFR)** up to **4 hectares** of cultivated land.
- Claims must prove occupation before **13 December 2005**.
- The **Gram Sabha** verifies and recommends claims.
- **Community Forest Resource (CFR)** (सामुदायिक वन संसाधन अधिकार) rights allow communities to manage and protect forest resources.
- FRA **recognizes rights** — it does **not** authorize forest diversion. That is **FCA's** role.

National Forest Policy, 1952

- The **National Forest Policy 1952** focused on **revenue and industrial use** of forests.
- Under NFP **1952**, forests were classed as **Protected/Conserved Forests, National Forests, Village Forests, and Tree Lands** — **National Park is not** a 1952 forest class.
- It was superseded in practice by the **National Forest Policy 1988**, which targets **33% forest/tree cover (वृक्ष आवरण) nationally** and **66% in hills**, with people's participation and ecology first.
- NFP **1988** includes afforestation of wastelands, reforestation, and wood substitutes — **not** wide promotion of insecticides/pesticides.
- Notes may cite **1952** from syllabus — know it as the **early revenue-oriented policy**.

Act	Purpose
FCA 1980	Stop forest **diversion** without Central approval
FRA 2006	Recognize forest **dweller rights**
NFP 1952	Early policy — revenue focus

17.6 Biological Diversity Act, 2002 & National Green Tribunal Act, 2010

Identity: BDA = CBD domestic law | NBA Chennai | NGT = appeals tribunal

BDA 2002

- The **Biological Diversity Act 2002** (जैव विविधता अधिनियम 2002) implements India's obligations under the **Convention on Biological Diversity (CBD)**.
- The **National Biodiversity Authority (NBA)** (राष्ट्रीय जैव विविधता प्राधिकरण) is headquartered at **Chennai**.
- **State Biodiversity Boards (SBB)** operate at state level.
- **Biodiversity Management Committees (BMC)** function at local panchayat level.
- **People's Biodiversity Registers (PBR)** document local biodiversity.
- The Act regulates **access and benefit sharing (ABS)** of biological resources and traditional knowledge.

NGT Act 2010

- The **NGT Act was passed in 2010** and the tribunal became operational in **2011**.
- NGT is a **specialized environmental court** for fast disposal of environmental cases.
- It draws inspiration from the right to a **healthy environment** read into **Article 21**.
- It has jurisdiction over civil matters under **EPA, WPA, FCA, Water Act, Air Act, and BDA**.
- NGT hears **appeals against Environmental Clearance** and pollution orders.
- The Act targets case disposal within **6 months**.
- NGT **does not grant EC** — MoEFCC/SEIAA do.
- NGT was created by an **Act**. CPCB was constituted under the **Water Act, 1974** — not by a mere executive order.
- NGT provides environmental justice and reduces the burden on higher courts. CPCB promotes cleanliness of streams and wells and improves air quality.

- The **National Environment Appellate Authority (NEAA, 1997)** was dissolved after NGT was established.

Act	Year	Key body
BDA	2002	NBA Chennai
NGT Act	2010	NGT Delhi

Complete PYQ Bank (Topic 17)

Q1. UPPCS Prelims 2022, Q11

When was the Environment (Protection) Act passed?

A. 1971

B. 1974

C. 1981

D. 1986

> 💡 **Show answer**
>
> **Logic:** Tests the EPA enactment year after Bhopal — separate from WPA 1972 and Water Act 1974. **Ans: D.** The **Environment (Protection) Act was enacted in 1986** as India's umbrella environmental law. **Trap:** Pairing EPA with **1974** (Water Act year) or **1981** (Air Act year).

Q2. UPPCS Prelims 2023, Q109

The Water (Prevention and Control of Pollution) Act was enacted in:

A. 1974

B. 1975

C. 1976

D. 1977

> 💡 **Show answer**
>
> **Logic:** Tests India's first major post-Stockholm pollution law and the year that created CPCB/SPCB. **Ans: A.** The **Water Act was passed in 1974** — not 1981. **Trap:** Reversing Water **1974** with Air **1981**.

Q3. UPPCS Prelims pattern — EIA Notification

Under which law is the Environmental Impact Assessment (EIA) Notification issued?

A. Water (Prevention and Control of Pollution) Act, 1974

B. Wildlife (Protection) Act, 1972

C. Environment (Protection) Act, 1986

D. Forest (Conservation) Act, 1980

> 💡 **Show answer**
>
> **Logic:** EIA, CRZ, and ESZ rules flow from EPA Section 3 — not from sectoral Water or WPA laws. **Ans: C.** The **EIA Notification is issued under the EPA, 1986**. **Trap:** Crediting EIA to the **Water Act 1974** because pollution boards handle consent.

Q4. UPPCS Prelims pattern — Bhopal trigger

The Environment (Protection) Act, 1986 was enacted primarily after which event exposed gaps in environmental regulation?

A. Stockholm Conference, 1972

B. Bhopal Gas Tragedy, 1984

C. Rio Earth Summit, 1992

D. Chernobyl nuclear accident, 1986

💡 **Show answer**

Logic: Bhopal (1984) is the standard legislative trigger cited for EPA 1986 in Indian environmental law chronology. **Ans: B.** The **Bhopal Gas Tragedy (1984)** led to the **EPA in 1986**. **Trap:** Picking Stockholm 1972 (which influenced Water Act 1974) instead of Bhopal.

Q5. UPPCS Prelims pattern — WPA Schedules

Under the Wildlife (Protection) Act, 1972, which schedule lists species with the **highest** protection and which schedule lists **vermin**?

A. Schedule I — vermin; Schedule V — highest protection

B. Schedule I — highest protection; Schedule V — vermin

C. Schedule VI — highest protection; Schedule II — vermin

D. Schedule IV — highest protection; Schedule I — vermin

💡 **Show answer**

Logic: WPA six-schedule ladder — I is strictest for animals; V explicitly allows vermin control. **Ans: B.** **Schedule I** species get the **highest protection**; **Schedule V** lists **vermin**. **Trap:** Swapping Schedule **V** (vermin) with Schedule **I** (highest protection).

Q6. UPPCS Prelims pattern — Forest diversion

Prior approval of the Central Government is required for diversion of forest land for non-forest use under:

A. Forest Rights Act, 2006

B. Forest (Conservation) Act, 1980

C. Environment (Protection) Act, 1986

D. Biological Diversity Act, 2002

💡 **Show answer**

Logic: FCA 1980 governs forest land diversion; FRA 2006 recognises community rights — different statutes. **Ans: B.** **Forest diversion needs Central approval under the FCA, 1980**. **Trap:** Mixing **FCA 1980** (diversion) with **FRA 2006** (tribal forest rights).

Q7. UPPCS Prelims pattern — Tiger Reserves

Tiger Reserves in India are notified under which section of the Wildlife (Protection) Act, 1972?

A. Section 18

B. Section 35

C. Section 38V

D. Section 48A

💡 **Show answer**

Logic: Section 35 = National Park; Section 18 = Sanctuary; Section 38V = Tiger Reserve (2006 amendment path). **Ans: C.** **Tiger Reserves are notified under Section 38V** with NTCA oversight. **Trap:** Section **35** (National Park) or **18** (Sanctuary) distractors.

Q8. UPPCS Prelims pattern — NGT Act

The National Green Tribunal was established under an Act passed in:

A. 1986

B. 2006

C. 2010

D. 2015

💡 **Show answer**

Logic: NGT Act 2010 is distinct from EPA 1986 and DM Act 2005 — year recall. **Ans: C.** The **NGT was created by the NGT Act, 2010**. **Trap:** **EPA 1986** or **2006** (FRA year) as false neighbours.

Q9. UPPCS Prelims pattern — Water vs Air Act years

Which pair is correctly matched?

A. Water Act — 1981; Air Act — 1974

B. Water Act — 1974; Air Act — 1981

C. Water Act — 1986; Air Act — 1974

D. Water Act — 1972; Air Act — 1986

💡 **Show answer**

Logic: Pollution law chronology — Water first (1974), then Air (1981), then EPA umbrella (1986). **Ans: B.** **Water Act 1974** and **Air Act 1981** is the correct pair. **Trap:** Reversing **1974** and **1981** — the most common UPPCS trap in this chapter.

Q10. UPPCS Prelims pattern — Bhopal–EPA chain

Consider the following statements:

1. The Bhopal Gas Tragedy occurred in 1984.
2. The Environment (Protection) Act was passed in 1986.

A. Only 1

B. Only 2

C. Both 1 and 2

D. Neither 1 nor 2

💡 **Show answer**

Logic: Standard cause–law sequence tested across environmental law and governance chapters. **Ans: C.** **Bhopal 1984** preceded **EPA 1986** — both statements are correct. **Trap:** Shifting EPA to **1972** (WPA year) or Bhopal to **1986**.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. Arrange the act-year chain:

WPA → Water → FCA → Air → EPA → BDA → FRA → NGT

A. 1974-1972-1981-1980-1986-2006-2002-2010 B. 1972-1974-1980-1981-1986-2002-2006-2010 C. 1986-1972-1974-1980-1981-2002-2006-2010 D. 1972-1980-1974-1981-1986-2002-2006-2010

> 💡 **Show answer**
>
> **Ans: B.** 1972→1974→1980→1981→1986→2002→2006→2010. **Logic:** Master law spine.

Q2. With reference to Articles, which of the following statements is/are correct?

1. Article 48A — State DPSP on environment.
2. Article 51A(g) is a Directive Principle, not a Fundamental Duty.
3. Article 48 — environment protection article.

A. Only 1 B. 1 and 2 C. Only 3 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1 is correct. **Logic:** Statement 1 is true (Art. 48A DPSP). Statement 2 is false — 51A(g) is the citizen's Fundamental Duty. Statement 3 is false — environment DPSP is 48A, not 48.

Q3. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Article 21 has been read to include a healthy environment.

Reason (R): The 42nd Amendment (1976) added 48A and 51A(g).

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; 42nd Amendment does not by itself explain Art. 21 expansion. **A/R logic:** Two constitutional tracks.

Q4. Which of the following pairs is/are correctly matched?

1. EPA 1986 — post-Bhopal umbrella under Art. 253
2. EIA/CRZ/ESZ/GEAC powers have no link to EPA Section 3.
3. EIA Notification — flows from Water Act 1974 only

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 is correct. **Logic:** Statement 1 is true (EPA 1986 umbrella / Art. 253). Statement 2 is false — those powers sit in the EPA Section 3 neighbourhood. Statement 3 is false — EIA Notification flows from EPA, not from the Water Act alone.

Q5. Match List-I with List-II:

List-I	List-II
A. NP	1. Sec. 18
B. Sanctuary	2. Sec. 35
C. Tiger Reserve	3. Sec. 38V

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** NP 35; Sanctuary 18; TR 38V.

Q6. With reference to WPA Schedules, which of the following statements is/are correct?

1. Schedules I–IV protect animals.
2. Schedule V is vermin teaching.
3. Schedule VI covers plants.

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

💡 **Show answer**

Ans: B. All three. **Logic:** Six-schedule map.

Q7. Which of the following pairs is/are NOT correctly matched?

1. FCA — prior Central approval to divert forest land
2. FRA — forest-dweller rights
3. FCA and FRA — one and the same Act

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** FCA ≠ FRA.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Biological Diversity Act, 2002 implements CBD through NBA–SBB–BMC.

Reason (R): PBRs are prepared at the BMC level.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R fits the local rung of the BDA ladder. **A/R logic:** NBA→SBB→BMC+PBR.

Q9. Arrange consent vs EC vs forest clearance:

A. All three identical under Water Act only B. All granted only by UNEP C. All granted only by NGT D. Consent SPCB Water/Air; EC MoEFCC/SEIAA EPA; forest clearance FCA Central

💡 **Show answer**

Ans: D. Three different clearance tracks. **Logic:** Consent ≠ EC ≠ FCA.

Q10. With reference to Water vs Air Acts, which of the following statements is/are correct?

1. Water Act 1974 created CPCB/SPCB consent regime.
2. Air Act 1981 covers air pollution and NAAQS.
3. Water is 1981 and Air is 1974.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Years swapped in statement 3.

Q11. Match List-I with List-II:

List-I	List-II
A. BDA	1. 2006
B. FRA	2. 2002
C. NGT Act	3. 2010

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** 2002 / 2006 / 2010.

Q12. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): National Park is stricter than Sanctuary under WPA.

Reason (R): Schedule VI plants may be cultivated freely without any licence.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Schedule VI needs CWLW licence route.

Q13. With reference to CAMPA/NPV, which of the following statements is/are correct?

1. They sit in the FCA forest-clearance world.
2. FCA requires prior Central approval for diversion.
3. CAMPA repealed WPA Schedules.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Does not repeal WPA.

Q14. Which one of the following correctly states FRA individual forest rights cap teaching neighbourhood?

A. FRA is the Montreal Protocol B. FRA is identical to EPA 1986 C. FRA creates UNEP D. FRA recognises rights of STs and OTFDs

💡 **Show answer**

Ans: D. ST/OTFD rights recognition. **Logic:** Keep FRA identity clear.

Q15. How many of the following article facts are correct?

1. 48A State DPSP env
2. 51A(g) duty
3. 21 healthy environment reading

A. All three B. Only two C. Only one D. None

Show answer

Ans: A. All three. **Logic:** Constitutional env trio.

Q16. With reference to NGT Act 2010, which of the following statements is/are correct?

1. Creates green tribunal for environmental appeals/compensation.
2. Six-month disposal target teaching.
3. NGT is the same as SEIAA.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: B. Only 1 and 2. **Logic:** NGT ≠ SEIAA.

Q17. Arrange EPA relative to Bhopal:

A. EPA 1986 before Bhopal 1984 B. EPA 1972 with Stockholm only C. EPA 1986 post-Bhopal 1984 D. EPA 2010 with NGT only

Show answer

Ans: C. Post-Bhopal umbrella 1986. **Logic:** 1984 tragedy → 1986 EPA.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Industry consent (CTE/CTO) is under Water/Air via SPCB.

Reason (R): Environmental Clearance is under EPA/EIA via MoEFCC/SEIAA.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: A. Both true; each describes a different track — R does not explain consent. **A/R logic:** Parallel regimes.

Q19. Which of the following pairs is/are correctly matched?

1. WPA 1972 — six Schedules
2. Air 1981 — after Water 1974
3. FRA 2006 — before BDA 2002

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

Show answer

Ans: D. Only 1 and 2. **Logic:** BDA 2002 before FRA 2006.

Q20. Match List-I with List-II:

List-I	List-II
A. Art. 48A	1. Citizen duty
B. Art. 51A(g)	2. State DPSP env
C. Art. 48	3. Agriculture/cattle

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** 48A DPSP; 51A(g) duty; 48 agri.

Q21. With reference to Tiger Reserve law, which of the following statements is/are correct?

1. Section 38V (2006 path).
2. Under NTCA governance neighbourhood.
3. Same as Community Reserve Section 36B exactly.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** 38V ≠ 36B.

Q22. Which of the following statements about Schedule I is/are correct?

1. Maximum protection among animal schedules teaching.
2. Free hunting with oral collector order is allowed.
3. Tiger/elephant/rhino often appear in Schedule I facts.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 3. **Logic:** No free hunting.

Q23. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): FCA controls diversion of forest land.

Reason (R): FRA recognises rights of forest dwellers.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; different Acts — R does not explain FCA diversion control. **A/R logic:** FCA ≠ FRA jobs.

Q24. How many of the following years are correctly paired?

WPA 1972, Water 1974, Air 1981, EPA 1986

A. Only one B. Only two C. Only three D. All four

💡 **Show answer**

Ans: D. All four. **Logic:** Core years.

Q25. With reference to GEAC/CRZ/ESZ, which of the following statements is/are correct?

1. They sit in EPA Section 3 power neighbourhood.
2. They are Water Act Schedule V categories.
3. ESZ is an EPA buffer idea around PAs.

A. 1 and 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 3. **Logic:** Not Water Act Schedule V.

Q26. Arrange BDA ladder:

A. BMC → NBA → SBB B. NBA → SBB → BMC (+ PBR) C. SPCB → CPCB → NGT only D. MoEF → MoEFCC only

💡 **Show answer**

Ans: B. NBA → SBB → BMC + PBR. **Logic:** Biodiversity institutions.

Q27. Which of the following pairs is/are NOT correctly matched?

1. EPA — umbrella post-Bhopal
2. Art. 253 — EPA neighbourhood teaching
3. Art. 48 — environment DPSP

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** 48A is env DPSP — not 48.

Q28. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NGT can award compensation under Polluter Pays teaching.

Reason (R): NGT Act creates the green tribunal for environmental appeals.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R frames why NGT can hear such matters. **A/R logic:** Tribunal + remedy.

Q29. With reference to Sanctuary vs NP, which of the following statements is/are correct?

1. Sanctuary Sec. 18 — limited regulated use possible.
2. NP Sec. 35 — stricter.
3. Sanctuary is always stricter than NP.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** NP stricter.

Q30. Which one of the following is the correct Water/Air year pair?

A. Water 1974; Air 1981 B. Water 1981; Air 1974 C. Both 1986 D. Both 2002

💡 **Show answer**

Ans: A. Water 1974; Air 1981. **Logic:** Classic swap trap.

Mains Answer Framework

Environmental laws (250 words): Constitutional base (48A, 51A(g), Art 21). Chronological chain — WPA 1972, Water 1974, FCA 1980, Air 1981, EPA 1986, BDA 2002, FRA 2006, NGT 2010 (एनजीटी 2010). EPA as umbrella + EIA. WPA schedules and protected areas. FCA vs FRA distinction. Pollution consent vs EC vs forest clearance. BDA three-tier structure. NGT role. NFP 1952/1988. Conclude with integrated compliance and Bhopal as law reform trigger.

Common Traps — Don't Fall For These

Trap	Correct fact
Water Act = 1981	**Water 1974; Air 1981**
EPA = 1972	**EPA 1986; WPA 1972**
FCA = FRA	**1980 diversion vs 2006 rights**
Art 48 = environment	**Art 48A**
Schedule V highest protection	**Schedule I**
EIA under Water Act	**EPA 1986**
NGT grants EC	**MoEFCC/SEIAA**
NBA HQ Delhi	**Chennai**
NFP 1988 in syllabus list	Syllabus says **1952** — know both
Consent = EC	**SPCB consent ≠ EC**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 18 — International Environmental Agreements & Conferences

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Conferences: Stockholm 1972 (स्टॉकहोम सम्मेलन) | Brundtland Commission | Rio 1992 | Agenda 21 | SD conferences **Treaties:** CBD | UNFCCC (यूएनएफसीसीसी) | Kyoto (क्योटो) | Paris | Montreal | Ramsar (रामसर) | CITES (साइट्स) | CMS | Vienna | Basel | Rotterdam | Stockholm POPs | Minamata

Sources baked in: UNEP, UNFCCC, CBD Secretariat, Ramsar, CITES, NCERT, MoEFCC, UPPCS PYQs 2018–2025
Weight: ★★★ High — Montreal ≠ Kyoto, Agenda 21 = 1992, treaty ↔ issue matching **Last verified:** August 2026

Current Affairs (this topic)

- COP outcomes and new listings refresh CA; treaty years and Montreal≠Kyoto stay fixed.
- Ramsar / CITES / CMS updates — verify live lists; UP Ramsar baseline remains **12**.

💡 🎯 **Consolidated — 30 Must-Score Facts (Click to Expand)**

1. Conference spine: **Stockholm 1972 → Brundtland 1987 → Rio 1992 → Johannesburg 2002 → Rio+20 2012 → SDGs and Paris 2015**. 2. **Stockholm 1972** (स्टॉकहोम सम्मेलन) was the first UN Conference on the Human Environment (पर्यावरण). It led to **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) (Nairobi). World Environment Day (विश्व पर्यावरण दिवस) is **5 June**. The slogan fact is "Only One Earth." 3. The **Brundtland Report** (**1987**, *Our Common Future*) popularised **sustainable development**. It is **not** *Limits to Growth* (**1972**, Club of Rome). 4. **Rio 1992** (UNCED) produced the Rio Declaration, **Agenda 21**, Forest Principles, and opened **UNFCCC** (यूएनएफसीसीसी) and **CBD** for signature. **CITES** is **not** a Rio product — CITES is **1973**. 5. **Agenda 21** is Rio's non-binding sustainable-development action plan for the twenty-first century. It was **not** signed in 1995. 6. Climate chain: **UNFCCC 1992 → Kyoto Protocol 1997 → Paris Agreement (पेरिस समझौता) 2015**. Paris uses universal **NDCs** and the **1.5°C** ambition language. 7. Ozone chain: **Vienna Convention 1985 → Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) 1987 → Kigali Amendment 2016** (HFCs). Montreal is **ozone**, not climate. 8. Treaty–issue fact: Montreal/Vienna = ozone; Kyoto/Paris/UNFCCC = climate; CBD = biodiversity; Ramsar = wetlands; CITES = wildlife **trade** (पण्याध्यक्ष). 9. **CMS (Bonn)** covers migratory species. **Basel** covers hazardous **waste**. **Rotterdam** covers hazardous **chemicals** (PIC). **Stockholm POPs** covers persistent organic pollutants. **Minamata** covers **mercury**. 10. **CBD** has three pillars: conserve biodiversity, sustainable use, and fair benefit-sharing. **Nagoya Protocol (2010)** details access and benefit-sharing. 11. **Ramsar** (रामसर) (**1971**) is the wetlands treaty under the Wise Use idea. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर). 12. **CITES** (**1973**, Washington) regulates trade through Appendices **I–III**. IUCN (आईयूसीएन)'s Red List assesses extinction risk; CITES regulates trade — different jobs. 13. **GEF** was established in **1991** on the eve of Rio as a finance mechanism for major environmental conventions. 14. **UNCCD (1994)** (मरुस्थलीकरण रोकथाम संधि) is the sole legally binding convention linking environment, development, and sustainable land management; it stresses bottom-up local participation against desertification (मरुस्थलीकरण). 15. The **New York Declaration on Forests (2014)** is a voluntary, non-legally binding political declaration (halve forest loss by 2020 / end by 2030 ambition language); India was **not** an original signatory. 16. Kyoto set binding

targets mainly for developed parties. Paris is a universal NDC architecture. Do not treat them as identical instruments. 17. Kigali amends Montreal to phase down **HFCs**. It is an ozone-protocol amendment path, not a UNFCCC protocol. 18. Basel is waste shipment control. Rotterdam is prior informed consent for chemicals. Stockholm POPs is about persistent toxics such as listed DDT/PCB classes. Minamata is mercury. 19. CMS is migratory species; CITES is trade; Ramsar is wetlands. Mixing those three is a classic match trap. 20. UNEP was born from Stockholm, not from Rio. Rio opened UNFCCC and CBD. 21. Paris is **2015**; Kyoto is **1997**; UNFCCC framework is **1992**. Keep the climate chronology straight. 22. Montreal successful ODS phase-out is often contrasted with slower climate politics — but stems still demand the correct issue tag, not slogans. 23. Rudrasagar is a **Tripura** Ramsar site. Sultanpur is **Haryana**. Neither is an Uttar Pradesh (उत्तर प्रदेश) site. 24. Uttar Pradesh Ramsar examples include sites such as Nawabganj, Samaspur, Sarsai Nawar, Upper Ganga (गंगा), Bakhira, and Haiderpur — verify the live list before the paper, but never park Rudrasagar or Sultanpur in UP. 25. CBD opened at Rio **1992**; Nagoya ABS is **2010**. Do not date CBD as 2010. 26. Forest Principles at Rio are non-binding soft law — not the same as CITES appendices. 27. Johannesburg **2002** and Rio+20 **2012** continue the sustainable-development conference line after Rio; they do not reopen CITES as a 1992 invention. 28. “Only One Earth” and UNEP creation fact to Stockholm **1972**, not to Paris **2015**. 29. Wise Use is the Ramsar philosophy. It is not the CBD’s three objectives word-for-word. 30. Master year table worth ratta: Ramsar 1971, CITES 1973, Vienna 1985, Montreal 1987, Basel 1989, UNFCCC/CBD 1992, UNCCD 1994, Kyoto 1997, Rotterdam 1998, Stockholm POPs 2001, Nagoya 2010, Minamata 2013, New York Forests Declaration 2014, Paris 2015, Kigali 2016.

💡 ⚡ Confused Pairs (Click to Expand)

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Stockholm Conference (1972)** | **Rio Earth Summit (1992)** | First UN global conference on the Human Environment that placed environment on the global political agenda and created UNEP vs UN Conference on Environment and Development (UNCED) that produced Rio Declaration, Agenda 21, CBD, UNFCCC, and UNCCD | स्टॉकहोम सम्मेलन 1972 (UNEP उद्भव) / रियो पृथ्वी शिखर सम्मेलन 1992 (रियो संधियां) | | **Kyoto Protocol (1997)** | **Paris Agreement (2015)** | Annex-I developed nations bound by legal emission targets with punitive compliance mechanisms vs universal bottom-up framework where all 196+ nations self-determine mitigation contributions (NDCs) with five-yearly global stocktakes | क्योटो प्रोटोकॉल (शीर्ष-से-नीचे बाध्यकारी) / पेरिस समझौता (स्वैच्छिक NDCs) | | **Basel Convention (1989)** | **Rotterdam Convention (1998)** | Restricts and regulates transboundary movements and disposal of hazardous wastes (especially preventing dumping in developing countries) vs establishes legally binding Prior Informed Consent (PIC) procedure for hazardous international trade chemicals and pesticides | बासेल कन्वेंशन (खतरनाक अपशिष्ट आवागमन) / रॉटरडैम कन्वेंशन (पूर्व सूचित सहमति) | | **Stockholm Convention (2001)** | **Minamata Convention (2013)** | Mandates the global elimination, restriction, and unintentional release reduction of Persistent Organic Pollutants (POPs - "Dirty Dozen") vs global treaty designed to protect human health and the environment from anthropogenic emissions and releases of toxic Mercury (Hg) | स्टॉकहोम संधि (POPs उन्मूलन) / मिनामाता संधि (पारा प्रदूषण नियंत्रण) | | **Ramsar Convention (1971)** | **Convention on Migratory Species (CMS / Bonn 1979)** | Dedicated specifically to the ecological conservation and sustainable wise use of internationally significant wetlands and waterfowl habitats vs treaty dedicated to the conservation of terrestrial, aquatic, and avian migratory wild animal species across range states | रामसर कन्वेंशन (आर्द्रभूमि संरक्षण) / बॉन कन्वेंशन (प्रवासी प्रजाति संरक्षण) | | **CITES (1973)** | **Convention on Biological Diversity (CBD 1992)** | International trade treaty strictly regulating commercial import/export of endangered wild flora and fauna through Appendices I, II, III vs comprehensive framework convention covering in-situ/ex-situ conservation, sustainable use, and fair benefit-sharing | साइट्स (CITES, वन्यजीव व्यापार) / जैव विविधता संधि (CBD, समग्र संरक्षण) |

💡 📋 Must-score facts — conferences, treaty–issue map (Click to Expand)

Conferences | Item | Lock | |---|---| | Spine | Stockholm 72 → Brundtland 87 → Rio 92 → Joburg 02 → Rio+20 12 → SDGs/Paris 15 | | Stockholm → UNEP | **1972**; UNEP Nairobi; WED **5 June** | | Rio products |

Agenda 21 · UNFCCC · CBD (CITES **not** Rio product) | | Climate ladder | UNFCCC 92 → Kyoto 97 → Paris 15 | | Ozone ladder | Vienna 85 → Montreal 87 → Kigali 16 | ### Issue tags | Item | Lock | |------|------| | Montreal / Kyoto–Paris | Ozone / climate | | CBD / Ramsar / CITES | Biodiversity / wetlands / wildlife trade | | Basel / Rotterdam / Stockholm POPs | Haz. waste / PIC chemicals / POPs | | CMS (Bonn) | Migratory species | | GEF / UNCCD | **1991** finance / **1994** desertification | --- ![Global Environmental Governance Summits Roadmap & MEA Clusters](images/env_ch18_01_global_environmental_treaties_roadmap.png)

18.1 Conferences — Stockholm, Brundtland, Rio, Agenda 21 & SD Summits

Identity: UNEP 1972 | SD definition 1987 | Rio triple outcome 1992 | Agenda 21 trap | Earth Summit +5 | Rio+20

1972 Stockholm Summit - The **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन) was the first major UN conference on the **human environment**. - Its theme was **"Only One Earth"** and it adopted (अंगीकृत) the **Stockholm Declaration** with 26 principles. - Stockholm led to creation of **UNEP** with headquarters in **Nairobi, Kenya**. - **World Environment Day** is celebrated on **5 June** every year, marking the Stockholm opening day. 1987 Brundtland - The **Brundtland Commission (WCED)** published ***Our Common Future*** in **1987** with the classic **sustainable development** definition. - The Brundtland Report is **not** *The Limits to Growth* (Club of Rome, 1972). - Chronology often tested: ***Limits to Growth*** **(1972) → Brundtland Report (1987)** (ब्रंटलैंड रिपोर्ट) → **Montreal Protocol enforcement (January 1989) → Rio Earth Summit (June 1992)**. - Montreal was **signed in September 1987**; papers that say "enforcement" mean **January 1989**. 1992 Rio Earth Summit - The **Rio Earth Summit (1992)** (रियो पृथ्वी शिखर सम्मेलन 1992) or **UNCED** produced the **Rio Declaration**, opened **UNFCCC** and **CBD** for signature, and adopted **Agenda 21**. - **Agenda 21** is a non-binding action plan for **sustainable development in the 21st century** — adopted at **Rio 1992**, not 1995. - Agenda 21 is a large multi-chapter blueprint (about **40 chapters** in **4 sections** / groupings) — bank stems sometimes say "4 agreements." - **Local Agenda 21** encourages cities and local bodies to prepare their own SD plans. - The **Commission on Sustainable Development (CSD)** was created to monitor Earth Summit implementation. - **Earth Summit +5** was the UN General Assembly special session (पंगु सत्र) held in **New York in June 1997** to review Rio progress. - **Johannesburg Summit (2002)** was Rio+10 on sustainable development and reaffirmed Agenda 21. - **Rio+20 (2012)** — the United Nations Conference on Sustainable Development — reviewed SD progress and produced the outcome document ***The Future We Want***. - **PAGE (Partnership for Action on Green Economy (हरित अर्थव्यवस्था))** was created as the UN response to Rio+20's call for green-economy transitions. - **SDGs were adopted in 2015** at the UN General Assembly. - The **Global Environment Facility (GEF)** (जीईएफ) was established in **1991** on the eve of Rio to finance global environmental projects. - GEF is a financial mechanism for five major conventions: **Minamata** (mercury), **Stockholm POPs**, **CBD**, **UNCCD**, and **UNFCCC**. - Key implementing partners include **UNDP** (यूएनडीपी), **UNEP**, and the **World Bank**. GEF also supports tools such as the **Least Developed Countries Fund (LDCF)** for National Adaptation Programmes of Action.

PYQ — UPPCS Prelims 2025, Q51

(A) SD should not compromise future generations' needs. (R) Agenda 21 was signed in 1995.

💡 **Show answer**

Ans: C — Agenda 21 = **Rio 1992**.

PYQ — UPPCS Prelims 2023, Q132

Agenda-21 of Rio Summit (1992) is related to:

A. Sustainable Development

B. Polluter Pays

C. Environmental Education (पर्यावरण शिक्षा)

D. Ozone Layer

💡 **Show answer**

Ans: A

PYQ — UPPCS Prelims 2024, Q51

Brundtland Report is also known as *The Limits to Growth*?

💡 **Show answer**

Ans: D (A true, R false) — Brundtland = *Our Common Future* (1987).

Conference	Year	Key output
Stockholm	1972	UNEP, WED 5 June
Limits to Growth	1972	Club of Rome study
Brundtland	1987	SD definition
Montreal enforcement	Jan 1989	ODS phase-out starts
Rio	1992	UNFCCC, CBD, Agenda 21
Earth Summit +5	1997	Rio progress review (NY)
Johannesburg	2002	Rio+10
Rio+20	2012	*The Future We Want*; PAGE
SDGs	2015	17 goals, 2030 deadline

18.2 Climate Treaties — UNFCCC, Kyoto Protocol & Paris Agreement

Identity: UNFCCC 1992 framework | Kyoto 1997 binding Annex-I | Paris 2015 NDCs

- The **UNFCCC** was opened for signature at **Rio 1992** to stabilize greenhouse gas concentrations.
- UNFCCC holds annual **COP** meetings. The secretariat is in **Bonn, Germany**.
- The **Kyoto Protocol (1997)** (क्योटो प्रोटोकॉल) set **legally binding emission reduction targets** for **Annex-I (developed) countries**. It entered into force on **16 February 2005**.
- Kyoto introduced **Clean Development Mechanism (CDM)** carbon credits and **Emissions Trading**. CDM projects run in **developing countries**; **1 CER ≈ 1 tCO_2**.
- The **Green Climate Fund (GCF)** (जीसीएफ) was established at **Cancun COP-16 (2010)** under UNFCCC — not at Durban.
- **Earth Hour** (अर्थ आवर) is a **WWF** (डब्ल्यूडब्ल्यूएफ) climate-awareness event (usually last Saturday of March). **GHG Protocol** = **WRI + WBCSD**. 2015 Paris & SDGs
- The **Paris Agreement (2015)** (पेरिस समझौता) at **COP21** applies to **all countries** through **Nationally Determined Contributions (NDCs)**.
- Paris aims to limit warming to **well below 2°C**, pursuing **1.5°C**.
- Paris entered into force in **2016**. It is **not identical** to Kyoto's binding annex-only model.

PYQ — UPPCS Prelims 2024, Q35

Which agreement relates to climate change (जलवायु परिवर्तन) mitigation?

A. Montreal Protocol

B. Ramsar

C. Kyoto Protocol

D. CITES

💡 **Show answer**

Ans: C — Montreal = **ozone**, not climate.

Treaty	Year	Model
UNFCCC	1992	Framework
Kyoto	1997	Binding Annex-I targets
Paris	2015	Universal NDCs

> **Key fact:Montreal** = ozone/ODS phase-out (this section). **UNFCCC–Paris** = climate/NDC framework (see climate-change chapters). India's **NAPCC** and **NDC** pledges sit under national climate policy, not the Montreal ozone regime.

18.3 Ozone Treaties — Vienna Convention & Montreal Protocol

Identity: Vienna 1985 framework | Montreal 1987 ODS phase-out | Kigali 2016 HFCs

- The **Vienna Convention (1985)** (वियना कन्वेंशन) created the framework to protect the **ozone layer**.
- The **Montreal Protocol (1987)** (मॉन्ट्रियल प्रोटोकॉल) set **binding schedules** to phase out **ozone-depleting substances (ODS)** like CFCs.
- Montreal is the **most successful environmental treaty** — universal ratification, ODS emissions down ~99%.
- **International Ozone Day** is **16 September** (Montreal signing date).
- The **Kigali Amendment (2016)** (किगाली संशोधन) phases down **HFCs** under the Montreal framework.
- Montreal controls **ozone/ODS** — **not climate change mitigation** trap logic.

> **Logic:** Full ozone chemistry and Antarctic hole → **Topic 11**.

18.4 Biodiversity & Species Treaties — CBD, CITES, CMS & Ramsar

Identity: CBD 1992 three objectives | CITES trade | CMS migration | Ramsar wetlands

CBD

- The **Convention on Biological Diversity (CBD)** was opened at **Rio 1992**.
- Its three objectives are **conservation**, **sustainable use**, and **fair benefit-sharing (ABS)**.
- The **Cartagena Protocol** covers **biosafety** for living modified organisms.
- **TRAFFIC** (WWF–IUCN alliance) monitors wildlife trade for sustainability — it is **not** a UNEP bureau.
- **BirdLife International** identifies Important Bird and Biodiversity Areas; biodiversity **hotspots** were framed by **Norman Myers**, not BirdLife.
- The **Nagoya Protocol (2010)** implements **Access and Benefit Sharing (ABS)**.
- India implemented CBD through the **Biological Diversity Act, 2002**.

CITES

- **CITES (1973, Washington)** regulates **international trade** in endangered wild fauna (प्राणीजात) and flora (वनस्पति).
- **Appendix I** — trade banned except in exceptional cases.
- **Appendix II** — regulated trade with permits.
- **Appendix III** — species listed by a member country requesting cooperation.

CMS (Bonn Convention) (बॉन कन्वेंशन)

- The **Convention on Migratory Species (1979)** protects **migratory species** and their flyways across borders.
- CMS focuses on **habitat (वास स्थान) and migration routes**, not commercial trade like CITES.

Ramsar

- The **Ramsar Convention (1971)** (रामसर कन्वेंशन) conserves **wetlands** through the **"Wise Use"** principle.
- **World Wetlands Day** is **2 February**.
- India has about **100** Ramsar sites; state ranks often tested are **Tamil Nadu (नाडु) 16**, **Uttar Pradesh 10**, **Punjab 6**.
- The **Montreux Record** lists Ramsar sites where ecological character has changed or may change from human interference.
- India has many Ramsar sites including several in **UP**.

PYQ — UPPCS Prelims 2025, Q58

Which Ramsar sites are **NOT** in Uttar Pradesh?

1. Sarsai Nawar 2. Samaspur 3. Rudrasagar 4. Sultanpur

> 💡 **Show answer**
>
> **Ans: B (3 and 4)** — Rudrasagar (Tripura); Sultanpur (Haryana).

Treaty	Issue	Year
CBD	Biodiversity	1992
CITES	Species **trade**	1973
CMS	**Migratory** species	1979
Ramsar	**Wetlands**	1971

18.5 Chemicals & Waste Treaties — Basel, Rotterdam, Stockholm POPs & Minamata

Identity: Basel waste | Rotterdam PIC chemicals | Stockholm POPs | Minamata mercury

- The **Basel Convention (1989)** (बासेल कन्वेंशन) controls **transboundary movement of hazardous waste**.
- It aims to prevent developed countries from dumping hazardous waste in developing countries.
- The **Rotterdam Convention (1998)** (रॉटरडैम कन्वेंशन) implements **Prior Informed Consent (PIC)** for trade in **hazardous chemicals**.
- The **Stockholm Convention on POPs (2001)** targets **persistent organic pollutants** like **DDT and PCBs** that remain in the environment.
- The **Minamata Convention (2013)** (मिनामाता संधि) controls **mercury** mining, use, emissions, and releases.
- It is named after **Minamata disease** from mercury poisoning in Japan.

Treaty	Targets
Basel	Hazardous **waste** shipment
Rotterdam	Hazardous **chemicals** trade (PIC)
Stockholm POPs	Persistent organic pollutants
Minamata	**Mercury**

Complete PYQ Bank (Topic 18)

Q1. UPPCS Prelims 2024, Q35

Which of the following international agreements is related to climate change **mitigation**?

A. Montreal Protocol

B. Ramsar Convention

C. Kyoto Protocol

D. CITES

💡 **Show answer**

Logic: Treaty–issue matching — Montreal controls ozone/ODS; Kyoto/Paris address greenhouse gases. **Ans: C.** The **Kyoto Protocol** sets binding GHG mitigation targets for developed countries. **Trap:** **Montreal Protocol** — ozone treaty, not climate mitigation.

Q2. UPPCS Prelims 2025, Q51

Assertion (A): Sustainable development should take place without harming the environment and without compromising the needs of future generations.

Reason (R): Agenda 21 was signed by world leaders in 1995.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)

B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)

C. (A) is true, but (R) is false

D. (A) is false, but (R) is true

💡 **Show answer**

A/R logic: (A) states the Brundtland-style SD principle (true). (R) tests Agenda 21 year — Rio **1992**, not 1995. **Ans: C.** (A) is true; (R) is false because **Agenda 21 was adopted at Rio 1992**. **Trap:** Accepting **1995** as the Agenda 21 year.

Q3. UPPCS Prelims 2023, Q132

Agenda-21 of the Rio Summit (1992) is related to:

A. Sustainable Development

B. Polluter Pays Principle

C. Environmental Education

D. Ozone Layer Protection

💡 **Show answer**

Logic: Agenda 21 is Rio's non-binding action plan for sustainable development in the 21st century. **Ans: A.** **Agenda 21** addresses **sustainable development**. **Trap:** Ozone layer (Montreal) or Polluter Pays (domestic principle) distractors.

Q4. UPPCS Prelims 2024, Q51

Assertion (A): Sustainable development was popularised by the Brundtland Report.

Reason (R): The Brundtland Report is also known as *The Limits to Growth*.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)

B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)

C. Both (A) and (R) are true and (R) explains (A)

D. (A) is true, but (R) is false

💡 **Show answer**

A/R logic: (A) is true — Brundtland (1987) defined SD. (R) is false — *Limits to Growth* is Club of Rome **1972**. **Ans: D.** Brundtland = ***Our Common Future* (1987)**; not *Limits to Growth*. **Trap:** Treating Brundtland and Club of Rome as the same report.

Q5. UPPCS Prelims 2025, Q58

Which of the following Ramsar sites are **NOT** located in Uttar Pradesh?

1. Sarsai Nawar Jheel
2. Samaspur Bird Sanctuary
3. Rudrasagar Lake
4. Sultanpur National Park

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Logic: UP Ramsar geography — Rudrasagar and Sultanpur sound plausible but lie outside UP. **Ans: B.** **Rudrasagar Lake** is in **Tripura**; **Sultanpur** is in **Haryana**. **Trap:** Assuming every bird sanctuary name in the list belongs to UP.

Q6. UPPCS Prelims 2025, Q145

With reference to the Global Environment Facility (GEF), consider the following statements:

1. GEF was established in 1991.
2. Only developed countries are donors to GEF.

A. Neither 1 nor 2

B. Only 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: GEF 1991 fact is standard; donor base is broader than developed-only. **Ans: D (Only 1).** **GEF was established in 1991**; statement 2 is false. **Trap:** Rejecting 1991 or accepting "developed-only donors."

Q7. UPPCS Prelims pattern — Montreal vs climate

Which of the following pairs is **NOT** correctly matched?

A. Montreal Protocol — ozone-depleting substances

B. Kyoto Protocol — climate change mitigation

C. Montreal Protocol — climate change mitigation

D. Paris Agreement — nationally determined contributions

💡 **Show answer**

Logic: Montreal = ozone/ODS phase-out; UNFCCC–Kyoto–Paris = climate framework. **Ans: C.** **Montreal Protocol does not mitigate climate change** — it protects the ozone layer. **Trap:** "Montreal = climate treaty" — the single most tested treaty confusion.

Q8. UPPCS Prelims pattern — CITES vs CMS

Which statement is correct?

A. CITES (1973) regulates international trade in endangered species; CMS (1979) protects migratory species and flyways.

B. CITES protects migratory flyways; CMS bans all wildlife trade.

C. Both CITES and CMS were opened for signature at Rio 1992.

D. CITES assesses extinction risk; CMS issues Red List categories.

💡 **Show answer**

Logic: CITES = trade permits; CMS = migration/habitat — separate 1970s treaties, not Rio products. **Ans: A.** **CITES regulates trade**; **CMS protects migration routes**. **Trap:** Rio 1992 origin for CITES (CITES is **1973**, pre-Rio).

Q9. UPPCS Prelims pattern — UNEP headquarters

The headquarters of the United Nations Environment Programme (UNEP) is located at:

A. Geneva, Switzerland

B. New York, USA

C. Nairobi, Kenya

D. Bonn, Germany

💡 **Show answer**

Logic: HQ matching — UNEP Nairobi; UNDP New York; IPCC Geneva; UNFCCC Bonn. **Ans: C.** **UNEP headquarters is in Nairobi**. **Trap:** **Geneva** (IPCC) or **New York** (UNDP).

Q10. UPPCS Prelims pattern — Rio 1992 outcomes

The Rio Earth Summit (1992) directly produced which of the following?

A. UNFCCC, CBD, and Agenda 21

B. CITES, Ramsar Convention, and Stockholm Declaration

C. Kyoto Protocol, Paris Agreement, and SDGs

D. Montreal Protocol, Vienna Convention, and Minamata Convention

💡 **Show answer**

Logic: Rio "triple outcome" — climate framework, biodiversity convention, and Agenda 21 action plan. **Ans: A.** Rio **1992** opened **UNFCCC** and **CBD** and adopted **Agenda 21**. **Trap:** CITES (**1973**) or Kyoto (**1997**) as Rio products.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. Arrange the conference spine:

Stockholm → Brundtland → Rio → Johannesburg → Rio+20 → SDGs/Paris

A. 1987-1972-1992-2012-2002-2015 B. 1972-1987-1992-2002-2012-2015 C. 1992-1972-1987-2002-2015-2012 D. 1972-1992-1987-2012-2002-2015

> 💡 **Show answer**
>
> **Ans: B.** 1972→1987→1992→2002→2012→2015. **Logic:** Full conference ladder.

Q2. With reference to Stockholm 1972, which of the following statements is/are correct?

1. Stockholm 1972 was the first UN climate COP under UNFCCC.
2. Led to UNEP at Nairobi.
3. World Environment Day is observed on 22 April.

A. Only 2 B. Only 1 C. 1 and 2 D. Only 3

> 💡 **Show answer**
>
> **Ans: A.** Only 2 is correct. **Logic:** Statement 2 is true (UNEP, Nairobi). Statement 1 confuses Stockholm/Human Environment with UNFCCC COPs. Statement 3 is false — World Environment Day is 5 June.

Q3. Which of the following pairs is/are NOT correctly matched?

1. Brundtland 1987 — Our Common Future
2. Limits to Growth 1972 — Club of Rome
3. Brundtland — identical to Limits to Growth

A. 1 and 2 only B. 1, 2 and 3 C. 2 and 3 only D. 3 only

> 💡 **Show answer**
>
> **Ans: D.** Only pair 3 is not correctly matched. **Logic:** Different books/years.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Rio 1992 produced Agenda 21 and opened UNFCCC and CBD for signature.

Reason (R): CITES is a Rio 1992 product identical to Agenda 21.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** CITES is 1973 Washington — not a Rio product.

Q5. Match issue tags:

List-I	List-II
A. Montreal/Vienna	1. Wetlands

List-I	List-II
B. Kyoto/Paris/UNFCCC	2. Ozone
C. Ramsar	3. Climate
D. CITES	4. Wildlife trade

Code:

A. A-2, B-3, C-1, D-4 B. A-3, B-2, C-1, D-4 C. A-2, B-1, C-3, D-4 D. A-1, B-3, C-2, D-4

💡 **Show answer**

Ans: A. A-2, B-3, C-1, D-4. **Logic:** Ozone/climate/wetlands/trade.

Q6. Arrange climate ladder:

A. Paris 1992 → Kyoto 2015 → UNFCCC 1997 B. UNFCCC 1992 → Kyoto 1997 → Paris 2015 C. Kyoto 1987 → Paris 1992 → UNFCCC 2015 D. UNFCCC 2015 → Kyoto 1992 → Paris 1997

💡 **Show answer**

Ans: B. 1992→1997→2015. **Logic:** UNFCCC–Kyoto–Paris.

Q7. With reference to ozone ladder, which of the following statements is/are correct?

1. Vienna 1985 → Montreal 1987 → Kigali 2016.
2. Kyoto is an ozone treaty, not a climate treaty.
3. Kigali phases down CFCs, not HFCs.

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Statement 1 is the Vienna–Montreal–Kigali ladder. Statement 2 is false — Kyoto is climate, not ozone. Statement 3 is false — Kigali phases down HFCs.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Agenda 21 is Rio's non-binding sustainable-development action plan.

Reason (R): Agenda 21 was signed in 1995.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** 1992 — not 1995.

Q9. Which of the following pairs is/are correctly matched?

1. CBD — conserve, sustainable use, fair benefit-sharing
2. Nagoya 2010 — ABS
3. Cartagena — biosafety/LMOs

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** CBD family.

Q10. Match List-I with List-II:

List-I	List-II
A. Basel	1. Mercury
B. Rotterdam	2. Hazardous waste
C. Stockholm POPs	3. PIC chemicals
D. Minamata	4. POPs

Code:

A. A-2, B-3, C-4, D-1 B. A-3, B-2, C-4, D-1 C. A-2, B-4, C-3, D-1 D. A-1, B-3, C-2, D-4

💡 **Show answer**

Ans: A. A-2, B-3, C-4, D-1. **Logic:** Waste/PIC/POPs/mercury.

Q11. With reference to CMS/Bonn, which of the following statements is/are correct?

1. Covers migratory species.
2. Is identical to CITES trade appendices.
3. Is different from Ramsar wetlands treaty.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 3. **Logic:** CMS ≠ CITES.

Q12. Arrange two Stockholms trap:

A. Stockholm 2001 conference created UNEP; Stockholm 1972 is POPs B. Both are identical ozone treaties C. Stockholm 1972 conference→UNEP; Stockholm 2001 POPs treaty D. Both are Paris Agreement annexes

💡 **Show answer**

Ans: C. 1972 conference vs 2001 POPs. **Logic:** Classic double-Stockholm trap.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): GEF was established in 1991 on the eve of Rio as a finance mechanism.

Reason (R): UNCCD (1994) is the desertification convention.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; UNCCD year does not explain GEF's 1991 creation. **A/R logic:** Two finance/land facts.

Q14. With reference to Paris Agreement, which of the following statements is/are correct?

1. Year 2015.
2. Universal NDCs.
3. 1.5°C ambition language teaching.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Paris package.

Q15. Which of the following pairs is/are correctly matched?

1. Ramsar 1971 — Wise Use
2. World Wetlands Day — 2 February
3. CITES — IUCN Red List assessment body

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** IUCN assesses; CITES regulates trade.

Q16. How many of the following issue tags are correct?

Montreal=ozone; CBD=biodiversity; Basel=hazardous waste

A. Only one B. All three C. Only two D. None

💡 **Show answer**

Ans: B. All three. **Logic:** Tag ratta.

Q17. Match List-I with List-II:

List-I	List-II
A. Kyoto	1. 2015
B. Paris	2. 1997
C. Montreal	3. 1987

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** 1997 / 2015 / 1987.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Kyoto set binding targets mainly for Annex-I parties.

Reason (R): Paris uses universal NDCs for all parties.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; Paris design contrasts Kyoto but does not explain Kyoto's Annex-I design as its cause. **A/R logic:** Kyoto vs Paris architectures.

Q19. With reference to UNCCD, which of the following statements is/are correct?

1. Adopted 1994.
2. Links environment, development, and sustainable land management.
3. Is the ozone phase-out treaty.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Ozone = Montreal.

Q20. Arrange CBD protocol chronology teaching:

A. CBD 1992 → Cartagena biosafety → Nagoya ABS 2010 B. Nagoya 1992 → CBD 2010 C. Cartagena 2015 → CBD 1972 D. Nagoya 1987 → Montreal 2010

💡 **Show answer**

Ans: A. CBD 1992 then protocols (Cartagena; Nagoya 2010). **Logic:** Biodiversity law family.

Q21. Which of the following statements about Rio+20 is/are correct?

1. Year 2012.
2. Sits after Johannesburg 2002 on the spine.
3. Rio+20 repealed the Wildlife (Protection) Act.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Does not repeal WPA.

Q22. With reference to Minamata, which of the following statements is/are correct?

1. Mercury treaty 2013.
2. Different from Minamata disease teaching history but shares the name hook.
3. Minamata is the Forest Rights Act.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Not FRA.

Q23. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): UNFCCC secretariat sits in Bonn.

Reason (R): UNFCCC was opened at Rio 1992.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; Rio opening does not explain Bonn secretariat location by itself. **A/R logic:** Opening vs HQ/secretariat.

Q24. Which of the following pairs is/are correctly matched?

1. Rotterdam — PIC
2. Basel — hazardous waste
3. Stockholm POPs — DDT/PCBs neighbourhood

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Chemicals/waste trio.

Q25. Arrange Ramsar / CITES years:

A. Ramsar 1971 → CITES 1973 B. CITES 1971 → Ramsar 1973 C. Both 1992 D. Both 2015

💡 **Show answer**

Ans: A. Ramsar 1971; CITES 1973. **Logic:** Wetlands then trade.

Q26. With reference to WED, which of the following statements is/are correct?

1. 5 June marks Stockholm opening.
2. Run by UNEP neighbourhood teaching.
3. WED is identical to Earth Day 22 April.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Different days.

Q27. How many of the following climate instruments are correctly dated?

UNFCCC 1992, Kyoto 1997, Paris 2015

A. Only one B. Only two C. All three D. None

💡 **Show answer**

Ans: C. All three. **Logic:** Climate ladder years.

Q28. Which one of the following is correct?

A. Montreal = climate NDCs B. Ramsar = mercury only C. Kyoto = ozone hole only D. Montreal = ozone ODS

💡 **Show answer**

Ans: D. Montreal = ozone ODS. **Logic:** Never swap with climate.

Q29. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Johannesburg 2002 sits on the sustainable development conference spine.

Reason (R): SDGs were adopted in 2015.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; SDG year does not explain Johannesburg 2002's place. **A/R logic:** Spine vs later SDGs.

Q30. With reference to Forest Principles at Rio, which of the following statements is/are correct?

1. Rio produced Forest Principles among outputs.
2. Rio Declaration and Agenda 21 were Rio products.
3. Rio 1992 is identical to Kyoto Protocol.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Kyoto is 1997 climate.

Mains Answer Framework

International environmental agreements (250 words): Conference timeline Stockholm→Brundtland→Rio→SDGs. Rio outcomes. Treaty clusters — climate (UNFCCC/Kyoto/Paris), ozone (Vienna/Montreal), biodiversity (CBD/Nagoya), species (CITES/CMS), wetlands (Ramsar), chemicals/waste (Basel/Rotterdam/Stockholm POPs/Minamata). India implementation examples (BDA 2002, Montreal compliance, Paris NDC). Trap distinctions Montreal≠climate, Agenda 21=1992. Conclude with multilateralism and GEF finance role.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q148

Match the COP session to its location:

COP session	Location
1. COP 27	A. Belem
2. COP 28	B. Baku
3. COP 29	C. Dubai
4. COP 30	D. Sharm El-Sheikh

Row order is not the answer code.

A. 1-A, 2-B, 3-C, 4-D B. 1-B, 2-C, 3-D, 4-A C. 1-C, 2-D, 3-A, 4-B D. 1-D, 2-C, 3-B, 4-A

💡 **Show answer**

Logic: Baku for COP29 anchors the match code. **Ans: D (Series B provisional key).** COP27 Sharm El-Sheikh; COP28 Dubai; COP29 Baku; COP30 Belem.

Q. UKPCS Prelims 2025, Q149

With reference to the Global Environment Facility, consider the following statements:

1. The pilot GEF was agreed to be set up in 1990.
2. UNDP, UNEP and the World Bank were the initial partners.
3. India is both a donor and a recipient of GEF.

A. Only 2 and 3 B. 2 and 3 C. 1 and 2 D. All of the above statements

💡 **Show answer**

Logic: All three GEF statements are standard — partial option is the trap. **Ans: D (Series B provisional key).** Pilot agreed 1990; UNDP–UNEP–World Bank partners; India is both donor and recipient.

Q. UKPCS Prelims 2025, Q117

As on April 2025, India is not a party to which of the following?

A. CITES B. World Heritage (विश्व धरोहर) Convention C. Occupational Safety and Health Convention D. International Whaling Commission

💡 **Show answer**

Logic: Among env/heritage options, the ILO labour convention is the odd one India has not ratified. **Ans: C (Series B provisional key).** India has not ratified the ILO Occupational Safety and Health Convention (1981). It is party to CITES and the World Heritage Convention and a member of the IWC.

Common Traps — Don’t Fall For These

Trap	Correct fact
Montreal = climate treaty	**Ozone/ODS**
Agenda 21 signed 1995	**Rio 1992**
Brundtland = Limits to Growth	***Our Common Future* 1987**
CITES born at Rio 1992	**1973 Washington**
Ramsar = forests	**Wetlands**
CMS = CITES	**Migration vs trade**
Vienna = binding ODS phase-out	**Framework; Montreal = binding**
Kyoto = Paris model	**Annex binding vs universal NDCs**
Stockholm POPs = Stockholm 1972	**2001 POPs treaty — different**
UNEP HQ Geneva	**Nairobi**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 19 — Climate & Environmental Institutions

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

IPCC (आईपीसीसी) | UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) | UNDP (यूएनडीपी) | IUCN (आईयूसीएन) | WWF (डब्ल्यूडब्ल्यूएफ) | FAO (Environment-related Role)

Sources baked in: IPCC AR6, UNEP reports, UNDP HDR, IUCN Red List, WWF Living Planet Report, FAO SOFO, UPPCS PYQs 2018–2025 **Weight:** ★★★ High — org ↔ report ↔ HQ matching **Last verified:** August 2026

Current Affairs (this topic)

- IPCC AR cycle and UNEP / WMO headlines are institutional CA.
- Keep UNFCCC (यूएनएफसीसीसी) Secretariat vs IPCC role split clear.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. **IPCC** sits in **Geneva**, issues Assessment Reports, and synthesises climate science. It does **not** set NDCs and is **not** the UNFCCC treaty body. 2. IPCC was established in **1988** by **WMO and UNEP** and shared the Nobel Peace Prize in **2007**. 3. **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) headquarters is **Nairobi**. Flagship products include the **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) and GEO. UNEP was created after Stockholm **1972**. 4. UNEP hosts several MEA secretariats (including Montreal, Basel, and Minamata lines) and works with GEF-linked convention finance. 5. **UNDP** headquarters is **New York**. It publishes the **Human Development Report** (from **1990**) and the **HDI**. HDR is UNDP, not UNEP. 6. HDI is linked to **Mahbub ul Haq** and **Amartya Sen**. Do not invent a wrong founding-name trap for HDR. 7. **IUCN** (आईयूसीएन) headquarters is **Gland**. It publishes the **Red List** of extinction risk. IUCN assesses risk; it does **not** regulate wildlife trade (पण्याध्यक्ष) — that is **CITES** (साइट्स). 8. On the Red List, **Threatened** means **Vulnerable + Endangered + Critically Endangered** (VU + EN + CR). "Endangered alone" is not the full threatened set. 9. **WWF** is a conservation NGO also linked to Gland, known for the panda logo and the **Living Planet Report / Living Planet Index**. 10. **FAO** headquarters is **Rome**. Environment-facing outputs include the **State of the World's Forests (SOFO)** and agriculture–forest links. FAO partners UNEP on some chemical-convention work such as Rotterdam. 11. HQ memory: Nairobi = UNEP; New York = UNDP; Geneva = IPCC; Gland = IUCN and WWF; Rome = FAO. 12. IPCC ≠ UNFCCC. Science assessment is not the same as the climate treaty and COP process. 13. UNEP ≠ UNDP. Environment (पर्यावरण) programme versus development agency — swap reports and you lose marks. 14. Emissions Gap Report → **UNEP**. Human Development Report → **UNDP**. Red List → **IUCN**. Living Planet → **WWF**. Assessment Reports → **IPCC**. 15. IUCN ≠ CITES. Risk categories versus trade appendices. 16. WWF ≠ IUCN. NGO Living Planet work versus Red List authority. 17. FAO is food and agriculture first; it is not the Red List body and not the Emissions Gap publisher. 18. GEF (**1991**) finances global environmental conventions; do not confuse GEF with IPCC authorship of Assessment Reports. 19. Red List codes to know: CR, EN, VU as the threatened band; also EX, EW, NT, LC, DD, NE in the wider ladder. 20. UNEP's Green Economy (हरित अर्थव्यवस्था) work and GEO sit in the environment-programme lane — not in UNDP's HDI lane. 21. IPCC reports inform negotiators; parties still adopt NDCs under the UNFCCC/Paris track. 22. Living Planet Index tracks vertebrate abundance trends; it is not a legal trade permit system. 23. SOFO is FAO's forest flagship — not FSI's India

State of Forest Report. 24. Match org → city → product before matching slogans. Geography of headquarters is a frequent prelims filter.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Intergovernmental Panel on Climate Change (IPCC)** | **UN Framework Convention on Climate Change (UNFCCC)** | Scientific assessment panel established in 1988 by WMO and UNEP that reviews peer-reviewed literature (does not conduct research or negotiate policy) vs international legal treaty and negotiating secretariat that coordinates annual Conference of Parties (COPs) | आईपीसीसी (वैज्ञानिक समीक्षा निकाय) / यूएनएफसीसीसी (वार्ता व संधि सचिवालय) | | **United Nations Environment Programme (UNEP)** | **United Nations Development Programme (UNDP)** | Apex global environmental authority coordinating UN environmental activities, hosting MEA secretariats (HQ: Nairobi, Kenya) vs UN global development network focusing on poverty reduction, sustainable growth, and publishing the Human Development Report (HQ: New York) | यूएनईपी (नैरोबी, पर्यावरण एजेंसी) / यूएनडीपी (न्यूयॉर्क, विकास एजेंसी) | | **International Union for Conservation of Nature (IUCN)** | **World Wide Fund for Nature (WWF)** | World's oldest and largest global conservation union composed of governments and civil society (1948, Gland) publishing the IUCN Red List vs independent international private conservation NGO (1961, Gland) famous for the Giant Panda symbol and Earth Hour | आईयूसीएन (रेड लिस्ट प्रकाशक) / डब्ल्यूडब्ल्यूएफ (पांडा प्रतीक एनजीओ) | | **Global Environment Facility (GEF)** | **Green Climate Fund (GCF)** | Multilateral financial partnership established in 1991 serving as financial mechanism for 5 MEAs (CBD, UNCCD, Stockholm, Minamata, UNFCCC) vs dedicated operating financial mechanism of the UNFCCC established at COP16 Cancun focusing specifically on climate projects in developing nations | जीईएफ (GEF, बहु-संधि कोष) / जीसीएफ (GCF, हरित जलवायु कोष) | | **Food and Agriculture Organization (FAO)** | **World Meteorological Organization (WMO)** | Specialized UN agency leading international efforts to defeat hunger, maintain agricultural biodiversity, and soil health (HQ: Rome) vs specialized UN agency dedicated to atmospheric science, climatology, and operational hydrology (HQ: Geneva) | एफएओ (रोम, खाद्य व कृषि) / डब्लूएमओ (जिनेवा, मौसम विज्ञान) | | **International Solar Alliance (ISA)** | **International Renewable Energy Agency (IRENA)** | Intergovernmental treaty-based organization launched jointly by India and France at COP21 Paris to promote solar energy deployment (HQ: Gurugram, India) vs intergovernmental organization supporting countries in their transition to all renewable energy types (HQ: Abu Dhabi, UAE) | अंतरराष्ट्रीय सौर गठबंधन (ISA, गुरुग्राम) / इरेना (IRENA, अबू धाबी) |

💡 📋 **Must-score facts — IPCC, UNEP, IUCN HQs & reports (Click to Expand)**

HQ / product | Item | Lock | |------|------| | IPCC | Geneva; WMO+UNEP **1988**; Nobel Peace **2007**; ARs | | UNEP | Nairobi; Emissions Gap + GEO | | UNDP | New York; HDR/HDI from **1990** | | IUCN | Gland; Red List (risk ≠ CITES trade) | | WWF | Gland; Living Planet Report/LPI | | FAO | Rome; SOFO | | Threatened | VU + EN + CR | | IPCC ≠ UNFCCC | Science assessment ≠ climate treaty/COP |

19.1 IPCC

Identity: 1988 WMO+UNEP | Geneva | Assessment Reports | Nobel 2007 | Not policy body

- The **IPCC (Intergovernmental Panel on Climate Change (जलवायु परिवर्तन))** was established in **1988** by **WMO** and **UNEP**.
- It provides governments with **scientific assessments** of climate change, impacts, and response options.
- **Headquarters** is in **Geneva, Switzerland**.
- IPCC publishes **Assessment Reports (AR)** roughly every 5–7 years. **AR6** was completed in **2023**.
- **Working Group I** covers physical science. **WG II** covers impacts and adaptation. **WG III** covers mitigation.
- The **Special Report on 1.5°C (2018)** was important for Paris Agreement (पेरिस समझौता) ambition.

- IPCC shared the **Nobel Peace Prize in 2007** with Al Gore.
- IPCC **synthesises existing science**. It does **not** conduct primary research or set emission targets.
- **UNFCCC** (यूएनएफसीसीसी) is the **treaty body** where countries negotiate **NDCs** at **COP** meetings. IPCC informs but does not decide policy.

> **Logic:** "IPCC sets Paris Agreement targets" = **false**.

Body	Role
IPCC	**Science assessment**
UNFCCC	**Treaty negotiations**

19.2 UNEP

Identity: Stockholm 1972 (स्टॉकहोम सम्मेलन) | Nairobi HQ | Emissions Gap | GEO | MEA secretariats

- **UNEP** was created by the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन) as the UN's principal environmental authority.
- **Headquarters** is in **Nairobi, Kenya** — the only major UN environment HQ in the Global South.
- UNEP publishes the annual **Emissions Gap Report** on the shortfall between NDC pledges and **1.5°C/2°C** pathways.
- **GEO (Global Environment Outlook)** is UNEP's comprehensive state-of-environment assessment.
- UNEP hosts or supports secretariats for treaties like **Montreal, Basel, Stockholm POPs, and Minamata**.
- UNEP co-created **IPCC** with WMO but the **Emissions Gap Report is UNEP's own product**, not an IPCC report.
- The **Global Environment Facility (GEF)** (जीईएफ) was established in **1991** and finances projects under UNFCCC, CBD, Montreal, and other conventions. UNEP is one implementing agency.

PYQ — UPPCS Prelims 2024, Q6

Emissions Gap Report → Options include UNEP (2), UNDP (3)

> 💡 **Show answer**
>
> **Ans: UNEP (2)** — Full match: HDR→UNDP(3), WEO→IMF(1), WIR→UNCTAD(4), Emissions Gap→UNEP(2) = **A (3 1 4 2)**.

PYQ — UPPCS Prelims 2025, Q145

1. GEF established in 1991.
2. Only developed countries are donors.

> 💡 **Show answer**
>
> **Ans: D (Only 1)** — GEF **1991** true; donors not **only** developed countries.

19.3 UNDP

Identity: New York | HDR 1990 | HDI | Mahbub ul Haq + Amartya Sen

- **UNDP** is the UN's global **development** network working on poverty, governance, and SDG implementation.
- **Headquarters** is in **New York**.
- The first **Human Development Report (HDR)** was published in **1990** and introduced the **Human Development Index (HDI)**.

- **HDI** combines **life expectancy**, **education**, and **GNI per capita (PPP)**.
- HDI was developed by **Mahbub ul Haq** and **Amartya Sen** — **not** S. Priesner.
- UNDP supports **SDG localisation** and climate/biodiversity projects but is **not** the Emissions Gap publisher.

PYQ — UPPCS Prelims 2025, Q132

1. UNDP published first HDR with HDI in 1990.
2. HDI team led by Mahbub-ul-Haq and S. Priesner.

> 💡 **Show answer**
>
> **Ans: D (Only 1)** — Statement 2 false; Sen not Priesner.

Report	Publisher
Human Development Report	**UNDP**
Emissions Gap Report	**UNEP**

19.4 IUCN

Identity: 1948 Gland | Red List | Threatened = VU+EN+CR | Not CITES

- **IUCN** is a global conservation union of states and NGOs, founded in **1948**.
- **Headquarters** is in **Gland, Switzerland**.
- The **IUCN Red List** is the standard system for assessing **extinction risk** of species.
- Categories include **LC, NT, VU, EN, CR, EW, EX** plus DD and NE.
- **Threatened species** (संकटग्रस्त प्रजातियां) means **Vulnerable + Endangered + Critically Endangered (VU + EN + CR)**.
- IUCN **assesses scientific risk**. It does **not** regulate international trade — **CITES** does that.
- IUCN provides scientific input to **CBD and CITES** but is a separate organisation.

> **Logic:** "IUCN bans wildlife trade" = **false** — **CITES** regulates trade.

19.5 WWF

Identity: NGO Gland | Living Planet Report | Panda logo | LPI

- **WWF (World Wide Fund for Nature)** is a global conservation **NGO**, not a UN agency.
- **Headquarters** is also in **Gland, Switzerland** — same city as IUCN but **different organisation and role**.
- WWF publishes the **Living Planet Report** with the **Living Planet Index (LPI)** tracking vertebrate population (जनसंख्या) trends.
- WWF is known for the **giant panda logo** and flagship-species fundraising campaigns.
- WWF **advocates and campaigns**. IUCN **assesses** via Red List. Complementary, not identical.

Org	Output
IUCN	**Red List**
WWF	**Living Planet Report**

19.6 FAO (Environment-related Role)

Identity: Rome HQ | SOFO | Rotterdam with UNEP | ag-environment link

- **FAO (Food and Agriculture Organization)** is a UN specialised agency for **food and agriculture**.
- **Headquarters** is in **Rome, Italy**.
- FAO's environment role links **agriculture, forests, biodiversity, and land use**.
- FAO co-publishes **State of the World's Forests (SOFO)** with UNEP.
- FAO administers the **Rotterdam Convention** jointly with **UNEP** for **Prior Informed Consent (PIC)** on hazardous chemicals in trade.
- FAO work on **pesticides, forestry, and ag-biodiversity** connects farming to deforestation and habitat (वास स्थान) loss.
- FAO is **not** the publisher of HDR, Emissions Gap, or Red List.

Complete PYQ Bank (Topic 19)

Q1. UPPCS Prelims 2024, Q6

Match List-I with List-II:

List-I (Report)	List-II (Organisation)
A. Human Development Report	1. IMF
B. World Economic Outlook	2. UNEP
C. World Investment Report	3. UNDP
D. Emissions Gap Report	4. UNCTAD

Row order is not the answer code.

A. 3 1 4 2

B. 3 4 1 2

C. 1 2 3 4

D. 2 3 4 1

💡 **Show answer**

Logic: Report–publisher matching — Emissions Gap is UNEP's product, not IPCC or UNDP. **Ans: A (3 1 4 2).** HDR→**UNDP**; WEO→**IMF**; WIR→**UNCTAD**; Emissions Gap→**UNEP**. **Trap:** Crediting the Emissions Gap Report to **IPCC** or **UNDP**.

Q2. UPPCS Prelims 2025, Q132

With reference to the Human Development Report (HDR), consider the following statements:

1. UNDP published the first HDR with HDI in 1990.
2. The HDI was developed by a team led by Mahbub ul Haq and S. Priesner.

A. Neither 1 nor 2

B. Only 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: HDR 1990 and HDI authorship — Mahbub ul Haq and **Amartya Sen**, not Priesner. **Ans: D (Only 1).** First HDR **1990** is correct; statement 2 is false. **Trap:** Accepting **Priesner** instead of **Sen** on the HDI team.

Q3. UPPCS Prelims 2025, Q145

With reference to the Global Environment Facility (GEF), consider the following statements:

1. GEF was established in 1991.
2. Only developed countries are donors to GEF.

A. Neither 1 nor 2

B. Only 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: GEF pre-Rio finance mechanism — 1991 establishment is tested; donor pool is not developed-only. **Ans: D (Only 1).** **GEF was established in 1991**; statement 2 is false. **Trap:** Choosing "Neither" by doubting the 1991 date.

Q4. UPPCS Prelims pattern — IPCC origin

The Intergovernmental Panel on Climate Change (IPCC) was established in 1988 by:

A. WMO and UNEP

B. UNFCCC and UNDP

C. UNEP and UNDP

D. WMO and FAO

💡 **Show answer**

Logic: IPCC parent bodies — science assessment under WMO + UNEP, not the climate treaty secretariat. **Ans: A.** **IPCC was created by WMO and UNEP in 1988**. **Trap:** UNFCCC (policy treaty body) as IPCC founder.

Q5. UPPCS Prelims pattern — UNEP headquarters

The headquarters of UNEP is located at:

A. Geneva

B. Nairobi

C. New York

D. Rome

💡 **Show answer**

Logic: Institution HQ ladder — Nairobi = UNEP; New York = UNDP; Rome = FAO. **Ans: B.** **UNEP HQ is Nairobi**. **Trap:** **Geneva** (IPCC) or **New York** (UNDP).

Q6. UPPCS Prelims pattern — Red List vs CITES

Which organisation prepares the **IUCN Red List**, and which treaty regulates international wildlife **trade**?

A. IUCN — Red List; CITES — trade

B. CITES — Red List; IUCN — trade

C. UNEP — Red List; CMS — trade

D. WWF — Red List; CBD — trade

💡 **Show answer**

Logic: IUCN assesses scientific extinction risk; CITES controls cross-border trade permits. **Ans: A.** **Red List = IUCN**; **trade regulation = CITES**. **Trap:** "IUCN bans wildlife trade" — IUCN does **not** regulate trade.

Q7. UPPCS Prelims pattern — IPCC vs UNFCCC

Which statement is correct?

A. IPCC sets national emission targets at COP meetings; UNFCCC publishes Assessment Reports.

B. IPCC synthesises climate science; UNFCCC is the treaty forum where countries negotiate NDCs.

C. IPCC and UNFCCC are the same body with two names.

D. UNFCCC was created in 1988; IPCC was opened for signature at Rio 1992.

💡 **Show answer**

Logic: Science (IPCC) versus policy treaty (UNFCCC/COP) — recurring institution trap. **Ans: B.** **IPCC assesses science**; **UNFCCC/COP handles NDC negotiations**. **Trap:** Swapping roles — "IPCC sets NDCs."

Q8. UPPCS Prelims pattern — Living Planet Report

The Living Planet Report and Living Planet Index are published by:

A. IUCN

B. UNEP

C. WWF

D. IPCC

💡 **Show answer**

Logic: WWF NGO product versus IUCN Red List — both in Gland but different roles. **Ans: C.** The **Living Planet Report is published by WWF**. **Trap:** **IUCN** (Red List) or **UNEP** (Emissions Gap).

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to IPCC, which of the following statements is/are correct?

1. IPCC headquarters is in Nairobi and it issues NDCs instead of Assessment Reports.
2. Established 1988 by WMO and UNEP.
3. Sets NDCs for all parties as the UNFCCC treaty body.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: A. Only 2 is correct. **Logic:** Statement 2 is true (1988, WMO+UNEP). Statement 1 is false — IPCC HQ is Geneva and it issues Assessment Reports. Statement 3 is false — parties set NDCs under UNFCCC; IPCC is not the treaty body that sets them.

Q2. Match List-I with List-II:

List-I	List-II
A. UNEP	1. New York; HDR/HDI
B. UNDP	2. Nairobi; Emissions Gap/GEO
C. FAO	3. Rome; SOFO

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Nairobi/NY/Rome.

Q3. Which of the following pairs is/are correctly matched?

1. IUCN — Gland; Red List
2. IUCN — Gland; Living Planet Report
3. IUCN — regulates CITES trade appendices as its only job

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Threatened on IUCN means VU + EN + CR.

Reason (R): Endangered alone equals the full Threatened set.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Threatened ≠ Endangered alone.

Q5. Arrange HQ memory:

Nairobi / New York / Geneva / Gland / Rome

A. UNEP / UNDP / IPCC / IUCN-WWF / FAO B. UNDP / UNEP / FAO / IPCC / IUCN C. FAO / IPCC / UNEP / UNDP / IUCN D. IUCN / FAO / UNDP / UNEP / IPCC

💡 **Show answer**

Ans: A. UNEP Nairobi; UNDP NY; IPCC Geneva; IUCN/WWF Gland; FAO Rome. **Logic:** HQ map.

Q6. With reference to Emissions Gap Report, which of the following statements is/are correct?

1. It is a UNDP product.
2. It is a UNEP flagship, not an IPCC Assessment Report.
3. It is identical to the Human Development Report.

A. Only 2 B. Only 1 C. 2 and 3 D. 1 and 2

> 💡 **Show answer**
>
> **Ans: A.** Only 2 is correct. **Logic:** Statement 2 is true (UNEP Emissions Gap ≠ IPCC AR). Statement 1 is false — it is UNEP, not UNDP. Statement 3 confuses it with HDR.

Q7. Which of the following pairs is/are NOT correctly matched?

1. HDR — UNDP from 1990
2. HDI — Mahbub ul Haq / Amartya Sen link
3. HDR — UNEP Nairobi flagship

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only pair 3 is not correctly matched. **Logic:** HDR ≠ UNEP.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): IPCC shared the Nobel Peace Prize in 2007.

Reason (R): IPCC was created by WMO + UNEP in 1988.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; creation year does not explain the 2007 prize. **A/R logic:** Related IPCC facts.

Q9. Match report → body:

List-I	List-II
A. Assessment Reports	1. WWF
B. Living Planet	2. IPCC
C. Red List	3. IUCN

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

> 💡 **Show answer**
>
> **Ans: D.** A-2, B-1, C-3. **Logic:** IPCC/WWF/IUCN.

Q10. With reference to UNEP, which of the following statements is/are correct?

1. Created after Stockholm 1972.
2. Hosts several MEA secretariats (Montreal/Basel/Minamata lines teaching).
3. Publishes ISFR for India.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** ISFR is FSI.

Q11. Arrange IPCC ≠ UNFCCC idea:

A. Same body sets NDCs and writes all ARs as one office B. IPCC = science assessment; UNFCCC = climate treaty/COP process C. UNFCCC publishes Red List D. IPCC grants India's forest clearances

💡 **Show answer**

Ans: B. Science vs treaty. **Logic:** Do not merge.

Q12. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): UNEP ≠ UNDP.

Reason (R): Environment programme versus development agency — swap reports and you lose marks.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both (A) and (R) are true, but (R) is not the correct explanation of (A). **A/R logic:** Both (A) and (R) are true, but (R) does NOT explain (A). Reason (R) states an exam distractor tip ("swap reports and you lose marks"), which is meta-commentary, not the legal/institutional charter reason why UNEP and UNDP are distinct UN bodies.

Q13. Which of the following statements about FAO is/are correct?

1. HQ Rome.
2. SOFO — State of the World's Forests.
3. FAO is identical to IUCN Red List office.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** FAO ≠ IUCN.

Q14. How many of the following HQ facts are correct?

UNEP Nairobi; UNDP New York; IPCC Geneva; IUCN Gland

A. Only one B. Only two C. Only three D. All four

💡 **Show answer**

Ans: D. All four. **Logic:** Core HQ set.

Q15. With reference to WWF, which of the following statements is/are correct?

1. Conservation NGO; panda logo.
2. Living Planet Report / LPI.
3. WWF sets Kyoto Annex-I legally binding targets.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not a Kyoto legal body.

Q16. Match List-I with List-II:

List-I	List-II
A. GEO	1. UNDP
B. HDR	2. UNEP
C. SOFO	3. FAO

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** GEO UNEP; HDR UNDP; SOFO FAO.

Q17. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): IUCN and WWF both link to Gland.

Reason (R): They perform identical jobs: both regulate wildlife trade appendices.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Assessment NGO vs trade regulation differ; CITES ≠ IUCN job.

Q18. Which of the following pairs is/are correctly matched?

1. IPCC — synthesises climate science
2. UNFCCC — COP negotiations
3. IPCC — issues India's PARIVESH clearances

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** PARIVESH is MoEFCC.

Q19. With reference to Red List categories risk order, which of the following statements is/are correct?

1. Among living threatened tiers, CR above EN above VU.
2. Threatened = VU+EN+CR.
3. LC means Critically Endangered.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** LC = Least Concern.

Q20. Arrange creation: UNEP moment → IPCC:

A. Stockholm/UNEP 1972 moment → IPCC 1988 B. IPCC 1972 → UNEP 1988 C. Both 2015 D. Both 1997

💡 **Show answer**

Ans: A. 1972 UNEP moment → 1988 IPCC. **Logic:** Institution timeline.

Q21. Which one of the following report facts is correct?

A. Emissions Gap → UNDP B. Emissions Gap → UNEP C. HDR → UNEP D. Red List → FAO

💡 **Show answer**

Ans: B. Emissions Gap → UNEP. **Logic:** Do not swap.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): GEF works with convention finance neighbourhood teaching.

Reason (R): UNEP interacts with GEF-linked convention finance.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; UNEP-GEF link supports but is not a full definition of GEF alone. **A/R logic:** Related finance facts.

Q23. With reference to UNESCO vs UNEP, which of the following statements is/are correct?

1. UNESCO — Paris; WH + MAB teaching.
2. UNEP — Nairobi environment coordination.
3. They are the same organisation.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Different bodies.

Q24. How many of the following are true?

1. IPCC Geneva
2. Threatened = VU+EN+CR
3. HDR from 1990 UNDP

A. Only one B. Only two C. None D. All three

💡 **Show answer**

Ans: D. All three. **Logic:** High-yield trio.

Q25. Match List-I with List-II:

List-I	List-II
A. WMO+UNEP	1. Living Planet
B. IUCN	2. Created IPCC
C. WWF	3. Red List

Code:

A. A-2, B-3, C-1 B. A-3, B-2, C-1 C. A-2, B-1, C-3 D. A-1, B-3, C-2

💡 **Show answer**

Ans: A. A-2, B-3, C-1. **Logic:** IPCC creators; Red List; Living Planet.

Q26. With reference to HDI names, which of the following statements is/are correct?

1. Linked to Mahbub ul Haq and Amartya Sen teaching.
2. Inventing a wrong founding-name trap loses marks.
3. HDI is published by IUCN Gland.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** HDI/HDR = UNDP.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): CITES regulates international wildlife trade.

Reason (R): IUCN Red List assesses extinction risk — different job.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R clarifies why IUCN ≠ CITES. **A/R logic:** Risk vs trade.

Q28. Which of the following pairs is/are correctly matched?

1. GEO — UNEP
2. Assessment Reports — IPCC
3. SOFO — UNDP

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** SOFO = FAO.

Q29. Arrange Gland organisations teaching:

A. Only FAO B. Only IPCC C. Only UNEP D. IUCN and WWF both link to Gland

💡 **Show answer**

Ans: D. IUCN and WWF — Gland. **Logic:** Shared city; different jobs.

Q30. With reference to IPCC products, which of the following statements is/are correct?

1. Assessment Reports synthesise climate science.
2. IPCC does not set NDCs.
3. IPCC is the same as MoEFCC.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not MoEFCC.

Mains Answer Framework

Environmental institutions (250 words): Map IPCC (science), UNFCCC (policy treaty), UNEP (environment coordination, Emissions Gap, GEO, Nairobi), UNDP (HDR/HDI, SDGs), IUCN (Red List), WWF (Living Planet), FAO (ag-forest-environment). HQ and report matching. Distinctions IPCC≠UNFCCC, UNEP≠UNDP, IUCN≠CITES. GEF finance role. India engagement (IPCC authors, MoEFCC-UNEP treaties, HDR rank). Conclude with science-policy-civil society division of labour.

Common Traps — Don't Fall For These

Trap	Correct fact
IPCC sets NDCs	**UNFCCC/COP**
IPCC = UNFCCC	**Science vs treaty**
UNEP HQ Geneva	**Nairobi**
Emissions Gap = IPCC/UNDP	**UNEP**
HDR = UNEP	**UNDP**
HDI team included Priesner	**Haq + Sen**
IUCN regulates trade	**CITES**
WWF = IUCN	**Same city, different roles**
FAO HQ Nairobi	**Rome**
GEF donors only developed	**Stmt 2 false in Q145**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 20 — Biodiversity Conservation Methods

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

In-situ (स्थान पर) Conservation (स्व-स्थाने संरक्षण) | Ex-situ (स्थानांतरित) Conservation (बाह्य-स्थाने संरक्षण) | Gene Bank | Seed Bank (बीज बैंक) | Botanical Garden (वनस्पति उद्यान) | Zoological Park

Sources baked in: NCERT Bio 12 Ch 15; CBD 1992; WPA 1972; BDA 2002; ICAR-NBPGR; NBAGR; CZA; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — in-situ vs ex-situ traps; NBPGR/NBAGR location; facility ↔ method matching **Last verified:** August 2026

Current Affairs (this topic)

- 30×30 / Kunming–Montreal CA sits here with in-situ vs ex-situ method traps.
- Sacred grove and seed-bank examples stay teaching baselines.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. **In-situ** (स्थान पर) conservation protects biodiversity **in the natural habitat (वास स्थान)**. It is the preferred primary strategy because the whole ecosystem (पारिस्थितिकी तंत्र) and natural selection stay intact. 2. **Ex-situ** (स्थानांतरित) conservation protects biodiversity **outside** the natural habitat — zoos, botanic gardens, seed banks, gene banks, cryobanks. It is a backup, not a replacement for in-situ. 3. In-situ examples: National Park, Wildlife Sanctuary, Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र), Conservation/Community Reserve (सामुदायिक आरक्षित क्षेत्र), sacred grove, Ramsar (रामसर) wetland, wildlife corridor. 4. Ex-situ examples: zoo, botanical garden, seed bank, gene bank, captive breeding centre, tissue culture, cryopreservation (क्रायो-संरक्षण). 5. Four-way pattern: National Park and sacred grove are **in-situ**; zoo, seed bank, and botanic garden are **ex-situ**. 6. A **gene bank** stores genetic material broadly — seeds, DNA, tissue, pollen, sperm, embryos. A **seed bank** stores **seeds only**. 7. **Orthodox seeds** (सामान्य (रूढ़िवादी) बीज) (wheat, rice) tolerate drying and cold storage around **−18°C**. **Recalcitrant seeds** (कठिन (असहनीय) बीज) (mango, coconut) cannot take simple dry seed-bank storage and need living collections or cryo routes. 8. **Cryopreservation** stores material at about **−196°C** in liquid nitrogen. 9. India's plant gene/seed flagship is **NBPGR**, New Delhi (**1976/1978** line). India's animal gene-bank flagship is **NBAGR**, Karnal (**1984**). 10. The **Indian Botanic Garden**, Howrah (**1787**), is India's classic large botanic garden, linked with BSI administration. The **National Zoological Park**, New Delhi (**1959**), is the national zoo flagship. 11. **CZA** is the statutory zoo regulator under **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय). 12. The **Svalbard Global Seed Vault** in Norway is a duplicate backup vault — not India's NBPGR. 13. NBPGR is plants in Delhi; NBAGR is animals in Karnal. BSI in Kolkata surveys plants; it is **not** the national gene bank. 14. Dudhwa (दुधवा) and Katerniaghat-style protected areas are **in-situ**. Kukrail crocodile centre and garden collections are **ex-situ**. 15. Sacred groves (पवित्र उपवन) are community in-situ pockets — do not park them with zoos. 16. Wildlife corridors keep in-situ populations linked; they are not ex-situ cages. 17. Captive breeding is ex-situ recovery support; releasing only works if habitat (in-situ) remains. 18. Seed banks fail for recalcitrant species if the stem assumes every crop seed stores like wheat. 19. Gene bank ⊃ seed bank. Calling every gene bank a "seeds-only room" is wrong. 20. Biosphere Reserves (जैवमंडल आरक्षित क्षेत्र) and National Parks are in-situ frameworks; botanic gardens are living ex-situ plant collections. 21. UP colour: Dudhwa/Katerniaghat = in-situ; NBRI Lucknow (लखनऊ) garden collections = ex-situ; NBPGR in Delhi still

serves national plant genetic resources including UP crops. 22. Community and Conservation Reserves are newer in-situ tools beside classic parks and sanctuaries. 23. Cryobanking is an ex-situ technology for germplasm that ordinary refrigeration cannot hold. 24. Priority line for prelims: **in-situ first, ex-situ supplementary** — never reverse the preference.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **In-situ Conservation** | **Ex-situ Conservation** | On-site conservation of wild populations within their natural ecosystems where evolutionary processes continue uninterrupted (National Parks, Wildlife Sanctuaries, Sacred Groves) vs off-site conservation of threatened components outside natural habitats in human custody (Botanical Gardens, Zoological Parks, Cryobanks) | स्व-स्थाने संरक्षण (प्राकृतिक आवास) / बाह्य-स्थाने संरक्षण (मानव-निर्मित व्यवस्था) | | **Botanical Garden** | **Herbarium** | Ex-situ facility maintaining curated collections of *living* plant species for scientific research, propagation, and conservation display vs reference repository of *dried, pressed, and mounted* plant specimens preserved on archival sheets for taxonomic identification | वनस्पति उद्यान (जीवित पादप संग्रह) / पादप संग्रहालय (हर्बेरियम, शुष्क नमूने) | | **Zoological Park (Zoo)** | **Safari / Wildlife Sanctuary** | Ex-situ facility where wild animals are maintained in artificial enclosures under human management for captive breeding and public education vs in-situ natural protected habitat where free-ranging wild animals live in untamed wild conditions | चिड़ियाघर (कृत्रिम बाड़ा) / वन्यजीव अभयारण्य / सफारी | | **Orthodox Seeds** | **Recalcitrant Seeds** | Desiccation-tolerant seeds that can survive dehydration to low moisture content (≤5%) and sub-zero freezing, ideal for conventional long-term seed banking (wheat, rice, legumes) vs desiccation-sensitive seeds that lose viability rapidly if dried or frozen, requiring cryopreservation (mango, rubber, jackfruit, tea) | सामान्य (रूढ़िवादी) बीज / कठिन (असहनीय) बीज | | **Cryopreservation** | **Seed Bank** | Ultra-low temperature preservation of living cells, gametes, embryos, and tissues suspended in liquid nitrogen at **-196°C** stopping all metabolic decay vs storage of dried orthodox seeds in vault freezers at standard temperatures of -18°C to -20°C | क्रायो-संरक्षण (-196°C द्रवित नाइट्रोजन) / बीज बैंक (-18°C शुष्क भंडारण) | | **Sacred Groves** | **Biosphere Reserves** | Community-protected virgin forest patches preserved through traditional spiritual beliefs and religious taboos (e.g. Deorais, Orans, Kavu, Law Kyntang) vs modern scientific statutory conservation landscape demarcated by government under UNESCO MAB | पवित्र उपवन (पारंपरिक संरक्षण) / जैवमंडल आरक्षित क्षेत्र (वैज्ञानिक प्रारूप) | | **National Bureau of Plant Genetic Resources (NBPGR)** | **National Bureau of Animal Genetic Resources (NBAGR)** | Premier ICAR institute managing the National Gene Bank for agricultural crop and wild relative germplasm (HQ: New Delhi) vs premier ICAR institute maintaining registrations and genetic databases of indigenous livestock breeds (HQ: Karnal, Haryana) | एनबीपीजीआर (नई दिल्ली, पादप जीन बैंक) / एनबीएजीआर (करनाल, पशु आनुवंशिकी) |

💡 📋 **Must-score facts — in-situ, ex-situ, gene banks (Click to Expand)**

Methods | Item | Lock | |---|---| | In-situ | Natural habitat — preferred primary | | Ex-situ | Zoo, botanic garden, seed/gene/cryobank — backup | | Gene bank vs seed bank | Broad genetic material / seeds only | | Orthodox / Recalcitrant | Dry+cold OK (wheat/rice) / cannot (mango/coconut) | | Cryopreservation | ~−196°C liquid nitrogen | | NBPGR / NBAGR | Plants Delhi / animals Karnal | | Indian Botanic Garden | Howrah **1787**; BSI link | | Svalbard | Norway backup vault (≠ NBPGR) | | CZA | Statutory zoo regulator, MoEFCC | --- ! [Biodiversity Conservation In-situ vs Ex-situ Taxonomy & IUCN Red List Hierarchy] (images/env_ch20_01_in_situ_ex_situ_conservation_taxonomy.png)

20.1 In-situ Conservation

Identity: On-site | Natural habitat | Preferred primary strategy | WPA 1972 protected areas

- **In-situ conservation** protects species **where they naturally live**, keeping habitat, food webs, pollinators, and competitors intact.

- NCERT and global policy treat in-situ as the **preferred primary strategy** because it preserves **ecosystem integrity**, **natural selection**, and **genetic diversity** (आनुवंशिक विविधता) in real conditions.
- **National Parks** offer strict habitat protection under **Wildlife (Protection) Act, 1972**, with human activity restricted. **Dudhwa NP (UP)** is a Terai (तराई) in-situ example.
- **Wildlife Sanctuaries** protect habitat with slightly more flexible human use than NPs but remain **in-situ**. **Katerniaghat WLS (UP)** protects gharial and Gangetic dolphin in natural rivers.
- **Biosphere Reserves** follow the UNESCO **MAB** model with **core**, **buffer**, and **transition** zones, combining strict protection with sustainable community use.
- **Conservation Reserves** and **Community Reserves** (WPA amendments) protect habitat patches with community participation.
- **Sacred groves** (deorai, sarna (सरना), kavu) are traditional **in-situ** community forests protected by local belief systems.
- **Ramsar wetlands** and **marine protected areas** are specialised **in-situ** tools for aquatic and coastal ecosystems.
- **Wildlife corridors** link fragmented habitats at landscape scale. The **Terai Arc Landscape** connects Dudhwa–Corbett habitats.
- **Limitation:** In-situ alone cannot save species whose habitat is completely destroyed or whose wild population (जनसंख्या) is critically small without ex-situ backup.

PYQ — UPPCS Prelims 2023, Q118

In National Parks, genetic diversity is maintained through—

A. In-situ conservation

B. Ex-situ conservation

C. Gene pool

D. None of the above

💡 **Show answer**

Ans: A — NPs protect wild populations **in natural habitat** = in-situ.

Method	Type	India example
National Park	In-situ	Dudhwa (UP)
Wildlife Sanctuary	In-situ	Katerniaghat (UP)
Biosphere Reserve	In-situ	Similipal
Sacred Grove	In-situ	Tribal (आदिवासी) UP belts
Seed Bank	**Ex-situ**	NBPGR Delhi

> **Logic:** UPPCS 2019 pattern — "Which is in-situ?" → only **NP/sanctuary/BR/sacred grove**. Zoo, seed bank, botanic garden = **ex-situ distractors**.

20.2 Ex-situ Conservation

Identity: Off-site | Supplementary backup | Insurance against extinction | Cannot replace in-situ

- **Ex-situ conservation** removes or copies genetic material **outside the natural ecosystem** — zoos, seed banks, gene banks, captive breeding, tissue culture.

- It is **supplementary**, not a substitute. Ex-situ cannot replicate full ecosystem complexity, predator–prey dynamics, or natural selection.
- **When essential:** critically small wild populations, **habitat already lost**, **reintroduction stock** needed (gharial, vulture), or **Extinct in the Wild (EW)** species surviving only in collections.
- **Major ex-situ types:** zoological parks, botanical gardens, seed banks, gene banks, captive breeding centres, tissue culture, cryopreservation.
- **Genetic bottleneck risk:** small captive groups lose diversity through inbreeding — zoos rotate breeding pairs and maintain studbooks.
- **Reintroduction link:** ex-situ breeding succeeds only if **in-situ habitat** is restored. Cheetah reintroduction required **Kuno NP** habitat preparation alongside captive stock.
- **IUCN (आईयूसीएन) Red List "Extinct in the Wild (EW)"** marks species surviving only in zoos, botanic gardens, or seed banks with no viable wild population.
- **CBD + BDA 2002:** India regulates access to genetic resources; ex-situ collections at NBPGR/NBAGR support national biodiversity obligations.

PYQ — UPPCS Prelims 2018, Q54

Which of the following is NOT an example of insitu conservation strategy?

A. Biosphere reserve

B. Botenical garden

C. National Park

D. Sacred groves

💡 **Show answer**

Ans: B — Botanical garden = **ex-situ** living collection. All others = in-situ.

Feature	In-situ	Ex-situ
Location	Natural habitat	Captive/stored
Priority	**Primary**	**Supplementary**
Examples	Dudhwa NP, sacred grove	Delhi Zoo, NBPGR
trap	NP, sanctuary, BR	Zoo, seed bank, botanic garden

> **Logic:** "Zoos are the best/primary form of biodiversity conservation" = **FALSE**. In-situ protected areas remain primary.

20.3 Gene Bank

Identity: Ex-situ genetic repository | NBPGR Delhi (plants) | NBAGR Karnal (animals) | Cryo at -196°C

- A **gene bank** stores **multiple forms of genetic material**: seeds, pollen, tissue cultures, DNA, animal semen, and embryos.
- **ICAR-NBPGR** at **Pusa Campus, New Delhi** (est. **1978**) is India's national **plant** gene bank. It holds one of the world's largest plant collections.
- The **National Gene Bank** at NBPGR (est. **1996** unit) is the long-term **base collection** stored at **-18°C to -20°C** for orthodox seeds, with cryo units for recalcitrant material.
- **NBAGR, Karnal (Haryana)** (est. **1984**) conserves **animal** genetic resources — semen, embryos, and DNA of indigenous livestock breeds (Sahiwal, Gir (गीर), Red Sindhi).

- **Cryopreservation** uses **liquid nitrogen at -196°C**, stopping biological activity for decades-long storage of embryos, sperm, pollen, and tissue.
- **Active collection** holds frequently accessed seeds for breeders. **Base collection** is long-term duplicate storage with minimal disturbance.
- Stored seeds lose viability over time and must be **periodically regenerated** by germination and re-collection.
- **Trap: NBPGR vs BSI** — NBPGR **stores** genetic resources (Delhi). **BSI** does plant **survey/taxonomy** (Kolkata).
- **Trap: NBPGR vs NBAGR** — **NBPGR = plants (Delhi)**; **NBAGR = animals (Karnal)**.

Institution	Location	Conserves
ICAR-NBPGR	New Delhi	Plant seeds, pollen, tissue, DNA
National Gene Bank	New Delhi (NBPGR)	Long-term base collection
NBAGR	Karnal, Haryana	Livestock semen, embryos, DNA
Svalbard Global Seed Vault	Norway	Duplicate global seed backup

> **Logic:** "National gene bank for animal genetic resources is NBPGR" = **FALSE** — animals = **NBAGR Karnal**.

20.4 Seed Bank

Identity: Ex-situ | Seeds only | Orthodox = cold storage | Recalcitrant = mango/coconut trap

- A **seed bank** stores **seeds only** (narrower than a gene bank). Seeds are dried to **3–7% moisture**, sealed airtight, and kept at **-18°C to -20°C**.
- **Orthodox seeds** (wheat, rice, maize, millets, pulses) tolerate desiccation and cold — suitable for **decades of storage**.
- **Recalcitrant seeds** (mango, jackfruit, coconut, sal) **cannot tolerate drying** and need **botanical garden living collections** or **cryopreservation** instead of conventional seed banking.
- **NBPGR** operates the **National Active Collection** (breeder access) and **National Gene Bank** (long-term base store) with **4 lakh+ accessions** including landraces and crop wild relatives.
- **Svalbard Global Seed Vault (Norway, 2008)** stores **duplicate copies** of national collections as Arctic "doomsday backup" — not ownership transfer.
- **Regeneration** is mandatory when germination rates fall — gene banking is dynamic, not store-and-forget.
- **Community seed banks** (on-farm) preserve landraces in active cultivation — different from **formal ex-situ cold storage** at NBPGR. Notes usually mean NBPGR-style cold storage.

Type	Storage	Examples
Orthodox	Conventional seed bank (-18°C)	Wheat, rice, chickpea
Recalcitrant	Living collection or cryo	Mango, coconut, jackfruit
Feature	Seed Bank	Gene Bank
---	---	---
Material	Seeds primarily	Seeds + DNA + tissue + sperm + embryos
Scope	Narrower	Broader
Both are	**Ex-situ**	**Ex-situ**

> **Logic:** "Seed bank is in-situ conservation" = **FALSE**. UPPCS 2019 NOT-in-situ MCQs use seed bank as the ex-situ answer.

20.5 Botanical Garden

Identity: Ex-situ living plants | Indian Botanic Garden Howrah 1787 | BSI administers | Not same as BSI mandate

- **Botanical gardens** maintain **living plant collections** outside wild habitat for conservation, research, education, and display.
- Unlike seed banks (dormant storage), gardens keep **living plants** — essential for **recalcitrant-seed species** and **Extinct in the Wild** plants.
- The **Acharya Jagadish Chandra Bose Indian Botanic Garden** at **Howrah, West Bengal** (est. **1787** by Colonel Robert Kyd) is the **largest and oldest** botanic garden in India, famous for the **Great Banyan Tree**.
- It is administered by **Botanical Survey of India (BSI)** (भारतीय वनस्पति सर्वेक्षण). BSI is a **survey organisation** — it is not synonymous with "botanical garden" as a conservation method.
- **Other major gardens:** Lalbagh (Bengaluru, 1760), **NBRI garden (Lucknow)**, Empress Garden (Pune (पुणे)).
- **BGCI** links **3500+ gardens** globally for coordinated ex-situ plant conservation.
- **Limitation:** living collections hold limited genetic diversity, artificial conditions differ from wild ecology (पारिस्थितिकी), and maintenance is costly.

Garden	Location	Notable
Indian Botanic Garden	Howrah	1787; Great Banyan; largest in India
Lalbagh	Bengaluru	1760; heritage trees
NBRI Garden	Lucknow	Medicinal plant research

> **Logic:** UPPCS in-situ MCQs always list **botanical garden** as an **ex-situ** distractor alongside zoo and seed bank.

20.6 Zoological Park

Identity: Ex-situ captive animals | CZA under MoEFCC | Delhi Zoo 1959 | Captive breeding for recovery

- **Zoological parks (zoos)** hold animals in **enclosed, managed environments** — classic **ex-situ** conservation removed from free-ranging wild habitat.
- **Roles:** conservation breeding, public education, scientific research, and genetic reservoir for reintroduction when habitat is secured.
- **Central Zoo Authority (CZA)** (केंद्रीय चिड़ियाघर प्राधिकरण) was created under the **Wildlife (Protection) Amendment Act, 1991** and functions under **MoEFCC**. It grants zoo recognition (मान्यता), sets standards, and coordinates breeding programmes.
- The **National Zoological Park (Delhi Zoo)** in **New Delhi** (est. **1959**) is a flagship ex-situ facility for endangered Indian species.
- **Species recovery centres** (also ex-situ): vulture breeding centres (Pinjore, Rani, Buxa), **Kukrail Crocodile Centre (Lucknow, UP)**, gharial breeding for river release.
- **Trap: zoo vs sanctuary** — zoo = **captive ex-situ**; sanctuary = **wild animals in natural habitat (in-situ)**.
- **Limitations:** inbreeding in small groups, loss of wild survival skills, stereotypic behaviour — zoos **cannot replace** habitat-level in-situ protection.

Facility	Location	Type
National Zoological Park	New Delhi	Zoo (ex-situ)
CZA	New Delhi	Regulator (MoEFCC)
Kukrail Crocodile Centre	Lucknow, UP	Captive breeding (ex-situ)

Facility	Location	Type
Katerniaghat WLS	UP	Sanctuary (**in-situ** contrast)

> **Logic:** Every UPPCS in-situ question lists **zoological park** as ex-situ. Match: Zoo = ex-situ | NP/Sanctuary = in-situ.

Complete PYQ Bank (Topic 20)

Q1. UPPCS Prelims 2018, Q54

Which of the following is **NOT** an example of an in-situ conservation strategy?

A. Biosphere reserve

B. Botanical garden

C. National Park

D. Sacred groves

💡 **Show answer**

Logic: Four-way in-situ vs ex-situ trap — botanic gardens are off-site living collections. **Ans: B.** A **botanical garden is ex-situ** conservation; all other options are in-situ. **Trap:** Treating managed gardens as "natural habitat" in-situ.

Q2. UPPCS Prelims 2023, Q118

In National Parks, genetic diversity is maintained through—

A. In-situ conservation

B. Ex-situ conservation

C. Gene pool

D. None of the above

💡 **Show answer**

Logic: NPs protect wild populations in natural habitat — primary in-situ strategy. **Ans: A.** **National Parks maintain genetic diversity through in-situ conservation**. **Trap:** **Ex-situ** (zoos/seed banks) as the answer for NP management.

Q3. UPPCS Prelims 2019 pattern — NOT in-situ

Which of the following is **NOT** an in-situ method of biodiversity conservation?

A. Wildlife Sanctuary

B. Seed bank

C. National Park

D. Sacred grove

💡 **Show answer**

Logic: Seed banks store germplasm off-site — classic ex-situ distractor in UPPCS stems. **Ans: B.** A **seed bank is ex-situ** conservation. **Trap:** Picking sanctuary or sacred grove (both in-situ).

Q4. UPPCS Prelims 2019 pattern — in-situ choice

Which of the following is an example of **in-situ** conservation?

A. Zoological park

B. National Park

C. Seed bank

D. Botanical garden

💡 **Show answer**

Logic: Only NP/sanctuary/BR/sacred grove among typical MCQ lists are in-situ. **Ans: B.** A **National Park is in-situ**; zoo, seed bank, and botanic garden are ex-situ. **Trap:** **Zoological park** — captive ex-situ, not wild in-situ.

Q5. UPPCS Prelims pattern — NBPGR location

The Indian national plant gene bank (ICAR-NBPGR) is located at:

A. Kolkata

B. New Delhi

C. Karnal

D. Lucknow

💡 **Show answer**

Logic: NBPGR Delhi vs BSI Kolkata — storage versus survey institutions. **Ans: B.** **NBPGR is at New Delhi** (Pusa Campus). **Trap:** **Kolkata** — BSI/ZSI headquarters, not NBPGR.

Q6. UPPCS Prelims pattern — NBAGR

India's National Bureau of Animal Genetic Resources (NBAGR) is located at:

A. New Delhi

B. Karnal, Haryana

C. Kolkata

D. Hyderabad

💡 **Show answer**

Logic: Plant gene bank = NBPGR Delhi; animal gene bank = NBAGR Karnal. **Ans: B.** **NBAGR is at Karnal** for livestock genetic resources. **Trap:** **New Delhi** (NBPGR) or assigning animals to NBPGR.

Q7. UPPCS Prelims pattern — Svalbard Vault

The Svalbard Global Seed Vault, a duplicate backup for national seed collections, is located in:

A. India

B. Norway

C. Switzerland

D. United Kingdom

💡 **Show answer**

Logic: Arctic "doomsday" vault — duplicate storage, not India's primary gene bank. **Ans: B.** **Svalbard Global Seed Vault is in Norway** (2008). **Trap:** Locating Svalbard in **India** (NBPGR) or confusing with

formal ownership transfer.

Q8. UPPCS Prelims pattern — CZA nodal ministry

The Central Zoo Authority (CZA) functions under which ministry?

A. Ministry of Agriculture

B. Ministry of Environment, Forest and Climate Change

C. Ministry of Jal Shakti

D. Ministry of Tribal Affairs

💡 **Show answer**

Logic: CZA is the statutory zoo regulator under MoEFCC, not agriculture or tribal affairs. **Ans: B.** **CZA operates under MoEFCC**. **Trap:** Agriculture ministry because zoos display animals/farming link distractor.

Q9. UPPCS Prelims 2014 pattern — most effective conservation

The most effective method for conserving biodiversity is:

A. Ex-situ zoos alone

B. In-situ protected areas in natural habitat

C. Seed banks without habitat protection

D. Botanical gardens only

💡 **Show answer**

Logic: NCERT/CBD priority — in-situ first; ex-situ is supplementary backup. **Ans: B.** **In-situ protected areas** are the primary effective strategy. **Trap:** "Zoos are the best/primary form" — false preference reversal.

Q10. UPPCS Prelims 2011 pattern — ex-situ mode

Which of the following is a mode of **ex-situ** conservation?

A. National Park

B. Wildlife Sanctuary

C. Seed banks

D. Sacred grove

💡 **Show answer**

Logic: Ex-situ = off-site storage or captive collections among mixed lists. **Ans: C.** **Seed banks** are ex-situ; parks, sanctuaries, and groves are in-situ. **Trap:** **Sacred grove** — community in-situ forest, not ex-situ.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to in-situ conservation, which of the following statements is/are correct?

1. Protects biodiversity in the natural habitat.
2. Preferred secondary strategy.
3. Zoo is the best example of in-situ.

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q2. Which of the following pairs is/are correctly matched?

1. National Park — in-situ
2. Sacred grove — ex-situ
3. Seed bank — in-situ

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q3. Match List-I with List-II:

List-I	List-II
A. Gene bank	1. Seeds only
B. Seed bank	2. Broad genetic material
C. Cryopreservation	3. ~−196°C liquid nitrogen

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

💡 **Show answer**

Ans: D. A-2, B-1, C-3. **Logic:** Gene broad; seed only; cryo LN2.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Orthodox seeds (wheat/rice) tolerate drying and cold storage around −18°C.

Reason (R): Recalcitrant seeds (mango/coconut) cannot take simple dry seed-bank storage.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; recalcitrant fact contrasts but does not explain orthodoxy's tolerance mechanism as its cause. **A/R logic:** Seed-type pair.

Q5. With reference to Indian flagships, which of the following statements is/are correct?

1. NBPGR — plant gene/seed flagship, New Delhi.
2. NBAGR is a plant gene bank at Kolkata, not an animal gene bank at Karnal.
3. NBPGR HQ is Kolkata BSI campus only.

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Statement 1 is true (NBPGR New Delhi plant gene/seed). Statement 2 is false — NBAGR is the animal gene bank at Karnal. Statement 3 is false — NBPGR HQ is not the Kolkata BSI campus.

Q6. Arrange four-way pattern teaching:

A. Zoo and seed bank are in-situ; NP is ex-situ B. NP and sacred grove in-situ; zoo, seed bank, botanic garden ex-situ C. All are ex-situ only D. All are in-situ only

💡 **Show answer**

Ans: B. NP/sacred grove in-situ; zoo/seed/botanic ex-situ. **Logic:** Classic pattern.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Indian Botanic Garden Howrah 1787 — BSI link
2. Svalbard Global Seed Vault — Norway backup
3. Svalbard — India's NBPGR headquarters

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Svalbard ≠ NBPGR.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): CZA is the statutory zoo regulator under MoEFCC.

Reason (R): Zoos are ex-situ conservation institutions.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; ex-situ nature does not by itself create CZA's statute. **A/R logic:** Regulator vs method.

Q9. With reference to Dudhwa vs Kukrail teaching, which of the following statements is/are correct?

1. Dudhwa-style PAs are in-situ.
2. Kukrail crocodile centre is ex-situ.
3. Both are identical UNESCO Biosphere Reserves of UP.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** UP has zero BRs; centres ≠ BR.

Q10. Match List-I with List-II:

List-I	List-II
A. In-situ	1. Captive breeding centre

List-I	List-II
B. Ex-situ	2. Wildlife Sanctuary
C. Cryobank	3. Liquid nitrogen storage

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Sanctuary in-situ; captive ex-situ; cryo LN2.

Q11. Arrange storage temperatures teaching:

A. Cryo −18°C; orthodox −196°C B. Orthodox ~−18°C; cryo ~−196°C C. Both +25°C only D. Both 100°C

💡 **Show answer**

Ans: B. Orthodox ~−18°C; cryo ~−196°C. **Logic:** Temperature facts.

Q12. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Ex-situ is a backup strategy.

Reason (R): In-situ conserves the whole ecosystem and natural processes better as primary strategy.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains why ex-situ is backup. **A/R logic:** Primary vs backup.

Q13. Which of the following is/are ex-situ examples?

1. Zoo
2. Botanical garden
3. Tissue culture
4. National Park

A. 2, 3 and 4 only B. 1 and 4 only C. 1, 2 and 3 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: C. 1, 2 and 3 only. **Logic:** NP is in-situ.

Q14. With reference to gene bank contents, which of the following statements is/are correct?

1. May include seeds, DNA, tissue, pollen, sperm, embryos.
2. Broader than seed bank.
3. Gene bank stores only living tigers in a sanctuary.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not a sanctuary substitute definition.

Q15. How many of the following are in-situ?

Wildlife Sanctuary, Biosphere Reserve, Conservation Reserve, sacred grove, zoo

A. Only four B. Only three C. Only two D. All five

💡 **Show answer**

Ans: A. Only four. **Logic:** Zoo out.

Q16. Which of the following pairs is/are correctly matched?

1. NBPGR — plants Delhi
2. NBAGR — animals Karnal
3. BSI Kolkata — national plant gene bank identical to NBPGR

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** BSI surveys — not NBPGR.

Q17. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Recalcitrant seeds cannot take simple dry cold seed-bank storage.

Reason (R): Mango and coconut are recalcitrant examples.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R exemplifies A. **A/R logic:** Examples support the rule.

Q18. With reference to National Zoological Park teaching neighbourhood, which of the following statements is/are correct?

1. Zoos are ex-situ.
2. CZA regulates zoos.
3. Every zoo is a Tiger Reserve core under Sec. 38V.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Zoo ≠ TR core.

Q19. Match List-I with List-II:

List-I	List-II
A. Howrah Botanic Garden	1. 1976/78 line Delhi plants
B. NBPGR	2. 1787
C. NBAGR	3. 1984 Karnal

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

💡 **Show answer**

Ans: D. A-2, B-1, C-3. **Logic:** 1787 / NBPGR Delhi / NBAGR Karnal.

Q20. Arrange preference teaching:

A. In-situ primary; ex-situ backup B. Ex-situ primary; in-situ never used C. Only cryobanks legally allowed D. Only match-list codes without habitats

💡 **Show answer**

Ans: A. In-situ primary; ex-situ backup. **Logic:** Strategy hierarchy.

Q21. With reference to Ramsar as conservation method neighbourhood, which of the following statements is/are correct?

1. Ramsar wetlands can be in-situ conservation landscapes.
2. Ramsar is a WPA Schedule VI plant list.
3. Wise Use allows sustainable human use teaching.

A. 1 and 2 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 3. **Logic:** Not Schedule VI.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Tissue culture is an ex-situ tool.

Reason (R): It conserves material outside the natural habitat.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R defines ex-situ. **A/R logic:** Outside habitat = ex-situ.

Q23. Which of the following statements about sacred groves is/are correct?

1. In-situ community/traditional conservation.
2. Natural habitat protection pattern.
3. Identical to Svalbard vault.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Not Svalbard.

Q24. How many of the following are ex-situ?

Zoo, botanic garden, seed bank, gene bank, cryobank

A. Only two B. Only three C. Only four D. All five

💡 **Show answer**

Ans: D. All five. **Logic:** Ex-situ set.

Q25. With reference to pollen/sperm/embryos in banks, which of the following statements is/are correct?

1. Can be part of gene-bank holdings.

2. Broader than “seeds only.”
3. Prove the site is a National Park automatically.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Bank ≠ NP.

Q26. Which of the following pairs is/are correctly matched?

1. Captive breeding — ex-situ
2. Community Reserve — in-situ WPA path
3. Aquarium gene vault — always in-situ forest

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Aquarium-style holding is ex-situ.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): BSI surveys plants; it is not the national gene bank.

Reason (R): NBPGR is the plant gene-bank flagship in Delhi.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R separates survey vs gene bank. **A/R logic:** BSI ≠ NBPGR.

Q28. Arrange NBPGR vs NBAGR:

A. Both animals in Delhi B. Both plants in Karnal C. Plants Delhi (NBPGR); animals Karnal (NBAGR) D. Both in Nairobi

💡 **Show answer**

Ans: C. NBPGR Delhi plants; NBAGR Karnal animals. **Logic:** Do not swap.

Q29. With reference to Conservation/Community Reserve, which of the following statements is/are correct?

1. They are in-situ WPA categories.
2. They differ from ex-situ zoos.
3. They replace cryopreservation entirely.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Do not replace cryo tools.

Q30. Which one of the following best states why in-situ is preferred?

A. Conserves species with natural ecosystem processes B. Because zoos are illegal everywhere C. Because seed banks are banned D. Because UNEP forbids ex-situ

💡 **Show answer**

Ans: A. Whole ecosystem + natural processes. **Logic:** Primary strategy rationale.

Common Traps

Trap	Correct
Zoo = in-situ	**Ex-situ** — captive, off-site
Seed bank = in-situ	**Ex-situ** — cold storage off-site
Botanical garden = in-situ	**Ex-situ** — managed living collection
NBPGR = Kolkata	**New Delhi** — Kolkata = BSI/ZSI
NBAGR = Delhi or NBPGR	**Karnal** — animals only
NBPGR stores animal semen	**NBAGR Karnal**
BSI = botanical garden	BSI **surveys** flora; garden is a **facility**
Zoos are primary conservation	**In-situ PAs preferred**
Mango in standard seed bank	**Recalcitrant** — living collection/cryo
Svalbard vault in India	**Norway** — duplicate backup only
Sanctuary = zoo	Sanctuary = **wild in-situ**; zoo = **captive ex-situ**
Gene bank = in-situ farmer field	Formal gene bank = **ex-situ** storage

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 21 — Species & Ecology (पारिस्थितिकी)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Red Data Book (रेड डेटा बुक) | IUCN (आईयूसीएन) Red List Categories | Wetlands | Mangroves | Coral Reefs | Grasslands | Himalayan Ecosystem (पारिस्थितिकी तंत्र) | Marine Ecosystem

Sources baked in: NCERT Bio 12 Ch 15–16; Geo 11; IUCN Red List; Ramsar 1971; BSI/ZSI; MoEFCC; UPPCS PYQs 2018–2025 **Weight:** ★★★ Very High — IUCN threatened trap; UP Ramsar geography; coral bleaching; Himalayan zones **Last verified:** August 2026

Current Affairs (this topic)

- New IUCN category moves for flagship species (प्रतीक (फ्लैगशिप) प्रजाति) are the usual CA refresh.
- Keystone / umbrella / flagship / indicator definitions stay concept-fixed.

💡 🎯 **Consolidated — 26 Must-Score Facts (Click to Expand)**

1. India's **Red Data Book** is a **national** catalogue — **BSI** for plants, **ZSI** for animals (both Kolkata). The **IUCN Red List** is the **global** extinction-risk system from IUCN (Gland). 2. IUCN has nine categories: EX, EW, CR, EN, VU, NT, LC, DD, NE. Risk order puts CR above EN above VU among living threatened tiers. 3. **Threatened** on IUCN means **VU + EN + CR** only — not "Endangered" alone. 4. IUCN assesses extinction risk. **CITES** (साइट्स) regulates international wildlife **trade** (पण्याध्यक्ष). Different institutions, different jobs. 5. **Ramsar (रामसर) (1971)** is the wetlands treaty under Wise Use. World Wetlands Day (विश्व आर्द्रभूमि दिवस) is **2 February** (रामसर). 6. Uttar Pradesh (उत्तर प्रदेश) Ramsar examples include Sarsai Nawar, Samaspur, Nawabganj, Upper Ganga (गंगा), Bakhira, and Haiderpur. **Rudrasagar is Tripura**; **Sultanpur is Haryana** — not UP. 7. **Mangroves** are salt-tolerant coastal forests with **pneumatophores** (वात-मूल) and often viviparous seeds. **Sundarbans** (सुंदरबन) is the classic largest mangrove belt; Rhizophora and Avicennia are common genera. 8. **MISHTI** is the Union Budget **2023–24** mangrove restoration push. Do not date it as 2019. 9. **Coral reefs** are built from calcium-carbonate skeletons and live with symbiotic **zooxanthellae** (ज़ूजैंथेली). **Bleaching** happens when heat stress expels the algae. 10. India's coral belts include Gulf of Mannar, Lakshadweep (लक्षद्वीप), Andaman (अंडमान) & Nicobar (निकोबार), and Gulf of Kutch. 11. **Grasslands** are grass-dominated systems maintained by fire and grazing. UP's Terai (तराई) is a foothill grassland–forest mosaic; **Dudhwa** (दुधवा) holds barasingha (swamp deer). Kaziranga (काजीरंगा) is the floodplain grassland rhino fact. 12. Himalayan vegetation follows altitude belts: tropical → subtropical → temperate → alpine → nival. Snow leopard is a high Himalaya (हिमालय) fact; Dudhwa is Terai foothill fringe, not alpine. 13. Marine **neritic** shelf waters are the highest fisheries-productivity zone; pelagic means open ocean. India's coast is about **7516 km**. 14. The **Gangetic dolphin** is a **freshwater river** dolphin — not a marine species trap. 15. Wetland ≠ mangrove ≠ coral. Inland marsh, saline (लवणीय) coastal forest, and reef builders are three different ecosystems. 16. Pneumatophores are mangrove breathing roots for gas exchange in waterlogged saline soil. 17. Amrit Dharohar (**2023**) is India's wetland stewardship framing beside Ramsar listing. 18. Coral bleaching (प्रवाल विरंजन) is a stress response, not a synonym for mangrove dieback. 19. Terai grasslands of UP are not Himalayan alpine meadows. 20. Red Data Book (national BSI/ZSI) is not identical to IUCN Red List categories, even when species appear on both tracks. 21. World Wetlands Day is February **2**, not Environment (पर्यावरण) Day (5 June). 22. Mangrove restoration schemes do

not make landlocked UP a mangrove state — UP links are scheme awareness and river–wetland ecology. 23. Zooxanthellae live inside coral polyps; losing them bleaches the coral's colour and energy supply. 24. Threatened = three IUCN tiers combined. Memorise VU, EN, CR as the band. 25. CMS covers migratory species; Ramsar covers wetlands; do not swap those treaty jobs when ecosystems are asked. 26. Dudhwa = Terai grassland–forest and barasingha; Sundarbans = mangrove; Lakshadweep/Mannar = coral — keep the flagship map clean.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **IUCN Red List** | **Red Data Book of India** | Global extinction risk assessments maintained by IUCN (HQ: Gland, Switzerland) categorizing global species from LC to EX vs national taxonomic inventories published by BSI (plants) and ZSI (animals) cataloguing Indian threatened flora/fauna | आईयूसीएन रेड लिस्ट (वैश्विक) / भारतीय रेड डेटा बुक (राष्ट्रीय - BSI/ZSI) | | **Threatened Species** | **Endangered Species (EN)** | Broader collective umbrella category encompassing three distinct risk levels: Vulnerable (VU) + Endangered (EN) + Critically Endangered (CR) vs a specific single IUCN conservation status indicating very high risk of extinction | संकटग्रस्त प्रजातियां (VU + EN + CR समग्र) / लुप्तप्राय प्रजाति (EN विशिष्ट वर्ग) | | **Mangrove Ecosystem** | **Coral Reef Ecosystem** | Salt-tolerant halophytic coastal wetland forests adapted to intertidal muddy sediments featuring pneumatophores (Rhizophora, Avicennia) vs marine calcium carbonate structures formed by symbiotic coral polyps and zooxanthellae in clear, shallow, warm waters | मैंग्रोव पारितंत्र (श्वसन मूल वन) / प्रवाल भित्ति पारितंत्र (उष्ण जलीय प्रवाल) | | **Pneumatophores** | **Stilt Roots** | Specialized vertical negative geotropic aerial roots that emerge above mud to facilitate oxygen gas exchange in waterlogged swamp soils (Avicennia) vs adventitious support roots emerging from lower trunk/branches providing mechanical anchor against coastal tidal surges (Rhizophora) | वात-मूल (श्वसन मूल, न्यूमेटोफोर) / अवस्तंभ मूल (सहारा जड़ें) | | **Coral Bleaching** | **Ocean Acidification** | Stress response where coral polyps expel their symbiotic photosynthetic zooxanthellae due to elevated sea surface temperatures turning white vs chemical lowering of ocean pH caused by oceanic absorption of excess atmospheric CO_2 dissolving calcium carbonate aragonite structures | प्रवाल विरंजन (ऊष्मा तनाव से शैवाल निष्कासन) / महासागरीय अम्लीकरण (pH ह्रास) | | **Neritic Marine Zone** | **Oceanic (Pelagic) Zone** | Shallow coastal marine zone extending from low-tide mark to the edge of the continental shelf (depth up to 200 m) with abundant light and high marine productivity vs deep open marine waters beyond the continental shelf edge where nutrients and light drop precipitously | नेरिटिक मंडल (महाद्वीपीय मग्नतट) / खुला महासागरीय मंडल (पेलाजिक) | | **Zooxanthellae** | **Coral Polyp** | Photosynthetic single-celled dinoflagellate microalgae living symbiotically within coral tissues providing up to 90% of their organic nutrients vs tiny, soft-bodied marine invertebrate animal belonging to Phylum Cnidaria that secretes a hard calcium carbonate skeleton | ज़ूजैंथेली (सहजीवी प्रकाश-संश्लेषी शैवाल) / प्रवाल पॉलीप (कैल्शियम स्रावी जंतु) |

💡 📋 **Must-score facts — Red List, Ramsar, biomes (Click to Expand)**

Lists / wetlands | Item | Lock | |---|---| | India Red Data | BSI plants + ZSI animals (Kolkata) | | IUCN 9 categories | EX EW CR EN VU NT LC DD NE | | Threatened | VU+EN+CR only | | IUCN vs CITES | Extinction risk / trade regulation | | Ramsar / Wetlands Day | **1971** / **2 February** | | Rudrasagar / Sultanpur | Tripura / Haryana | | MISHTI | Budget **2023–24** mangroves | | India corals | Mannar · Lakshadweep · A&N · Kutch (**not** Sundarbans) | | Gangetic dolphin | Freshwater river (≠ marine trap) |

21.1 Red Data Book

Identity: National endangered-species catalogue | BSI plants | ZSI animals | Kolkata HQ | ≠ IUCN Red List

- A **Red Data Book** lists **endangered, rare, and threatened species (संकटग्रस्त प्रजातियां)** of a country or region — "red" signals conservation urgency.
- In **India**, **BSI (Botanical Survey of India)** publishes the **Red Data Book of Indian Plants**.

- **ZSI (Zoological Survey of India)** publishes the **Red Data Book of Indian Animals**.
- Both **BSI and ZSI** are headquartered in **Kolkata**.
- IUCN pioneered the early Red Data Book concept in the **1960s**, which evolved into the modern **IUCN Red List** with standard global categories.
- **Red Data Book ≠ IUCN Red List:** Red Data Book is **national documentation**; IUCN Red List is a **global standardised assessment** (EX through NE).
- **Red Data Book ≠ CITES:** Red Data Book documents status; **CITES** regulates **international trade** in listed species.
- Species may appear in Red Data Books, **IUCN Red List**, and **WPA Schedules** simultaneously — three parallel systems (documentation, global assessment, legal protection).

Feature	Red Data Book (India)	IUCN Red List
Scope	National	Global
Publisher	BSI (plants), ZSI (animals)	IUCN (Gland)
Function	National threat inventory	Standardised risk categories

> **Logic:** "Red Data Book of Indian plants published by ZSI" = **FALSE** — plants = **BSI**, animals = **ZSI**.

21.2 IUCN Red List Categories

Identity: 9 categories | Threatened = VU+EN+CR | IUCN Gland | Assesses risk ≠ CITES trade

- The **IUCN Red List** assigns assessed species to **nine categories** based on population (जनसंख्या), decline rate, range, and extinction probability.
- **Extinct (EX):** No reasonable doubt the last individual has died.
- **Extinct in the Wild (EW):** Survives only in **captivity, cultivation, or seed banks** — no viable wild population.
- **Critically Endangered (CR) (अति संकटग्रस्त):** Extremely high extinction risk — e.g., **gharial**, **Great Indian Bustard**.
- **Endangered (EN) (लुप्तप्राय):** Very high risk — e.g., **Gangetic dolphin**, **tiger** (improved from CR in some assessments).
- **Vulnerable (VU):** High risk but lower urgency than EN/CR — e.g., **sloth bear**.
- **Near Threatened (NT):** Close to qualifying as threatened but not yet.
- **Least Concern (LC):** Widespread, low extinction risk.
- **Data Deficient (DD):** Insufficient data — not the same as safe.
- **Not Evaluated (NE):** Not yet assessed — majority of world's species.
- **Threatened = VU + EN + CR** — the most tested formula. **Endangered alone is only one-third of threatened.**
- **Risk hierarchy** among living categories: **CR > EN > VU > NT > LC**.
- **IUCN assesses extinction risk.CITES** regulates **trade** — separate bodies, separate functions.

Code	Category
EX	Extinct
EW	Extinct in Wild
CR	Critically Endangered
EN	Endangered
VU	Vulnerable

Code	Category
NT	Near Threatened
LC	Least Concern
DD	Data Deficient
NE	Not Evaluated

> **Logic:** "All threatened species are Endangered" = **FALSE**. Threatened = **VU + EN + CR**.

21.3 Wetlands

Identity: Ramsar 1971 | Wise Use | 2 February | UP Ramsar geography trap

- **Wetlands** are ecosystems where **water saturates the soil** — marshes, swamps, bogs, floodplains, lake margins — permanent or seasonal.
- They are among the **most productive ecosystems**, providing flood control, groundwater recharge, water purification, fisheries, and migratory bird habitat (वास स्थान).
- The **Ramsar Convention (1971)** (रामसर कन्वेंशन) promotes **conservation and wise use** of wetlands. **World Wetlands Day = 2 February**.
- A **Ramsar Site** is a wetland of **international importance** nominated by the contracting country — separate from WPA national park status.
- **India** has **75+ Ramsar sites** (2024 milestone). **Amrit Dharohar Scheme (2023)** supports wetland conservation and livelihoods.
- **UP Ramsar sites:Sarsai Nawar** (Etawah (इटावा)), **Samaspur Bird Sanctuary** (Raebareli), **Nawabganj** (Unnao), **Upper Ganga River**, **Bakhira**, **Haiderpur**, **Parvati Agra Bird Sanctuary**.
- **NOT in UP:Rudrasagar Lake** = **Tripura**. **Sultanpur National Park** = **Gurugram, Haryana**. Both are Ramsar sites outside UP.
- **Threats:** drainage for agriculture, pollution (eutrophication (सुपोषण)), invasive species (आक्रामक विदेशी प्रजाति) (**water hyacinth** chokes UP ponds), hydrological alteration.

PYQ — UPPCS Prelims 2025, Q58

Which Ramsar sites are NOT located in Uttar Pradesh?

1. Sarsai Nawar Jheel 2. Samaspur Bird Sanctuary 3. Rudrasagar Lake 4. Sultanpur National Park

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Ans: B (3 and 4) — Rudrasagar = **Tripura**; Sultanpur = **Haryana**. 1 and 2 are in UP.

Type	India example
Freshwater marsh	Keoladeo (केवलादेव) Ghana (घन)
Riverine wetland	Upper Ganga (UP)
High-altitude wetland	Chandra Taal, Tso Moriri
Coastal/mangrove wetland	Sundarbans

> **Logic:** Ramsar = **wetlands**, not climate or forests.

21.4 Mangroves

Identity: Halophytes | Coastal tidal | Pneumatophores | Sundarbans | MISHTI 2023-24

- **Mangroves** are **salt-tolerant** trees and shrubs growing in **intertidal mudflats** of tropical and subtropical coasts.
- **Pneumatophores** (vertical aerial breathing roots) and **prop roots** (Rhizophora) enable gas exchange in waterlogged anaerobic soil.
- **Viviparous germination:** seeds germinate **while attached to the parent tree** and drop as ready seedlings (propagules).
- **Ecological roles:** coastal protection against cyclones/tsunamis, fish nursery grounds, sediment trapping, shoreline stability, **blue carbon** (नीला कार्बन) sequestration.
- Mangroves act as a **safety hedge** because trees **do not get uprooted easily** — they have **extensive, deep root systems** (not because of a wide uninhabited buffer alone).
- **Coastal Odisha** is India's most cyclone-prone Bay of Bengal coast; large mangrove loss in the **Mahanadi (महानदी) delta (डेल्टा)** increases cyclone risk (classic A/R pair).
- **Sundarbans** (West Bengal + Bangladesh) is the **largest mangrove forest** — **Sundari** tree is the flagship associate; royal Bengal tiger and estuarine crocodile.
- **Bhitarkanika** (Kendrapara, Odisha) sits on the **Brahmani–Baitarni** delta — **not** Vansadhara–Subarnarekha, and **not** West Bengal. Ramsar site (**2002**).
- **Coringa** mangrove (Andhra Pradesh) lies in the **Godavari** (गोदावरी) delta — do not confuse with Coondapur/Honnavar (Karnataka coast names).
- Other Indian mangrove areas: **Pichavaram** (TN), **Gulf of Kutch / Khambhat** (Gujarat), **Andaman** (अंडमान) coasts, **Goa** (गोवा) (e.g. Chorao protected mangrove).
- Mangrove area rank (ISFR 2021 style): **West Bengal > Gujarat > Andaman & Nicobar (~616–617 km^2) > Andhra Pradesh**.
- **Andaman & Nicobar** uniquely combine **mangrove + evergreen + deciduous** forests in one geography.
- **UP is landlocked** — no natural mangrove coast; questions test national ecology and schemes.
- **MISHTI** (Mangrove Initiative for Shoreline Habitats and Tangible Income) was announced in **Union Budget 2023–24** to restore mangrove forests and support livelihoods.

PYQ — UPPCS Prelims 2025, Q93

MISHTI: (1) It was initiated in 2019. (2) It aims to restore mangrove forests.

A. Only 2

B. Neither

C. Both

D. Only 1

> 💡 **Show answer**
>
> **Ans: A** — Restoration aim is correct. MISHTI = **2023–24 Budget**, not 2019.

Species	Feature
Rhizophora	Stilt/prop roots
Avicennia	Salt-secreting leaves

Species	Feature
Sonneratia	Pneumatophores

> **Logic:** Mangroves = **coastal saline**, not freshwater marshes alone.

21.5 Coral Reefs

Identity: $CaCO_3$ polyps + zooxanthellae | Warm shallow seas | Bleaching = heat | India four (चातुर्याम) reef regions

- **Coral reefs** are underwater **calcium carbonate structures** built by **coral polyps** (Cnidaria) in warm, shallow, clear seas.
- **Zooxanthellae** (symbiotic dinoflagellate algae) live inside coral tissue and supply up to **90% of coral energy** via photosynthesis.
- **Coral bleaching:** under stress (especially **high temperature**), corals **expel zooxanthellae** and turn white. Prolonged stress causes mortality.
- Growth needs warm water (**18–30°C**), shallow depth (**<30 m** for light), low turbidity, normal salinity.
- **Reef types:fringing** (attached to coast), **barrier** (lagoon between reef and shore), **atoll** (ring around submerged volcano).
- **India's coral reef areas:Gulf of Mannar** (first marine biosphere reserve (जैवमंडल आरक्षित क्षेत्र)), **Lakshadweep** (लक्षद्वीप), **Andaman & Nicobar**, **Gulf of Kutch** (northernmost Indian reefs).
- **Threats:** ocean warming (primary bleaching driver), acidification, sedimentation, destructive fishing, pollution.

PYQ — UPPCS Prelims 2024, Q19

Coral bleaching in Andaman & Nicobar caused by:

A. Increased industrial activity

B. Increase of 2°C in Indian Ocean

C. Heavy marine pollution

D. Decrease of 4°C in Indian Ocean

💡 **Show answer**

Ans: B — Heat stress from **+2°C warming** bleaches coral. Cooling does not cause bleaching.

> **Logic:** Bleaching = **temperature rise**, not pollution alone or **−4°C cooling**.

21.6 Grasslands

Identity: Grass-dominated | Fire + grazing | Terai UP | Dudhwa barasingha | Kaziranga rhino

- **Grasslands** are ecosystems **dominated by grasses** with few trees — maintained by **grazing, fire, and climate**.
- **Tropical savanna (सवाना):** warm climate, wet-dry seasons, scattered trees — Indian **semi-arid Deccan (दक्कन)** and **Vidarbha**.
- **Terai grasslands (UP):** Himalayan foothills with **tall elephant grass** (Saccharum) and sal forest mosaic. **Dudhwa NP** protects tiger, rhino, **barasingha (swamp deer)**.
- **Floodplain grassland:Kaziranga** (Assam) — tall grass supports one-horned rhino.
- **Alpine meadow:** above **~3,500 m treeline** in Himalaya — short summer herb growth.

- **Shola-grassland mosaic (Western Ghats (पश्चिमी घाट)):** montane grass patches alternating with shola forest — Nilgiri (नीलगिरि) tahr habitat.
- **Threats:** conversion to cropland, overgrazing, invasive woody species (lantana), fire suppression.

Type	Location	Key species
Terai grassland	UP foothills	Barasingha, tiger — Dudhwa
Floodplain grassland	Assam	Rhino — Kaziranga
Semi-arid savanna	Deccan	Blackbuck
Alpine meadow	High Himalaya	Medicinal herbs

> **Logic:** Terai = UP grassland-forest mosaic**. Puszta (पुस्ज़टा) =** Hungary** (world geography distractor), not Indian grassland.

21.7 Himalayan Ecosystem

Identity: Altitude → climate → vegetation belts | Snow leopard | Trans-Himalaya cold desert | Dudhwa Terai fringe

- The **Himalayan ecosystem** spans from tropical foothills to **nival (permanent snow) zone** — biodiversity changes with **altitude and climate**.
- **Altitudinal vegetation belts** (general sequence upward): **Tropical** (<1,000 m — sal, shisham) → **Subtropical** (chir pine) → **Temperate** (deodar, fir) → **Sub-alpine** (rhododendron) → **Alpine meadow** → **Nival** (>4,500 m — lichens, snow).
- Temperature drops roughly **6.5°C per 1,000 m** (lapse rate), driving vegetation zonation.
- **Flagship fauna (प्राणीजात):** Terai/foothills — elephant, tiger; mid-Himalaya — musk deer, monal; high Himalaya — **snow leopard**, blue sheep.
- **Trans-Himalaya** (Ladakh (लद्दाख)/Spiti rain-shadow): cold desert, sparse vegetation — radically different from southern wet slopes.
- Himalayas are the **source of perennial rivers** (Ganga, Yamuna (यमुना), Brahmaputra (ब्रह्मपुत्र) tributaries) — snow and glacier melt feeds base flow.
- **UP link:** Terai and Shivalik (शिवालिक) foothills (Dudhwa) form the transition between Gangetic plain and Himalaya.

PYQ — UPPCS Prelims 2025, Q21

Assertion (A): In the Himalayan mountains different types of vegetation are found. Reason (R): In Himalayas, there are variations in climate with change in altitude.

A. Both true; R does not explain A

B. A false, R true

C. A true, R false

D. Both true; R explains A

> 💡 **Show answer**
>
> **Ans: D** — Altitude-driven climate change (जलवायु परिवर्तन) explains vegetation zonation — standard NCERT chain.

Zone	Altitude (approx.)	Vegetation
Tropical foothills	<1,000 m	Sal, riverine forest

Zone	Altitude (approx.)	Vegetation
Temperate	2,000–3,000 m	Deodar, fir
Alpine meadow	3,600–4,500 m	Herbs, dwarf shrubs
Nival	>4,500 m	Lichens, permanent snow

> **Logic:** Snow leopard = **high Himalayan** zones, not Terai or Sundarbans.

21.8 Marine Ecosystem

Identity: ~71% Earth surface | Neritic = productive | India 7516 km coast | Gangetic dolphin ≠ marine

- **Marine ecosystems** cover **~71% of Earth's surface** — estuaries, continental shelves, coral reefs, open ocean, deep sea.
- **Neritic zone** (continental shelf, **0–200 m**) has the **highest fisheries productivity** — light, nutrients, upwelling.
- **Pelagic zone** = open ocean water; **benthic zone** = seafloor organisms.
- **Euphotic zone** (0–200 m) allows photosynthesis; **aphotic** zone is dark.
- **Upwelling** (Arabian Sea west coast during SW monsoon (दक्षिण-पश्चिम)) brings nutrient-rich deep water — supports sardines, mackerel.
- **Estuaries** (river meets sea) are highly productive nurseries but pollution-sensitive.
- **India:~7,516 km coastline**, **Arabian Sea** (west) and **Bay of Bengal** (east), EEZ ~2.02 million km^2.
- **Gangetic dolphin** is a **freshwater river** mammal — **NOT marine** — common trap.
- **Threats:** plastic pollution, oil spills, overfishing, bottom trawling, warming/acidification affecting reefs.

Zone	Character	Productivity
Intertidal	Tide-exposed shore	Moderate (नरम दल)
Neritic	Continental shelf	**High**
Pelagic	Open ocean	Low–moderate
Benthic	Seafloor	Variable

> **Logic:** Highest marine fisheries productivity = **neritic/continental shelf**, not open deep ocean.

Complete PYQ Bank (Topic 21)

Q1. UPPCS Prelims 2025, Q21

Assertion (A): In the Himalayan mountains, different types of vegetation are found.

Reason (R): In the Himalayas, there are variations in climate with change in altitude.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)

B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)

C. (A) is true, but (R) is false

D. (A) is false, but (R) is true

💡 **Show answer**

A/R logic: (A) states altitudinal vegetation zonation (true). (R) gives the standard NCERT cause — altitude-driven climate change. **Ans: A.** Both true; **R explains A** through lapse-rate-driven vegetation belts.

Trap: Choosing "R does not explain A" — altitude–climate–vegetation is the textbook chain.

Q2. UPPCS Prelims 2025, Q58

Which of the following Ramsar sites are **NOT** located in Uttar Pradesh?

1. Sarsai Nawar Jheel
2. Samaspur Bird Sanctuary
3. Rudrasagar Lake
4. Sultanpur National Park

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Logic: UP wetland geography — Tripura and Haryana sites masquerade as UP names. **Ans: B (3 and 4).** **Rudrasagar = Tripura**; **Sultanpur = Haryana**. **Trap:** Sarsai Nawar and Samaspur **are** in UP — do not pick 1 or 2 as wrong.

Q3. UPPCS Prelims 2025, Q93

With reference to MISHTI, consider the following statements:

1. It was initiated in 2019.
2. It aims to restore mangrove forests.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: MISHTI = Union Budget **2023–24** mangrove scheme — not 2019. **Ans: A (Only 2).** Restoration aim is correct; **2019 start is false**. **Trap:** Accepting **2019** launch year.

Q4. UPPCS Prelims 2024, Q19

Coral bleaching in the Andaman & Nicobar region was mainly caused by:

A. Increased industrial activity alone

B. Increase of about 2°C in the Indian Ocean

C. Heavy marine pollution alone

D. Decrease of 4°C in the Indian Ocean

💡 **Show answer**

Logic: Bleaching = heat stress expelling zooxanthellae — not cooling or pollution alone. **Ans: B.** **+2°C ocean warming** drives mass bleaching events. **Trap:** **−4°C cooling** — corals bleach from **heat**, not cold stress in this framing.

Q5. UPPCS Prelims 2023, Q147

World Wetlands Day is observed on:

A. 2 February

B. 22 March

C. 5 June

D. 16 September

💡 **Show answer**

Logic: Ramsar calendar — 2 February wetlands; 22 March water; 5 June environment; 16 September ozone. **Ans: A.** **World Wetlands Day = 2 February**. **Trap:** **22 March** (World Water Day) or **5 June** (WED).

Q6. UPPCS Prelims 2018, Q54

Which of the following is **NOT** an example of in-situ conservation strategy?

A. Biosphere reserve

B. Botanical garden

C. National Park

D. Sacred groves

💡 **Show answer**

Logic: Cross-topic in-situ trap — botanic gardens are ex-situ living collections. **Ans: B.** **Botanical garden = ex-situ**; others are in-situ. **Trap:** Biosphere reserve or sacred grove — both remain in-situ.

Q7. UPPCS Prelims 2018, Q133

Which of the following statements are true?

1. Natural disasters cause maximum damage in developing countries.
2. Bhopal gas tragedy was man-made.
3. India is a disaster-free country.
4. Mangroves reduce the impact of cyclones.

A. 1, 2 and 3

B. 2, 3 and 4

C. 1, 2 and 4

D. 1, 3 and 4

💡 **Show answer**

Logic: Multi-statement disaster–environment link — India is **not** disaster-free; mangroves **do** buffer cyclones. **Ans: C (1, 2 and 4).** Statements **1, 2, and 4** are true; **3 is false**. **Trap:** Accepting "India is disaster-free" or rejecting mangrove protection role.

Q8. UPPCS Prelims 2018, Q122

In which ecosystem is species diversity relatively **highest**?

A. Deep sea

B. Tropical rain forest

C. Coral reefs

D. Desert

💡 **Show answer**

Logic: Among given options, coral reefs often top local species richness per unit area. **Ans: C.** **Coral reefs** show exceptionally high species diversity among the choices. **Trap:** **Tropical rainforest** — globally rich, but this stem keys **coral reefs** among the four.

Q9. UPPCS Prelims 2020, Q48

How many more Ramsar sites were added to Uttar Pradesh in January 2020?

A. 3

B. 6

C. 9

D. 12

💡 **Show answer**

Logic: UP Ramsar expansion CA fact — January 2020 batch count. **Ans: B.** **Six** Ramsar sites were added to UP in **January 2020**. **Trap:** Over-counting (9 or 12) or under-counting (3).

Q10. UPPCS Prelims pattern — threatened species

On the IUCN Red List, “threatened species” includes which categories?

A. Endangered only

B. Vulnerable, Endangered, and Critically Endangered

C. Near Threatened and Endangered

D. Extinct and Extinct in the Wild

💡 **Show answer**

Logic: Threatened band = **VU + EN + CR** — not Endangered alone. **Ans: B.** **Threatened = VU + EN + CR**. **Trap:** "Threatened = Endangered only" — drops Vulnerable and CR.

Q11. UPPCS Prelims pattern — Red Data Book publisher

The Red Data Book of **Indian plants** is published by:

A. ZSI

B. BSI

C. IUCN

D. MoEFCC

💡 **Show answer**

Logic: National Red Data Books — BSI plants, ZSI animals (Kolkata). **Ans: B.** **BSI publishes the plant Red Data Book**. **Trap:** **ZSI** (animals) or **IUCN** (global Red List, not national book).

Q12. UPPCS Prelims pattern — zooxanthellae

Zooxanthellae are symbiotic algae associated with:

A. Mangrove pneumatophores

B. Coral polyps

C. Grassland rhizomes

D. Deep-sea anglerfish only

> 💡 **Show answer**
>
> **Logic:** Coral energy symbiosis — zooxanthellae supply photosynthate; bleaching = their expulsion. **Ans: B.** **Zooxanthellae live inside coral polyps**. **Trap:** Mangroves or deep-sea-only distractors.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to Red Data sources, which of the following statements is/are correct?

1. India's Red Data Book — BSI plants, ZSI animals (Kolkata).
2. IUCN Red List is the global extinction-risk catalogue.
3. BSI publishes the IUCN Red List for the whole world.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** IUCN is global — not BSI's world list.

Q2. Which of the following pairs is/are correctly matched?

1. Threatened — VU + EN + CR
2. Endangered alone — full Threatened set
3. Among living threatened tiers the risk order is VU > EN > CR.

A. Only 3 B. Only 1 C. 1 and 2 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 is correct. **Logic:** Statement 1 is true (Threatened = VU+EN+CR). Statement 2 is false — Endangered alone is not the full Threatened set. Statement 3 reverses CR > EN > VU.

Q3. Match List-I with List-II:

List-I	List-II
A. IUCN	1. Wildlife trade
B. CITES	2. Extinction risk
C. Ramsar	3. Wetlands / Wise Use

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-2, B-1, C-3

> 💡 **Show answer**

Ans: D. A-2, B-1, C-3. **Logic:** Risk / trade / wetlands.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Rudrasagar is in Tripura and Sultanpur is in Haryana.

Reason (R): Both are Uttar Pradesh Ramsar sites.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Neither is a UP Ramsar site.

Q5. With reference to mangroves, which of the following statements is/are correct?

1. Mangroves are freshwater-only inland forests without pneumatophores.
2. Often viviparous seeds.
3. Sundarbans is a coral-reef hotspot identical to Lakshadweep atolls.

A. Only 1 B. Only 2 C. 1 and 2 D. 2 and 3

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Statement 2 is true (vivipary is common). Statement 1 denies salt tolerance/pneumatophores. Statement 3 confuses Sundarbans mangroves with Lakshadweep coral atolls.

Q6. Arrange MISHTI year trap:

A. 2019 mangrove mission B. Union Budget 2023–24 mangrove push C. 1972 Stockholm statute D. 2006 FRA title

💡 **Show answer**

Ans: B. Budget 2023–24. **Logic:** Not 2019.

Q7. Which of the following coral belts is/are correctly listed for India?

1. Gulf of Mannar
2. Lakshadweep
3. Andaman & Nicobar
4. Gulf of Kutch

A. 1 and 2 only B. 1, 2 and 3 only C. 1, 2, 3 and 4 D. 2, 3 and 4 only

💡 **Show answer**

Ans: C. All four. **Logic:** Standard India coral set.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Coral bleaching is linked to heat stress expelling zooxanthellae.

Reason (R): Corals are built from calcium-carbonate skeletons with symbiotic zooxanthellae.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R supplies the biology behind bleaching. **A/R logic:** Symbiosis + heat stress.

Q9. With reference to UP Ramsar examples, which of the following statements is/are correct?

1. Sarsai Nawar, Samaspur, Nawabganj, Upper Ganga, Bakhira, Haiderpur appear in teaching lists.
2. Sultanpur is a UP Ramsar site.
3. World Wetlands Day is 2 February.

A. 1, 2 and 3 B. 1 and 2 only C. 2 and 3 only D. 1 and 3 only

💡 **Show answer**

Ans: D. Only 1 and 3. **Logic:** Sultanpur = Haryana.

Q10. Match List-I with List-II:

List-I	List-II
A. Gangetic dolphin	1. High Himalaya
B. Snow leopard	2. Freshwater river dolphin
C. Barasingha	3. Dudhwa Terai teaching

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** River dolphin / Himalaya / Dudhwa.

Q11. Arrange IUCN nine categories memory idea:

A. Only three categories exist B. EX EW CR EN VU NT LC DD NE C. Only Threatened = all nine equally D. Only LC and NE

💡 **Show answer**

Ans: B. Nine categories including EX…NE. **Logic:** Full IUCN set.

Q12. With reference to grasslands, which of the following statements is/are correct?

1. Grass-dominated; maintained by fire and grazing.
2. UP Terai is a foothill grassland–forest mosaic.
3. Dudhwa holds barasingha teaching.

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: C. All three. **Logic:** Terai grassland facts.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Marine neritic shelf waters are the highest fisheries-productivity zone teaching.

Reason (R): Pelagic means open ocean.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; pelagic definition does not explain neritic productivity. **A/R logic:** Neritic vs pelagic.

Q14. Which of the following pairs is/are NOT correctly matched?

1. Himalayan vegetation — altitude belts tropical→…→nival
2. India's coast — about 7516 km teaching
3. Gangetic dolphin — marine open-ocean whale

A. 1 and 2 only B. 1, 2 and 3 C. 2 and 3 only D. 3 only

💡 **Show answer**

Ans: D. Only pair 3 is not correctly matched. **Logic:** Freshwater river dolphin.

Q15. How many of the following UP negative facts are correct?

Rudrasagar≠UP; Sultanpur≠UP; dolphin≠marine open ocean

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Trap trio.

Q16. With reference to Ramsar 1971, which of the following statements is/are correct?

1. Wetlands of international importance.
2. Wise Use idea.
3. Identical to CITES Appendices.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Ramsar ≠ CITES.

Q17. Match List-I with List-II:

List-I	List-II
A. MISHTI	1. 2 February
B. Wetlands Day	2. Budget 2023–24
C. Coral bleaching	3. Warming / zooxanthellae loss

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** MISHTI/Wetlands Day/bleaching.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): IUCN assesses extinction risk.

Reason (R): CITES regulates international wildlife trade — different institutions, different jobs.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R clarifies the separation. **A/R logic:** Risk vs trade.

Q19. Which one of the following correctly states LC on IUCN?

A. Not Evaluated identical to CR B. Critically Endangered C. Extinct in Wild only D. Least Concern

💡 **Show answer**

Ans: D. Least Concern. **Logic:** LC ≠ CR.

Q20. With reference to altitude belts, which of the following statements is/are correct?

1. Tropical → subtropical → temperate → alpine → nival.
2. Snow leopard is high Himalaya.
3. Snow leopard is a Dudhwa Terai plains endemic only.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not Dudhwa plains endemic.

Q21. Arrange Wetlands Day / Wildlife Day trap:

A. Both 22 March B. Wetlands 2 Feb; Wildlife 3 March C. Both 5 June D. Both 16 September

💡 **Show answer**

Ans: B. 2 Feb / 3 March. **Logic:** Do not put Wildlife on 22 March.

Q22. Which of the following statements about pneumatophores is/are correct?

1. Associated with mangroves.
2. Help in gaseous exchange in waterlogged soils.
3. Prove a site is a desert dune only.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Mangrove adaptation.

Q23. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): India's coast length is often taught near 7516 km.

Reason (R): Neritic waters over the shelf are high fisheries-productivity zones.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; productivity does not explain the coast-length figure. **A/R logic:** Two marine facts.

Q24. How many of the following are IUCN Threatened tiers?

VU, EN, CR, LC

A. Only one B. Only two C. All four D. Only three

💡 **Show answer**

Ans: D. Only three. **Logic:** LC is not Threatened.

Q25. With reference to Haiderpur/Bakhira teaching, which of the following statements is/are correct?

1. They appear among UP Ramsar examples.
2. They prove UP has marine national parks.
3. They are wetland sites.

A. 1 and 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 3. **Logic:** UP landlocked — no marine NPs.

Q26. Which of the following pairs is/are correctly matched?

1. EX — Extinct
2. EW — Extinct in the Wild
3. DD — Data Deficient

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

💡 **Show answer**

Ans: B. All three. **Logic:** Category expansions.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): MISHTI should not be dated as 2019.

Reason (R): It is a Budget 2023–24 mangrove initiative.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the year trap. **A/R logic:** 2023–24 not 2019.

Q28. With reference to viviparous mangrove seeds, which of the following statements is/are correct?

1. Seeds may germinate while still on the parent plant teaching.
2. Helps establishment in tidal habitats.
3. Means mangroves are identical to alpine conifers only.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Not alpine conifers.

Q29. Arrange risk severity among CR/EN/VU:

A. VU > EN > CR B. All equal always C. EN > CR > VU D. CR > EN > VU

💡 **Show answer**

Ans: D. CR > EN > VU. **Logic:** Living threatened order.

Q30. Which one of the following correctly separates IUCN and CITES?

A. IUCN risk assessment; CITES trade regulation B. Same job; same appendices only C. Both set India's NDCs D. Both publish ISFR

💡 **Show answer**

Ans: A. Risk vs trade. **Logic:** Different jobs.

Common Traps

Trap	Correct
Threatened = Endangered only	**VU + EN + CR**
Red Data Book = IUCN Red List	**National vs global**
BSI = animal Red Data Book	**BSI = plants; ZSI = animals**
IUCN = CITES	**Assessment vs trade**
Sultanpur Ramsar in UP	**Haryana (Gurugram)**
Rudrasagar Ramsar in UP	**Tripura**
MISHTI launched 2019	**2023–24 Budget**
Coral bleaching from cooling	**Heat (+2°C) stress**
Mangroves = freshwater wetland only	**Coastal saline**
Gangetic dolphin = marine	**Freshwater river**
Zoo/sanctuary same as wetland	Different ecosystems
EW = EX	**EW = captive survival remains**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 22 — Renewable Energy (नवीकरणीय ऊर्जा)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Solar Energy | Wind Energy | Biomass Energy (बायोमास ऊर्जा) | Geothermal Energy (भू-तापीय ऊर्जा) | Tidal Energy | Small Hydro Projects | Green Hydrogen

Sources baked in: NCERT Geo 11–12; MNRE; NAPCC; National Solar Mission; Green Hydrogen Mission 2023; ISA; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — biogas composition; ISA traps; Himalayan hydro; green vs grey hydrogen **Last verified:** August 2026

Current Affairs (this topic)

- India's renewable capacity and green-hydrogen headlines refresh energy CA.
- Keep renewable vs clean vs conventional classification traps stable.

💡 🎯 **Consolidated — 34 Must-Score Facts (Click to Expand)**

1. **Renewable** energy replenishes on a human timescale — sun, wind, water, biomass, geothermal. **Nuclear** is low-carbon but usually **not** counted as renewable MCQs. 2. **Solar PV** turns sunlight directly into electricity. **CSP** (सीएसपी) uses mirrors to make heat, then steam and turbines. 3. The **National Solar Mission** began in **2010** under NAPCC (JNNSM from Jan 2010). India chased about **100 GW** solar by 2022 and **500 GW** non-fossil capacity by **2030**. 4. Solar has the **greatest RE potential** in India among common options and is the most eco-friendly among solar cell vs fossils/nuclear fission options. 5. **Bhadla** in Rajasthan is India's mega solar-park fact (not Gujarat as "largest"). **PM-KUSUM** supports farmer solar pumps. **OSOWOG** is One Sun One World One Grid. 6. **Domestic Content Requirement (DCR)** in solar news means mandating domestically manufactured PV cells/modules. 7. The **International Solar Alliance** began in **2015** (India–France, COP21). Headquarters is **Gurugram**. Not all UN members — originally sun-belt countries between Tropics. First Summit **New Delhi, March 2018**. Members target mobilising about **USD 1 trillion** for solar by **2030**. 8. **Cochin International Airport** = world's first fully solar-powered airport. **Diu** = first Indian UT / smart city running **100% solar** (daytime / UT stems). **Surat** = first district with **100% solar-powered PHCs**. 9. Wind power converts kinetic energy of air; power scales roughly with the **cube** of wind speed. Leading states include **Tamil Nadu (नाडु), Gujarat, and Karnataka**. 10. **Biomass** is energy from organic matter. **Biogas** (बायोगैस) from anaerobic digestion is mainly **methane (CH_4) and carbon dioxide (CO_2) (कार्बन डाइऑक्साइड)** — not a butane/carbon-monoxide pair. 11. Bagasse cogeneration, **GOBAR-Dhan**, and **SATAT** (bio-CNG) are biomass-policy neighbours. 12. **Geothermal** uses Earth's internal heat. Indian provinces include **Himalaya (हिमालय), Cambay, West Coast, SONATA (Son–Narmada (नर्मदा)–Tapi (तापी)), Godavari (गोदावरी), Mahanadi (महानदी)** — **not Ganga (गंगा) Delta (डेल्टा)**. Sites: **Puga, Manikaran, Tattapani**. 13. **Tidal** energy comes from Moon-driven sea-level change. India's best potential is often locked to the **Gulf of Khambhat**, then Gulf of Kutch. 14. **Small hydro** under MNRE is **≤25 MW**. Above that is treated as large hydro. Himalayan streams hold large SHP potential. 15. The **Ramganga (रामगंगा) Project** in Garhwal (गढ़वाल), Uttarakhand (उत्तराखंड), is about **198 MW** — a hydro fact that also matters for irrigation links toward UP. 16. **Green hydrogen** is H_2 from water electrolysis powered by **renewable** electricity. Grey hydrogen comes from natural gas; blue adds CCS (सुरक्षा कैबिनेट समिति) to gas routes. 17. The **National Green Hydrogen Mission** (January **2023**) targets

about **5 MMT** per year by **2030**. National Hydrogen Energy Board dates to **3 Oct 2003**. 18. Hydrogen can power **IC engines**, be **blended with natural gas**, and run **fuel-cell vehicles** — all three are correct in recent stems. 19. Fuel cells with pure H_2 emit **heat and water**; they produce **DC**; they can power buildings and portable devices. 20. Solar ≠ wind ≠ tidal ≠ geothermal. Source and physics differ even when all are “clean.” 21. ISA HQ is Gurugram, not Paris and not New Delhi as a casual guess. 22. Biogas major pair is CH_4 + CO_2. Wrong gas pairs are frequent traps. 23. Geothermal and nuclear are not “derived from the Sun” in the renewable-sun family sense used in many stems — though fossil-fuel energy ultimately traces to the Sun. 24. India still lacks a large commercial tidal plant even where potential maps look strong. 25. Green / grey / blue hydrogen colours mark production route and carbon intensity — not the colour of the gas. 26. UP colour: Mirzapur (मिर्ज़ापुर) solar collaboration history (इतिहास), western UP bagasse cogeneration, limited wind, Ramganga water–power links. 27. CSP is heat-first; PV is photon-to-electron. Do not describe CSP as a simple rooftop panel. 28. Wind leadership is southern and western coastal/plateau states — not a default UP fact. 29. Small hydro ceiling **25 MW** is the MNRE number; do not invent a different cutoff. 30. OSOWOG is a grid-connectivity vision around solar; it is not a biomass scheme. 31. Non-fossil **500 GW by 2030** is a capacity ambition fact beside older Solar Mission milestones. 32. Tariffs for solar are set by **CERC / SERCs**, not by SECI alone; India is **not** a top silicon-wafer manufacturer (panels often assembled from imported cells). 33. Microbial fuel cells and bioasphalt appear as advanced sustainable-energy neighbours (wastewater electricity; non-petroleum road binder). 34. Algae biofuel limits for developing countries: high expertise + large facilities with ecological/social concerns — **not** “seas only.”

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Solar Photovoltaic (PV)	**Concentrated Solar Power (CSP)**	Semiconductor solar cells that convert incident sunlight photon energy directly into direct-current electricity via the photoelectric effect vs optical mirrors/lenses that concentrate solar thermal energy to heat a fluid and drive a steam turbine	सौर फोटोवोल्टिक (सीधा विद्युत उत्पादन) / संकेंद्रित सौर ताप (CSP, टरबाइन चालित)
Green Hydrogen	**Grey Hydrogen / Blue Hydrogen**	Produced by zero-carbon electrolysis of water powered entirely by renewable electricity (solar/wind) vs produced via steam methane reforming of natural gas releasing unabated CO_2 (Grey) or with carbon capture and storage (Blue)	हरित हाइड्रोजन (शून्य कार्बन जल अपघटन) / धूसर व नीला हाइड्रोजन (जीवाश्म स्रोत)
Biomass Energy	**Biogas**	Direct thermal or electrical energy produced by combusting solid organic matter (bagasse, rice husk, wood pellets) vs gaseous fuel mixture consisting primarily of methane (50–70%) and CO_2 generated through anaerobic microbial digestion of organic waste	बायोमास ऊर्जा (ठोस जैविक ईंधन) / बायोगैस (अवायवीय पाचन, मीथेन प्रधान)
Geothermal Energy	**Ocean Thermal Energy (OTEC)**	Harnesses natural high-temperature hydrothermal heat and steam from Earth's deep crustal magma reservoirs (e.g. Puga Valley, Ladakh; Manikaran, HP) vs exploits the temperature differential between warm tropical surface ocean water and freezing deep seawater (≥20°C gradient)	भू-तापीय ऊर्जा (पृथ्वी का आंतरिक ताप) / महासागरीय तापीय ऊर्जा (OTEC, सतही-गहन ताप भेद)
Small Hydro Power (SHP)	**Large Hydro Power**	Hydropower installations with individual installed capacity up to and including **25 MW** administered under the Ministry of New and Renewable Energy (MNRE) vs large dam-based hydropower plants >25 MW administered under the Ministry of Power	लघु जल विद्युत (≤25 MW, MNRE) / बड़ी जल विद्युत परियोजनाएं (>25 MW)
International Solar Alliance (ISA)	**International Renewable Energy Agency (IRENA)**	Treaty-based intergovernmental solar coalition initiated by India and France at COP21 Paris with global HQ in Gurugram, India vs global renewable energy intergovernmental agency covering all renewables founded in 2009 with HQ in Abu Dhabi, UAE	अंतरराष्ट्रीय सौर गठबंधन (ISA, गुरुग्राम) / इरेना (IRENA, अबू धाबी)

💡 📋 Must-score facts — solar, wind, biomass, tidal (Click to Expand)

Solar / wind

Item	Lock
NSM / JNNSM	**2010** under NAPCC
Targets class	~100 GW solar by 2022; **500 GW** non-fossil by 2030
Bhadla	Rajasthan mega solar park
ISA	**2015** India–France COP21; HQ **Gurugram**
PM-KUSUM / OSOWOG	Farmer solar pumps / One Sun One World

One Grid | | Wind leaders | TN, Gujarat (power ∝ speed^3) | | Biogas majors | **CH_4 + CO_2** (≠ butane/CO) | | Tidal best India | Gulf of Khambhat → Gulf of Kutch | | SHP (MNRE) | ≤**25 MW** |

22.1 Solar Energy

Identity: PV + CSP | National Solar Mission 2010 | ISA Gurugram 2015 | 500 GW non-fossil 2030

- **Solar energy** is radiant light and heat from the Sun — harnessed via **photovoltaic (PV)** panels or **concentrated solar power (CSP)** (संकेंद्रित सौर ताप).
- **PV cells** (silicon semiconductors) convert photons to DC electricity; an inverter converts to AC for the grid.
- **CSP** uses mirrors to heat fluid, produce steam, and drive a turbine — can store heat for after-sunset generation.
- Highest insolation (सूर्यताप) in **Rajasthan, Gujarat, Madhya Pradesh, Maharashtra**. **Bhadla Solar Park (Rajasthan)** is among the world's largest.
- **National Solar Mission (2010)** is a pillar of **NAPCC (2008)** — also called **JNNSM** from January 2010 — target scaled to **100 GW solar**, achieved around **2022–2024**.
- Solar has the **greatest renewable potential** in India among common options; solar cells are the most eco-friendly among petroleum / forest products / nuclear fission / solar options.
- **500 GW non-fossil fuel capacity by 2030** (COP26, 2021) includes solar, wind, hydro, nuclear, and biomass — solar remains the largest growth component.
- **International Solar Alliance (ISA)** (अंतरराष्ट्रीय सौर गठबंधन) launched **30 November 2015** at **COP21 Paris** by **India and France** — treaty-based IGO, HQ **Gurugram, Haryana**.
- ISA membership is **not** "all UN members" — historically focused on countries between the Tropics of Cancer and Capricorn (membership has since widened). First Summit: **New Delhi, 11 March 2018**. Declaration ambition: mobilise about **USD 1 trillion** for solar by **2030**.
- **Domestic Content Requirement (DCR)** in solar RfS means using domestically manufactured PV cells and modules.
- Silicon wafers / cells are largely imported or assembled — India is **not** among the top silicon-wafer manufacturers; solar tariffs are regulated by **CERC/SERCs**, not set solely by SECI.
- **OSOWOG (One Sun One World One Grid)** and **World Solar Bank** concepts are linked to ISA's cross-border solar grid vision.
- **PM-KUSUM (2019):** solar pumps for farmers and grid-connected solar on barren/fallow land — MNRE nodal.
- **Cochin International Airport** was the **first airport in the world** fully powered by solar energy.
- **Diu** became the first Indian UT / smart-city example to run on **100% solar** (daytime / UT stems). **Surat** became the first district with **100% solar-powered Primary Health Centres**.
- **Bhadla (Rajasthan)** is India's mega solar park fact — not "Gujarat has the largest solar park" in newer stems. India's large floating solar example: **Ramagundam (Telangana)**.

PYQ — UPPCS Prelims 2025, Q109

ISA: (1) First treaty-based IGO in India with 38 member countries. (2) World Solar Bank and OSOWOG under ISA.

A. Only 2

B. Neither

C. Both

D. Only 1

💡 **Show answer**

Ans: A — OSOWOG/World Solar Bank linked to ISA is correct. **"38 members" is outdated** — signatories far exceed 38.

Scheme	Year	Fact
National Solar Mission	2010	NAPCC pillar / JNNSM
ISA	2015	HQ Gurugram; Summit 2018 Delhi
PM-KUSUM	2019	Solar pumps
PLI Solar PV	2021	Domestic manufacturing

> **Logic:** ISA HQ = **Gurugram** — trap: New Delhi or Paris. DCR = domestic solar manufacturing.

22.2 Wind Energy

Identity: Kinetic air → turbines | Power ∝ v^3 | Tamil Nadu leader | Onshore dominant

- **Wind energy** converts the **kinetic energy of moving air** into electricity using **wind turbines**.
- Power output depends on **wind speed cubed (v^3)** — doubling wind speed gives roughly **8× power**.
- Utility-scale sites need consistent annual average wind **>6 m/s** — coastal and open plateau areas preferred.
- **Tamil Nadu** is historically India's leading wind state (**Muppandal wind farm**). **Gujarat, Karnataka, Maharashtra, Rajasthan** follow.
- **Onshore** turbines dominate India. **Offshore wind** potential on **Gujarat and Tamil Nadu coasts** — stronger winds but higher cost.
- **Intermittency:** wind is variable — needs **grid integration**, **hybrid wind+solar**, or **storage** (batteries, pumped hydro).
- **Capacity factor:** wind farms typically produce **~25–35%** of nameplate capacity annually — location-dependent.
- **UP has limited wind potential** compared to western and coastal states — focus is national leaders.

Feature	Onshore	Offshore
Cost	Lower	Higher
Wind quality	Moderate (नरम दल)	Stronger, steadier
India status	Mature	Early policy stage

> **Logic:** Rajasthan leads **solar**, not wind. **Tamil Nadu** = wind leader trap.

22.3 Biomass Energy

Identity: Organic matter → energy | Biogas = CH_4 + CO_2 | Bagasse UP sugar belt | Anaerobic digestion

- **Biomass energy** is stored **chemical energy in organic matter** — plants, crop residue, animal waste, wood — released by burning, fermentation, or **anaerobic digestion**.
- **Anaerobic digestion:** bacteria break organic matter **without oxygen** in a digester → **biogas** + **slurry** (organic fertiliser).
- **Biogas composition:Methane (CH_4) ~50–70%** + **Carbon dioxide (CO_2) ~25–45%** + traces of H_2S and moisture.

- **Butane** is an LPG component — **not** a major biogas constituent. **Carbon monoxide** is incomplete combustion product — not main biogas component.
- **Bagasse** (sugarcane crush residue) is burned in **UP sugar mills** (Meerut (मेरठ), Muzaffarnagar belt) for **cogeneration** — electricity + process heat.
- **GOBAR-Dhan / SATAT:** government push for community biogas and **bio-CNG** — oil marketing companies procure compressed biogas.
- **E20 ethanol blending target by 2025** — ethanol from sugarcane, maize, and agri-waste reduces petrol imports.
- **Stubble burning** wastes biomass and causes **PM2.5 pollution** — not sustainable energy use.

PYQ — UPPCS Prelims 2025, Q28

Major constituents of biogas: (1) Methane (2) Butane (3) Carbon dioxide (4) Carbon monoxide

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Ans: A — **Methane + Carbon dioxide**. Butane is the trap.

> **Logic:** Biogas = **CH_4 + CO_2** — not butane.

22.4 Geothermal Energy

Identity: Earth's interior heat | Puga Ladakh (लद्दाख) | Baseload potential | NOT from Sun

- **Geothermal energy** harnesses **heat from Earth's interior** — hot rocks, steam, or underground water — for electricity or direct heating.
- Heat sources: **radioactive decay** of minerals + residual heat from Earth's formation — temperature rises **~25–30°C per km** depth.
- Wells tap hot reservoirs → steam/hot water spins **turbine** — or **direct use** for heating (greenhouses, spas).
- **India's prospects / provinces:** Himalayas, Cambay, West Coast, **SONATA (Son–Narmada–Tapi)**, Godavari, Mahanadi, Sohana — **Ganga Delta is not** a listed geothermal province.
- Key sites: **Puga Valley (Ladakh)** — highest potential; **Manikaran (HP)**; **Tattapani (Chhattisgarh)**; **Cambay Basin (Gujarat)**.
- **No large-scale commercial geothermal power plant** operational in India yet — pilot exploration only (GSI/MNRE).
- **Advantages:baseload renewable** — can run 24/7 unlike solar/wind at a suitable site.
- **Global leaders:** Iceland, USA, Indonesia, Philippines — India remains early stage.
- **NOT derived from the Sun** in the "which is not from the Sun" MCQ sense — unlike biomass, wind, or solar.

> **Logic:** Geothermal = **Earth heat**, not sunlight. Iceland benchmark ≠ India primary source.

22.5 Tidal Energy

Identity: Moon's gravity | Khambhat highest range | Predictable | No major Indian plant yet

- **Tidal energy** harnesses the **rise and fall of sea levels** caused primarily by **gravitational pull of the Moon** (and Sun).
- **Tidal barrage:** dam (दाम -) across estuary (ज्वारनदमुख) — gates allow water in at high tide, release through **turbines** at low tide.
- **Tidal stream turbines:** underwater turbines in fast tidal channels — no large dam required.
- **India's potential:Gulf of Khambhat (Cambay)** — tidal range up to **~11 m**, highest in India. **Gulf of Kutch** is second.
- **No major commercial tidal power plant** operational in India — papers test **potential**, not installed leadership.
- **Advantages:highly predictable** (lunar calendar) — unlike variable wind/solar.
- **Disadvantages:** high capital cost, estuary ecosystem (पारिस्थितिकी तंत्र) disruption, siltation in Khambhat.
- **Tidal ≠ wave energy:** tidal = gravity-driven sea level change; wave = wind-driven surface waves.

PYQ — UPPCS Prelims 2022

Largest tidal energy producing area in India:

A. North Circars

B. Mannar

C. Khambhat Coast

D. Kerala

💡 **Show answer**

Ans: C — **Gulf of Khambhat (Cambay)** has the highest tidal range.

> **Logic:** Tidal potential = **Gujarat coast** (Khambhat/Kutch), not Thar (थार) Desert.

22.6 Small Hydro Projects

Identity: ≤25 MW MNRE | Himalayan potential | Ramganga 198 MW Garhwal | Run-of-river vs storage

- **Hydropower** converts **potential/kinetic energy of falling water** into electricity — renewable, rainfall-dependent.
- **MNRE definition:Small hydro project (SHP) = ≤25 MW**. Above 25 MW = **large hydro**.
- **Run-of-river:** minimal large reservoir — lower submergence, preferred in sensitive Himalaya.
- **Storage/pondage:** dam + reservoir (Tehri (टिहरी), Bhakra model) — higher head but displacement/seismic risks.
- **Himalayan rivers** have **enormous hydroelectric potential** — steep gradient, snow-fed perennial flow.
- **Ramganga Multipurpose Project:Garhwal (Uttarakhand)**, dam on Ramganga tributary of Ganga, **~198 MW** hydropower — irrigates western **UP** plains.
- **Pumped storage hydro:** uses off-peak electricity to pump water uphill, releases at peak demand — grid balancing with solar/wind.

PYQ — UPPCS Prelims 2025, Q31

Hydroelectric power in Northern India: (1) Himalayan region has enormous hydroelectric potential. (2) Eastern UP and Bihar cannot utilize this because Himalayas to their north lie in Nepal.

A. Only 2

B. Neither

C. Both

D. Only 1

> 💡 **Show answer**
>
> **Ans: D** — Stmt 1 true. Stmt 2 false — major Indian hydro in **Uttarakhand, HP, Sikkim, Arunachal**; UP benefits via Ganga/Yamuna (यमुना) systems.

PYQ — UPPCS Prelims 2024, Q72

Ramganga Project — which pair is NOT correctly matched?

Options include: Dam length 825.8 m | Hydropower **198 MW** | Location Garhwal | Irrigated area 6 lakh ha

> 💡 **Show answer**
>
> **Verify all four (चातुर्याम) stats** — **198 MW** and **Garhwal** (गढ़वाल) are standard correct facts. Irrigated area figure is the common mismatch trap in options.

Category	Capacity
Small hydro (SHP)	≤25 MW
Large hydro	>25 MW
Ramganga	~198 MW (large)

> **Logic:** SHP cap = **25 MW**. Singrauli = **thermal**, not hydro.

22.7 Green Hydrogen

Identity: Electrolysis + renewable power | Mission 2023 | 5 MMT/yr 2030 | Green vs grey vs blue

- **Hydrogen (H_2)** is an **energy carrier** — must be produced from water, gas, or biomass. Burns to water vapour with no carbon at point of use.
- **Green hydrogen:electrolyser** splits **H_2O → H_2 + O_2** powered by **renewable electricity** (solar/wind/hydro) — near-zero lifecycle CO_2.
- **Grey hydrogen:** ~98% of current global production — **natural gas steam reforming** — releases large **CO_2**. **NOT green.**
- **Blue hydrogen:** grey production + **carbon capture and storage (CCS)** — lower emissions, still fossil-dependent.
- **Black/brown hydrogen:** from **coal gasification** — very high emissions.
- **National Green Hydrogen Mission (January 2023):** target **5 MMT green hydrogen per annum by 2030** — MNRE coordinates.
- **National Hydrogen Energy Board** was set up on **3 October 2003** to promote hydrogen energy (older bank fact; Mission 2023 is the current flagship).
- Green hydrogen can be used **directly in internal combustion**, **blended with natural gas** for heat/power, and in **hydrogen fuel cells** for vehicles — all three statements are correct in recent stems.
- **Fuel cells:** electrochemical conversion of H_2 + O_2 → electricity; by-products **heat and water**; generate **Direct Current (DC)**; usable for buildings **and** portable devices (laptops etc.).
- **Microbial fuel cells (MFC):** living microbes as catalysts convert organic substrates to electricity; attractive in **wastewater treatment** (clean water + power). Substrates are mainly organic (glucose, etc.), not "inorganic-only."
- **Sectors:** refineries, **ammonia/fertiliser (urea)**, **green steel**, heavy transport, shipping.

- **Challenges:** high cost vs grey H_2, energy conversion losses, storage/transport infrastructure (compress, liquefy, pipeline).
- **Algae biofuels:** possible on continents as well as seas; developing-country limits = high expertise + large facilities with ecological/social concerns.
- **Bioasphalt:** non-petroleum renewable binder for roads — lowers fossil dependence and can reduce surface temperature versus conventional asphalt.

Type	Source	CO_2
Green	Renewable electrolysis	Very low
Grey	Natural gas	High
Blue	Gas + CCS	Moderate

> **Logic:Green = renewable electrolysis only.** Mission launched **2023**, not 2019. Fuel cell output = **DC**.

Complete PYQ Bank (Topic 22)

Q1. UPPCS Prelims 2025, Q28

Major constituents of biogas include:

1. Methane
2. Butane
3. Carbon dioxide
4. Carbon monoxide

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Logic: Biogas from anaerobic digestion — CH_4 + CO_2; butane is an LPG trap. **Ans: A (1 and 3).** **Methane and carbon dioxide** are the major biogas constituents. **Trap:** **Butane** or **carbon monoxide** as main components.

Q2. UPPCS Prelims 2025, Q31

With reference to hydroelectric power in Northern India, consider the following statements:

1. The Himalayan region has enormous hydroelectric potential.
2. Eastern Uttar Pradesh and Bihar cannot utilise Himalayan hydro because the Himalayas to their north lie in Nepal.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: Himalayan potential is real; statement 2 falsely excuses UP/Bihar — Indian hydro sits in Uttarakhand, HP, Sikkim, etc. **Ans: D (Only 1).** **Himalayan potential is enormous**; the Nepal excuse in statement 2 is false. **Trap:** Accepting statement 2 — UP benefits via Ganga/Yamuna systems from Indian Himalayan projects.

Q3. UPPCS Prelims 2025, Q109

With reference to the International Solar Alliance (ISA), consider the following statements:

1. It is the first treaty-based intergovernmental organization headquartered in India and has 38 member countries.
2. The World Solar Bank and One Sun One World One Grid (OSOWOG) initiatives have been undertaken under it.

A. Only 1

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 2

💡 **Show answer**

Logic: OSOWOG/World Solar Bank linkage to ISA is correct; "exactly 38 members" is outdated. **Ans: A (Only 2).** **Only statement 2 is correct** — member count in statement 1 is obsolete. **Trap:** Rejecting OSOWOG because statement 1 sounds authoritative.

Q4. UPPCS Prelims 2024, Q72

With reference to the Ramganga Multipurpose Project, which of the following pairs is **correctly** matched?

A. Location — Garhwal (Uttarakhand)

B. Hydropower capacity — 198 MW

C. Both A and B

D. Neither A nor B

💡 **Show answer**

Logic: Ramganga teaching facts — Garhwal site and ~198 MW are standard correct anchors. **Ans: C.** **Garhwal location** and **~198 MW** are both correctly matched. **Trap:** Irrigated-area or dam-length figures in other booklet options — verify against ~198 MW + Garhwal core pair.

Q5. UPPCS Prelims 2022, Q20

The largest tidal energy potential in India is associated with the:

A. North Circars coast

B. Mannar coast

C. Khambhat (Cambay) coast

D. Kerala coast

💡 **Show answer**

Logic: Gulf of Khambhat has India's highest tidal range (~11 m) — potential, not installed capacity leader. **Ans: C.** **Gulf of Khambhat (Cambay)** has the largest tidal energy potential. **Trap:** Kerala or eastern coast names — Gujarat/Khambhat is the tested fact.

Q6. UPPCS Prelims 2022, Q69

Which of the following is **NOT** a biomass energy source?

A. Coal

B. Nuclear fuel

C. Gobar (cow dung)

D. Wood

💡 **Show answer**

Logic: Biomass = organic matter; coal is fossilised carbon; nuclear is uranium/thorium. **Ans: A.** **Coal is not biomass** — it is a fossil fuel. **Trap:** Nuclear is also non-biomass, but coal is the standard keyed distractor in this stem.

Q7. UPPCS Prelims 2022, Q132

Solar energy is converted into electric energy through—

A. Dry cells

B. Leclanché cells

C. Voltaic cells

D. Photovoltaic cells

💡 **Show answer**

Logic: PV direct conversion vs CSP/tidal/geothermal distractors. **Ans: D.** **Photovoltaic cells** convert solar energy into electricity. **Trap:** Dry cells, Leclanché, or Voltaic cell distractors — not solar-specific.

Q8. UPPCS Prelims 2022, Q139

Most sources of energy we use represent stored solar energy. Which of the following is **NOT** ultimately derived from the Sun's energy?

A. Biomass energy

B. Nuclear energy

C. Wind energy

D. Geothermal energy

💡 **Show answer**

Logic: Solar-linked = biomass, wind; Earth's interior heat and nuclear fission are non-solar. **Ans: D.** **Geothermal energy** is not derived from the Sun. **Trap:** **Nuclear energy** is also non-solar — paper may list either; geothermal is the keyed answer here.

Q9. UPPCS Prelims 2022, Q27

The first airport powered by solar energy in India is:

A. Chennai

B. Cochin

C. Ahmedabad

D. New Delhi

💡 **Show answer**

Logic: Cochin International Airport = world's first fully solar-powered airport fact. **Ans: B.** **Cochin** was India's first solar-powered airport. **Trap:** **Diu** (100% solar UT) or **Ahmedabad** — different solar milestones.

Q10. UPPCS Prelims 2020, Q46

In which district of Uttar Pradesh was a major solar power plant established in collaboration with France in 2018?

A. Agra

B. Ballia

C. Deoria

D. Mirzapur

💡 **Show answer**

Logic: UP–France solar collaboration maps to Mirzapur belt in standard PYQ framing. **Ans: D.** The major UP–France solar plant is linked with **Mirzapur**. **Trap:** Eastern UP districts (Ballia/Deoria) with no solar–France association.

Q11. UPPCS Prelims 2019, Q63

The Secretariat of the International Solar Alliance (ISA) is located at:

A. Paris

B. New Delhi

C. Gurugram

D. Nairobi

💡 **Show answer**

Logic: ISA HQ Gurugram — not Paris (launch city) or generic New Delhi. **Ans: C.** **ISA Secretariat is at Gurugram, Haryana**. **Trap:** **Paris** (COP21 launch) or **New Delhi** casual guess.

Q12. UPPCS Prelims pattern — National Solar Mission

The National Solar Mission is a pillar of which national climate/action framework?

A. NAPCC (National Action Plan on Climate Change)

B. Montreal Protocol

C. Wildlife (Protection) Act, 1972

D. Ramsar Convention

💡 **Show answer**

Logic: JNNSM/National Solar Mission (2010) sits under NAPCC (2008). **Ans: A.** The **National Solar Mission is under NAPCC**. **Trap:** Montreal (ozone) or WPA (wildlife) cross-topic distractors.

Extra Drill — Solar, Hydrogen & Geothermal (Q31–Q40)

Q31. International Solar Alliance — which is correct?

A. Launched at UN Climate Change Conference 2015; includes all UN members

B. Launched 2015; not all UN members; HQ Gurugram

C. HQ Paris; launched 2010

D. First Summit only in New York

💡 **Show answer**

Logic: ISA membership, DCR, firsts (Kochi/Diu), and hydrogen/fuel-cell physics dominate recent stems.
Ans: B. First Summit New Delhi **2018**. USD ~1 trillion solar finance ambition by **2030**.

Q32. Domestic Content Requirement (DCR) refers to:

A. Foreign TV channel licences

B. Developing domestic solar PV cell/module manufacturing

C. Food export quotas only

D. Foreign university campuses

💡 **Show answer**

Ans: B. MNRE solar RfS manufacturing condition.

Q33. Correct solar "firsts" set:

A. Kochi airport fully solar; Diu 100% solar UT/daytime; Surat 100% solar PHCs

B. Gujarat always has India's largest solar park (vs Bhadla)

C. India is #3 silicon-wafer manufacturer

D. SECI alone sets all solar tariffs

💡 **Show answer**

Ans: A. Bhadla (Rajasthan) mega park; CERC/SERCs regulate tariffs; wafers largely not Indian-made.

Q34. Geothermal energy sources have **not** been found in which area among options?

A. Godavari Delta

B. Ganga Delta

C. Himalayas

D. West Coast

💡 **Show answer**

Ans: B. Provinces include Himalaya, Cambay, West Coast, SONATA, Godavari, Mahanadi — not Ganga Delta.

Q35. Green hydrogen uses — how many are correct? 1. Direct fuel for IC engines 2. Blend with natural gas 3. Fuel-cell vehicles

A. Only one

B. Only two

C. All three

D. None

💡 **Show answer**

Ans: C. Mission **2023**; Hydrogen Energy Board **2003**.

Q36. Most eco-friendly among common options:

A. Petroleum products

B. Forest products

C. Nuclear fission

D. Solar cells

💡 **Show answer**

Ans: D. Solar has greatest RE potential in India among common options.

Q37. Pongamia pinnata is preferred for biodiesel because:

A. It grows only in Antarctica

B. Grows in arid regions; seeds rich in lipids (~half oleic acid)

C. It is a nuclear fuel crop

D. It produces only biogas

💡 **Show answer**

Ans: B. Pair with Jatropha in dual-statement stems.

Q38. Assertion (A): Coal-based thermal plants contribute to acid rain. Reason (R): Oxides of carbon are emitted when coal burns.

A. Both true; R explains A

B. Both true; R does not explain A

C. A false; R true

D. A true; R false

💡 **Show answer**

Ans: B. Acid rain driver is mainly **SO_2** from sulphur in coal — CO_2 emission is true but does not explain acid rain.

Q39. Microbial fuel cells — correct set:

A. Living microbes as catalysts; useful in wastewater plants; substrates mainly organic

B. Only inorganic substrates forever

C. Cannot cleanse wastewater

D. Same as nuclear reactors

💡 **Show answer**

Ans: A. "Inorganic-only substrates" is the false statement in classic stems.

Q40. Algae-based biofuel limits in developing countries:

A. Possible only in seas, never continents

B. High expertise/tech + large facilities may raise ecological/social concerns

C. No expertise needed

D. Always cheaper than petrol with zero land use

> 💡 **Show answer**
>
> **Ans: B.** Statement “seas only” is false.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to renewable energy, which of the following statements is/are correct?

1. Renewable energy excludes sun, wind, water, biomass and geothermal on a human timescale.
2. Nuclear is low-carbon but usually not counted as renewable in Indian prelims framing.
3. Coal is a renewable source.

A. Only 1 B. 1 and 2 C. 2 and 3 D. Only 2

> 💡 **Show answer**
>
> **Ans: D.** Only 2 is correct. **Logic:** Statement 2 is true (nuclear is low-carbon but usually not counted as renewable in Indian prelims framing). Statement 1 wrongly excludes the classic renewable sources. Statement 3 is false — coal is non-renewable.

Q2. Match List-I with List-II:

List-I	List-II
A. Solar PV	1. Mirrors → heat → steam
B. CSP	2. Direct sunlight → electricity
C. NSM/JNNSM	3. 2010 under NAPCC

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-1, C-3. **Logic:** PV/CSP/NSM 2010.

Q3. Which of the following pairs is/are correctly matched?

1. Bhadla — Rajasthan mega solar park
2. ISA — 2007 India–France COP21; HQ Gurugram
3. Bhadla — Gujarat as the only correct “largest” fact

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Ans: D.** Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): PM-KUSUM supports farmer solar pumps.

Reason (R): OSOWOG is One Sun One World One Grid.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; OSOWOG does not explain PM-KUSUM. **A/R logic:** Two solar policy facts.

Q5. With reference to targets, which of the following statements is/are correct?

1. ~100 GW solar by 2022 chase teaching.
2. 500 GW non-fossil capacity by 2030 teaching.
3. Solar has negligible RE potential in India among common options.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Solar has greatest RE potential teaching.

Q6. Arrange ISA facts:

A. HQ Nairobi; began 1972 B. Began 2015 COP21; HQ Gurugram C. HQ Rome; began 2006 D. HQ Geneva; began 1988

💡 **Show answer**

Ans: B. 2015; Gurugram. **Logic:** ISA fact.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Cochin airport — world's first fully solar-powered airport teaching
2. Diu — first Indian UT/smart city 100% solar daytime teaching
3. DCR — means only imported PV mandatory

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** DCR = domestic content mandate.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Wind power roughly scales with the cube of wind speed.

Reason (R): Leading wind states include Tamil Nadu and Gujarat teaching.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; state leadership does not explain the cube law. **A/R logic:** Physics vs geography.

Q9. With reference to biogas, which of the following statements is/are correct?

1. From anaerobic digestion.
2. Major constituents CH_4 + CO_2.

3. Major pair is butane + carbon monoxide only.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Not butane/CO major pair.

Q10. Match List-I with List-II:

List-I	List-II
A. Tidal	1. ≤25 MW MNRE
B. SHP	2. Moon-driven sea level
C. Geothermal India	3. Himalaya, Cambay, West Coast, SONATA...

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: A.** A-2, B-1, C-3. **Logic:** Tidal/SHP/geothermal.

Q11. Arrange tidal potential India teaching:

A. Only Ganga plains B. Gulf of Khambhat → Gulf of Kutch best potential teaching C. Only Ladakh lakes D. Only Chilika as tidal barrage capital

> 💡 **Show answer**
>
> **Ans: B.** Khambhat then Kutch. **Logic:** Tidal geography.

Q12. With reference to biomass policy neighbours, which of the following statements is/are correct?

1. Bagasse cogeneration.
2. GOBAR-Dhan.
3. SATAT (bio-CNG).

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

> 💡 **Show answer**
>
> **Ans: C.** All three. **Logic:** Biomass policy cluster.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Small hydro under MNRE is ≤25 MW.

Reason (R): Above that is treated as large hydro in Indian classification teaching.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains the SHP cut-off. **A/R logic:** ≤25 MW rule.

Q14. Which one of the following correctly states geothermal energy?

A. Only nuclear fission B. Only solar PV panels C. Only wind turbines D. Earth's internal heat

💡 **Show answer**

Ans: D. Earth's internal heat. **Logic:** Geothermal definition.

Q15. How many of the following are renewable in standard lists?

Sun, wind, biomass, geothermal, coal

A. Only four B. Only three C. Only two D. All five

💡 **Show answer**

Ans: A. Only four. **Logic:** Coal out.

Q16. With reference to OTEC, which of the following statements is/are correct?

1. Uses temperature difference between warm surface and cold deep water.
2. Not the same as tidal barrage power.
3. Identical to PM-KUSUM pumps.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** OTEC ≠ KUSUM.

Q17. Match List-I with List-II:

List-I	List-II
A. ISA	1. Farmer solar pumps
B. PM-KUSUM	2. Gurugram HQ
C. Bhadla	3. Rajasthan

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** ISA/KUSUM/Bhadla.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): DCR means mandating domestically manufactured PV cells/modules.

Reason (R): DCR means banning all solar generation in India.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Domestic content — not ban.

Q19. Which of the following pairs is/are correctly matched?

1. Wind leaders — TN, Gujarat teaching
2. Solar greatest RE potential among common options
3. Nuclear — always counted as renewable in Indian prelims

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Nuclear usually not renewable.

Q20. Arrange NAPCC solar mission start:

A. 2010 (JNNSM) B. 2023 only C. 1972 D. 2006

💡 **Show answer**

Ans: A. 2010. **Logic:** NSM/JNNSM.

Q21. With reference to ocean energy, which of the following statements is/are correct?

1. Wider set includes wave, OTEC, tidal.
2. Tidal is Moon-driven.
3. Ocean energy means only coal offshore platforms.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not coal platforms.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Himalayan streams hold large SHP potential teaching.

Reason (R): SHP under MNRE is ≤25 MW.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; MW cut-off does not explain Himalayan potential geography. **A/R logic:** Potential vs definition.

Q23. Which one of the following is correct about CSP?

A. Direct PV only without heat B. Only biogas digesters C. Concentrating Solar Power — mirrors to heat then turbines D. Only geothermal wells

💡 **Show answer**

Ans: C. Mirrors → heat → steam/turbines. **Logic:** CSP vs PV.

Q24. How many of the following ISA facts are correct?

2015 start; India–France COP21; HQ Gurugram

A. Only one B. Only two C. None D. All three

💡 **Show answer**

Ans: D. All three. **Logic:** ISA trio.

Q25. With reference to SATAT, which of the following statements is/are correct?

1. Bio-CNG neighbourhood.

2. Biomass/policy neighbour of GOBAR-Dhan.
3. Identical to Project Tiger.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not Project Tiger.

Q26. Which of the following pairs is/are correctly matched?

1. Geothermal provinces — Himalaya/Cambay/West Coast/SONATA teaching
2. Bagasse — sugar industry cogeneration
3. Wind power — independent of wind speed

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Power $\propto$ speed^3.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): 500 GW non-fossil by 2030 is a standard teaching target.

Reason (R): It is the same as India's net-zero year 2070.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Capacity target $\neq$ net-zero year.

Q28. Arrange Cochin / Diu solar firsts teaching:

A. Both false always B. Both are coal plants C. Cochin airport fully solar first teaching; Diu UT/smart city solar first teaching D. Both are nuclear only

💡 **Show answer**

Ans: C. Cochin + Diu firsts. **Logic:** Solar firsts.

Q29. With reference to large vs small hydro, which of the following statements is/are correct?

1. SHP $\leq$25 MW under MNRE.
2. Large hydro sits on conventional side in many Indian classifications.
3. All hydro below 1000 MW is solar PV.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Statement 3 false.

Q30. Which one of the following best states India's mega solar-park fact among common options?

A. Bhadla, Rajasthan B. Only Antarctica C. Only Greenland ice D. Only Nicobar coral

💡 **Show answer**

Ans: A. Bhadla, Rajasthan. **Logic:** Not Gujarat largest trap.

Common Traps

Trap	Correct
Biogas = butane + CO_2	**Methane + CO_2**
ISA HQ Paris/New Delhi	**Gurugram**
ISA exactly 38 members / all UN	**Outdated / false**
Rajasthan leads wind	**Tamil Nadu wind; Rajasthan solar**
Geothermal from Sun	**Earth interior heat**
Geothermal in Ganga Delta	**Not** a listed province
Tidal = wave energy	**Moon gravity vs wind waves**
Tidal potential in Thar	**Khambhat/Kutch coast**
SHP = 50 MW	**≤25 MW**
Nepal border = zero Indian hydro	**False — Q31**
Grey hydrogen = green	**Grey = fossil gas**
Largest solar park = always Gujarat	**Bhadla, Rajasthan**
Fuel cell = AC	**DC**; heat + water
DCR = TV licences	**Domestic solar PV content**
India #3 silicon wafers	**Not** a wafer leader
Green H_2 Mission 2019	**January 2023**
Nuclear = renewable in MCQs	**Usually non-renewable category**
Singrauli = hydro	**Thermal power hub**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 23 — Disaster & Environment (पर्यावरण)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Floods | Drought (सूखा) | Landslides | Cyclones | Heat Waves | Forest Fires

Sources baked in: NCERT Geo 11–12; NDMA; IMD; Sendai Framework 2015; FSI; MoEFCC; UPPCS PYQs 2018–2025
Weight: ★★★ High — disaster–environment chain; UP flood/drought/heat; DPAP year; Piral Lao **Last verified:** August 2026

Current Affairs (this topic)

- Flood / landslide / heat-wave seasons refresh disaster–environment CA for UP–Himalaya (हिमालय).
- Sendai Framework year (**2015**) stays fixed — link hazards to land-use and climate stress.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. Environmental degradation (पर्यावरणीय क्षरण) raises disaster risk (deforestation worsens floods and landslides). Disasters also damage ecosystems through erosion, pollution, and forest loss. 2. Flood types include riverine floods on the Ganga (गंगा) plains, flash floods/cloudbursts in Himalaya–Terai (तराई), and urban (नगरीय) waterlogging. UP river facts include Ganga, Yamuna (यमुना), Ghaghara (घाघरा), Rapti (राप्ती), and Gandak (गंडक). 3. **NDMA** (आपदा प्राधिकरण) is under the Disaster Management Act **2005**. The **Prime Minister** chairs NDMA. **NDRF** is the response force. Sendai Framework for disaster risk reduction is **2015**. 4. Drought progresses from **meteorological** (low rain) to **agricultural** (soil moisture) to **hydrological** (rivers and wells). Bundelkhand's seven UP districts are a chronic drought hotspot. 5. **DPAP** (Drought Prone Areas Programme) is **1973**. Do not date DPAP as **1952**. A rough programme spine is CDP (सीडीपी) 1952 (सामुदायिक विकास) → DPAP 1973 → FWP 1977 → TRYSEM 1979. 6. Landslides need gravity, water, weak slopes, and often vegetation loss. Himalaya and north-eastern hills are core belts. Kedarnath (केदारनाथ) **2013** is the cloudburst–landslide–flood chain fact. Roads, mining, and deforestation are human triggers. 7. Tropical cyclones need warm seas around **26–27°C**. The **Bay of Bengal** produces more than the Arabian Sea. The **eye** is calm; the **eyewall** has the strongest winds; **storm surge** (तूफानी महोर्मि) is the coastal flood killer. 8. Mangroves reduce storm-surge damage. Named cyclone memory often uses Fani (**2019**) and Amphan (**2020**). 9. IMD plains heat-wave idea: about **≥40°C** with departure **+4.5°C to +6.4°C** as heat wave; severe when departure is **≥ +6.4°C** or temperature **≥47°C**. Peak is May–June in North India. Urban heat island amplifies cities but is not identical to a heat-wave definition. 10. Forest fires need fuel, oxygen, and ignition. Peak season is roughly March–June. Chir pine needles are a Himalayan fuel fact. FSI issues satellite fire alerts. 11. **Piral Lao–Paise Pao** in Uttarakhand (उत्तराखंड) pays for pine-needle collection to cut fire risk and raise income. It is an Uttarakhand flagship, not a default UP scheme name. 12. NDMA Chair = Prime Minister — a recurring institution fact. 13. Eye ≠ strongest winds. Eyewall holds the violence; eye is the calm core. 14. Landslides are not "purely natural" when options ignore slope cutting and deforestation. 15. Heat wave (IMD criteria) ≠ urban heat island (city microclimate effect), though cities worsen heat stress. 16. DPAP year is **1973**, not the older CDP **1952** tag. 17. Cloudburst is extreme local rain in a short time — often the Himalayan flood trigger neighbour to landslides. 18. Storm surge is sea pushed ashore by cyclone winds — distinct from ordinary astronomical tide. 19. Meteorological drought (मौसमी (मौसम विज्ञान) सूखा) is rainfall failure; agricultural drought (कृषि सूखा) is crop-moisture failure; hydrological drought (जल-वैज्ञानिक सूखा)

is stream/groundwater failure. Keep the sequence. 20. Sendai **2015** is the global DRR framework fact beside India's NDMA architecture. 21. Forest-fire season and chir-pine fuel matter more in Uttarakhand hills than in deep Gangetic floodplain districts. 22. Cyclone naming and Bay of Bengal dominance are coastal facts; UP's disaster map is flood–drought–heat first. 23. NDRF is the specialised response force; NDMA is the apex policy body chaired by the PM. 24. Bundelkhand drought and Terai/Ganga floods are the two UP geography disaster anchors on this sheet.

💡 ⚡ **Confused Pairs (Click to Expand)**

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Meteorological Drought	**Agricultural Drought**	Situation where actual seasonal rainfall over an area is significantly less than the long-term climatological mean (>25% deficient = moderate; >50% deficient = severe) vs condition where soil moisture deficiency severely impedes crop growth, triggering wilting and agrarian loss	मौसमी (मौसम विज्ञान) सूखा / कृषि सूखा (मृदा नमी संकट)
Hydrological Drought	**Socio-Economic Drought**	Prolonged meteorological drought leading to depletion of surface water reservoirs, dried-up streams, and falling groundwater aquifer tables vs severe water scarcity disrupting economic goods, public water supply, and livelihoods	जल-वैज्ञानिक सूखा (जलाशय ह्रास) / सामाजिक-आर्थिक सूखा (आजीविका संकट)
Cloudburst	**Urban Flash Flood**	Extreme, highly localized torrential rainfall event exceeding **100 mm per hour** over a small geographic area (~20–30 km^2) causing catastrophic flash torrents vs rapid inundation of built-up city areas caused by impermeable concrete surfaces and choked storm drains	बादल फटना (≥100 मिमी/घंटा अतिवृष्टि) / नगरीय आकस्मिक बाढ़
Storm Surge	**Tsunami**	Abnormal coastal sea-level rise and massive sea surge pushed onshore by the fierce winds and extreme low atmospheric pressure of a tropical cyclone vs series of immense sea waves generated by underwater seaquake, submarine volcanic eruption, or seafloor tectonic dislocation	तूफानी महोर्मि (चक्रवात जनित उभार) / सुनामी (अंतःसमुद्री भूकंप जनित तरंग)
National Disaster Management Authority (NDMA)	**National Institute of Disaster Management (NIDM)**	Apex statutory policy-making and executive body for disaster management established under DM Act 2005, chaired ex-officio by the Prime Minister vs premier national statutory institute responsible for capacity building, research, training, and documentation in disaster management	राष्ट्रीय आपदा प्रबंधन प्राधिकरण (NDMA, नीति पीठ) / राष्ट्रीय आपदा प्रबंधन संस्थान (NIDM, प्रशिक्षण)
Hazard	**Vulnerability**	Dangerous physical event, phenomenon, or human activity that has the potential to cause loss of life, injury, or property damage (e.g. earthquake, cyclone) vs conditions determined by physical, social, economic, and environmental factors increasing susceptibility to hazards	आपदा जोखिम/खतरा (Hazard) / सुभेद्यता (Vulnerability)

💡 📋 **Must-score facts — NDMA, drought, cyclone, fire (Click to Expand)**

Institutions / hazards

Item	Lock
NDMA	DM Act **2005**; Chair = **PM**; NDRF response
Sendai	Disaster risk reduction framework
Drought ladder	Meteorological → agricultural → hydrological
DPAP	**1973** (≠ 1952)
Cyclone needs	Warm sea ~26–27°C; BoB > Arabian Sea
Eye / Eyewall	Calm core / strongest winds
Heat wave (plains)	≥40°C + departure rules; severe ≥47°C or +6.4°C
Forest fire season	Roughly Mar–Jun; chir pine fuel
Piral Lao–Paise Pao	UK pine-needle scheme (≠ Delhi)

23.1 Floods

Identity: Riverine / flash / urban | UP Ganga–Yamuna–Terai | NDMA 2005 | Sendai 2015

- A **flood** occurs when water overflows onto normally dry land — river channel capacity or drainage is exceeded.
- **Riverine floods** swell slowly over days–weeks on plains. **Flash floods** follow cloudbursts in hills within hours. **Urban flooding** is waterlogging from impervious surfaces and clogged drains.
- **UP flood rivers:** Ganga, Yamuna, Ghaghara (Saryu), Rapti, Gandak, Gomti** — monsoon inundation of the Gangetic floodplain.

- **Human amplification:** floodplain encroachment, embankments that raise silted beds, and **deforestation** in catchments that speeds runoff.
- **Terai flash risk:** Himalayan/Nepal-side catchments can inundate **Pilibhit, Lakhimpur Kheri, Bahraich** rapidly after extreme rain.
- **Natural flood benefit:** controlled flooding deposits **alluvial (जलोढ़) silt** that renews soil fertility — the problem is uncontrolled flooding of encroached zones.
- **Institutions:NDMA (Disaster Management Act 2005)** — **Prime Minister** is Ex-officio Chair; **NDRF** for response; **IMD/CWC (केंद्रीय जल आयोग)** for forecasting; **Sendai Framework 2015** for disaster risk reduction.
- **Mitigation:** floodplain zoning, wetland restoration (natural sponges), catchment afforestation, early warning.

Type	Speed	UP example
Riverine	Days–weeks	Ganga at Prayagraj (प्रयागराज)/Varanasi (वाराणसी)
Flash	Hours	Terai foothills
Urban	Hours	Lucknow (लखनऊ), Kanpur (कानपुर) waterlogging

> **Logic:** Floods are India's most frequent disaster by affected population (जनसंख्या) — not limited to coastal states.

23.2 Drought

Identity: Three types | Bundelkhand hotspot | DPAP 1973 | Creeping (slow-onset) disaster

- **Drought** develops slowly — weeks to months — unlike sudden floods.
- **Meteorological drought:** rainfall below long-term normal.
- **Agricultural drought:** soil moisture insufficient for crops (even if some rain occurs).
- **Hydrological drought:** rivers, lakes, and groundwater below normal after prolonged dry spell.
- Sequence often runs meteorological → agricultural → hydrological.
- **Bundelkhand (UP):** seven districts — **Jhansi (झांसी), Lalitpur, Jalaun, Banda, Hamirpur, Mahoba, Chitrakoot (चित्रकूट)** — chronic drought–desertification (मरुस्थलीकरण) hotspot.
- **Drought Prone Areas Programme (DPAP)** launched in **1973** — soil conservation, water harvesting, afforestation in dryland districts.
- Chronology fact s: **CDP 1952 → DPAP 1973 → Food for Work 1977 → TRYSEM 1979**.
- **Feedback loop:** drought kills vegetation → soil exposed → erosion → worse water retention → next drought worse.
- Other schemes: Watershed Development, **MGNREGA** water works, Bundelkhand Package, Jal Jeevan Mission.

PYQ — UPPCS Prelims 2025, Q134

Arrange chronologically: (1) Food for Work (2) Community Development Programme (3) TRYSEM (4) Drought Prone Areas Programme

A. 4, 2, 3, 1

B. 2, 4, 1, 3

C. 2, 4, 3, 1

D. 4, 2, 1, 3

> 💡 **Show answer**
>
> **Ans: B** — CDP **1952** → DPAP **1973** → FWP **1977** → TRYSEM **1979**.

> **Logic:** DPAP = **1973**, not 1952 (that is CDP).

23.3 Landslides

Identity: Gravity mass wasting | Himalaya/NE | Kedarnath 2013 | Human triggers matter

- A **landslide** is downhill movement of rock, soil, or debris when gravity overcomes slope stability.
- Needs **steep slope + weak geology + water lubrication + loss of root binding**.
- Types include slides, rock falls, **debris flows** (most deadly after cloudburst), and rotational slumps.

Kedarnath 2013 — Cause, Course, Result

Cause: Extreme rainfall / cloudburst over fragile Himalayan slopes, plus prior slope weakening from roads, settlement, and vegetation stress, set up a mass-wasting cascade near Kedarnath (Uttarakhand).

Course: Cloudburst triggered landslides and debris flows that choked and then released flash flood water through the Mandakini (मंदाकिनी) corridor — a hydrometeorological + mass-movement chain, not a pure earthquake or tsunami (सुनामी) frame.

Result: Catastrophic loss of life and infrastructure; became the national teaching case for environment–disaster coupling in the Himalaya. Downstream landslide-dam and flood risk also matters for Terai plains including UP.

- **Human triggers:** deforestation (roots bind soil), road cutting/blasting, unplanned hillside construction, mining — landslides are **not purely natural**.
- **Earthquake link:** Himalayan seismic shaking can trigger widespread mass failures.
- **Downstream risk:** landslide debris can dam (दाम -) rivers; dam-break floods then hit plains including UP Terai.
- **Mitigation:** bio-engineering, retaining walls, afforestation, land-use zoning on steep slopes, GSI hazard zonation maps.

> **Logic:** "Landslides are purely natural — humans play no role" = **FALSE**.

23.4 Cyclones

Identity: Warm ocean ≥26–27°C | Bay of Bengal nursery | Eye calm / eyewall strongest | Storm surge

- A **tropical cyclone** is an intense rotating storm over warm tropical oceans with a low-pressure centre.
- Formation needs **sea surface temperature ≥26–27°C**, moist air, low vertical wind shear, and Coriolis force (not on the equator).
- **Eye:** calm, clear, low-pressure centre. **Eyewall:** ring of **strongest winds and rain** around the eye. Trap: strongest winds are **not** in the eye.
- **Storm surge:** cyclone winds push seawater onshore — often deadlier than wind on flat coasts (Odisha, WB, Bangladesh).
- **Bay of Bengal** produces **more frequent and intense** cyclones than the **Arabian Sea** (warmer, shallower, moister waters).
- East-coast states most hit: Odisha, West Bengal, Andhra Pradesh, Tamil Nadu (नाडु). UP gets rain remnants, not landfall.
- Regional names: **cyclone** (Indian Ocean), **hurricane** (Atlantic), **typhoon** (NW Pacific), **Willy-Willies** (Australia), **Baguios** (Philippines).
- **Mangroves** (e.g. Sundarbans (सुंदरबन)) absorb surge energy — coastal buffer. Deforestation of mangroves increases damage.

- Recent named storms: **Amphan (2020)**, **Fani (2019)**, **Hudhud (2014)**, **Yaas (2021)** — IMD names North Indian Ocean cyclones.

PYQ — UPPCS Prelims 2018 pattern

Natural disasters cause max damage in developing countries; Bhopal was man-made; India is disaster-free; mangroves reduce cyclone impact.

💡 **Show answer**

True statements: 1, 2, and 4. India is **not** disaster-free. Mangroves **do** reduce cyclone impact.

Feature	Eye	Eyewall
Winds	Calm	**Strongest**
Pressure	Lowest	Rising outward

> **Logic:** Baguios–Philippines; Hurricanes–USA; Typhoons–China/NW Pacific; Willy-Willies–Australia.

23.5 Heat Waves

Identity: IMD thresholds | Urban heat island | May–June peak | Climate frequency rise

- A **heat wave** is a prolonged period of abnormally high temperature relative (सापेक्ष) to the location's normal.
- **IMD plains criteria:** maximum temperature **≥40°C** with departure **+4.5°C to +6.4°C** = heat wave; departure **≥+6.4°C** or absolute (निरपेक्ष) temp **≥47°C** = **severe heat wave**.
- **Urban heat island:** cities stay hotter than surrounding rural areas due to concrete, asphalt, waste heat, and lost greenery — amplifies heat-wave mortality.
- UP hotspots: **Kanpur, Lucknow, Agra, Ghaziabad, Noida** — peak stress in **May–June** pre-monsoon.
- **Climate link:** warming makes heat waves more frequent, longer, and hotter (IPCC (आईपीसीसी) AR6).
- **Compound risk:** heat + drought raises crop failure and forest-fire fuel dryness; heat + power surge risks blackouts.
- **Health:** heatstroke, dehydration — elderly, outdoor workers, children most vulnerable.
- **Adaptation:** cool roofs, urban greening, IMD heat warnings, Heat Action Plans (Ahmedabad (अहमदाबाद) model).

Feature	Heat wave	Urban heat island
Scale	Regional weather event	Local city microclimate
Duration	Days–weeks	Persistent year-round bias
Cause	Atmospheric pattern	Built surfaces, lost greenery

> **Logic:** Heat wave ≠ urban heat island — different scale and cause.

23.6 Forest Fires

Identity: Chir pine fuel | March–June peak | Piral Lao–Paise Pao | FSI satellite alerts

- A **forest fire** is uncontrolled fire through forest vegetation — needs **fuel + oxygen + ignition**.
- **Surface fire** burns litter/grass; **crown fire** burns canopy (most destructive); **ground fire** smoulders in organic soil.

- **Chir pine** forests (Uttarakhand, Himachal (हिमाचल)) accumulate **dry pine needles** — major summer fuel load.
- Peak season: **March–June** before monsoon rains.
- Human causes dominate: pasture burning, negligence, arson — not only lightning.
- **Environmental damage:** biodiversity loss, soil organic matter destruction, erosion, **PM2.5 smoke**, CO_2 release; post-fire slopes raise landslide/flood risk.
- **FSI** uses satellite (MODIS/VIIRS) for real-time forest fire monitoring and state alerts.
- **"Piral Lao – Paise Pao"**: Uttarakhand campaign — villagers collect pine needles (**piral**), reducing fire fuel and earning income; launched under Uttarakhand CM.

PYQ — UPPCS Prelims 2024, Q86

Piral Lao–Paise Pao: (1) reduces forest fires and gives income to villagers (2) launched by Uttarakhand CM

A. Only 2

B. Neither

C. Only 1

D. Both 1 and 2

💡 **Show answer**

Ans: D — Both statements correct — fire prevention + livelihood; Uttarakhand CM campaign.

> **Logic:** Piral Lao = **Uttarakhand** (उत्तराखंड) pine-needle campaign — not a UP state flagship scheme.

Complete PYQ Bank (Topic 23)

Q1. UPPCS Prelims 2025, Q134

Arrange the following programmes in chronological order of commencement:

1. Food for Work Programme
2. Community Development Programme
3. TRYSEM
4. Drought Prone Areas Programme

A. 4, 2, 3, 1

B. 2, 4, 1, 3

C. 2, 4, 3, 1

D. 4, 2, 1, 3

💡 **Show answer**

Logic: Chronology stem — CDP 1952 → DPAP 1973 → FWP 1977 → TRYSEM 1979. **Ans: B (2, 4, 1, 3).** Order: **CDP → DPAP → FWP → TRYSEM**. **Trap:** DPAP **1973** swapped with CDP **1952**.

Q2. UPPCS Prelims 2024, Q86

With reference to the "Piral Lao – Paise Pao" campaign, consider the following statements:

1. It reduces forest fires and gives income to villagers.
2. It was launched by the Chief Minister of Uttarakhand.

A. Only 1

B. Neither 1 nor 2

C. Only 2

D. Both 1 and 2

💡 **Show answer**

Logic: Uttarakhand pine-needle collection — fire fuel reduction plus livelihood. **Ans: D (Both 1 and 2).** Both fire-prevention and CM-launch statements are correct. **Trap:** Assigning the scheme to **UP** instead of Uttarakhand.

Q3. UPPCS Prelims 2021, Q26

Who is the ex-officio Chairman of the National Disaster Management Authority (NDMA)?

A. The Prime Minister

B. The Home Minister

C. The Defence Minister

D. The Health and Family Welfare Minister

💡 **Show answer**

Logic: NDMA under DM Act 2005 — PM is ex-officio Chair; NDRF handles response. **Ans: A.** The **Prime Minister** is the ex-officio **NDMA Chairman**. **Trap:** **Home Minister** — common false association with disaster response.

Q4. UPPCS Prelims 2020, Q74

Match List-I with List-II:

List-I (Tropical cyclones)	List-II (Country/region)
A. Baguios	1. Australia
B. Hurricanes	2. China
C. Typhoons	3. Philippines
D. Willy-Willies	4. United States of America

Row order is not the answer code.

A. 3 4 1 2

B. 3 4 2 1

C. 2 3 4 1

D. 2 1 3 4

💡 **Show answer**

Logic: Regional cyclone naming — Baguios–Philippines; Hurricanes–USA; Typhoons–China/NW Pacific; Willy-Willies–Australia. **Ans: B (3 4 2 1).** **Baguios–Philippines; Hurricanes–USA; Typhoons–China; Willy-Willies–Australia**. **Trap:** Swapping **Typhoons** (China) with **Hurricanes** (USA).

Q5. UPPCS Prelims 2019, Q78

Match List-I with List-II:

List-I (Cyclone name)	List-II (Country)
A. Willy-Willies	1. Philippines
B. Taifu	2. Australia
C. Baguio	3. Japan
D. Hurricanes	4. U.S.A.

Row order is not the answer code.

A. 3 4 1 2

B. 2 3 4 1

C. 1 3 2 4

D. 2 3 1 4

💡 **Show answer**

Logic: Same regional naming set as 2020 Q74 with variant list labels. **Ans: D (2 3 1 4).** Willy-Willies–Australia; Taifu–Japan; Baguio–Philippines; Hurricanes–USA. **Trap:** **Taifu** (Japan) confused with **Typhoon** generic label only.

Q6. UPPCS Prelims 2018, Q133

Which of the following statements are true?

1. Natural disasters cause maximum damage in developing countries.
2. Bhopal gas tragedy was man-made.
3. India is a disaster-free country.
4. Mangroves reduce the impact of cyclones.

A. 1, 2 and 3

B. 2, 3 and 4

C. 1, 2 and 4

D. 1, 3 and 4

💡 **Show answer**

Logic: Developing-country vulnerability + man-made Bhopal + mangrove buffer; India is **not** disaster-free. **Ans: C (1, 2 and 4).** Statements 1, 2, and 4 are true. **Trap:** Statement 3 — India faces floods, cyclones, droughts, landslides, and earthquakes.

Q7. UPPCS Prelims pattern — Sendai Framework

The Sendai Framework (2015) primarily relates to:

A. Disaster risk reduction

B. Ozone layer protection

C. Wildlife trade regulation

D. Carbon credit trading

💡 **Show answer**

Logic: Sendai 2015 = global DRR framework parallel to India's NDMA architecture. **Ans: A.** **Sendai Framework is for disaster risk reduction**. **Trap:** Montreal (ozone) or Kyoto (carbon) cross-topic distractors.

Q8. UPPCS Prelims pattern — Kedarnath 2013

The Kedarnath disaster of 2013 is best described as a cascade involving:

A. Cloudburst leading to landslides and flash floods

B. Purely tectonic earthquake with no rainfall link

C. Tsunami after submarine earthquake

D. Volcanic eruption in Garhwal Himalaya

💡 **Show answer**

Logic: Classic environment–disaster chain — extreme rain → slope failure → downstream flood. **Ans: A.** **Cloudburst–landslide–flash flood** cascade. **Trap:** Pure earthquake or tsunami framing — Kedarnath was hydrometeorological + mass movement.

Q9. UPPCS Prelims pattern — storm surge

Storm surge during a tropical cyclone is caused mainly by:

A. Cyclone winds pushing seawater onshore

B. Calm eye of the cyclone drawing water inward only

C. Lunar tides alone without wind

D. Ozone depletion over the ocean

💡 **Show answer**

Logic: Surge = wind-driven coastal flooding — often deadlier than wind on flat coasts. **Ans: A.** **Cyclone winds push seawater onshore** as storm surge. **Trap:** **Eye** is calm — strongest winds and surge occur in the **eyewall**, not the eye.

Q10. UPPCS Prelims pattern — cyclone structure

In a mature tropical cyclone, the strongest winds are found in the:

A. Eye

B. Eyewall

C. Outer rainbands only

D. Stratosphere above the storm

💡 **Show answer**

Logic: Eye = calm low pressure; eyewall = maximum wind and rain ring. **Ans: B.** The **eyewall** has the **strongest winds**. **Trap:** **Eye** — calm centre, not maximum wind zone.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to environment and disasters, which of the following statements is/are correct?

1. Degradation raises disaster risk (e.g., deforestation worsens floods/landslides).
2. Disasters never damage ecosystems; only ecosystems cause disasters.

3. Deforestation always prevents landslides.

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Ans: D.** Only 1 is correct. **Logic:** Statement 1 is true (degradation raises risk). Statement 2 is false — disasters also damage ecosystems. Statement 3 is false — deforestation worsens, not prevents, many landslides.

Q2. Match List-I with List-II:

List-I	List-II
A. NDMA	1. Response force
B. NDRF	2. DM Act 2005; PM chairs
C. Sendai	3. DRR framework

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-1, C-3. **Logic:** NDMA/NDRF/Sendai.

Q3. Arrange drought ladder:

A. Hydrological → meteorological → agricultural B. No sequence C. Agricultural → meteorological only D. Meteorological → agricultural → hydrological

> 💡 **Show answer**
>
> **Ans: D.** Met → agri → hydro. **Logic:** Standard teaching order.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): DPAP began in 1973.

Reason (R): DPAP began in 1952 as the first Five-Year Plan only.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** DPAP 1973 — not 1952 (CDP trap).

Q5. With reference to cyclones, which of the following statements is/are correct?

1. Tropical cyclones form best over cold seas below about 15°C.
2. Bay of Bengal produces more than Arabian Sea.
3. Eye has the strongest winds; eyewall is calm.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 2 is correct. **Logic:** Statement 2 is true (BoB > Arabian Sea frequency). Statement 1 is false — warm seas ~26–27°C are needed. Statement 3 swaps eye (calm) and eyewall (strongest winds).

Q6. Which of the following pairs is/are correctly matched?

1. Fani — 2019
2. Amphan — 2020
3. Mangroves — reduce storm-surge damage

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

> 💡 **Show answer**
>
> **Ans: B.** All three. **Logic:** Named cyclones + mangroves.

Q7. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Landslides need gravity, water, weak slopes, and often vegetation loss.

Reason (R): Kedarnath 2013 is a Himalayan disaster teaching landmark.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; Kedarnath year does not define landslide mechanics. **A/R logic:** Mechanics vs case.

Q8. With reference to heat waves (plains), which of the following statements is/are correct?

1. About ≥40°C with departure rules for heat wave teaching.
2. Severe when departure ≥+6.4°C or temperature ≥47°C teaching.
3. Heat waves occur only at −10°C.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** Statement 3 false.

Q9. Match List-I with List-II:

List-I	List-II
A. Forest fire season	1. UK pine-needle scheme
B. Chir pine needles	2. Roughly Mar–Jun peak
C. Piral Lao–Paise Pao	3. Himalayan fuel fact

Code:

A. A-3, B-1, C-2 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-2, B-3, C-1

> 💡 **Show answer**
>
> **Ans: D.** A-2, B-3, C-1. **Logic:** Season/fuel/UK scheme.

Q10. Arrange NDMA Chair fact:

A. Prime Minister B. Chief Justice of India only C. UNEP Executive Director only D. State Forest Officer only

> 💡 **Show answer**
>
> **Ans: A.** Prime Minister. **Logic:** Recurring institution fact.

Q11. With reference to floods, which of the following statements is/are correct?

1. Riverine floods on Ganga plains.
2. Flash floods/cloudbursts in Himalaya–Terai.
3. Urban waterlogging is never a flood type.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** Urban waterlogging is taught.

Q12. Which of the following pairs is/are NOT correctly matched?

1. Eye — calm core
2. Eyewall — strongest winds
3. Eye — strongest winds always

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only pair 3 is not correctly matched. **Logic:** Eye ≠ strongest.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): FSI issues satellite-based forest-fire alerts teaching.

Reason (R): Forest fires need fuel, oxygen, and ignition.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; fire triangle does not explain FSI's alert role specifically. **A/R logic:** Institution vs chemistry.

Q14. With reference to Bundelkhand, which of the following statements is/are correct?

1. Drought–degradation hotspot including seven UP districts teaching.
2. Links meteorological to agricultural stress.
3. Bundelkhand is a coastal cyclone capital of India.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Not coastal cyclone capital.

Q15. How many of the following landslide belts are core teaching?

Himalaya, north-eastern hills, deep ocean abyssal plain as primary Indian landslide belt

A. Only two B. Only one C. All three D. None

> 💡 **Show answer**
>
> **Ans: A.** Only two. **Logic:** Abyssal plain not the Indian landslide belt.

Q16. Arrange programme spine teaching trap:

A. DPAP 1952 → CDP 1973 B. CDP 1952 → DPAP 1973 → … C. Both identical 2015 D. Both identical 2005 only

💡 **Show answer**

Ans: B. CDP 1952 → DPAP 1973. **Logic:** Year traps.

Q17. With reference to Sendai Framework, which of the following statements is/are correct?

1. Disaster risk reduction framework.
2. Sits with NDMA/DM Act neighbourhood teaching.
3. Identical to Montreal Protocol ODS list.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Not Montreal.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Mangroves reduce storm-surge damage.

Reason (R): Coastal vegetation can buffer surge energy.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Buffering mechanism.

Q19. Which one of the following correctly states NDRF?

A. National Dobson Radiation Force B. National Desert Research Forest C. National Dairy Reform Fund D. National Disaster Response Force

💡 **Show answer**

Ans: D. Response force. **Logic:** NDRF expansion.

Q20. With reference to cloudbursts, which of the following statements is/are correct?

1. Linked to Himalaya–Terai flash floods teaching.
2. Can worsen with steep slopes and debris.
3. Occur only in Thar as the sole Indian setting.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not Thar-only.

Q21. Match List-I with List-II:

List-I	List-II
A. Meteorological drought	1. Rivers/wells
B. Agricultural drought	2. Low rain
C. Hydrological drought	3. Soil moisture

Code:

A. A-1, B-2, C-3 B. A-2, B-3, C-1 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-3, C-1. **Logic:** Rain/soil/rivers.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Landslides are not "purely natural" when options ignore slope cutting and deforestation.

Reason (R): Human slope modification can raise landslide risk.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Anthropogenic trigger.

Q23. How many of the following cyclone facts are correct?

Warm sea ~26–27°C; BoB > Arabian Sea; mangroves buffer surge

A. Only one B. Only two C. All three D. None

💡 **Show answer**

Ans: C. All three. **Logic:** Cyclone trio.

Q24. With reference to Piral Lao–Paise Pao, which of the following statements is/are correct?

1. Uttarakhand pine-needle collection scheme teaching.
2. Cuts fire risk and raises income teaching.
3. It is a Delhi NCR metro statute only.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** UK flagship — not Delhi metro.

Q25. Arrange fire triangle:

A. Fuel + oxygen + ignition B. Only ignition C. Only argon D. Only nitrogen

💡 **Show answer**

Ans: A. Fuel, oxygen, ignition. **Logic:** Fire needs all three.

Q26. Which of the following pairs is/are correctly matched?

1. DM Act — 2005
2. NDMA Chair — PM
3. NDRF — grants Environmental Clearance Category A

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** NDRF ≠ EC granter.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Urban waterlogging is a flood-related teaching type.

Reason (R): Poor drainage and intense rain interact in cities.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains A. **A/R logic:** Urban flood mechanism.

Q28. With reference to Ganga plains floods, which of the following statements is/are correct?

1. Riverine floods are classic.
2. Link to monsoon discharge.
3. Ganga plains never flood in any year in teaching.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** Statement 3 false.

Q29. Arrange eyewall vs eye:

A. Eyewall calm; eye violent B. Neither exists C. Both identical always D. Eye calm; eyewall violent

> 💡 **Show answer**
>
> **Ans: D.** Eye calm; eyewall violent. **Logic:** Classic trap.

Q30. Which one of the following correctly states Sendai's theme neighbourhood?

A. Disaster risk reduction B. Ozone ODS phase-out only C. Eighth Schedule languages only D. GST Council only

> 💡 **Show answer**
>
> **Ans: A.** DRR framework. **Logic:** Sendai identity.

Common Traps

Trap	Correct
DPAP = 1952	**1973** (CDP = 1952)
Strongest winds in cyclone eye	**Eyewall**
Landslides purely natural	**Deforestation/road cutting trigger**
Heat wave = urban heat island	**Different scale/cause**
All drought = only rainfall deficit	Also **agricultural/hydrological**
Piral Lao = UP scheme	**Uttarakhand**
India disaster-free	**False**
Arabian Sea > Bay for cyclones	**Bay of Bengal more**
Floods only coastal	**UP riverine/Terai major**
Mangroves worsen cyclones	**Reduce surge/wind damage**

Trap	Correct
NDMA Chair = Home Minister	**Prime Minister**
Sendai = ozone treaty	**Disaster risk reduction 2015**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 24 — Current Environmental Issues

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 Covers syllabus (click to expand)

Microplastics | Carbon Credit (कार्बन क्रेडिट) | Carbon Trading | Net Zero Emissions (नेट जीरो उत्सर्जन) | Green Finance (हरित वित्त) | Blue Economy (नीली अर्थव्यवस्था) | LiFE Mission

Sources baked in: UNEP; MoEFCC; SEBI; RBI; IPCC; UN Plastics Treaty; Green Credit Rules 2023; CCTS 2023; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — LiFE COP trap; carbon credit vs trading; India net zero 2070; microplastic size **Last verified:** August 2026

Current Affairs (this topic)

- This chapter is the CA layer — keep NbS, blue economy, and plastics stems current.
- UP is landlocked: do not invent statewide blue-economy projects.

💡 🎯 Consolidated — 26 Must-Score Facts (Click to Expand)

1. **Microplastics** are plastic fragments **smaller than 5 mm**. Primary sources include microbeads; secondary sources come from larger plastic breaking down. They are not "fast biodegradable." 2. Microplastics enter food webs from plankton to fish to humans. India's identified single-use plastic bans tightened around **2022**. UN plastics treaty talks (INC) also run from that period. 3. One **carbon credit** equals **1 tonne of CO_2-equivalent** reduced, removed, or avoided. Kyoto (क्योटो)'s CDM produced CERs; voluntary markets use standards such as Verra and Gold Standard. 4. **Carbon trading** is the market that buys and sells allowances or credits (cap-and-trade). The **EU ETS** is the largest compliance market. India's **CCTS (2023)** under the Energy Conservation Act line is steered with **BEE**. 5. **PAT** creates energy-efficiency certificates (**ESCerts**). PAT is **not** identical to carbon credits under CCTS. 6. **Green Credit Programme (2023)** rewards eco-actions. It is **not** the same instrument as a carbon credit. 7. **Net zero** means anthropogenic emissions balanced by removals. India's target is **2070** (COP26 Glasgow **2021**). China is often locked at **2060**; EU/UK at **2050**. Net zero is not gross zero. 8. India's NDC track includes cutting emission intensity and raising non-fossil capacity share toward the **2030** horizon (including the ~50% non-fossil capacity language in later updates). 9. **Green finance** is capital for climate and environment (पर्यावरण) projects — green bonds, climate funds, GCF. India issued **Sovereign Green Bonds** in FY **2022–23**; SEBI's Green Bond Framework is **2023**. 10. The **100 billion** a year climate-finance pledge is the developed-to-developing finance politics fact. Loss and Damage Fund operationalisation is linked to **COP28 (2023)**. 11. **Blue economy** is sustainable ocean and coastal use — fisheries, ports, tourism, offshore renewables — aligned with **SDG 14**. **Blue carbon** (नीला कार्बन) is carbon stored in mangroves and seagrass. 12. **MISHTI (2023 Budget)** restores mangroves and sits beside Sagarmala and Deep Ocean Mission in the blue–coastal policy neighbourhood. 13. **LiFE** (Lifestyle for Environment (जीवनशैली पर्यावरण)) is a MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) behaviour-change mission. The idea was pitched at **COP26 Glasgow (2021)** and launched in **June 2022**. It is **not** a COP-25 Madrid product. 14. LiFE catalogues pro-planet actions for citizens; it is lifestyle awareness, not a carbon-market statute. 15. Carbon credit (unit) ≠ carbon trading (market) ≠ Green Credit (eco-action incentive) ≠ green finance (capital). 16. India net zero is **2070**, not 2050. 17. Microplastics are **<5 mm**, not 5 cm. 18. Kyoto flexible mechanisms are climate tools; Montreal is ozone — do not swap treaty families in "current issues"

stems. 19. Nature-based credits often come from forestry and mangrove (blue carbon) projects. 20. UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम)'s Emissions Gap Report (उत्सर्जन अंतराल रिपोर्ट) tracks the gap between pledges and pathways — neighbour to net-zero politics. 21. SUP item bans are domestic plastic regulation; they do not by themselves equal a global plastics treaty. 22. CCTS is India's carbon market architecture; PAT remains an efficiency-certificate neighbour under BEE. 23. Blue economy is ocean wealth with sustainability; it is not a synonym for inland wetland policy alone. 24. Glasgow **2021** facts both India's 2070 net-zero announcement track and the LiFE idea pitch. 25. Offset means using credits to compensate one's own emissions — related to, but not identical with, "trading" as a market design. 26. Keep MISHTI dated **2023**, not 2019, when mangrove restoration is asked.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Primary Microplastics** | **Secondary Microplastics** | Microscopic plastic particles (<5 mm) intentionally manufactured for commercial applications (microbeads in facial scrubs, toothpaste, virgin industrial resin pellets) vs microplastics formed by physical weathering, photo-degradation, and fragmentation of macro-plastics in the environment | प्राथमिक माइक्रोप्लास्टिक (निर्मित मनके) / द्वितीयक माइक्रोप्लास्टिक (अपघटित कण) | | **Carbon Credit** | **Green Credit (2023)** | Tradable compliance unit representing 1 metric tonne of CO_2 equivalent emission reduction or removal registered under carbon markets vs incentives under India's Green Credit Programme (2023) rewarding diverse eco-actions (afforestation, water harvesting, mangrove conservation) beyond carbon | कार्बन क्रेडिट (1 टन CO_2e मानक) / ग्रीन क्रेडिट (पर्यावरण कार्य प्रोत्साहन) | | **Net Zero Emissions** | **Gross Zero Emissions** | Balancing residual greenhouse gas emissions across all sectors with equivalent anthropogenic carbon removals and sequestration by 2050/2070 vs absolute elimination of all greenhouse gas emissions to absolute zero without relying on offsets or sinks | नेट जीरो (उत्सर्जन-अवशोषण संतुलन) / सकल शून्य (पूर्ण उत्सर्जन शून्यता) | | **Blue Economy** | **Blue Carbon** | Sustainable utilization of ocean, coastal, and marine resources for economic growth, improved livelihoods, and jobs while preserving marine ecosystem health vs organic carbon captured, sequestered, and stored over centuries by coastal marine ecosystems (mangroves, seagrasses, salt marshes) | नीली अर्थव्यवस्था (महासागरीय संधारणीय विकास) / नीला कार्बन (तटीय मैंग्रोव संचित कार्बन) | | **Green Finance** | **Greenwashing** | Structured financial investments, bonds, and banking loans directed specifically towards environmentally sustainable development and clean energy projects vs deceptive marketing, publicity, or PR spin falsely exaggerating a company's environmental credentials and sustainability | हरित वित्त (टिकाऊ निवेश) / ग्रीनवॉशिंग (झूठा पर्यावरण प्रचार) | | **Mission LiFE (2022)** | **Lifestyle for Environment** | Global citizen movement launched in October 2022 at Kevadia by PM Modi and UN Secretary-General mobilizing individuals to become 'Pro-Planet People' (P3) through 75 actionable behavioral changes vs abstract corporate environmental pledges | मिशन लाइफ (LiFE, P3 नागरिक आंदोलन) / जीवनशैली पर्यावरण |

💡 📋 **Must-score facts — carbon markets, net zero, LiFE (Click to Expand)**

Markets / targets | Item | Lock | |---|---| | Microplastics | <**5 mm**; primary beads / secondary fragments | | 1 carbon credit | 1 tCO_2e reduced/removed/avoided | | PAT | ESCerts — ≠ carbon credit under CCTS | | Green Credit | **2023** eco-action rewards (≠ carbon credit) | | India net zero | **2070** (COP26 Glasgow) | | LiFE | Idea COP26 **2021**; launch Jun **2022** | | MISHTI | Budget **2023** mangroves | | Blue economy | Sustainable ocean use; SDG **14** |

24.1 Microplastics

Identity: <5 mm | Primary vs secondary | Persistent | Ganga (गंगा)/food-chain contamination

- **Microplastics** are plastic particles **smaller than 5 mm** — primary (manufactured small) or secondary (fragmented from larger plastic).

- **Primary:** microbeads in cosmetics, plastic **nurdles**, industrial abrasives.
- **Secondary:** bags, bottles, fishing nets, and **tyre wear** breaking down — most environmental mass is secondary.
- **Major sources:** single-use packaging, **synthetic textile fibres** (laundry effluent), tyre abrasion, fishing gear, agricultural plastic mulch.
- **Pathways:** land → rivers → sea. **Ganga and Yamuna (यमुना)** studies show microplastic contamination downstream of urban (नगरीय) centres (Kanpur (कानपुर), Varanasi (वाराणसी), Lucknow (लखनऊ)).
- **Health concern:** detected in blood, lungs, placenta — potential carrier for toxins; research ongoing.
- **Persistence trap:** polyethylene/polypropylene persist **decades to centuries** — photodegradation makes **smaller** particles, not harmless minerals.
- **Global policy:** UNEA **2022** resolution launched **Global Plastics Treaty** negotiations (INC process).
- **India:** Plastic Waste Management Rules (EPR); **ban on identified single-use plastic items (July 2022)**; National Dashboard for EPR.

Type	Origin	Example
Primary	Manufactured small	Microbeads, nurdles
Secondary	Large plastic breaks down	Bag fragments, tyre particles

> **Logic:** Microplastics = **<5 mm**, not 5 cm. "Biodegrade within weeks" = **FALSE**.

24.2 Carbon Credit

Identity: 1 credit = 1 tCO_2e | Kyoto CDM/CERs | Nature-based credits | Green Credit ≠ carbon credit

- **Carbon credit** is a tradable certificate representing **one tonne of CO_2 equivalent (tCO_2e)** reduced, avoided, or removed.
- **Project-based generation:** verified emission-reduction project (solar displacing coal, biogas (बायोगैस), afforestation) → accredited body issues credits → sold to buyer.
- **Kyoto CDM (Clean Development Mechanism):** developed countries fund projects in developing countries → **Certified Emission Reductions (CERs)**. India hosted many wind/solar CDM projects.
- **Voluntary Carbon Market:** companies buy credits for net-zero pledges — **Verra (VCS), Gold Standard** standards.
- **Nature-based credits:** afforestation, REDD+, **mangrove restoration** — permanence risk if forest burns or is cleared.
- **Avoidance vs removal:** avoidance prevents emissions that would occur; removal pulls CO_2 from atmosphere (afforestation, DAC).
- **Additionality:** project must prove reduction would not happen without credit revenue.
- **Green Credit Programme (2023)** rewards tree plantation, water conservation, waste management — **NOT the same as carbon credit** (broader eco-action incentive, not strictly 1 tCO_2e unit).

PYQ — UPPCS Prelims 2021, Q125

The concept of 'carbon credit' originated from:

A. Earth Summit, Rio

B. Kyoto Protocol

C. Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल)

D. G-8 Summit

💡 **Show answer**

Ans: B — Kyoto Protocol's **CDM and flexible mechanisms** established carbon credit architecture.

> **Logic:** Green Credit Programme **2023 ≠ carbon credit**. Double counting avoided under Paris Article 6.

24.3 Carbon Trading

Identity: Cap-and-trade | EU ETS largest | Kyoto origin | India CCTS 2023 | PAT = ESCerts

- **Carbon trading** is the **market mechanism** to buy and sell emission **allowances or credits** under a regulatory cap.
- **Cap-and-trade:** regulator sets total emissions cap → firms get allowances → efficient firms sell surplus to high emitters → cap met at lowest cost.
- **Carbon credit vs carbon trading:** credit = **unit** (1 tCO_2e); trading = **market** exchanging those units.
- **EU Emissions Trading System (EU ETS)** is the world's **largest compliance carbon market**.
- **Kyoto flexible mechanisms:** (1) Emissions Trading among Annex I countries, (2) **CDM**, (3) Joint Implementation.
- **India Carbon Credit Trading Scheme (CCTS), 2023** under **Energy Conservation Act 2001** — **Bureau of Energy Efficiency (BEE)** (ऊर्जा दक्षता ब्यूरो) as designated authority.
- **PAT (Perform, Achieve and Trade (पण्याध्यक्ष)), 2012:** trades **energy efficiency certificates (ESCerts)** — parallel market-based compliance, **not identical to carbon credits**.
- **Compliance vs voluntary:** compliance markets are legally mandated (EU ETS, future CCTS); voluntary markets are optional corporate purchases.

PYQ — UPPCS Prelims 2024, Q35

Which international agreement is related to climate change (जलवायु परिवर्तन) mitigation?

A. Montreal Protocol

B. Ramsar (रामसर) Convention

C. Kyoto Protocol

D. CITES (साइट्स)

💡 **Show answer**

Ans: C — Kyoto = GHG targets + flexible mechanisms. Montreal = **ozone**, not climate.

Concept	What it is
Carbon credit	Unit = 1 tCO_2e
Carbon trading	Market exchanging credits/allowances
Carbon offset (कार्बन ऑफसेट)	Using credits to compensate own emissions

> **Logic:** Montreal = ozone. Kyoto/Paris = climate. PAT = **energy efficiency**, not carbon directly.

24.4 Net Zero Emissions

Identity: Emissions = removals | India 2070 COP26 | NDC 2030 interim | Emissions Gap = UNEP

- **Net zero:** anthropogenic GHG emissions are **balanced by anthropogenic removals** over a specified period — not absolute (निरपेक्ष) gross zero.
- **Gross zero** means eliminating virtually all emissions — harder and costlier than net zero.
- **India's target: net zero by 2070** — announced by PM at **COP26 Glasgow, November 2021**.
- **Interim NDC 2030 (updated 2022):** (1) reduce emissions **intensity of GDP by 45%** from 2005, (2) **50% cumulative electric capacity from non-fossil sources**, (3) additional **2.5–3 $GtCO_2e$ carbon sink (कार्बन सिंक)** through forests.
- **Carbon neutral vs net zero:** carbon neutral often means **CO_2 only**; full GHG net zero includes methane and N_2O.
- **Country comparison:** EU/UK **2050**, China **2060**, India **2070**, US **2050** (pledge).
- **UNEP Emissions Gap Report** measures gap between NDC pledges and **1.5°C/2°C pathways** — published by **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम).

Country	Net zero year
EU / UK	2050
China	2060
India	**2070**

> **Logic:** India net zero = **2070**, not 2050. Net zero ≠ no fossil fuels from tomorrow.

24.5 Green Finance

Identity: Green bonds | Sovereign Green Bonds FY 2022–23 | SEBI framework | GCF | Loss and Damage COP28

- **Green finance** mobilises capital for investments with environmental benefits — clean energy, pollution prevention, biodiversity, climate adaptation (जलवायु अनुकूलन).
- **Green bonds:** debt where proceeds fund **eligible green projects** only — renewables, clean transport, water, adaptation.
- **India's first Sovereign Green Bonds** issued in **FY 2022–23** (January 2023 tranches) for solar, wind, green hydrogen, EV public transport.
- **SEBI Green Bond Framework (2023)** defines eligible categories and disclosure — aligns with ICMA Green Bond Principles.
- **Green Climate Fund (GCF)** (जीसीएफ) operates under **UNFCCC** (यूएनएफसीसीसी) — established at **Cancun COP-16 (2010)** to support developing-country adaptation and mitigation. It is **not** a Durban creation and not founded under UNEP/OECD/ADB/World Bank.
- **100 billion/year climate finance pledge** by developed countries — committed at **COP15 Copenhagen (2009)**, reaffirmed Paris 2015.
- **Loss and Damage Fund** operationalised at **COP28 UAE (2023)** — finances recovery from climate impacts exceeding adaptation capacity.
- **Green finance ≠ Green Credit Programme** — finance = capital markets; Green Credit = eco-action incentive certificates.

Instrument	India milestone
Sovereign Green Bonds	FY **2022–23**
SEBI Green Bond Framework	**2023**
Loss and Damage Fund	**COP28 2023**

> **Logic:** Sovereign Green Bonds = **FY 2022–23**. Loss and Damage Fund = **COP28**, not COP26.

24.6 Blue Economy

Identity: SDG 14 | Sustainable ocean use | MISHTI 2023 | Sagarmala | Blue carbon

- **Blue economy** is the **sustainable use of ocean resources** for economic growth and livelihoods while preserving marine and coastal ecosystem (पारिस्थितिकी तंत्र) health.
- **SDG 14** = **Life Below Water** — conserve and sustainably use oceans, seas, and marine resources.
- India has **~2.37 million sq km EEZ** — fisheries, ports, shipping, coastal tourism, offshore wind, marine biotechnology.
- **Blue carbon:** mangroves, seagrass, salt (लवणाध्यक्ष) marshes sequester carbon faster per hectare than many terrestrial forests.
- **MISHTI** (Mangrove Initiative for Shoreline Habitats and Tangible Income) announced in **Union Budget 2023–24** — restore mangrove forests for cyclone protection and livelihoods.
- **Sagarmala:** port-led coastal development — MoPSW nodal.
- **Deep Ocean Mission:** manned submersible (Samudrayaan), deep-sea mining exploration, biodiversity survey.
- **UP is landlocked** — blue economy questions are national (MISHTI, Sagarmala), not UP coastal schemes.

PYQ — UPPCS Prelims 2025, Q93

MISHTI: (1) It was initiated in 2019. (2) It aims to restore mangrove forests.

A. Only 2

B. Neither

C. Both

D. Only 1

> 💡 **Show answer**
>
> **Ans: A** — Restoration aim correct. MISHTI = **2023–24 Budget**, not 2019.

Sector	Scheme
Mangrove restoration	MISHTI 2023
Ports/shipping	Sagarmala
Deep-sea exploration	Deep Ocean Mission

> **Logic:** MISHTI = **coastal mangroves**, not Ganga basin (गंगा). Blue economy = **SDG 14**.

24.7 LiFE Mission

Identity: Lifestyle for Environment | MoEFCC | COP26 idea 2021 | Launched June 2022 | 75 actions

- **LiFE (Lifestyle for Environment)** is an India-led movement for **individual and community pro-planet behaviour change** — mindful consumption over destructive consumption.
- **Idea announced at COP26, Glasgow, November 2021** by the PM — complement to government climate policy; individuals as change agents for SDG 12 and SDG 13.
- **Mission LiFE formally launched in June 2022** — **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is nodal ministry.

- **75 pro-planet actions:** save energy, reduce waste, save water, sustainable food, e-waste disposal, tree plantation — via **Mission LiFE app/portal**.
- **Mechanism:** nudge and awareness (not regulatory law) — reduces lifestyle carbon footprint from food, transport, household energy.
- Supports **net zero 2070** through demand-side emission reduction alongside technology deployment.

PYQ — UPPCS Prelims 2025, Q139

LiFE: (1) It was launched in June 2022. (2) The idea was proposed at COP-25 in Madrid, Spain.

A. Only 2

B. Neither

C. Both

D. Only 1

💡 **Show answer**

Ans: D — Launch **June 2022** true. Idea at **COP26 Glasgow 2021**, not COP-25 Madrid 2019.

Field	Detail
Idea	COP26 Glasgow **2021**
Launch	**June 2022**
Nodal	**MoEFCC**
Actions	**75** pro-planet behaviours

> **Logic:** LiFE = **COP26 2021 / launch 2022** — COP-25 Madrid trap is **FALSE**.

Complete PYQ Bank (Topic 24)

Q1. UPPCS Prelims 2025, Q139

With reference to the LiFE (Lifestyle for Environment) Mission, consider the following statements:

1. It was launched in June 2022.
2. The idea was proposed at COP-25 in Madrid, Spain.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: LiFE launch June 2022 is true; idea came at **COP26 Glasgow 2021**, not COP-25 Madrid. **Ans: D (Only 1).** **Launch June 2022** is correct; COP-25 Madrid claim is false. **Trap:** **COP-25 Madrid 2019** instead of **COP26 Glasgow 2021**.

Q2. UPPCS Prelims 2025, Q93

With reference to MISHTI, consider the following statements:

1. It was initiated in 2019.

2. It aims to restore mangrove forests.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

Show answer

Logic: MISHTI = Union Budget **2023–24** mangrove scheme — restoration aim is correct. **Ans: A (Only 2).** Only the **mangrove restoration** aim is correct. **Trap:** **2019** start year — MISHTI is a **2023–24 Budget** announcement.

Q3. UPPCS Prelims 2024, Q35

Which international agreement is related to climate change mitigation?

A. Montreal Protocol

B. Ramsar Convention

C. Kyoto Protocol

D. CITES

Show answer

Logic: Kyoto/Paris = GHG mitigation; Montreal = ozone/ODS. **Ans: C.** The **Kyoto Protocol** addresses climate change mitigation. **Trap:** **Montreal Protocol** — ozone treaty, not climate.

Q4. UPPCS Prelims 2024, Q6

Match List-I with List-II:

List-I (Report)	List-II (Organisation)
A. Human Development Report	1. IMF
B. World Economic Outlook	2. UNEP
C. World Investment Report	3. UNDP
D. Emissions Gap Report	4. UNCTAD

Row order is not the answer code.

A. 3 1 4 2

B. 3 4 1 2

C. 1 2 3 4

D. 2 3 4 1

Show answer

Logic: Emissions Gap = UNEP product in net-zero/emissions-gap teaching block. **Ans: A (3 1 4 2).** Emissions Gap Report → **UNEP**; HDR → **UNDP**. **Trap:** IPCC or UNDP as Emissions Gap publisher.

Q5. UPPCS Prelims 2021, Q125

The concept of ‘carbon credit’ originated from:

A. Earth Summit, Rio

B. Kyoto Protocol

C. Montreal Protocol

D. G-8 Summit

💡 **Show answer**

Logic: Kyoto flexible mechanisms (CDM/CERs) established tradable carbon credit architecture. **Ans: B.** **Carbon credits originated from the Kyoto Protocol** mechanisms. **Trap:** **Rio 1992** (UNFCCC framework) without Kyoto's CDM detail.

Q6. UPPCS Prelims pattern — carbon credit unit

One carbon credit typically represents reduction or removal of:

A. One kilogram of CO_2

B. One tonne of CO_2 equivalent (tCO_2e)

C. One megatonne of methane only

D. One hectare of forest cover

💡 **Show answer**

Logic: Standard market unit = 1 tCO_2e per credit certificate. **Ans: B.** **One carbon credit = one tonne of CO_2 equivalent**. **Trap:** Kilogram scale or hectare land unit confusion.

Q7. UPPCS Prelims pattern — CDM

The Clean Development Mechanism (CDM) was introduced under:

A. Montreal Protocol

B. Kyoto Protocol

C. Ramsar Convention

D. CITES

💡 **Show answer**

Logic: CDM = Kyoto flexible mechanism for project-based credits in developing countries. **Ans: B.** **CDM operates under the Kyoto Protocol**. **Trap:** Paris Agreement (CDM era is Kyoto-linked in standard stems).

Q8. UPPCS Prelims pattern — PAT scheme

Perform, Achieve and Trade (PAT) is a market-based scheme for trading:

A. Carbon credits identical to EU ETS

B. Energy efficiency certificates (ESCerts)

C. Wildlife trade permits

D. Ozone-depleting substance quotas

💡 **Show answer**

Logic: PAT = BEE-administered energy efficiency trading — parallel but not identical to carbon credits. **Ans: B.** **PAT trades energy efficiency certificates (ESCerts)**. **Trap:** Calling PAT a **carbon credit**

scheme directly.

Q9. UPPCS Prelims pattern — India net zero

India announced a target to achieve net-zero emissions by:

A. 2050

B. 2060

C. 2070

D. 2030

💡 **Show answer**

Logic: COP26 Glasgow 2021 announcement — India net zero **2070**. **Ans: C.** **India's net-zero target year is 2070**. **Trap:** **2050** (EU/UK/US pledge timeline) applied to India.

Q10. UPPCS Prelims pattern — SDG 14

Sustainable Development Goal 14 (SDG 14) focuses on:

A. Life on Land

B. Life Below Water

C. Climate Action

D. Responsible Consumption

💡 **Show answer**

Logic: Blue economy chapter links MISHTI/Sagarmala to SDG 14 oceans goal. **Ans: B.** **SDG 14 = Life Below Water**. **Trap:** SDG 15 (Life on Land) or SDG 13 (Climate Action).

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to microplastics, which of the following statements is/are correct?

1. Plastic fragments smaller than 5 mm.
2. Primary microplastics come only from fragmentation of larger plastics; secondary ones are only factory microbeads.
3. They are always rapidly biodegradable in hours.

A. Only 1 B. 1 and 2 C. Only 3 D. 2 and 3

💡 **Show answer**

Ans: A. Only 1 is correct. **Logic:** Statement 1 is true (<5 mm). Statement 2 swaps primary (microbeads/pellets) and secondary (fragmentation). Statement 3 is false — they are not rapidly biodegradable in hours.

Q2. Match List-I with List-II:

List-I	List-II
A. 1 carbon credit	1. 2023 eco-action rewards
B. PAT	2. 1 tCO_2e
C. Green Credit	3. ESCerts

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-3, C-1. **Logic:** Credit/PAT/Green Credit.

Q3. Which of the following pairs is/are correctly matched?

1. India net zero — 2070 (COP26 Glasgow)
2. China often — 2052 teaching
3. Green Credit — identical to Kyoto CER always

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

💡 **Show answer**

Ans: B. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): PAT creates energy-efficiency certificates (ESCerts).

Reason (R): PAT is identical to carbon credits under CCTS.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** PAT ≠ carbon credit.

Q5. With reference to LiFE, which of the following statements is/are correct?

1. Idea pitched at COP26 Glasgow 2021.
2. Launched June 2022.
3. Idea first at COP25 Madrid.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not COP25.

Q6. Arrange MISHTI year:

A. 2019 B. Budget 2023 C. 1972 D. 2006

💡 **Show answer**

Ans: B. Budget 2023. **Logic:** Mangrove initiative.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Blue economy — sustainable ocean use; SDG 14
2. Blue carbon — coastal/ocean sinks
3. Blue economy — identical to Schedule VI plants

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Not Schedule VI.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Net zero balances remaining emissions with removals.

Reason (R): India's year fact is 2070.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; 2070 does not define the meaning of net zero globally. **A/R logic:** Definition vs India year.

Q9. With reference to carbon markets, which of the following statements is/are correct?

1. Carbon trading buys/sells allowances or credits.
2. EU ETS is the largest compliance market teaching.
3. Kyoto CDM produced CERs.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Market facts.

Q10. Match List-I with List-II:

List-I	List-II
A. SUP ban push	1. Lifestyle mission
B. LiFE	2. ~2022
C. Loss & Damage Fund	3. Climate finance politics

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** SUP/LiFE/L&D.

Q11. Arrange Scope 1/2/3 idea:

A. Scope 3 direct only B. Scope 1 direct; 2 purchased energy; 3 value chain C. All scopes mean only methane from cows in one farm D. Scopes are IUCN categories

💡 **Show answer**

Ans: B. 1 direct / 2 energy / 3 chain. **Logic:** Footprint scopes.

Q12. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Green Credit Programme (2023) rewards eco-actions.

Reason (R): It is the same instrument as a Kyoto-style carbon credit.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Different instruments.

Q13. With reference to climate finance, which of the following statements is/are correct?

1. $100 billion a year pledge is developed-to-developing politics fact.
2. Loss and Damage Fund operationalisation is current teaching.
3. Climate finance means only GST compensation.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Not GST.

Q14. Which one of the following correctly states India's NDC neighbourhood teaching?

A. Only Article 370 restoration B. Only ozone hole targets C. Only Eighth Schedule expansion D. Intensity cut + non-fossil capacity share toward 2030

💡 **Show answer**

Ans: D. Intensity + non-fossil share. **Logic:** NDC track.

Q15. How many of the following are true?

Microplastics <5 mm; enter food webs; SUP bans ~2022 push

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Plastics trio.

Q16. With reference to green finance, which of the following statements is/are correct?

1. Capital for climate/environment projects.
2. Includes green bonds / climate funds / GCF neighbourhood.
3. India issued Sovereign Green Bonds teaching in recent years.

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

💡 **Show answer**

Ans: B. All three. **Logic:** Green finance set.

Q17. Match List-I with List-II:

List-I	List-II
A. Net zero India	1. 2023
B. Green Credit	2. 2070
C. MISHTI Budget	3. Mangroves

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** 2070 / 2023 / mangroves.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Circular economy closes material loops through 3R/9R and EPR teaching.

Reason (R): It links to SDG 12 neighbourhood.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; SDG 12 association does not fully define circular loops by itself. **A/R logic:** Related SD framing.

Q19. With reference to climate refugees, which of the following statements is/are correct?

1. Mostly internally displaced teaching.
2. Not a separate legal category under 1951 Refugee Convention teaching.
3. Automatically get UN Security Council permanent seats.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Statement 3 false.

Q20. Arrange LiFE chronology:

A. Idea COP26 2021 → launch June 2022 B. Launch 2021 → idea 2022 C. Both 2019 D. Both 2005

💡 **Show answer**

Ans: A. 2021 idea → 2022 launch. **Logic:** LiFE years.

Q21. With reference to circular economy neighbourhood, which of the following statements is/are correct?

1. Closes material loops through 3R/9R and EPR teaching.
2. Links to SDG 12 neighbourhood.
3. Replaces the Wildlife (Protection) Act.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Circular economy ≠ WPA repeal.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Microplastics enter food webs from plankton to fish to humans.

Reason (R): Primary microplastics include microbeads.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; microbead source does not fully explain food-web transfer. **A/R logic:** Source vs pathway.

Q23. With reference to Sagarmala / Deep Ocean Mission neighbourhood, which of the following statements is/are correct?

1. Sit beside blue–coastal policy teaching with MISHTI.
2. Link to ocean/coastal economy.
3. They repeal the Wildlife (Protection) Act.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Do not repeal WPA.

Q24. How many of the following instruments are distinct?

Carbon credit, PAT ESCert, Green Credit

A. Only one B. Only two C. None exist D. All three distinct

💡 **Show answer**

Ans: D. All three distinct. **Logic:** Do not merge.

Q25. Which one of the following correctly states 1 carbon credit?

A. 1 tCO_2e reduced/removed/avoided B. 1 hectare forest only always C. 1 IUCN category D. 1 Dobson Unit

💡 **Show answer**

Ans: A. 1 tCO_2e. **Logic:** Credit definition.

Q26. With reference to EU ETS, which of the following statements is/are correct?

1. Largest compliance carbon market teaching.
2. Cap-and-trade neighbourhood.
3. Identical to India's Forest Rights Act.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not FRA.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): India's identified SUP bans tightened around 2022 teaching.

Reason (R): SUP feeds microplastic pathways.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains policy push rationale. **A/R logic:** SUP → microplastics.

Q28. Arrange China / India net-zero teaching years:

A. India 2060; China 2070 B. Both 2050 only always C. China ~2060; India 2070 D. Both 1992

> 💡 **Show answer**
>
> **Ans: C.** China ~2060; India 2070. **Logic:** Year pair.

Q29. With reference to MoEFCC LiFE, which of the following statements is/are correct?

1. Behaviour-change / lifestyle mission.
2. Catalogues pro-planet citizen actions.
3. Is a carbon-market statute identical to EU ETS.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Not ETS statute.

Q30. Which of the following statements about voluntary vs compliance markets is/are correct?

1. Compliance markets include ETS teaching.
2. Voluntary markets use standards/credits outside mandatory caps.
3. Both are identical to Biosphere Reserve cores.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1 and 2. **Logic:** Not BR cores.

Common Traps

Trap	Correct
LiFE idea at COP-25 Madrid	**COP26 Glasgow 2021**
LiFE launched 2019	**June 2022**
MISHTI launched 2019	**2023–24 Budget**
India net zero 2050	**2070**
Microplastics <5 cm	**<5 mm**
Microplastics biodegrade quickly	**Persist decades**
Carbon credit = carbon trading	**Unit vs market**
Green Credit = carbon credit	**Different schemes**
Green finance = Green Credit	**Capital vs incentive certificate**
Montreal = climate treaty	**Ozone — Kyoto/Paris = climate**

Trap	Correct
PAT trades carbon credits directly	**ESCerts — energy efficiency**
Net zero = gross zero tomorrow	**Balanced with removals**
MISHTI for Ganga/UP	**Coastal mangroves**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 25 — Global Environmental Geography

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Global Environmental Geography (biomes · climate zones · world deserts · floral regions · hotspots)

Sources baked in: NCERT Geo 11; Bio 12 Ch 15–16; Köppen; Myers hotspots; UPPCS PYQs 2018–2025 **Weight:** ★★ Medium–High — biome matching; desert countries; Brazil Selva; tribe–habitat traps **Last verified:** August 2026

Current Affairs (this topic)

- Biome (जीवोम) and realm stems stay geographic; CA rarely moves the Köppen (कोपेन) or biome map.
- Bridge biome teaching with Ecology (पारिस्थितिकी) & Ecosystem (पारिस्थितिकी तंत्र) without duplicating full cards.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. Global environmental geography asks **where and why** Earth's climate–biome patterns occur. Climate (temperature and rainfall) shapes biomes, biodiversity, and degradation belts. 2. Major biomes to ratta: tropical rainforest (Amazon **Selva** (सेल्वा)), tropical deciduous (Indian monsoon forests), desert (**<25 cm** rain), savanna (सवाना) (scattered trees with wet–dry seasons), temperate grassland (**Prairie (प्रेयरी), Steppe (स्टेपी), Pampas, Puszta (पुस्ज़टा)**), taiga/boreal, tundra (टुंड्रा), Mediterranean chaparral, and coastal mangrove as a transitional belt. 3. **Köppen** letters: **A** tropical, **B** arid, **C** temperate, **D** cold, **E** polar. India is largely monsoon (**Am** type language); Thar (थार) is hot desert (**BWh**); Gobi is cold desert (**BWk**). 4. **Sahara** is the largest hot desert. **Atacama** is among the driest. **Gobi** spans **Mongolia and China**. 5. Australia's interior desert is explained mainly by **subtropical high pressure and continentality** — not by calling northern Australia "temperate." 6. Brazil facts: **Selva** = Amazon equatorial rainforest name; **Terra Roxa** = fertile dark volcanic soil. Both can be true together in match stems. 7. Floral matching classics: Planting Hearth → **C.O. Sauer**; World Floral Regions → **Adolf Engler**; Cryophyte Steppe → Arctic tundra setting; **Puszta** → **Hungary**. 8. Tribe–region facts: Bushman–Kalahari; Eskimo–northern Canada; Pygmy–Congo. **Masai are East Africa**, not West Africa. 9. **Permafrost** (स्थायी हिम मृदा) appears in **taiga and tundra**. It is **not** a savanna feature. 10. Global biome counts (often **14** in textbook lists) are **not** the same as India's **ten** biogeographic zones (Rodgers and Panwar). 11. Biome is a large climate-defined vegetation–fauna (प्राणीजात) region. Ecosystem is a local biotic–abiotic functional unit. Do not treat them as identical scales. 12. Temperate grassland names: Pampas (Argentina), Prairie (North America), Steppe (Eurasia), Puszta (Hungary). 13. Taiga is boreal conifer forest; tundra is moss–lichen cold treeless belt. Savanna is tropical grassland with trees. 14. Mangrove is coastal transitional vegetation — not a purely inland terrestrial biome. 15. Hotspot (Myers) means high endemism plus severe habitat (वास स्थान) loss (often taught as **≥70%** loss). It is not merely "any rich forest." 16. Gobi is cold desert Asia — not Sahara Africa and not Atacama Chile. 17. Selva ≠ Terra Roxa. Forest name versus soil name. 18. Masai wrong-option trap is West Africa; correct region is East Africa. 19. India's monsoon deciduous belt is not Amazon Selva. 20. Permafrost continuous cover is strongest in tundra; taiga may show patches — still not savanna. 21. Mediterranean biome is winter-rain shrubland (chaparral type), distinct from monsoon India. 22. Desert rainfall fact **<25 cm** is the quick arid filter in many MCQs. 23. Engler floral regions and Sauer planting hearth are

cultural–biogeography matchs, not climate-classification letters. 24. Keep “14 biomes vs 10 Indian biogeographic zones” as a numbers trap pair.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Biome** | **Ecosystem** | Continental-scale geographical region characterized by distinct climate regimes, soil types, and climax vegetation communities (e.g. Tundra, Taiga, Savanna) vs localized structural and functional ecological unit where living organisms interact with abiotic components (e.g. a pond, forest patch) | जीवोम (बायोम, महाद्वीपीय स्तर) / पारिस्थितिकी तंत्र (स्थानीय कार्यात्मक इकाई) | | **Taiga (Boreal Forest)** | **Tundra** | World's largest terrestrial biome dominated by evergreen needleleaf coniferous trees (Spruce, Fir, Pine, Larch) in subarctic latitudes south of the tree-line vs cold, treeless arctic and alpine biome dominated by mosses, lichens, dwarf shrubs, underlain by permafrost | टैगा (बोरियल शंकुधारी वन) / टुंड्रा (वृक्षविहीन शीत मरुस्थल) | | **Savanna** | **Steppe / Prairie** | Tropical grassland biome characterized by continuous tall grass cover interspersed with scattered, drought-deciduous drought-resistant trees (e.g. Acacia, Baobab) vs temperate mid-latitude semi-arid grassland biome with treeless expanses of short to tall grasses | सवाना (उष्णकटिबंधीय सवृक्ष घास मैदान) / स्टेपी/प्रेयरी (शीतोष्ण घास मैदान) | | **Selva** | **Puszta** | Dense equatorial tropical rainforest biome of the Amazon River basin characterized by evergreen multi-tiered canopy and extreme biodiversity vs temperate flat grassland plain of the Pannonian Basin in Hungary characterized by continental climate and fertile chernozem soils | सेल्वा (अमेज़न वर्षावन) / पुस्ज़टा (हंगरी शीतोष्ण घास मैदान) | | **Permafrost** | **Mollisols (Chernozem)** | Subsurface ground (soil, sediment, or rock) that remains continuously frozen below 0°C for at least two consecutive years, trapping massive ancient methane and organic carbon deposits vs deep, dark, humus-rich organic soils characteristic of temperate grasslands with highest agricultural fertility | स्थायी हिम मृदा (पर्माफ्रॉस्ट) / चेर्नोज़म (काली उपजाऊ घास मृदा) | | **Biodiversity Hotspot** | **Ecological Island** | Large biogeographic region meeting Norman Myers' criteria (≥1,500 endemic plant species and ≥70% primary habitat lost) vs isolated terrestrial habitat surrounded by completely different ecosystems or human urbanization functioning like an oceanic island | जैव विविधता हॉटस्पॉट (ह्रासग्रस्त समृद्ध क्षेत्र) / पारिस्थितिक द्वीप (पृथक वास) | ### Biome ↔ ecosystem bridge (read with Ecology chapter) - A **biome** is the large climate-defined vegetation–fauna belt (Amazon Selva, taiga, savanna). - An **ecosystem** (पारिस्थितिकी तंत्र) is the local functional unit of energy flow and nutrient cycling inside that belt. - Global textbook biome counts (often **~14**) are **not** India’s **ten** biogeographic zones (Rodgers and Panwar). - UP plains sit mainly in the **Gangetic Plain** biogeographic colour with Terai (तराई)–deciduous transitions — not a tropical rainforest biome answer.

💡 📋 **Must-score facts — biomes, Köppen, grasslands (Click to Expand)**

Biomes / climate | Item | Lock | |---|---| | Köppen letters | A tropical · B arid · C temperate · D cold · E polar | | India / Thar / Gobi | Monsoon Am-type / BWh hot desert / Mongolia–China | | Largest hot / driest | Sahara / Atacama | | Selva / Terra Roxa | Amazon rainforest name / fertile volcanic soil (Brazil) | | Temperate grasslands | Pampas AR · Prairie NA · Steppe Eurasia · Puszta HU | | Taiga / Tundra / Savanna | Boreal conifer / moss–lichen treeless / tropical grass+trees | | Permafrost | Taiga–tundra (≠ savanna) | | Biome ≠ ecosystem | Large climate region / local functional unit | | India biogeographic zones | **10** (Rodgers & Panwar) ≠ global biome count | --- ![Global Biomes Distribution, Climatic Pressure Belts & World Deserts](images/env_ch25_01_global_biomes_climatic_zones_geography.png)

25.1 Scope & Climate Controls

Identity: Spatial environment (पर्यावरण) patterns | Climate primary control | Latitude + continentality + altitude

- **Global environmental geography** studies *where* environmental systems occur and *why* — linking physical geography (climate, relief, oceans) to biomes, biodiversity, and degradation.
- **Climate** (temperature + precipitation) is the primary control on biome distribution.

- **Latitude:** equatorial regions receive direct sunlight year-round → high productivity and biodiversity; polar regions receive oblique rays → low productivity.
- **Continentality:** interior continents (central Australia, central Asia) have extreme temperatures and aridity.
- **Altitude:** Himalaya (हिमालय), Andes, Alps show vertical zonation — tropical forest at base, alpine/nival at summit.
- **Ocean currents:** warm currents (Gulf Stream) mild climate; cold currents (Humboldt) support fisheries and coastal deserts (Atacama).
- **Subtropical highs (horse latitudes (हॉर्स अक्षांश) ~20–35° N/S)** suppress rainfall — major desert belt (Sahara, Arabian, Thar, Australian interior).
- **Latitudinal biodiversity gradient:** species richness highest near equator, decreases toward poles.

PYQ — UPPCS Prelims 2025, Q1

Assertion (A): The interior part of Australia is desert and semi-desert. Reason (R): Northern Australia is situated in the temperate zone.

Options: A. Both true; R does not explain A B. A false, R true C. A true, R false D. Both true; R explains A

<details>
<summary>💡 Show answer</summary>

Ans: C — Interior aridity from **subtropical high + continentality**. Northern Australia is largely **tropical**, not temperate.
</details>

> **Logic:** Australia desert ≠ temperate-zone explanation.

25.2 Major Global Biomes

Identity: Climate-defined regions | Rainforest → desert → grassland → taiga → tundra | Mangrove transitional

- A **biome** is a large region with characteristic vegetation and fauna shaped by climate.
- **Tropical rainforest:** hot, wet year-round; layered canopy; highest biodiversity — Amazon (**Selva**), Congo, SE Asia, Western Ghats (पश्चिमी घाट).
- **Tropical deciduous:** wet–dry monsoon — sal/teak forests of peninsular India.
- **Desert:** typically **<25 cm** rain; sparse xerophytes — Sahara, Arabian, Gobi, Thar, Atacama.
- **Savanna:** tropical grassland with **scattered trees** and distinct wet/dry seasons — East Africa ("Land of Big Games"); Gir (गीर) (Gujarat) is India patch.
- **Temperate grassland:** few trees; fertile soils — Prairies (N America), Steppes (Eurasia), **Pampas (Argentina)**, **Puszta (Hungary)**.
- **Taiga (boreal):** cold winters; spruce/fir/pine; **permafrost patches** — Siberia, Canada, Scandinavia.
- **Tundra:** short summer; moss/lichen; **continuous permafrost** — Arctic coasts; no trees.
- **Mediterranean (chaparral):** wet winters, dry summers; fire-adapted shrubs — California, Mediterranean Basin, Cape Town.
- **Mangrove / wetland:transitional** terrestrial–aquatic — not purely terrestrial.

Biome	Climate	Global example
Tropical rainforest	Hot wet year-round	Amazon Selva
Desert	Arid <25 cm	Sahara, Gobi, Thar
Savanna	Wet–dry grassland + trees	East Africa
Temperate grassland	Moderate (नरम दल) rain	Pampas, Puszta

Biome	Climate	Global example
Taiga	Cold coniferous	Siberia, Canada
Tundra	Very cold, permafrost	Arctic

> **Logic:** Permafrost = Taiga + Tundra**, not Savannah.** 14 global biomes ≠ 10 Indian biogeographic zones**.

25.3 Climate Zones (Köppen)

Identity: A–B–C–D–E | India Am | Horse latitudes = deserts | ITCZ drives monsoon

- **Köppen Group A (Tropical):** all months ≥18°C — Af rainforest, **Am monsoon**, Aw savanna. India mainly **Am**.
- **Group B (Arid):** evaporation > precipitation — BW desert, BS steppe. Thar **BWh** (hot); Gobi **BWk** (cold).
- **Group C (Temperate):** mild winters — Mediterranean Csa, humid subtropical Cwa.
- **Group D (Cold/Continental):** cold winters — taiga belt Siberia/Canada.
- **Group E (Polar):** always cold — tundra ET, ice cap EF (Antarctica).
- **ITCZ** seasonal shift drives monsoon and equatorial rain; failure causes Sahel drought (सूखा).
- Same latitude can differ: cold west-coast currents → aridity; warm east coasts → wetter.

Group	Name	India / example
A	Tropical	India Am
B	Arid	Thar BWh, Gobi BWk
C	Temperate	Mediterranean climates
D	Cold	Siberia taiga
E	Polar	Antarctica

> **Logic:** "India has Af rainforest climate everywhere" = **FALSE** — India is mainly **Am**.

25.4 World Deserts & Critical Ecosystems

Identity: Sahara largest hot | Atacama driest | Gobi Mongolia+China | Brazil Selva + Terra Roxa

- **Sahara (Africa):** largest **hot desert**.
- **Atacama (Chile):** among the **driest** deserts on Earth — cold Humboldt Current + rain shadow (छाया).
- **Gobi:** cold desert extending mainly into **Mongolia and China**.
- **Kalahari:** southern Africa — **Bushman (San)** habitat.
- **Thar:** India–Pakistan hot desert — subtropical high + rain shadow.
- **Australian interior:** desert/semi-desert under subtropical high and continentality.
- **Amazon (Brazil):** equatorial rainforest called **Selva** — among Earth's richest biodiversity zones.
- **Terra Roxa:** fertile dark volcanic (basalt) soil of Brazil — coffee/soy region (often tested with Selva).

PYQ — UPPCS Prelims 2025, Q27

The Gobi Desert extends into: (1) Mongolia (2) China (3) Russia (4) Kazakhstan

Options: A. Only 1 and 2 B. Only 1, 2 and 4 C. Only 1, 2 and 3 D. Only 1

💡 **Show answer**

Ans: A — Primary extent = **Mongolia + China**.

PYQ — UPPCS Prelims 2025, Q63

Brazil: (1) Equatorial rainforests called “Selva”. (2) Fertile dark soil known as “Terra Roxa”.

Options: A. Only 2 B. Neither C. Both D. Only 1

💡 **Show answer**

Ans: C — Both statements correct.

PYQ — UPPCS Prelims 2025, Q23

Tribe–habitat NOT correctly matched: Bushman–Kalahari; Eskimo–N Canada; Pygmy–Congo; Masai–West Africa

💡 **Show answer**

Ans: B (Only 4) — Masai live in **East Africa** (Kenya/Tanzania), not West Africa.

Desert	Region	Fact
Sahara	N Africa	Largest hot desert
Atacama	Chile	Driest
Gobi	Mongolia + China	Cold desert
Kalahari	S Africa	Bushman/San
Thar	India–Pakistan	Hot desert

> **Logic:** Pampas = **Argentina** (2018). Land of Big Games = **Tropical Savanna**.

25.5 Floral Regions & Vegetation Matching

Identity: Engler floral regions | Sauer planting hearth | Puszta Hungary | Vegetation–region pairs

- **World Floral Regions** classification is associated with **Adolf Engler**.
- **Planting Hearth** (agricultural origin centres) is associated with **C.O. Sauer**.
- **Cryophyte Steppe** = cold-region steppe vegetation of the **Arctic Tundra** belt.
- **Puszta Grassland** = temperate grassland of **Hungary**.
- Classic vegetation–region pairs: **Epiphytes–Equatorial**; **Acacia–Savanna**; **Baobab–Savanna**; **Cedars–Mediterranean**.

PYQ — UPPCS Prelims 2024, Q29

Match: A. Planting Hearth B. Cryophyte Steppe C. World Floral Regions D. Puszta Grassland With: 1. Hungary 2. Adolf Engler 3. C.O. Sauer 4. Arctic Tundra

Options: A. 3 4 2 1 B. 2 1 4 3 C. 3 4 1 2 D. 2 3 4 1

💡 **Show answer**

Ans: A — Sauer (3), Arctic Tundra (4), Engler (2), Hungary (1).

Pair	Fact
Planting Hearth	C.O. Sauer
World Floral Regions	Adolf Engler

Pair	Fact
Cryophyte Steppe	Arctic Tundra
Puszta	Hungary
Pampas	Argentina

> **Logic:** Do not swap Engler ↔ Sauer.

Eco-Geographical Adaptation Rules & Ecotone Edge Effect

Eco-Geographical Rules & Ecotone Dynamics

- **Bergmann's rule:** Body size in endotherms increases with colder climate (higher latitude) to reduce heat loss per unit volume (e.g., Polar Bear > Sun Bear).
- **Allen's rule:** Extremities and appendages (ears, limbs, snouts) are shorter and more compact in colder climates (e.g., Arctic Fox short ears vs Desert Fennec Fox large ears).
- **Gloger's rule:** Endotherms in warm, humid climates develop darker melanin pigmentation (eumelanin) than those in cold, arid regions.
- **Jordan's rule:** Marine fish developing in colder, high-latitude waters have more vertebrae than relatives in warm tropical waters.
- **Ecotone (संक्रमिका):** A transition zone where two biomes overlap (e.g., Mangrove between sea and land; Estuary between river and ocean).
- **Edge effect (किनारा प्रभाव):** Greater species diversity and population density occur in the ecotone than in either adjacent community alone, harboring unique "edge species".

25.6 Biodiversity Hotspots & Degradation Belts

Identity: Myers criteria | 36 hotspots | 4 in India | Pollution corridors

- **Biodiversity hotspot (जैव विविधता हॉटस्पॉट) (Myers):** high **endemism** + ≥**70%** natural habitat loss.
- About **36** global hotspots. **Four (चातुर्याम) in India:** Himalaya, Western Ghats–Sri Lanka, Indo-Burma (NE), Sundaland (Nicobar (निकोबार)).
- **Megadiversity countries:** 17 globally including India, Brazil, Indonesia, China, Australia.
- **Global degradation belts:** Indo-Gangetic air pollution corridor; Amazon deforestation arc; Niger Delta (डेल्टा) oil pollution; plastic ocean gyres (Great Pacific Garbage Patch).
- Environmental problems are often **transboundary** — Saharan dust to Europe, Amazon fires affecting global CO_2, plastic gyres crossing EEZs.

India hotspot	Region
Himalaya	Mountain endemism
Western Ghats	Peninsular rainforest
Indo-Burma	NE India
Sundaland	Nicobar

> **Logic:** Hotspot needs **both** endemism and habitat loss — not richness alone.

Complete PYQ Bank (Topic 25)

Q1. UPPCS Prelims 2025, Q1

Assertion (A): The interior part of Australia is desert and semi-desert.

Reason (R): Northern Australia is situated in the temperate zone.

A. Both (A) and (R) are true and (R) is the correct explanation of (A)

B. Both (A) and (R) are true, but (R) is not the correct explanation of (A)

C. (A) is true, but (R) is false

D. (A) is false, but (R) is true

💡 **Show answer**

A/R logic: (A) on interior aridity is true. (R) is false — northern Australia is largely **tropical**, not temperate. **Ans: C.** Interior desert from **subtropical high + continentality**, not temperate-zone location. **Trap:** Accepting northern Australia as **temperate** to explain desert.

Q2. UPPCS Prelims 2025, Q23

Consider the following tribe–habitat pairs:

1. Bushman — Kalahari
2. Eskimo — Northern Canada
3. Pygmy — Congo basin
4. Masai — West Africa

Which pair(s) is/are **NOT** correctly matched?

A. Only 1

B. Only 4

C. Only 2

D. Only 3

💡 **Show answer**

Logic: Masai are East African (Kenya/Tanzania), not West Africa — only the fourth pair is wrong. **Ans: B (Only 4).** **Masai–West Africa** is the incorrectly matched pair. **Trap:** Bushman–Kalahari or Pygmy–Congo — both are standard correct pairs.

Q3. UPPCS Prelims 2025, Q27

The Gobi Desert extends into:

1. Mongolia
2. China
3. Russia
4. Kazakhstan

A. Only 1 and 2

B. Only 1, 2 and 4

C. Only 1, 2 and 3

D. Only 1

💡 **Show answer**

Logic: Gobi cold desert spans Mongolia–China core; Russia/Kazakhstan are marginal distractors. **Ans: A.** **Gobi extends mainly into Mongolia and China**. **Trap:** Adding **Russia** or **Kazakhstan** as primary extent countries.

Q4. UPPCS Prelims 2025, Q63

With reference to Brazil, consider the following statements:

1. Equatorial rainforests there are called “Selva”.
2. Fertile dark soil known as “Terra Roxa” is found there.

A. Only 1

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 2

💡 **Show answer**

Logic: Amazon Selva + basaltic Terra Roxa coffee/soy belt — both standard Brazil geography facts. **Ans: C (Both 1 and 2).** **Selva** and **Terra Roxa** statements are both correct. **Trap:** Terra Roxa confused with **latosol/terra firme** generic labels only.

Q5. UPPCS Prelims 2024, Q29

Match List-I with List-II:

List-I	List-II
A. Planting Hearth	1. Hungary
B. Cryophyte Steppe	2. Adolf Engler
C. World Floral Regions	3. C.O. Sauer
D. Puszta Grassland	4. Arctic Tundra

Row order is not the answer code.

A. 3 4 2 1

B. 2 1 4 3

C. 3 4 1 2

D. 2 3 4 1

💡 **Show answer**

Logic: Engler = floral regions; Sauer = planting hearth; Puszta = Hungary; cryophyte steppe = Arctic tundra. **Ans: A (3 4 2 1).** Sauer (3), Arctic Tundra (4), Engler (2), Hungary (1). **Trap:** Swapping **Engler** and **Sauer**.

Q6. UPPCS Prelims 2019, Q10

Which one of the following natural regions is known as the “Land of Big Games”?

A. Temperate Grassland

B. Tropical Monsoon Region

C. Hot Desert Region

D. Tropical Savanna Region

💡 **Show answer**

Logic: African savanna (Serengeti-type) = big-game grassland–woodland mosaic. **Ans: D.** **Tropical Savanna** is the Land of Big Games. **Trap:** **Temperate grassland** (Pampas/Prairie) or **hot desert** safari confusion.

Q7. UPPCS Prelims 2019, Q3

Match List-I with List-II:

List-I (Natural Vegetation)	List-II (Region)
A. Epiphytes	1. Mediterranean
B. Acacia	2. Equatorial
C. Baobab	3. Sahara
D. Cedars	4. Savanna

Row order is not the answer code.

A. 2 3 4 1

B. 2 3 1 4

C. 2 4 3 1

D. 2 4 1 3

💡 **Show answer**

Logic: Epiphytes–equatorial; acacia–Sahara fringe; baobab–savanna; cedars–Mediterranean. **Ans: A (2 3 4 1).** Code **2 3 4 1** for equatorial–Sahara–savanna–Mediterranean pairs. **Trap:** Baobab on Sahara or acacia on equatorial rainforest.

Q8. UPPCS Prelims 2018, Q26

Pampas grassland is located mainly in:

A. Argentina

B. Hungary

C. Australia

D. Kazakhstan

💡 **Show answer**

Logic: Temperate grassland names — Pampas = Argentina; Puszta = Hungary; Prairie = North America. **Ans: A.** **Pampas grassland is in Argentina**. **Trap:** **Hungary** (Puszta) or **Kazakhstan** (steppe).

Q9. UPPCS Prelims pattern — permafrost

Permafrost is characteristically found in which biomes?

A. Tropical rainforest and monsoon forest

B. Taiga and tundra

C. Hot desert and Mediterranean scrub

D. Tropical savanna only

💡 **Show answer**

Logic: Cold high-latitude/altitude belts — boreal forest with frozen ground patches and continuous tundra permafrost. **Ans: B.** **Permafrost occurs in taiga and tundra**. **Trap:** **Savanna** or **desert** — no continuous permafrost.

Q10. UPPCS Prelims pattern — biodiversity gradient

Species diversity generally _____ from the equator towards the poles.

A. increases

B. decreases

C. remains constant

D. peaks at mid-latitudes only

💡 **Show answer**

Logic: Latitudinal diversity gradient — richest near equator, poorer toward poles. **Ans: B.** **Species diversity decreases from equator to poles**. **Trap:** Statement "diversity increases toward poles" — false stem in 2019 Q58 trap family.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to biomes, which of the following statements is/are correct?

1. Large climate-defined vegetation–fauna regions.
2. Ecosystem means only a continent-scale climate zone identical to a biome.
3. Biome and ecosystem are identical terms always.

A. Only 1 B. 1 and 2 C. Only 3 D. 2 and 3

💡 **Show answer**

Ans: A. Only 1 is correct. **Logic:** Statement 1 is true (biome = large climate-defined region). Statement 2 is false — ecosystem is the local biotic–abiotic functional unit. Statement 3 is false — biome ≠ ecosystem always.

Q2. Match Köppen letters:

List-I	List-II
A. A	1. Arid
B. B	2. Tropical
C. C	3. Temperate
D. E	4. Polar

Code:

A. A-2, B-3, C-1, D-4 B. A-2, B-1, C-3, D-4 C. A-1, B-2, C-3, D-4 D. A-3, B-1, C-2, D-4

💡 **Show answer**

Ans: B. A-2, B-1, C-3, D-4. **Logic:** Tropical/arid/temperate/polar.

Q3. Which of the following pairs is/are correctly matched?

1. Sahara — largest hot desert
2. Atacama — among the driest
3. Gobi — Mongolia and China

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Desert classics.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Australia's interior desert is explained mainly by subtropical high pressure and continentality.

Reason (R): Northern Australia is situated in the temperate zone.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** UPPCS-style Australia A/R trap — north is tropical, not temperate.

Q5. With reference to Brazil, which of the following statements is/are correct?

1. Selva = Amazon equatorial rainforest name.
2. Terra Roxa = fertile dark volcanic soil.
3. Both can be true together in match stems.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** Selva + Terra Roxa.

Q6. Match List-I with List-II:

List-I	List-II
A. Planting Hearth	1. Adolf Engler
B. World Floral Regions	2. C.O. Sauer
C. Puszta	3. Hungary grassland

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** Sauer/Engler/Puszta.

Q7. Which of the following tribe–region pairs is/are correctly matched?

1. Bushman — Kalahari

2. Eskimo — northern Canada
3. Pygmy — Congo
4. Masai — West Africa

A. 2, 3 and 4 only B. 1, 2 and 4 only C. 1, 2 and 3 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: C. 1, 2 and 3 only. **Logic:** Masai = East Africa — not West.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Permafrost appears in taiga and tundra.

Reason (R): Permafrost is a classic savanna feature.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Not savanna.

Q9. Arrange temperate grassland names:

A. Only one name globally B. Only Sahara C. Only Selva D. Pampas (AR) · Prairie (NA) · Steppe (Eurasia) · Puszta (HU)

💡 **Show answer**

Ans: D. Pampas/Prairie/Steppe/Puszta. **Logic:** Grassland synonyms.

Q10. With reference to taiga/tundra/savanna, which of the following statements is/are correct?

1. Taiga — boreal conifer forest.
2. Tundra — moss–lichen cold treeless belt.
3. Savanna — tropical grassland with trees.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** Biome trio.

Q11. Which of the following pairs is/are NOT correctly matched?

1. India monsoon — Am-type language teaching
2. Thar — BWh hot desert teaching
3. Mangrove — purely inland terrestrial biome only

A. 1 and 2 only B. 3 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only pair 3 is not correctly matched. **Logic:** Mangrove = coastal transitional.

Q12. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Global biome counts (often ~14 in textbook lists) are not the same as India's ten biogeographic zones.

Reason (R): India's zones follow Rodgers and Panwar teaching.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the India-specific scheme. **A/R logic:** Biome ≠ 10 zones.

Q13. Match List-I with List-II:

List-I	List-II
A. Tropical rainforest	1. <25 cm rain
B. Desert	2. Amazon Selva
C. Savanna	3. Tropical grass with trees

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Selva/desert/savanna.

Q14. With reference to Cryophyte Steppe, which of the following statements is/are correct?

1. Linked to Arctic tundra setting teaching.
2. Cold environment vegetation idea.
3. Identical to Sahara hot desert only.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not Sahara.

Q15. How many of the following desert facts are correct?

Sahara largest hot; Atacama driest neighbourhood; Gobi Mongolia–China

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Desert map.

Q16. Arrange Köppen D climate:

A. Tropical B. Cold C. Polar only D. Arid only

💡 **Show answer**

Ans: B. Cold. **Logic:** D = cold.

Q17. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Climate (temperature and rainfall) shapes biomes.

Reason (R): Biomes are random with no climate control.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Climate controls biomes.

Q18. With reference to Indian monsoon forests, which of the following statements is/are correct?

1. Tropical deciduous neighbourhood teaching.
2. Linked to monsoon climate.
3. Identical to polar ice cap biome.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Not polar.

Q19. Which of the following pairs is/are correctly matched?

1. Prairie — North America
2. Steppe — Eurasia
3. Pampas — Argentina

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Grassland geography.

Q20. Match List-I with List-II:

List-I	List-II
A. Masai	1. Congo
B. Pygmy	2. East Africa
C. Bushman	3. Kalahari

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** East Africa/Congo/Kalahari.

Q21. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Thar is hot desert (BWh) teaching.

Reason (R): Gobi is a hot Sahara-identical desert in Africa only.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Gobi = Mongolia–China.

Q22. With reference to mangrove biome placement, which of the following statements is/are correct?

1. Coastal transitional vegetation.
2. Not a purely inland terrestrial biome.
3. Identical to taiga boreal forest.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** Not taiga.

Q23. How many of the following floral matches are correct?

Sauer–planting hearths; Engler–world floral regions

A. Only one B. None C. Both D. Three including Montreal

> 💡 **Show answer**
>
> **Ans: C.** Both. **Logic:** Classic matches.

Q24. Arrange Selva meaning:

A. Hot desert dune only B. Temperate grassland only C. Polar ice only D. Amazon equatorial rainforest name

> 💡 **Show answer**
>
> **Ans: D.** Amazon rainforest name. **Logic:** Selva fact.

Q25. Which of the following statements about continentality is/are correct?

1. Helps explain Australia's interior dryness with subtropical highs.
2. Means coastal areas are always wetter inland identically.
3. Is irrelevant to desert geography.

A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1. **Logic:** Interior dryness mechanism.

Q26. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Eskimo–northern Canada is a standard tribe–region fact.

Reason (R): Masai are West African rainforest pygmies.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** Masai East Africa; Pygmy Congo.

Q27. With reference to India's biogeographic zones, which of the following statements is/are correct?

1. Ten zones (Rodgers and Panwar).
2. Not the same count as global biome textbook lists.
3. Identical to Köppen letter A only.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not Köppen-A only.

Q28. Match List-I with List-II:

List-I	List-II
A. Taiga	1. Tropical grass+trees
B. Tundra	2. Boreal conifer
C. Savanna	3. Moss–lichen treeless

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-3, C-1. **Logic:** Boreal/tundra/savanna.

Q29. Arrange Atacama teaching:

A. Wettest place on Earth B. Only mangrove swamp C. Only polar ice D. Among the driest deserts

💡 **Show answer**

Ans: D. Among the driest. **Logic:** Atacama fact.

Q30. Which one of the following correctly states Puszta?

A. Hungary temperate grassland name B. Sahara dune type only C. Antarctic ice sheet only D. Coral atoll only

💡 **Show answer**

Ans: A. Hungary grassland. **Logic:** Puszta fact.

Common Traps

Trap	Correct
Northern Australia temperate	**Tropical**
Gobi in Russia/Kazakhstan as main	**Mongolia + China**
Masai = West Africa	**East Africa**
Selva false / Terra Roxa false	**Both TRUE**
Engler = Planting Hearth	**Sauer = Hearth; Engler = Floral**
Puszta = Argentina	**Hungary** (Pampas = Argentina)
Permafrost in savanna	**Taiga + Tundra**
Mangrove purely terrestrial	**Transitional coastal**
India has 14 biogeographic zones	**10 zones**
Hotspot = richness alone	**Endemism + habitat loss**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 26 — Water Resources & Water Conservation

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Water Resources | IWRM | Rainwater Harvesting | Watershed Management (वाटरशेड (जलग्रहण) प्रबंधन) | River Conservation | Wetland Conservation | Groundwater Conservation | Drinking Water Quality

Sources baked in: NCERT Geo 11–12; CGWB; CPCB; Jal Shakti; Namami Gange/NMCG; Jal Jeevan Mission; Wetland Rules 2017; IS 10500:2012; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — world water order; CGWB ≠ CPCB; Jal Jeevan year; UP Ramsar traps **Last verified:** August 2026

Current Affairs (this topic)

- Jal Jeevan / Namami Gange (नमामि गंगे) / wetland CA refresh water chapters.
- Keep watershed vs command-area and BOD water-quality traps stable.

💡 🎯 **Consolidated — 38 Must-Score Facts (Click to Expand)**

1. World water rough shares: oceans about **97.2%**, ice/glaciers about **2%**, groundwater about **0.6%**, rivers about **0.01%**. Descending order is **Oceans > Polar ice > Groundwater > Rivers**. 2. India holds roughly **4%** of world freshwater for about **18%** of world population (जनसंख्या). Per-capita availability around **1400 m^3** marks water stress in many readings. 3. The **water cycle** is driven primarily by **solar radiation** through evaporation from oceans, lakes, and rivers. Humidity is measured with a **hygrometer** or psychrometer. 4. Amount of rainfall depends strongly on **humidity** in the atmosphere — more vapour means greater chance of cloud and rain when temperature falls. 5. Many transplanted seedlings fail because most **root hairs** are lost during transplantation; root hairs absorb water and mineral nutrients. 6. **Freshwater is not a biotic resource.** Biotic resources come from living/organic material (forests, animals, fish); **coal and petroleum** are classed biotic because they form from decayed organic matter. 7. **World Water Day (विश्व जल दिवस) = 22 March** (UN observance from **1993**; UN-Water since **2003**). Theme **2024** = *Leveraging Water for Peace* (also taught as prosperity and peace). 8. **Rajendra Singh** ("Jal Purush" / Waterman of India; **Tarun Bharat Sangh**, Alwar) won **Ramon Magsaysay 2001** and **Stockholm Water Prize 2015**. 9. **IWRM** follows Dublin Principles (**1992**) and treats the **river basin** as the planning unit. National Water Policy **2012** priority language puts drinking first, then ecology (पारिस्थितिकी), agriculture, and industry. 10. **National Water Mission** (NAPCC, from **30 June 2008**) aims at conservation, less wastage, equitable distribution via IWRM. Five goals include public water database + climate impact assessment, citizen/state conservation action, focus on over-exploited areas, **+20% water-use efficiency**, and basin-level IWRM. Urban (नगरीय) wastewater recycling and ocean-water tech for coastal cities fit NWM; **Himalayan–peninsular river linking and free borewell reimbursement do not**. 11. The **Ministry of Jal Shakti (शक्ति)** (**2019**) merged water resources and drinking-water departments. 12. **Rainwater harvesting** captures rain for use or recharge — rooftops, pits, check dams, johads. Tamil Nadu (नाडु)'s mandatory RWH (**2003**) is a state flagship fact. 13. A **watershed** is land draining to a common outlet. **Hariyali** is the GOI watershed project for rural water conservation (≠ Pani Sansad / Jal Kranti as the 2022 key). 14. Desert plants conserve water via hard/waxy leaves, tiny/absent leaves, and thorns instead of leaves. 15. River cleaning spine: **Central Ganga (गंगा) Authority / GAP 1985 → NRCA 1995 → NGRBA 20 Feb 2009 (EPA; PM chair) → Namami Gange 2014** (NMCG). **STP** = domestic; **CETP** = industrial. 16.

Yamuna (यमुना) Action Plan launched **1993** (YAP-I to ~2003; YAP-II Dec 2004; **YAP-III 7 May 2016** under Namami Gange). 17. **Ganges River Dolphin** (*Platanista gangetica*) = **National Aquatic Animal (2009)**; dams, nets, agrochemicals threaten it — crocodile increase is **not** a standard decline cause in UPSC (संघ लोक सेवा आयोग) framing. 18. Kanpur (कानपुर) is the Ganga tannery pollution fact. **Gomti** is Lucknow (लखनऊ)'s river — do not put Gomti's main story on Kanpur. 19. **NLCP (June 2001)** restored urban/semi-urban lakes; merged with NWCP into **NPCA (Feb 2013)**. NLCP lakes include Bhoj, Sukhna, Pichola, Bhimtal — **Chilika is not NLCP**. 20. Wetlands follow **Ramsar (रामसर) 1971**, Wise Use, World Wetlands Day (विश्व आर्द्रभूमि दिवस) on **2 February** (रामसर), and Wetland Rules **2017**. UP Ramsar sites exist; **Rudrasagar** and **Sultanpur** are **not** UP. 21. **CGWB** prepares the Annual Groundwater Quality Report. **CGWA** (groundwater regulation) is under **EPA 1986** (पर्यावरण संरक्षण 1986). India has the **largest groundwater-irrigated area** in the world. Arsenic drinking-water teaching often points to **Kolkata**. 22. Disinfection: **chlorine**, chloramine, chlorine dioxide, ozone; **UV** kills microbes without changing taste/odour — UV does **not** remove odour or settle turbidity. 23. Water Act **1974** (amended **1988**); Water Cess Act **1977** (enforced **1 Apr 1978**). 24. **Atal Bhujal Yojana (2019)** is a groundwater management scheme tag. 25. **Jal Jeevan Mission** launched in **August 2019** for rural household taps (Har Ghar Jal). Do not date JJM as **2014**. 26. Drinking-water standard fact **IS 10500:2012**: fluoride about **≤1.5 mg/L**, arsenic about **≤0.01 mg/L**, pH about **6.5–8.5**. 27. Heavy sand mining lowers water table and can pollute groundwater — it does **not** decrease river salinity as a standard consequence. 28. **Lisbon Declaration** (UN Ocean Conference) is about **oceans**, not freshwater glaciers. 29. Rivers are **not** second after oceans in global freshwater ranking — ice/glaciers are. 30. CGWB ≠ CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) on the groundwater quality report; CGWA ≠ CGWB. 31. Namami Gange is **2014**; GAP is **1985**. NGRBA chair is the **Prime Minister**, not rotating CMs. 32. IWRM is basin coordination; watershed is the local drainage unit used in ridge-to-valley works. 33. Aquifer is the underground storage layer; it is not a surface canal. 34. BOD measures organic pollution strength in water (mg/L) — a quality metric, not a volume share. 35. Bundelkhand scarcity, Kanpur tanneries, Gomti–Lucknow, and arsenic plains are the UP colour set. 36. JJM is rural tap connectivity; it is not Namami Gange's river-cleaning mission. 37. STP vs CETP: domestic vs industrial effluent trains — swap them and consent logic breaks. 38. National Water Policy priority order starts with **drinking water**, not industry.

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Integrated Water Resources Management (IWRM)	**Watershed Management**	Coordinated macro-framework promoting the balanced development of water, land, and related resources across an entire river basin to maximize economic and social welfare equitably vs micro-level land and water conservation technique focusing on a ridge-to-valley catchment draining to a single outlet	एकीकृत जल संसाधन प्रबंधन (IWRM) / वाटरशेड (जलग्रहण) प्रबंधन
Unconfined Aquifer	**Confined (Artesian) Aquifer**	Permeable water-bearing geological formation whose upper boundary is the free atmospheric water table directly recharged by infiltrating rainfall vs deep aquifer bounded above and below by impermeable aquitards where groundwater is stored under hydrostatic artesian pressure	अप्रतिबंधित जलभृत (मुक्त भूजल स्तर) / प्रतिबंधित (आर्टेशियन) जलभृत
Sewage Treatment Plant (STP)	**Common Effluent Treatment Plant (CETP)**	Treatment facility designed specifically to clean domestic municipal sewage and blackwater from residential settlements vs collective facility built to treat hazardous industrial toxic effluents generated by a cluster of small- and medium-scale industrial factories	मलजल उपचार संयंत्र (STP, घरेलू सीवेज) / साझा उत्प्रवाह उपचार संयंत्र (CETP, औद्योगिक अपशिष्ट)
Rainwater Harvesting (RWH)	**Artificial Groundwater Recharge**	Collection, conveyance, and storage of runoff rainwater from rooftops or ground surfaces into surface storage tanks for immediate usage vs purposeful diversion and injection of harvested surface runoff deep into depleted subterranean aquifers through recharge shafts/borewells	वर्षा जल संचयन (सतही उपयोग) / कृत्रिम भूजल पुनर्भरण (भूजल रीचार्ज)
Central Ground Water Board (CGWB)	**Central Ground Water Authority (CGWA)**	Apex national scientific multidisciplinary organization under Ministry of Jal Shakti assessing groundwater resources, aquifer mapping, and monitoring vs statutory regulatory authority constituted under Section 3(3) of EPA 1986	

issuing mandatory NOCs and regulating industrial groundwater extraction | केंद्रीय भूजल बोर्ड (CGWB, वैज्ञानिक आकलन) / केंद्रीय भूजल प्राधिकरण (CGWA, कानूनी नियमन) | | **Jal Jeevan Mission (JRM / Har Ghar Jal)** | **Atal Bhujal Yojana (ABHY)** | Centrally sponsored flagship mission launched in August 2019 ensuring 55 lpcd functional household tap connections (FHTC) to all rural households by 2024 vs World Bank assisted community-led groundwater management scheme focused on water-stressed districts across 7 states | जल जीवन मिशन (हर घर जल, नल कनेक्शन) / अटल भूजल योजना (सामुदायिक भूजल प्रबंधन) |

💡 📋 **Must-score facts — water shares, IWRM, days, schemes (Click to Expand)**

Shares / policy | Item | Lock | |------|------| | World water order | Oceans ~97.2% > ice ~2% > GW ~0.6% > rivers ~0.01% | | India freshwater share | ~**4%** water for ~**18%** population | | Water cycle driver | Solar radiation / evaporation | | World Water Day | **22 March** | | Rajendra Singh | Jal Purush; Magsaysay 2001; Stockholm Water Prize 2015 | | IWRM / Dublin | **1992**; basin as planning unit | | NWM | NAPCC; from **30 Jun 2008** | | Jal Shakti Ministry | **2019** merger | | Freshwater resource type | **Abiotic** (≠ biotic) | --- ! [Global Water Partitioning, Aquifer Hydrogeology & Water Stress] (images/env_ch26_01_global_water_partitioning_aquifers.png)

26.1 Water Resources

Identity: Oceans dominate | Rivers tiny fraction | India water-stressed | Surface + groundwater

- Earth's water: oceans ~**97.2%** (saline (लवणीय)); polar ice/glaciers ~**2%**; groundwater ~**0.6%**; rivers/lakes ~**0.01%**.
- India holds ~**4%** of world renewable freshwater but ~**18%** of population — per capita availability ~**1,400 m^3/year** (stress below **1,700 m^3**).
- **Surface water:** rivers, lakes, reservoirs — highly seasonal under monsoon.
- **Groundwater:** India's largest **extractable** freshwater source (~45% irrigation, ~80% rural domestic).
- **Ganga–Brahmaputra (ब्रह्मपुत्र)** system carries about **40%** of India's river water.
- Drivers of stress: population, urbanisation, inefficient irrigation, pollution reducing usable supply.
- The **hydrological / water cycle** moves water through evaporation, condensation, precipitation, infiltration, and runoff.
- **Solar radiation** plays the main role in driving the water cycle by evaporating water from oceans, lakes, and rivers.
- The water cycle is essentially a **closed system** for planetary water volume on human timescales.
- **Humidity** measures water-vapour content of air and is measured with a **hygrometer** or psychrometer.
- Higher humidity means more vapour available for clouds, so **amount of rain** depends strongly on atmospheric humidity when cooling occurs.
- Many transplanted seedlings fail because most **root hairs** are lost during transplantation. Root hairs absorb water and mineral nutrients from soil.
- **Biotic resources** come from living or organic material (forests, animals, fish) and from materials obtained from them. **Coal and petroleum** are counted as biotic because they form from decayed organic matter.
- **Freshwater / pure water is not a biotic resource** — it is an abiotic natural resource, and usable drinking freshwater is scarce, so conservation techniques matter.
- **World Water Day (22 March)** highlights freshwater and sustainable management (UN designation; first observed **1993**; **UN-Water** coordination from **2003**). Theme **2024**: *Leveraging Water for Peace*.
- **Rajendra Singh** of Alwar (Rajasthan), called **Jal Purush / Waterman of India**, runs **Tarun Bharat Sangh** for community water harvesting; awards include **Ramon Magsaysay (2001)** and **Stockholm Water Prize (2015)**.
- The **Lisbon Declaration** (*Our Ocean, Our Future, Our Responsibility*) from the UN Ocean Conference is about **ocean** conservation and marine pollution — not freshwater glaciers.

PYQ — UPPCS Prelims 2025, Q88

Arrange descending: (1) Oceans (2) Rivers (3) Polar ice and glaciers (4) Groundwater

Options: A. 1, 3, 4, 2 B. 1, 2, 4, 3 C. 1, 2, 3, 4 D. 1, 4, 3, 2

💡 **Show answer**

Ans: A — Oceans > Ice > Groundwater > Rivers.

> **Logic:** Rivers are the **smallest** fraction — not second after oceans. Freshwater ≠ biotic.

26.2 Integrated Water Resources Management (IWRM) (एकीकृत जल संसाधन प्रबंधन)

Identity: Dublin 1992 | Basin unit | National Water Policy 2012 | Jal Shakti 2019

- **IWRM** coordinates water, land, and related resources within a **river basin** to balance economy, society, and ecosystems.
- **Dublin Principles (1992):** freshwater is finite; participatory management; women central; water has **economic value**.
- **National Water Policy 2012:** river basin as planning unit; allocation priority **drinking water > ecology > agriculture > industry**.
- **Ecological flows (e-flows):** minimum release below dams for river ecology.
- **Conjunctive use:** coordinated surface water + groundwater management.
- **Ministry of Jal Shakti (May 2019)** merged Water Resources with Drinking Water & Sanitation — institutional home for IWRM.
- IWRM ≠ only building dams — includes demand management, quality, ecology, and stakeholder participation.
- **National Water Mission (NWM)** is one of eight **NAPCC** missions (plan launched **30 June 2008**). Goal: conserve water, cut wastage, and ensure more equitable distribution across and within states through integrated water resource development and management.
- NWM's five goals: (1) comprehensive water database in the public domain and assessment of climate-change impacts on water; (2) citizen and state action for conservation, augmentation, and preservation; (3) focused attention to vulnerable / over-exploited areas; (4) increasing water-use efficiency by **20%**; (5) promotion of basin-level IWRM.
- Proper NWM implementation supports urban **wastewater recycling** and, for water-scarce coastal cities, technologies that allow use of **ocean water**. It does **not** include linking all Himalayan rivers to peninsular rivers or fully reimbursing farmers' borewell costs.

Traditional	IWRM
Departmental silos	River basin planning
Supply focus	Supply + demand + quality
Environment (पर्यावरण) ignored	Ecological flows included

> **Logic:** "IWRM means maximum irrigation extraction" = **FALSE**. River linking ≠ NWM package.

26.3 Rainwater Harvesting

Identity: Capture + store/recharge | Tamil Nadu 2003 mandate | Dual flood + aquifer benefit

- **Rainwater harvesting** collects rooftop or surface runoff for **direct use** or **aquifer recharge** before it drains away.
- **Rooftop:** roof → filter → tank (or recharge pit).
- **Surface:** check dams, percolation tanks, farm ponds, contour trenches.
- **Traditional:** Rajasthan **johad** (community pond) and **taanka** (cistern).
- **Benefits:** reduces urban flood peaks, recharges groundwater, supplements municipal supply.
- **Tamil Nadu (2003)** first made RWH **mandatory** in new buildings — Chennai model widely cited.
- First-flush diversion improves stored water quality; RWH supplements, does not replace, bulk supply.

Method	Function
Rooftop tank	Direct use
Recharge pit/well	Aquifer recharge
Check dam (दाम -)	Slow runoff + percolation
Johad/taanka	Traditional arid storage

> **Logic:** RWH = both storage **and** recharge — not tanks alone.

Ridge-to-Valley Watershed Architecture & Traditional Rainwater Harvesting Geography

26.4 Watershed Management

Identity: Ridge-to-valley | Drainage basin | IWMP / Hariyali | Collects-stores-releases water

- A **watershed** is the land area draining to a common outlet — also called catchment/drainage basin.
- It **collects, stores, and releases** water — UPPCS 2023 definition fact.
- **Ridge-to-valley treatment:** afforestation and contour bunds at top → check dams mid-slope → farm ponds at bottom.
- Measures: contour bunding, terracing, nala bunding, gully plugs, percolation tanks.
- **IWMP** (Integrated Watershed Management Programme) — central watershed scheme; converges with MGNREGA.
- **Hariyali** — Government of India watershed project enabling rural water conservation (2022 PYQ).
- Classic success models: **Sukhomajri (Haryana)**, **Ralegan Siddhi (Maharashtra)**.
- Outcomes: less erosion, higher water table, drought (सूखा) resilience, lower flood peaks.
- Desert plants reduce water loss through **hard and waxy leaves**, **tiny or absent leaves**, and **thorns instead of leaves** — all three modifications inhibit water loss.
- Heavy **sand mining** in riverbeds can pollute groundwater and lower the water table (sand acts as a sponge/filter). It does **not** typically decrease river salinity as a listed consequence.

PYQ — UPPCS Prelims 2023, Q110

A geographical unit that collects, stores and releases water is called—

Options: A. Watershed B. Wasteland C. Wetland D. Water Reserve

> 💡 **Show answer**
>
> **Ans: A** — Watershed / drainage basin.

PYQ — UPPCS Prelims 2022

Watershed Development Project enabling rural population to conserve water:

Options: A. Jal Kranti B. Pani Sansad C. Hariyali D. None

💡 **Show answer**

Ans: C — **Hariyali**.

> **Logic:** Watershed management **reduces** soil erosion — does not increase it.

26.5 River Conservation

Identity: GAP 1985 → NGRBA 2009 → Namami Gange 2014 | YAP | Dolphin | STP/CETP

- River degradation drivers: untreated sewage, industrial effluent, sand mining, floodplain encroachment, flow alteration by dams.
- **Ganga** (गंगा) was declared a **National River** on **4 November 2008**.
- **Central Ganga Authority (Feb 1985)** under Rajiv Gandhi (गांधी) launched the **Ganga Action Plan** (गंगा एक्शन प्लान); renamed **National River Conservation Authority (NRCA)** in **September 1995** (PM as chair).
- **GAP Phase I = 1985**. Phase II (**1993**) brought in **Yamuna Action Plan** and **Gomti Action Plan**.
- **Yamuna Action Plan:** formally launched **1993** (YAP-I ~1993–2003; YAP-II Dec **2004**; **YAP-III from 7 May 2016** under Namami Gange). *Maili Se Nirmal Yamuna* revitalisation framing from **2017**.
- **NGRBA (National Ganga River Basin Authority)** constituted **20 February 2009** under **EPA 1986**. The **Prime Minister** chairs it; CMs of Ganga states are members — chair is **not** rotational among CMs. It plans, finances, monitors, and coordinates basin-scale Ganga conservation.
- **Namami Gange, 2014** (Budget outlay teaching): integrated Ganga rejuvenation — STPs/CETPs, river-front/ghat works, surface cleaning, biodiversity, afforestation, industrial effluent monitoring, Ganga Gram. Pillars include **Nirmal Dhara (धारा)** and **Aviral Dhara**.
- **NMCG (National Mission for Clean Ganga)** under **Jal Shakti** implements Namami Gange.
- **NRCP** works: intercept/divert raw (रॉ) sewage, STPs, low-cost sanitation on banks, electric/improved wood crematoria, public awareness.
- **STP** treats domestic sewage; **CETP** treats combined industrial effluent (e.g. Kanpur tanneries — chromium).
- Clean stretch targets: BOD **<3 mg/L**, DO **>6 mg/L**.
- **UP facts: Kanpur** (कानपुर) = Ganga tannery stretch; **Gomti** = Lucknow municipal sewage; **Varanasi** (वाराणसी) = sewage + ghat waste.
- **Ganges River Dolphin** (*Platanista gangetica*) is one of four (चातुर्याम) obligate freshwater dolphins; declared India's **National Aquatic Animal in 2009**. Decline drivers include dams/barrages, accidental fishing-net capture, and agrochemicals — **not** mainly rising crocodile numbers.
- Jharkhand State Ganga River Conservation Authority was formed in **2009** (CM as ex-officio chair).

Programme	Year	Fact
Central Ganga Authority / GAP	**1985**	First Ganga STPs
NRCA rename	**1995**	Broader river mandate
YAP launch	**1993**	Yamuna bilateral Japan–India track
NGRBA	**2009**	PM-chaired basin authority
Namami Gange	**2014**	NMCG nodal
YAP-III	**2016**	Under Namami Gange
NRCP	1995	Other rivers

PYQ — UPPCS Prelims 2022

Main purpose of Ganga Action Plan (1985):

Options: A. Make water pollution-free B. Irrigation only C. Build new dams D. None

💡 **Show answer**

Ans: A — Pollution abatement / make water pollution-free.

> **Logic:** Namami Gange = **2014**, not 1985. NGRBA chair = **PM**, not rotating CM. Gomti ≠ Kanpur.

26.6 Wetland Conservation

Identity: Ramsar 1971 | NLCP 2001 → NPCA 2013 | Wetland Rules 2017 | UP Ramsar geography

- Wetlands provide flood absorption, groundwater recharge, purification, fisheries, and bird habitat (वास स्थान).
- **Ramsar Convention (1971) (रामसर कन्वेंशन):** wetland conservation and **wise use** — World Wetlands Day = **2 February**.
- India joined Ramsar **1982**; Wetland Rules **2017** (MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय)) regulate notified wetlands.
- **National Lake Conservation Plan (NLCP)** launched **June 2001** to restore degraded urban and semi-urban lakes via an integrated ecosystem (पारिस्थितिकी तंत्र) approach (pollution control from wastewater inflows).
- In **February 2013**, CCEA approved merger of NLCP and the **National Wetlands Conservation Programme** into the **National Plan for Conservation of Aquatic Ecosystems (NPCA)**.
- Classic NLCP teaching lakes: **Bhoj (MP)**, **Sukhna (Chandigarh)**, **Pichola (Udaipur (उदयपुर))**, **Bhimtal / Nainital (नैनीताल) set (Uttarakhand (उत्तराखंड))**, Powai, Pushkar, Ooty, Ramgarh Tal (Gorakhpur (गोरखपुर)), Mansi Ganga (Mathura (मथुरा)), Lakshmi Tal (Jhansi (झांसी)). **Chilika (Odisha) is not under NLCP**.
- **UP Ramsar sites** include Sarsai Nawar, Samaspur, Nawabganj, Upper Ganga, Bakhira, Haiderpur.
- **NOT in UP: Rudrasagar Lake (Tripura)** and **Sultanpur NP (Haryana)** — classic state-trap pair.

> **Logic:** Full wetland ecology depth sits in Topic 21 — here fact **conservation schemes + UP geography**. Chilika ≠ NLCP.

26.7 Groundwater Conservation

Identity: CGWB | Atal Bhujal 2019 | Over-extraction belt | Arsenic & fluoride

- Groundwater is India's main **extractable** freshwater stock — aquifers recharged by monsoon infiltration.
- **CGWB (Central Ground Water Board)** under Jal Shakti assesses quantity and quality.
- **Annual Groundwater Quality Report** is prepared by **CGWB** — not CPCB or CWC (केंद्रीय जल आयोग).
- **CGWA (Central Ground Water Authority)** was formed under the **Environment (Protection) Act, 1986** to regulate and control groundwater development — **CGWA ≠ CGWB**.
- India has the **largest area under groundwater irrigation** in the world (followed by China and the USA).
- **Over-extraction hotspots:** Punjab–Haryana–western UP Green Revolution (हरित क्रांति) belt — falling water tables.
- **Contamination: arsenic** in Ganga–Brahmaputra alluvium (urban drinking-water teaching often cites (साइट्स) **Kolkata**); **fluoride** in parts of Rajasthan and UP.
- **Atal Bhujal Yojana (2019):** community-led groundwater management in water-stressed blocks (7 states).
- Conservation tools: recharge structures, crop diversification, regulated tubewell electricity, conjunctive use.

- Legal stack reminder: **Water (Prevention and Control of Pollution) Act, 1974** (amended **1988**); **Water Cess Act, 1977** (enforced **1 April 1978**).

PYQ — UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 prepared by: (1) CPCB (2) CWC (3) CGWB

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: B — **CGWB only**.

Body	Role
CGWB	Groundwater quantity/quality assessment
CGWA	Groundwater regulation (EPA 1986)
CPCB	Surface water & air pollution
CWC	Surface water/irrigation data

> **Logic:** CGWB ≠ CPCB; CGWA ≠ CGWB.

26.8 Drinking Water Quality

Identity: Jal Jeevan Mission 2019 | Disinfection | IS 10500:2012 | SDG 6

- **Jal Jeevan Mission (JJM):** announced **August 2019** / Union Budget **FY 2019–20** — functional household tap connection (**Har Ghar Jal**) in rural India.
- Nodal: Ministry of Jal Shakti; aligns with **SDG 6** (Clean Water and Sanitation).
- **IS 10500:2012 (BIS)** drinking water standard: pH **6.5–8.5**; fluoride **≤1.5 mg/L**; arsenic **≤0.01 mg/L**; faecal coliform ideally **zero**.
- Disinfection / purification tools: **chlorine** gas and bleaching powder (classic primary bactericidal step); also **chloramine (NH_2Cl)**, **chlorine dioxide (ClO_2)**, **ozone**, iodine compounds, heat, and **ultraviolet radiation**.
- **UV purification** inactivates/kills harmful microorganisms without adding chemicals and without changing taste, smell, or colour. UV does **not** remove all undesirable odours or quicken sedimentation/turbidity clearance — those claims are false in UPSC framing.
- Quality risks: microbial contamination from sewage; fluoride/arsenic geogenic contamination (Kolkata arsenic teaching); industrial toxins near polluted stretches.
- Swachh Bharat (2014) + JJM (2019) together address sanitation and drinking water access.

PYQ — UPPCS Prelims 2022, Q145

Jal Jeevan Mission was declared in Union Budget of which financial year?

Options: A. 2021–22 B. 2022–23 C. 2020–21 D. 2019–20

💡 **Show answer**

Ans: D — **2019–20**.

Parameter	Limit (IS 10500)
pH	6.5–8.5
Fluoride	≤1.5 mg/L

Parameter	Limit (IS 10500)
Arsenic	≤0.01 mg/L

> **Logic:** JJM = **2019**, not 2014 (Swachh Bharat year). UV = microbial kill only among the three classic claims.

Complete PYQ Bank (Topic 26)

Q1. UPPCS Prelims 2025, Q88

Arrange the following freshwater reservoirs in **descending** order of their share of total global freshwater:

1. Oceans
2. Rivers
3. Polar ice and glaciers
4. Groundwater

A. 1, 3, 4, 2

B. 1, 2, 4, 3

C. 1, 2, 3, 4

D. 1, 4, 3, 2

> 💡 **Show answer**
>
> **Logic:** Freshwater fraction ranking — ice caps > groundwater > rivers; oceans are saline (often listed as distractor in stem). **Ans: A (1-3-4-2).** Among freshwater stores: **ice > groundwater > rivers**. **Trap:** Rivers second after oceans — rivers are the **smallest** usable freshwater fraction.

Q2. UPPCS Prelims 2025, Q70

The Annual Groundwater Quality Report 2024 was prepared by:

1. Central Pollution Control Board (CPCB)
2. Central Water Commission (CWC)
3. Central Ground Water Board (CGWB)

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

> 💡 **Show answer**
>
> **Logic:** CGWB assesses groundwater quantity/quality; CPCB handles surface pollution; CWC surface water data. **Ans: B (Only 3).** **CGWB** publishes the Annual Groundwater Quality Report. **Trap:** **CPCB** (surface/air pollution board) or **CWC** confusion.

Q3. UPPCS Prelims 2025, Q58

Which of the following Ramsar sites are **NOT** located in Uttar Pradesh?

1. Sarsai Nawar Jheel
2. Samaspur Bird Sanctuary

3. Rudrasagar Lake
4. Sultanpur National Park

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Logic: Wetland conservation + UP geography — Tripura and Haryana sites outside UP. **Ans: B (3 and 4).** **Rudrasagar = Tripura**; **Sultanpur = Haryana**. **Trap:** Assuming all four names are UP Ramsar sites.

Q4. UPPCS Prelims 2023, Q110

A geographical unit that collects, stores and releases water is called—

A. Watershed

B. Wasteland

C. Wetland

D. Water Reserve

💡 **Show answer**

Logic: Watershed/catchment definition — drainage basin hydrology. **Ans: A.** A **watershed** collects, stores, and releases water. **Trap:** **Wetland** — stores water but is an ecosystem type, not the drainage-unit definition.

Q5. UPPCS Prelims 2022 pattern — Hariyali

The watershed development project enabling rural populations to conserve water is known as:

A. Jal Kranti

B. Pani Sansad

C. Hariyali

D. Atal Bhujal Yojana

💡 **Show answer**

Logic: Hariyali = GoI watershed project for rural water conservation. **Ans: C.** **Hariyali** is the watershed development project. **Trap:** **Jal Kranti** or **Atal Bhujal** (groundwater-specific, 2019).

Q6. UPPCS Prelims 2022 pattern — GAP 1985

The main purpose of the Ganga Action Plan (1985) was to:

A. Make Ganga water pollution-free

B. Build large irrigation dams only

C. Privatise river water rights

D. Divert Ganga water to southern states

💡 **Show answer**

Logic: GAP/NRCP lineage — sewage interception and pollution abatement on Ganga. **Ans: A.** GAP aimed to make Ganga water **pollution-free**. **Trap:** River-linking or dam-building distractors — GAP is **pollution control**.

Q7. UPPCS Prelims 2022, Q145

Jal Jeevan Mission was declared in the Union Budget of which financial year?

A. 2021–22

B. 2022–23

C. 2020–21

D. 2019–20

💡 **Show answer**

Logic: JJM launch August 2019 / Budget FY 2019–20 — Har Ghar Jal rural tap connections. **Ans: D.** **Jal Jeevan Mission was announced in FY 2019–20**. **Trap:** Post-COVID years (2020–21 onward) as launch window.

Q8. UPPCS Prelims 2023 pattern — Water Act year

The Water (Prevention and Control of Pollution) Act was enacted in:

A. 1974

B. 1981

C. 1986

D. 1972

💡 **Show answer**

Logic: Water 1974 created CPCB/SPCB — separate from Air 1981 and EPA 1986. **Ans: A.** The **Water Act year is 1974**. **Trap:** **1981** (Air Act) or **1986** (EPA).

Q9. UPPCS Prelims pattern — Tamil Nadu RWH

Which state first made rooftop rainwater harvesting mandatory in new buildings (2003)?

A. Rajasthan

B. Tamil Nadu

C. Kerala

D. Gujarat

💡 **Show answer**

Logic: Tamil Nadu 2003 mandatory RWH — Chennai model cited in water conservation teaching. **Ans: B.** **Tamil Nadu (2003)** mandated RWH in new buildings. **Trap:** **Rajasthan** (traditional johad/taanka) without mandatory urban law date.

Q10. UPPCS Prelims pattern — Sukhomajri

Sukhomajri (Haryana) is a classic success story of:

A. Community watershed management

B. Offshore desalination

C. Nuclear-powered irrigation

D. Interlinking of Himalayan rivers

💡 **Show answer**

Logic: Community watershed + afforestation + water harvesting model before IWMP scaling. **Ans: A.** **Sukhomajri exemplifies community watershed management**. **Trap:** National river-linking megaproject distractor.

Extra Drill (Ghatnachakra Water Resources map)

Q31. Which is not a biotic resource?

A. Coal

B. Petroleum

C. Fish

D. Freshwater

💡 **Show answer**

Logic: Bank traps on biotic vs freshwater, NWM scope, disinfection, lakes, Ganga institutions, and personalities. **Ans: D.** Freshwater is abiotic; coal/petroleum counted biotic via organic origin.

Q32. 'Jal Purush' / Waterman of India is—

A. Sunderlal Bahuguna

B. Rajendra Singh (Tarun Bharat Sangh)

C. Salim Ali

D. Medha Patkar

💡 **Show answer**

Ans: B. Magsaysay 2001; Stockholm Water Prize 2015.

Q33. National Water Mission correctly includes—

A. Only full Himalaya–peninsula river linking as a core goal

B. Wastewater recycling for urban needs and +20% water-use efficiency framing

C. Full reimbursement of all farmer borewell costs nationwide

D. Only ozone treaty compliance

💡 **Show answer**

Ans: B. Statements 1–2 of the classic UPSC NWM stem; 3–4 false.

Q34. UV in water purification primarily—

A. Removes all odours and settles all turbidity

B. Inactivates/kills harmful microorganisms without changing taste/odour

C. Adds chlorine taste deliberately

D. Only measures BOD

💡 **Show answer**

Ans: B. Only the microbial-kill claim is correct among the three UPSC statements.

Q35. NLCP urban lakes set correctly excludes—

A. Bhoj (MP)

B. Sukhna (Chandigarh)

C. Pichola (Rajasthan)

D. Chilika (Odisha)

💡 **Show answer**

Ans: D. Chilika is not under the NLCP list in bank teaching.

Q36. NGRBA was constituted in—

A. 1985

B. February 2009 under EPA; PM is chair

C. 2014 only as Namami Gange rename

D. 1974 Water Act alone

💡 **Show answer**

Ans: B. Basin planning authority; chair is not rotating CM.

Q37. Yamuna Action Plan was formally launched in—

A. 1985

B. 1993

C. 2014

D. 2009

💡 **Show answer**

Ans: B. YAP-III (2016) sits under Namami Gange.

Q38. National Aquatic Animal of India—

A. Saltwater crocodile

B. Ganges River Dolphin (2009)

C. Olive Ridley only

D. Gharial only

💡 **Show answer**

Ans: B. *Platanista gangetica*; dams/nets/chemicals threaten populations.

Q39. CGWA was formed under—

A. Wildlife Act 1972 only

B. Environment (Protection) Act, 1986

C. Forest Act 1927 only

D. Montreal Protocol

💡 **Show answer**

Ans: B. Regulatory groundwater authority — not the same as CGWB.

Q40. Lisbon Declaration headlines are associated with—

A. Air quality only

B. Mountains only

C. Oceans

D. Freshwater glaciers only

💡 **Show answer**

Ans: C. UN Ocean Conference framing — marine/coastal protection.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to world water shares, which of the following statements is/are correct?

1. Oceans hold only about 2% of Earth's water; ice holds about 97%.
2. Ice/glaciers about 2% — ahead of rivers in freshwater ranking.
3. Rivers are second after oceans in global freshwater ranking.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: A. Only 2 is correct. **Logic:** Statement 2 is true among freshwater stores (ice/glaciers ahead of rivers). Statement 1 reverses ocean vs ice shares. Statement 3 is false — rivers are a tiny freshwater share, not second after oceans.

Q2. Match List-I with List-II:

List-I	List-II
A. World Water Day	1. 2 February
B. Wetlands Day	2. 22 March
C. JJM launch	3. August 2019

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Water Day / Wetlands Day / JJM.

Q3. Which of the following pairs is/are correctly matched?

1. India holds about 40% of global freshwater for about 18% of world population.
2. Per-capita ~1400 m^3 — water-stress neighbourhood teaching

3. Freshwater — biotic resource identical to forests

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Ans: B.** Only 2 is correct. **Logic:** Statement 2 is the water-stress per-capita neighbourhood teaching. Statement 1 exaggerates India's freshwater share (usual teaching ~4% water / ~18% population). Statement 3 is false — freshwater is abiotic, not identical to forests.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): The water cycle is driven primarily by solar radiation through evaporation.

Reason (R): Amount of rainfall depends strongly on atmospheric humidity.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; humidity–rain link does not explain what drives the cycle. **A/R logic:** Driver vs rainfall control.

Q5. With reference to Rajendra Singh, which of the following statements is/are correct?

1. Called Jal Purush / Waterman of India.
2. Tarun Bharat Sangh, Alwar.
3. He chairs NGRBA as Prime Minister.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1 and 2. **Logic:** PM chairs NGRBA — not Rajendra Singh.

Q6. Arrange Ganga cleaning spine years:

A. Namami Gange 1985 → GAP 2014 B. GAP 1985 → NGRBA 2009 → Namami Gange 2014 C. Both 2020 only D. JJM 1985 only

> 💡 **Show answer**
>
> **Ans: B.** GAP → NGRBA → Namami Gange. **Logic:** Year ladder.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Kanpur — Ganga tannery pollution
2. Gomti — Lucknow's river
3. Gomti's main tannery story — Kanpur

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only pair 3 is not correctly matched. **Logic:** Do not put Gomti on Kanpur.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): JJM launched in August 2019 for rural household taps.

Reason (R): JJM is identical to Namami Gange's river-cleaning mission.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** JJM ≠ Namami Gange.

Q9. With reference to National Water Policy 2012, which of the following statements is/are correct?

1. Priority language starts with drinking water.
2. IWRM treats the river basin as the planning unit.
3. Industry is first priority over drinking water.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Drinking water first.

Q10. Match List-I with List-II:

List-I	List-II
A. CGWB	1. Groundwater regulation under EPA
B. CGWA	2. Annual Groundwater Quality Report
C. Atal Bhujal	3. 2019 groundwater management

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** CGWB ≠ CGWA.

Q11. Arrange IS 10500:2012 teaching limits idea:

A. Fluoride ≤100 mg/L only B. Fluoride ≤1.5 mg/L; arsenic ≤0.01 mg/L; pH ~6.5–8.5 C. Arsenic ≤50 mg/L only D. No pH range taught

💡 **Show answer**

Ans: B. F ≤1.5; As ≤0.01; pH 6.5–8.5. **Logic:** Drinking-water standards.

Q12. With reference to rainwater harvesting, which of the following statements is/are correct?

1. Captures rain for use or recharge.
2. Tamil Nadu's mandatory RWH (2003) is a classic state fact.
3. RWH means only desalination of ocean water.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Not desalination.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): National Aquatic Animal (2009) is the Ganges River Dolphin.

Reason (R): Crocodile is the National Aquatic Animal.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Dolphin — not crocodile.

Q14. Which of the following pairs is/are correctly matched?

1. Watershed — land draining to a common outlet
2. Hariyali — GOI watershed project neighbourhood
3. Aquifer — surface canal identical always

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Aquifer ≠ surface canal.

Q15. How many of the following are correct?

Ministry of Jal Shakti 2019 merger; Water Act 1974; Water Cess Act 1977

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Institution/law trio.

Q16. With reference to disinfection, which of the following statements is/are correct?

1. UV kills microbes without changing taste/odour teaching.
2. UV does not remove organic load like BOD treatment.
3. UV alone is identical to BOD as a pollution metric.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** UV ≠ BOD metric.

Q17. Match List-I with List-II:

List-I	List-II
A. Yamuna Action Plan	1. 2014
B. Namami Gange	2. 1993
C. NLCP start	3. June 2001

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** YAP / Namami / NLCP.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Heavy sand mining can lower water table and pollute groundwater.

Reason (R): A standard consequence is that sand mining decreases river salinity.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Salinity decrease is not the standard fact.

Q19. Which one of the following correctly states BOD?

A. Identical to aquifer volume B. Ocean volume share only C. Glacier mass share only D. Organic pollution strength metric (mg/L)

💡 **Show answer**

Ans: D. Organic pollution strength. **Logic:** Quality metric — not volume.

Q20. With reference to National Water Mission, which of the following statements is/are correct?

1. Under NAPCC from 30 June 2008.
2. Aims conservation, less wastage, equitable distribution via IWRM.
3. Identical to Lisbon Declaration on oceans.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Lisbon = oceans — not NWM.

Q21. Arrange STP vs CETP:

A. Both identical always B. STP domestic; CETP industrial effluent trains C. STP industrial only; CETP domestic only D. Neither treats wastewater

💡 **Show answer**

Ans: B. STP domestic; CETP industrial. **Logic:** Swap trap.

Q22. Which of the following pairs is/are correctly matched?

1. Transplanted seedlings often fail — root-hair loss
2. Root hairs absorb water and mineral nutrients
3. Humidity measured with a thermometer only always

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Humidity uses hygrometer neighbourhood — not thermometer-only.

Q23. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NGRBA chair is the Prime Minister.

Reason (R): NGRBA is under EPA neighbourhood teaching with PM chair fact.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the chair fact. **A/R logic:** PM chairs NGRBA.

Q24. How many of the following UP colour facts are correct?

Bundelkhand scarcity; Kanpur tanneries; Gomti–Lucknow; arsenic plains

A. Only one B. Only two C. Only three D. All four

💡 **Show answer**

Ans: D. All four. **Logic:** UP water colour set.

Q25. With reference to wetlands, which of the following statements is/are correct?

1. Ramsar 1971 and Wise Use.
2. World Wetlands Day on 2 February.
3. Rudrasagar is a UP Ramsar site.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Rudrasagar ≠ UP.

Q26. Match List-I with List-II:

List-I	List-II
A. Dublin Principles	1. Local drainage unit
B. Watershed	2. 1992 IWRM neighbourhood
C. NPCA	3. NLCP+NWCP merger 2013

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** Dublin / watershed / NPCA.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Desert plants conserve water via hard/waxy leaves, tiny leaves, or thorns.

Reason (R): These are adaptations that reduce water loss.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Xeric adaptations.

Q28. Arrange JJM year trap:

A. 2014 Namami identical B. 1985 GAP C. August 2019 Har Ghar Jal D. 1974 Water Act only

💡 **Show answer**

Ans: C. August 2019. **Logic:** Not 2014.

Q29. With reference to groundwater institutions, which of the following statements is/are correct?

1. CGWB prepares Annual Groundwater Quality Report.
2. CGWA regulates groundwater under EPA 1986 neighbourhood.
3. CPCB publishes that same CGWB groundwater quality report as its only job.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** CGWB ≠ CPCB on that report.

Q30. Which one of the following correctly states IWRM vs watershed?

A. IWRM = basin coordination; watershed = local drainage unit B. Both identical surface canals only C. IWRM = only rooftop RWH D. Watershed = only ocean Lisbon

💡 **Show answer**

Ans: A. Basin vs local drainage. **Logic:** Scale distinction.

Common Traps

Trap	Correct
Rivers second after oceans	**Ice then GW then rivers**
Groundwater report = CPCB	**CGWB**
CGWA = CGWB	**EPA regulator ≠ assessment board**
JJM launched 2014	**2019**
Namami Gange = 1985	**2014** (GAP = 1985)
Freshwater = biotic resource	**Not** — abiotic
NWM = river linking + free borewells	**Recycling / coastal tech / +20% efficiency** yes; linking/reimbursement **no**
UV removes odour + turbidity	**Only microbial kill** among classic claims
Chilika under NLCP	**Not**
NGRBA chair = rotating CM	**Prime Minister**
Dolphin decline = more crocodiles	**Dams, nets, agrochemicals**
Lisbon = freshwater glaciers	**Oceans**
Hariyali = Jal Kranti	**Watershed programme**
Namami Gange = 1985	**2014** (GAP = 1985)
Gomti = Kanpur	**Lucknow**
Sultanpur in UP	**Haryana**
Watershed increases erosion	**Reduces erosion**
IWRM = dams only	**Basin + quality + ecology**
CWC = groundwater quality	**Surface water data**

Trap	Correct
Drinking water unlimited fluoride	**≤1.5 mg/L**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 27 — Renewable & Non-Renewable Energy (अनवीकरणीय ऊर्जा)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Renewable | Non-renewable | Conventional | Non-conventional | Solar | Wind | Biomass | Biofuels | Biogas (बायोगैस) | Bioethanol (बायो-इथेनॉल) | Biodiesel (बायो-डीजल) | Small Hydro | Tidal | Ocean | Energy Security | India Scenario | National RE Programme | National Solar Mission | Energy Conservation | Energy Efficiency

Sources baked in: NCERT Geo 11–12; MNRE; Energy Conservation Act 2001; BEE/PAT; National Policy on Biofuels 2018; National Solar Mission; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — renewable vs non-renewable; biogas CH_4+CO_2; conventional/non-conventional; BEE/PAT; E20 **Last verified:** August 2026 **Overlap note:** Solar/wind/biomass/biogas/SHP/tidal detail also lives in the Renewable Energy (नवीकरणीय ऊर्जा) chapter — this file keeps those as one-home facts and expands **non-renewable, biofuels, ocean, security, conservation/efficiency**.

Current Affairs (this topic)

- Capacity and fuel-mix CA overlaps the Renewable Energy chapter — keep classification traps consistent.
- Nuclear is low-carbon but conventionally non-renewable in many Prelims frames.

💡 🎯 **Consolidated — 36 Must-Score Facts (Click to Expand)**

1. **Renewable** sources replenish on a human timescale (sun, wind, water, biomass, geothermal). **Non-renewable** sources are finite stocks (coal, oil, gas, uranium). Notes usually treat **nuclear as non-renewable** even though it is low-carbon. 2. In Indian usage, **conventional** energy means coal, oil, gas, **large hydro**, and nuclear. **Non-conventional** means solar, wind, biomass, biogas, tidal, geothermal, ocean, and **small hydro ≤25 MW** — treated as roughly equal to **renewable** in many stems. 3. Non-conventional is treated as roughly equal to renewable in many prelims stems — but large hydro sits on the conventional side of that Indian classification. 4. **Energy crisis** in bank language means the danger of exhaustion of **fossil fuels** (coal, oil, gas) — not merely "short hydro" or malnutrition. 5. UNGA declared **2014–2024** the **Decade of Sustainable Energy for All** (supports SDG 7). Initiative is of the **United Nations**, not World Bank alone. 6. Ultimate source of energy stored in fossil fuels is the **Sun** (ancient photosynthesis). 7. Among coal / oil-gas / nuclear / hydro options, **hydroelectricity** is often the best answer for electricity generation under **sustainable development** (water not depleted by the generation cycle). 8. **Uranium is not a fossil fuel** — nuclear fuel, non-renewable. Coal / petroleum / natural gas are fossils. 9. **Biogas** major constituents are **methane and carbon dioxide**. Butane or carbon monoxide as the major pair is wrong. 10. **Bioethanol** (maize, sugarcane, etc.) blends with petrol. **Biodiesel** (Jatropha, Pongamia, UCO) blends with diesel. Biofuels are eco-friendly and can ease crisis but are often **not cost-effective** vs fossils. 11. Biomass gasification of shells/husk yields **producer gas** (CO, H_2, traces of CH_4 — not CO_2-only); can run IC engines. 12. Solar facts include National Solar Mission **2010**, ISA at **Gurugram**, **500 GW** non-fossil by **2030**, and **PM-KUSUM**. 13. Wind leadership often names Tamil Nadu (नाडु) and Gujarat. **SHP ≤25 MW**. **Tidal** is Moon-driven. **Ocean energy** is a wider set that includes wave, OTEC, and tidal. 14. **OTEC** uses the temperature difference between warm surface water and cold deep water — not the same as tidal barrage power. 15. Energy security: India imports a very large share of crude oil (often taught near **~85%**), while coal is mostly domestic. Strategic Petroleum Reserves

sit at **Vizag, Mangalore, and Padur**. 16. **Energy Conservation Act 2001** created **BEE**. **PAT (2012)** trades **ESCerts**. Star labels and UJALA LEDs are efficiency/conservation tools. PAT is **not** CCTS carbon credits. 17. Coal is fossil non-renewable; biomass is renewable organic matter — do not call coal "biomass." Nuclear is **not** a biomass energy (बायोमास ऊर्जा) source. 18. Tidal ≠ wave ≠ OTEC, even though all are ocean-related. 19. Conventional includes large hydro; calling all hydro "non-conventional" is wrong under the Indian split. 20. E20 is the higher ethanol blend ambition beyond E10 — do not freeze policy forever at E10. 21. BEE sits under the EC Act **2001** efficiency architecture. 22. SPR locations are Vizag, Mangalore, Padur — three coastal storage facts. 23. Western UP sugarcane supports bioethanol; bagasse supports cogeneration; UP wind remains limited; rooftop solar is the distributed play. 24. **Hydrogen** is the least-polluting common fuel option (burns to water); called a **future fuel**. National Hydrogen Energy Board set up **3 October 2003**. 25. Fuel cells: pure H_2 → electricity + **heat and water**; can power buildings **and** small devices; output is **DC** (not AC). 26. Microbial fuel cells use living microbes as catalysts; useful in wastewater plants — substrates are mainly **organic**, not "only inorganic." 27. **Bioasphalt** is non-petroleum / renewable-based paving binder — eco-friendlier road surfacing. 28. Green hydrogen can be used in IC engines, blended with natural gas, and in fuel-cell vehicles — all three routes appear in stems. 29. Ocean energy is the umbrella; tidal is one member of that set. 30. ISA is solar diplomacy/IGO; BEE is domestic efficiency regulation — different lanes. 31. PAT certificates measure energy savings trading; carbon markets measure CO_2 units — related climate politics, different instruments. 32. Small hydro cutoff **≤25 MW** is the recurring MNRE number. 33. Nuclear's classification is usually non-renewable/conventional, not "renewable because low carbon." 34. Energy conservation (using less) and energy efficiency (more service per unit energy) are neighbours under BEE/PAT, not synonyms for renewable capacity addition alone. 35. **SATAT** promotes **compressed biogas (CBG)** for transport; oil-marketing companies procure CBG. **GOBAR-Dhan** is the community/dung biogas neighbour scheme — do not invent butane as a major biogas gas. 36. Biogas forms by **anaerobic** digestion; slurry is useful bio-fertiliser after gas capture.

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Renewable Energy	**Non-Renewable Energy**	Energy derived from natural regenerative sources that replenish naturally on a human timescale without depleting stocks (solar, wind, small hydro, geothermal, biomass) vs finite geological energy reserves that take millions of years to form and cannot be replenished once consumed (coal, petroleum, natural gas, uranium)	नवीकरणीय ऊर्जा (अक्षय स्रोत) / अनवीकरणीय ऊर्जा (परिमित जीवाश्म भंडार)
Conventional Energy	**Non-Conventional Energy**	Traditional energy sources that have been historically established and extensively utilized for commercial power generation for centuries (coal, oil, gas, large hydro, commercial nuclear) vs alternative, modern clean renewable sources developed to replace fossil fuels (solar, wind, tidal, ocean thermal, fuel cells)	पारंपरिक ऊर्जा स्रोत / गैर-पारंपरिक (वैकल्पिक) ऊर्जा स्रोत
Bioethanol	**Biodiesel**	Pure ethanol fuel produced through the biochemical fermentation of sugary and starchy feedstocks (sugarcane juice, molasses, damaged grains, corn) blended with petrol (E20 mandate) vs fatty acid methyl esters (FAME) produced via the transesterification of non-edible plant vegetable oils (Jatropha curcas, Pongamia, animal fats) blended with diesel	बायो-इथेनॉल (किण्वन, पेट्रोल संवर्धन) / बायो-डीजल (ट्रांसएस्टरीकरण, डीजल संवर्धन)
Perform, Achieve and Trade (PAT)	**Renewable Purchase Obligation (RPO)**	Market-based regulatory mechanism under the National Mission for Enhanced Energy Efficiency (BEE / EC Act 2001) issuing tradable Energy Saving Certificates (ESCerts) to designated energy-intensive industries vs statutory mandate under Electricity Act 2003 requiring power discoms to purchase a minimum percentage of electricity from renewable sources	निष्पादन, उपलब्धि और व्यापार (PAT, ऊर्जा बचत प्रमाण पत्र) / नवीकरणीय खरीद दायित्व (RPO)
Bureau of Energy Efficiency (BEE)	**Petroleum Conservation Research Association (PCRA)**	Statutory body established under the Energy Conservation Act 2001 (Ministry of Power) promoting star rating standards for electric appliances and energy conservation codes vs registered society under the Ministry of Petroleum and Natural Gas promoting conservation and fuel efficiency in the petroleum sector	ऊर्जा दक्षता ब्यूरो (BEE, स्टार लेबलिंग) / पेट्रोलियम संरक्षण अनुसंधान संघ (PCRA)
Biogas	**Compressed Biogas (CBG)**	Unrefined raw gaseous fuel mixture generated from anaerobic

digestion of cattle dung and agro-waste containing ~55–60% methane and ~40% CO_2 vs purified, upgraded methane gas (≥90% CH_4) scrubbed of CO_2 and H_2S and compressed to ~250 bar for use as clean green automotive transport fuel (SATAT initiative) | बायोगैस (कच्ची गोबर गैस) / संपीडित बायोगैस (CBG, परिष्कृत 90%+ मीथेन) |

💡 📋 **Must-score facts — conventional vs non-conventional (Click to Expand)**

Classifications | Item | Lock | |------|------| | Renewable | Sun, wind, water, biomass, geothermal | | Non-renewable | Coal, oil, gas, uranium (nuclear fuel ≠ fossil) | | Indian conventional | Coal, oil, gas, **large hydro**, nuclear | | Non-conventional | Solar, wind, biomass, biogas, tidal, SHP… | | Fossil ultimate source | **Sun** (ancient photosynthesis) | | UN Decade Sustainable Energy | **2014–2024** (UNGA; SDG 7) | | Biogas majors | CH_4 + CO_2 | | Bioethanol / Biodiesel | Petrol blend / diesel blend (Jatropha, UCO…) | | OTEC vs tidal | Warm–cold water ΔT / Moon-driven sea level | | SHP | ≤**25 MW** |

27.1 Renewable Energy

Identity: Human-timescale replenishment | MNRE sources | Cleaner than fossils | Intermittency challenge

- **Renewable energy** comes from sources replenished continuously on a human timescale — sun, wind, flowing water, tides, biomass, geothermal heat.
- India types: solar, wind, biomass/biogas, small hydro, tidal, geothermal, waste-to-energy.
- Advantages: lower GHG than coal/oil; cuts oil/gas import pressure; decentralised rural supply.
- Limits: intermittency (night/no-wind); needs storage, hybrids, or backup; high upfront cost.
- **Energy crisis** commonly means the danger of exhaustion of **fossil fuels** such as coal and petroleum.
- UNGA declared **2014–2024** the **Decade of Sustainable Energy for All** (UN initiative supporting SDG 7).
- Ultimate energy in fossil fuels traces to the **Sun** via ancient photosynthesis.
- Among common options, **hydroelectricity** is often preferred for sustainable electricity generation versus coal, oil-gas, or nuclear stock depletion.
- India target: **500 GW non-fossil capacity by 2030** (COP26) — supports net zero **2070**.
- **Nuclear** is usually classified as **non-renewable** (finite uranium + waste).

> **Logic:** "Nuclear is renewable" = usually **FALSE** in UPPCS/UPSC (संघ लोक सेवा आयोग). Non-conventional ≈ renewable in many stems.

27.2 Non-renewable Energy

Identity: Coal, oil, gas, uranium | Finite stock | Oil import risk | Coal domestic

- **Non-renewable** sources form over geological time — use depletes the stock.
- **Coal:** India's primary electricity fuel; large domestic reserves in Jharkhand, Odisha, Chhattisgarh, West Bengal — eastern belt.
- **Petroleum:** India imports ~**85%** of crude — main energy-security vulnerability.
- **Natural gas:** cleaner than coal; used in power, fertiliser, CNG; partly imported (LNG).
- **Uranium/thorium:** nuclear fuel — low operational carbon, finite ore, radioactive waste — **not a fossil fuel**.
- Coal is **not** a biomass energy source — fossilised carbon, not living organic matter.
- Nuclear energy is **not** a biomass energy source.

PYQ — UPPCS Prelims 2021, Q124

A non-renewable source of energy is:

Options: A. Solar B. Petroleum C. Wind D. Biogas

💡 **Show answer**

Ans: B — Petroleum is finite fossil fuel.

Logic: India is **coal-rich domestically** but **oil-import dependent** — do not swap.

27.3 Conventional Sources of Energy

Identity: Coal, oil, gas, large hydro, nuclear | Traditional grid backbone

- In Indian usage, **conventional** means long-established grid sources: **coal, oil, natural gas, large hydropower (>25 MW), and nuclear**.
- Conventional ≠ only fossil — **large hydro and nuclear** are included.
- Advantages: baseload reliability and established infrastructure.
- Disadvantages: GHG (fossils), air pollution, mining/displacement, waste (nuclear).

Conventional	Non-conventional
Coal, oil, gas	Solar, wind, biomass
Large hydro	Small hydro ≤25 MW
Nuclear	Tidal, geothermal, ocean

27.4 Non-conventional Sources of Energy

Identity: Solar, wind, biomass, tidal, SHP, geothermal, ocean | ≈ renewable

- **Non-conventional** sources are newer/alternative sources — solar, wind, biomass, biogas, tidal, geothermal, ocean energy, small hydro.
- In MCQs, non-conventional sources are treated as **renewable**.

PYQ — UPPCS Prelims 2023, Q102

Non-conventional energy (गैर-पारंपरिक (वैकल्पिक) ऊर्जा स्रोत) sources are those that are—

Options: A. Non-renewable B. Renewable C. Produced from electricity D. Produced from heat

💡 **Show answer**

Ans: B — Renewable.

Logic: Non-conventional ≈ renewable; conventional includes fossils + large hydro + nuclear.

27.5 Solar Energy

Identity: PV + CSP (सीएसपी) | National Solar Mission 2010 | ISA Gurugram | 500 GW non-fossil 2030

- **PV** converts sunlight directly to electricity; **CSP** uses mirrors to heat fluid for steam turbines.
- **National Solar Mission (2010)** under NAPCC — solar capacity scaled toward ~100 GW by early 2020s.
- **ISA (2015):** India–France treaty IGO, HQ **Gurugram**; OSOWOG linked to ISA.
- **PM-KUSUM (2019):** solar pumps and farmer solarisation.

- Flagship park: **Bhadla (Rajasthan)**.

> **Logic:** ISA HQ = Gurugram. "38 members forever" outdated; OSOWOG under ISA is the tested true fact.

27.6 Wind Energy

Identity: Kinetic air → turbines | Power ∝ v^3 | Tamil Nadu leader | Onshore dominant

- Wind turbines convert kinetic energy of air to electricity; output rises with **cube of wind speed**.
- Leading states: **Tamil Nadu, Gujarat, Karnataka, Maharashtra**.
- Onshore mature; offshore potential on Gujarat/TN coasts — stronger winds, higher cost.
- Intermittent — needs grid balancing or hybrid solar–wind systems.

> **Logic:** Rajasthan leads **solar**, not wind.

27.7 Biomass Energy

Identity: Organic matter → heat/power | Bagasse UP sugar belt | Not coal/nuclear

- Biomass energy uses plant/animal organic matter — wood, dung, crop residue, bagasse.
- Direct combustion or conversion to biogas/biofuels.
- **Bagasse** cogeneration is major in western UP sugar mills.
- **Coal and nuclear are NOT biomass** sources.

PYQ — UPPCS Prelims 2022, Q69

NOT a biomass energy source:

Options: A. Coal B. Nuclear C. Gobar D. Wood

> 💡 **Show answer**
>
> **Ans: A** (also nuclear is not biomass) — coal is fossil; gobar/wood are biomass.

27.8 Biofuels

Identity: Liquid/gaseous fuels from biomass | Policy 2018 | E20 target | Food vs fuel debate

- **Biofuels** are fuels produced from biomass — bioethanol, biodiesel, biogas/CBG, advanced (2G) fuels.
- **National Policy on Biofuels, 2018** sets blending roadmap and feedstock categories.
- **1st generation:** sugarcane molasses ethanol; edible-oil biodiesel — food vs fuel concern.
- **2G:** agri-waste, residues — reduces food competition.
- Benefits: lower net GHG than fossils; domestic feedstock; energy security. Trap: biofuels are often **not cost-effective** versus fossil fuels.
- **Maize** is a classic **ethanol** crop; **Jatropha / Pongamia / sunflower** are classic **biodiesel** crops — do not swap.
- Biomass **gasification** (coconut/groundnut shells, rice husk) produces combustible **producer gas** (CO + H_2 + traces of CH_4) usable for heat and **internal combustion engines** — not "CO_2 only" and not "heat only."
- UP western sugarcane belt is a major **bioethanol** feedstock zone.

Fuel	Feedstock	Blend use
Bioethanol	Molasses, maize, 2G waste	Petrol E10/E20

Fuel	Feedstock	Blend use
Biodiesel	Jatropha, Pongamia, UCO	Diesel B5/B10
Biogas/CBG	Dung, organic waste	Cooking / transport

> **Logic:** Biofuels ⊃ ethanol + biodiesel + biogas — terms are not interchangeable.

27.9 Biogas

Identity: Anaerobic digestion | CH_4 + CO_2 | GOBAR-Dhan / SATAT

- Biogas forms when organic matter digests **without oxygen**.
- Major constituents: **methane (~50–70%)** + **carbon dioxide (~25–45%)**.
- **Butane** and **carbon monoxide** are not major biogas components.
- Slurry is useful bio-fertiliser; SATAT promotes compressed biogas (CBG) (संपीडित बायोगैस).

PYQ — UPPCS Prelims 2025, Q28

Major constituents of biogas: (1) Methane (2) Butane (3) Carbon dioxide (4) Carbon monoxide

Options: A. 1 and 3 B. 3 and 4 C. 2 and 3 D. 1 and 2

> 💡 **Show answer**
>
> **Ans: A** — Methane + CO_2.

27.10 Bioethanol

Identity: C_2H_5OH from fermentation | Petrol blend | E10 achieved | E20 by 2025–26

- **Bioethanol** is alcohol produced mainly by fermenting sugarcane molasses (also maize, 2G waste).
- Blended with petrol: **E10** (10%) largely achieved; **E20 by 2025–26** under Biofuels Policy 2018.
- Reduces petrol import bill and transport CO_2 intensity.
- Western UP sugar belt distilleries are key producers.

> **Logic:** Bioethanol = **petrol** engines; biodiesel = **diesel** engines.

27.11 Biodiesel

Identity: Transesterification | Jatropha/Pongamia/UCO | Diesel blend B5/B10

- **Biodiesel** is fatty acid methyl ester (**FAME**) made by **transesterification** of vegetable oils + methanol + catalyst → biodiesel + glycerol.
- Feedstocks: **Jatropha**, **Pongamia pinnata (karanj)**, used cooking oil (**RUCO** initiative).
- **Pongamia** grows in arid / saline (लवणीय) / alkaline soils; seeds are lipid-rich with nearly half **oleic acid** — classic dual-statement biodiesel stem with Jatropha.
- Blended with diesel as **B5/B10**.
- Biodiesel ≠ biogas (liquid ester vs methane gas).
- First Indian airline biofuel ATF flight (Aug **2018**): **SpiceJet** (Dehradun (देहरादून)–Delhi).

> **Logic:** Jatropha was promoted as non-edible biodiesel crop — yield challenges limited scale.

27.12 Small Hydropower

Identity: ≤25 MW MNRE | Himalayan potential | Run-of-river preferred

- MNRE defines **small hydro as ≤25 MW**; above that = large hydro (conventional).
- Himalayan rivers have **enormous** hydro potential — Indian projects in UK, HP, Sikkim, Arunachal.
- Run-of-river SHP has lower submergence than large storage dams.
- **Ramganga (रामगंगा) (~198 MW)** is **large hydro**, not SHP.

> **Logic:** Himalayan potential true; "Nepal border means zero Indian hydro" false.

27.13 Tidal Energy

Identity: Moon's gravity | Khambhat highest | Predictable | No major Indian plant yet

- Tidal energy uses rise/fall of sea level driven mainly by the **Moon**.
- India potential: **Gulf of Khambhat** (highest range), **Gulf of Kutch**.
- Highly predictable vs wind/solar; capital cost and estuary (ज्वारनदमुख) impact are high.
- Tidal ≠ wave energy (waves = wind-driven surface motion).

27.14 Ocean Energy

Identity: Wave + OTEC + salinity + tidal stream | Broader than tidal alone

- **Ocean energy** includes **wave**, **OTEC**, **salinity gradient**, and **tidal stream** technologies.
- **OTEC:** uses ≥~20°C temperature difference between warm surface and cold deep water — tropical seas (Lakshadweep (लक्षद्वीप)/Andaman (अंडमान) potential).
- Wave energy: wind-driven surface waves — R&D stage on Indian coasts.
- Trap: "Ocean energy means only tidal" = **FALSE**.

Type	Driver
Tidal	Moon gravity
Wave	Wind on sea surface
OTEC	Warm–cold water ΔT

> **Logic:** Ocean energy ⊃ tidal, but is not limited to it.

27.15 Energy Security

Identity: Reliable affordable supply | Oil import risk | SPR | Renewables + biofuels buffer

- **Energy security** means reliable, affordable energy supply without crippling import or shortage risk.
- India imports ~**85%** crude oil — price and geopolitics shock risk.
- Coal is largely **domestic** — electricity security anchor.
- **Strategic Petroleum Reserve (SPR):** Visakhapatnam, Mangalore, Padur — crisis oil buffer.
- Tools: diversify suppliers, expand renewables, biofuel blending, gas pipelines, grid reliability.
- UP has little fossil production — depends on inter-state power/fuel; local solar + ethanol help.

> **Logic:** Energy security ≠ oil alone — includes coal, gas, electricity, and strategic stocks.

27.16 Renewable Energy Scenario in India

Identity: MNRE nodal | Solar largest growth | Wind second | 500 GW non-fossil 2030

- MNRE is nodal for new and renewable energy (large hydro usually under Power Ministry).
- Rough installed picture (mid-2020s order): solar leading growth; wind second; biomass/SHP smaller shares.
- Earlier 175 GW by 2022 ambition expanded into **500 GW non-fossil by 2030**.
- Challenges: land, grid evacuation, storage, seasonal mismatch, state DISCOM finances.

Source	Key fact
Solar	Fastest capacity growth
Wind	TN/Gujarat strongholds
SHP	Himalayan streams ≤25 MW
Biomass	Bagasse, biogas, waste-to-energy

27.17 National Renewable Energy Programme

Identity: MNRE umbrella | NSM + wind + biomass + SHP + waste | PLI manufacturing

- National renewable push is an **MNRE programme cluster**, not one single Act.
- Includes National Solar Mission, wind policies, biomass/waste-to-energy, SHP support, offshore wind policy.
- **PLI for solar PV** supports domestic manufacturing (Atmanirbhar).
- Convergence with Power Ministry grid reforms and state renewable purchase obligations (RPOs).

27.18 National Solar Mission

Identity: 2010 NAPCC mission | Solar scale-up | Part of 8 National Missions

- Launched **2010** under **NAPCC (2008)** as one of eight National Missions.
- Aimed to make India a global solar leader — capacity, cost reduction, domestic manufacturing.
- Target path: early 20 GW ambition → revised to **100 GW solar** → contributes to 500 GW non-fossil 2030.
- Complements ISA (2015) and PM-KUSUM (2019).

> **Logic:** NSM = **2010**, under NAPCC — not 2015 (ISA year).

27.19 Energy Conservation

Identity: EC Act 2001 | BEE | Star labeling | Avoided demand

- **Energy conservation** means using less energy for the same service — avoid waste.
- **Energy Conservation Act 2001** created **Bureau of Energy Efficiency (BEE)** (ऊर्जा दक्षता ब्यूरो) under Ministry of Power.
- BEE sets standards, labels, and norms for designated consumers.
- **Star labeling (1–5 stars)** on ACs, fridges, fans, TVs, geysers — higher stars = lower consumption.
- Building energy codes and awareness campaigns cut demand without new generation.

Item	Year/body
Energy Conservation Act	**2001**
BEE	Under Power Ministry
Star labeling	Appliance efficiency

> **Logic:** BEE = efficiency; MNRE = renewables — different mandates.

27.20 Energy Efficiency

Identity: Same output, less input | PAT ESCerts | UJALA LEDs | PAT ≠ CCTS

- **Energy efficiency** means delivering the same service with less energy input.
- **PAT (Perform, Achieve and Trade (पण्याध्यक्ष)), 2012:** market scheme for energy-intensive industries — surplus efficiency becomes tradable **ESCerts**.
- PAT is **not** carbon credit (कार्बन क्रेडिट) trading (**CCTS 2023** is the carbon market).
- **UJALA:** mass LED distribution — large residential efficiency gain.
- Efficiency + renewables together cut emissions faster than either alone.

Scheme	What it trades/does
PAT	ESCerts (energy efficiency)
CCTS	Carbon credits (CO_2e)
UJALA	LED bulbs at scale

> **Logic:** PAT = **energy efficiency certificates**, not carbon credits.

Complete PYQ Bank (Topic 27)

Q1. UPPCS Prelims 2025, Q28

Major constituents of biogas include:

1. Methane
2. Butane
3. Carbon dioxide
4. Carbon monoxide

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Logic: Biogas = anaerobic digestion product — CH_4 + CO_2 dominant. **Ans: A (1 and 3).** **Methane and carbon dioxide** are major biogas constituents. **Trap:** **Butane** (LPG component) as biogas constituent.

Q2. UPPCS Prelims 2025, Q31

With reference to hydroelectric power in Northern India, consider the following statements:

1. The Himalayan region has enormous hydroelectric potential.
2. Eastern Uttar Pradesh and Bihar cannot utilise Himalayan hydro because the Himalayas to their north lie in Nepal.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: Himalayan hydro potential true; Nepal excuse in statement 2 is false. **Ans: D (Only 1).** **Only statement 1 is correct**. **Trap:** Accepting statement 2 — Indian projects in Uttarakhand/HP benefit the plain states.

Q3. UPPCS Prelims 2025, Q109

With reference to the International Solar Alliance (ISA), consider the following statements:

1. It is the first treaty-based intergovernmental organization headquartered in India and has 38 member countries.
2. The World Solar Bank and One Sun One World One Grid (OSOWOG) initiatives have been undertaken under it.

A. Only 1

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 2

💡 **Show answer**

Logic: OSOWOG under ISA is correct; fixed "38 members" count is outdated. **Ans: A (Only 2).** **Only statement 2 is correct**. **Trap:** Rejecting statement 2 because statement 1 sounds precise.

Q4. UPPCS Prelims 2023, Q102

Non-conventional energy sources are those that are—

A. Non-renewable

B. Renewable

C. Produced from electricity

D. Produced from heat only

💡 **Show answer**

Logic: Indian usage treats non-conventional ≈ renewable (solar, wind, biomass, tidal, SHP). **Ans: B.** **Non-conventional sources are renewable**. **Trap:** **Non-renewable** — that describes conventional fossils, not solar/wind.

Q5. UPPCS Prelims 2022, Q69

Which of the following is **NOT** a biomass energy source?

A. Coal

B. Nuclear fuel

C. Gobar (cow dung)

D. Wood

💡 **Show answer**

Logic: Biomass = organic matter; coal is fossilised carbon; nuclear is uranium/thorium. **Ans: A.** **Coal is not biomass**. **Trap:** Nuclear is also non-biomass, but coal is the standard keyed answer.

Q6. UPPCS Prelims 2021, Q124

A non-renewable source of energy is:

A. Solar energy

B. Petroleum

C. Wind energy

D. Biogas

💡 **Show answer**

Logic: Petroleum is a finite fossil fuel; solar, wind, and biogas are renewable. **Ans: B.** **Petroleum is non-renewable**. **Trap:** Confusing **biogas** (renewable) with fossil gas.

Q7. UPPCS Prelims 2022, Q20

The largest tidal energy potential in India is associated with the:

A. North Circars coast

B. Mannar coast

C. Khambhat (Cambay) coast

D. Kerala coast

💡 **Show answer**

Logic: Gulf of Khambhat has India's highest tidal range — potential fact. **Ans: C.** **Khambhat (Cambay) coast** has the largest tidal energy potential. **Trap:** Kerala or eastern coast names without highest range.

Q8. UPPCS Prelims 2022, Q132

Solar energy is converted into electric energy through—

A. Dry cells

B. Leclanché cells

C. Voltaic cells

D. Photovoltaic cells

💡 **Show answer**

Logic: PV cells directly convert sunlight to electricity in solar applications. **Ans: D.** **Photovoltaic cells** convert solar to electric energy. **Trap:** Generic **voltaic/dry cell** chemistry distractors.

Q9. UPPCS Prelims pattern — EC Act 2001

The Energy Conservation Act, 2001 established which body as the nodal authority for energy efficiency?

A. Bureau of Energy Efficiency (BEE)

B. Central Electricity Authority only

C. NITI Aayog

D. MNRE

💡 **Show answer**

Logic: BEE under EC Act 2001 — administers PAT and standards/star labelling. **Ans: A.** **BEE was established under the Energy Conservation Act, 2001**. **Trap:** **MNRE** (renewable promotion) vs **BEE** (efficiency regulation).

Q10. UPPCS Prelims pattern — PAT scheme

Perform, Achieve and Trade (PAT) is a scheme for:

A. Trading energy efficiency certificates among designated consumers

B. Trading wildlife permits

C. Trading CFC quotas under Montreal Protocol

D. Trading Ramsar wetland credits

💡 **Show answer**

Logic: PAT = market-based energy efficiency compliance under BEE — ESCerts trading. **Ans: A.** **PAT trades energy efficiency certificates (ESCerts)**. **Trap:** Confusing PAT with **carbon credit** markets directly.

Extra Drill — Energy Classification & Biofuels (Q31–Q40)

Q31. Energy crisis commonly means:

A. Only shortage of hydroelectricity

B. Malnutrition in the human body

C. Danger of exhaustion of fossil fuels like coal and petroleum

D. Only shortage of thermal power plants

💡 **Show answer**

Logic: Classification swaps (renewable / fossil / nuclear / biomass) and biofuel feedstock pairs are the densest traps. **Ans: C.** UN Decade of Sustainable Energy for All = **2014–2024** (UN initiative).

Q32. Which is **not** a fossil fuel?

A. Coal

B. Petroleum

C. Natural gas

D. Uranium

💡 **Show answer**

Ans: D. Uranium is nuclear fuel — non-renewable but not fossil.

Q33. Best electricity source among options for sustainable development:

A. Coal

B. Mineral oil and gas

C. Hydroelectricity

D. Atomic energy

💡 **Show answer**

Ans: C. Water is not depleted by generation the way fossil/nuclear stocks are.

Q34. Ultimate source of energy stored in fossil fuels:

A. Earth interior only

B. Sun

C. Sea

D. Moon

💡 **Show answer**

Ans: B. Ancient photosynthesis. Geothermal heat itself is **not** "from the Sun" in MCQ sense.

Q35. Maize is primarily cultivated among energy crops for:

A. Biodiesel only

B. Ethanol

C. Nuclear fuel

D. Tidal power

💡 **Show answer**

Ans: B. Jatropha / Pongamia / sunflower → biodiesel; maize → ethanol.

Q36. Biomass gasification — which is correct?

A. Only CO_2 is produced; cannot run engines

B. Shells/husk usable; producer gas has CO + H_2 (+ traces CH_4); can run IC engines

C. Only hydrogen and carbon dioxide; heat only

D. Nuclear waste is a biomass feedstock

💡 **Show answer**

Ans: B. Statement "CO_2 only" / "heat only" pairs are false.

Q37. Which statement about biofuels is **incorrect**?

A. Biofuel is eco-friendly

B. Biofuel is always cost-effective versus fossils

C. Biofuel can help remedy energy crisis

D. Biofuel can be made from corn

💡 **Show answer**

Ans: B. Cost remains a classic limitation.

Q38. Fuel cells with pure hydrogen:

A. Emit heat and water; produce DC; can power buildings and small devices

B. Produce only AC; cannot power laptops

C. Emit CO_2 as the main by-product

D. Are identical to coal boilers

> 💡 **Show answer**
>
> **Ans: A.** Statement pairs claiming AC-only or buildings-only are false.

Q39. Least polluting common fuel among options:

A. Diesel

B. Coal

C. Hydrogen

D. Kerosene

> 💡 **Show answer**
>
> **Ans: C.** Future fuel / National Hydrogen Energy Board **2003**; Green Hydrogen Mission **2023**.

Q40. Non-conventional energy sources are those that are:

A. Only produced from heat

B. Only produced from electricity

C. Non-renewable electricity

D. Renewable

> 💡 **Show answer**
>
> **Ans: D.** Non-conventional ≈ renewable in Indian prelims usage (solar, wind, etc.).

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to renewable vs non-renewable, which of the following statements is/are correct?

1. Renewable replenishes on a human timescale.
2. Non-renewable sources are infinite stocks that replenish within years.
3. Coal is renewable biomass.

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Ans: D.** Only 1 is correct. **Logic:** Statement 1 is true. Statement 2 is false — coal/oil/gas/uranium are finite non-renewable stocks. Statement 3 is false — coal is not renewable biomass.

Q2. Match List-I with List-II:

List-I	List-II
A. Conventional (India)	1. Solar, wind, SHP, biomass…
B. Non-conventional	2. Coal, oil, gas, large hydro, nuclear
C. SHP cut-off	3. ≤25 MW

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Indian split + SHP.

Q3. Which of the following pairs is/are correctly matched?

1. Energy crisis in teaching means only temporary grid-billing disputes, never fossil exhaustion risk.
2. Ultimate fossil energy source — Sun
3. Uranium — fossil fuel identical to coal

A. Only 3 B. Only 2 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Statement 2 is true (ultimate fossil energy source = Sun). Statement 1 denies the fossil-exhaustion teaching of energy crisis. Statement 3 is false — uranium is nuclear fuel, not a fossil fuel identical to coal.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Large hydro sits on the conventional side in Indian classification.

Reason (R): Calling all hydro "non-conventional" is wrong under that split.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R restates the classification fact. **A/R logic:** Large hydro = conventional.

Q5. With reference to biogas/biofuels, which of the following statements is/are correct?

1. Biogas major pair CH_4 + CO_2.
2. Bioethanol blends with petrol; biodiesel with diesel.
3. Major biogas pair is butane + CO.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not butane/CO.

Q6. Arrange UN Decade of Sustainable Energy for All:

A. 2001–2011 World Bank only B. 2014–2024 UNGA (SDG 7 neighbourhood) C. 1972–1982 UNEP only D. 1992–2002 WTO only

💡 **Show answer**

Ans: B. 2014–2024 UNGA. **Logic:** Not World Bank.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Producer gas — CO, H_2, traces CH_4
2. OTEC — warm surface vs cold deep water
3. OTEC — identical to tidal barrage

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** OTEC ≠ tidal.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): EC Act 2001 created BEE.

Reason (R): PAT (2012) trades ESCerts under the efficiency architecture.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; PAT does not explain BEE's creation under the 2001 Act. **A/R logic:** Creation vs later PAT.

Q9. With reference to energy security, which of the following statements is/are correct?

1. India imports a very large share of crude (~85% teaching).
2. Coal is mostly domestic.
3. SPR locations include Vizag, Mangalore, Padur.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Import + SPR trio.

Q10. Match List-I with List-II:

List-I	List-II
A. ISA	1. Domestic efficiency
B. BEE	2. Solar diplomacy / Gurugram
C. E20	3. Higher ethanol blend ambition

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** ISA/BEE/E20.

Q11. Arrange hydroelectricity vs sustainability option set:

A. Coal always best B. Among coal/oil-gas/nuclear/hydro, hydro often best sustainability answer for electricity C. Nuclear always renewable D. Oil always zero carbon

💡 **Show answer**

Ans: B. Hydro often best among that set. **Logic:** Bank language.

Q12. With reference to hydrogen/fuel cells, which of the following statements is/are correct?

1. H_2 burns to water — least-polluting common fuel option teaching.
2. Fuel cell output is DC.
3. Fuel cell output is always AC only.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** DC — not AC-only.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): PAT certificates measure energy-savings trading.

Reason (R): Carbon markets measure CO_2 units — related politics, different instruments.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; carbon-market definition does not explain what PAT certificates measure. **A/R logic:** PAT ≠ carbon credit.

Q14. Which one of the following correctly states conservation vs efficiency?

A. Conservation = only building more coal plants B. Both identical always C. Efficiency = only importing more oil D. Conservation = using less; efficiency = more service per unit energy

💡 **Show answer**

Ans: D. Using less vs more service per unit. **Logic:** Neighbours — not synonyms.

Q15. How many of the following ocean facts are distinct?

Tidal, wave, OTEC

A. All three distinct B. Only two C. Only one D. None exist

💡 **Show answer**

Ans: A. All three distinct. **Logic:** Umbrella ≠ identity.

Q16. With reference to wind/solar, which of the following statements is/are correct?

1. Wind leaders often TN and Gujarat.
2. NSM 2010; 500 GW non-fossil by 2030 teaching.
3. UP wind is the dominant national wind leader always.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** UP wind limited.

Q17. Match List-I with List-II:

List-I	List-II
A. Microbial fuel cell	1. Non-petroleum paving
B. Bioasphalt	2. Living microbes as catalysts
C. National Hydrogen Board	3. Set up teaching neighbourhood

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** MFC / bioasphalt / H_2 board.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Nuclear is usually non-renewable/conventional in Indian prelims framing.

Reason (R): Low-carbon status automatically makes nuclear renewable.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Low-carbon ≠ renewable.

Q19. Which of the following pairs is/are correctly matched?

1. Green hydrogen routes — IC engines, NG blend, FCEVs
2. Western UP sugarcane — bioethanol neighbourhood
3. Bagasse — cogeneration

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** H_2 + UP biofuel colour.

Q20. Arrange E10 vs E20:

A. E20 is higher ethanol blend ambition beyond E10 B. Freeze forever at E10 only C. E20 means only diesel biodiesel D. E10 means nuclear blend

💡 **Show answer**

Ans: A. E20 beyond E10. **Logic:** Do not freeze at E10.

Q21. With reference to UJALA / Star labels, which of the following statements is/are correct?

1. Efficiency/conservation tools under BEE neighbourhood.
2. Link to EC Act architecture.
3. They create IUCN Red List categories.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not IUCN.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Tidal energy is Moon-driven.

Reason (R): Ocean energy is a wider set that includes wave, OTEC, and tidal.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; umbrella set does not explain Moon-driven mechanism. **A/R logic:** Member vs umbrella.

Q23. How many of the following are fossils?

Coal, petroleum, natural gas, uranium

A. Only one B. Only two C. Only three D. All four

💡 **Show answer**

Ans: C. Only three. **Logic:** Uranium not fossil.

Q24. Which one of the following correctly states biomass gasification producer gas?

A. Only N_2 always B. Only pure O_2 C. Only argon D. CO, H_2, traces CH_4 — not CO_2-only

💡 **Show answer**

Ans: D. CO/H_2/traces CH_4. **Logic:** Not CO_2-only.

Q25. With reference to PM-KUSUM / ISA, which of the following statements is/are correct?

1. Sit in solar policy neighbourhood with NSM 2010.
2. ISA HQ Gurugram teaching.
3. ISA is identical to BEE Star labels.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** ISA ≠ BEE.

Q26. Match List-I with List-II:

List-I	List-II
A. Bioethanol feedstocks	1. Jatropha, Pongamia, UCO
B. Biodiesel feedstocks	2. Maize, sugarcane…
C. SPR	3. Vizag, Mangalore, Padur

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** Ethanol/diesel/SPR.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Microbial fuel cells can be useful in wastewater plants.

Reason (R): Substrates are mainly organic — not "only inorganic salts."

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains the wastewater link. **A/R logic:** Organic substrates.

Q28. Arrange SHP MNRE cut-off:

A. ≤250 MW only B. ≤2.5 MW only C. ≤25 MW D. No cut-off taught

> 💡 **Show answer**
>
> **Ans: C.** ≤25 MW. **Logic:** Recurring number.

Q29. With reference to non-conventional ≈ renewable, which of the following statements is/are correct?

1. Often treated as roughly equal in prelims stems.
2. Large hydro exception sits on conventional side.
3. All nuclear is therefore renewable.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Nuclear not renewable.

Q30. Which one of the following best states the Sun–fossil link?

A. Ultimate source of energy in fossils is ancient photosynthesis (Sun) B. Fossils store only lunar tidal energy C. Fossils are abiotic rocks with no solar past D. Fossils are identical to geothermal magma only

> 💡 **Show answer**
>
> **Ans: A.** Ancient photosynthesis. **Logic:** Sun ultimate source.

Common Traps

Trap	Correct
Nuclear = renewable	Usually **non-renewable**
Non-conventional = non-renewable	**= renewable**
Coal = biomass	**Fossil, not biomass**
Uranium = fossil fuel	**Nuclear fuel**, not fossil
Biogas = butane	**CH_4 + CO_2**
Tidal = OTEC/wave	**Different drivers**
PAT = carbon credits	**ESCerts**
BEE = MNRE	**Power vs Renewables**
Large hydro = non-conventional	Usually **conventional**

Trap	Correct
E20 already forever since 2010	**Target ~2025–26**
SPR only in Delhi	**Vizag, Mangalore, Padur**
Energy crisis = only hydro shortage	**Fossil fuel exhaustion risk**
Biofuels always cheap	Often **not cost-effective**
Maize = biodiesel crop	**Ethanol**; Jatropha = biodiesel
Fuel cell = AC output	**DC**; by-products heat + water
Gasification gas = CO_2 only	**Producer gas** CO + H_2 (+ CH_4)

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 28 — Environmental Research & Institutions

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

BSI | ZSI | FSI | FRI | WII | NBA | SBB | BMC | NBPGR

Sources baked in: MoEFCC; ICAR; ICFRE; Biological Diversity Act 2002; ISFR 2023; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — BSI/ZSI trap; ISFR→FSI; NBA Chennai; BMC→PBR; NBPGR Delhi **Last verified:** August 2026

Current Affairs (this topic)

- FRI / FSI / WII / IIRS function traps are evergreen Dehradun (देहरादून) CA colour.
- Do not swap ISFR (FSI) with FRI research mandate.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. Headquarters map: **Kolkata** — BSI and ZSI; **Dehradun** (देहरादून) — FSI, FRI, and WII; **New Delhi** — NBPGR; **Chennai** — NBA. 2. **BSI (1890, Kolkata)** surveys **plants** (*Flora* (वनस्पति), plant Red Data Book (रेड डेटा बुक)). **ZSI (1916, Kolkata)** surveys **animals** (*Fauna* (प्राणीजात), animal Red Data Book). 3. **FSI (1981, Dehradun)** maps forest cover (वन आवरण) and publishes **ISFR every two years**. **FRI (1906, Dehradun)** does forestry **research** under the ICFRE world — it does **not** publish ISFR. 4. **WII (1982, Dehradun)** does wildlife research, IFS/officer training, and tiger-census protocols. 5. **NBA** under the Biological Diversity Act sits in **Chennai**, not Delhi. It handles foreign access and ABS at the national level. 6. Biodiversity pyramid (पिरामिड): **NBA (national) → SBB (state) → BMC (local)**. BMCs prepare the **People's Biodiversity Register (PBR)**. 7. **NBPGR (1978, New Delhi, ICAR)** is the plant gene bank. It is **not** Kolkata BSI and **not** Karnal **NBAGR** (animals). 8. ISFR **2023** class figures often cited: India about **21.76%** forest cover; Uttar Pradesh (उत्तर प्रदेश) about **6.24%**. UP districts with **>20%** forest cover in that reading (रीडिंग) were **Chandauli and Shravasti only** — **Bahraich** was not in that pair. 9. BSI ≠ animals; ZSI ≠ plants; FRI ≠ ISFR; NBA ≠ Delhi; PBR ≠ NBA headquarters function. 10. SBB is the state biodiversity board — not the State Pollution Control Board. 11. NBPGR ≠ Kolkata. Gene bank is Delhi; plant survey is Kolkata. 12. ISFR is biennial, not annual. 13. UPSBB sits in the Lucknow (लखनऊ) state-biodiversity lane; WII supports Terai (तराई) wildlife science links. 14. BMC is local and owns PBR documentation of local biodiversity knowledge. 15. SBB supervises state-level commercial access and BMC work under BDA — domestic track beside NBA's foreign-access focus. 16. FRI research ≠ FSI satellite forest-cover accounting, even though both are Dehradun forestry institutions. 17. WII is wildlife science/training — not the ISFR publisher and not the plant gene bank. 18. Red Data Books from BSI/ZSI are national catalogues; IUCN (आईयूसीएन) Red List remains the global risk ladder (different chapter neighbour). 19. NBA year fact rides on BDA **2002** institutionalisation — headquarters city remains Chennai. 20. Gene-bank NBPGR serves crop genetic resources for all states, including UP, from New Delhi. 21. Forest-cover percentage traps for UP hinge on Chandauli + Shravasti, not a long distractor list of Terai districts. 22. PBR is prepared locally by BMCs; it is not an NBA HQ publication series. 23. Match institution → city before matching function. Wrong city is enough to kill a match list. 24. Dehradun trio: FSI (cover/ISFR), FRI (research), WII (wildlife). Keep the three jobs separate.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Botanical Survey of India (BSI)** | **Zoological Survey of India (ZSI)** | Premier taxonomic institution (HQ: Kolkata, established 1890) undertaking floristic surveys, documenting plant species, and managing the Central National Herbarium vs premier institution (HQ: Kolkata, established 1916) surveying faunal diversity, animal taxonomy, and systematic collections | भारतीय वनस्पति सर्वेक्षण (BSI, 1890) / भारतीय प्राणी सर्वेक्षण (ZSI, 1916) | | **Forest Survey of India (FSI)** | **Forest Research Institute (FRI)** | Subordinate organization under MoEFCC (HQ: Dehradun, established 1981) conducting national forest resource inventory via satellite imagery and publishing biennial ISFR vs deemed university and premier research institution under ICFRE (HQ: Dehradun, established 1906 as Imperial Forest Research Institute) | भारतीय वन सर्वेक्षण (FSI, द्विवार्षिक वन रिपोर्ट) / वन अनुसंधान संस्थान (FRI, वानिकी अनुसंधान) | | **National Biodiversity Authority (NBA)** | **State Biodiversity Board (SBB)** | Autonomous statutory regulatory authority established in 2003 under Biological Diversity Act 2002 (HQ: Chennai) regulating foreign nationals' access to Indian bioresources and advising Union vs state-level statutory bodies regulating commercial utilization of bioresources by Indian citizens | राष्ट्रीय जैव विविधता प्राधिकरण (NBA, चेन्नई) / राज्य जैव विविधता बोर्ड (SBB) | | **National Bureau of Plant Genetic Resources (NBPGR)** | **National Bureau of Agricultural Insect Resources (NBAIR)** | Nodal ICAR organization (HQ: New Delhi) managing the National Gene Bank preserving orthodox seeds and germplasm of food and agricultural crops vs premier ICAR research bureau (HQ: Bengaluru) dedicated to insect taxonomy, molecular characterization, and biocontrol insect resources | राष्ट्रीय पादप आनुवंशिक संसाधन ब्यूरो (NBPGR, दिल्ली) / कृषि कीट संसाधन ब्यूरो (NBAIR, बेंगलुरु) | | **Wildlife Institute of India (WII)** | **Indira Gandhi National Forest Academy (IGNFA)** | Autonomous scientific research and training institute under MoEFCC (HQ: Dehradun) conducting wildlife ecological research and Tiger Census vs premier apex staff college (HQ: Dehradun) providing two-year professional induction training to directly recruited Indian Forest Service (IFS) officers | भारतीय वन्यजीव संस्थान (WII, अनुसंधान) / इंदिरा गांधी राष्ट्रीय वन अकादमी (IGNFA, IFS प्रशिक्षण) | | **India State of Forest Report (ISFR)** | **Basic Animal Husbandry Statistics** | Biennial statutory publication by Forest Survey of India (FSI, Dehradun) reporting state-wise forest cover, tree cover, mangrove cover, and forest carbon stocks vs annual publication by Ministry of Fisheries, Animal Husbandry & Dairying reporting livestock census, milk, egg, and meat production | भारत वन स्थिति रिपोर्ट (ISFR, द्विवार्षिक) / बुनियादी पशुपालन सांख्यिकी (वार्षिक) |

💡 📋 **Must-score facts — BSI ZSI FSI NBA NBPGR (Click to Expand)**

HQ map | Item | Lock | |---|---| | Kolkata | BSI **1890** plants; ZSI **1916** animals | | Dehradun | FSI **1981** ISFR; FRI **1906**; WII **1982** | | New Delhi | NBPGR plant gene bank | | Chennai | NBA (≠ Delhi) | | Karnal | NBAGR animals | | ISFR | Biennial (≠ annual); FSI product | | BDA ladder | NBA → SBB → BMC → PBR | | SBB ≠ SPCB | Biodiversity board ≠ pollution board |

28.1 Botanical Survey of India (BSI) (भारतीय वनस्पति सर्वेक्षण)

Identity: 1890 | Kolkata | Plants only | Flora of India | Red Data Book (plants)

- **BSI** is the national organisation for survey, taxonomy, and documentation of **Indian plant diversity**.
- Established **1890**; headquarters **Kolkata** (Central National Herbarium (पादप संग्रहालय), Shibpur, Howrah); under **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय).
- Publishes ***Flora of India*** and state/regional floras; maintains herbarium network.
- Publishes the **Red Data Book of Indian Plants** — national plant threat list (distinct from IUCN Red List).
- Discovers and describes new plant species — especially Western Ghats (पश्चिमी घाट), Northeast, Andamans.
- **Does not** survey animals (ZSI), map forest cover (FSI), or store crop gene banks (NBPGR).

> **Logic:** BSI = **plants only**. "BSI surveys animals" = **FALSE**.

28.2 Zoological Survey of India (ZSI) (भारतीय प्राणी सर्वेक्षण)

Identity: 1916 | Kolkata | Animals only | Fauna of India | Red Data Book (animals)

- **ZSI** documents **Indian animal (faunal) diversity** — taxonomy and inventories.
- Established **1916**; HQ **Kolkata**; under **MoEFCC**.
- Publishes ***Fauna of India*** series and the **Red Data Book of Indian Animals**.
- Maintains type specimens; scientific data informs WPA Schedule amendments.
- **ZSI ≠ WII:** ZSI = species inventory/taxonomy; WII = applied wildlife ecology (पारिस्थितिकी) and training.

> **Logic:** ZSI = **animals only**. Younger than BSI (1890 vs 1916).

28.3 Forest Survey of India (FSI) (भारतीय वन सर्वेक्षण)

Identity: 1981 | Dehradun | ISFR biennial | Forest cover ≥10% canopy

- **FSI** monitors India's **forest and tree cover (वृक्ष आवरण)** using satellite imagery and ground verification.
- Established **1981**; HQ **Dehradun**; under **MoEFCC**.
- Publishes **India State of Forest Report (ISFR)** every **two years** — not annual.
- Forest cover = land **≥1 ha** with canopy density **≥10%**. Classes: VDF (≥70%), MDF (40–70%), OF (10–40%).
- **ISFR 2023:** India forest cover **21.76%**; UP **6.24%**.
- UP districts with **>20%** forest: **Chandauli** and **Shravasti** only — **Bahraich is not**.
- Also runs satellite **forest fire alerts**. Does **not** set pollution standards (CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड)) or do taxonomy (BSI/ZSI).

PYQ — UPPCS Prelims 2025, Q8

ISFR 2023 — UP districts with forest over >20% of area: (1) Bahraich (2) Chandauli (3) Shravasti

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Ans: C** — Chandauli + Shravasti. Bahraich ~10% despite Terai location.

> **Logic:** ISFR = **FSI only**. Bahraich Terai ≠ automatically >20% cover.

28.4 Forest Research Institute (FRI) (वन अनुसंधान संस्थान)

Identity: 1906 | Dehradun | Forestry research | ICFRE | ≠ FSI

- **FRI** is India's premier **forestry research** institute — silviculture, pathology, entomology, wood science, NTFPs.
- Established **1906** (Imperial Forest Research Institute); now under **ICFRE**; HQ **Dehradun**.
- Trains forestry scientists; iconic (साकार) Dehradun campus.
- **FRI ≠ FSI:** both in Dehradun — FRI = **how to grow/manage forests**; FSI = **how much forest exists on the map**.
- FRI does **not** publish ISFR.

Feature	FRI	FSI
Year	1906	1981
Role	Research	Cover survey

Feature	FRI	FSI
Output	Silviculture protocols	**ISFR**

> **Logic:** "FRI publishes ISFR" = **FALSE**.

28.5 Wildlife Institute of India (WII) (भारतीय वन्यजीव संस्थान)

Identity: 1982 | Dehradun | Wildlife research + IFS training | Tiger census protocols

- **WII** is MoEFCC's autonomous institute for **wildlife research, training, and conservation science**.
- Established **1982** at **Dehradun**.
- Trains **IFS** and state forest officers in wildlife management.
- Designs **All India Tiger Estimation** scientific protocols (with NTCA).
- Studies population (जनसंख्या) ecology, human–wildlife conflict, corridors (including UP Terai).
- **Does not** do national plant/animal taxonomy (BSI/ZSI) or publish ISFR (FSI).

> **Logic:** Dehradun trio — **FSI** (cover), **FRI** (forestry research), **WII** (wildlife).

28.6 National Biodiversity Authority (NBA) (राष्ट्रीय जैव विविधता प्राधिकरण)

Identity: BDA 2002 | Chennai HQ | Foreign access + ABS | ≠ PBR preparer

- **NBA** is the national statutory body under the **Biological Diversity Act, 2002**.
- Headquarters: **Chennai, Tamil Nadu (नाडु)** — **not New Delhi**.
- Approves **foreign** access to Indian biological resources; regulates biopiracy; ensures **Access and Benefit-Sharing (ABS)**.
- Implements India's CBD / Nagoya Protocol obligations.
- Advises the Centre on biodiversity policy; can support **Biodiversity Heritage Sites**.
- **Does not** prepare People's Biodiversity Registers — that is **BMC**.

Level	Body	Seat	Role
National	NBA	**Chennai**	Foreign access, ABS
State	SBB	State capital	Domestic commercial access
Local	BMC	Panchayat/ward	**PBR**

> **Logic:** NBA HQ = **Chennai**. "NBA prepares PBR" = **FALSE**.

28.7 State Biodiversity Boards (SBB)

Identity: BDA 2002 | One per state | Domestic access | BMC supervision | ≠ SPCB

- Every state/UT has an **SBB** under BDA 2002.
- **Uttar Pradesh State Biodiversity Board (UPSBB)** sits at **Lucknow** (लखनऊ).
- Regulates **Indian** commercial access to bio-resources within the state; foreign cases go to NBA.
- Guides BMCs; receives and validates **PBRs**; maintains state biodiversity database.
- Can identify/recommend (सिफारिश) **Biodiversity Heritage Sites**.

- **SBB ≠ SPCB:** SBB = biodiversity (BDA); SPCB = pollution (Water/Air Acts).

> **Logic:** Do not confuse State Biodiversity Board with State Pollution Control Board.

28.8 Biodiversity Management Committees (BMC)

Identity: Local BDA body | Prepares PBR | Panchayat/ward level

- **BMC** is the **local** committee under BDA 2002 — at panchayat, municipality, or cantonment level.
- Core duty: prepare and update the **People's Biodiversity Register (PBR)** (जन जैव विविधता रजिस्टर) — local species, habitats, and traditional knowledge.
- Supports local conservation and channels ABS benefits when bio-resources are commercially used.
- Works under SBB guidance; national reporting links upward to NBA.
- **PBR is not prepared by NBA or BSI**.

> **Logic:** BMC → PBR**. National apex ≠ local register.

28.9 National Bureau of Plant Genetic Resources (NBPGR)

Identity: 1978 | New Delhi (Pusa) | ICAR | Plant gene bank | Ex-situ (स्थानांतरित)

- **NBPGR** conserves **plant genetic resources** — seeds, tissue, DNA — for food and agriculture.
- Established **1978**; HQ **New Delhi (Pusa Campus)**; under **ICAR**.
- Operates the **National Gene Bank** (long-term base collection) and active collections for breeders.
- **Ex-situ** (स्थानांतरित) conservation — complementary to in-situ (स्थान पर) protected areas.
- **NBPGR ≠ BSI:** BSI surveys/taxonomy (Kolkata); NBPGR stores germplasm (Delhi).
- **NBPGR ≠ NBAGR:** animals/livestock gene bank is **NBAGR, Karnal**.

> **Logic:** NBPGR = **New Delhi**, plants. Not Kolkata. Not animal gene bank.

Complete PYQ Bank (Topic 28)

Q1. UPPCS Prelims 2025, Q8

ISFR 2023 — Uttar Pradesh districts with forest over more than 20% of geographical area:

1. Bahraich
2. Chandauli
3. Shravasti

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

💡 **Show answer**

Logic: ISFR 2023 UP district trap — Terai Bahraich is not automatically above 20% cover. **Ans: C.** Chandauli and Shravasti crossed 20%; Bahraich stays near ~10% despite Terai location. **Trap:** Listing every Terai border district as high-cover.

Q2. UPPCS Prelims 2022, Q61

As per the India State of Forest Report–2021, which State registered the maximum increase in forest cover during the last two years?

A. Kerala

B. Assam

C. Arunachal Pradesh

D. Andhra Pradesh

💡 **Show answer**

Logic: ISFR 2021 absolute gain leader among the four options — Andhra Pradesh, not percentage-leader Mizoram. **Ans: D.** Andhra Pradesh led absolute gain in that ISFR cycle (~647 km^2 class), with Telangana and Odisha also high. **Trap:** Picking Mizoram or Madhya Pradesh because they lead on percentage or total area respectively.

Q3. UPPCS / UPSC pattern

Which pair is correctly matched?

A. Botanical Survey of India — national animal survey, Kolkata

B. Zoological Survey of India — national plant survey, Kolkata

C. Botanical Survey of India — national plant survey, Kolkata

D. Zoological Survey of India — forest cover mapping, Dehradun

💡 **Show answer**

Logic: BSI/ZSI split — plants vs animals; both Kolkata, not Dehradun. **Ans: C.** BSI (1890) documents plants; ZSI (1916) documents animals — both HQ Kolkata. **Trap:** Swapping plant/animal mandates or placing survey institutes in Dehradun with FSI/FRI/WII.

Q4. UPPCS / UPSC pattern

India State of Forest Report (ISFR) is published by:

A. Forest Survey of India, Dehradun, every two years

B. Forest Research Institute, Dehradun, annually

C. Botanical Survey of India, Kolkata, every two years

D. CPCB, New Delhi, annually

💡 **Show answer**

Logic: ISFR publisher, city, and frequency — FSI Dehradun, every two years. **Ans: A.** FSI (1981, Dehradun) maps forest and tree cover and releases ISFR biennially. **Trap:** FRI publishes ISFR, or ISFR is annual/CPCB product.

Q5. UPPCS / UPSC pattern

Assertion (A): Forest Research Institute publishes the India State of Forest Report. Reason (R): Both FRI and Forest Survey of India are headquartered at Dehradun.

A. Both true; R explains A

B. Both true; R does not explain A

C. A false; R true

D. Both false

💡 **Show answer**

Logic: Dehradun duo — FRI researches forestry; FSI maps cover; same city, different output. **A/R logic:** A tests ISFR publisher; R tests shared Dehradun HQ — same city does not mean same mandate. **Ans: C.** A is false — **FSI** publishes ISFR; FRI (1906) does silviculture/forestry research under ICFRE. R is true. **Trap:** "Both in Dehradun → same body."

Q6. UPPCS / UPSC pattern

Wildlife Institute of India was established in:

A. 1982 at Dehradun

B. 1890 at Kolkata

C. 2002 at Chennai

D. 1978 at New Delhi

💡 **Show answer**

Logic: WII founding year and wildlife-science role. **Ans: A.** WII (1982, Dehradun) trains IFS officers and designs tiger-estimation protocols — not BSI/ZSI taxonomy or ISFR mapping. **Trap:** Confusing WII with ZSI (1916) or NBA (BDA 2002).

Q7. UPPCS / UPSC pattern

National Biodiversity Authority under the Biological Diversity Act, 2002 is headquartered at:

A. New Delhi

B. Chennai

C. Kolkata

D. Lucknow

💡 **Show answer**

Logic: NBA seat under Biological Diversity Act 2002 — Chennai, not Delhi. **Ans: B.** NBA regulates foreign access and ABS from **Chennai**. **Trap:** Placing NBA in Delhi with MoEFCC, or in Kolkata with BSI/ZSI.

Q8. UPSC pattern

People's Biodiversity Register (PBR) is prepared by:

A. Biodiversity Management Committee at local level

B. National Biodiversity Authority, Chennai

C. Forest Survey of India

D. State Pollution Control Board

💡 **Show answer**

Logic: PBR is a local BMC duty under BDA — not NBA headquarters work. **Ans: A.** BMCs at panchayat/municipality level document local species and traditional knowledge in the PBR. **Trap:** "Biodiversity authority → national register."

Q9. UPSC pattern

National Bureau of Plant Genetic Resources (NBPGR) is located at:

A. New Delhi (Pusa Campus)

B. Kolkata

C. Karnal

D. Chennai

💡 **Show answer**

Logic: NBPGR = ICAR plant gene bank, New Delhi (Pusa) — ex-situ germplasm. **Ans: A.** NBPGR (1978, ICAR) conserves crop genetic resources ex-situ. **Trap:** Kolkata because BSI is there, or Karnal because that is the animal gene bank.

Q10. Topic 20 crossover

Which statement is correct?

A. NBPGR at New Delhi stores plant germplasm; NBAGR at Karnal stores animal/livestock germplasm

B. Both NBPGR and NBAGR are headquartered at Kolkata

C. NBPGR surveys wild fauna for Red Data Book

D. NBAGR publishes ISFR

💡 **Show answer**

Logic: Plant gene bank (NBPGR Delhi) vs livestock gene bank (NBAGR Karnal). **Ans: A.** NBPGR = plants, Delhi; NBAGR = animals/livestock, Karnal — do not interchange. **Trap:** Treating any "gene bank" as one Kolkata institute.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to HQ map, which of the following statements is/are correct?

1. Kolkata — BSI and ZSI.
2. Dehradun — FSI, FRI and WII.
3. NBA — New Delhi.

A. Only 3 B. Only 1 C. 1 and 2 D. 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2 are correct. **Logic:** Statements 1–2 match the HQ map (Kolkata BSI/ZSI; Dehradun FSI/FRI/WII). Statement 3 is false — NBA HQ is Chennai, not New Delhi.

Q2. Match List-I with List-II:

List-I	List-II
A. BSI	1. Animals / Fauna
B. ZSI	2. Plants / Flora
C. NBPGR	3. New Delhi plant gene bank

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Plants/animals/gene bank.

Q3. Which of the following pairs is/are correctly matched?

1. FSI — ISFR every two years
2. FSI — forestry research (ICFRE neighbourhood)
3. FRI — publishes ISFR

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NBA sits in Chennai under the Biological Diversity Act.

Reason (R): NBA headquarters is New Delhi like NBPGR.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Chennai — not Delhi.

Q5. With reference to biodiversity pyramid, which of the following statements is/are correct?

1. NBA → SBB → BMC.
2. BMCs prepare People's Biodiversity Register.
3. PBR is an NBA HQ publication series only.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** PBR is local BMC work.

Q6. Arrange ISFR frequency:

A. Weekly B. Biennial (every two years) C. Every decade only D. Never published

💡 **Show answer**

Ans: B. Every two years. **Logic:** Not annual.

Q7. Which of the following pairs is/are NOT correctly matched?

1. NBPGR — New Delhi ICAR plant gene bank
2. NBAGR — Karnal animals neighbourhood
3. NBPGR — Kolkata BSI identical

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Gene bank ≠ BSI Kolkata.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): WII does wildlife research, IFS training, and tiger-census protocols.

Reason (R): WII publishes ISFR.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** ISFR = FSI.

Q9. With reference to ISFR 2023 class figures, which of the following statements is/are correct?

1. India ~21.76% forest cover teaching.
2. UP ~6.24% teaching.
3. UP districts >20% often hinge on Chandauli + Shravasti teaching.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Cover figures + UP trap.

Q10. Match List-I with List-II:

List-I	List-II
A. SBB	1. Local PBR
B. BMC	2. State biodiversity board
C. UPSBB	3. Lucknow state lane

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** SBB/BMC/UPSBB.

Q11. Arrange Dehradun trio jobs:

A. All three publish ISFR B. FSI cover/ISFR; FRI research; WII wildlife C. All three are plant gene banks D. All three sit in Chennai

💡 **Show answer**

Ans: B. Cover / research / wildlife. **Logic:** Keep jobs separate.

Q12. With reference to BSI/ZSI years, which of the following statements is/are correct?

1. BSI 1890; ZSI 1916 — both Kolkata.
2. BSI animals; ZSI plants.
3. National Red Data Books vs IUCN global list differ.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 3. **Logic:** BSI plants; ZSI animals.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): SBB is the state biodiversity board.

Reason (R): SBB is identical to the State Pollution Control Board.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** SBB ≠ SPCB.

Q14. Which one of the following correctly states NBA's national role?

A. Only FSI satellite mapping B. Publishes ISFR C. Only Karnal animal gene bank D. Foreign access and ABS at national level

💡 **Show answer**

Ans: D. Foreign access / ABS. **Logic:** BDA national body.

Q15. How many of the following HQ facts are correct?

BSI Kolkata; FSI Dehradun; NBPGR New Delhi; NBA Chennai

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** City map.

Q16. With reference to domestic vs foreign biodiversity access, which of the following statements is/are correct?

1. NBA focuses foreign access.
2. SBB supervises state-level commercial access and BMC work.
3. Both are identical to CPCB consent orders.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not CPCB.

Q17. Match List-I with List-II:

List-I	List-II
A. FSI	1. 1906
B. FRI	2. 1981
C. WII	3. 1982

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** 1981 / 1906 / 1982.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Match institution → city before function in match lists.

Reason (R): Wrong city alone can kill the key.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the method. **A/R logic:** City-first method.

Q19. Which of the following pairs is/are correctly matched?

1. Gene bank NBPGR serves crop genetic resources including UP
2. WII supports Terai wildlife science links
3. NBA sits in Lucknow UPSBB office only

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** NBA = Chennai.

Q20. Arrange BDA institutionalisation neighbourhood:

A. NBA year rides on BDA 2002; HQ Chennai B. NBA 1890 Kolkata only C. NBA = FSI 1981 D. NBA = NBPGR Karnal

💡 **Show answer**

Ans: A. BDA 2002; Chennai. **Logic:** Institution year + city.

Q21. With reference to Red Data Books, which of the following statements is/are correct?

1. BSI/ZSI national catalogues.
2. IUCN Red List remains global risk ladder.
3. BSI publishes the global IUCN list for all countries.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** National ≠ global IUCN.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): FRI research is not the same as FSI satellite forest-cover accounting.

Reason (R): Both are Dehradun forestry institutions with different jobs.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the separation. **A/R logic:** Same city, different jobs.

Q23. How many of the following traps are valid?

BSI≠animals; ZSI≠plants; FRI≠ISFR; NBA≠Delhi

A. Only one B. Only two C. All four D. Only three

💡 **Show answer**

Ans: C. All four. **Logic:** Classic trap set.

Q24. Which one of the following correctly states BMC?

A. Animal gene bank Karnal B. National ABS only in Chennai C. ISFR publisher D. Local body preparing PBR

💡 **Show answer**

Ans: D. Local PBR owner. **Logic:** BMC identity.

Q25. With reference to UP forest-cover traps, which of the following statements is/are correct?

1. 20% districts often hinge on Chandauli + Shravasti.
2. Long distractor lists of Terai districts can mislead.
3. UP forest cover is identical to India's 21.76% always.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** UP ~6.24% — not India figure.

Q26. Match List-I with List-II:

List-I	List-II
A. BSI	1. 1916
B. ZSI	2. 1890
C. NBPGR	3. 1978

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** 1890 / 1916 / 1978.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): PBR documents local biodiversity knowledge.

Reason (R): BMCs prepare PBR at local level.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains who prepares PBR. **A/R logic:** Local documentation.

Q28. Arrange NBPGR vs NBAGR:

A. Both Kolkata plants only B. Both Chennai NBA C. NBPGR plants Delhi; NBAGR animals Karnal neighbourhood D. Both publish ISFR

💡 **Show answer**

Ans: C. Plants Delhi / animals Karnal. **Logic:** Gene-bank pair.

Q29. With reference to WII, which of the following statements is/are correct?

1. Wildlife science and training.
2. Not the plant gene bank.
3. Not the ISFR publisher.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** WII job boundaries.

Q30. Which one of the following correctly states FSI's core product?

A. Forest cover mapping / ISFR B. People's Biodiversity Register only C. Foreign ABS approvals only D. Animal Red Data Book only

💡 **Show answer**

Ans: A. Cover / ISFR. **Logic:** FSI identity.

Common Traps

Trap	Correct
BSI = animals	**Plants**
ZSI = plants	**Animals**
FRI publishes ISFR	**FSI publishes ISFR**
NBA HQ Delhi	**Chennai**
NBA prepares PBR	**BMC prepares PBR**
SBB = SPCB	**Biodiversity ≠ pollution**
NBPGR = Kolkata	**New Delhi**
NBPGR = animal bank	**NBAGR Karnal**
Bahraich >20% forest	**False**
ISFR annual	**Biennial (2 years)**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 29 — Indian Environmental Movements

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Chipko (चिपको) | Appiko (अप्पिको) | Silent Valley (साइलेंट वैली) | Narmada (नर्मदा) Bachao (नर्मदा बचाओ) Andolan (नर्मदा बचाओ) | Bishnoi (बिश्नोई) | Jungle Bachao Andolan | Save Western Ghats (पश्चिमी घाट)

Sources baked in: NCERT Bio/Geo; Gadgil 2011; Kasturirangan 2013; UPPCS/UPSC PYQs **Weight:** ★★★ High — movement ↔ state/year/leader matching **Last verified:** August 2026

Current Affairs (this topic)

- Movement years and states are fixed history (इतिहास); CA rarely renames Chipko / Appiko / NBA.
- Keep NBA (Andolan) ≠ National Biodiversity Authority acronym trap ready.

💡 🎯 **Consolidated — 22 Must-Score Facts (Click to Expand)**

1. **Bishnoi** (बिश्नोई) resistance at **Khejarli, Rajasthan (1730)** is led in memory by **Amrita Devi** (अमृता देवी) and the **363** martyrs who died protecting khejri trees. It is centuries older than Chipko. 2. **Chipko (1973)** began at **Reni, Uttarakhand** (then Uttar Pradesh (उत्तर प्रदेश)). Leaders include **Sunderlal Bahuguna, Chandi Prasad Bhatt (चंडी प्रसाद भट्ट), and Gaura Devi**. Method: embracing trees against logging. 3. **Silent Valley (1970s)** (शांत घाटी) in **Kerala (Palakkad)** opposed a hydro project in rainforest. **KSSP** and scientists were central. It is **not** a Tamil Nadu (नाडु) movement. 4. **Jungle Bachao Andolan (1982)** (जंगल बचाओ) in **Singhbhum, Jharkhand** was tribal (आदिवासी) resistance for forest rights. 5. **Appiko (1983)** in **Karnataka (Sirsi)** was led by **Panduranga Hegde**. It is the southern tree-embrace movement. Appiko is **Karnataka only** — not Uttarakhand, UP, or Kerala. 6. **Narmada Bachao Andolan (from 1985)** spans **Madhya Pradesh, Gujarat, and Maharashtra**, with **Medha Patkar** (मेधा पाटकर) and **Baba Amte** as leader facts. It is anti-large-dam displacement politics. 7. **Save Western Ghats (from about 2011)** is multi-state ecology (पारिस्थितिकी) politics around the **Gadgil** and **Kasturirangan** reports, mining, and ESZ questions. 8. Method families: Chipko, Appiko, and Bishnoi = forest/tree protection. Silent Valley and Narmada Bachao = anti-dam. Save Western Ghats = report/policy ecology protests. 9. Chipko ≠ Appiko. Chipko is Uttarakhand; Appiko is Karnataka. 10. Bishnoi **1730** ≠ Chipko **1973**. Century trap matters. 11. “NBA” as Narmada Bachao Andolan is **not** the National Biodiversity Authority. Same letters, different bodies. 12. Silent Valley is Kerala rainforest–dam (दाम -) conflict, not a Western Ghats mining-only slogan. 13. Jungle Bachao is Jharkhand Singhbhum tribal forest rights — not Chipko’s Garhwal (गढ़वाल) story. 14. Narmada Bachao is multi-state Narmada valley politics — not a single-state Kerala story. 15. Save Western Ghats leans on expert committee reports; Chipko leans on village tree hugging. 16. UP link: Chipko’s Reni lay in erstwhile UP until Uttarakhand statehood (**2000**). Appiko and Silent Valley are **not** UP movements. 17. Gaura Devi is a Chipko village leadership fact; Panduranga Hegde is Appiko; Medha Patkar is Narmada. 18. Khejarli martyrdom is Rajasthan Bishnoi memory — not Uttarakhand Chipko. 19. Anti-dam movements still differ: Silent Valley saved a Kerala valley rainforest project path; Narmada contested a large multi-state dam-displacement model. 20. Gadgil report is often read as stronger ecological protection; Kasturirangan as a later recalibration — both sit inside Save Western Ghats politics. 21. Tree-hugging method travelled from Chipko

north to Appiko south — same method family, different states and years. 22. Match movement → year → state → leader before answering "which state" stems; state alone often decides the key.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Bishnoi Movement (1730)** | **Chipko Movement (1973)** | Amrita Devi / Khejri tree sacrifice in Khejarli, Rajasthan vs Gaura Devi / Sunderlal Bahuguna anti-commercial logging in Chamoli (Uttarakhand) | बिश्नोई आंदोलन (1730) / चिपको आंदोलन (1973) | | **Chipko Movement (1973)** | **Appiko Movement (1983)** | Himalayan forest conservation in Chamoli (Bahuguna/Chandi Prasad Bhatt) vs Western Ghats forest hugging in Uttara Kannada, Karnataka (Pandurang Hegde) | चिपको (उत्तराखंड) / अप्पिको (कर्नाटक) | | **Silent Valley Movement (1970s)** | **Narmada Bachao Andolan (1985+)** | Kunthipuzha river anti-hydro dam protecting lion-tailed macaque in Kerala vs anti-large dam displacement on Narmada (Medha Patkar, Baba Amte) | साइलेंट वैली (केरल) / नर्मदा बचाओ आंदोलन | | **Jungle Bachao Andolan (1982)** | **Chipko Movement (1973)** | Singhbhum (Jharkhand) tribals opposing replacement of natural sal with commercial teak ("Greed Game Political Populism") vs tree-hugging in UP hills | जंगल बचाओ (सिंहभूम) / चिपको आंदोलन | | **Appiko Movement (1983)** | **Silent Valley (1970s)** | Forest conservation by tree embrace in Karnataka vs grassroots campaign stopping hydro-electric project in Kerala rainforest | अप्पिको (कर्नाटक) / शांत घाटी (केरल) | | **Tehri Dam Protest** | **Narmada Bachao Andolan** | High-seismic zone anti-dam fast by Sunderlal Bahuguna on Bhagirathi vs multi-state anti-submergence movement across MP/Gujarat/Maharashtra | टिहरी बांध विरोध / नर्मदा बचाओ | | **Gadgil Commission (2011)** | **Kasturirangan Committee (2013)** | WGEEP designated entire Western Ghats as ecologically sensitive (ESZ 1, 2, 3) vs HLWG recommending only 37% area as Eco-Sensitive Area (ESA) | गाडगिल समिति / कस्तूरीरंगन समिति |

💡 📋 **Must-score facts — Chipko, Appiko, Narmada, Silent Valley (Click to Expand)**

Movement ↔ place / year | Item | Lock | |---|---| | Bishnoi / Khejarli | **1730** Rajasthan; Amrita Devi; 363 martyrs | | Chipko | **1973** Reni, Uttarakhand; Bahuguna / Bhatt / Gaura Devi | | Silent Valley | 1970s **Kerala** rainforest–dam; KSSP | | Jungle Bachao | **1982** Singhbhum, Jharkhand | | Appiko | **1983** Sirsi, **Karnataka**; Panduranga Hegde | | Narmada Bachao | From **1985**; MP–GJ–MH; Medha Patkar / Baba Amte | | Chipko ≠ Appiko | UK / Karnataka | | NBA letters trap | Narmada Bachao ≠ National Biodiversity Authority | --- ![Cartographic Spatial Map of Historic Indian Environmental Movements] (images/env_ch29_01_indian_environmental_movements_map.png)

29.1 Chipko Movement (चिपको आंदोलन)

Identity: 1973 | Reni, Uttarakhand | Tree hugging | Bahuguna, Bhatt, Gaura Devi

Cause, Course, Result

Cause: Commercial logging threatened village forests, soil, and water security in Garhwal hills.

Course: Chipko began in **March 1973** at **Reni, Chamoli (चमोली)** (then UP, now Uttarakhand). Villagers hugged trees to stop contractors. On **26 March 1974**, **Gaura Devi** led women who blocked ash-tree felling. Leaders include **Sunderlal Bahuguna**, **Chandi Prasad Bhatt** (चंडी प्रसाद भट्ट), and **Gaura Devi**. Method = non-violent passive resistance (निष्क्रिय प्रतिरोध).

Result: National symbol of community forest protection. Influenced the conservation tilt behind **Forest (Conservation) Act 1980** and **National Forest Policy 1988**. Bahuguna's line — "ecology is permanent economy" — entered coaching memory.

- **Maiti Movement** (मैती आंदोलन) (Uttarakhand; pioneered by **Kalyan Singh Rawat**, mid-1990s) asks newly married couples to plant a sapling at the bride's parental home — separate from Chipko.
- Related movements: **Appiko** (अप्पिको) (Karnataka), **Narmada Bachao Andolan** (anti-large dam on Narmada).

> **Logic:** Chipko = **Uttarakhand 1973**, not Karnataka (Appiko) or Kerala (Silent Valley).

29.2 Appiko Movement

Identity: 1983 | Karnataka | Panduranga Hegde | Southern Chipko

Cause, Course, Result

Cause: Commercial clear-felling and monoculture plantations threatened Western Ghats watersheds and local livelihoods in **Uttara Kannada**.

Course: Appiko ("to embrace" in **Kannada**) launched in **1983** at **Sirsi / Kalase**, Karnataka. **Panduranga Hegde** led tree-embrace protests for sustainable forestry. The fight protected high-rainfall catchments of the **Kali** (काली), **Aghanashini**, and **Sharavathi**.

Result: Became the southern tree-embrace counterpart of Chipko. It remains a **Karnataka-only** state trap and is **not** the same campaign as Save Western Ghats (2011+ report politics).

PYQ — UPPCS Prelims 2025, Q122

Appiko Movement is associated with: (1) Uttarakhand (2) Uttar Pradesh (3) Kerala (4) Karnataka

Options: A. 1 and 2 B. Only 4 C. 3 and 4 D. Only 1

> 💡 **Show answer**
>
> **Logic:** Appiko ≠ Chipko state. Answer is **only Karnataka**. **Ans: B.** **Karnataka only**. UK/UP = Chipko geography; Kerala = Silent Valley.

29.3 Silent Valley Movement

Identity: 1970s–83 | Kerala | Anti-hydro dam | KSSP | SVNP 1984

Cause, Course, Result

Cause: A Kerala State Electricity Board hydro project on the **Kunthipuzha** threatened to submerge **Silent Valley** (साइलेंट वैली) tropical evergreen rainforest in **Palakkad**.

Course: Peak mobilisation ran **1978–1983**. **Kerala Sastra Sahitya (साहित्य) Parishad (KSSP)** and scientists led petitions, media, and science-based advocacy — not primarily tree-hugging. Pressure reached the Centre.

Result: Indira Gandhi (गांधी) declared protection in **1983**. **Silent Valley National Park** was notified in **1984** and later included in the **Nilgiri (नीलगिरि) Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र)**. The case stands as India's classic anti-dam rainforest save — different from Chipko/Appiko anti-logging.

> **Logic:** Silent Valley = **Kerala**, not Tamil Nadu.

29.4 Narmada Bachao Andolan

Identity: 1985+ | Medha Patkar | Sardar Sarovar (सरदार सरोवर) | MP–Gujarat–Maharashtra

Cause, Course, Result

Cause: Large dams on the **Narmada**, especially **Sardar Sarovar** (सरदार सरोवर), threatened mass displacement of adivasi (आदिवासी) and peasant communities with weak rehabilitation and river-ecology loss.

Course: The movement formalised around **1985**. Leaders include **Medha Patkar** and **Baba Amte**. Methods mixed satyagraha (सत्याग्रह), jal yatras, court petitions, and drowning protests (jalsamarpan). **World Bank withdrew** funding in **1993** after NBA pressure. Supreme Court rulings (**2000**, **2005**) balanced construction with R&R conditions.

Result: Raised national rehabilitation standards and made displacement a permanent large-dam debate. The project continued under court conditions; the movement's lasting mark is the **R&R + ecology** frame, not a total dam stop.

> **Logic:** In movement questions, **NBA = Narmada Bachao Andolan** — not National Biodiversity Authority (Chennai).

29.5 Bishnoi Movement

Identity: 1730 | Khejarli, Rajasthan | Amrita Devi | 363 martyrs | Khejri trees

Cause, Course, Result

Cause: Maharaja (महाराज) Abhay Singh ordered felling of **khejri** trees near **Khejarli (Jodhpur)** for kiln fuel, hitting Bishnoi sacred ecology.

Course: In **September 1730**, **Amrita Devi** and villagers hugged trees; **363 Bishnois** were martyred defending them. The **Bishnoi sect** (founded by **Guru (गुरु) Jambheshwar, 1485**) already taught flora (वनस्पति)–fauna (प्राणीजात) protection (blackbuck, chinkara) among its **29** principles.

Result: The Maharaja later banned tree felling in Bishnoi villages. The episode is the oldest syllabus forest-protection memory and a conceptual precursor to Chipko — different century and desert ecology.

> **Logic:** Bishnoi = **1730 Rajasthan**, not 1973 Uttarakhand.

29.6 Jungle Bachao Andolan

Identity: 1982 | Singhbhum, Jharkhand | Tribal forest resistance

Cause, Course, Result

Cause: Commercial forest exploitation and plantation policies in **Singhbhum** threatened customary forest rights of adivasi communities.

Course: Around **1982**, tribal resistance in Singhbhum (then Bihar, now **Jharkhand**) opposed contractor timber interests and state plantation frames that displaced forest-dependent people.

Result: Became part of broader Jharkhand tribal ecology politics — distinct from Himalayan Chipko or Kerala Silent Valley.

> **Logic:** Jungle Bachao = **Singhbhum/Jharkhand**, not Garhwal or Karnataka.

29.7 Save Western Ghats Movement

Identity: 2011+ | Multi-state | Gadgil 2011 | Kasturirangan 2013 | ESZ/mining politics

Cause, Course, Result

Cause: Mining, hydropower, and infrastructure pressure on the **Western Ghats** (पश्चिमी घाट) biodiversity hotspot (जैव विविधता हॉटस्पॉट) across Kerala, Karnataka, Maharashtra, Goa (गोवा), and Tamil Nadu raised demands for stronger ecological zoning.

Course: The **Western Ghats Ecology Expert Panel (Madhav Gadgil), 2011** recommended strong ecological sensitivity zoning. The **Kasturirangan High-Level Working Group, 2013** moderated ESA extent. Farmer, mining, and conservation protests followed ESZ and report politics.

Result: Save Western Ghats became a multi-state **policy/ecology** conflict distinct from **Appiko 1983** (local Karnataka tree-felling protest). Gadgil vs Kasturirangan remains the report-pair trap.

Report	Year	Tone
Gadgil WGEEP	**2011**	Stronger ESZ / conservation
Kasturirangan HLWG	**2013**	Narrower ESA / moderated

> **Logic:** Do not equate Appiko 1983 with Save Western Ghats 2011+.

Complete PYQ Bank (Topic 29)

Q1. UPPCS Prelims 2025, Q122

Appiko Movement is associated with:

1. Uttarakhand
2. Uttar Pradesh
3. Kerala
4. Karnataka

A. 1 and 2

B. Only 4

C. 3 and 4

D. Only 1

💡 **Show answer**

Logic: Appiko state trap — Karnataka only; UK/UP = Chipko; Kerala = Silent Valley. **Ans: B.** Appiko (1983, Sirsi/Uttara Kannada) is **Karnataka only**. **Trap:** Pairing Appiko with Garhwal because both used tree-embrace methods.

Q2. UPSC Prelims 2012 — pattern

Chipko Movement and Dandi March are both examples of:

A. Passive resistance / non-violent protest

B. Armed rebellion against the State

C. International treaty negotiations

D. Judicial public-interest litigation only

💡 **Show answer**

Logic: Chipko and Dandi March grouped as non-violent passive resistance in UPSC stems. **Ans: A.** Chipko villagers hugged trees; Dandi March broke the salt law — both are classic **passive resistance**. **Trap:** Calling Chipko armed or purely legal petitioning.

Q3. UPPCS / UPSC pattern

Chipko Movement began in:

A. 1973 at Reni, Garhwal Himalaya (then Uttar Pradesh, now Uttarakhand)

B. 1983 in Sirsi, Karnataka

C. 1730 at Khejarli, Rajasthan

D. 1970s in Silent Valley, Kerala

💡 **Show answer**

Logic: Chipko origin — 1973 Reni, Garhwal; Gaura Devi and Bahuguna are anchor names. **Ans: A.** Chipko = **1973**, Reni, leaders include **Gaura Devi**, **Bahuguna**, **Bhatt**. **Trap:** 1983 Appiko year or 1730 Bishnoi date applied to Chipko.

Q4. UPPCS / UPSC pattern

Silent Valley Movement was primarily:

A. Anti-hydroelectric dam struggle to save tropical evergreen forest in Kerala

B. Anti-logging tree-hug in Uttarakhand

C. Mining protest only in Jharkhand Singhbhum

D. Western Ghats policy report movement of 2011

💡 **Show answer**

Logic: Silent Valley = Kerala rainforest anti-dam; Kunthipuzha river; KSSP science campaign. **Ans: A.** Silent Valley (Palakkad, Kerala) opposed KSEB dam on **Kunthipuzha**; KSSP led science-based advocacy. **Trap:** Tamil Nadu or Karnataka placement.

Q5. UPPCS / UPSC pattern

Narmada Bachao Andolan is most closely linked to:

A. Medha Patkar and anti-large-dam displacement in the Narmada valley (MP, Gujarat, Maharashtra)

B. Panduranga Hegde and Karnataka logging

C. Amrita Devi and 1730 khejri martyrdom

D. Gadgil Committee report on Western Ghats mining

💡 **Show answer**

Logic: Narmada Bachao = Medha Patkar; multi-state Narmada valley anti-large-dam displacement. **Ans: A.** NBA (movement) spans **MP–GJ–MH** with **Medha Patkar** and **Baba Amte** as leader facts — not the National Biodiversity Authority. **Trap:** Medha Patkar = Chipko.

Q6. UPPCS / UPSC pattern

Amrita Devi and the Khejarli martyrdom (363 lives) are associated with:

A. Bishnoi Movement, Rajasthan, 1730

B. Chipko Movement, 1973

C. Appiko Movement, 1983

D. Save Western Ghats, 2011

💡 **Show answer**

Logic: Bishnoi 1730 Khejarli — centuries before Chipko; Amrita Devi memory. **Ans: A.** Bishnoi resistance at **Khejarli (1730)** protected **khejri** trees — oldest movement in the syllabus set. **Trap:** 1973 Chipko date or Amrita Devi Smriti Award confused with Chipko itself.

Q7. UPPCS / UPSC pattern

Appiko Movement (1983) was led by:

A. Panduranga Hegde in Karnataka

B. Sunderlal Bahuguna in Garhwal

C. Medha Patkar on the Narmada

D. Gaura Devi at Reni

💡 **Show answer**

Logic: Appiko leader Hegde, year 1983, Kannada "embrace" — southern Chipko. **Ans: A.** **Hegde** launched Appiko in **Sirsi/Kalase, Karnataka** to stop clear-felling. **Trap:** Bahuguna/Hegde swap or Appiko dated 1973.

Q8. Pattern

Jungle Bachao Andolan (1982) originated in:

A. Singhbhum, Jharkhand

B. Reni, Uttarakhand

C. Palakkad, Kerala

D. Sirsi, Karnataka

💡 **Show answer**

Logic: Jungle Bachao = Singhbhum tribal forest rights, 1982 — not Garhwal. **Ans: A.** Tribal communities in **Singhbhum** resisted forest exploitation for livelihood rights. **Trap:** Mapping every "Jungle Bachao" slogan to Chipko Garhwal.

Q9. Pattern

Save Western Ghats Movement (from about 2011) centred on:

A. Gadgil (2011) and Kasturirangan (2013) expert reports on mining, ESZ, and ecology

B. Tree-hugging at Reni village in 1973

C. Khejarli martyrdom in Rajasthan

D. Floating phumdis of Loktak Lake

💡 **Show answer**

Logic: Save Western Ghats ecology politics — Gadgil 2011 vs Kasturirangan 2013 reports. **Ans: A.** Multi-state **policy/ecology** protest around Western Ghats expert committees — distinct from 1983 Appiko tree embrace. **Trap:** Treating Appiko and Save Western Ghats as the same campaign.

Q10. Pattern

Which pair is **NOT** correctly matched?

A. Silent Valley Movement — Kerala

B. Chipko Movement — Uttarakhand (Garhwal)

C. Appiko Movement — Karnataka

D. Silent Valley Movement — Tamil Nadu

> 💡 **Show answer**
>
> **Logic:** Silent Valley state — Kerala, not Tamil Nadu. **Ans: D.** Silent Valley is in **Kerala (Palakkad)**; A, B, C are correct. **Trap:** Palakkad proximity to TN border used to push Tamil Nadu as the state.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to Bishnoi, which of the following statements is/are correct?

1. Khejarli, Kerala, 1730.
2. Amrita Devi and 363 martyrs memory.
3. It began at Reni, Uttarakhand in 1973.

A. 1 and 2 B. Only 3 C. Only 2 D. Only 1

> 💡 **Show answer**
>
> **Ans: C.** Only 2 is correct. **Logic:** Only statement 2 remains true among the numbered claims. Statement 1 is false. Statement 2 is true. Statement 3 is false.

Q2. Match List-I with List-II:

List-I	List-II
A. Chipko	1. Sirsi, Karnataka 1983
B. Appiko	2. Reni 1973
C. Jungle Bachao	3. Singhbhum 1982

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-1, C-3. **Logic:** Chipko/Appiko/Jungle Bachao.

Q3. Which of the following pairs is/are correctly matched?

1. Silent Valley — Kerala Palakkad rainforest–dam
2. Narmada Bachao Andolan has no base in Madhya Pradesh, Gujarat or Maharashtra.
3. Appiko — Uttarakhand only

A. Only 1 B. 1 and 2 C. Only 3 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1 is correct. **Logic:** Statement 1 is true. Statement 2 is false — Narmada Bachao spans Madhya Pradesh, Gujarat and Maharashtra. Statement 3 is false — Appiko is Karnataka (Western Ghats), not Uttarakhand-only.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Chipko began at Reni in then Uttar Pradesh (now Uttarakhand).

Reason (R): Appiko and Silent Valley are UP movements.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Appiko Karnataka; Silent Valley Kerala.

Q5. With reference to method families, which of the following statements is/are correct?

1. Chipko, Appiko, Bishnoi — forest/tree protection.
2. Silent Valley, Narmada Bachao — anti-dam.
3. Save Western Ghats — Gadgil/Kasturirangan ESZ politics neighbourhood.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** Method family map.

Q6. Arrange Chipko leaders teaching:

A. Only Medha Patkar B. Sunderlal Bahuguna, Chandi Prasad Bhatt, Gaura Devi C. Only Panduranga Hegde D. Only Amrita Devi 1730 as Chipko

💡 **Show answer**

Ans: B. Bahuguna / Bhatt / Gaura Devi. **Logic:** Chipko leadership.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Panduranga Hegde — Appiko
2. Medha Patkar — Narmada Bachao
3. Gaura Devi — Appiko Karnataka

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Gaura Devi = Chipko.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): "NBA" as Narmada Bachao Andolan is not the National Biodiversity Authority.

Reason (R): Same letters can mean different bodies.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the acronym trap. **A/R logic:** NBA letters trap.

Q9. With reference to Save Western Ghats, which of the following statements is/are correct?

1. From about 2011 multi-state ecology politics.

2. Around Gadgil and Kasturirangan reports.
3. Identical to Chipko village tree-hugging only.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Expert-committee politics ≠ Chipko method.

Q10. Match List-I with List-II:

List-I	List-II
A. Bishnoi	1. 1973
B. Chipko	2. 1730
C. Appiko	3. 1983

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: A.** A-2, B-1, C-3. **Logic:** 1730 / 1973 / 1983.

Q11. Arrange century trap:

A. Bishnoi 1973; Chipko 1730 B. Bishnoi 1730; Chipko 1973 C. Both 2011 D. Both 1985

> 💡 **Show answer**
>
> **Ans: B.** 1730 vs 1973. **Logic:** Century matters.

Q12. With reference to Jungle Bachao, which of the following statements is/are correct?

1. Singhbhum, Jharkhand, 1982.
2. Tribal resistance for forest rights.
3. It is Chipko's Garhwal story.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** Not Garhwal Chipko.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Tree-hugging method travelled from Chipko north to Appiko south.

Reason (R): Same method family, different states and years.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains A. **A/R logic:** Method migration.

Q14. Which one of the following correctly states Silent Valley?

A. Only Narmada multi-state dam politics B. Only Western Ghats mining slogan in Jharkhand C. Only Bishnoi Khejarli 1730 D. Kerala rainforest opposed hydro project; KSSP/scientists central

💡 **Show answer**

Ans: D. Kerala rainforest–dam. **Logic:** Silent Valley identity.

Q15. How many of the following leader facts are correct?

Gaura Devi–Chipko; Hegde–Appiko; Patkar–Narmada

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Leader–movement map.

Q16. With reference to anti-dam movements, which of the following statements is/are correct?

1. Silent Valley saved a Kerala valley rainforest project path.
2. Narmada contested large multi-state valley politics.
3. Both are identical single-state Kerala stories.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Narmada ≠ Kerala-only.

Q17. Match List-I with List-II:

List-I	List-II
A. Medha Patkar	1. Appiko
B. Panduranga Hegde	2. Narmada
C. Amrita Devi	3. Bishnoi/Khejarli

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Patkar/Hegde/Amrita.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Gadgil is often read as stronger ecological protection; Kasturirangan as later recalibration.

Reason (R): Both sit inside Save Western Ghats politics.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; SWG umbrella does not itself define the Gadgil vs Kasturirangan reading. **A/R logic:** Reports vs movement frame.

Q19. Which of the following pairs is/are correctly matched?

1. Chipko ≠ Appiko by state
2. Khejarli ≠ Chipko by century
3. Silent Valley = Tamil Nadu only always

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Silent Valley = Kerala.

Q20. Arrange Narmada Bachao start neighbourhood:

A. From 1985 B. 1730 C. 1973 only D. 2011 only

> 💡 **Show answer**
>
> **Ans: A.** From 1985. **Logic:** NBA year.

Q21. With reference to Baba Amte, which of the following statements is/are correct?

1. Leader fact with Medha Patkar for Narmada Bachao.
2. Multi-state Narmada valley politics.
3. He led Appiko at Sirsi only.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** Appiko = Hegde.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Match movement → year → state → leader before answering "which state" stems.

Reason (R): State alone often decides the key.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains the method. **A/R logic:** State decides many keys.

Q23. How many of the following are forest/tree-protection family?

Chipko, Appiko, Bishnoi, Narmada Bachao

A. Only one B. Only two C. Only three D. All four

> 💡 **Show answer**
>
> **Ans: C.** Only three. **Logic:** Narmada = anti-dam.

Q24. Which one of the following correctly states Appiko?

A. Singhbhum 1982 only B. Reni Garhwal 1973 only C. Khejarli 1730 only D. Southern tree-embrace; Sirsi Karnataka; Hegde

> 💡 **Show answer**
>
> **Ans: D.** Karnataka Appiko. **Logic:** Southern Chipko twin.

Q25. With reference to UP link, which of the following statements is/are correct?

1. Chipko's Reni lay in erstwhile UP until Uttarakhand statehood (2000).
2. Appiko is not a UP movement.
3. Silent Valley is a UP Terai movement.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Silent Valley = Kerala.

Q26. Match List-I with List-II:

List-I	List-II
A. Silent Valley	1. 1982
B. Jungle Bachao	2. 1970s
C. Save Western Ghats	3. ~2011

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** 1970s / 1982 / 2011.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Khejarli martyrdom is Rajasthan Bishnoi memory.

Reason (R): It is Uttarakhand Chipko's founding event.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Not Chipko founding.

Q28. Arrange Chipko vs Appiko states:

A. Both Kerala only B. Both Jharkhand only C. Chipko Uttarakhand; Appiko Karnataka D. Both Rajasthan 1730

💡 **Show answer**

Ans: C. UK vs Karnataka. **Logic:** North–south twin.

Q29. With reference to KSSP, which of the following statements is/are correct?

1. Central in Silent Valley resistance neighbourhood.
2. Kerala science movement link.
3. Led Appiko in Karnataka Sirsi.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Appiko = Hegde.

Q30. Which one of the following best separates Save Western Ghats from Chipko?

A. SWG leans on expert committee reports; Chipko on village tree hugging B. Both are identical 1730 Bishnoi events C. Both are only Singhbhum tribal cases D. Both are only Narmada multi-state dam politics

💡 **Show answer**

Ans: A. Reports vs hugging. **Logic:** Method/politics split.

Common Traps

Trap	Correct
Appiko = Uttarakhand/UP	**Karnataka**
Chipko = Karnataka	**Uttarakhand 1973**
Silent Valley = Tamil Nadu	**Kerala**
Bishnoi = 1973 Chipko	**1730 Rajasthan**
Medha Patkar = Chipko	**Narmada Bachao**
Appiko = Save W Ghats	**1983 vs 2011+**
NBA = Biodiversity Authority	Context: movement = Narmada
Jungle Bachao = Garhwal	**Singhbhum Jharkhand**
Gaura Devi = Appiko	**Chipko Reni**
Hegde = Chipko	**Appiko**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 30 — Environmental Literature & Awareness

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Environmental Literature | Environmental Education (पर्यावरण शिक्षा) | Environmental Awareness (पर्यावरण जागरूकता) | Environmental Communication

Sources baked in: NCERT; MoEFCC schemes; NEP 2020; UNESCO ESD; Tbilisi 1977; UPPCS/UPSC PYQs **Weight:** ★★★ High — Carson/Hardin/Brundtland traps; ECO Club; Art 51A(g); LiFE **Last verified:** August 2026

Current Affairs (this topic)

- Book–author pairs (Carson, Fukuoka, etc.) stay fixed; new award themes are light CA.
- Silent Spring ≠ Silent Valley (साइलेंट वैली) (movement).

💡 🎯 **Consolidated — 28 Must-Score Facts (Click to Expand)**

1. Four (चातुर्याम) pillars on this sheet: **literature** (landmark books/reports), **education** (formal EE), **awareness** (public campaigns), and **communication** (science to citizens). 2. ***Silent Spring* (1962)** by **Rachel Carson** exposed pesticide/DDT harms. It is not Hardin's commons essay. 3. **"Tragedy of the Commons" (1968)** by **Garrett Hardin** is about overuse of shared resources. 4. ***Limits to Growth* (1972)** is the **Club of Rome** systems study. ***Our Common Future* (1987)** is the **Brundtland Report** that popularised sustainable development. Brundtland ≠ Limits to Growth. 5. ***Small Is Beautiful* (1973)** by **E.F. Schumacher** critiques mass-production economics — "economics as if people mattered." It is not Carson's pesticide book. 6. Assertion–reason traps often make Brundtland and Limits to Growth look interchangeable — they are different books/years/authors. 7. Environmental education landmarks include **Tbilisi (1977)**, MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) **NGC/ECO Clubs**, **GSDP (2017)**, and **NEP 2020** ESD language. 8. Environmental awareness hooks include **Article 51A(g)**, **NEAC**, **Van Mahotsav** (July tree planting), and **LiFE** (launched **June 2022** from a COP26 idea). 9. LiFE is **not** a COP-25 product. Idea pitch is Glasgow COP26; launch is June 2022. 10. Environmental communication flagships include CSE's ***Down To Earth***, Gobar Times, and IEC components inside schemes. **KSSP** also sits in science–public communication memory from Kerala. 11. Carson ≠ Hardin. Pesticides book versus commons essay. 12. Brundtland ≠ Limits to Growth. SD definition report versus Club of Rome growth study. 13. Literature shapes thought; education is curriculum and clubs; awareness is campaigns and duties; communication is media/IEC — keep the four jobs distinct. 14. NGC/ECO Clubs are MoEFCC school eco-action programmes — not a Bharat Nirman infrastructure scheme. 15. Agenda 21 is a sustainable-development action plan from Rio; it is not "only environmental education." 16. Article **51A(g)** is the constitutional citizen duty to protect the environment (पर्यावरण) — an awareness/duty fact. 17. Van Mahotsav is the July greening festival fact for schools and public tree planting. 18. NEP 2020 supports education for sustainable development language; it does not replace Tbilisi as the classic EE conference fact. 19. Down To Earth is CSE journalism for environment — a communication pillar example. 20. **Eco-mark (1991)** is the BIS earthen-pot label for certified environment-friendly consumer goods. It is **not** AGMARK and not a FAO mark. 21. **Indira Gandhi Paryavaran Puraskar (1987, MoEF)** honours measurable contribution to environment protection. **Rajiv Gandhi Environment Award** recognises clean technology and industrial pollution reduction. 22. **Tyler Prize** (University of Southern California) is often called the "Nobel of Environment" for

environmental science, health, and energy. 23. UNEP's **Global 500 Award** (from **1987**) was later succeeded in spirit by **Champions of the Earth** (from **2005**). Narendra Modi received Champions of the Earth in **2018**. 24. The **Borlaug Award** is for outstanding agricultural research. Norman Borlaug's Nobel was the **Peace** Prize (Green Revolution (हरित क्रांति)), not medicine. 25. UP colour: NEAC district campaigns, Van Mahotsav in schools, NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) city awareness (Kanpur (कानपुर)), and ECO Clubs in UP schools. 26. Silent Spring's year is **1962**; Limits to Growth is **1972**; Brundtland is **1987** — three different decades. 27. Hardin's commons logic is shared-resource overuse; it is not Carson's pesticide narrative. 28. Match pillar → flagship before matching author names: Carson/Hardin/Brundtland/Schumacher for literature; Tbilisi/ECO Club for education; 51A(g)/LiFE for awareness; Down To Earth for communication; awards for recognition traps.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Environmental Education** | **Environmental Awareness** | Structured pedagogical curriculum in schools/colleges (Eco-Clubs, NGC) vs general public sensitization & behavioral change (Mission LiFE) | पर्यावरण शिक्षा / पर्यावरण जागरूकता | | **Silent Spring (1962)** | **Tragedy of the Commons (1968)** | Rachel Carson's book exposing chemical pesticide (DDT) bioaccumulation vs Garrett Hardin's economic essay on shared resource degradation | साइलेंट स्प्रिंग / ट्रैजेडी ऑफ द कॉमन्स | | **Limits to Growth (1972)** | **Brundtland Report (1987)** | Club of Rome computer model projecting economic/ecological collapse vs UN WCED report (*Our Common Future*) coining Sustainable Development | लिमिट्स टू ग्रोथ (1972) / ब्रंटलैंड रिपोर्ट (1987) | | **Small Is Beautiful (1973)** | **Silent Spring (1962)** | E.F. Schumacher's critique of mass production ("economics as if people mattered") vs Carson's chemical toxicity expose | स्मॉल इज ब्यूटीफुल / साइलेंट स्प्रिंग | | **Eco-mark (1991)** | **Greenwashing** | Official BIS earthen pot label for certified environment-friendly consumer goods vs deceptive corporate marketing claiming false eco-credentials | इको-मार्क (मिट्टी का घड़ा) / ग्रीनवॉशिंग | | **Greendex** | **Ecological Footprint** | National Geographic consumer sustainability index vs Global Footprint Network measure of nature's demand vs biocapacity | ग्रीनडेक्स / पारिस्थितिक पदचिह्न | | **National Green Corps (NGC)** | **Mission LiFE (2022)** | MoEFCC Eco-club programme in school networks vs international citizen-movement promoting lifestyle for environment | नेशनल ग्रीन कॉर्प्स / मिशन लाइफ (LiFE) |

💡 📋 **Must-score facts — books, EE, LiFE (Click to Expand)**

Books / campaigns | Item | Lock | |---|---| | Silent Spring | Rachel Carson **1962**; pesticides/DDT | | Tragedy of the Commons | Garrett Hardin **1968** | | Limits to Growth | Club of Rome **1972** | | Our Common Future | Brundtland **1987** | | Tbilisi | EE landmark **1977** | | LiFE | COP26 idea **2021**; launch Jun **2022** | | NGC / ECO Clubs | MoEFCC school programmes | | CSE flagship | *Down To Earth* / Gobar Times |

30.1 Environmental Literature

Identity: Carson / Hardin / Limits / Brundtland | Author ↔ work matching

- **Environmental literature** includes books and landmark reports that shaped ecological thought and policy.
- **Rachel Carson, *Silent Spring* (1962):** exposed DDT/pesticide harm to birds, food chains, and health — triggered the modern environmental movement.
- **Garrett Hardin, "Tragedy of the Commons" (1968):** argued individuals overuse shared resources (pasture, fisheries, air) for private gain.
- **Club of Rome, *The Limits to Growth* (1972):** modelled exponential growth hitting planetary resource limits — **not** the Brundtland Report.
- **Brundtland Commission, *Our Common Future* (1987):** defined and **popularised Sustainable Development** — meeting present needs without compromising future generations.

- **W.M. Adams, *Green Development* (first edition 1990):** environment and sustainability in the developing world.
- **Masanobu Fukuoka** was a Japanese farmer-philosopher who pioneered **natural farming** and re-vegetation of desertified lands.
- **Aldo Leopold, *A Sand County Almanac* (1949):** "Land Ethic" — moral duty to the biotic community.
- **Paul Ehrlich, *The Population (जनसंख्या) Bomb* (1968):** population growth vs resource limits debate.
- **Indian voices:** Gandhi (गांधी) *Hind Swaraj (स्वराज)* (हिंद स्वराज) (1909) — restraint vs reckless industrialism; Tagore/Shantiniketan — nature-linked education; Bahuguna — "ecology (पारिस्थितिकी) is permanent economy"; CSE (Anil Agarwal, Sunita Narain) — *Down To Earth*.

PYQ — UPPCS Prelims 2024, Q51

Assertion (A): Sustainable Development was popularised by the Brundtland Report. Reason (R): The Brundtland Report is also known as "The Limits to Growth".

Options: A. Both true; R does not explain A B. A false, R true C. Both true; R explains A D. A true, R false

💡 **Show answer**

Ans: D — Brundtland = *Our Common Future* 1987. Limits to Growth = Club of Rome 1972.

PYQ — UPPCS Prelims 2020, Q17

Silent Spring (1962) was written by:

Options: A. Carolyn Merchant B. Karl Marx C. Rachel Carson D. Rajgopalan

💡 **Show answer**

Ans: C

Work	Author/Body	Year
Silent Spring	Rachel Carson	1962
Tragedy of the Commons	Garrett Hardin	1968
Limits to Growth	Club of Rome	1972
Our Common Future	Brundtland / WCED	1987
Green Development	W.M. Adams	1990

> **Logic:** Carson ≠ Hardin. Brundtland ≠ Limits to Growth. Fukuoka = natural farming.

30.2 Environmental Education

Identity: Tbilisi 1977 | NGC/ECO Club | NEP 2020 ESD | Formal + non-formal

- **Environmental Education (EE)** builds knowledge, attitudes, skills, and participation for environmental protection.
- **Stockholm 1972** (स्टॉकहोम सम्मेलन) put EE on the global agenda; **Tbilisi Declaration (1977)** became the first intergovernmental EE framework.
- **Rio 1992 Agenda 21 Chapter 36** called for reorienting education toward sustainable development.
- **NEP 2020** integrates environmental awareness and ESD across stages through experiential, multidisciplinary learning.

- **National Green Corps (NGC) (नेशनल ग्रीन कॉर्प्स) / ECO Clubs (MoEFCC):** school/college clubs for plantation, waste management, biodiversity surveys, energy audits.
- **Paryavaran Mitra (मित्र -):** CEE Ahmedabad (अहमदाबाद) + MoEFCC — student/teacher green ambassadors.
- **Green Skill Development Programme (GSDP), 2017:** MoEFCC + Skill Ministry — vocational green skills.
- Formal EE = curriculum; non-formal = ECO Clubs, camps, community programmes.

PYQ — UPPCS Prelims 2023, Q132

Agenda-21 of Rio Summit (1992) is related to—

Options: A. Sustainable Development B. Polluter Pays C. Environmental Education D. Ozone Layer

💡 **Show answer**

Ans: A — Agenda 21 is the SD action plan (Chapter 36 covers EE as one part).

Year	Milestone
1972	Stockholm — EE on agenda
1977	**Tbilisi Declaration**
1992	Agenda 21 Ch. 36
~2001–02	NGC / ECO Club national push
2017	GSDP
2020	NEP 2020 ESD integration

> **Logic:** ECO Club = MoEFCC NGC. Tbilisi = EE; Stockholm = UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) birth — different milestones.

30.3 Environmental Awareness

Identity: Art 51A(g) | NEAC | Van Mahotsav | LiFE | Behaviour change

- **Environmental awareness** is public understanding and concern that drives behaviour change and civic action.
- **Article 51A(g)** (42nd (42वां) Amendment, 1976): Fundamental Duty to protect and improve forests, lakes, rivers, and wildlife — pairs with **Art 48A** (state duty).
- **National Environment Awareness Campaign (NEAC):** MoEFCC-funded NGO/school campaigns since the 1980s.
- **Van Mahotsav:** tree-plantation week in **July**, popularised by **K.M. Munshi (1950)**.
- **LiFE Mission:** idea at **COP26 (2021)**; launched **June 2022** — 75 pro-planet lifestyle actions (awareness/nudge, not school curriculum).
- International awareness anchors: **World Environment Day (विश्व पर्यावरण दिवस) (5 June)**, **Earth Day (पृथ्वी दिवस) (22 April)** — full date list in Topic 31.
- EE = structured institutional learning; awareness = broad public campaigns and duties.

Eco-mark, greenwashing (ग्रीनवॉशिंग), and related awareness facts

- **Eco-mark / ECOMARC** is India's eco-labelling scheme launched in **1991** by the Government of India and administered by the **Bureau of Indian Standards (BIS)**.
- Eco-mark identifies **environment-friendly consumer products** that cause least ecosystem (पारिस्थितिकी तंत्र) impact — it is **not** Agmark or a general quality-only mark. Drugs/antibiotics are typically **not** on the Eco-mark notified consumer list in bank teaching.

- **AGMARK** is a quality certification mark for agricultural produce under the **Agricultural Produce (Grading and Marking) Act, 1937**, issued through the Directorate of Marketing and Inspection (Ministry of Agriculture) — **not FAO**.
- **BIS / ISI** marks cover many industrial and consumer products (including mandatory certification for automotive tyres and tubes in classic stems).
- **Greenwashing** means conveying a **false or misleading impression** that a company (कंपनी)'s products or practices are eco-friendly.
- **Greendex** (ग्रीनडेक्स) (National Geographic / GlobeScan) measures **environmentally sustainable consumer behaviour** across countries — not carbon-credit sales volume.
- **Plachimada** (Palakkad, **Kerala**) became a landmark local struggle against groundwater contamination linked to a Coca-Cola bottling plant.
- Australia's **Green Army (सेना)** was a youth environment-works programme; it closed on **30 June 2018**.
- **GACSA (Global Alliance for Climate-Smart Agriculture)** has **non-binding** membership. It is **not** an outcome of the Paris 2015 Climate Summit, and India was **not** instrumental in creating it.
- **Conservation Agriculture** principles are **minimum tillage**, **soil cover with crop residues**, and **crop rotation / sequencing**.

PYQ — UPPCS Prelims 2025, Q139

LiFE: (1) launched June 2022 (2) idea proposed at COP-25 Madrid

Options: A. Only 2 B. Neither C. Both D. Only 1

💡 **Show answer**

Ans: D — Launch true. Idea = **COP26 Glasgow 2021**, not COP-25.

> **Logic:** LiFE = awareness/behaviour campaign. COP-25 Madrid trap is false.

30.4 Environmental Communication

Identity: Science → public | Down To Earth | IEC | Risk communication

- **Environmental communication** transmits environmental science, policy, and risk information accurately and accessibly.
- Bridges scientists/policymakers and citizens — translates AQI, BOD, IPCC (आईपीसीसी) summaries into usable messages.
- **CSE *Down To Earth*** (Anil Agarwal / Sunita Narain): leading Indian env journalism; **Gobar Times** for children.
- **Official channels:** MoEFCC/PIB releases, ISFR summaries, NAPCC documents.
- **KSSP model:** people's science communication during Silent Valley — data → public pressure → policy win.
- **IEC (Information, Education, Communication)** components in Swachh Bharat, Namami Gange (नमामि गंगे), NCAP, plastic-waste rules.
- Digital tools: Mission LiFE app, Sameer (CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) AQI), DIKSHA eco-modules.
- **IPCC** communicates climate science assessments — it is not an MoEFCC-style public awareness campaign body.

Channel	Example
Magazine	Down To Earth
Children's	Gobar Times
Scheme IEC	Namami Gange / Swachh Bharat

Channel	Example
People's science	KSSP

> **Logic:** Communication ≠ awareness alone — accurate transmission of science/policy is the core.

30.5 Environment Awards (high-yield)

Identity: Indira Gandhi Paryavaran | Tyler | Global 500 → Champions of the Earth | Rajiv Gandhi | Borlaug

- **Indira Gandhi Paryavaran Puraskar (1987, MoEF)** honours significant measurable contribution to environment protection and improvement — not only wildlife research or wasteland alone.
- **Tyler Prize for Environmental Achievement** (University of Southern California) is often called the "Nobel of Environment" — environmental science, health, and energy. Notable awardees include Madhav Gadgil (2015 pairing), Partha Dasgupta (2016), Pavan Sukhdev & Gretchen Daily (2020).
- **Rajiv Gandhi Environment Award** recognises clean/new technology and industrial pollution-reduction practice (MoEFCC track).
- **Global 500 Award** (UNEP, from **1987**) honoured environmental achievement; later succeeded in spirit by **Champions of the Earth** (from **2005**). Narendra Modi received Champions of the Earth **2018**.
- **Borlaug Award** is for outstanding agricultural research/contribution. Norman Borlaug's Nobel was the **Peace** Prize (Green Revolution (हरित क्रांति)), not a medicine Nobel.

> **Logic:** Tyler ≠ drama/child-rights prize. Global 500 = environment, not anti-terror. AGMARK ≠ Eco-mark; AGMARK ≠ FAO mark.

Complete PYQ Bank (Topic 30)

Q1. UPPCS Prelims 2024, Q51

Assertion (A): Sustainable Development was popularised by the Brundtland Report. Reason (R): The Brundtland Report is also known as "The Limits to Growth".

A. Both true; R does not explain A

B. A false; R true

C. Both true; R explains A

D. A true; R false

> 💡 **Show answer**
>
> **A/R logic:** A tests Brundtland → SD popularisation; R wrongly names Limits to Growth as Brundtland Report. **Ans: D.** A is true — *Our Common Future* (1987) popularised SD. R is false — *Limits to Growth* (1972) is **Club of Rome**, not Brundtland. **Trap:** Club of Rome title glued to Brundtland name.

Q2. UPPCS Prelims 2020, Q17

Silent Spring (1962) was written by:

A. Carolyn Merchant

B. Karl Marx

C. Rachel Carson

D. Rajgopalan

💡 **Show answer**

Logic: Silent Spring authorship — Rachel Carson 1962, not Hardin or Club of Rome. **Ans: C.** Carson exposed pesticide harm (especially DDT) and catalysed modern environmental awareness. **Trap:** Garrett Hardin (*Tragedy of the Commons*, 1968) or Brundtland for Carson works.

Q3. UPPCS Prelims 2020 — pattern

The 1987 UN report that defined and popularised sustainable development is:

A. Silent Spring

B. Our Common Future

C. Limits to Growth

D. Hind Swaraj

💡 **Show answer**

Logic: 1987 WCED report title — Our Common Future = Brundtland Report. **Ans: B.** Brundtland Commission / WCED produced ***Our Common Future* (1987)**. **Trap:** Limits to Growth (1972) or Silent Spring (1962) year-title swap.

Q4. UPPCS Prelims 2023, Q132

Agenda-21 of Rio Summit (1992) is related to—

A. Sustainable Development

B. Polluter Pays

C. Environmental Education alone

D. Ozone Layer

💡 **Show answer**

Logic: Agenda 21 primary frame — Rio 1992 sustainable-development action plan (Chapter 36 covers EE as one part). **Ans: A.** Agenda 21 is Rio's non-binding **SD** action plan; EE appears inside it but is not the sole headline. **Trap:** Picking "Environmental Education only" because Tbilisi/EE syllabus overlap.

Q5. UPPCS Prelims 2025, Q139

With reference to Mission LiFE:

1. It was launched in June 2022.
2. The idea was first proposed at COP-25 Madrid.

A. Only 2

B. Neither

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: LiFE launch June 2022 true; COP-25 Madrid idea false — idea at COP-26 Glasgow 2021. **Ans: D.** Launch **June 2022** is correct; the idea came at **COP-26 Glasgow (2021)**, not COP-25 Madrid. **Trap:** Any "COP-25 LiFE" distractor.

Q6. UPSC pattern

"The Tragedy of the Commons" (1968) was written by:

A. Garrett Hardin

B. Rachel Carson

C. Brundtland Commission

D. Club of Rome

💡 **Show answer**

Logic: Tragedy of the Commons essay — Garrett Hardin 1968. **Ans: A.** Hardin's essay on overuse of shared resources — distinct from Carson (1962) and Club of Rome *Limits* (1972). **Trap:** Hardin credited with *Silent Spring*.

Q7. UPSC pattern

The Tbilisi Declaration (1977) is chiefly associated with:

A. Environmental Education

B. Ozone layer protection

C. Wetlands wise use

D. Carbon credit trading

💡 **Show answer**

Logic: Tbilisi 1977 — landmark Environmental Education declaration; not Stockholm UNEP birth (1972). **Ans: A.** **Tbilisi (1977)** formalised global EE principles after Stockholm (1972) put EE on the agenda. **Trap:** Stockholm 1972 = UNEP creation, not Tbilisi EE milestone.

Q8. Pattern

Protection of the natural environment is a duty of every citizen under:

A. Article 51A(g) of the Constitution

B. Article 368 only

C. Schedule 7 List I alone

D. Wildlife Protection Act 1972 Section 35

💡 **Show answer**

Logic: Art 51A(g) — Fundamental Duty to protect environment; distinct from Art 48A State policy. **Ans: A.** **Art 51A(g)** is the citizen's fundamental duty; **Art 48A** is the State's directive principle. **Trap:** Mixing fundamental duty with DPSP or WPA sections.

Q9. Pattern

ECO Clubs under the National Green Corps programme are run by:

A. Ministry of Environment, Forest and Climate Change

B. Central Pollution Control Board alone

C. Reserve Bank of India

D. International Solar Alliance secretariat only

💡 **Show answer**

Logic: ECO Club / National Green Corps under MoEFCC — not CPCB pollution enforcement. **Ans: A.** NGC/ECO Club is MoEFCC's school EE network (~2001-02 national push). **Trap:** CPCB because "environment" appears in both names.

Q10. Pattern

Van Mahotsav, the tree-plantation campaign associated with K.M. Munshi, is observed in:

A. July

B. February

C. 5 June

D. 22 April

💡 **Show answer**

Logic: Van Mahotsav — July plantation week; K.M. Munshi; not February or WED June. **Ans: A.** **Van Mahotsav = July** mass planting drive in India. **Trap:** 5 June (WED), 22 April (Earth Day), or 21 March (International Day of Forests).

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to the four pillars, which of the following statements is/are correct?

1. The four pillars are only IUCN Red List categories, not literature/education/awareness/communication.
2. Keep pillar → flagship distinct.
3. All four are identical to IUCN categories.

A. Only 1 B. 1 and 2 C. Only 3 D. Only 2

💡 **Show answer**

Ans: D. Only 2 is correct. **Logic:** Statement 2 keeps pillar distinct from flagship. Statement 1 mislabels the four pillars. Statement 3 wrongly equates them with IUCN categories.

Q2. Match List-I with List-II:

List-I	List-II
A. Silent Spring	1. Garrett Hardin 1968
B. Tragedy of the Commons	2. Rachel Carson 1962
C. Limits to Growth	3. Club of Rome 1972

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Carson/Hardin/Club of Rome.

Q3. Which of the following pairs is/are correctly matched?

1. Our Common Future — Brundtland 1997
2. Limits to Growth — Club of Rome 1972
3. Brundtland — identical to Limits to Growth

A. Only 2 B. Only 1 C. 1 and 2 D. Only 3

> 💡 **Show answer**
>
> **Ans: A.** Only 2 is correct. **Logic:** Only statement 2 remains true among the numbered claims. Statement 1 is false. Statement 2 is true. Statement 3 is false.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Silent Spring exposed pesticide/DDT harms.

Reason (R): It is Hardin's commons essay.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** Carson ≠ Hardin.

Q5. With reference to EE landmarks, which of the following statements is/are correct?

1. Tbilisi 1977.
2. MoEFCC NGC/ECO Clubs.
3. GSDP 2017 and NEP 2020 ESD language.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: A.** All three. **Logic:** EE spine.

Q6. Arrange LiFE years:

A. Idea COP25; launch 2019 B. Idea COP26 2021; launch June 2022 C. Both 1962 D. Both 1977 Tbilisi

> 💡 **Show answer**
>
> **Ans: B.** 2021 idea → 2022 launch. **Logic:** Not COP25.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Article 51A(g) — citizen duty to protect environment
2. Van Mahotsav — July tree planting
3. NGC/ECO Clubs — Bharat Nirman infrastructure scheme

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only pair 3 is not correctly matched. **Logic:** ECO Clubs ≠ Bharat Nirman.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Brundtland popularised sustainable development definition neighbourhood.

Reason (R): Limits to Growth is the same Brundtland report.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Different decades/authors.

Q9. With reference to communication pillar, which of the following statements is/are correct?

1. CSE's Down To Earth.
2. Gobar Times / IEC in schemes.
3. KSSP also sits in communication/awareness neighbourhood teaching.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Media/IEC set.

Q10. Match List-I with List-II:

List-I	List-II
A. Literature	1. 51A(g), Van Mahotsav, LiFE
B. Education	2. Carson, Hardin, Brundtland
C. Awareness	3. Tbilisi, ECO Clubs

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-3, C-1. **Logic:** Pillar map.

Q11. Arrange year ladder Carson → Limits → Brundtland:

A. 1987 → 1972 → 1962 B. 1962 → 1972 → 1987 C. All 2022 D. All 1977

💡 **Show answer**

Ans: B. 1962 / 1972 / 1987. **Logic:** Three decades.

Q12. With reference to Hardin, which of the following statements is/are correct?

1. Tragedy of the Commons 1968.
2. About overuse of shared resources.
3. Identical to Carson's pesticide narrative.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Commons ≠ pesticides.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NEP 2020 supports ESD language.

Reason (R): NEP 2020 replaces Tbilisi as the classic EE conference fact.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Tbilisi remains classic EE conference.

Q14. Which one of the following correctly states Agenda 21?

A. Only Van Mahotsav July B. Only Carson 1962 book C. Only Hardin essay D. Rio sustainable-development action plan — not "only EE"

💡 **Show answer**

Ans: D. Rio SD action plan. **Logic:** Not EE-only.

Q15. How many of the following UP colour facts are correct?

NEAC district campaigns; Van Mahotsav in schools; NCAP city awareness (Kanpur); ECO Clubs

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** UP awareness colour.

Q16. With reference to LiFE, which of the following statements is/are correct?

1. MoEFCC behaviour-change / lifestyle mission.
2. Not a COP-25 product.
3. Identical to Brundtland Report 1987.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** LiFE ≠ Brundtland.

Q17. Match List-I with List-II:

List-I	List-II
A. Tbilisi	1. June 2022 launch
B. LiFE	2. 1977 EE conference
C. GSDP	3. 2017

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** 1977 / 2022 / 2017.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Literature shapes thought; education is curriculum/clubs; awareness is campaigns/duties; communication is media/IEC.

Reason (R): The four pillars should not be collapsed into one label.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains why the map matters. **A/R logic:** Pillar separation.

Q19. Which of the following pairs is/are correctly matched?

1. Down To Earth — CSE journalism
2. Van Mahotsav — July greening festival
3. Silent Spring — 1987 Brundtland

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Silent Spring = 1962 Carson.

Q20. Arrange Article 51A(g) role:

A. Constitutional citizen duty to protect environment B. Only Union List entry on mines C. Only Finance Commission grant D. Only ISFR publisher

💡 **Show answer**

Ans: A. Citizen duty. **Logic:** Awareness/duty fact.

Q21. With reference to A/R traps on this sheet, which of the following statements is/are correct?

1. Brundtland ≠ Limits to Growth.
2. Carson ≠ Hardin.
3. Both pairs are interchangeable always.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Classic separations.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): LiFE idea pitch is Glasgow COP26.

Reason (R): LiFE launch is June 2022.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; launch year does not explain the COP26 pitch itself. **A/R logic:** Pitch vs launch.

Q23. How many of the following year facts are correct?

Silent Spring 1962; Commons 1968; Limits 1972; Brundtland 1987

A. Only one B. Only two C. All four D. Only three

💡 **Show answer**

Ans: C. All four. **Logic:** Literature years.

Q24. Which one of the following correctly states NGC/ECO Clubs?

A. Hardin 1968 essay only B. Bharat Nirman roads only C. Club of Rome 1972 only D. MoEFCC school eco-action programmes

💡 **Show answer**

Ans: D. School eco-action. **Logic:** Education pillar.

Q25. With reference to pillars vs books, which of the following statements is/are correct?

1. Match pillar → flagship before author names when useful.
2. Carson/Hardin/Brundtland for literature.
3. Tbilisi/ECO Club for education.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** Match method.

Q26. Match List-I with List-II:

List-I	List-II
A. Our Common Future	1. 1962
B. Silent Spring	2. 1987
C. Tragedy of the Commons	3. 1968

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** 1987 / 1962 / 1968.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Hardin's commons logic is shared-resource overuse.

Reason (R): It is Carson's pesticide narrative.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Commons ≠ DDT book.

Q28. Arrange NEAC neighbourhood:

A. Only nuclear energy board B. Only ISFR publisher C. Environmental awareness campaigns (district/public) D. Only BSI Kolkata

💡 **Show answer**

Ans: C. Awareness campaigns. **Logic:** NEAC fact.

Q29. With reference to Club of Rome, which of the following statements is/are correct?

1. Limits to Growth 1972 systems study.
2. Different from Brundtland 1987.
3. Identical to Down To Earth CSE magazine.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not CSE magazine.

Q30. Which one of the following correctly states Van Mahotsav?

A. July tree-planting / greening festival B. Only COP26 LiFE launch C. Only Stockholm 1972 conference D. Only Montreal Protocol

💡 **Show answer**

Ans: A. July greening festival. **Logic:** Awareness hook.

Common Traps

Trap	Correct
Silent Spring = Hardin	**Carson 1962**
Brundtland = Limits to Growth	**Our Common Future 1987**
Limits to Growth = Brundtland	**Club of Rome 1972**
LiFE idea at COP-25	**COP26 2021**
Agenda 21 = only EE	**Primarily Sustainable Development**
ECO Club = CPCB	**MoEFCC NGC**
Awareness = only school EE	**Also campaigns + Art 51A(g)**
Bharat Nirman = NEP scheme	**Infrastructure, not NEP EE**
IPCC = MoEFCC awareness campaign	**Science assessment body**
Van Mahotsav = February	**July**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 31 — Environmental Days

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

World Environment (पर्यावरण) Day (विश्व पर्यावरण दिवस) | World Wetlands Day (विश्व आर्द्रभूमि दिवस) | Earth Day (पृथ्वी दिवस) | International Day for Biological Diversity | International Day of Forests | World Water Day (विश्व जल दिवस) | World Ozone Day (विश्व ओजोन दिवस) | World Wildlife Day (विश्व वन्यजीव दिवस)

Sources baked in: UNEP; UN; Ramsar; CITES; CBD; Montreal Protocol; MoEFCC; UPPCS PYQs **Weight:** ★★★ High — date ↔ day matching; March cluster; World Wildlife Day trap **Last verified:** August 2026

Current Affairs (this topic)

- Annual themes for WED / Earth Day / IDF refresh; calendar dates stay fixed (WED = **5 June**).
- Do not invent UP-only international day dates.

💡 🎯 **Consolidated — 28 Must-Score Facts (Click to Expand)**

1. **World Wetlands Day** falls on **2 February** (रामसर) and marks the **Ramsar (रामसर) Convention (1971)** (रामसर कन्वेंशन). It is not 2 March or 3 March. 2. **World Wildlife Day** falls on **3 March** and marks the adoption of **CITES (साइट्स) (1973)** (साइट्स). It is **not** 22 March. 3. **International Day of Forests** falls on **21 March**. It was proclaimed by the UN General Assembly in **2012**. 4. **World Water Day** falls on **22 March**. The classic trap is pairing Wildlife Day with 22 March. 5. **Earth Day** falls on **22 April**. It began in the United States in **1970** as a civic teach-in movement. 6. **International Day for Biological Diversity** falls on **22 May** and commemorates the CBD text adopted (अंगीकृत) at Nairobi in **1992**. 7. **World Environment Day** falls on **5 June**. It marks the opening of the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन) and is run by **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम). 8. **World Ozone Day** falls on **16 September** and marks the **Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) (1987)** (मॉन्ट्रियल प्रोटोकॉल). 9. In calendar order the eight syllabus days run **2 Feb → 3 Mar → 21 Mar → 22 Mar → 22 Apr → 22 May → 5 Jun → 16 Sep**. 10. The March ladder is **Wildlife (3) → Forests (21) → Water (22)**. Never put Wildlife on 22 March. 11. World Environment Day is **not** Earth Day and **not** Wetlands Day. The dates are **5 June**, **22 April**, and **2 February**. 12. India hosted World Environment Day in **2011** (Forests theme) and **2018** (Beat Plastic Pollution). 13. Biological Diversity Day shifted from an older **29 December** link to **22 May** from **2000**; papers use **22 May** only. 14. World Water Day is coordinated under **UN-Water**. Wetlands Day is a **Ramsar** (रामसर) day, not a Water Day synonym. 15. World Wildlife Day was proclaimed by the UN in **2013**, but the date itself is the **1973 CITES** anniversary. 16. Ozone Day belongs to the **Montreal Protocol** (मॉन्ट्रियल प्रोटोकॉल) family. Do not date it as 2 October, 21 April, or 30 January. 17. Earth Day is a civic global day. It is **not** the same UN treaty day as World Environment Day. 18. Biodiversity Day is broader than Wildlife Day. Wildlife Day is fauna (प्राणीजात)–flora (वनस्पति) trade (पण्याध्यक्ष) and crime awareness under the CITES date. 19. Van Mahotsav is a separate **July** plantation campaign in India. It is **not** International Day of Forests. 20. Earth Hour (अर्थ आवर) (last Saturday of March, WWF (डब्ल्यूडब्ल्यूएफ)) is **not** on this syllabus day list and is not Earth Day. 21. Uttar Pradesh (उत्तर प्रदेश) has **10** Ramsar sites — useful Wetlands Day geography, but the day itself stays **2 February**. 22. A correct Ozone–WED pair is **16 September** and **5 June**. A Wildlife–**22 March** pair is always wrong. 23. Stockholm **1972** anchors WED. Ramsar **1971** anchors Wetlands. Montreal **1987** anchors Ozone. CITES **1973** anchors Wildlife.

24. Same-number traps: **22** March is Water, **22** April is Earth, **22** May is Biodiversity — three different months. 25. **World Toilet Day = 19 November** (UN, from 2013). **World No Tobacco Day = 31 May** (not 5 May). **World Habitat (वास स्थान) Day** = first Monday of October (not a fixed 5 October). 26. **Swachh Bharat Mission** launched **2 October 2014** (Gandhi (गांधी) Jayanti). Sierra Leone observes national cleanliness on the last Saturday of every month. 27. Earth Day theme teaching: **2024 = Planet vs. Plastics**. WED **2023 = Beat Plastic Pollution**. 28. **International Tiger Day** falls on **29 July** and marks the **St Petersburg Tiger Summit (2010)**. It is **not** World Wildlife Day (3 March).

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **World Wetlands Day** | **World Wildlife Day** | **2 February** (signing of Ramsar Convention 1971) vs **3 March** (adoption of CITES 1973) | विश्व आर्द्रभूमि दिवस (2 फरवरी) / विश्व वन्यजीव दिवस (3 मार्च) | | **World Forestry Day** | **World Water Day** | **21 March** (UN International Day of Forests) vs **22 March** (UN World Water Day) | विश्व वानिकी दिवस (21 मार्च) / विश्व जल दिवस (22 मार्च) | | **World Earth Day** | **World Environment Day** | **22 April** (Gaylord Nelson, 1970) vs **5 June** (Stockholm Conference opening, 1972) | पृथ्वी दिवस (22 अप्रैल) / विश्व पर्यावरण दिवस (5 जून) | | **International Biodiversity Day** | **World Earth Day** | **22 May** (Nairobi agreed text of CBD 1992) vs **22 April** (Earth Day) | जैव विविधता दिवस (22 मई) / पृथ्वी दिवस (22 अप्रैल) | | **World Ozone Day** | **World Environment Day** | **16 September** (signing of Montreal Protocol 1987) vs **5 June** (Stockholm UNEP foundation) | विश्व ओजोन दिवस (16 सितंबर) / पर्यावरण दिवस (5 जून) | | **Earth Day** | **Earth Hour** | Fixed annual civic observance on **22 April** vs WWF voluntary **last Saturday of March** 1-hour lights-off event | पृथ्वी दिवस (22 अप्रैल) / अर्थ आवर (मार्च अंतिम शनिवार) | | **World Soil Day** | **World Day to Combat Desertification** | **5 December** (FAO celebration of King Bhumibol's birthday) vs **17 June** (UNCCD adoption anniversary) | विश्व मृदा दिवस (5 दिसंबर) / मरुस्थलीकरण रोकथाम दिवस (17 जून) |

💡 📋 **Must-score facts — calendar order of key days (Click to Expand)**

Day ↔ date | Item | Lock | |---|---| | Wetlands | **2 February** (Ramsar) | | Wildlife | **3 March** (CITES) — ≠ 22 Mar | | Forests | **21 March** | | Water | **22 March** | | Earth | **22 April** (US 1970) | | Biological Diversity | **22 May** (CBD) | | Environment | **5 June** (Stockholm/UNEP) | | Ozone | **16 September** (Montreal) | | March ladder | Wildlife 3 → Forests 21 → Water 22 | | India WED hosts | **2011** Forests; **2018** Beat Plastic |

31.1 World Environment Day

Identity: 5 June | UNEP | Stockholm 1972 (स्टॉकहोम सम्मेलन) | Host country rotates

- **World Environment Day (WED)** is the UN's main annual day for environmental awareness (पर्यावरण जागरूकता) and action.
- Date **5 June** marks the opening of the **Stockholm Conference on the Human Environment (1972)**; first observed in **1973**.
- Administered by **UNEP**; host country rotates and sets the year's theme.
- India hosted in **2011** ("Forests: Nature at Your Service") and **2018** ("Beat Plastic Pollution").
- Recent host/theme facts: **2019** China — Air Pollution; **2020** Colombia (+Germany) — Biodiversity; **2021** Pakistan — Ecosystem (पारिस्थितिकी तंत्र) Restoration; **2022** Only One Earth; **2023** Solutions to plastic pollution.
- **2022** also marked **50 years** since Stockholm 1972, which designated **5 June** as World Environment Day.
- WED is **not** Earth Day (**22 April**) and **not** Wetlands Day (**2 February**).

PYQ — UPPCS Prelims 2025, Q62

Which pair(s) NOT correctly matched? 1. International Ozone Day — 16 September 2. World Environment Day — 5 June 3. World Wildlife Day — 22 March

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: B — Pairs 1 and 2 are correct. Wildlife Day is **3 March**, not 22 March (that is Water Day).

> **Logic:** WED = **5 June** — Stockholm opening. Pair "WED — 2 February" is false.

31.2 World Wetlands Day

Identity: 2 February | Ramsar 1971 | Wise use

- **World Wetlands Day** marks adoption of the **Ramsar Convention on Wetlands** at Ramsar, Iran (**2 February 1971**).
- Focus is wetland conservation and **wise use** — sustainable human use, not a total ban.
- India has about **100** Ramsar sites; **Uttar Pradesh has 10**; **Tamil Nadu (नाडु)** leads with **16** (geography overlap).
- Date traps: **not 2 March**, **not 3 March** (Wildlife), **not 22 March** (Water).

PYQ — UPPCS Prelims 2023, Q147

World Wetlands Day is observed on—

Options: A. 10 January B. 15 January C. 1 February D. 2 February

💡 **Show answer**

Ans: D

> **Logic:** Wetlands Day = **2 February only**.

31.3 Earth Day

Identity: 22 April | First 1970 (USA) | Civic global movement

- **Earth Day** promotes environmental protection, climate action, and sustainability every **22 April**.
- First observed **22 April 1970** in the **United States** (Senator Gaylord Nelson / mass teach-ins).
- Globally recognised civic movement — **not** the same UN treaty day as WED.
- Trap pairs: ≠ WED (**5 June**); ≠ Water Day (**22 March**); ≠ Biodiversity Day (**22 May**); ≠ **Earth Hour** (last Saturday of March, WWF — not on this syllabus).

> **Logic:** Earth Day = **22 April** — same "22" number as Water and Biodiversity, different months.

31.4 International Day for Biological Diversity

Identity: 22 May | CBD 1992 | Nairobi Final Act

- **International Day for Biological Diversity (IDB)** promotes biodiversity conservation awareness every **22 May**.

- Commemorates adoption of the **Convention on Biological Diversity** text at Nairobi (**22 May 1992**); Rio Earth Summit signed CBD the same year.
- Originally linked to **29 December** (CBD entry into force 1993); shifted to **22 May** from **2000** — papers use **22 May** only.
- The UN declared **2011–2020** the **United Nations Decade on Biodiversity**.
- India implements CBD through the **Biological Diversity Act 2002** (जैव विविधता अधिनियम 2002) (NBA / SBBs / BMCs).
- Scope is broader than World Wildlife Day (**3 March**, CITES fauna focus).

> **Logic:** IDB = **22 May** — not 22 March (Water) and not 5 June (WED). Decade 2011–20 ≠ climate-decade branding.

31.5 International Day of Forests

Identity: 21 March | UNGA 2012 | Between Wildlife and Water

- **International Day of Forests** raises awareness of forests and sustainable forest management every **21 March**.
- It was proclaimed by the **UN General Assembly in 2012**.
- India's **Wildlife Week** is celebrated **2–8 October** (national awareness week — not the same as World Wildlife Day).
- Theme examples: IDF 2018 = Forests and Sustainable Cities; IDF 2024 = Forests and innovation.
- Proclaimed by the **UN General Assembly in 2012**.
- Sits in the March ladder between **Wildlife (3 Mar)** and **Water (22 Mar)** — easy to drop in match lists.
- India observance links to ISFR messaging and plantation drives (Van Mahotsav is a separate July campaign).

> **Logic:** Forests = **21 March** — not 22 March (Water) and not 3 March (Wildlife).

31.6 World Water Day

Identity: 22 March | UN 1993 | Freshwater focus

- **World Water Day** focuses on freshwater resources, access, and sustainable water management every **22 March**.
- Designated by the **United Nations** (observed from **1993**); coordinated under **UN-Water** (programme continuity from **2003**).
- Theme **2024**: *Leveraging Water for Peace* (also phrased as prosperity and peace in some bank notes).
- Most dangerous trap: confuse with **World Wildlife Day (3 March)** — used this exact false pair.
- Also ≠ Wetlands Day (**2 February**) and ≠ Earth Day (**22 April**).

> **Logic:** Water = **22 March**. Wildlife = **3 March**. Never swap.

31.7 World Ozone Day

Identity: 16 September | Montreal Protocol 1987 | UNEP Ozone Secretariat

- **World Ozone Day** (International Day for the Preservation of the Ozone Layer) is observed every **16 September**.
- Marks signing of the **Montreal Protocol on Substances that Deplete the Ozone Layer (16 September 1987)**; Montreal entered into force on **1 January 1989**.

- The UN General Assembly declared the day in **1994**.
- Linked to the Vienna Convention family; administered with UNEP Ozone Secretariat support.
- India observance often highlights HCFC phase-out and cooling-sector compliance.

PYQ — UPPCS Prelims 2022, Q82

On which date is Ozone Day observed?

Options: A. October 2 B. September 16 C. April 29 D. January 30

💡 **Show answer**

Ans: B

PYQ — UPPCS Prelims 2021, Q114

'World Ozone Day' is celebrated on

Options: A. 25 December B. 21 April C. 16 September D. 30 January

💡 **Show answer**

Ans: C

> **Logic:** Ozone Day = **16 September** — Montreal Protocol date. Do not swap it with World Wildlife Day (**3 March**).

31.8 World Wildlife Day

Identity: 3 March | CITES 1973 | UN + CITES

- **World Wildlife Day** celebrates wild fauna and flora and fights wildlife crime every **3 March**.
- Date marks adoption of **CITES** (**Convention on International Trade in Endangered Species**) on **3 March 1973**.
- UN proclaimed the day in **2013**; observance involves UN and CITES Secretariat.
- **Highest-yield trap:** listing Wildlife Day as **22 March** (that is Water Day) — wrong pair in **

> **Logic:** Wildlife = **3 March only**. Water = **22 March**. Wildlife Week in India = **2–8 October**. **International Tiger Day = 29 July** (St Petersburg Summit **2010**).

31.9 International Tiger Day

Identity: 29 July | St Petersburg 2010 | Awareness day

- **International Tiger Day** is observed every **29 July**.
- It was founded at the **Saint Petersburg Tiger Summit (2010)** to raise awareness for tiger conservation.
- Do not confuse with World Wildlife Day (**3 March**) or Wildlife Week (**2–8 October**).

31.10 Other High-Yield Days

Identity: Toilet | No Tobacco | Habitat | Cities | Population (जनसंख्या) | Health | Overshoot

- **World Toilet Day = 19 November** (UNGA 2013; sanitation for all / end open defecation messaging).
- **World No Tobacco Day = 31 May** (WHO) — not 5 May. Themes rotate (e.g. tobacco–environment / grow food not tobacco).

- **World Habitat Day** = **first Monday of October** (UN; not a fixed calendar date like 5 October).
- **World Cities Day = 31 October**. **World Population Day = 11 July**. **World Health Day = 7 April**.
- **Earth Overshoot Day** marks when humanity's annual demand exceeds Earth's regenerative capacity — an environmental-degradation awareness marker (Global Footprint Network partnership framing).
- **Swachh Bharat Mission** launched **2 October 2014**. Sierra Leone's national cleanliness day is the last Saturday of each month.

> **Logic:** Anti-tobacco ≠ 5 May. Habitat Day ≠ fixed 5 Oct.

Complete PYQ Bank (Topic 31)

Q1. UPPCS Prelims 2025, Q62

Which of the following pair(s) is/are **NOT** correctly matched?

1. International Ozone Day — 16 September
2. World Environment Day — 5 June
3. World Wildlife Day — 22 March

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

💡 **Show answer**

Logic: Wildlife Day on 22 March is the classic wrong pair — correct date is 3 March (CITES); Water Day is 22 March. **Ans: B.** Pairs 1 and 2 are correct. **World Wildlife Day = 3 March** (CITES 1973); **22 March = World Water Day**. **Trap:** Same "22" number chain — Wildlife vs Water vs Biodiversity (22 May).

Q2. UPPCS Prelims 2023, Q147

World Wetlands Day is observed on—

A. 10 January

B. 15 January

C. 1 February

D. 2 February

💡 **Show answer**

Logic: Ramsar adoption date — World Wetlands Day is 2 February only. **Ans: D.** Marks Ramsar Convention adoption at Ramsar, Iran (**2 February 1971**). **Trap:** 2 March, 3 March (Wildlife), or 22 March (Water).

Q3. UPPCS Prelims 2022, Q82

On which date is Ozone Day observed?

A. October 2

B. September 16

C. April 29

D. January 30

💡 **Show answer**

Logic: Ozone Day = Montreal signing date, 16 September — not WED 5 June. **Ans: B.** **16 September** — Montreal Protocol signed **1987**; entered into force **1 January 1989**. **Trap:** Gandhi Jayanti (2 Oct), Earth Day (22 Apr), or WED (5 Jun).

Q4. UPPCS Prelims 2021, Q114

'World Ozone Day' is celebrated on

A. 25 December

B. 21 April

C. 16 September

D. 30 January

💡 **Show answer**

Logic: Same Ozone Day fact — UPPCS 2021 stem wording variant. **Ans: C.** International Day for the Preservation of the Ozone Layer = **16 September**. **Trap:** 21 April (Earth Day) or 5 June (WED).

Q5. UPPCS Prelims 2018 — pattern

World Population Day is observed on:

A. 11 July

B. 5 June

C. 22 April

D. 2 February

💡 **Show answer**

Logic: World Population Day = 11 July — 5 June is WED distractor in 2018-style stems. **Ans: A.** **Population Day = 11 July**. **Trap:** 5 June looks "environmental" but is **World Environment Day (Stockholm 1972)**.

Q6. UPSC / pattern

World Environment Day is observed on:

A. 5 June (Stockholm Conference 1972)

B. 22 April (Earth Day 1970 USA)

C. 2 February (Ramsar Wetlands Day)

D. 16 September (Montreal Protocol)

💡 **Show answer**

Logic: WED anchor — 5 June, Stockholm Conference opening 1972, run by UNEP. **Ans: A.** WED = **5 June** — UNEP-led UN day from Stockholm **1972**. **Trap:** Earth Day (22 Apr) or Ozone Day (16 Sep) as "the" global environment day.

Q7. Pattern

Earth Day is observed every:

A. 22 April (first celebrated 1970 in the United States)

B. 5 June

C. 22 March

D. 21 March

💡 **Show answer**

Logic: Earth Day civic origin — 22 April 1970 USA teach-ins; not a UN treaty day like WED. **Ans: A.** **Earth Day = 22 April** — global civic movement from 1970 US origin. **Trap:** 22 March (Water), 22 May (Biodiversity), or 5 June (WED).

Q8. Pattern

International Day for Biological Diversity commemorates CBD adoption on:

A. 22 May

B. 22 March

C. 22 April

D. 3 March

💡 **Show answer**

Logic: CBD text adoption — Biological Diversity Day = 22 May (not March/April). **Ans: A.** **22 May** marks Nairobi **1992** CBD text adoption (observed from 2000 onward in UN calendar). **Trap:** 22 March (Water) or 3 March (Wildlife/CITES).

Q9. Pattern

World Water Day falls on:

A. 22 March

B. 3 March

C. 21 March

D. 2 February

💡 **Show answer**

Logic: UN-Water coordinates World Water Day — 22 March in the March ladder after Wildlife (3) and Forests (21). **Ans: A.** **Water = 22 March** — between International Day of Forests (**21 March**) and Earth Day (**22 April**). **Trap:** 3 March (Wildlife) — the 2025 Q62 killer pair.

Q10. Pattern

World Wildlife Day (wild fauna and flora) is observed on:

A. 3 March (CITES adoption 1973)

B. 22 March

C. 2–8 October (Wildlife Week India)

D. 29 July (International Tiger Day)

💡 **Show answer**

Logic: CITES anniversary — Wildlife Day 3 March; proclaimed by UN 2013. **Ans: A.** **Wildlife Day = 3 March** — CITES **1973**; UN proclaimed **2013**. **Trap:** 22 March (Water), October Wildlife Week, or 29 July Tiger Day.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to environmental days, which of the following statements is/are correct?

1. World Wetlands Day — 2 February (Ramsar 1971).
2. World Wildlife Day — 3 March (CITES adoption anniversary).
3. World Wildlife Day — 22 March.

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

Show answer

Ans: C. Only 1 and 2 are correct. **Logic:** Wetlands Day is 2 February; Wildlife Day is 3 March (CITES). Statement 3 is the Water Day trap (22 March).

Q2. Match List-I with List-II:

List-I	List-II
A. World Water Day	1. 22 April
B. Earth Day	2. 22 March
C. Biological Diversity Day	3. 22 May

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

Show answer

Ans: C. A-2, B-1, C-3. **Logic:** Same-number 22 traps across months.

Q3. Which of the following pairs is/are correctly matched?

1. International Day of Forests is observed on 22 May.
2. World Environment Day is observed on 22 April.
3. World Ozone Day — 16 September

A. Only 2 B. Only 3 C. Only 1 D. 2 and 3

Show answer

Ans: B. Only 3 is correct. **Logic:** Statement 3 is true (Ozone Day 16 September). Statement 1 is false — Forests Day is 21 March. Statement 2 is false — WED is 5 June (22 April is Earth Day).

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): The March ladder is Wildlife (3) → Forests (21) → Water (22).

Reason (R): Wildlife Day falls on 22 March.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Never put Wildlife on 22 March.

Q5. With reference to treaty anchors, which of the following statements is/are correct?

1. Stockholm 1972 anchors WED.
2. Montreal 1987 anchors Ozone Day.
3. Ramsar 1971 anchors Wetlands Day.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** Treaty → day map.

Q6. Arrange calendar order of the eight syllabus days (first three):

A. 5 Jun → 2 Feb → 3 Mar B. 2 Feb → 3 Mar → 21 Mar C. 16 Sep → 22 Apr → 2 Feb D. 22 May → 2 Feb → 5 Jun

💡 **Show answer**

Ans: B. Wetlands → Wildlife → Forests. **Logic:** Calendar spine start.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Earth Day — 22 April (began USA 1970)
2. WED — 5 June (UNEP / Stockholm opening)
3. Earth Day — identical to World Environment Day

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Earth Day ≠ WED.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Biodiversity Day shifted to 22 May from 2000; papers use 22 May.

Reason (R): It commemorates CBD text adopted at Nairobi in 1992.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; Nairobi 1992 does not itself explain the 2000 date shift. **A/R logic:** Commemoration vs date shift.

Q9. With reference to India hosting WED, which of the following statements is/are correct?

1. 2011 — Forests theme.
2. 2018 — Beat Plastic Pollution.
3. India hosted WED every year since 1972.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not every year.

Q10. Match List-I with List-II:

List-I	List-II
A. Ozone Day	1. CITES 1973
B. Wildlife Day	2. Montreal 1987
C. Wetlands Day	3. Ramsar 1971

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Montreal/CITES/Ramsar.

Q11. Arrange Van Mahotsav vs Forests Day:

A. Identical dates always B. Van Mahotsav = July India plantation; Forests Day = 21 March UN C. Both 16 September D. Both 2 February

💡 **Show answer**

Ans: B. July vs 21 March. **Logic:** Do not merge.

Q12. With reference to Water Day, which of the following statements is/are correct?

1. 22 March under UN-Water neighbourhood.
2. Wetlands Day is a Ramsar day — not a Water Day synonym.
3. Water Day is identical to Wildlife Day.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Water ≠ Wildlife.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Earth Hour (last Saturday of March, WWF) is not Earth Day.

Reason (R): Earth Day is 22 April.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R clarifies the date separation. **A/R logic:** Hour ≠ Day.

Q14. Which one of the following correctly states World Toilet Day?

A. 2 February B. 5 June C. 22 April D. 19 November (UN from 2013)

💡 **Show answer**

Ans: D. 19 November. **Logic:** Extra calendar fact.

Q15. How many of the following same-number facts are correct?

22 Mar Water; 22 Apr Earth; 22 May Biodiversity

A. All three B. Only two C. Only one D. None

Show answer

Ans: A. All three. **Logic:** Three different months.

Q16. With reference to Wildlife Day proclamation, which of the following statements is/are correct?

1. UN proclaimed World Wildlife Day in 2013.
2. The date itself is the 1973 CITES anniversary.
3. The date is Stockholm 1972 opening day.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: B. Only 1 and 2. **Logic:** Stockholm anchors WED.

Q17. Match List-I with List-II:

List-I	List-II
A. World No Tobacco Day	1. 2 October 2014
B. Swachh Bharat launch	2. 31 May
C. World Habitat Day	3. First Monday of October

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

Show answer

Ans: C. A-2, B-1, C-3. **Logic:** Tobacco / SBM / Habitat.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): A Wildlife–22 March pair is always wrong.

Reason (R): 22 March is World Water Day.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: D. Both true and R explains A. **A/R logic:** Classic March trap.

Q19. Which of the following pairs is/are correctly matched?

1. Biodiversity Day — broader than Wildlife Day
2. Wildlife Day — fauna–flora trade/crime under CITES date
3. Ozone Day — 2 October

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Ozone = 16 September.

Q20. Arrange Forests Day proclamation:

A. UNGA 2012; date 21 March B. Ramsar 1971 only C. CITES 1973 only D. Montreal 1987 only

💡 **Show answer**

Ans: A. UNGA 2012; 21 March. **Logic:** Forests Day fact.

Q21. With reference to UP Wetlands Day geography, which of the following statements is/are correct?

1. UP has 10 Ramsar sites teaching.
2. The day itself stays 2 February.
3. Wetlands Day moves to 5 June in UP only.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Date does not move.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Correct Ozone–WED pair is 16 September and 5 June.

Reason (R): Ozone Day belongs to the Montreal Protocol family.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; Montreal family explains Ozone date, not why WED is 5 June. **A/R logic:** Pair vs Ozone anchor.

Q23. How many of the following are on the eight syllabus-day list?

Wetlands, Wildlife, Forests, Water, Earth, Biodiversity, WED, Ozone

A. Only four B. Only six C. All eight D. Only two

💡 **Show answer**

Ans: C. All eight. **Logic:** Syllabus calendar set.

Q24. Which one of the following correctly states Earth Day theme 2024 teaching?

A. Only Ramsar Wise Use B. Only Stockholm opening C. Only CITES adoption D. Planet vs. Plastics

💡 **Show answer**

Ans: D. Planet vs. Plastics. **Logic:** Theme teaching.

Q25. With reference to No Tobacco Day, which of the following statements is/are correct?

1. 31 May.
2. Not 5 May.
3. Identical to World Environment Day.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not WED.

Q26. Match List-I with List-II:

List-I	List-II
A. WED	1. 22 April
B. Earth Day	2. 5 June
C. Ozone Day	3. 16 September

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** WED/Earth/Ozone.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): World Water Day is coordinated under UN-Water.

Reason (R): Wetlands Day is a Ramsar day, not a Water Day synonym.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; Ramsar identity does not explain UN-Water coordination of Water Day. **A/R logic:** Two different days.

Q28. Arrange full eight-day spine end:

A. Ends at 2 Feb only B. … → 16 Sep → 5 Jun C. … → 5 Jun → 16 Sep D. Ends at 3 Mar only

💡 **Show answer**

Ans: C. WED then Ozone. **Logic:** Calendar end.

Q29. With reference to Swachh Bharat, which of the following statements is/are correct?

1. Launched 2 October 2014 (Gandhi Jayanti).
2. Neighbourhood cleanliness awareness teaching.
3. It replaces World Wetlands Day date.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Does not replace Wetlands Day.

Q30. Which one of the following correctly separates Biodiversity Day from Wildlife Day?

A. Biodiversity broader CBD date; Wildlife is CITES fauna–flora trade/crime awareness B. Both identical 22 March only C. Both identical 16 September only D. Both identical Van Mahotsav July only

💡 **Show answer**

Ans: A. CBD vs CITES frames. **Logic:** Breadth vs trade day.

Common Traps

Trap	Correct
Wildlife Day = 22 March	**3 March** (Water is 22 March)
Wetlands Day = 2 / 3 March	**2 February**
Earth Day = 5 June	**22 April**; WED = 5 June
Earth Day = 22 March	**22 April**; Water = 22 March
Biodiversity Day = 22 March / April	**22 May**
Forests Day = 22 March	**21 March**
Ozone Day = 5 June	**16 September**
WED = Earth Day	Different dates and origins
Earth Hour = Earth Day	Hour ≠ Day
Population Day = 5 June	**11 July**; 5 June = WED

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 32 — National Parks & Protected Areas (Advanced)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

National Parks of Uttar Pradesh (उत्तर प्रदेश) | Wildlife Sanctuaries of Uttar Pradesh | National Parks of India | Marine National Parks | Important National Parks of the World | National Parks as Natural Heritage | Coastal National Parks

Sources baked in: NWDC/WII; MoEFCC; WPA 1972; UNESCO WH; NCERT; UPPCS/UPSC PYQs **Weight:** ★★★ High — NP ↔ state; UP negative geography; marine first; Yellowstone vs Corbett **Last verified:** August 2026

Current Affairs (this topic)

- New NP / TR notifications refresh advanced PA CA — cross-check with the Protected Areas hierarchy.
- Location-match traps stay high-yield; UP has no marine PA.

💡 🎯 **Consolidated — 34 Must-Score Facts (Click to Expand)**

1. Uttar Pradesh has **one National Park — Dudhwa (दुधवा)** — in **Lakhimpur Kheri** on the Indo-Nepal Terai (तराई). 2. Dudhwa moved sanctuary (**1958**) → National Park (**1977**) → core of **Dudhwa Tiger Reserve (1987)** with Kishanpur and Katarniaghat. Rhino rehabilitation began **1984** (later phases including ~**2018**). 3. **Valmiki** (वाल्मीकि) National Park is in **Bihar**. **Sultanpur** National Park is in **Haryana**. **Corbett** is in **Uttarakhand** (उत्तराखंड). None is a UP national park. 4. Uttar Pradesh is **landlocked**, so it has **zero** marine or coastal national parks. 5. UP has **25 Wildlife Sanctuaries**. **Hastinapur** (हस्तिनापुर) is the largest. **Chandraprabha** is in **Chandauli**. **Katarniaghat** was established in **1975**. 6. **Rudrasagar** Lake is in **Tripura**. **Sultanpur** NP is in **Haryana**. **Samaspur** and **Sarsai Nawar** are in Uttar Pradesh. 7. India's first National Park is **Hailey / Jim Corbett (जिम कॉर्बेट) (1936)** in Uttarakhand. The world's first is **Yellowstone (USA, 1872)** — Old Faithful geyser; Wyoming–Montana–Idaho. 8. India's **largest** National Park is **Hemis** in **Ladakh** (लद्दाख). **Madhya Pradesh** has the most national parks (~**11**). **Andaman (अंडमान) & Nicobar (निकोबार)** has the most wildlife sanctuaries. 9. **Gir (गीर) (Gujarat)** holds the only wild **Asiatic lions**. **Kaziranga (काजीरंगा) (Assam)** is the classic one-horned rhino stronghold. 10. A National Park is the strictest **Wildlife Protection (वन्यजीव सुरक्षा) Act** category (Section **35**). Boundary change needs the **State Legislature**. 11. A Wildlife Sanctuary (Section **18**) is more flexible than a National Park. National Parks are **in-situ** (स्थान पर) conservation. 12. India's first Marine National Park is **Gulf of Kutch (Gujarat)**. Others include **Gulf of Mannar (Tamil Nadu (नाडु))** and the Andaman marine parks. 13. Marine NP facts: **Mahatma Gandhi (महात्मा गांधी) Marine** and **Rani Jhansi (झांसी) Marine (1996)** are in the **Andaman** (अंडमान) group. 14. Natural World Heritage (विश्व धरोहर) overlaps include Kaziranga, Keoladeo (केवलादेव), Manas, Sundarbans (सुंदरबन), Nanda Devi (नंदा देवी)–Valley of Flowers (फूलों की घाटी), Western Ghats (पश्चिमी घाट), Great Himalayan (**2014**), and Khangchendzonga (mixed **2016**). 15. Uttar Pradesh has **no** natural UNESCO World Heritage site of its own. 16. Classic match set: **Dachigam–J&K–Hangul**, **Keoladeo–Rajasthan–birds**, **Kanha (कान्हा)–MP–hard-ground barasingha / Bhoorsingh mascot**, **Periyar (पेरियार)–Kerala–elephant/tiger**. 17. **Pilibhit (2014)** and **Ranipur (2023)** are tiger reserves. They are **not** additional full state national parks of UP. 18. National Chambal (चंबल) Sanctuary spans UP–MP–Rajasthan and is famous for **gharial** and dolphin — **not Haryana**. 19. River–park traps: Corbett = **Ramganga (रामगंगा) + Kosi (कोसी)**; Kaziranga = **Brahmaputra (ब्रह्मपुत्र) / Diphlu**;

Silent Valley (साइलेंट वैली) = **Kunthipuzha**. 20. Species–park traps: **Keoladeo Ghana (घन) ≠ lion (सिंह)**; **Eravikulam = Nilgiri (नीलगिरि) Tahr** (not Hoolock); **Hoolock = NE Assam**; **Bhitarkanika = saltwater crocodile**; **Desert NP = Great Indian Bustard** (Jaisalmer–Barmer; human habitation exists). 21. Dudhwa flagship fauna (प्राणीजात) include tiger, reintroduced one-horned rhino, and **barasingha** (swamp deer). 22. Terai Arc pieces with Dudhwa are **Kishanpur** (Lakhimpur Kheri) and **Katarniaghat** (Bahraich). 23. **Rajaji (राजाजी) NP (UK)** = Dehradun (देहरादून)–Haridwar (हरिद्वार)–Pauri (पौड़ी) (Rajaji + Motichur + Chilla merged **1983**); Tiger Reserve **2015**. Named after C. Rajagopalachari. 24. **Pin Valley = Himachal (हिमाचल) (Lahaul–Spiti)**, not J&K. **Van Vihar = Bhopal**. **Salim Ali NP = Srinagar (ex–City Forest)**. **Nagarhole = Rajiv Gandhi NP (Karnataka)**. 25. **Indravati = Bastar (Chhattisgarh)** NP 1981 / TR 1983. **Bandhavgarh (Umaria, MP)** famous for white tiger association. **Betla/Palamu = Jharkhand**. 26. **Keibul Lamjao (Manipur)** on **Loktak** is the only floating NP (**phumdis**; Sangai (सांगाइ) deer). **Valley of Flowers** (फूलों की घाटी) lies wholly in temperate alpine zone. 27. **Govind NP (Uttarkashi (उत्तरकाशी))** includes the source region of the **Tons** (टौंस) (Yamuna (यमुना) tributary). **Great Himalayan NP = Kullu (HP)**. 28. **Wild Ass Sanctuary (Rann of Kachchh, Gujarat)** is India's largest wildlife sanctuary. **Kuno–Palpur (MP)** is the Asiatic lion second-home site. 29. UK west→east sanctuary line: **Kedarnath (केदारनाथ) → Nanda Devi → Binsar → Askot** (Askot/Kedarnath = musk deer country). 30. **Sathyamangalam TR** links Eastern and Western Ghats. **Nagarjunsagar–Srisailam** holds the largest critical tiger habitat (वास स्थान) among common options. 31. Bird/butterfly facts: first butterfly park = **Bannerghatta (Karnataka)**; first State butterfly = **Blue Mormon (Maharashtra)**; Great Indian Hornbill (हॉर्नबिल) = **Western Ghats** (पश्चिमी घाट). 32. World-first versus India-first is a fixed pair: **Yellowstone 1872** versus **Corbett 1936**. 33. Early world peers: **Royal National Park, Australia (1879)**; **Banff NP, Canada (1885)**. 34. Continent icons: **Serengeti (Tanzania)**, **Kruger (South Africa)**, **Kakadu (Australia)**, **Galápagos (Ecuador)**, plus USA icons Yosemite / Everglades / Grand Canyon.

💡 Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
National Park (NP)	**Wildlife Sanctuary (WLS)**	WPA Sec 35; highest protection, no human settlement/grazing, boundaries altered only by State Legislature on NBWL advice vs WPA Sec 18; regulated grazing/traditional rights allowed	राष्ट्रीय उद्यान (NP) / वन्यजीव अभयारण्य (WLS)
Community Reserve	**Conservation Reserve**	Established on private or community-owned lands under WPA Sec 36C vs established on government-owned contiguous lands under WPA Sec 36A	सामुदायिक रिज़र्व / संरक्षण रिज़र्व
First Marine NP	**First National Park in India**	**Gulf of Kutch Marine NP** (1982, Gujarat) vs **Hailey / Jim Corbett NP** (1936, Uttarakhand)	प्रथम समुद्री उद्यान (कच्छ की खाड़ी) / प्रथम राष्ट्रीय उद्यान (कॉर्बेट 1936)
Dudhwa National Park	**Valmiki National Park**	Only National Park in **Uttar Pradesh** (Lakhimpur Kheri) vs only National Park in **Bihar** (West Champaran)	दुधवा राष्ट्रीय उद्यान (यूपी) / वाल्मीकि राष्ट्रीय उद्यान (बिहार)
Keibul Lamjao NP	**Khangchendzonga NP**	World's only floating park on Loktak Lake, Manipur (Sangai deer) vs India's only Mixed World Heritage Site in Sikkim	केइबुल लामजाओ (मणिपुर) / कंचनजंगा (सिक्किम)
Hemis National Park	**South Button Island NP**	India's largest National Park (Ladakh, snow leopard) vs India's smallest National Park (Andaman & Nicobar, 0.03 km^2)	हेमिस (लद्दाख, सबसे बड़ा) / साउथ बटन (अंडमान, सबसे छोटा)
Biosphere Reserve Core	**National Park**	Untouched strictly protected sanctuary within UNESCO MAB zoning vs statutory strictly protected area under Wildlife Protection Act, 1972	बायोस्फीयर कोर क्षेत्र / राष्ट्रीय उद्यान

💡 📋 Must-score facts — UP PAs, firsts, marine (Click to Expand)

UP / India firsts

Item	Lock
UP National Park	**Only Dudhwa** (Lakhimpur Kheri)
Dudhwa path	Sanctuary 1958 → NP **1977** → TR core **1987**
Valmiki / Sultanpur / Corbett	Bihar / Haryana / Uttarakhand (**not** UP NP)
UP marine NP	**Zero** (landlocked)
UP sanctuaries	**25**; Hastinapur largest
India first NP	Hailey/Corbett **1936** (UK)
World first NP	Yellowstone **1872**

Largest India NP | Hemis (Ladakh) | | Most NPs / most WLS | MP / A&N | | First Marine NP (प्रथम समुद्री उद्यान) | Gulf of Kutch **1980** | | Gir / Kaziranga | Only Asiatic lions / one-horned rhino classic |

32.1 National Parks of Uttar Pradesh

Identity: Dudhwa only | Lakhimpur Kheri | Terai | Tiger + rhino + barasingha

- **Uttar Pradesh has one National Park — Dudhwa** — in **Lakhimpur Kheri** on the Indo-Nepal Terai border.
- Sanctuary **1958** → National Park **1977** → core of **Dudhwa Tiger Reserve (1987)** with **Kishanpur** and **Katarniaghat** WLS.
- Habitat: Sal forest, grassland, swamp along the **Sharda (शारदा) (Mahakali)** system.
- Flagship fauna: **Bengal tiger**, reintroduced **one-horned rhino**, **barasingha** (swamp deer).
- **Pilibhit TR (2014)** and **Ranipur TR (2023)** are tiger reserves — not additional full state NPs.
- Negative geography: **Valmiki NP = Bihar**; **Sultanpur NP = Haryana**; **Corbett = Uttarakhand**.
- UP is **landlocked** — no marine or coastal NP.

PYQ — UPPCS Prelims 2022, Q56

Dudhwa National Park (दुधवा राष्ट्रीय उद्यान) is situated in which district of Uttar Pradesh?

Options: A. Bahraich B. Lakhimpur Kheri C. Pilibhit D. Sitapur

💡 **Show answer**

Ans: B

> **Logic:** "UP has two NPs — Dudhwa and Valmiki" is **false**.

32.2 Wildlife Sanctuaries of Uttar Pradesh

Identity: 25 WLS | Hastinapur largest | Chandraprabha Chandauli | Katarniaghat 1975

- UP has **25 Wildlife Sanctuaries** (official PA network) — most of the state's in-situ protection outside Dudhwa.
- **Terai trio with Dudhwa:Katarniaghat** (Bahraich) and **Kishanpur** (Lakhimpur Kheri) — tiger, gharial, dolphin corridor.
- **Hastinapur WLS** (~2,073 sq km) is UP's **largest**, spanning Meerut (मेरठ)–Muzaffarnagar–Ghaziabad–Bijnor–Amroha.
- **National Chambal WLS** (Agra–Etawah (इटावा) + MP–Rajasthan) — gharial, dolphin, Indian skimmer.
- **Chandraprabha WLS** is in **Chandauli** (1957; among earliest in UP).
- **Katarniaghat** was established in **1975**.
- Bird/Ramsar (रामसर) overlap includes Nawabganj, Samaspur, Sandi, Sur Sarovar, Parvati Arga, Bakhira, Saman, Sarsai Nawar, Shekha Jheel, and related wetlands.

PYQ — UPPCS Prelims 2023, Q56

Chandraprabha Wildlife Sanctuary is situated in which district of Uttar Pradesh?

Options: A. Sonbhadra (सोनभद्र) B. Chandauli C. Bahraich D. Lakhimpur Kheri

💡 **Show answer**

Ans: B

PYQ — UPPCS Prelims 2022, Q36

Which Wildlife Sanctuary of Uttar Pradesh was established in 1975?

Options: A. Chambal B. Katarniaghat C. Raipur D. Kishanpur

💡 **Show answer**

Ans: B

PYQ — UPPCS Prelims 2025, Q58

Which Ramsar sites are NOT located in Uttar Pradesh? 1. Sarsai Nawar Jheel 2. Samaspur Bird Sanctuary 3. Rudrasagar Lake 4. Sultanpur National Park

Options: A. 1 and 3 B. 3 and 4 C. 2 and 3 D. 1 and 2

💡 **Show answer**

Ans: B — Sarsai Nawar and Samaspur **are in UP**. Rudrasagar = **Tripura**; Sultanpur NP = **Haryana**.

All 25 Wildlife Sanctuaries of UP

#	Sanctuary	District(s)	Key fact
1	Bakhira	Sant Kabir (कबीर) Nagar	Ramsar waterfowl
2	Chandraprabha	**Chandauli**	Early WLS (1957)
3	Hastinapur	Meerut–Muzaffarnagar–Ghaziabad–Bijnor–Amroha	**Largest**
4	Kaimur	Mirzapur (मिर्ज़ापुर), Sonbhadra	Vindhyan forest
5	Katarniaghat	Bahraich	**1975**; Terai Arc
6	Kishanpur	Lakhimpur Kheri	Dudhwa corridor
7	Lakh Bahosi	Kannauj (कन्नौज)	Lake birds
8	Mahavir Swami (स्वामी)	Lalitpur	Bundelkhand
9	National Chambal	Agra, Etawah (+ MP, RJ)	Gharial/dolphin
10	Nawabganj	Unnao	Ramsar birds
11	Okhla	GB Nagar, Ghaziabad	NCR wetland
12	Parvati Arga	Gonda	Ramsar
13	Patna (पटना) Bird	Etah	Waterfowl
14	Ranipur	Chitrakoot (चित्रकूट), Banda	Ranipur TR 2023
15	Saman	Mainpuri	Ramsar
16	Samaspur	Rae Bareli	**IN UP** (Q58)
17	Sandi	Hardoi	Ramsar
18	Shekha Jheel	Aligarh (अलीगढ़)	Ramsar
19	Sohagibarwa	Maharajganj	Indo-Nepal
20	Sohelwa	Shravasti, Balrampur, Gonda	Bhabar (भाबर)–Terai
21	Sur Sarovar (Keetham)	Agra	Ramsar
22	Surha Tal	Ballia	Eastern wetland
23	Turtle	Varanasi (वाराणसी)	Gangetic turtles
24	Vijai Sagar	Mahoba	Bundelkhand lake

#	Sanctuary	District(s)	Key fact
25	Sarsai Nawar	Etawah	**IN UP** (Q58)

> **Logic:** Samaspur and Sarsai Nawar are **in UP** — do not mark them "NOT in UP."

32.3 National Parks of India

Identity: Corbett 1936 | ~106–110 NPs | Match-list heavy | In-situ

- India's first NP is **Hailey / Jim Corbett NP (1936)**, **Uttarakhand**.
- NWDC count ~**106** NPs (~1.35% of India's area); newer notifications keep the total near **106–110**.
- **Largest NP:Hemis**, Ladakh (~3,350 sq km) — snow leopard country.
- **Most NPs:Madhya Pradesh** (~11).
- Unique facts: **Gir (Gujarat)** = only wild Asiatic lions; **Kaziranga (Assam)** = one-horned rhino stronghold; **Keoladeo (Rajasthan)** = migratory birds + Ramsar + WH.
- National Parks maintain genetic diversity (आनुवंशिक विविधता) through **in-situ conservation** (स्व-स्थाने संरक्षण).
- Negative traps: **Simlipal = Odisha** (not MP); **Sultanpur = Haryana**; zero-NP UTs include Delhi/Lakshadweep-type distractors in some lists.
- Uttarakhand has **6 NPs** vs UP's **1** (post-2000 bifurcation trap).

PYQ — UPPCS Prelims 2025, Q90

Match: A. Dachigam B. Keoladeo C. Kanha D. Periyar with 1. Madhya Pradesh 2. Rajasthan 3. Kerala 4. Jammu and Kashmir

Options: A. 4 2 3 1 B. 2 4 1 3 C. 2 4 3 1 D. 4 2 1 3

> 💡 **Show answer**
>
> **Ans: D** — Dachigam=J&K; Keoladeo=Rajasthan; Kanha=MP; Periyar=Kerala.

PYQ — UPPCS Prelims 2023, Q118

In National Parks, genetic diversity is maintained through—

Options: A. In-situ conservation B. Ex-situ (स्थानांतरित) conservation (बाह्य-स्थाने संरक्षण) C. Gene pool D. None

> 💡 **Show answer**
>
> **Ans: A**

PYQ — UPPCS Prelims 2022, Q73

Keibul Lamjao National Park (केइबुल लामजाओ) is located in which lake?

Options: A. Loktak B. Dal C. Wular D. Kolleru

> 💡 **Show answer**
>
> **Ans: A** — Manipur; only floating national park.

PYQ — UPPCS Prelims 2020, Q62

Match: A. Indravati B. Mollem C. Kalesar D. Betla with 1. Jharkhand 2. Haryana 3. Goa (गोवा) 4. Chhattisgarh

Options: A. 4 3 2 1 B. 4 2 3 1 C. 4 1 3 2 D. 3 4 2 1

💡 **Show answer**

Ans: A — Indravati=Chhattisgarh; Mollem=Goa; Kalesar=Haryana; Betla=Jharkhand.

PYQ — UPPCS Prelims 2018, Q49 / Q58

Dachigam National Park is associated with—

💡 **Show answer**

Hangul / Kashmir Stag

Which pair is NOT correctly matched?

💡 **Show answer**

Simlipal National Park — Madhya Pradesh is wrong → Simlipal = **Odisha** (ओडिशा). Bandipur (बांदीपुर)–Karnataka, Manas–Assam, Periyar–Kerala are correct.

High-yield NP ↔ Leading states

National Park	State/UT	Extra fact
Jim Corbett	Uttarakhand	India first (1936); rivers **Ramganga + Kosi**
Dudhwa	Uttar Pradesh	Only UP NP; rhino rehab **1984+**
Valmiki	Bihar	NOT UP
Sultanpur / Kalesar	Haryana	Sultanpur trap
Dachigam / Salim Ali	J&K	Hangul; Salim Ali = ex–City Forest
Keoladeo Ghana	Rajasthan	Birds + WH — **not lion**
Kanha / Bandhavgarh / Panna / Van Vihar	Madhya Pradesh	Kanha hard-ground barasingha / **Bhoorsingh**; Bandhavgarh white tiger; Van Vihar = **Bhopal**
Gir	Gujarat	Asiatic lion
Kaziranga / Manas	Assam	Rhino / WH; Kaziranga rivers **Brahmaputra–Diphlu**
Periyar / Silent Valley / Eravikulam	Kerala	Elephant–tiger; **Kunthipuzha**; Nilgiri Tahr ("cloud goat")
Bandipur / Nagarhole (Rajiv Gandhi)	Karnataka	Western Ghats
Hemis	Ladakh	Largest NP
Keibul Lamjao	Manipur	Loktak floating / Sangai
Mollem	Goa	—
Indravati	Chhattisgarh	Bastar; NP 1981 / TR 1983
Betla (Palamu)	Jharkhand	Project Tiger (प्रोजेक्ट टाइगर)
Simlipal / Bhitarkanika	Odisha	Simlipal ≠ MP; saltwater crocodile
Sundarbans / Buxa	West Bengal	Mangrove + WH; Buxa Jalpaiguri
Desert / Ranthambore (रणथंभौर) / Sariska	Rajasthan	GIB; both tiger reserves; Sariska Alwar
Great Himalayan / Pin Valley	Himachal Pradesh	GHNP = **Kullu** WH 2014; Pin = Lahaul–Spiti (**not J&K**)
Rajaji / Govind / Valley of Flowers	Uttarakhand	Rajaji TR **2015**; Govind = **Tons** source; VoF alpine

National Park	State/UT	Extra fact
Khangchendzonga	Sikkim	Mixed WH 2016
Namdapha	Arunachal Pradesh	Eastern Himalaya (हिमालय) hotspot; tropical→arctic climate belts
Gulf of Mannar	Tamil Nadu	Marine NP **1980**
Intanki / Ntangki	Nagaland	—
Guindy	Tamil Nadu	—
Papikonda	Andhra Pradesh	—
Phawngpui	Mizoram	Blue Mountain

River / species / wrong-pair keys

Stem	Correct lock
Corbett river	**Ramganga + Kosi** — not Ganga (गंगा)
Kaziranga river	**Brahmaputra / Diphlu** — not Manas
Silent Valley river	**Kunthipuzha** — not Kaveri (कावेरी)
Ghana / Bharatpur	Birds — **not lion**
Eravikulam	**Nilgiri Tahr** — not Hoolock
Hoolock Gibbon	**NE India / Assam**
Periyar	Elephant / tiger — **not Hangul**
Desert NP	GIB; **two districts**; habitation **exists**
Nokrek / Loktak / Namdapha	Garo (गारो) Hills / Manipur lake / Arunachal — only Nokrek–Garo matched among classic three-pair stems

> **Logic:** Kanha = **MP**. Keoladeo = **Rajasthan** (थार), not UP. Pin Valley ≠ Jammu & Kashmir.

32.4 Marine National Parks

Identity: 4 marine NPs | Gulf of Kutch first | UP = none

- India's **first Marine National Park** is **Gulf of Kutch Marine NP, Gujarat (1980/82)** — coral, mangrove, islands.
- **Gulf of Mannar Marine NP, Tamil Nadu** — coral islands, dugong; sits inside Gulf of Mannar Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र).
- **Mahatma Gandhi Marine NP (Wandoor)** and **Rani Jhansi Marine NP** are in the **Andaman** Islands.
- Do not call **Sundarbans** (सुंदरबन) or **Gahirmatha** India's first marine NP — Sundarbans is coastal mangrove NP; Gahirmatha is a **marine sanctuary** (olive ridley), not an NP.

#	Marine NP	State/UT	Fact
1	Gulf of Kutch	Gujarat	**First**
2	Gulf of Mannar	Tamil Nadu	Dugong / GoM BR
3	Mahatma Gandhi Marine	Andaman	Wandoor
4	Rani Jhansi Marine	Andaman	Ritchie's Archipelago

> **Logic:** First Marine NP = **Gulf of Kutch**, not Mannar or Sundarbans.

32.5 Important National Parks of the World

Identity: Yellowstone 1872 | Continent icons | ≠ Corbett

- **World's first National Park** is **Yellowstone NP, USA (1872)** — signed under President **Ulysses S. Grant**.
- Early peers: **Royal National Park, Australia (1879)**; **Banff NP, Canada (1885)**.
- icons: **Serengeti (Tanzania)**, **Kruger (South Africa)**, **Kakadu (Australia)**, **Galápagos (Ecuador)**, **Yosemite / Everglades / Grand Canyon (USA)**.
- Never reverse: **Yellowstone = world first**; **Corbett = India first**.

Park	Country	Fact
Yellowstone	USA	World first **1872**
Banff	Canada	**1885**
Serengeti	Tanzania	Savanna (सवाना) mega-fauna
Kruger	South Africa	Classic African NP
Kakadu	Australia	Wetland/rock art landscape
Galápagos	Ecuador	Evolution / WH

> **Logic:** World first ≠ India's first.

32.6 National Parks as Natural Heritage

Identity: UNESCO Natural WH | IUCN (आईयूसीएन) Cat II | UP has none

- National Parks protect ecosystems and species as **natural heritage** for future generations; IUCN **Category II** is the NP standard.
- UNESCO **World Heritage Convention (1972)** designates natural sites under criteria **vii–x**.
- India natural WH landscapes overlapping NPs: **Kaziranga, Keoladeo, Manas, Sundarbans, Nanda Devi + Valley of Flowers, Western Ghats (serial), Great Himalayan NP, Khangchendzonga**.
- **Keoladeo** (केवलादेव) often carries **NP + Ramsar + UNESCO WH** together.
- **Manas** was removed from the WH danger list in **2011** after recovery.
- **Uttar Pradesh has no natural UNESCO World Heritage Site** — nearest are Valley of Flowers / Nanda Devi in Uttarakhand.
- Dudhwa is an NP but **not** a UNESCO WH site.

Site	State	WH note
Kaziranga / Manas	Assam	1985 batch
Keoladeo	Rajasthan	Also Ramsar
Sundarbans	West Bengal	Mangrove WH
Nanda Devi + VoF	Uttarakhand	Alpine
Western Ghats	Multi-state	Serial **2012**
Great Himalayan	Himachal	**2014**
Khangchendzonga	Sikkim	Mixed **2016**

> **Logic:** "All UNESCO WH in India are cultural" is **false**.

32.7 Coastal National Parks

Identity: Mangrove / estuary (ज्वारनदमुख) / marine coast | East vs West | CRZ link

- **Coastal NPs** protect mangrove, estuary, delta (डेल्टा), intertidal, and reef ecosystems; all marine NPs are coastal, but not every coastal NP is a "Marine NP" label.
- Key coastal NPs: **Sundarbans (WB)**, **Bhitarkanika (Odisha)**, plus the **four (चातुर्याम) Marine NPs**.
- East coast cluster: Sundarbans, Bhitarkanika, Gulf of Mannar. West coast marine flagship: **Gulf of Kutch**.
- Coastal Regulation Zone (**CRZ** under EPA 1986 (पर्यावरण संरक्षण 1986)) restricts harmful development near sensitive coastal stretches.
- **Gahirmatha** = marine **sanctuary** (turtle rookery), not NP.
- UP has **zero** coastal NPs.

Coastal / marine park	Coast	Fact
Sundarbans NP	East	Mangrove + WH
Bhitarkanika NP	East	Mangrove / croc
Gulf of Mannar Marine NP	East	Coral islands
Gulf of Kutch Marine NP	West	First Marine NP
MG / Rani Jhansi Marine	Andaman	Island marine

> **Logic:** Coastal ≠ automatically "first marine NP."

Complete PYQ Bank (Topic 32)

Q1. UPPCS Prelims 2025, Q90

Match List-I with List-II:

List-I	List-II
A. Dachigam	1. Madhya Pradesh
B. Keoladeo	2. Rajasthan
C. Kanha	3. Kerala
D. Periyar	4. Jammu and Kashmir

Row order is not the answer code.

A. 4 2 3 1

B. 2 4 1 3

C. 2 4 3 1

D. 4 2 1 3

Show answer

Logic: Four-NP state match — Dachigam J&K, Keoladeo Rajasthan, Kanha MP, Periyar Kerala. **Ans: D.** Dachigam=J&K (4); Keoladeo=Rajasthan (2); Kanha=MP (1); Periyar=Kerala (3). **Trap:** Swapping Kanha–

Periyar states or placing Keoladeo in UP.

Q2. UPPCS Prelims 2025, Q58

Which Ramsar sites are **NOT** located in Uttar Pradesh?

1. Sarsai Nawar Jheel
2. Samaspur Bird Sanctuary
3. Rudrasagar Lake
4. Sultanpur National Park

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

💡 **Show answer**

Logic: UP Ramsar trap — Sarsai Nawar and Samaspur are in UP; Rudrasagar Tripura and Sultanpur Haryana are not. **Ans: B.** **Sarsai Nawar** and **Samaspur** are in UP. **Rudrasagar = Tripura**; **Sultanpur NP = Haryana**. **Trap:** Assuming every bird wetland in the Gangetic plain is UP.

Q3. UPPCS Prelims 2023, Q56

Chandraprabha Wildlife Sanctuary is situated in which district of Uttar Pradesh?

A. Sonbhadra

B. Chandauli

C. Bahraich

D. Lakhimpur Kheri

💡 **Show answer**

Logic: Chandraprabha WLS district — Chandauli, among earliest UP sanctuaries (1957). **Ans: B.** Chandraprabha lies in **Chandauli** district. **Trap:** Sonbhadra (Kaimur forests) or Bahraich (Terai trio with Katarniaghat).

Q4. UPPCS Prelims 2023, Q118

In National Parks, genetic diversity is maintained through—

A. In-situ conservation

B. Ex-situ conservation

C. Gene pool

D. None

💡 **Show answer**

Logic: National Parks conserve wild populations in natural habitat — in-situ genetic diversity maintenance. **Ans: A.** NPs are strict in-situ sites under WPA. **Trap:** "Gene pool" label or botanical garden (ex-situ) distractors.

Q5. UPPCS Prelims 2022, Q36

Which Wildlife Sanctuary of Uttar Pradesh was established in 1975?

A. Chambal

B. Katarniaghat

C. Raipur

D. Kishanpur

💡 **Show answer**

Logic: Katarniaghat establishment year among UP Terai sanctuaries — 1975. **Ans: B.** **Katarniaghat (Bahraich)** was established **1975** in the Dudhwa Terai Arc. **Trap:** Kishanpur (same corridor, different year) or National Chambal (multi-state).

Q6. UPPCS Prelims 2022, Q56

Dudhwa National Park is situated in which district of Uttar Pradesh?

A. Bahraich

B. Lakhimpur Kheri

C. Pilibhit

D. Sitapur

💡 **Show answer**

Logic: Dudhwa NP district — Lakhimpur Kheri, not Pilibhit or Bahraich. **Ans: B.** Dudhwa — UP's only full NP — sits in **Lakhimpur Kheri**. **Trap:** Pilibhit Tiger Reserve (adjacent Terai, not Dudhwa district).

Q7. UPPCS Prelims 2022, Q73

Keibul Lamjao National Park is located in which lake?

A. Loktak

B. Dal

C. Wular

D. Kolleru

💡 **Show answer**

Logic: Keibul Lamjao — world's only floating NP on Loktak, Manipur (phumdis, Sangai deer). **Ans: A.** Keibul Lamjao floats on **Loktak (Manipur)**. **Trap:** Dal/Wular (J&K) or Kolleru (AP) lake-NP pairs.

Q8. UPPCS Prelims 2020, Q62

Match List-I with List-II:

List-I	List-II
A. Indravati	1. Jharkhand
B. Mollem	2. Haryana
C. Kalesar	3. Goa
D. Betla	4. Chhattisgarh

Row order is not the answer code.

A. 4 3 2 1

B. 4 2 3 1

C. 4 1 3 2

D. 3 4 2 1

> 💡 **Show answer**
>
> **Logic:** 2020 four-NP state code — Indravati CG, Mollem Goa, Kalesar Haryana, Betla Jharkhand. **Ans: A.** Indravati=Chhattisgarh (4); Mollem=Goa (3); Kalesar=Haryana (2); Betla=Jharkhand (1). **Trap:** Betla/Betva OCR confusion or Mollem placed in Karnataka.

Q9. UPPCS Prelims 2018, Q49

Dachigam National Park is associated with which of the following?

A. Musk Deer

B. Golden Oriole

C. Yellow-throated Marten

D. Hangul or Kashmir Stag

> 💡 **Show answer**
>
> **Logic:** Dachigam flagship species — Hangul (Kashmir stag), not musk deer or Gir lion. **Ans: D.** **Hangul** is the Kashmir stag of **Dachigam (J&K)**. **Trap:** Musk deer (Himalayan sanctuaries) or Sangai (Loktak).

Q10. UPPCS Prelims 2018, Q58

Which of the following pairs is **NOT** correctly matched?

A. Bandipur National Park — Karnataka

B. Manas Wildlife Sanctuary — Assam

C. Periyar Wildlife Sanctuary — Kerala

D. Simlipal National Park — Madhya Pradesh

> 💡 **Show answer**
>
> **Logic:** Simlipal state trap — Odisha, not Madhya Pradesh (Bandipur/Kanha are MP/Karnataka). **Ans: D.** **Simlipal = Odisha**; A, B, C are correct. **Trap:** Simlipal grouped with central-Indian tiger landscapes (MP).

Extra Drill — Sanctuary / NP Match Bank (Q31–Q40)

Q31. Which river–park pair is correct?

A. Corbett — Ganga

B. Kaziranga — Manas

C. Silent Valley — Kunthipuzha

D. Corbett — Kaveri

> 💡 **Show answer**
>
> **Logic:** River–park, species–park, and state–park mismatches are the densest traps in this chapter. **Ans: C.** Corbett = Ramganga + Kosi; Kaziranga = Brahmaputra / Diphlu system; Silent Valley = Kunthipuzha.

Q32. Which sanctuary–species pair is **wrong**?

A. Jim Corbett — Tiger

B. Keoladeo Ghana — Lion

C. Kaziranga — Rhinoceros

D. Periyar — Elephant

💡 **Show answer**

Ans: B. Ghana / Bharatpur is a bird park — not lion habitat. Gir = Asiatic lion.

Q33. Pin Valley National Park is in:

A. Jammu and Kashmir

B. Himachal Pradesh (Lahaul–Spiti)

C. Uttarakhand

D. Sikkim

💡 **Show answer**

Ans: B. Great Himalayan NP = Kullu (HP). Van Vihar = Bhopal (MP).

Q34. Which statements about National Park vs Sanctuary / BR are correct? 1. NP boundaries are defined by legislation; alteration needs State Legislature. 2. Biosphere Reserves exist only to conserve a few specific species. 3. Limited biotic interference may be permitted in a Wildlife Sanctuary.

A. 1 only

B. 2 and 3

C. 1 and 3

D. 1, 2 and 3

💡 **Show answer**

Ans: C. Statement 2 is false — BRs conserve all forms of life with support systems and sustainable use.

Q35. Rajaji National Park is correctly described as:

A. Only in Dehradun district; never a tiger reserve

B. Merger of Rajaji–Motichur–Chilla (1983); TR in 2015; three UK districts

C. Located in Rajasthan like Sariska

D. Same as Corbett Tiger Reserve

💡 **Show answer**

Ans: B. Named after C. Rajagopalachari. Corbett remains India's first NP (1936).

Q36. Keibul Lamjao National Park is unique because it is:

A. India's first marine NP

B. A floating park on Loktak Lake (phumdis)

C. The hangul stronghold of Kashmir

D. Located in Odisha mangrove deltas

💡 **Show answer**

Ans: B. Sangai (Manipur dancing deer). Hangul = Dachigam. Bhitarkanika = saltwater crocodile.

Q37. Desert National Park — which is correct?

A. Spread over one district only; no human habitation; no GIB

B. Two districts (Jaisalmer–Barmer); GIB habitat; human habitation exists

C. Located in Gujarat Rann only

D. Famous for Asiatic lion

💡 **Show answer**

Ans: B. GIB / Godawan. Gir = lion. Wild Ass Sanctuary = largest WLS in Kachchh.

Q38. Nokrek Biosphere / Loktak / Namdapha — correctly matched?

A. All three: Garo Hills / Barail / Dafla Hills

B. Only Nokrek — Garo Hills

C. Only Loktak — Barail Range

D. Only Namdapha — Dafla Hills

💡 **Show answer**

Ans: B. Loktak = Manipur freshwater lake; Namdapha = Arunachal Eastern Himalaya hotspot — classic wrong-pair stem.

Q39. Hard-ground barasingha / "Bhoorsingh" mascot is associated with:

A. Manas

B. Kanha

C. Mudumalai

D. Tal Chhapar

💡 **Show answer**

Ans: B. Kanha (MP) — exclusively graminivorous hard-ground swamp deer subspecies.

Q40. Sathyamangalam Tiger Reserve is important because it:

A. Is India's first Marine NP

B. Links Eastern Ghats and Western Ghats

C. Is the hangul habitat of J&K

D. Lies entirely in Andhra Pradesh

💡 **Show answer**

Ans: B. Tamil Nadu. Nagarjunsagar–Srisailam = largest critical tiger habitat among common options. International Tiger Day = **29 July**.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to UP protected areas, which of the following statements is/are correct?

1. UP's sole National Park is Valmiki in West Champaran.
2. UP is landlocked — zero marine/coastal NPs.
3. Valmiki NP is a UP national park.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: A. Only 2 is correct. **Logic:** Statement 2 is true (landlocked UP → no marine NP). Statement 1 is false — Dudhwa (Lakhimpur Kheri) is UP's classic National Park fact. Statement 3 is false — Valmiki is in Bihar.

Q2. Match List-I with List-II:

List-I	List-II
A. Corbett	1. Haryana
B. Sultanpur	2. Uttarakhand
C. Rudrasagar	3. Tripura

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** None is UP NP.

Q3. Which of the following pairs is/are correctly matched?

1. World's first NP — Yellowstone 1872
2. India's first NP — Corbett/Hailey 1946 (not 1936)
3. India's largest NP is Jim Corbett, not Hemis (Ladakh).

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Statement 1 is true (Yellowstone 1872). Statement 2 is false — Hailey/Corbett is 1936, not 1946. Statement 3 is false — Hemis (Ladakh) is the usual largest-NP teaching fact.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Dudhwa moved sanctuary (1958) → NP (1977) → Tiger Reserve core (1987).

Reason (R): Pilibhit (2014) and Ranipur (2023) are additional full state national parks of UP.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Pilibhit/Ranipur = TRs — not extra NPs.

Q5. With reference to UP sanctuaries, which of the following statements is/are correct?

1. UP has 25 Wildlife Sanctuaries.
2. Hastinapur is the largest.
3. Chandraprabha is in Chandauli.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** Sanctuary map.

Q6. Arrange WPA categories idea:

A. Sanctuary stricter than NP always B. NP strictest (Sec 35); Sanctuary more flexible (Sec 18) C. Both identical Section 18 only D. NP needs no legislature for boundary change

💡 **Show answer**

Ans: B. NP stricter; boundary change needs legislature. **Logic:** Category fact.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Gir — only wild Asiatic lions
2. Kaziranga — one-horned rhino stronghold
3. Keoladeo Ghana — Asiatic lion

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Keoladeo = birds.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): India's first Marine NP is Gulf of Kutch (Gujarat).

Reason (R): UP has multiple marine national parks.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** UP landlocked — zero marine NPs.

Q9. With reference to Dudhwa fauna, which of the following statements is/are correct?

1. Tiger, reintroduced one-horned rhino, barasingha.
2. Terai Arc pieces include Kishanpur and Katarniaghat.
3. Dudhwa is a marine national park.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not marine.

Q10. Match List-I with List-II:

List-I	List-II
A. Dachigam	1. MP — barasingha / Bhoorsingh
B. Kanha	2. J&K — Hangul
C. Keoladeo	3. Rajasthan — birds

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Classic match set.

Q11. Arrange MP parks / A&N teaching:

A. A&N has most NPs among states always false B. MP has most NPs (~11); A&N has most among UT/island set teaching C. UP has most NPs in India D. Haryana has most NPs

💡 **Show answer**

Ans: B. MP most; A&N island lead teaching. **Logic:** Count traps.

Q12. With reference to river–park traps, which of the following statements is/are correct?

1. Corbett — Ramganga + Kosi.
2. Kaziranga — Brahmaputra / Diphlu.
3. Silent Valley — Kunthipuzha.
4. Corbett — Brahmaputra only.

A. 2, 3 and 4 only B. 1, 2 and 4 only C. 1, 2 and 3 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: C. 1, 2 and 3 only. **Logic:** Corbett ≠ Brahmaputra.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): UP has no natural UNESCO World Heritage site of its own.

Reason (R): National Parks automatically equal World Heritage listing.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** WH is a separate overlay.

Q14. Which one of the following correctly states Keibul Lamjao?

A. Hemis Ladakh only B. Gujarat lion park only C. UP Dudhwa twin only D. Only floating NP on Loktak; Sangai / phumdis

💡 **Show answer**

Ans: D. Floating NP Manipur. **Logic:** Sangai fact.

Q15. How many of the following are correct UP negative facts?

Valmiki≠UP; Sultanpur≠UP; Rudrasagar≠UP

A. All three B. Only two C. Only one D. None

> 💡 **Show answer**
>
> **Ans: A.** All three. **Logic:** State traps.

Q16. With reference to species–park traps, which of the following statements is/are correct?

1. Eravikulam — Nilgiri Tahr (not Hoolock).
2. Hoolock — NE Assam neighbourhood.
3. Bhitarkanika — saltwater crocodile neighbourhood.

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

> 💡 **Show answer**
>
> **Ans: B.** All three. **Logic:** Species–park map.

Q17. Match List-I with List-II:

List-I	List-II
A. National Chambal	1. Gujarat largest WLS
B. Wild Ass Sanctuary	2. UP–MP–Rajasthan; gharial/dolphin
C. Kuno–Palpur	3. Asiatic lion second-home neighbourhood

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-1, C-3. **Logic:** Chambal / Wild Ass / Kuno.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Samaspur and Sarsai Nawar are in Uttar Pradesh.

Reason (R): Sultanpur NP is also in Uttar Pradesh.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** Sultanpur = Haryana.

Q19. Which of the following pairs is/are correctly matched?

1. Pin Valley — Himachal (Lahaul–Spiti)
2. Van Vihar — Bhopal
3. Pin Valley — J&K only

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Pin ≠ J&K.

Q20. Arrange Katarniaghat year teaching:

A. 1975 B. 1936 C. 2014 D. 2023

💡 **Show answer**

Ans: A. 1975. **Logic:** Katarniaghat established.

Q21. With reference to Rajaji, which of the following statements is/are correct?

1. UK — Dehradun–Haridwar–Pauri merge 1983.
2. Tiger Reserve 2015.
3. Named after C. Rajagopalachari neighbourhood teaching.
4. It is UP's second national park.

A. 1, 2 and 4 only B. 1, 2 and 3 only C. 2, 3 and 4 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: B. 1, 2 and 3 only. **Logic:** Rajaji = UK — not UP NP.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Nagarjunsagar–Srisailam holds the largest critical tiger habitat among common options.

Reason (R): Sathyamangalam TR links Eastern and Western Ghats.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; Sathyamangalam link does not explain NSTR size. **A/R logic:** Two TR facts.

Q23. How many of the following Andaman marine NP facts are correct?

Mahatma Gandhi Marine; Rani Jhansi Marine (1996)

A. Only one B. None C. Both D. Three including Dudhwa

💡 **Show answer**

Ans: C. Both. **Logic:** Andaman marine pair.

Q24. Which one of the following correctly states first butterfly park teaching?

A. Yellowstone only B. Dudhwa only C. Hemis only D. Bannerghatta (Karnataka)

💡 **Show answer**

Ans: D. Bannerghatta. **Logic:** Butterfly park first.

Q25. With reference to Bandhavgarh / Indravati, which of the following statements is/are correct?

1. Bandhavgarh (Umaria, MP) — white tiger association teaching.
2. Indravati — Bastar Chhattisgarh NP 1981 / TR 1983.
3. Both are UP national parks.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not UP.

Q26. Match List-I with List-II:

List-I	List-II
A. Govind NP	1. Kullu HP
B. Great Himalayan NP	2. Tons source region (UK)
C. Salim Ali NP	3. Srinagar

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** Govind/GHNP/Salim Ali.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): National Parks are in-situ conservation.

Reason (R): Boundary change of an NP needs the State Legislature.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; legislature rule does not define in-situ meaning. **A/R logic:** In-situ vs boundary rule.

Q28. Arrange UK west→east sanctuary line teaching:

A. Askot → Kedarnath only reversed always B. Only Gir → Kaziranga C. Kedarnath → Nanda Devi → Binsar → Askot D. Only Dudhwa → Corbett east of Assam

💡 **Show answer**

Ans: C. Kedarnath → Nanda Devi → Binsar → Askot. **Logic:** Musk deer country ends.

Q29. With reference to natural WH overlaps, which of the following statements is/are correct?

1. Include Kaziranga, Keoladeo, Manas, Sundarbans, Nanda Devi–VoF, Western Ghats...
2. UP has its own natural UNESCO WH site.
3. WH overlay ≠ automatic NP identity.

A. 1, 2 and 3 B. 1 and 2 only C. 2 and 3 only D. 1 and 3 only

💡 **Show answer**

Ans: D. Only 1 and 3. **Logic:** UP has no natural WH of its own.

Q30. Which one of the following correctly states Valley of Flowers setting?

A. Wholly temperate alpine neighbourhood teaching B. Only Rann of Kachchh C. Only Gulf of Kutch marine D. Only Loktak phumdis

💡 **Show answer**

Ans: A. Temperate alpine. **Logic:** VoF fact.

Common Traps

Trap	Correct
Valmiki / Sultanpur in UP	**Bihar / Haryana**
Dudhwa = Pilibhit district	**Lakhimpur Kheri**
First Marine NP = Sundarbans	**Gulf of Kutch**
World first NP = Corbett	**Yellowstone 1872**
Simlipal = MP	**Odisha**
Keoladeo = UP	**Rajasthan**
Samaspur / Sarsai Nawar not in UP	**Both are in UP**
Gahirmatha = Marine NP	**Marine sanctuary**
UP has UNESCO natural WH	**None**
Hangul = Gir	**Dachigam**
NP boundary needs Parliament	**State Legislature**
Corbett river = Ganga	**Ramganga + Kosi**
Silent Valley river = Kaveri	**Kunthipuzha**
Ghana = lion	**Birds** (Keoladeo)
Pin Valley = J&K	**Himachal (Lahaul–Spiti)**
Eravikulam = Hoolock	**Nilgiri Tahr**
Periyar = Hangul	**Elephant / tiger**
Desert NP = no people	Habitation **exists**; GIB yes
Nagarhole = Andhra	**Karnataka** (Rajiv Gandhi NP)
Bandipur = Tamil Nadu	**Karnataka**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 33 — Biosphere Reserves (जैवमंडल आरक्षित क्षेत्र)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र) | Major Biosphere Reserves of India | Biosphere Reserves of Uttar Pradesh (उत्तर प्रदेश) | Biosphere Reserves of the World

Sources baked in: UNESCO MAB; MoEFCC; NWDC; NCERT; UPPCS 2021–2025 PYQs **Weight:** ★★★ High — BR ↔ state; Nokrek ≠ Manipur; UP = zero BR **Last verified:** August 2026

Current Affairs (this topic)

- WNBR new listings drift the UNESCO count; India national BR list stays the teaching spine.
- Biosphere ≠ WPA statutory category.

💡 🎯 **Consolidated — 25 Must-Score Facts (Click to Expand)**

1. A Biosphere Reserve is a **UNESCO MAB (1971)** landscape with **core → buffer → transition**. It is **not** a Wildlife Protection (वन्यजीव सुरक्षा) Act category. 2. India has **18** notified national Biosphere Reserves. About **13** sit on the UNESCO World Network. 3. India's **first** Biosphere Reserve is **Nilgiri (नीलगिरि) (1986)**. The **largest** is **Great Rann of Kutch (Gujarat)**. The **smallest** is **Nokrek (Meghalaya (मेघालय))**. 4. **Nokrek** is in **Meghalaya** (मेघालय). Pairing Nokrek with **Manipur** is always wrong. 5. **Manas (Assam)** remains a national Biosphere Reserve but was **removed from the UNESCO World Network in 2011**. 6. **Uttar Pradesh has zero Biosphere Reserves.** Nanda Devi (नंदा देवी) BR is in **Uttarakhand** (उत्तराखंड), not UP. 7. Dudhwa (दुधवा) is a National Park. Valmiki (वाल्मीकि) is a Bihar National Park. Neither is a UP Biosphere Reserve. 8. Match fact: **Simlipal–Odisha**, **Nokrek–Meghalaya**, **Agasthyamalai–Kerala**, **Manas–Assam**. 9. Match fact: **Simlipal–Odisha**, **Pachmarhi–Madhya Pradesh**, **Nanda Devi–Uttarakhand**, **Nokrek–Meghalaya**. 10. **Panna (Madhya Pradesh, 2011)** is India's newest notified Biosphere Reserve. 11. Madhya Pradesh has **three** Biosphere Reserves: Pachmarhi, Achanakmar–Amarkantak (with Chhattisgarh), and Panna. 12. Multi-state classics: **Nilgiri** (नीलगिरि) (Tamil Nadu (नाडु)–Karnataka–Kerala), **Agasthyamalai** (Kerala–Tamil Nadu), **Achanakmar–Amarkantak** (MP–Chhattisgarh). 13. Core allows **no exploitation**. Buffer allows limited research and education use. Transition supports livelihoods. 14. A National Park is the strictest Wildlife Protection Act site. A Biosphere Reserve adds sustainable use and research under MAB. 15. World Network membership is a UNESCO subset of national Biosphere Reserves. National listing alone is not enough for WNBR. 16. Bihar and Rajasthan also have **zero** Biosphere Reserves. Valmiki remains an NP, not a BR. 17. Gulf of Mannar is a **marine** Biosphere Reserve in Tamil Nadu. Sundarbans (सुंदरबन) BR is in West Bengal. 18. Cold Desert BR is in **Himachal (हिमाचल) Pradesh**. Khangchendzonga BR is in **Sikkim**. 19. Seshachalam Hills BR is in **Andhra Pradesh**. Great Nicobar (निकोबार) BR is in the Andaman (अंडमान) and Nicobar Islands. 20. Periodic UNESCO review can delist a site from the World Network — India's example is **Manas 2011**. 21. Yellowstone, Serengeti–Ngorongoro, Galápagos, Lake Baikal, Doñana, and Mount Kenya are classic world Biosphere icons. 22. Do not place Nanda Devi in Uttar Pradesh on a match list. After bifurcation it is an **Uttarakhand** site. 23. Agasthyamalai paired only with Kerala is accepted as correct in recent papers even though the landscape also spans Tamil Nadu. 24. Biosphere Reserves can overlap National Parks and World Heritage (विश्व धरोहर) tags, but the MAB label is still a separate tool. 25. Among common options, **biosphere reserves** are the best strategy for conserving

biodiversity **together with traditional human life / cultural diversity**. **Gulf of Mannar** is often cited as India's richest marine biodiversity BR mosaic.

💡 ⚡ **Confused Pairs (Click to Expand)**

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Biosphere Reserve (BR)	**National Park (NP)**	Landscape-level conservation integrating human livelihoods (MAB 3-tier zoning) vs species/habitat preservation strictly excluding human activity	जैवमंडल आरक्षित क्षेत्र / राष्ट्रीय उद्यान
National BRs in India	**WNBR (UNESCO) Listed**	Total **18** Biosphere Reserves designated by MoEFCC vs **12** inducted into UNESCO's World Network of Biosphere Reserves	18 राष्ट्रीय बायोस्फीयर / 12 यूनेस्को WNBR सूची
First BR in India	**Latest BR added to WNBR**	**Nilgiri** (1986, TN/KER/KAR) vs **Panna** (2020, Madhya Pradesh)	प्रथम बायोस्फीयर (नीलगिरि 1986) / नवीनतम यूनेस्को (पन्ना 2020)
Core Zone	**Buffer Zone**	Legally protected untouched sanctum sanctorum vs zone surrounding core where research, education, and limited tourism are permitted	कोर क्षेत्र (पूर्णतः सुरक्षित) / बफर क्षेत्र (सीमित गतिविधि)
Buffer Zone	**Transition Zone**	Managed area buffering the core vs outermost area where local communities, agriculture, and settlements sustainably coexist	बफर जोन / संक्रमण क्षेत्र (बस्तियां/खेती)
Nokrek Biosphere Reserve	**Loktak Lake**	Located in Garo Hills, **Meghalaya** (Citrus indica) vs freshwater wetland in **Manipur**	नोकरेक (मेघालय) / लोकटक (मणिपुर)
Uttar Pradesh BR count	**Madhya Pradesh BR count**	**Zero** Biosphere Reserves in UP vs **3** Biosphere Reserves in MP (Pachmarhi, Achanakmar-Amarkantak, Panna)	यूपी (शून्य बायोस्फीयर) / म.प्र. (3 बायोस्फीयर)

💡 📋 **Must-score facts — MAB zones, India list tags (Click to Expand)**

Frame | Item | Lock |
|---|---|
| BR model | UNESCO MAB **1971**; core → buffer → transition |
| India count | **18** national; ~**13** UNESCO WNBR |
| First / largest / smallest | Nilgiri **1986** / Great Rann of Kutch / Nokrek |
| Nokrek state | **Meghalaya** (≠ Manipur) |
| Manas UNESCO | Removed from WNBR **2011** (still national BR) |
| UP BRs | **Zero**; Nanda Devi = Uttarakhand |
| Newest notified | Panna **2011** (MP) |
| MP trio | Pachmarhi · Achanakmar–Amarkantak · Panna |
| Multi-state | Nilgiri TN–KA–KL; Agasthyamalai KL–TN |

33.1 Biosphere Reserve

Identity: UNESCO MAB 1971 | Core–Buffer–Transition | MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) executive | No WPA category

- A **Biosphere Reserve (BR)** (जैवमंडल आरक्षित क्षेत्र) is a large landscape that conserves biodiversity while allowing regulated sustainable community use and research.
- UNESCO launched the **Man and the Biosphere (MAB) Programme in 1971** (same year as Ramsar (रामसर)).
- India notifies BRs through **MoEFCC executive action** — BRs are **not** a separate WPA 1972 statutory category; cores usually overlap an existing NP/sanctuary.
- **Three zones: Core** (no exploitative use / genetic diversity (आनुवंशिक विविधता)) → **Buffer** (research, education, regulated eco-tourism and grazing) → **Transition** (settlements, crop cultivation, livelihoods).
- **Triple mandate:** conservation + sustainable development + logistic support (research/monitoring/education).
- Edward Suess is classically linked to the **biosphere** concept; **biodiversity** as a contracted term is linked to **W.G. Rosen (1985)** after **Raymond F. Dasmann** used "biological diversity" (1968).
- Among common Prelims options, a **biosphere reserve** (जैवमंडल आरक्षित क्षेत्र) is the strongest answer for conserving biodiversity **along with traditional human life and cultural diversity**.
- India's first BR is **Nilgiri (1986)** — Tamil Nadu, Karnataka, Kerala.
- India has **18 notified BRs**; about **12–13** are in UNESCO's **World Network of Biosphere Reserves (WNBR)**.
- **Agasthyamalai** includes Neyyar, Peppara, Shendurney WLS and **Kalakad–Mundanthurai Tiger Reserve**.

- **Gulf of Mannar** is often the largest **UNESCO-certified** Indian BR by area; **Great Rann of Kutch** is the largest **national** BR.
- **Bundala** Biosphere Reserve is in **Sri Lanka**, not India.
- Area decreasing order among common options: **Nanda Devi > Pachmarhi > Agasthyamalai > Panna > Khangchendzonga**.
- Funding pattern under the Centrally Sponsored Scheme is typically **60:40** (Centre–State), **90:10** for NE/Himalayan states.

PYQ — UPPCS Prelims 2018, Q54

Which is NOT an example of in-situ (स्थान पर) conservation (स्व-स्थाने संरक्षण)?

Options: A. Biosphere reserve B. Botanical garden (वनस्पति उद्यान) C. National Park D. Sacred groves (पवित्र उपवन)

💡 **Show answer**

Ans: B — Botanical garden is **ex-situ** (स्थानांतरित). BR / NP / sacred groves are in-situ.

> **Logic:** "Biosphere Reserves allow only absolute (निरपेक्ष) ban on all human use" is **false** — buffer/transition allow regulated use.

33.2 Major Biosphere Reserves of India

Identity: 18 national BRs | Match-list heavy | Nokrek / Manas / Simlipal traps

- **Largest BR:Great Rann of Kutch**, Gujarat. **Smallest:Nokrek**, Meghalaya.
- **Newest notified:Panna**, Madhya Pradesh (2011).
- **Manas (Assam)** remains a **national BR** but was **removed from UNESCO WNBR in 2011**.
- **Madhya Pradesh** has three BRs: Pachmarhi, Achanakmar–Amarkantak (with Chhattisgarh), Panna.
- Multi-state classics: **Nilgiri** (TN–KA–KL), **Agasthyamalai** (KL–TN), **Achanakmar–Amarkantak** (MP–CG).

PYQ — UPPCS Prelims 2025, Q144

Which pair(s) NOT correctly matched? 1. Nokrek — Manipur 2. Manas — Assam 3. Agasthyamalai — Kerala

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D — Only stmt 1 is wrong. Nokrek = **Meghalaya**.

PYQ — UPPCS Prelims 2024, Q10

Match: A. Simlipal B. Nokrek C. Agasthyamalai D. Manas with 1. Kerala 2. Odisha 3. Assam 4. Meghalaya

Options: A. 1 2 4 3 B. 3 1 4 2 C. 2 4 3 1 D. 2 4 1 3

💡 **Show answer**

Ans: D — Simlipal=Odisha; Nokrek=Meghalaya; Agasthyamalai=Kerala; Manas=Assam.

PYQ — UPPCS Prelims 2021, Q140

Match: A. Similipal B. Pachmarhi C. Nandadevi D. Nokrek with 1. Uttarakhand 2. Meghalaya 3. Madhya Pradesh 4. Odisha

Options: A. 4 3 2 1 B. 4 3 1 2 C. 1 3 4 2 D. 3 4 1 2

💡 **Show answer**

Ans: B — Simlipal=Odisha; Pachmarhi=MP; Nanda Devi=Uttarakhand; Nokrek=Meghalaya.

PYQ — UPPCS Prelims 2022, Q110

Match: A. Manas B. Sundarbans C. Nanda Devi D. Pachmarhi with 1. Madhya Pradesh 2. Uttar Pradesh 3. Assam 4. West Bengal

💡 **Show answer**

Intended Ans: D (A-3, B-4, C-2, D-1) — paper's List-II labelled Nanda Devi as "Uttar Pradesh." **Geography fact:** Nanda Devi BR is in **Uttarakhand**. UP itself has **zero** BRs.

All 18 Biosphere Reserves of India

#	Biosphere Reserve	State(s)	Notified	WNBR note
1	Nilgiri	TN, KA, KL	**1986**	First; WNBR 2000
2	Nanda Devi	**Uttarakhand**	1988	WNBR 2004
3	Nokrek	**Meghalaya**	1988	Smallest; WNBR 2009
4	Gulf of Mannar	Tamil Nadu	1989	Marine; WNBR 2001
5	Sundarbans	West Bengal	1989	WNBR 2001
6	Manas	Assam	1989	**Removed UNESCO 2011**
7	Great Nicobar	A&N	1989	WNBR 2013
8	Simlipal	**Odisha** (ओडिशा)	1994	WNBR 2009
9	Dibru-Saikhowa	Assam	1997	National only
10	Dihang-Dibang	Arunachal Pradesh	1998	National only
11	Pachmarhi	Madhya Pradesh	1999	WNBR 2009
12	Khangchendzonga	Sikkim	2000	WNBR 2018
13	Agasthyamalai	Kerala, Tamil Nadu	2001	WNBR 2016
14	Achanakmar–Amarkantak	MP, Chhattisgarh	2005	WNBR 2012
15	Great Rann of Kutch	Gujarat	2008	**Largest**
16	Cold Desert	Himachal Pradesh	2009	Recent WNBR addition
17	Seshachalam Hills	Andhra Pradesh	2010	National only
18	Panna	Madhya Pradesh	**2011**	Newest; WNBR 2020

> **Logic: Agasthyamalai** spans **Kerala and Tamil Nadu** — pairing it with Kerala alone is still accepted when Tamil Nadu is not an option.

33.3 Biosphere Reserves of Uttar Pradesh

Identity: Zero BRs | Nearest = Nanda Devi (UK) | Dudhwa ≠ BR

- **Uttar Pradesh has no notified Biosphere Reserve.**
- Nearest Himalayan BR is **Nanda Devi (Uttarakhand)** — not in UP after the 2000 bifurcation.
- Do not treat **Dudhwa NP**, **Pilibhit TR**, or **Valmiki NP (Bihar)** as UP biosphere reserves.

- Terai (तराई) Arc (Dudhwa–Kishanpur–Katarniaghat) is landscape-conservation geography, but it is **not** a notified BR.
- Neighbour contrast: MP has **three** BRs; Bihar and Rajasthan also have **zero** BRs (Valmiki is an NP).

State	BR count	Fact
Uttar Pradesh (उत्तर प्रदेश)	**0**	Any "UP BR" option is false
Uttarakhand	1	Nanda Devi
Madhya Pradesh	3	Pachmarhi, Achanakmar–Amarkantak, Panna
Bihar	0	Valmiki = NP, not BR

> **Logic:** "Which BR is in UP?" → **None**.

33.4 Biosphere Reserves of the World

Identity: UNESCO WNBR | Global MAB sites | Living laboratories

- UNESCO's **World Network of Biosphere Reserves** links hundreds of sites worldwide under MAB for conservation and sustainable development.
- Famous icons: **Yellowstone (USA)**, **Serengeti–Ngorongoro (Tanzania)**, **Galápagos (Ecuador)**, **Lake Baikal (Russia)**, **Doñana (Spain)**, **Mount Kenya (Kenya)**.
- World BRs follow the same **core–buffer–transition** design as India's national BRs.
- A site can carry **BR + National Park + World Heritage** together (e.g., Sundarbans-type landscapes).
- Periodic review (~10 years) can lead to **delisting** from WNBR (India example: **Manas 2011**).

World BR / landscape	Country	Fact
Yellowstone	USA	Iconic (साकार) MAB / NP landscape
Serengeti–Ngorongoro	Tanzania	Savanna (सवाना) mega-fauna
Galápagos	Ecuador	Evolution / WH overlap
Lake Baikal	Russia	Freshwater biodiversity
Doñana	Spain	Wetland BR classic
Mount Kenya	Kenya	Afro-alpine BR

> **Logic:** World BR network ≠ World Heritage list — related but different UNESCO instruments.

Complete PYQ Bank (Topic 33)

Q1. UPPCS Prelims 2025, Q144

Which of the following pair(s) is/are **NOT** correctly matched?

1. Nokrek — Manipur
2. Manas — Assam
3. Agasthyamalai — Kerala

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

Show answer

Logic: Nokrek–Manipur is the sole wrong pair — Nokrek is in Meghalaya (smallest Indian BR). **Ans: D.** Only statement 1 is wrong — **Nokrek = Meghalaya**. Manas–Assam and Agasthyamalai–Kerala are correct. **Trap:** Nokrek confused with North-East state names (Manipur/Nagaland).

Q2. UPPCS Prelims 2024, Q10

Match List-I with List-II:

List-I	List-II
A. Simlipal	1. Kerala
B. Nokrek	2. Odisha
C. Agasthyamalai	3. Assam
D. Manas	4. Meghalaya

Row order is not the answer code.

A. 1 2 4 3

B. 3 1 4 2

C. 2 4 3 1

D. 2 4 1 3

Show answer

Logic: Four-BR match code 2-4-1-3 — Simlipal Odisha, Nokrek Meghalaya, Agasthyamalai Kerala, Manas Assam. **Ans: D.** Simlipal=Odisha (2); Nokrek=Meghalaya (4); Agasthyamalai=Kerala (1); Manas=Assam (3). **Trap:** Simlipal in MP or Agasthyamalai only in Tamil Nadu (it spans KL–TN but Kerala is the keyed match).

Q3. UPPCS Prelims 2022, Q110

Match List-I with List-II:

List-I	List-II
A. Manas	1. Madhya Pradesh
B. Sundarbans	2. Uttar Pradesh
C. Nanda Devi	3. Assam
D. Pachmarhi	4. West Bengal

Row order is not the answer code.

A. A-2, B-1, C-3, D-4

B. A-4, B-3, C-2, D-1

C. A-1, B-2, C-3, D-4

D. A-3, B-4, C-2, D-1

Show answer

Logic: 2022 paper keyed Nanda Devi to Uttar Pradesh in List-II — geography fact is Uttarakhand; UP has zero BRs. **Ans: D (paper code).** Manas=Assam (3); Sundarbans=WB (4); Pachmarhi=MP (1); Nanda Devi

paired with UP (2) in the paper. **Geography fact:** Nanda Devi BR is in **Uttarakhand**; **UP has no BR**. **Trap:** Post-2000 statehood — never mark Nanda Devi as UP on a fact card.

Q4. UPPCS Prelims 2021, Q140

Match List-I with List-II:

List-I	List-II
A. Similipal	1. Uttarakhand
B. Pachmarhi	2. Meghalaya
C. Nandadevi	3. Madhya Pradesh
D. Nokrek	4. Odisha

Row order is not the answer code.

A. 4 3 2 1

B. 4 3 1 2

C. 1 3 4 2

D. 3 4 1 2

💡 **Show answer**

Logic: 2021 four-BR code 4-3-1-2 — Similipal Odisha, Pachmarhi MP, Nanda Devi Uttarakhand, Nokrek Meghalaya. **Ans: B.** Similipal=Odisha (4); Pachmarhi=MP (3); Nanda Devi=Uttarakhand (1); Nokrek=Meghalaya (2). **Trap:** Nanda Devi listed as UP or Nokrek as Manipur.

Q5. UPPCS Prelims 2018, Q54

Which of the following is **NOT** an example of in-situ conservation?

A. Biosphere reserve

B. Botanical garden

C. National Park

D. Sacred groves

💡 **Show answer**

Logic: In-situ vs ex-situ — botanical garden is ex-situ; BR/NP/sacred groves are in-situ. **Ans: B.** Botanical gardens conserve plants **outside** natural wild habitat (ex-situ). **Trap:** Treating any garden under biodiversity law as in-situ.

Q6. Pattern

India's first Biosphere Reserve (notified 1986) is:

A. Nilgiri (Tamil Nadu–Karnataka–Kerala)

B. Nanda Devi (Uttarakhand)

C. Nokrek (Meghalaya)

D. Sundarbans (West Bengal)

💡 **Show answer**

Logic: First notified Indian BR — Nilgiri 1986, not Nanda Devi. **Ans: A.** **Nilgiri (1986)** was India's first BR. **Trap:** Nanda Devi WH fame or Nokrek smallest label used to pick wrong first.

Q7. Pattern

In a Biosphere Reserve under UNESCO MAB, regulated human settlements and livelihoods belong mainly to the:

A. Core zone

B. Buffer zone

C. Transition (manipulation) zone

D. Exosphere

💡 **Show answer**

Logic: MAB zonation — core strict protection; transition sustainable human use. **Ans: C.** **Transition zone** supports sustainable use and settlements; **core** bans exploitative activity. **Trap:** No people anywhere in a BR or farming allowed in core.

Q8. Pattern

Manas Biosphere Reserve (Assam):

A. Was removed from UNESCO World Network of Biosphere Reserves in 2011 but remains a national BR

B. Is no longer any kind of Biosphere Reserve in India

C. Was never part of the World Network

D. Is located in Manipur

💡 **Show answer**

Logic: Manas UNESCO delisting 2011 — remains national BR. **Ans: A.** UNESCO **delisted Manas in 2011**; it is still a **national BR** in Assam. **Trap:** Removed from UNESCO means ceased to be a BR entirely.

Q9. Pattern

Which statement about Biosphere Reserves in Uttar Pradesh is correct?

A. Uttar Pradesh has no notified Biosphere Reserve

B. Dudhwa is UP's only Biosphere Reserve

C. Valmiki is a UP Biosphere Reserve

D. Nanda Devi lies in Uttar Pradesh

💡 **Show answer**

Logic: Uttar Pradesh has zero notified Biosphere Reserves. **Ans: A.** **UP = zero BRs**. Dudhwa is an NP; Valmiki is Bihar; Nanda Devi is Uttarakhand. **Trap:** Any UP Terai name assumed to carry a BR tag.

Q10. Pattern

Which pair is correctly matched?

A. Largest Biosphere Reserve of India — Great Rann of Kutch

B. Largest Biosphere Reserve of India — Nokrek

C. Smallest Biosphere Reserve — Great Rann of Kutch

D. Smallest Biosphere Reserve — Nilgiri

Show answer

Logic: Great Rann largest national BR; Nokrek smallest. **Ans: A.** **Great Rann of Kutch** is the largest national BR; **Nokrek (Meghalaya)** is the smallest. **Trap:** Gulf of Mannar largest marine confused with overall largest.

Extra Drill — Biosphere Zones & Match (Q31–Q40)

Q31. Correct zone match for a Biosphere Reserve:

A. Core — crop cultivation and settlements

B. Buffer — no human activity of any kind forever

C. Transition — settlements and livelihood use allowed

D. Core — industrial mining encouraged

Show answer

Logic: Zone purposes, largest/smallest, and state–BR mismatches dominate BR stems. **Ans: C.** Core = no exploitation; Buffer = research/tourism/regulated use; Transition = settlements/cultivation.

Q32. Which is **not** a UNESCO WNBR site among common options historically?

A. Nilgiri

B. Gulf of Mannar

C. Rann of Kutch (often national-only in older stems)

D. Nanda Devi

Show answer

Ans: C. Great Rann of Kutch is India's largest **national** BR; older stems treat it as outside UNESCO network. Nilgiri/Mannar/Nanda Devi are classic WNBR sites.

Q33. Agasthyamalai Biosphere Reserve includes:

A. Kawal and Srisailam only

B. Neyyar, Peppara, Shendurney WLS and Kalakad–Mundanthurai TR

C. Mudumalai and Silent Valley only

D. Only Andhra Papikonda

Show answer

Ans: B. Kerala–Tamil Nadu Western Ghats landscape.

Q34. Decreasing area order (common set):

A. Pachmarhi > Nanda Devi > Agasthyamalai > Panna > Khangchendzonga

B. Nanda Devi > Pachmarhi > Agasthyamalai > Panna > Khangchendzonga

C. Khangchendzonga > Panna > Nanda Devi

D. Panna > Gulf of Mannar > Sundarbans

Show answer

Ans: B. Among UNESCO area comparisons, Gulf of Mannar > Sundarbans > Pachmarhi > Khangchendzonga.

Q35. Bundala Biosphere Reserve is in:

A. India

B. Bangladesh

C. Sri Lanka

D. Nepal

💡 **Show answer**

Ans: C. Southeast coast of Sri Lanka — migratory bird wintering ground.

Q36. Cold Desert / Seshachalam / Panna notification order facts:

A. Cold Desert = 16th (HP, 2009); Seshachalam = 17th (AP, 2010); Panna = 18th (MP, 2011)

B. Panna was first Indian BR

C. Cold Desert is in Ladakh only as a national BR

D. Seshachalam is in Kerala

💡 **Show answer**

Ans: A. Panna is newest notified national BR; also WNBR 2020.

Q37. "Biosphere reserve" term / biodiversity coinage — best teaching pair:

A. Ecosystem = Edward Suess; Biodiversity = A.G. Tansley

B. Biosphere concept classically linked to Edward Suess; "biodiversity" contraction linked to Rosen after Dasmann's biological diversity

C. MAB launched in 1991

D. BR = WPA Section 35 category

💡 **Show answer**

Ans: B. MAB = **1971**. BR ≠ WPA statutory category.

Q38. Which BR–state pair is wrong?

A. Nokrek — Meghalaya

B. Manas — Assam

C. Khangchendzonga — Himachal Pradesh

D. Agasthyamalai — Kerala

💡 **Show answer**

Ans: C. Khangchendzonga = **Sikkim**. Cold Desert = Himachal.

Q39. Nallamalai is:

A. An Indian Biosphere Reserve

B. Not a notified Biosphere Reserve (classic distractor)

C. Same as Nilgiri BR

D. UP's only BR

> 💡 **Show answer**
>
> **Ans: B.** Nallamalai appears as a trap option against Agasthyamalai/Nilgiri/Pachmarhi.

Q40. Purpose of Biosphere Reserves among options:

A. Preserve only grasslands

B. Conserve genetic diversity / all forms of life with sustainable use

C. Only atmospheric balance engineering

D. Only agricultural produce storage

> 💡 **Show answer**
>
> **Ans: B.** Core protects genetic variation; transition supports people.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to Biosphere Reserves, which of the following statements is/are correct?

1. UNESCO MAB (1971) landscape: core → buffer → transition.
2. Biosphere Reserve is a category created under Section 35 of the Wildlife (Protection) Act, identical to a National Park.
3. Identical to Section 35 National Park only.

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Ans: D.** Only 1 is correct. **Logic:** Statement 1 is true (UNESCO MAB zonation). Statement 2 is false — BR is not a WPA statutory category. Statement 3 is false — BR ≠ National Park only.

Q2. Match List-I with List-II:

List-I	List-II
A. First BR (India)	1. Great Rann of Kutch
B. Largest BR	2. Nilgiri 1986
C. Smallest BR	3. Nokrek

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-1, C-3. **Logic:** Nilgiri / Kutch / Nokrek.

Q3. Which of the following pairs is/are correctly matched?

1. India has only 8 notified national Biosphere Reserves.

2. About 13 on UNESCO World Network
3. UP — three Biosphere Reserves

A. Only 1 B. 1 and 2 C. 2 and 3 D. Only 2

> 💡 **Show answer**
>
> **Ans: D.** Only 2 is correct. **Logic:** Statement 2 is true (~13 on the UNESCO World Network class fact). Statement 1 is false — India has 18 notified national BRs. Statement 3 is false — UP does not have three BRs.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Nokrek is in Meghalaya.

Reason (R): Pairing Nokrek with Manipur is correct.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** Nokrek ≠ Manipur.

Q5. With reference to Manas, which of the following statements is/are correct?

1. Remains a national Biosphere Reserve.
2. Removed from UNESCO World Network in 2011.
3. National listing alone guarantees WNBR membership.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1 and 2. **Logic:** WNBR is a UNESCO subset.

Q6. Arrange zone permissions:

A. Core = livelihoods first B. Core no exploitation; buffer limited research/education; transition livelihoods C. Transition = no human use ever D. Buffer = identical to core ban

> 💡 **Show answer**
>
> **Ans: B.** Core / buffer / transition uses. **Logic:** MAB zoning.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Nanda Devi BR — Uttarakhand
2. Dudhwa — UP Biosphere Reserve
3. Valmiki — Bihar NP (not UP BR)

A. 1 and 3 only B. 2 and 3 only C. 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only pair 2 is not correctly matched. **Logic:** Dudhwa = NP — not UP BR.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Panna (MP, 2011) is India's newest notified Biosphere Reserve.

Reason (R): Madhya Pradesh has three BRs: Pachmarhi, Achanakmar–Amarkantak, Panna.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; MP count does not itself prove Panna is newest. **A/R logic:** Newest vs MP trio.

Q9. With reference to multi-state BRs, which of the following statements is/are correct?

1. Nilgiri — TN–Karnataka–Kerala.
2. Agasthyamalai — Kerala–TN.
3. Achanakmar–Amarkantak — MP–Chhattisgarh.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Multi-state classics.

Q10. Match List-I with List-II:

List-I	List-II
A. Simlipal	1. Meghalaya
B. Nokrek	2. Odisha
C. Agasthyamalai	3. Kerala (also TN landscape)

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Simlipal/Nokrek/Agasthyamalai.

Q11. Arrange NP vs BR tool:

A. Identical WPA labels always B. NP = strictest WPA site; BR adds sustainable use/research under MAB C. BR replaces all NPs D. BR is only Section 18 sanctuary

💡 **Show answer**

Ans: B. WPA vs MAB tools. **Logic:** Separate instruments.

Q12. With reference to zero-BR states, which of the following statements is/are correct?

1. UP has zero BRs.
2. Bihar and Rajasthan also have zero BRs.
3. Valmiki is a Bihar Biosphere Reserve.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Valmiki remains NP.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Do not place Nanda Devi in Uttar Pradesh on a match list.

Reason (R): After bifurcation it is an Uttarakhand site.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Statehood trap.

Q14. Which one of the following correctly states Gulf of Mannar BR?

A. Sikkim only B. Cold Desert HP only C. UP Terai BR only D. Marine BR in Tamil Nadu

💡 **Show answer**

Ans: D. Marine BR TN. **Logic:** Gulf of Mannar.

Q15. How many of the following state–BR facts are correct?

Cold Desert–HP; Khangchendzonga–Sikkim; Seshachalam–AP; Great Nicobar–A&N

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** State map.

Q16. With reference to world BR icons, which of the following statements is/are correct?

1. Yellowstone, Serengeti–Ngorongoro, Galápagos appear in classic lists.
2. Lake Baikal, Doñana, Mount Kenya also appear.
3. Dudhwa is a UNESCO World Network BR in UP.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Dudhwa ≠ UP BR.

Q17. Match List-I with List-II:

List-I	List-II
A. Pachmarhi	1. Assam
B. Manas	2. Madhya Pradesh
C. Sundarbans BR	3. West Bengal

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Pachmarhi/Manas/Sundarbans.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Among common options, BRs best conserve biodiversity together with traditional human life / sustainable use.

Reason (R): Transition zone supports livelihoods under MAB design.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the strategy edge. **A/R logic:** Livelihood zoning.

Q19. Which of the following pairs is/are correctly matched?

1. Agasthyamalai paired only with Kerala — accepted as correct in recent papers
2. Landscape also spans Tamil Nadu
3. Agasthyamalai is only in Assam

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not Assam.

Q20. Arrange UNESCO review idea:

A. Periodic UNESCO review can delist from World Network — Manas 2011 example B. National BR can never be reviewed C. Delisting deletes the national BR always D. Only Yellowstone can be delisted

💡 **Show answer**

Ans: A. Manas 2011 delist example. **Logic:** WNBR ≠ national forever.

Q21. With reference to overlaps, which of the following statements is/are correct?

1. BRs can overlap NPs and World Heritage tags.
2. MAB label remains a separate tool.
3. Overlap means BR is a WPA Section 35 synonym.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Separate tool.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): World Network membership is a UNESCO subset of national BRs.

Reason (R): National listing alone is not enough for WNBR.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R restates the subset rule. **A/R logic:** National ≠ WNBR auto.

Q23. How many of the following match facts are correct?

Simlipal–Odisha; Pachmarhi–MP; Nanda Devi–UK; Nokrek–Meghalaya

A. Only one B. Only two C. All four D. Only three

💡 **Show answer**

Ans: C. All four. **Logic:** Match set.

Q24. Which one of the following correctly states India's first BR year?

A. Dudhwa 1977 as BR first B. Panna 2011 only as first C. Manas 2011 as first D. Nilgiri 1986

💡 **Show answer**

Ans: D. Nilgiri 1986. **Logic:** First BR.

Q25. With reference to Cold Desert / Khangchendzonga, which of the following statements is/are correct?

1. Cold Desert BR — Himachal Pradesh.
2. Khangchendzonga BR — Sikkim.
3. Both are UP Biosphere Reserves.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Not UP.

Q26. Match List-I with List-II:

List-I	List-II
A. Core	1. Livelihoods
B. Buffer	2. No exploitation
C. Transition	3. Limited research/education

Code:

A. A-1, B-2, C-3 B. A-2, B-3, C-1 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-3, C-1. **Logic:** Zone uses.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Great Rann of Kutch is India's largest BR.

Reason (R): Nokrek is the smallest.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; smallest fact does not explain why Kutch is largest. **A/R logic:** Largest vs smallest.

Q28. Arrange Seshachalam / Great Nicobar:

A. Both UP only B. Both Meghalaya only C. Seshachalam AP; Great Nicobar A&N D. Both Kerala only

💡 **Show answer**

Ans: C. AP / A&N. **Logic:** State placement.

Q29. With reference to MP BRs, which of the following statements is/are correct?

1. Pachmarhi, Achanakmar–Amarkantak, Panna.
2. Achanakmar–Amarkantak shared with Chhattisgarh.
3. MP has zero Biosphere Reserves.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** MP has three.

Q30. Which one of the following correctly states why BR ≠ NP alone?

A. BR adds sustainable use and research under MAB beyond strict protection B. BR deletes all cores C. BR is only a city municipal park D. BR is only a carbon credit

💡 **Show answer**

Ans: A. MAB sustainable-use design. **Logic:** Strategy difference.

Common Traps

Trap	Correct
Nokrek = Manipur	**Meghalaya**
Simlipal = Madhya Pradesh	**Odisha**
UP has a Biosphere Reserve	**Zero**
Nanda Devi = UP	**Uttarakhand**
Manas no longer a BR	Still **national** BR; UNESCO removed 2011
BR = WPA category	**Executive / MAB**, not WPA Sec 35
Core allows farming	**No** exploitative use in core
Dudhwa / Valmiki = UP BR	NP / Bihar NP — not UP BR
Agasthyamalai = Assam	**Kerala (+ TN)**
First BR = Nanda Devi	**Nilgiri 1986**
BR only for a few species	Conserves **all forms of life** + sustainable use
Largest BR = Nokrek	**Great Rann of Kutch** (national); Gulf of Mannar often largest UNESCO-certified
Khangchendzonga = HP	**Sikkim**; Cold Desert = HP
Bundala = India	**Sri Lanka**
Nallamalai = Indian BR	**Not** a notified BR
Transition = no people	Transition = **settlements / livelihoods**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 34 — Climate Change (जलवायु परिवर्तन) (Advanced)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Climate Change | Atmospheric Studies | Global Climate Monitoring | Greenhouse Gas Emissions | Sources of Greenhouse Gases | Greenhouse Effect | Global Warming (भूमंडलीय तापन) | Carbon Cycle | Carbon Budget | Carbon Sink (कार्बन सिंक) | Carbon Credit (कार्बन क्रेडिट) | Carbon Trading | Climate Change Mitigation | Climate Change Adaptation | Climate Change Policy of India | IPCC (आईपीसीसी) Assessment Reports

Sources baked in: IPCC AR6; WMO; UNEP; Global Carbon Project; MoEFCC; NAPCC; India NDC; CCTS 2023; UPPCS PYQs **Weight:** ★★★ High — Kyoto vs Montreal; Emissions Gap=UNEP; carbon credit; IPCC≠UNFCCC **Last verified:** August 2026

Current Affairs (this topic)

- Advanced climate CA mirrors Climate Change — NDC / net zero / extreme-event science updates.
- Keep adaptation vs mitigation and Montreal≠Paris traps identical to the core climate chapter.

💡 🎯 **Consolidated — 32 Must-Score Facts (Click to Expand)**

1. **Climate change** is a long-term shift in temperature, rainfall, and extremes. **Global warming** is the temperature-rise part only. 2. IPCC AR6 places warming near **1.1°C** above **1850–1900**, with human influence dominant. Past-century rise ≈ **1°C / 1.8°F**. 3. The **Keeling Curve** records atmospheric **CO_2** at **Mauna Loa** since **1958**. 4. The **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) is an annual **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) product. It is **not** published by the IPCC. 5. Main greenhouse gases are **CO_2, CH_4, N_2O, and F-gases**. **Argon is not a greenhouse gas**. **Joseph Fourier (1820s)** framed the greenhouse-effect idea. 6. WMO-style relative (सापेक्ष) shares often tested: **CO_2 ~64% > CH_4 ~19% > CFCs+others ~11% > N_2O ~6%**. Water vapour dominates the **natural** effect. 7. India ranks near the top in **absolute** (निरपेक्ष) emissions (**China #1**, India ~**3rd**) but stays **low per capita**. **Bhutan** is often carbon-negative. 8. A **carbon sink** absorbs net CO_2. A **carbon budget** is the cumulative CO_2 allowed for a temperature target. **Blue carbon** (नीला कार्बन) = ocean/coastal sinks. 9. A **carbon credit** equals **1 tonne of CO_2 equivalent**. Carbon trading is the market for credits or allowances (Kyoto (क्योटो) origin). 10. **Green Credit (2023)** (ग्रीन क्रेडिट) and **PAT ESCert** are **not** the same thing as a Kyoto-style carbon credit. 11. **Mitigation** cuts emissions and grows sinks. **Adaptation** adjusts society and systems to climate impacts. 12. **Kyoto** (क्योटो) is climate mitigation (जलवायु शमन). **Montreal** is ozone protection. Do not swap the two treaties. 13. India's **NAPCC** launched in **2008** with **eight** missions (**not** nuclear power). 14. India's updated NDC path includes about **45%** emission-intensity cut and about **50%** non-fossil electricity capacity share. 15. India's **net-zero** year fact is **2070**. Net zero = balance emissions with removals, not absolute zero every emission. 16. **LiFE** was floated as an idea at **COP-26 (2021)** and launched in **June 2022**. COP-25 Madrid is the false date trap. 17. India's **Carbon Credit Trading Scheme (2023)** sits with **BEE** under the Power Ministry. 18. The **IPCC** does science assessment under **WMO + UNEP** parents. It does **not** set national targets or write the Emissions Gap Report. 19. The **greenhouse effect** is infrared trapping by greenhouse gases. Without it Earth would be about **−18°C**. 20. The **carbon cycle** moves carbon among atmosphere, oceans, land, and fossil pools. Sequestration is the storing process. 21. **Milankovitch** = eccentricity, obliquity, precession — **not** solar

irradiance. **Ice cores** = cryogenic climate archive. 22. **GCF** = Cancun **2010**. **CDM/CERs** = Kyoto. **GHG Protocol** = WRI + WBCSD. **Earth Hour** (अर्थ आवर) = WWF (डब्ल्यूडब्ल्यूएफ). 23. Carbon trading grew from Kyoto-era market tools; Paris-era systems use national NDCs and domestic markets. 24. Do not treat IPCC Assessment Reports as UNFCCC (यूएनएफसीसीसी) treaty text. Science reports and the climate convention are different homes. 25. Adaptation examples include early warning, climate-resilient crops, coastal defence, and heat-action plans — not only emission cuts. 26. **Methane hydrates**, paddy+fertiliser (CH_4+N_2O), and wetland CH_4 are high-yield source traps. 27. **Carbon fertilization** ≠ ocean acidification (महासागरीय अम्लीकरण). 28. **Forest Carbon Partnership Facility** and **BioCarbon Fund ISFL** are **World Bank** programmes — FCPF does **not** fund universities/individuals for research grants. 29. **Climate Action Tracker** = Climate Analytics + New Climate Institute (not IPCC/UNEP wing). 30. **Deep Carbon Observatory** maps deep-Earth carbon (physics, reservoirs, deep energy, deep life). 31. **Common Carbon Metric** (UNEP-linked) measures building-operation carbon footprints. 32. Primary recent warming drivers are fossil fuels, autos, deforestation — **not** solar flares.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Global Warming** | **Climate Change** | Long-term rise in Earth's average surface temperature vs broader spectrum of changes including shifts in precipitation, weather extremes, and sea levels | भूमंडलीय तापन / जलवायु परिवर्तन | | **Climate Mitigation** | **Climate Adaptation** | Actions to reduce or prevent GHG emissions (renewable energy, carbon taxes) vs actions to adjust and build resilience to actual or expected climate impacts (seawalls, drought crops) | शमन (उत्सर्जन कटौती) / अनुकूलन (अनुकूलनशीलता) | | **Carbon Sink** | **Carbon Sequestration** | Natural or artificial reservoir absorbing more carbon than it releases (oceans, forests) vs the process/technique of capturing and storing carbon dioxide | कार्बन सिंक (अवशोषक) / कार्बन पृथक्करण (भंडारण प्रक्रिया) | | **Carbon Credit** | **Carbon Tax** | Tradable permit allowing emission of 1 tonne of CO_2 equivalent under cap-and-trade vs direct price levied by government per tonne of GHG emitted | कार्बन क्रेडिट (1 टन CO_2e) / कार्बन कर | | **IPCC** | **UNFCCC** | Scientific assessment body established in 1988 by WMO/UNEP (doesn't conduct research) vs multilateral legal treaty framework adopted at Rio 1992 | आईपीसीसी (वैज्ञानिक निकाय) / यूएनएफसीसीसी (कानूनी संधि) | | **Emissions Gap Report** | **Adaptation Gap Report** | Annual UNEP report comparing actual NDC pledges vs Paris 1.5°C targets vs annual UNEP assessment of adaptation finance and planning shortfalls | उत्सर्जन अंतराल रिपोर्ट / अनुकूलन अंतराल रिपोर्ट | | **Kyoto Protocol (1997)** | **Paris Agreement (2015)** | Top-down binding targets on developed countries (Annex-I) vs bottom-up voluntary Nationally Determined Contributions (NDCs) applicable to all | क्योटो प्रोटोकॉल (1997) / पेरिस समझौता (2015) |

💡 📋 **Must-score facts — AR6, sinks, NAPCC, NDC (Click to Expand)**

Advanced tags | Item | Lock | |---|---| | AR6 warming | ~**1.1°C** above 1850–1900 | | Keeling Curve | Mauna Loa CO_2 since **1958** | | Emissions Gap Report | **UNEP** (≠ IPCC) | | Mitigation / Adaptation | Cut emissions+sinks / adjust to impacts | | Montreal ≠ Kyoto | Ozone / climate | | NAPCC | **2008**; **eight** missions (no nuclear) | | NDC class facts | ~45% intensity cut; ~50% non-fossil capacity | | 1 carbon credit | 1 tCO_2e | | Green Credit / PAT | **2023** / ESCerts — ≠ Kyoto credit |

34.1 Climate Change

Identity: Long-term climate system change | Anthropogenic + natural | Broader than warming

- **Climate change** is a long-term change in the climate system — temperature, precipitation, extremes, sea level — lasting decades or longer.
- UNFCCC stresses human-driven change that alters atmospheric composition, beyond natural variability.
- IPCC AR6 finds human influence is the **dominant cause** of observed warming.

- Global warming is one part of climate change; monsoon shifts and glacier retreat also count.

> **Logic:** Warming ⊂ climate change — do not treat them as identical.

34.2 Atmospheric Studies

Identity: Radiative forcing (विकिरण दबाव) | Keeling Curve | Warming troposphere (क्षोभमंडल) / cooling stratosphere (समतापमंडल)

- **Radiative forcing** (W/m^2) measures the net energy imbalance driving warming.
- The **Keeling Curve** records continuous atmospheric **CO_2** at **Mauna Loa, Hawaii**, since **1958** — rising trend is the classic graph.
- GHG warming typically **warms the troposphere** and **cools the stratosphere** — a fingerprint distinct from solar-only warming.
- Water vapour is a strong greenhouse gas but mainly a **feedback**, not the primary anthropogenic driver.

> **Logic:** Mauna Loa = Keeling CO_2 monitoring, not ozone hole (ओजोन छिद्र) location (ozone hole = Antarctica).

34.3 Global Climate Monitoring

Identity: WMO / NOAA / NASA | UNEP Emissions Gap | Global Carbon Project

- **WMO** publishes the **State of the Global Climate** report.
- **NOAA** and **NASA GISS** maintain key global temperature datasets.
- **Global Carbon Project** issues the annual **Global Carbon Budget**.
- **Emissions Gap Report** is published by **UNEP** — it measures the gap between NDCs and 1.5–2°C pathways.

PYQ — UPPCS Prelims 2024, Q6

Match: A. Human Development Report B. World Economic Outlook C. World Investment Report D. Emissions Gap Report with 1. IMF 2. UNEP 3. UNDP (यूएनडीपी) 4. UNCTAD

Options: A. 3 1 4 2 B. 3 4 1 2 C. 1 2 3 4 D. 2 3 4 1

💡 **Show answer**

Ans: A — HDR=UNDP; WEO=IMF; WIR=UNCTAD; Emissions Gap=**UNEP**.

> **Logic:** Emissions Gap Report ≠ IPCC Assessment Report.

34.4 Greenhouse Gas Emissions

Identity: CO_2-led basket | Absolute vs per capita | India rank traps

- Major anthropogenic GHGs: **CO_2, CH_4, N_2O**, and fluorinated gases (HFCs, PFCs, SF_6).
- Emissions are often reported as **CO_2-equivalent (CO_2e)** using **GWP**.
- India is among the **top absolute emitters** (~**3rd** globally) but has **low per capita** emissions.
- ** **PYQ — UPPCS Prelims 2023, Q109**

What was India's rank in the world in greenhouse gas emissions in 2022?

Options: A. 9th B. 10th C. 11th D. 12th

💡 **Show answer**

Ans: B — Paper answer **10th**. Absolute total emitter rank is commonly ~**3rd** — do not mix the two frames.

PYQ — UPPCS Prelims 2018, Q132

Largest CO_2 emitter in 2015:

Options: A. China B. USA C. India D. France

💡 **Show answer**

Ans: A

> **Logic:** Absolute emitter rank ≠ CCPI / "performance" rank.

34.5 Sources of Greenhouse Gases

Identity: Energy / industry / AFOLU / transport / waste | Methane hydrates | Rice CH_4+N_2O

- **Energy and electricity** (coal, oil, gas) dominate CO_2.
- **Industry** adds process CO_2 (cement, steel) plus F-gases.
- **AFOLU** (agriculture, forestry, land use): **CH_4** from livestock/rice; **N_2O** from fertilisers; CO_2 from deforestation.
- **Rice** is a major anthropogenic source of **both methane and nitrous oxide**.
- Natural **wetlands** dominate natural methane; **methane hydrates** under Arctic tundra (टुंड्रा) and seafloor can release CH_4 if warmed.
- **Transport** and **buildings** add fossil CO_2; **waste** adds landfill methane.
- Crop/biomass residue burning can release **CO, SO_2, CH_4, and ozone-related pollutants**.
- **Argon is not a greenhouse gas.**

PYQ — UPPCS Prelims 2022, Q102

Which is NOT a greenhouse gas?

Options: A. Argon B. CFC C. Methane D. Carbon dioxide

💡 **Show answer**

Ans: A

> **Logic:** Detailed gas-by-gas chemistry also appears in Topic 37. Solar flares are **not** a primary recent climate driver.

34.6 Greenhouse Effect

Identity: Fourier | Natural vs enhanced | IR trapping | −18°C without it

- **Joseph Fourier (1820s)** postulated the greenhouse-effect concept.
- The **natural greenhouse effect** (प्राकृतिक ग्रीनहाउस प्रभाव) keeps Earth habitable (~**+15°C** average; without it ≈ **−18°C**).
- Shortwave solar energy reaches Earth; the surface re-emits **longwave IR**, which GHGs absorb and re-radiate.
- The **enhanced greenhouse effect** (मानवजनित संवर्धित प्रभाव) from extra anthropogenic GHGs traps more outgoing infrared radiation.

- Main natural greenhouse gases include water vapour, CO_2, CH_4, and N_2O; CFCs/HFCs are human-made.
- **Astronomical (Milankovitch) climate theory** uses eccentricity, obliquity, and precession — **solar irradiance is not** listed in that set. **Ice cores** are the classic cryogenic climate indicator.

> **Logic:** Greenhouse effect itself is not the problem — **enhanced** anthropogenic forcing is.

34.7 Global Warming

Identity: Surface temperature rise | AR6 ~1.1°C | Impacts subset

- **Global warming** means long-term rise in global mean surface temperature.
- IPCC AR6: about **1.1°C** above the **1850–1900** baseline.
- Impacts include heatwaves, glacier melt, sea-level rise, coral bleaching (प्रवाल विरंजन), and monsoon extremes (UP heat / Bundelkhand drought (सूखा) / Ganga (गंगा) floods).
- Warming is necessary but not sufficient to describe all climate-change impacts.

> **Logic:** Paris aims to hold warming well below **2°C** and pursue **1.5°C**.

34.8 Carbon Cycle

Identity: Fast vs slow cycle | Atmosphere–ocean–biosphere–lithosphere | CO_2 sources trap

- The **carbon cycle** moves carbon among atmosphere, biosphere, oceans, soils, and rocks.
- **Fast cycle:** atmosphere ↔ plants ↔ soils ↔ surface ocean (years to decades).
- **Slow cycle:** rock weathering, sediment burial, fossil carbon (millions of years).
- Burning fossil fuels transfers slow-cycle carbon into the fast atmospheric pool.
- Processes that **add CO_2** to the atmosphere include **respiration**, **decay of organic matter**, **volcanic action**, and **combustion of fossil fuels**.
- **Photosynthesis consumes** CO_2 and releases oxygen. It does **not** add carbon dioxide to the carbon cycle on Earth.
- Elemental carbon occurs as crystalline allotropes such as **graphite**, **diamond**, and **fullerene**.
- In compound form, carbon appears as CO_2, carbonates, and hydrogen carbonates.

> **Logic:** Photosynthesis removes CO_2; respiration/decay/volcanoes add it.

34.9 Carbon Budget

Identity: Cumulative CO_2 cap | 1.5°C / 2°C pathways | Net zero link

- A **carbon budget** is the cumulative CO_2 that can still be emitted while limiting warming to a chosen target.
- Smaller remaining budgets for **1.5°C** than for **2°C**.
- Net zero means residual emissions are balanced by removals — not "zero gross emissions overnight."
- India links its long-term budget pathway to **net zero by 2070**.

> **Logic:** Carbon budget ≠ annual government fiscal budget.

34.10 Carbon Sink

Identity: Forests / oceans / soils | Net absorption | Blue carbon

- A **carbon sink** absorbs more CO_2 than it releases over a period.
- Major sinks: **forests**, **oceans**, **soils**, and **mangroves/blue carbon**.
- India's NDC includes a **2.5–3 $GtCO_2e$** additional forest/tree-cover sink by 2030.
- Sinks can weaken (deforestation, ocean warming) and become sources.

> **Logic:** Sink = reservoir/net absorber; sequestration = the storage process.

34.11 Carbon Credit

Identity: 1 tCO_2e | Kyoto CDM/CERs | ≠ Green Credit 2023 | Market price

- A **carbon credit** is a tradeable unit equal to **one tonne of CO_2 equivalent (1 tCO_2e)** reduced, avoided, or removed.
- The concept originated under the **Kyoto Protocol** flexible mechanisms (especially **CDM** → **CERs**). Projects are implemented in **developing (Non-Annex I)** countries for Annex-I buyers — CDM is **not** Annex-I-only projects.
- Credits are **traded at market prices**, not a UNEP-fixed price.
- India's **Green Credit Programme (2023)** rewards broader eco-actions — it is **not identical** to a 1 tCO_2e carbon credit.
- Related tools: **GHG Protocol** (WRI + WBCSD) for accounting; **Common Carbon Metric** for building-operation footprints; **blue carbon** for coastal sinks.

PYQ — UPPCS Prelims 2021, Q125

The concept of 'carbon credit' originated from

Options: A. Earth Summit Rio B. Kyoto Protocol C. Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) D. G-8 Summit

💡 Show answer

Ans: B

> **Logic:** Carbon credit ≠ Montreal (ozone) and ≠ Green Credit Programme.

34.12 Carbon Trading

Identity: Market for credits/allowances | Kyoto CDM | India CCTS 2023

- **Carbon trading** is buying and selling emission allowances or carbon credits.
- **Cap-and-trade** sets an emissions cap and lets the market discover the price.
- Kyoto enabled international trading via **CDM, JI, and Emissions Trading**.
- India's **Carbon Credit Trading Scheme (CCTS, 2023)** is under the **Energy Conservation Act** framework, with **BEE** as a key authority (Ministry of Power).
- **PAT** trades **ESCerts** for energy efficiency — related market tool, **not** the same as carbon credits.

> **Logic:** Credit = unit; trading = market. Paris **Article 6** governs cooperative carbon markets.

34.13 Climate Change Mitigation

Identity: Cut emissions + enhance sinks | Kyoto / Paris | Renewables

- **Mitigation** reduces GHG emissions and/or enhances sinks to limit future warming.
- Tools: renewables, efficiency, electrification, afforestation, CCS (सुरक्षा कैबिनेट समिति), carbon markets, demand-side LiFE actions.
- Key treaties: **UNFCCC 1992**, **Kyoto Protocol 1997**, **Paris Agreement (पेरिस समझौता) 2015**.
- **Montreal Protocol is ozone**, not climate mitigation.

PYQ — UPPCS Prelims 2024, Q35

Which international agreement is related to climate change mitigation?

Options: A. Montreal Protocol B. Ramsar (रामसर) Convention C. Kyoto Protocol D. CITES (साइट्स)

💡 **Show answer**

Ans: C

> **Logic:** Kyoto = climate. Montreal = ODS/ozone.

34.14 Climate Change Adaptation

Identity: Adjust to impacts | NAPCC missions | Disaster/heat plans

- **Adaptation** means adjusting natural or human systems to actual or expected climate impacts.
- Examples: heat action plans, drought-resistant crops, flood forecasting, coastal protection, water efficiency.
- India's adaptation spine includes NAPCC missions on water, Himalaya (हिमालय), agriculture, and sustainable habitat (वास स्थान), plus state action plans (SAPCC).
- Adaptation does not replace mitigation — both are required.

> **Logic:** Adapt = adjust; mitigate = reduce.

34.15 Climate Change Policy of India

Identity: NAPCC 2008 | NDC 2022 | Net zero 2070 | LiFE | CCTS

- **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is the nodal ministry for climate policy and UNFCCC engagement.
- **NAPCC (2008)** has **eight national missions** (solar, efficiency, water, Himalaya, Green India, agriculture, habitat, strategic knowledge). **Nuclear power is not** a listed mission.
- **Updated NDC (2022): 45%** emissions-intensity cut vs 2005 by 2030; **50%** non-fossil electricity capacity; **2.5–3 $GtCO_2e$** forest sink.
- **Net zero by 2070** announced at **COP-26 Glasgow (2021)**, with **500 GW** non-fossil capacity by 2030 as a related pledge. Net zero means emissions are **compensated by absorption/removal**, not absolute zero every emission.
- **LiFE Mission** launched **June 2022**; idea floated at **COP-26**, not COP-25 Madrid.
- Finance/institutions often tested beside policy: **GCF (Cancun 2010)**, Kyoto **CDM**, **Earth Hour (WWF)**, **GHG Protocol (WRI+WBCSD)**.
- Diplomacy vehicles: **International Solar Alliance**, **CDRI**.

PYQ — UPPCS Prelims 2025, Q139

LiFE: (1) launched June 2022 (2) idea at COP-25 Madrid

Options: A. Only 2 B. Neither C. Both D. Only 1

💡 **Show answer**

Ans: D — Idea = **COP-26 Glasgow**.

Year	Fact
2008	NAPCC
2015	Paris + first NDC + ISA
2021	Net zero 2070 (COP-26)
2022	Updated NDC; LiFE launch
2023	CCTS / Green Credit

> **Logic:** India net zero = **2070**, not 2050.

34.16 IPCC Assessment Reports

Identity: Science assessment | AR1→AR6 | Not a treaty body

- **IPCC** (WMO + UNEP parents) assesses climate science; it does **not** negotiate treaties or set emission targets.
- Three Working Groups: **I** physical science, **II** impacts/adaptation, **III** mitigation; plus a **Synthesis Report**.
- Cycle: **AR1 (1990)** → **AR2 (1995)** → **AR3 (2001)** → **AR4 (2007)** → **AR5 (2014)** → **AR6 (2021–23)**.
- **AR6 Synthesis Report (March 2023)** consolidates ~**1.1°C** warming and urgency of action.
- Special reports: **SR1.5 (2018)**, Ocean/Cryosphere (2019), Climate and Land (2019).
- **AR4** shared the **2007 Nobel Peace Prize** with Al Gore.

Report	Year	Fact
AR1	1990	Helped lead to UNFCCC
AR4	2007	Nobel year
AR5	2014	Informed Paris
AR6 SYR	**2023**	Latest synthesis

> **Logic:** IPCC ≠ UNFCCC ≠ UNEP Emissions Gap Report.

Complete PYQ Bank (Topic 34)

Q1. UPPCS Prelims 2024, Q6

Match List-I with List-II:

List-I	List-II
A. Human Development Report	1. IMF
B. World Economic Outlook	2. UNEP
C. World Investment Report	3. UNDP

List-I	List-II
D. Emissions Gap Report	4. UNCTAD

Row order is not the answer code.

A. 3 1 4 2

B. 3 4 1 2

C. 1 2 3 4

D. 2 3 4 1

💡 **Show answer**

Logic: Report–agency match — Emissions Gap = UNEP; HDR = UNDP; WEO = IMF; WIR = UNCTAD. **Ans: A.** HDR=UNDP (3); WEO=IMF (1); WIR=UNCTAD (4); Emissions Gap=**UNEP (2)**. **Trap:** Emissions Gap credited to IPCC or UNDP.

Q2. UPPCS Prelims 2024, Q35

Which international agreement is related to climate change mitigation?

A. Montreal Protocol

B. Ramsar Convention

C. Kyoto Protocol

D. CITES

💡 **Show answer**

Logic: Kyoto = climate mitigation; Montreal = ozone/ODS — never swap treaty families. **Ans: C.** **Kyoto Protocol (1997)** sets GHG mitigation for developed parties. **Trap:** Montreal (ozone) or Ramsar (wetlands) as climate treaties.

Q3. UPPCS Prelims 2023, Q109

What was India's rank in the world in greenhouse gas emissions in 2022?

A. 9th

B. 10th

C. 11th

D. 12th

💡 **Show answer**

Logic: 2023 Q109 keyed 10th — distinct from absolute total-emissions rank (~3rd). **Ans: B.** Paper answer **10th** (performance-index framing). India is ~**3rd** in **absolute** total emissions — do not mix frames. **Trap:** Picking 3rd when options stop at 12th without reading the stem metric.

Q4. UPPCS Prelims 2022, Q102

Which is **NOT** a greenhouse gas?

A. Argon

B. CFC

C. Methane

D. Carbon dioxide

Show answer

Logic: Noble gases — argon is not a greenhouse gas; CO2, CH4, CFCs are. **Ans: A.** **Argon** is inert air (~0.93%) and not a GHG. **Trap:** Confusing N2 or Ar with N2O (which is a GHG).

Q5. UPPCS Prelims 2021, Q125

The concept of 'carbon credit' originated from

A. Earth Summit Rio

B. Kyoto Protocol

C. Montreal Protocol

D. G-8 Summit

Show answer

Logic: Carbon credit market origin — Kyoto CDM/JI/ET, not Montreal or Rio alone. **Ans: B.** Kyoto (1997) created **CDM/JI/Emissions Trading** — **1 credit ≈ 1 tCO2e**. **Trap:** Montreal (ozone) or Green Credit Programme (2023) as Kyoto twin.

Q6. UPPCS Prelims 2018, Q132

Largest CO2 emitter in 2015:

A. China

B. USA

C. India

D. France

Show answer

Logic: 2015 largest CO2 emitter — China ahead of USA and India. **Ans: A.** **China** led absolute CO2 emissions in 2015. **Trap:** India because of recent growth narrative, or USA historical cumulative confusion in a single-year stem.

Q7. UPPCS Prelims 2025, Q139

With reference to Mission LiFE:

1. It was launched in June 2022.
2. The idea was first proposed at COP-25 Madrid.

A. Only 2

B. Neither

C. Both 1 and 2

D. Only 1

Show answer

Logic: LiFE June 2022 launch true; COP-25 Madrid idea false — COP-26 Glasgow 2021. **Ans: D.** Launch **June 2022** correct; idea at **COP-26 Glasgow (2021)**, not COP-25. **Trap:** Any Madrid COP-25 LiFE distractor.

Q8. Pattern

Which statement is correct?

A. IPCC publishes scientific assessment reports; UNFCCC is the climate treaty framework

B. IPCC sets legally binding national emission targets for all parties

C. UNFCCC publishes the Emissions Gap Report annually

D. IPCC and UNEP Emissions Gap Report are the same document

💡 **Show answer**

Logic: IPCC assesses science; UNFCCC hosts treaty/NDC negotiations — different mandates. **Ans: A.** **IPCC = science (WMO+UNEP parents)**; **UNFCCC = treaty/COP/NDCs**. **Trap:** IPCC sets targets or publishes Emissions Gap (that is **UNEP**).

Q9. Pattern

Under Kyoto-era carbon markets, one carbon credit generally represents:

A. One tonne of CO2 equivalent reduced or avoided

B. One kilogram of ozone-depleting substance

C. One hectare of forest cover on a map

D. One megawatt of solar capacity only

💡 **Show answer**

Logic: One carbon credit = one tonne CO2 equivalent under Kyoto market tools. **Ans: A.** Standard unit = **1 tCO2e**. **Trap:** Confusing with PAT **ESCerts** (energy efficiency) or green credits (2023).

Q10. Pattern

India's net-zero target year and NAPCC launch year are:

A. Net zero 2070; NAPCC 2008

B. Net zero 2050; NAPCC 1997

C. Net zero 2030; NAPCC 2015 only

D. Net zero 2070; NAPCC 2023

💡 **Show answer**

Logic: India net-zero year 2070 announced at COP-26; NAPCC launched 2008. **Ans: A.** **Net zero = 2070** (COP-26, 2021); **NAPCC = 2008** with eight missions. **Trap:** 2050 net-zero or Paris 2015 mistaken as NAPCC birth year.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to climate change vs warming, which of the following statements is/are correct?

1. Climate change means only a one-day weather anomaly.
2. Global warming = temperature-rise part only.

3. Both terms are identical always.

A. Only 1 B. 1 and 2 C. 2 and 3 D. Only 2

> 💡 **Show answer**
>
> **Ans: D.** Only 2 is correct. **Logic:** Statement 2 correctly narrows global warming to the temperature-rise component. Statement 1 is false — climate change is a long-term shift in temperature, rainfall, and extremes. Statement 3 wrongly treats the terms as always identical.

Q2. Match List-I with List-II:

List-I	List-II
A. Keeling Curve	1. Annual UNEP
B. Emissions Gap Report	2. Mauna Loa CO_2 since 1958
C. IPCC	3. Science assessment; WMO+UNEP parents

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-1, C-3. **Logic:** Keeling / Gap / IPCC.

Q3. Which of the following pairs is/are correctly matched?

1. The main GHGs exclude CO_2, CH_4, N_2O and F-gases.
2. Argon — greenhouse gas
3. N_2 gas — not a greenhouse gas; N_2O is

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Ans: B.** Only 3 is correct. **Logic:** Statement 3 is true (N_2 ≠ GHG; N_2O is). Statement 1 wrongly excludes the main GHGs. Statement 2 is false — argon is not a GHG.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Emissions Gap Report is an annual UNEP product.

Reason (R): It is published by the IPCC.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** UNEP — not IPCC.

Q5. With reference to relative GHG shares (WMO-style), which of the following statements is/are correct?

1. CO_2 ~64% > CH_4 ~19% > CFCs+others ~11% > N_2O ~6%.
2. Water vapour dominates the natural greenhouse effect.
3. Solar flares are the primary recent warming driver.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Primary drivers = fossils/autos/deforestation.

Q6. Arrange India emissions teaching:

A. Highest per capita always B. Top absolute (China #1, India ~3rd) but low per capita; Bhutan often carbon-negative C. Zero absolute emissions D. Only methane hydrates count

💡 **Show answer**

Ans: B. High absolute / low per capita. **Logic:** Rank trap.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Carbon credit — 1 tCO_2e
2. Green Credit 2023 — identical to Kyoto carbon credit
3. PAT ESCert — energy-efficiency certificate (≠ carbon credit)

A. 1 and 3 only B. 2 and 3 only C. 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 2 is not correctly matched. **Logic:** Green Credit ≠ carbon credit.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Kyoto is climate mitigation; Montreal is ozone protection.

Reason (R): The two treaties should not be swapped.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R restates the separation. **A/R logic:** Kyoto ≠ Montreal.

Q9. With reference to India's climate path, which of the following statements is/are correct?

1. NAPCC 2008 — eight missions (not nuclear power).
2. Net-zero year fact 2070.
3. Updated NDC — ~45% intensity cut; ~50% non-fossil capacity share teaching.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** NAPCC / NZ / NDC.

Q10. Match List-I with List-II:

List-I	List-II
A. Mitigation	1. Adjust to impacts
B. Adaptation	2. Cut emissions / grow sinks
C. Net zero	3. Balance emissions with removals

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Mitigation/adaptation/NZ.

Q11. Arrange LiFE chronology:

A. COP25 idea; 2019 launch B. COP26 2021 idea; June 2022 launch C. Both 2008 NAPCC D. Both 1958 Keeling

💡 **Show answer**

Ans: B. 2021 → 2022. **Logic:** Not COP25.

Q12. With reference to carbon concepts, which of the following statements is/are correct?

1. Carbon sink absorbs net CO_2.
2. Carbon budget = cumulative CO_2 for a temperature target.
3. Blue carbon = ocean/coastal sinks.
4. Carbon fertilization = ocean acidification.

A. 2, 3 and 4 only B. 1, 2 and 4 only C. 1, 2 and 3 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: C. 1, 2 and 3 only. **Logic:** Fertilization ≠ acidification.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): IPCC does science assessment; it does not set national targets.

Reason (R): IPCC Assessment Reports are identical to UNFCCC treaty text.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Science reports ≠ convention text.

Q14. Which one of the following correctly states greenhouse effect without GHGs teaching?

A. Only argon controls temperature B. Earth would be +50°C always C. No temperature change possible D. Earth would be about −18°C

💡 **Show answer**

Ans: D. About −18°C. **Logic:** Natural GE importance.

Q15. How many of the following institution facts are correct?

GCF Cancun 2010; CDM/CERs Kyoto; GHG Protocol WRI+WBCSD; Earth Hour WWF

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** Institution map.

Q16. With reference to India's CCTS, which of the following statements is/are correct?

1. Carbon Credit Trading Scheme 2023.

2. Sits with BEE under Power Ministry neighbourhood.
3. Identical to Montreal Protocol ODS list.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not Montreal.

Q17. Match List-I with List-II:

List-I	List-II
A. Milankovitch	1. Cryogenic climate archive
B. Ice cores	2. Eccentricity, obliquity, precession
C. Deep Carbon Observatory	3. Deep-Earth carbon mapping

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Orbital / cores / deep carbon.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Adaptation examples include early warning, resilient crops, coastal defence, heat-action plans.

Reason (R): Adaptation means only emission cuts.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Adaptation ≠ only mitigation.

Q19. Which of the following pairs is/are correctly matched?

1. Climate Action Tracker — Climate Analytics + New Climate Institute
2. FCPF / BioCarbon Fund ISFL — World Bank programmes
3. Climate Action Tracker — IPCC/UNEP wing

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not IPCC/UNEP wing.

Q20. Arrange AR6 warming neighbourhood:

A. ~1.1°C above 1850–1900; human influence dominant B. Cooling of 5°C C. No human influence D. Only argon-driven

💡 **Show answer**

Ans: A. ~1.1°C; human-dominant. **Logic:** AR6 fact.

Q21. With reference to methane sources, which of the following statements is/are correct?

1. Methane hydrates.
2. Paddy + fertiliser (CH_4 + N_2O) neighbourhood.
3. Wetland CH_4.
4. Argon from dry air as main GHG.

A. 1, 2 and 4 only B. 1, 2 and 3 only C. 2, 3 and 4 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: B. 1, 2 and 3 only. **Logic:** Argon not GHG.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Carbon trading grew from Kyoto-era market tools.

Reason (R): Paris-era systems use national NDCs and domestic markets.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; Paris design does not explain Kyoto origin of trading. **A/R logic:** Kyoto origin vs Paris era.

Q23. How many of the following are primary recent warming drivers in teaching?

Fossil fuels, autos, deforestation, solar flares

A. Only one B. Only two C. Only three D. All four

💡 **Show answer**

Ans: C. Only three. **Logic:** Not solar flares.

Q24. Which one of the following correctly states Common Carbon Metric?

A. Only Article 51A(g) B. Only IUCN Red List C. Only Montreal ODS list D. UNEP-linked building-operation carbon footprints

💡 **Show answer**

Ans: D. Building carbon metric. **Logic:** CCM fact.

Q25. With reference to sequestration / carbon cycle, which of the following statements is/are correct?

1. Carbon cycle moves C among atmosphere, oceans, land, fossils.
2. Sequestration is the storing process.
3. Fourier (1820s) framed greenhouse-effect idea neighbourhood.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** Cycle + history.

Q26. Match List-I with List-II:

List-I	List-II
A. India net zero	1. 2008

List-I	List-II
B. NAPCC	2. 2070
C. CCTS	3. 2023

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** 2070 / 2008 / 2023.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): FCPF does not fund universities/individuals as its classic design rule.

Reason (R): FCPF and BioCarbon Fund ISFL are World Bank programmes.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; World Bank parentage does not explain the university/individual funding negation.
A/R logic: Programme identity vs funding rule.

Q28. Arrange Milankovitch contents:

A. Only solar irradiance synonym B. Only argon cycles C. Eccentricity, obliquity, precession — not solar irradiance D. Only Keeling Curve stations

💡 **Show answer**

Ans: C. Orbital parameters. **Logic:** Not irradiance.

Q29. With reference to net zero meaning, which of the following statements is/are correct?

1. Balance remaining emissions with removals.
2. Not absolute zero every emission.
3. India's year fact 2070.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Definition + India year.

Q30. Which one of the following correctly states Earth's air heating after surface warming?

A. Mainly long-wave terrestrial radiation B. Only direct shortwave alone always C. Only argon conduction D. Only solar flares

💡 **Show answer**

Ans: A. Long-wave terrestrial radiation. **Logic:** Lower-atmosphere heating — shared with Topic 35.

Common Traps

Trap	Correct
Emissions Gap = IPCC	**UNEP**
Montreal = climate mitigation	**Ozone**; Kyoto = climate
Carbon credit = Montreal	**Kyoto**
Green Credit = carbon credit	**Not identical**
IPCC sets emission targets	**Science only**
Absolute emitter rank = 10th	~**3rd** absolute; 2023 Q109 keyed **10th**
Argon is a GHG	**Not**
Fourier missing / modern invent	**Joseph Fourier (1820s)**
GCF = Durban	**Cancun 2010**
Earth Hour = UNEP	**WWF**
Net zero = absolute zero	Balance with **removals**
Nuclear = NAPCC mission	**Not** listed
Milankovitch includes solar irradiance	**No** — eccentricity/obliquity/precession
CDM projects only in Annex-I	Projects in **developing** countries
Argon is a GHG	**Not**
India net zero 2050	**2070**
LiFE idea at COP-25	**COP-26**
PAT = carbon credit trading	**ESCerts** (efficiency)

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 35 — Atmosphere

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Structure of the Atmosphere | Atmospheric Composition | Atmospheric Layers | Troposphere (क्षोभमंडल) | Stratosphere (समतापमंडल) | Mesosphere (मध्यमंडल) | Thermosphere | Exosphere

Sources baked in: NCERT Geo 11 Ch 8; MoEFCC/CPCB context; UPPCS/UPSC PYQs **Weight:** ★★★ High — ozone = stratosphere; heating by longwave; N_2 ≠ GHG **Last verified:** August 2026

Current Affairs (this topic)

- Atmosphere layer and composition facts are stable; WMO GHG / ozone bulletins are light CA.
- Troposphere weather vs stratosphere ozone remains the classic layer trap.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. From bottom to top the layers run **Troposphere → Stratosphere → Mesosphere → Thermosphere → Exosphere**. 2. The **troposphere** (क्षोभमंडल) holds weather. Temperature **falls** with height in this layer. 3. The **stratosphere** (समतापमंडल) holds the **ozone layer**. Temperature **rises** with height here. 4. The **mesosphere** is the **coldest** layer and is where most meteors burn. 5. The **thermosphere** is very hot but thin. Auroras, the ISS path, and much of the ionosphere sit here. 6. The **exosphere** is the outermost fringe and merges into space. 7. Dry air composition is about **N_2 78%**, **O_2 21%**, **argon ~0.93%**, and **CO_2 ~0.04%**. 8. **Nitrogen gas (N_2) is not a greenhouse gas.Nitrous oxide (N_2O)** is a greenhouse gas. Do not confuse the two. 9. **Argon is not a greenhouse gas.** Oxygen and helium are also not greenhouse gases. 10. Earth's air is heated mainly by **long-wave terrestrial radiation (भौमिक विकिरण)** from the warmed surface, not by direct shortwave alone. 11. The Sun sends **shortwave** insolation (सूर्यताप). The surface emits **longwave** radiation that warms the air. 12. Maximum ozone concentration is in the **stratosphere**, not the troposphere. 13. The **homosphere** (सममंडल) is well mixed to about **80 km**. Above that the **heterosphere** (विषममंडल) separates gases by mass. 14. Tropospheric ozone (क्षोभमंडलीय ओजोन) is a pollutant and greenhouse gas. Stratospheric ozone (समतापमंडलीय ओजोन) is the UV shield. 15. Temperature inversion and weather systems live in the troposphere. Commercial jets often cruise near the tropopause. 16. Do not place the ozone layer in the troposphere on a match list. 17. Mesosphere cold and thermosphere hot can both be true because density differs — thin hot air is not the same as surface heat. 18. Exosphere atoms can escape to space. It is not a weather layer. 19. Composition percentages are for **dry air**. Water vapour varies strongly by place and season. 20. Atmospheric studies for climate start from this layer ladder plus greenhouse-gas composition facts. 21. Ionosphere radio reflection is mainly a thermosphere–upper-atmosphere feature, not a troposphere fact. 22. Stratospheric heating with height comes largely from ozone absorbing ultraviolet radiation. 23. Never reverse the order of troposphere and stratosphere in a bottom-to-top ladder. 24. Long-wave terrestrial radiation is the standard answer for how the lower atmosphere is heated after surface warming.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Troposphere** | **Stratosphere** | Lowest weather layer, temperature decreases with altitude (normal lapse rate 6.5°C/km) vs calm dry layer where temperature increases with height due to UV absorption by ozone | क्षोभमंडल (मौसम परत) / समतापमंडल (ओजोन परत) | | **Stratosphere** | **Mesosphere** | Contains protective ozone layer; stable air suited for jet aircraft vs coldest atmospheric layer (-90°C) where incoming meteors burn upon entry | समतापमंडल / मध्यमंडल (उल्कापिंड दहन) | | **Mesosphere** | **Thermosphere (Ionosphere)** | Temperature plummets to lowest levels vs temperature surges past 1500°C; contains ionized gas reflecting terrestrial radio waves and hosting auroras | मध्यमंडल / तापमंडल (आयनमंडल) | | **Homosphere** | **Heterosphere** | Lower ~85 km where atmospheric gases remain uniformly mixed in proportion vs upper layer where gases stratify by molecular weight (N_2, O, He, H) | सममंडल (समान मिश्रण) / विषममंडल (गुरुत्वाकर्षण परतें) | | **Insolation** | **Terrestrial Radiation** | Shortwave electromagnetic solar radiation received by Earth vs outgoing longwave infrared radiation re-radiated by Earth's heated surface | सूर्यताप (लघुतरंग) / भौमिक विकिरण (दीर्घतरंग) | | **Good Ozone** | **Bad Ozone** | Naturally occurring stratospheric ozone shielding Earth from solar UV-B vs ground-level tropospheric ozone acting as secondary pollutant and respiratory irritant | अच्छा ओजोन (समतापमंडल) / बुरा ओजोन (क्षोभमंडल) |

💡 📋 **Must-score facts — layers, composition, radiation (Click to Expand)**

Structure | Item | Lock | |---|---| | Order (bottom→top) | Tropo → Strato → Meso → Thermo → Exo | | Weather / ozone layer | Troposphere / Stratosphere | | Coldest / meteors | Mesosphere | | Aurora / ISS | Thermosphere | | Dry air | N_2 ~78% · O_2 ~21% · Ar ~0.93% · CO_2 ~0.04% | | N_2 vs N_2O | Not GHG / is GHG | | Not GHGs | Ar, N_2, O_2, He | | Air heating | Mainly longwave from warmed surface | | Homosphere | Well mixed to ~**80 km** |

35.1 Structure of the Atmosphere

Identity: Gaseous envelope | Temp-based layers | Heated from below

- The **atmosphere** is the gaseous envelope held by Earth's gravity.
- About **99% of atmospheric mass** lies within roughly **32 km** of the surface.
- Structure is mainly classified by the **vertical temperature profile** into five layers (see §35.3–35.8).
- By composition: **homosphere** (well mixed to ~**80–90 km**) and **heterosphere** (gases separate by mass above that).
- Homosphere thermal layers are **troposphere + stratosphere + mesosphere**. The **ionosphere** is **not** included in the homosphere set in UKPCS-style stems.
- Sea-level pressure ≈ **1013 mb (1 atm)**; pressure and density fall rapidly with height.
- The Sun's **shortwave** energy mainly heats the **surface**; the surface emits **longwave infrared** that heats the lower atmosphere from below.

PYQ — UPPCS Prelims 2022, Q138

The Earth's atmosphere is mainly heated by which one of the following?

Options: A. Long-wave terrestrial radiation B. Scattered solar radiation C. Reflected solar radiation D. Short-wave solar radiation

💡 **Show answer**

Ans: A

> **Logic:** "Atmosphere heated mainly by direct shortwave" is **false**.

35.2 Atmospheric Composition

Identity: N_2 78% | O_2 21% | Ar ~1% | CO_2 ~0.04% | Variable gases | Mixture | Radon trap

- Air is a **mixture** of gases, not a single compound or element.
- **Dry air near the surface: Nitrogen ~78.08%, Oxygen ~20.95%, Argon ~0.93%, CO_2 ~0.04%.**
- **Permanent gases** (N_2, O_2, Ar) stay roughly constant in the homosphere.
- **Variable gases** include water vapour (0–4%), CO_2, CH_4, O_3, and aerosols.
- Noble gases present in air include **helium**, **neon**, and **argon**. **Radon** is **not** treated as a normal constituent of atmospheric air in this composition set.
- Plants release oxygen through photosynthesis. If plants and trees vanished, atmospheric **oxygen** would decline.
- The **Amazon rainforest** is estimated to produce about **20 percent** of the oxygen generated by terrestrial photosynthesis, which is why it is often called the "lungs of the Earth."
- **N_2 is not a greenhouse gas**; **N_2O** is. Do not confuse N_2 with nitrogen oxides or nitrous oxide.
- **Argon is not a GHG** (also covered in Topic 34 greenhouse-gas questions).
- **Water vapour** is the strongest natural tropospheric greenhouse gas by volume, but mainly a climate **feedback**.
- Dust/aerosols scatter light (blue sky / red sunset).

PYQ — UPPCS Prelims 2020, Q139

Which is NOT a greenhouse gas found naturally in the atmosphere?

Options: A. Nitrogen oxide B. Carbon dioxide C. Methane D. Ozone

💡 **Show answer**

Ans: A — CO_2, CH_4, and O_3 are natural GHGs here. Nitrogen oxides are not treated as that major natural GHG set.

PYQ — UPPCS Prelims 2021, Q134

Diffusion of light in the atmosphere takes place due to

Options: A. Carbon dioxide B. Dust particles C. Helium D. Water vapours

💡 **Show answer**

Ans: B

Gas	Approx. %	Key fact
N_2	78%	Not GHG
O_2	21%	Respiration
Ar	0.93%	Inert; not GHG
CO_2	0.04%	GHG
H_2O	0–4%	Variable

Logic: Composition % ≠ greenhouse importance (CO_2 is tiny but powerful).

35.3 Atmospheric Layers

Identity: Five temperature layers | Pauses reverse the trend

- Five layers from surface outward: **Troposphere** → **Stratosphere** → **Mesosphere** → **Thermosphere** → **Exosphere**.
- Boundaries: **tropopause, stratopause, mesopause** — where the temperature trend reverses.
- Temperature does **not** fall continuously with height through all layers.

Layer	Approx. height	Temp trend	Signature
Troposphere	0–8/18 km	↓ upward	Weather
Stratosphere	~12–50 km	↑ upward	Ozone
Mesosphere	~50–85 km	↓ upward	Meteors; coldest
Thermosphere	~85–600 km	↑ upward	Aurora / ISS
Exosphere	~600 km+	Transition	Space merge

> **Logic:** Learn **feature** ↔ **layer** matching, not only names.

35.4 Troposphere

Identity: Lowest layer | All weather | Lapse rate ~6.5°C/km

- The **troposphere** is the lowest layer and holds almost all weather, clouds, and storms.
- Thickness: about **8 km at poles** to **16–18 km at the equator**.
- Temperature generally **decreases with height** (environmental lapse rate ≈ **6.5°C/km**).
- Contains most of the atmosphere's mass and nearly all water vapour.
- Ends at the **tropopause**, where the fall in temperature stops.
- Thickness is greatest in the tropics (~**18–20 km**) and least near the poles; average height is often cited near **~13 km**.
- UP Indo-Gangetic plains sit in the dense lower troposphere — winter **inversions** trap smog (Kanpur (कानपुर), Lucknow (लखनऊ), Ghaziabad).

> **Logic:** Weather = troposphere only. Ozone shield ≠ troposphere. Homosphere excludes **ionosphere**.

35.5 Stratosphere

Identity: Ozone layer | Temp increases with height | UV shield

- The **stratosphere** lies roughly **12–50 km** above the surface.
- Temperature **increases with height** because ozone absorbs ultraviolet radiation.
- The **ozone layer (ozonosphere)** of maximum concentration sits mainly around **15–30/35 km** in the stratosphere (~**90%** of atmospheric ozone; ~**10%** in troposphere).
- Stratospheric ozone is naturally regulated in part by **nitrogen dioxide (NO_2)** and protects life from harmful **UV-B/C**; tropospheric ozone is a pollutant.
- Ends at the **stratopause**.

PYQ — UPPCS Prelims 2023, Q52

The ozone layer, which absorbs ultraviolet radiation, exists in which layer of the atmosphere?

Options: A. Troposphere B. Mesosphere C. Stratosphere D. Thermosphere

> 💡 **Show answer**
>
> **Ans: C**

PYQ — UPPCS Prelims 2018, Q52

The maximum concentration of Ozone is found in which of the following?

Options: A. Troposphere B. Mesosphere C. Stratosphere D. Exosphere

> 💡 **Show answer**
>
> **Ans: C**

PYQ — UPPCS Prelims 2018, Q57

Which pair is NOT correctly matched?

> 💡 **Show answer**
>
> **Ozone layer — Troposphere** is the wrong pair. Ozone layer = **stratosphere**.

PYQ — UPPCS Prelims 2025, Q45

With reference to the ozone layer: 1. Protects Earth's surface from solar ultraviolet radiation. 2. Ozone depletion (ओजोन क्षरण) has been linked to CFCs.

> 💡 **Show answer**
>
> **Both correct** — full ODS/Montreal depth in Topic 36.

> **Logic:** Ozone max = **stratosphere**. Pair "ozone–troposphere" is always wrong.

35.6 Mesosphere

Identity: ~50–85 km | Coldest layer | Meteors burn

- The **mesosphere** sits above the stratosphere, roughly **50–85 km**.
- Temperature **decreases with height**; the **mesopause** is the **coldest** part of the atmosphere (~**−90°C**).
- Most **meteors burn up** here due to friction with sparse air.
- Noozone-layer role and little weather activity compared with the troposphere.

> **Logic:** Meteors = mesosphere; ozone ≠ mesosphere.

35.7 Thermosphere

Identity: Very high temperature | Low density | Aurora / ISS / ionosphere

- The **thermosphere** extends roughly **85–600 km**.
- Kinetic temperature is very high, but density is so low that you would not "feel" hot.
- Contains much of the **ionosphere**, which reflects radio waves and hosts **auroras**.
- The **International Space Station** orbits in the lower thermosphere region.
- Temperature generally **increases with height** due to absorption of high-energy solar radiation.

> **Logic:** Hot ≠ dense. Thermosphere heat is not like tropospheric heat you feel.

35.8 Exosphere

Identity: Outermost layer | Merges into space | Light gases escape

- The **exosphere** is the outermost atmospheric region (~**600 km** upward, gradually fading).
- Gas molecules can travel long paths and eventually **escape** to space (especially H and He).
- It merges into interplanetary space rather than ending at a sharp boundary.
- Highest satellite orbits relate to this outer region matching.

> **Logic:** Exosphere = space transition, not the ozone layer.

Complete PYQ Bank (Topic 35)

Q1. UPPCS Prelims 2023, Q52

The ozone layer, which absorbs ultraviolet radiation, exists in which layer of the atmosphere?

A. Troposphere

B. Mesosphere

C. Stratosphere

D. Thermosphere

💡 **Show answer**

Logic: Ozone layer location — ~90% of atmospheric ozone sits in the stratosphere, not troposphere. **Ans: C.** Protective ozone maximum is in the **stratosphere** (~15–35 km). **Trap:** Troposphere because of ground-level photochemical ozone pollution.

Q2. UPPCS Prelims 2022, Q138

The Earth's atmosphere is mainly heated by which one of the following?

A. Long-wave terrestrial radiation

B. Scattered solar radiation

C. Reflected solar radiation

D. Short-wave solar radiation

💡 **Show answer**

Logic: Atmospheric heating — surface emits longwave IR that warms the lower atmosphere from below. **Ans: A.** The surface absorbs shortwave solar energy and re-emits **longwave terrestrial radiation** that heats the troposphere. **Trap:** Direct shortwave heats the ground first, not the main lower-air heating mechanism.

Q3. UPPCS Prelims 2018, Q52

The maximum concentration of Ozone is found in which of the following?

A. Troposphere

B. Mesosphere

C. Stratosphere

D. Exosphere

💡 **Show answer**

Logic: Maximum ozone concentration layer — stratosphere in both 2018 and 2023 stems. **Ans: C.** **Stratosphere** holds peak ozone column. **Trap:** Mesosphere (meteors) or exosphere (satellites).

Q4. UPPCS Prelims 2018, Q57

Which of the following pairs is **NOT** correctly matched?

A. Renneting — Cheese

B. Genetic Engineering — Plasmids

C. Golden rice — Vitamin A

D. Ozone layer — Troposphere

💡 **Show answer**

Logic: Wrong pair trap — ozone layer is never in troposphere on match lists. **Ans: D.** Ozone layer = **stratosphere**. A, B, C are correctly matched distractors. **Trap:** Accepting ozone anywhere in the atmosphere as the protective layer location.

Q5. UPPCS Prelims 2025, Q45

With reference to the ozone layer:

1. It protects Earth's surface from solar ultraviolet radiation.
2. Ozone depletion has been linked to CFCs.

A. Only 2

B. Neither

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: Stratospheric ozone protects from UV; CFCs linked to depletion — both statements true. **Ans: C.** Good stratospheric ozone shields **UV**; **CFCs/ODS** drive anthropogenic depletion. **Trap:** Denying CFC link or treating all ozone as harmful ground pollutant.

Q6. UPPCS Prelims 2021, Q134

Diffusion of light in the atmosphere takes place due to

A. Carbon dioxide

B. Dust particles

C. Helium

D. Water vapours

💡 **Show answer**

Logic: Light scattering — dust aerosols scatter sunlight (blue sky, red sunset). **Ans: B.** **Dust/aerosols** scatter visible light. **Trap:** CO_2 (GHG IR trapping) or water vapour alone without aerosol context.

Q7. UPPCS Prelims 2020, Q139

Which is **NOT** a greenhouse gas found naturally in the atmosphere?

A. Nitrogen oxide

B. Carbon dioxide

C. Methane

D. Ozone

💡 **Show answer**

Logic: Natural GHG set — CO2, CH4, O3 yes; nitrogen oxides not in the classic natural GHG list tested here. **Ans: A.** **CO2, CH4, O3** are natural GHGs in this stem set; **NOx** is not treated as that major natural GHG basket. **Trap:** Confusing N2 (not GHG) with N2O (is GHG).

Q8. Pattern

Weather (clouds, storms, rainfall) occurs mainly in the:

A. Troposphere

B. Stratosphere

C. Mesosphere

D. Thermosphere

💡 **Show answer**

Logic: Weather phenomena — confined to troposphere where water vapour and convection dominate. **Ans: A.** **Troposphere** alone has day-to-day weather. **Trap:** Stratosphere because ozone or jet streams are taught nearby.

Q9. Pattern

Which pair is correctly matched?

A. Meteors burning up — Mesosphere

B. Meteors burning up — Troposphere

C. Aurora borealis — Troposphere only

D. International Space Station orbit — Exosphere only

💡 **Show answer**

Logic: Meteors burn in mesosphere; aurora/ISS in thermosphere — layer-function pairing. **Ans: A.** **Meteors ~50–85 km (mesosphere)**; aurora/ISS sit in **thermosphere** band. **Trap:** Putting meteors in troposphere because shooting stars appear in the sky below.

Q10. Pattern

Near-surface dry air composition includes:

A. Nitrogen ~78%, Oxygen ~21%, Argon ~0.93%

B. Oxygen ~78%, Nitrogen ~21%, CO2 ~21%

C. Argon ~21%, Nitrogen ~0.04%, CO2 ~78%

D. Nitrogen and oxygen both major greenhouse gases

💡 **Show answer**

Logic: Dry air composition — N2 ~78%, O2 ~21%, Ar ~0.93%; N2 is not a GHG. **Ans: A.** Permanent gases **N2/O2/Ar** dominate by volume; trace **CO2 ~0.04%** still drives enhanced greenhouse effect. **Trap:** Percentage dominance mistaken for radiative importance, or argon listed as GHG.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to atmospheric layers (bottom to top), which of the following statements is/are correct?

1. Troposphere → Stratosphere → Mesosphere → Thermosphere → Exosphere.
2. Weather lives in the stratosphere.
3. Ozone layer maximum is in the troposphere.

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q2. Match List-I with List-II:

List-I	List-II
A. Troposphere	1. Ozone layer; T rises with height
B. Stratosphere	2. Weather; T falls with height
C. Mesosphere	3. Coldest; most meteors burn

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Tropo/strato/meso.

Q3. Which of the following pairs is/are correctly matched?

1. Thermosphere is a cold dense weather layer with no link to auroras or the ionosphere.
2. Exosphere — outermost fringe merges into space
3. Exosphere — main weather layer

A. Only 1 B. Only 2 C. 1 and 2 D. 2 and 3

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Statement 2 is true (exosphere = outermost fringe). Statement 1 is false — thermosphere is hot but thin and hosts aurora/ISS/ionosphere neighbourhood facts. Statement 3 is false — weather is tropospheric.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Dry air is about N_2 78%, O_2 21%, argon ~0.93%, CO_2 ~0.04%.

Reason (R): Argon is a major greenhouse gas.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Argon is not a GHG.

Q5. With reference to N_2 vs N_2O, which of the following statements is/are correct?

1. Nitrogen gas (N_2) is not a greenhouse gas.
2. Nitrous oxide (N_2O) is a greenhouse gas.
3. N_2 and N_2O are identical greenhouse gases.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Do not confuse N_2 with N_2O.

Q6. Arrange how lower atmosphere is heated:

A. Only direct shortwave alone B. Mainly long-wave terrestrial radiation from warmed surface C. Only exosphere escape D. Only meteor burns

💡 **Show answer**

Ans: B. Long-wave terrestrial radiation. **Logic:** Standard heating answer.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Sun — shortwave insolation
2. Surface — longwave emission warming air
3. Maximum ozone — troposphere

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Ozone max = stratosphere.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Stratospheric temperature rises with height largely because ozone absorbs UV.

Reason (R): The ozone layer's maximum concentration is in the stratosphere.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R supplies the location for the UV-absorption heating. **A/R logic:** Ozone heating mechanism.

Q9. With reference to tropospheric vs stratospheric ozone, which of the following statements is/are correct?

1. Tropospheric ozone — pollutant and GHG.
2. Stratospheric ozone — UV shield.

3. Both are identical in role always.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Different roles.

Q10. Match List-I with List-II:

List-I	List-II
A. Homosphere	1. Separates gases by mass
B. Heterosphere	2. Well mixed to ~80 km
C. Ionosphere radio	3. Thermosphere–upper-atmosphere feature

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: A.** A-2, B-1, C-3. **Logic:** Mixing / mass / radio.

Q11. Arrange temperature inversion location:

A. Only exosphere B. Troposphere (with weather systems) C. Only mesosphere meteors D. Only heterosphere

> 💡 **Show answer**
>
> **Ans: B.** Troposphere. **Logic:** Inversion + weather.

Q12. With reference to mesosphere vs thermosphere, which of the following statements is/are correct?

1. Mesosphere is coldest; thermosphere can be very hot but thin.
2. Both can be true because density differs.
3. Thin hot air equals surface heat sensation always.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** Density caveat.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Commercial jets often cruise near the tropopause.

Reason (R): The tropopause sits atop the troposphere neighbourhood.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R places the cruise level. **A/R logic:** Tropopause cruise.

Q14. Which one of the following correctly states water vapour in composition teaching?

A. Absent from atmosphere always B. Fixed at 78% always C. Identical to argon always D. Varies strongly by place and season; dry-air % exclude it

💡 **Show answer**

Ans: D. Variable; dry-air basis. **Logic:** H_2O variability.

Q15. How many of the following are NOT greenhouse gases?

N_2, O_2, argon, helium

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** None of these are GHGs.

Q16. With reference to layer order traps, which of the following statements is/are correct?

1. Never reverse troposphere and stratosphere bottom-to-top.
2. Exosphere atoms can escape to space.
3. Exosphere is the main weather layer.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Weather = troposphere.

Q17. Match List-I with List-II:

List-I	List-II
A. N_2	1. ~0.04%
B. O_2	2. ~78%
C. CO_2	3. ~21%

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-3, C-1. **Logic:** 78 / 21 / 0.04.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Do not place the ozone layer in the troposphere on a match list.

Reason (R): Maximum ozone concentration is in the stratosphere.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Ozone location trap.

Q19. Which of the following pairs is/are correctly matched?

1. Auroras — thermosphere neighbourhood
2. Most meteors burn — mesosphere
3. ISS path — thermosphere neighbourhood teaching

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Upper-atmosphere facts.

Q20. Arrange ionosphere radio reflection:

A. Mainly thermosphere–upper-atmosphere feature B. Mainly troposphere fog C. Only soil conductivity D. Only ocean salinity

💡 **Show answer**

Ans: A. Upper atmosphere / thermosphere. **Logic:** Radio reflection.

Q21. With reference to atmospheric studies for climate, which of the following statements is/are correct?

1. Start from layer ladder.
2. Plus greenhouse-gas composition facts.
3. Ignore long-wave terrestrial heating entirely.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Heating mechanism matters.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Temperature falls with height in the troposphere.

Reason (R): Temperature rises with height in the stratosphere.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; stratosphere lapse does not explain troposphere lapse. **A/R logic:** Two layer thermal rules.

Q23. How many of the following layer jobs are correct?

Troposphere weather; Stratosphere ozone; Mesosphere meteors; Exosphere escape fringe

A. Only one B. Only two C. All four D. Only three

💡 **Show answer**

Ans: C. All four. **Logic:** Job map.

Q24. Which one of the following correctly states argon share teaching?

A. Zero in dry air B. ~78% and major GHG C. ~21% and UV shield D. ~0.93% of dry air; not a GHG

💡 **Show answer**

Ans: D. ~0.93%; not GHG. **Logic:** Argon fact.

Q25. With reference to Sun vs Earth radiation, which of the following statements is/are correct?

1. Sun sends shortwave insolation.

2. Surface emits longwave that warms air.
3. Lower air heated mainly by direct shortwave alone in standard answer.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Standard = terrestrial long-wave.

Q26. Match List-I with List-II:

List-I	List-II
A. Tropospheric O_3	1. UV shield
B. Stratospheric O_3	2. Pollutant / GHG
C. Heterosphere	3. Above ~80 km mass separation

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** O_3 roles + heterosphere.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Oxygen and helium are not greenhouse gases.

Reason (R): Only CO_2 among all atmospheric gases can ever affect radiation.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Other GHGs exist (CH_4, N_2O, etc.).

Q28. Arrange five-layer ladder (bottom first two):

A. Stratosphere → Troposphere B. Exosphere → Troposphere C. Troposphere → Stratosphere D. Mesosphere → Exosphere as first

💡 **Show answer**

Ans: C. Troposphere then Stratosphere. **Logic:** Never reverse.

Q29. With reference to mesosphere, which of the following statements is/are correct?

1. Coldest layer.
2. Most meteors burn here.
3. Holds the ozone-layer maximum.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Ozone max = stratosphere.

Q30. Which one of the following correctly states the exosphere?

A. Outermost fringe; atoms can escape to space; not a weather layer B. Main monsoon cloud factory C. Only ozone UV shield D. Only surface inversion layer

💡 **Show answer**

Ans: A. Escape fringe — not weather. **Logic:** Exosphere identity.

Common Traps

Trap	Correct
Ozone layer in troposphere	**Stratosphere**
Atmosphere heated mainly by shortwave	**Longwave from surface**
N_2 is a GHG	**Not**; N_2O is
Temp falls in every layer	Tropo↓ Strato↑ Meso↓ Thermo↑
Weather in stratosphere	**Troposphere only**
Meteors burn in troposphere	**Mesosphere**
Max ozone in mesosphere/exosphere	**Stratosphere**
Argon is a GHG	**Not**
Thermosphere “feels” very hot	High T, **low density**
Composition % = climate importance	Trace CO_2 still critical

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 36 — Ozone Layer

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Ozone Layer | Ozone Depletion (ओजोन क्षरण) | Ozone Hole (ओजोन छिद्र) | Ozone Depleting Substances (ODS) | Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) | Vienna Convention

Sources baked in: NCERT; UNEP Ozone Secretariat; Vienna 1985; Montreal 1987 (force 1989); Kigali 2016; MoEFCC Ozone Cell; UPPCS/UPSC PYQs **Weight:** ★★★ High — Montreal ≠ Kyoto; ozone = stratosphere; Ozone Day 16 Sept; CFC link **Last verified:** September 2026

Current Affairs (this topic)

- Twin of Ozone Layer core — keep Montreal / Kigali as ozone-family CA only.
- Prefer one consistent ODS story across both ozone chapters.

💡 🎯 **Consolidated — 34 Must-Score Facts (Click to Expand)**

1. **Ozone (O_3)** is triatomic oxygen. **Good ozone** (अच्छा ओजोन) sits in the **stratosphere** (समतापमंडल) (about **15–30/35 km**). Ground-level ozone is a harmful pollutant. 2. About **90%** of atmospheric ozone is in the **stratosphere**; about **10%** is in the **troposphere** (क्षोभमंडल). 3. Stratospheric ozone (समतापमंडलीय ओजोन) forms when **UV** acts on **O_2** (Chapman cycle). Ozone absorbs **ultraviolet**, not IR as its main protective role. 4. Ozone amount is measured in **Dobson Units**. **1 DU ≈ 0.01 mm** of pure ozone at **0°C and 1 atm**. Average column ≈ **300 DU**. An ozone hole is often taken as below **220 DU**. 5. Stratospheric ozone is naturally regulated in part by **nitrogen dioxide (NO_2)**. Column thickness is often highest in **spring** and lowest in the **rainy season** (general seasonal pattern — distinct from the Antarctic spring hole). 6. Ozone depletion is driven by **ODS** such as CFCs through chlorine and bromine catalytic cycles. 7. One chlorine atom can destroy on the order of **100,000** ozone molecules before it is removed. 8. The classic **ozone hole** forms over **Antarctica** in spring (**September–October**) with polar stratospheric clouds, polar front/vortex, and CFCs. 9. Joe **Farman** (फ़रमान), Brian **Gardiner**, and Jonathan **Shanklin** (British Antarctic Survey) reported the Antarctic ozone hole in **Nature (May 1985)**; TOMS mapping was used in that era of monitoring. 10. ODS include **CFCs, Halon-1211/1301, carbon tetrachloride, methyl chloroform, methyl bromide, HCFCs**, and related chlorine/bromine carriers. **HFCs** are **not** classic ODS. 11. Refrigerant trade (पण्याध्यक्ष) name **Mafron** (halocarbon coolants) is common in domestic refrigerators; **ammonia** is used in many large plants. CFCs are **not** primarily "automobile exhaust" gases. 12. CFCs are used in refrigerators/AC, foam blowing, aerosol propellants, and cleaning some electronic components — **not** tubeless-tyre production as a classic CFC use. 13. **HFCs** are high-GWP climate gases under **Kigali**. They are **not** classic ozone-depleting substances. 14. The **Vienna Convention (1985)** (वियना कन्वेंशन) is the framework treaty. The **Montreal Protocol (16 September 1987; force 1 January 1989)** is the binding ODS phase-out. 15. World Ozone Day (विश्व ओजोन दिवस) is **16 September** (Montreal signing; UN observance from **1994**). 16. The **Kigali Amendment (2016; force 1 January 2019)** phases down **HFCs** for climate reasons under the Montreal family. 17. **Montreal = ozone. Kyoto (क्योटो) = climate.** Carbon trading ≠ Montreal. 18. ODP measures ozone-damage potential. GWP measures heat-trapping potential. 19. Stratospheric ozone absorbs harmful UV (**~93–99%** of the most damaging wavelengths). Tropospheric ozone (क्षोभमंडलीय ओजोन) is a pollutant and greenhouse gas. 20.

UV-A (315–400 nm) largely reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is blocked and does not reach the surface under normal conditions. 21. Extra UV-B raises **skin cancer**, cataracts, crop damage, and plankton stress — not "liver/brain/oral cancer" as the classic stem key. 22. India's Ozone Cell under MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) tracks HCFC phase-out and cooling-sector compliance. 23. Do not place the protective ozone layer in the troposphere. 24. CFCs are both ODS and strong greenhouse gases — "CFC not a GHG" is false. 25. Polar spring timing matters: the deepest Antarctic loss is not midsummer. 26. Vienna does not itself list every control measure; Montreal carries the binding phase-out schedules. 27. **G.W. Kent Moore** (University of Toronto) led the **2005** report of an ozone "halo" / low-ozone feature over the **Tibetan Plateau**. 28. Homosphere (सममंडल) (uniform composition) ≈ troposphere + stratosphere + mesosphere (मध्यमंडल); **ionosphere** is **not** part of the homosphere set in UKPCS-style stems. 29. Troposphere is deepest in the tropics (~**18–20 km**) and shallower near the poles. 30. Dobson Unit is a column measure, not a surface concentration in ppm. 31. A stem pairing Montreal with climate mitigation (जलवायु शमन) is usually the Kyoto/Paris trap. 32. Ozone as a gas near the surface is poisonous to lungs/respiratory systems; as a stratospheric layer it is essential for biotic life via UV shielding. 33. Formation of ozone is slower over polar regions, so depletion effects are strongest there. 34. Acid rain (अम्ल वर्षा) is driven by **SO_2 / NOx**, not "nitric acid alone" as a matched pair with ozone–CFC style keys.

💡 ⚡ **Confused Pairs (Click to Expand)**

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Stratospheric Ozone	**Tropospheric Ozone**	Beneficial atmospheric filter blocking lethal solar UV radiation vs harmful photochemical smog component damaging human lungs and crop yields	समतापमंडलीय ओजोन (सुरक्षा कवच) / क्षोभमंडलीय ओजोन (प्रदूषक)
Ozone Depletion	**Ozone Hole**	Global thinning of the stratospheric ozone layer by ODS vs severe seasonal Antarctic drop in total column ozone below 220 Dobson Units (DU)	ओजोन क्षरण / ओजोन छिद्र (<220 DU)
Vienna Convention (1985)	**Montreal Protocol (1987)**	Non-binding framework agreement establishing ozone research/cooperation vs binding international protocol mandating specific phase-out schedules for ODS	वियना कन्वेंशन (1985) / मॉन्ट्रियल प्रोटोकॉल (1987)
Chlorofluorocarbons (CFCs)	**Hydrofluorocarbons (HFCs)**	Chlorine-bearing potent ozone depleters + GHGs (phased out under Montreal) vs zero ODP non-ozone-depleting refrigerants that are super greenhouse gases (phased down under Kigali)	सीएफसी (ओजोन क्षयकारी) / एचएफसी (ग्रीनहाउस गैस)
Ozone Depletion Potential (ODP)	**Global Warming Potential (GWP)**	Measure of substance's ability to destroy O_3 relative to CFC-11 (=1.0) vs measure of heat-trapping efficiency relative to CO_2 (=1.0) over 100 years	ओजोन रिक्तीकरण क्षमता (ODP) / वैश्विक तापन क्षमता (GWP)
Dobson Unit (DU)	**Parts Per Million (ppm)**	Standard measure of atmospheric column ozone thickness (1 DU = 0.01 mm at STP; normal = ~300 DU) vs concentration ratio of gases in volume	डॉबसन यूनिट (DU) / पीपीएम (ppm)

💡 📋 **Must-score facts — DU, ODS, Vienna–Montreal–Kigali (Click to Expand)**

Science / treaties

Item	Lock
Good / bad	Stratospheric UV shield / tropospheric pollutant
Share	~90% / ~10%
1 DU	≈0.01 mm pure O_3; hole often <**220 DU**
Antarctic season	Sep–Oct spring
1985 discovery	Farman–Gardiner–Shanklin (BAS)
ODS set	CFCs, Halons, CCl_4, CH_3CCl_3, CH_3Br, HCFCs…
HFCs	High GWP; **not** classic ODS → **Kigali**
Vienna / Montreal / Kigali	**1985** / **1987** (force 1989) / **2016**
Ozone Day	**16 Sep**
Montreal ≠ Kyoto	Ozone / climate

36.1 Ozone Layer

Identity: Stratosphere ~15–30 km | UV shield | Dobson Unit | Chapman cycle | NO_2 regulation

- **Ozone (O_3)** is triatomic oxygen that absorbs harmful solar ultraviolet radiation.

- The protective **ozone layer** sits mainly in the lower **stratosphere** (~**15–30 km**, sometimes cited to ~**35 km**; peak near ~**25 km**). The stratosphere itself spans roughly **10–50 km**.
- About **90%** of atmospheric ozone is stratospheric; about **10%** is tropospheric.
- Natural formation/destruction follows the **Chapman cycle**: UV splits $O_2 \rightarrow O$; $O + O_2 \rightarrow O_3$; UV also breaks O_3.
- Stratospheric ozone is naturally regulated in part by **nitrogen dioxide (NO_2)**.
- Column thickness varies seasonally — often **highest in spring** and **lowest in the rainy season** (general pattern; the Antarctic "hole" is a separate spring-extreme story).
- **UV-A (315–400 nm)** largely reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is blocked by O_2/O_3 and does not reach the surface under normal conditions.
- Stratospheric ozone blocks roughly **93–99%** of the most damaging solar UV.
- Near the surface, ozone is a **poisonous pollutant** (lungs/respiratory harm). In the stratosphere it is essential because it shields life from UV.
- Ozone absorption heats the stratosphere, so temperature **rises with height** there.
- Column ozone is measured in **Dobson Units (DU)**; **1 DU ≈ 0.01 mm** of pure ozone at **0°C and 1 atm**. Global average ≈ **300 DU**. Total atmospheric ozone mass is on the order of **~3 billion tonnes**.

PYQ — UPPCS Prelims 2025, Q45

1. Ozone layer protects Earth's surface from solar UV.
2. Ozone depletion has been linked to CFCs.

Options: A. Only 2 B. Neither C. Both 1 and 2 D. Only 1

💡 **Show answer**

Ans: C

PYQ — UPPCS Prelims 2018, Q52 / Q57

Maximum ozone concentration / wrong pair:

💡 **Show answer**

Stratosphere. Pair **Ozone layer — Troposphere** is NOT correctly matched.

PYQ — UPPCS Prelims 2018, Q55

Ultra-violet radiation from sunlight causes the reaction that produces:

Options: A. CO B. SO_2 C. O_3 D. Fluorides

💡 **Show answer**

Ans: C

> **Logic:** UV-shield ozone = **stratosphere**. Ground O_3 = bad. Ozone absorbs **UV**, not IR as its protective role.

36.2 Ozone Depletion

Identity: ODS → Cl·/Br· | Catalytic destruction | Rowland–Molina 1974 | CFC uses

- **Ozone depletion** is a sustained fall in stratospheric ozone below natural levels.
- Stable **ODS** (especially CFCs) reach the stratosphere, where UV releases **chlorine** (or bromine) radicals.
- One chlorine atom can destroy on the order of **~100,000** ozone molecules in a catalytic cycle.

- **Rowland–Molina (1974)** linked CFCs to ozone loss; Nobel Prize in Chemistry **1995**.
- CFCs are non-toxic, non-flammable, long-lived (~**80–100 years**) coolants/solvents — also strong **greenhouse gases**. Saying "CFC is not responsible for the greenhouse effect" is **false**.
- Domestic fridge coolants are often sold under names like **Mafron** (halocarbon refrigerants). Large industrial plants may use **ammonia**.
- Classic CFC uses: refrigerators/AC, foam blowing, aerosol pressurising agents, cleaning some electronic components — **not** tubeless-tyre manufacture as a standard CFC use, and **not** automobile exhaust as the primary CFC source.
- Extra UV-B raises risks of **skin cancer**, cataracts, crop damage, and marine plankton stress.
- Depletion is not the same as the greenhouse effect, though some substitutes (HFCs) are strong climate gases.

> **Logic:** Depletion link to **CFCs** is a direct UPPCS fact. Skin cancer is the classic UV-harm key — not liver/brain/oral.

36.3 Ozone Hole

Identity: < 220 DU | Antarctic spring | PSCs + polar vortex | Farman 1985 | Tibet 2005

- The **ozone hole** is a region of severely thinned column ozone, defined as **below 220 DU**.
- It is **not** a literal hole in the sky.
- Classic location: **Antarctic spring** (**September–October**, Southern Hemisphere). Ozone forms slowly over poles, so depletion effects are strongest there.
- Extreme cold forms **Polar Stratospheric Clouds (PSCs)** (ध्रुवीय समतापमंडलीय बादल); nitric-acid chemistry on PSC surfaces activates chlorine from CFCs; the **polar front/vortex** isolates air; chlorine becomes highly active when sunlight returns.
- Discovered in **1985** by **Farman, Gardiner, and Shanklin** (British Antarctic Survey) in *Nature*; total-ozone mapping (TOMS-era) supported monitoring.
- In **2005**, **G.W. Kent Moore** (University of Toronto) and colleagues reported a low-ozone / "ozone halo" feature over the **Tibetan Plateau**.
- Arctic thinning occurs but is usually weaker/less persistent than the Antarctic hole.
- There is **no ozone hole over Uttar Pradesh (उत्तर प्रदेश) / India** as an Antarctic-style seasonal feature.

> **Logic:** Hole = Antarctic spring + DU threshold + PSC/CFC chemistry — not "everywhere CFCs are used."

36.4 Ozone Depleting Substances (ODS)

Identity: CFCs / Halon-1211·1301 / CCl_4 / CH_3Br / HCFCs | ODP scale

- **ODS** are chemicals controlled under the Montreal Protocol because they destroy stratospheric ozone.
- **CFCs:** refrigerants, aerosol propellants, foam blowing, electronics cleaning — long-lived chlorine sources.
- **Halons (e.g. Halon-1211, Halon-1301):** fire suppressants — bromine; often higher ozone-depleting impact per molecule.
- Also controlled: **carbon tetrachloride (CCl_4)**, **methyl chloroform (1,1,1-trichloroethane)**, **methyl bromide (CH_3Br)**, and transitional **HCFCs**.
- **ODP (Ozone Depleting Potential)** ranks destructiveness relative (सापेक्ष) to **CFC-11 = 1**.
- **HFCs** were ozone-friendly replacements but have high **GWP**; controlled later by the **Kigali Amendment** (climate, not classic ODS).
- **Nitrous oxide (N_2O)** also contributes to ozone chemistry / long-term ozone loss discussions, but CFCs/halons dominate classic "ozone hole" stems.

ODS family	Typical use	Fact
CFCs	Cooling, aerosols, foam, electronics cleaners	Main Cl source
Halons (1211, 1301)	Fire fighting	Br; high ODP
CCl_4 / methyl chloroform / CH_3Br	Solvent / fumigant	Controlled ODS
HCFCs	Transitional coolants	Lower ODP; being phased out
HFCs	Coolants	Not ODS; Kigali / GWP

> **Logic:** HFC ≠ ODS, but still relevant under Montreal's Kigali track. Methyl chloroform **is** an ODS under Montreal (ignore any bank key that treats it as non-ODS).

36.5 Montreal Protocol

Identity: Signed 16 Sep 1987 | Force 1 Jan 1989 | Binding ODS phase-out | Ozone Day

- The **Montreal Protocol on Substances that Deplete the Ozone Layer (1987)** is the binding global treaty to phase out ODS (CFCs, halons, CCl_4, etc.).
- It operates under the **Vienna Convention** framework.
- Signed on **16 September 1987**; entered into force on **1 January 1989**.
- **16 September** is observed as **International Day for the Preservation of the Ozone Layer (World Ozone Day)** — UN observance from **1994**.
- Key amendments: London 1990, Copenhagen 1992, Montreal 1997, Beijing 1999, **Kigali 2016 (HFCs; force 1 January 2019)**.
- Often called the most successful environmental treaty — near-universal ratification; ODS production/consumption cut dramatically from peak.
- India has an **Ozone Cell under MoEFCC**; CFC phase-out completed on developing-country schedule; HCFC phase-out continues; Kigali HFC schedule applies later.

PYQ — UPPCS Prelims 2024, Q35

Which agreement is related to climate change (जलवायु परिवर्तन) mitigation?

Options: A. Montreal Protocol B. Ramsar (रामसर) C. Kyoto Protocol D. CITES (साइट्स)

> 💡 **Show answer**
>
> **Ans: C** — Montreal = **ozone**, not climate mitigation.

PYQ — UPPCS Prelims 2019, Q132

Which is NOT correctly matched?

Options: A. Cartagena — Biosafety B. Stockholm — POPs C. Montreal — Ozone layer D. Kyoto — Save water

> 💡 **Show answer**
>
> **Ans: D** — Kyoto = climate/GHG, not "save water." Montreal–ozone is correct.

PYQ — UPPCS Prelims 2021, Q114 / 2022, Q82 / 2025, Q62

World / International Ozone Day date:

> 💡 **Show answer**
>
> **16 September** (Montreal signing date).

> **Logic:** Montreal = ozone. Kyoto = climate. Never swap. Carbon trading ≠ Montreal.

36.6 Vienna Convention

Identity: 1985 | Framework treaty | Research + cooperation | Parent of Montreal

- The **Vienna Convention for the Protection of the Ozone Layer (1985)** is the **framework** treaty.
- It commits parties to protect the ozone layer through research, monitoring, information exchange, and cooperation.
- It does **not** itself set the detailed binding ODS phase-out schedules — that is Montreal's job.
- Sequence fact: **Vienna 1985 → Montreal 1987**.
- Together they form the global ozone regime administered with UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) support (Ozone Secretariat).

Treaty	Year	Role
Vienna Convention	**1985**	Framework / cooperation
Montreal Protocol	**1987**	Binding ODS controls
Kigali Amendment	**2016** (force **1 Jan 2019**)	HFC phase-down under Montreal

> **Logic:** Vienna alone ≠ CFC ban list; Montreal adds the teeth.

Complete PYQ Bank (Topic 36)

Q1. UPPCS Prelims 2025, Q45

With reference to the ozone layer:

1. It protects Earth's surface from solar ultraviolet radiation.
2. Ozone depletion has been linked to CFCs.

A. Only 2

B. Neither

C. Both 1 and 2

D. Only 1

💡 **Show answer**

Logic: Both UV-shield role and CFC–depletion link are standard facts — answer both statements. **Ans: C.** Stratospheric ozone absorbs **UV-B/C**; **CFCs** release stratospheric chlorine radicals. **Trap:** Treating ground-level O3 smog as the protective layer.

Q2. UPPCS Prelims 2025, Q62 — Ozone Day pair

International Day for the Preservation of the Ozone Layer is observed on:

A. 5 June

B. 16 September

C. 22 March

D. 2 February

💡 **Show answer**

Logic: International Ozone Day = Montreal signing date, 16 September. **Ans: B.** **16 September** marks Montreal Protocol signature (**1987**). **Trap:** 5 June (WED) or 22 March (Water Day).

Q3. UPPCS Prelims 2024, Q35

Which international agreement is related to climate change mitigation?

A. Montreal Protocol

B. Ramsar Convention

C. Kyoto Protocol

D. CITES

💡 **Show answer**

Logic: Montreal = ozone/ODS; Kyoto = climate — 2024 Q35 mitigation stem. **Ans: C.** **Kyoto** mitigates GHGs; **Montreal** phases out **ODS**. **Trap:** Montreal because it is the headline ozone treaty in this chapter.

Q4. UPPCS Prelims 2023, Q52

The ozone layer exists mainly in which atmospheric layer?

A. Troposphere

B. Mesosphere

C. Stratosphere

D. Thermosphere

💡 **Show answer**

Logic: Ozone layer stratosphere location — overlaps Topic 35 but anchored here for ozone science. **Ans: C.** ~**90%** of ozone is stratospheric. **Trap:** Troposphere pair in NOT-matched lists.

Q5. UPPCS Prelims 2022, Q82

On which date is Ozone Day observed?

A. October 2

B. September 16

C. April 29

D. January 30

💡 **Show answer**

Logic: Ozone Day date recurrence — 2022 wording variant. **Ans: B.** **16 September** — Montreal Protocol date. **Trap:** Gandhi Jayanti or Earth Day month swaps.

Q6. UPPCS Prelims 2022, Q46

Photochemical smog always contains—

A. Phosphorus

B. Ozone

C. Methane

D. Aluminium ion

💡 **Show answer**

Logic: Photochemical smog always contains tropospheric ozone — secondary pollutant, distinct from stratospheric shield. **Ans: B.** **NOx + VOCs + sunlight → O3 + PAN** in photochemical smog. **Trap:** Confusing this bad ground O3 with stratospheric good ozone location.

Q7. UPPCS Prelims 2021, Q114

'World Ozone Day' is celebrated on

A. 25 December

B. 21 April

C. 16 September

D. 30 January

💡 **Show answer**

Logic: 2021 Ozone Day stem — same 16 September key. **Ans: C.** **16 September** worldwide. **Trap:** 21 April (Earth Day).

Q8. UPPCS Prelims 2019, Q132

Which is **NOT** correctly matched?

A. Cartagena — Biosafety

B. Stockholm — POPs

C. Montreal — Ozone layer

D. Kyoto — Save water

💡 **Show answer**

Logic: Treaty match trap — Kyoto is climate/GHG, not save water; Montreal–ozone is correct. **Ans: D.** **Kyoto = climate/GHG mitigation**, not a water treaty. Montreal–ozone is correct. **Trap:** Kyoto paired with ozone or water because of generic environment label.

Q9. UPPCS Prelims 2018, Q52 / Q55 / Q57 — set

Consider the following:

1. Ultra-violet radiation from sunlight helps produce ozone (O3).
2. The ozone layer is located in the troposphere.

A. Only 1

B. Only 2

C. Both 1 and 2

D. Neither 1 nor 2

💡 **Show answer**

Logic: UV radiation produces ozone photochemically; wrong pair = ozone layer in troposphere. **Ans: A.** Statement 1 is true (UV-driven O3 chemistry). Statement 2 is false — ozone **layer = stratosphere**. **Trap:**

Accepting pair ozone layer — troposphere on match lists (2018 Q57).

Q10. Pattern

The global ozone treaty sequence is:

A. Vienna Convention 1985 (framework) → Montreal Protocol 1987 (ODS phase-out; in force 1 January 1989)

B. Montreal 1985 → Vienna 1987

C. Kyoto 1987 → Montreal 1992

D. Vienna alone banned all CFCs without Montreal schedules

💡 **Show answer**

Logic: Vienna 1985 framework → Montreal 1987 binding ODS phase-out; force date 1 Jan 1989. **Ans: A.** **Vienna = framework/cooperation**; **Montreal = binding ODS cuts** (signed 16 Sep 1987, force **1 Jan 1989**). **Trap:** Vienna alone as full CFC ban, or reversing treaty order.

Extra Drill — Ghatnachakra-pattern (Q31–Q40)

Logic (set): Location, measurement, ODS identity, Antarctic chemistry, treaties, and UV harm.

Q31. Ozone layer absorbing UV exists mainly in the:

A. Mesosphere B. Troposphere C. Stratosphere D. Thermosphere

💡 **Show answer**

Ans: C. About **90%** of ozone is stratospheric.

Q32. Stratospheric ozone is naturally regulated in large part by:

A. Nitrous oxide only B. Nitrogen dioxide C. CFC alone as a natural regulator D. Water vapour alone

💡 **Show answer**

Ans: B. **NO_2** helps regulate stratospheric ozone chemically.

Q33. Which statement about Dobson Unit is correct?

A. It measures ground-level ppm of ozone pollution only B. 1 DU ≈ 0.01 mm pure ozone at 0°C and 1 atm C. Hole threshold is always 500 DU D. It measures only UV-A

💡 **Show answer**

Ans: B. Column measure; hole often **< 220 DU**.

Q34. Which is **NOT** correctly matched?

A. Renneting — Cheese B. Golden rice — Vitamin A C. Genetic engineering — Plasmids D. Ozone layer — Troposphere

💡 **Show answer**

Ans: D. Ozone layer = **stratosphere**.

Q35. Antarctic ozone-hole chemistry is best linked to:

A. Tropospheric turbulence only B. Polar front + stratospheric clouds + CFCs C. Absence of polar front and PSCs D. Warm polar midsummer only

💡 **Show answer**

Ans: B. PSC chlorine activation + vortex isolation.

Q36. Who discovered the Antarctic ozone hole reported in 1985?

A. Russian team only B. German team only C. American team only D. British team (Farman et al.)

💡 **Show answer**

Ans: D. Farman, Gardiner, Shanklin — *Nature* 1985.

Q37. 'Ozone halo' over the Tibetan Plateau (2005) is linked to work led by:

A. M. Molina alone B. Joseph Farman alone C. G.W. Kent Moore D. Marcus Rex alone

💡 **Show answer**

Ans: C. University of Toronto team led by **Kent Moore**.

Q38. Montreal Protocol entered into force on:

A. 16 September 1987 only as force date B. 1 January 1989 C. 1 January 1992 D. 16 February 2005

💡 **Show answer**

Ans: B. Signed **16 Sep 1987**; force **1 Jan 1989**.

Q39. Harmful UV reaching the surface mainly raises risk of:

A. Liver cancer B. Brain cancer C. Oral cancer D. Skin cancer

💡 **Show answer**

Ans: D. UV-B exposure → **skin cancer** (classic key).

Q40. Assertion (A): Stratospheric ozone is essential for biotic life. Reason (R): The ozone layer protects Earth's surface from high-energy UV radiation.

A. Both true; R explains A B. Both true; R does not explain A C. A true; R false D. A false; R true

💡 **Show answer**

A/R logic: A tests "good ozone" necessity; R tests the UV-shield mechanism (not ground-level toxicity).
Ans: A

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to ozone, which of the following statements is/are correct?

1. O_3 is diatomic oxygen identical to O_2.
2. About 90% is stratospheric; about 10% tropospheric.
3. Protective ozone layer sits mainly in the troposphere.

A. Only 1 B. Only 2 C. 1 and 2 D. 2 and 3

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Statement 2 is true (~90% stratospheric / ~10% tropospheric). Statement 1 is false — ozone is triatomic oxygen. Statement 3 is false — protective ozone is mainly stratospheric.

Q2. Match List-I with List-II:

List-I	List-II
A. Dobson Unit	1. 16 September
B. World Ozone Day	2. Column measure (~300 DU avg)
C. Ozone hole season	3. Antarctic spring Sep–Oct

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** DU / Ozone Day / spring hole.

Q3. Which of the following pairs is/are correctly matched?

1. Vienna Convention — 1985 framework
2. Kyoto Protocol — 16 Sep 1987; force 1 Jan 1989
3. Kigali Amendment — HFC phase-down for climate (2008; force 2019)

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

💡 **Show answer**

Ans: B. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Montreal = ozone; Kyoto = climate.

Reason (R): Carbon trading belongs to the Montreal Protocol.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Carbon trading ≠ Montreal.

Q5. With reference to ODS, which of the following statements is/are correct?

1. CFCs, Halons, CCl_4, methyl chloroform, methyl bromide, HCFCs…
2. HFCs are classic ODS identical to CFCs.
3. One Cl atom can destroy ~100,000 O_3 molecules teaching.

A. 1 and 3 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 3. **Logic:** HFCs = Kigali climate gases — not classic ODS.

Q6. Arrange Chapman / UV role:

A. Ozone absorbs mainly IR as UV shield B. Stratospheric O_3 forms via UV on O_2; absorbs UV (not IR as main shield role) C. Only tropospheric formation D. Only argon catalysis

> 💡 **Show answer**
>
> **Ans: B.** UV formation + UV absorption. **Logic:** Not IR as main protective role.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Farman, Gardiner, Shanklin — Antarctic ozone hole Nature 1985
2. ODP — ozone-damage potential; GWP — heat-trapping
3. CFCs — never greenhouse gases

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only pair 3 is not correctly matched. **Logic:** CFCs are also strong GHGs.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Antarctic ozone hole needs PSCs, polar vortex, and spring sunlight chemistry neighbourhood.

Reason (R): Deepest Antarctic loss is midsummer.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** Spring — not midsummer.

Q9. With reference to UV bands, which of the following statements is/are correct?

1. UV-A largely reaches surface.
2. UV-B mostly blocked by ozone.
3. UV-C blocked / does not reach surface teaching.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: D.** All three. **Logic:** UV-A/B/C map.

Q10. Match List-I with List-II:

List-I	List-II
A. Stratospheric O_3	1. Pollutant / GHG
B. Tropospheric O_3	2. UV shield
C. India's Ozone Cell	3. MoEFCC HCFC/cooling compliance

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Good/bad ozone + Ozone Cell.

Q11. Arrange 1 DU meaning:

A. Surface ppm only B. ≈0.01 mm pure ozone at 0°C and 1 atm (column) C. Only river BOD D. Only AQI band

💡 **Show answer**

Ans: B. Column thickness unit. **Logic:** Not surface ppm.

Q12. With reference to health/ecology of extra UV-B, which of the following statements is/are correct?

1. Skin cancer, cataracts, crop damage, plankton stress.
2. Classic stem does not centre "liver/brain/oral cancer."
3. Extra UV-B only cools the planet.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Statement 3 false.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Vienna does not itself list every control measure.

Reason (R): Montreal carries the binding phase-out schedules.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Framework vs controls.

Q14. Which one of the following correctly states Mafron / ammonia teaching?

A. Both banned under UNFCCC only B. Both only used in tubeless tyres C. Both identical to methane hydrates D. Mafron common in domestic fridges; ammonia in many large plants

💡 **Show answer**

Ans: D. Domestic vs large-plant coolants. **Logic:** Refrigerant facts.

Q15. How many of the following CFC use facts are correct?

Refrigerators/AC; foam blowing; aerosol propellants; cleaning some electronics

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** Not tubeless-tyre centre.

Q16. With reference to Tibetan ozone feature, which of the following statements is/are correct?

1. G.W. Kent Moore led 2005 ozone "halo" / low-ozone report over Tibetan Plateau.
2. Formation of ozone is slower over polar regions — depletion strongest there.
3. Protective ozone sits only at ground level in cities.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Ground-level O_3 is pollutant.

Q17. Match List-I with List-II:

List-I	List-II
A. ODP	1. Heat-trapping
B. GWP	2. Ozone-damage potential
C. NO_2	3. Helps regulate stratospheric ozone naturally

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** ODP/GWP/NO_2.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Surface ozone is poisonous to lungs/respiratory systems.

Reason (R): Stratospheric ozone is essential as a UV shield for biotic life.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; stratospheric benefit does not explain surface toxicity. **A/R logic:** Good vs bad ozone.

Q19. Which of the following pairs is/are correctly matched?

1. Homosphere ≈ troposphere + stratosphere + mesosphere teaching
2. Ionosphere is part of that homosphere set in usual prelims framing
3. Troposphere deepest in tropics (~18–20 km)

A. 1, 2 and 3 B. 1 and 2 only C. 2 and 3 only D. 1 and 3 only

💡 **Show answer**

Ans: D. Only 1 and 3. **Logic:** Ionosphere not in that homosphere set.

Q20. Arrange Kigali vs Montreal jobs:

A. Montreal ODS phase-out; Kigali HFC phase-down for climate under Montreal family B. Kigali phases out ODS chlorine only as Montreal clone C. Both are Kyoto carbon markets D. Both are acid-rain treaties

💡 **Show answer**

Ans: A. ODS vs HFC climate. **Logic:** Family but different targets.

Q21. With reference to column thickness seasonality, which of the following statements is/are correct?

1. Often highest in spring and lowest in autumn teaching neighbourhood.
2. DU is a column measure.
3. DU equals river BOD mg/L.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** DU ≠ BOD.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): A stem pairing Montreal with climate mitigation is usually the Kyoto/Paris trap.

Reason (R): Montreal's core job is ozone ODS control.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the trap. **A/R logic:** Montreal ≠ climate treaty.

Q23. How many of the following are ODS neighbourhood substances?

CFCs, Halon-1211/1301, methyl bromide, HFCs as classic ODS

A. Only one B. Only two C. Only three D. All four

💡 **Show answer**

Ans: C. Only three. **Logic:** HFCs not classic ODS.

Q24. Which one of the following correctly states acid rain vs ozone keys?

A. Acid rain = only Chapman cycle B. Acid rain = only Montreal ODS C. Acid rain = only Dobson Units D. Acid rain driven by SO_2/NOx — not nitric-acid-alone matched to CFC–ozone keys

💡 **Show answer**

Ans: D. SO_2/NOx pathway. **Logic:** Different problem.

Q25. With reference to refrigerants, which of the following statements is/are correct?

1. CFCs used in foam/aerosols/AC.
2. HFCs high-GWP under Kigali.
3. HFCs are the main Antarctic PSC crystals.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** PSCs = polar stratospheric clouds.

Q26. Match List-I with List-II:

List-I	List-II
A. Vienna	1. 2016
B. Montreal signing	2. 1985
C. Kigali	3. 1987

Code:

A. A-1, B-2, C-3 B. A-2, B-3, C-1 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-3, C-1. **Logic:** 1985 / 1987 / 2016.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Stratospheric ozone absorbs ~93–99% of the most damaging UV wavelengths teaching.

Reason (R): Tropospheric ozone is a pollutant and greenhouse gas.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; tropospheric role does not explain stratospheric UV absorption. **A/R logic:** Two ozone roles.

Q28. Arrange chlorine catalytic idea:

A. One Cl harmless always B. Only argon destroys ozone C. One Cl can destroy ~10^5 O_3 molecules before removal D. Only CO_2 destroys ozone

💡 **Show answer**

Ans: C. ~100,000 molecules. **Logic:** Catalytic cycle.

Q29. With reference to World Ozone Day, which of the following statements is/are correct?

1. 16 September (Montreal signing).
2. UN observance from 1994.
3. Identical to World Environment Day 5 June.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not WED.

Q30. Which one of the following correctly states polar depletion strength?

A. Ozone formation slower over poles — depletion effects strongest there B. Poles always have thickest ozone with no loss C. Only tropics have ozone holes D. Only oceans form ozone holes

💡 **Show answer**

Ans: A. Slower formation → strongest depletion effects. **Logic:** Polar vulnerability.

Common Traps

Trap	Correct
Montreal = climate mitigation	**Ozone / ODS**; Kyoto = climate
Ozone layer in troposphere	**Stratosphere** (~90% of O_3)
Ozone hole over India like Antarctica	**Antarctic spring** feature
Vienna = detailed CFC ban alone	**Framework**; Montreal = phase-out
HFC = classic ODS	**High GWP**; Kigali track

Trap	Correct
Ozone Day = 5 June	**16 September**
Hole = physical hole	Column O_3 **< 220 DU**
Good ozone at ground level	Ground O_3 = **bad** pollutant
CFC not a greenhouse gas	CFCs **are** GHGs and ODS
CFC main source = automobiles	Refrigerants / foam / aerosols / electronics cleaners
CFC use = tubeless tyres	**Not** a classic CFC use
Fridge gas = methane	**Mafron**/halocarbon; large plants may use **ammonia**
Ozone absorbs IR as UV-shield role	Protective role = **UV**
UV harm key = liver/brain cancer	Classic key = **skin cancer**
Carbon trading = Montreal	**Kyoto**
Agenda 21 = ozone treaty	**Sustainable Development**
Kyoto = save water / ozone	**GHG / climate**
Ionosphere in homosphere	Homosphere ≈ tropo + strato + meso; **ionosphere not included**
Methyl chloroform not an ODS	It **is** Montreal-controlled ODS

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 37 — Greenhouse Gases

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Carbon Dioxide (CO_2) (कार्बन डाइऑक्साइड) | Methane (CH_4) | Nitrous Oxide (N_2O) | Water Vapour | Ozone | Chlorofluorocarbons (CFCs) (सीएफसी) | Hydrofluorocarbons (HFCs) (एचएफसी)

Sources baked in: IPCC AR6; NCERT; Kyoto basket; Montreal/Kigali; UPPCS PYQs **Weight:** ★★★ High — Argon≠GHG; paddy→CH_4; laughing gas=N_2O; CFC dual role; biogas CH_4+CO_2 **Last verified:** August 2026

Current Affairs (this topic)

- WMO GHG Bulletin and methane CA refresh this twin of Climate Change (जलवायु परिवर्तन).
- GWP order traps (CO_2 / CH_4 / N_2O / fluorinated) stay core.

💡 🎯 **Consolidated — 30 Must-Score Facts (Click to Expand)**

1. A greenhouse gas absorbs outgoing infrared radiation and traps heat in the atmosphere. 2. Without the natural greenhouse effect (प्राकृतिक ग्रीनहाउस प्रभाव) Earth would average about **−18°C**. With it the surface averages about **+15°C**. **Joseph Fourier (1820s)** framed the idea. 3. **CO_2** has **GWP = 1**. It dominates long-term forcing from fossil fuels, cement, and deforestation. 4. Pre-industrial CO_2 was about **280 ppm (~0.03%)**. Present levels are above **415–420 ppm**. 5. Rough WMO-style relative (सापेक्ष) shares: **CO_2 ~64%, CH_4 ~19%, N_2O ~6%, CFCs + others ~11%**. Order: **CO_2 > CH_4 > CFCs > N_2O**. 6. **Methane (CH_4)** has GWP about **28–36**, a shorter lifetime near **12 years**, and strong links to paddies, cattle, landfills, termites, and wetlands (**marsh gas**). 7. **Nitrous oxide (N_2O)** is "laughing gas," has very high GWP (about **265**), and is strongly linked to **fertilizer** use. **Rice** emits both CH_4 and N_2O. 8. **Water vapour** is the most abundant natural greenhouse gas (~half of natural effect), but it mainly acts as a **feedback**, not the primary anthropogenic driver. 9. **Tropospheric ozone** (क्षोभमंडलीय ओजोन) is a greenhouse gas and pollutant. **Stratospheric ozone** (समतापमंडलीय ओजोन) is the UV shield. 10. **CFCs** are both greenhouse gases and ozone-depleting substances and fall under Montreal controls. 11. **HFCs** are greenhouse gases only (not classic ODS) and are phased down under **Kigali 2016**. 12. Gases that are **not** greenhouse gases include **argon, N_2, O_2, helium, hydrogen, and propane**. 13. The Kyoto (क्योटो) basket covers **CO_2, CH_4, N_2O, HFCs, PFCs, and SF_6** (later also **NF_3**). 14. **GWP** compares heat-trapping power to CO_2 over a stated horizon, usually **100 years**. Methane's short-horizon GWP is much higher than CO_2 — CH_4 is more potent per molecule. 15. **CO_2e** converts all greenhouse gases into carbon-dioxide equivalent using GWP. 16. Biogas (बायोगैस) is mainly **methane plus CO_2**. Do not treat biogas as CO_2-only. 17. Paddy fields are a classic **methane** source fact in Indian papers. 18. **Direct** GHGs: CO_2, CH_4, N_2O, CFCs, SF_6, NF_3. **Indirect:** NO_x, CO, NMVOCs, SO_2. 19. CFC versus HFC is dual-role ODS+GHG versus climate-only substitute under Kigali. 20. N_2O is not the same as N_2. Molecular nitrogen is inert as a greenhouse gas. 21. Water vapour rises as Earth warms; that feedback amplifies warming but does not replace fossil CO_2 as the driver. 22. F-gases (HFCs, PFCs, SF_6) are small by mass but can be huge by GWP. 23. Cement process emissions are a major industrial CO_2 source beyond fuel combustion. 24. Do not list argon as a greenhouse gas in a multi-statement stem. 25. Methane's shorter lifetime still matters because near-term warming impact is strong. 26. Ozone's greenhouse role is mainly **tropospheric**; do not call stratospheric ozone the main GHG form. 27. **Carbon fertilization** = higher plant growth from elevated CO_2. 28. **Methane hydrates** in

Arctic/seafloor can release CH_4 if warmed; CH_4 oxidises to CO_2 in about a decade or two. 29. **China** is the largest CO_2 emitter; India ~3rd absolute (निरपेक्ष); **Bhutan** often carbon-negative. 30. CO_2 is both **useful** (photosynthesis) and **harmful** (enhanced greenhouse effect (मानवजनित संवर्धित प्रभाव)) to life on Earth.

💡 ⚡ **Confused Pairs (Click to Expand)**

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Global Warming Potential (GWP)	**Radiative Forcing**	Heat absorption efficiency of 1 kg of a gas over 100 years relative to CO_2 (=1) vs net change in energy balance of Earth (W/m^2) caused by that gas	वैश्विक तापन क्षमता (GWP) / विकिरण दबाव
Carbon Dioxide (CO_2)	**Water Vapour (H_2O)**	Primary anthropogenic driver of long-term climate change (atmospheric residence 100+ yrs) vs dominant natural greenhouse gas acting as a fast temperature feedback	कार्बन डाइऑक्साइड (मानवजनित) / जलवाष्प (प्राकृतिक फीडबैक)
Methane (CH_4)	**Nitrous Oxide (N_2O)**	GWP ~28–30 (residence ~12 yrs; rice paddies, enteric livestock, wetlands) vs GWP ~265–273 (residence ~114 yrs; chemical fertilizers, adipic acid)	मीथेन (धान/पशुपालन) / नाइट्रस ऑक्साइड (उर्वरक)
Sulfur Hexafluoride (SF_6)	**Carbon Dioxide (CO_2)**	Most potent GHG known with GWP >23,500 (used in electrical switchgear) vs benchmark baseline GHG with GWP = 1	सल्फर हेक्साफ्लोराइड (GWP 23,500) / CO_2 (मानक 1)
Kyoto Six GHGs	**Montreal Phase-Out Gases**	CO_2, CH_4, N_2O, HFCs, PFCs, SF_6 (later NF_3 added) targeted for climate reduction vs CFCs, Halons, CCl_4, HCFCs targeted for stratospheric ozone protection	क्योटो 6 ग्रीनहाउस गैसें / मॉन्ट्रियल ओजोन गैसें

💡 📋 **Must-score facts — GWP, shares, Kyoto basket (Click to Expand)**

Gases

Item	Lock
Natural effect temps	~−18°C without; ~+15°C with
CO_2 GWP / levels	**1**; pre-ind ~280 ppm; now >415–420
Shares order	CO_2~64% > CH_4~19% > CFCs+~11% > N_2O~6%
CH_4 GWP / lifetime	~28–36; ~12 years; paddies–cattle–landfills
N_2O	GWP ~**265**; fertilizers; “laughing gas”
Water vapour	Most abundant natural GHG; mainly **feedback**
CFCs / HFCs	GHG+ODS (Montreal) / GHG only (Kigali)
Not GHGs	Ar, N_2, O_2, He, H_2, propane
Kyoto basket	CO_2 CH_4 N_2O HFCs PFCs SF_6 (+NF_3)

37.1 Carbon Dioxide (CO_2)

Identity: Dominant by volume | GWP = 1 | Fossil fuels / cement / deforestation

- **CO_2** is the main anthropogenic GHG by total forcing (~**76%** IPCC-style). Many Indian papers also use WMO-style relative shares: **~64% CO_2, ~19% CH_4, ~6% N_2O, ~11% CFCs + others**.
- **GWP = 1** — the reference gas for converting other gases to **CO_2e**.
- Pre-industrial level ~**280 ppm (~0.03%)**; now **415–420+ ppm**.
- Main human sources: **fossil fuel combustion**, **cement**, **deforestation / land-use change**.
- Persists in the climate system for a very long time compared with methane.
- India ranks among the top absolute emitters (~**3rd**; **China** #1) but has **low per capita** emissions.
- **Carbon fertilization** increases plant growth from higher CO_2 — distinct from warming and ocean acidification (महासागरीय अम्लीकरण).
- CO_2 forms weak carbonic acid in rain (natural pH ~5.6) but is **not** the main driver of damaging acid rain (अम्ल वर्षा) (SO_2/NOx dominate).
- CO_2 is both **beneficial** (photosynthesis) and **harmful** (enhanced greenhouse warming).

PYQ — UPPCS Prelims 2018, Q132

Largest CO_2 emitter in 2015:

Options: A. China B. USA C. India D. France

💡 **Show answer**

Ans: A

> **Logic:** Highest GWP ≠ most total warming. **CO_2 wins on volume.**

37.2 Methane (CH_4)

Identity: Marsh gas | GWP ~28–36 | Paddies / cattle / wetlands | Hydrates

- **Methane** is the second major anthropogenic GHG by forcing (~**16%** IPCC-style; ~**19%** in WMO-style relative-share stems).
- Also called **marsh gas**.
- **GWP ≈ 28–36** (100-year); atmospheric lifetime ~**12 years**. Short-horizon GWP is much higher than CO_2.
- Flooded **rice paddies** create anaerobic soils that emit CH_4. With nitrogen fertiliser, **rice** is also a major **N_2O** source.
- **Livestock** enteric fermentation (cattle/buffalo) is a major Indian source.
- Other sources: **landfills**, coal mines, natural **wetlands** (~**75–76%** of natural CH_4), and **termites**.
- **Methane hydrates** under Arctic tundra (टुंड्रा) and seafloor can release CH_4 if warmed; CH_4 oxidises to CO_2 in roughly a decade or two.
- **CNG** is mainly **methane** (~90%).
- **Biogas** is mainly **methane + carbon dioxide**.

PYQ — UPPCS Prelims 2019, Q62

Gas emitted in paddy fields that increases Earth's temperature:

Options: A. Nitrogen B. Carbon dioxide C. Carbon monoxide D. Methane

💡 **Show answer**

Ans: D

PYQ — UPPCS Prelims 2018, Q63

Methane is emitted from:

Options: A. Paddy fields only B. Termitary only C. Both D. None

💡 **Show answer**

Ans: C

PYQ — UPPCS Prelims 2022, Q113

CNG is mainly composed of—

Options: A. Methane B. Propane C. Ethane D. Butane

💡 **Show answer**

Ans: A

PYQ — UPPCS Prelims 2025, Q28

Major constituents of biogas:

1. Methane 2. Butane 3. Carbon dioxide 4. Carbon monoxide

Options: A. 1 and 3 B. 3 and 4 C. 2 and 3 D. 1 and 2

💡 **Show answer**

Ans: A — **CH_4 + CO_2**.

Logic: Paddy temperature gas = **methane**, not CO or N_2.

37.3 Nitrous Oxide (N_2O)

Identity: Laughing gas | GWP ~265 | Fertilizer soils

- **N_2O** is a long-lived GHG with **GWP ≈ 265** (100-year) and lifetime ~**100+ years**.
- It contributes about **~6%** of anthropogenic GHG forcing.
- Main human source: **nitrogen fertilisers** in soils (nitrification/denitrification).
- Also from combustion, biomass burning, and some industry.
- Common name: **laughing gas** (medical anaesthetic use).
- Do not confuse with **NO/NO_2** (air pollutants) or **ammonia (NH_3)**.
- In Kyoto climate basket; **not** a classic Montreal ODS.

PYQ — UPPCS Prelims 2023, Q105

Which is known as Laughing Gas?

Options: A. Ammonia B. Nitrous Oxide C. Carbon Monoxide D. Chlorofluorocarbon

💡 **Show answer**

Ans: B

Logic: Laughing gas = **N_2O only**.

37.4 Water Vapour

Identity: Most abundant natural GHG | Feedback, not primary driver

- **Water vapour** is the strongest contributor to the **natural** greenhouse effect by volume.
- It is mainly a **feedback**: warming increases evaporation → more vapour → more warming.
- Humans do not control water vapour the way they control CO_2/CH_4/N_2O emission inventories.
- Concentration varies widely (about **0–4%**) with place and weather; most is in the troposphere (क्षोभमंडल).
- Condenses into clouds; linked to the hydrological cycle.

Logic: "Largest anthropogenic GHG" is **not** water vapour — that label belongs to **CO_2** by forcing share.

37.5 Ozone (as Greenhouse Gas)

Identity: Tropospheric O_3 = GHG + pollutant | Stratospheric O_3 = UV shield

- **Tropospheric (ground-level) ozone** is both an air pollutant and a greenhouse gas.
- It forms in photochemical smog from **NOx + VOCs + sunlight**.
- **Stratospheric ozone** is the UV shield — its ma role is protection, not climate forcing.

- "Good up, bad down" remains the safest fact.

PYQ — UPPCS Prelims 2022, Q46

Photochemical smog always contains—

Options: A. Phosphorus B. Ozone C. Methane D. Aluminium ion

> 💡 **Show answer**
>
> **Ans: B**

PYQ — UPPCS Prelims 2020, Q139

NOT a greenhouse gas found naturally (among options):

Options: A. Nitrogen oxide B. CO_2 C. Methane D. Ozone

> 💡 **Show answer**
>
> **Ans: A** — CO_2, CH_4, and O_3 are treated as natural GHGs here.

> **Logic:** Ozone can be a GHG **and** a pollutant — layer decides the story.

37.6 Chlorofluorocarbons (CFCs)

Identity: Synthetic GHG + ODS | Very high GWP | Montreal phase-out

- **CFCs** are human-made compounds once used in refrigeration, aerosols, and foam blowing.
- They are **both** greenhouse gases (**very high GWP**) and **ozone-depleting substances**.
- Controlled and largely phased out under the **Montreal Protocol** (मॉन्ट्रियल प्रोटोकॉल).
- In the atmosphere they are long-lived; UV in the stratosphere (समतापमंडल) releases chlorine that destroys O_3.
- Still appear in "NOT a GHG?" questions as a **true GHG** distractor next to argon.

PYQ — UPPCS Prelims 2022, Q102

Which is NOT a greenhouse gas?

Options: A. Argon B. CFC C. Methane D. Carbon dioxide

> 💡 **Show answer**
>
> **Ans: A** — CFC **is** a GHG.

PYQ — UPPCS Prelims 2025, Q45

Ozone depletion (ओजोन क्षरण) has been linked to CFCs — **TRUE** (with UV-protection statement).

> **Logic:** CFC = climate warmer **and** ozone destroyer.

37.7 Hydrofluorocarbons (HFCs)

Identity: CFC substitutes | GHG only | Kigali 2016

- **HFCs** replaced many CFCs/HCFCs in air-conditioning and refrigeration.
- They **do not deplete ozone** (no chlorine/bromine ODS role) but have **very high GWP**.

- Included in the **Kyoto** (क्योटो) climate basket; phased down under the **Kigali Amendment (2016)** (किगाली संशोधन) to the Montreal Protocol.
- Key fact: **HFC = hot but no hole**.

Gas	ODS?	Climate role	Treaty home
CFC	Yes	High GWP GHG	Montreal
HFC	No	High GWP GHG	Kyoto + Kigali
HCFC	Yes (lower)	Transitional	Montreal phase-out

> **Logic:** Replacing CFCs with HFCs fixed ozone risk but created a climate problem.

Complete PYQ Bank (Topic 37)

Q1. UPPCS Prelims 2025, Q28

Major constituents of biogas:

1. Methane
2. Butane
3. Carbon dioxide
4. Carbon monoxide

A. 1 and 3 B. 3 and 4 C. 2 and 3 D. 1 and 2

Show answer

Logic: Tests biogas composition — methane plus carbon dioxide, not butane or CO alone. **Ans: A.** Biogas is mainly **CH_4 + CO_2** from anaerobic digestion. **Trap:** Butane and carbon monoxide are distractors; paddy gas is methane, not CO.

Q2. UPPCS Prelims 2025, Q45

Consider the following statements:

1. Chlorofluorocarbons (CFCs) are linked to stratospheric ozone depletion.
2. CFCs have no role in the greenhouse effect.

A. Only 1 B. Only 2 C. Both 1 and 2 D. Neither 1 nor 2

Show answer

Logic: CFC dual role — greenhouse gas and ozone-depleting substance under Montreal. **Ans: A (Only 1).** CFCs destroy stratospheric ozone **and** are potent GHGs — statement 2 is false. **Trap:** Treating CFC as ODS-only or as "not a GHG."

Q3. UPPCS Prelims 2023, Q105

Which is known as Laughing Gas?

A. Ammonia B. Nitrous Oxide C. Carbon Monoxide D. Chlorofluorocarbon

Show answer

Logic: Common-name recall — laughing gas is nitrous oxide, not ammonia or CO. **Ans: B.** **N_2O** is laughing gas (GWP ~265; fertiliser soils). **Trap:** Ammonia (NH_3) or CFC as distractors.

Q4. UPPCS Prelims 2022, Q102

Which is NOT a greenhouse gas?

A. Argon B. CFC C. Methane D. Carbon dioxide

> 💡 **Show answer**
>
> **Logic:** Noble gases and dry-air major components are not greenhouse gases; CFC is a GHG. **Ans: A.** **Argon** does not absorb outgoing infrared; CFC, CH_4, and CO_2 are GHGs. **Trap:** Picking CFC because it is "synthetic/industrial."

Q5. UPPCS Prelims 2022, Q46

Photochemical smog always contains—

A. Phosphorus B. Ozone C. Methane D. Aluminium ion

> 💡 **Show answer**
>
> **Logic:** Photochemical smog chemistry always yields tropospheric ozone as a secondary product. **Ans: B.** NOx + VOCs + sunlight → **O_3** (and PAN). **Trap:** Confusing with London SO_2 smog or stratospheric ozone shield.

Q6. UPPCS Prelims 2022, Q113

CNG is mainly composed of—

A. Methane B. Propane C. Ethane D. Butane

> 💡 **Show answer**
>
> **Logic:** CNG fuel composition — compressed natural gas is mainly methane. **Ans: A.** **CNG ≈ 90% methane.** **Trap:** Propane/butane as LPG distractors.

Q7. UPPCS Prelims 2019, Q62

Gas emitted in paddy fields that increases Earth's temperature:

A. Nitrogen B. Carbon dioxide C. Carbon monoxide D. Methane

> 💡 **Show answer**
>
> **Logic:** Flooded paddy fields emit marsh gas (methane) under anaerobic soils. **Ans: D.** Anaerobic rice paddies release **CH_4**. **Trap:** CO_2 or CO — paddies are classic methane stems.

Q8. UPPCS Prelims 2018, Q63

Methane is emitted from:

A. Paddy fields only B. Termitary only C. Both D. None

> 💡 **Show answer**
>
> **Logic:** Methane sources include wetlands, paddies, and termite mounds — not "paddy only." **Ans: C.** **Both** paddies and termites emit CH_4 (also landfills, cattle, wetlands). **Trap:** "Only one source" options.

Q9. UPPCS Prelims 2018, Q132

Largest CO_2 emitter in 2015:

A. China B. USA C. India D. France

💡 **Show answer**

Logic: Absolute CO_2 emissions ranking — China was largest emitter in 2015 framing. **Ans: A.** **China** #1 by absolute CO_2; India ~3rd with low per capita. **Trap:** Picking India because of population alone.

Q10. UPPCS Prelims 2020, Q139

Which is NOT a greenhouse gas found naturally (among the options)?

A. Nitrogen oxide B. Carbon dioxide C. Methane D. Ozone

💡 **Show answer**

Logic: Among listed options, nitrogen oxides are pollutants — CO_2, CH_4, and O_3 are treated as natural GHGs in the stem. **Ans: A.** **NO/NO_2** are air pollutants, not classic GHGs in this list; CO_2, CH_4, and tropospheric O_3 qualify. **Trap:** Confusing N_2O (GHG) with nitrogen oxide.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to the greenhouse effect, which of the following statements is/are correct?

1. GHGs absorb outgoing IR and trap heat.
2. Without the natural greenhouse effect Earth would be about +15°C; with it about −18°C.
3. Argon is the main greenhouse gas.

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

💡 **Show answer**

Ans: B. Only 1 is correct. **Logic:** Statement 1 is the GHG heat-trap mechanism. Statement 2 reverses the −18°C / +15°C teaching pair. Statement 3 is false — argon is not a main GHG.

Q2. Match List-I with List-II:

List-I	List-II
A. CO_2 GWP	1. ~28–36; lifetime ~12 yr
B. CH_4 GWP	2. 1
C. N_2O	3. Very high GWP (~265); fertiliser link

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** GWP ladder.

Q3. Which of the following pairs is/are correctly matched?

1. Pre-industrial CO_2 was about 415–420 ppm.
2. Present >415–420 ppm teaching
3. WMO-style share order is N_2O > CFCs > CH_4 > CO_2.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

Show answer

Ans: A. Only 2 is correct. **Logic:** Statement 2 is the present >415–420 ppm teaching. Statement 1 swaps pre-industrial (~280 ppm) with present levels. Statement 3 reverses the usual CO_2 > CH_4 > CFCs > N_2O share order.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Water vapour is the most abundant natural GHG (~half of natural effect).

Reason (R): It mainly acts as a feedback, not the primary anthropogenic driver.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: D. Both true and R clarifies its role vs fossil CO_2. **A/R logic:** Feedback ≠ primary driver.

Q5. With reference to CFC vs HFC, which of the following statements is/are correct?

1. CFCs = ODS + GHG (Montreal).
2. HFCs = GHG only; Kigali phase-down.
3. HFCs are classic ODS identical to CFCs.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: A. Only 1 and 2. **Logic:** HFC ≠ classic ODS.

Q6. Arrange Kyoto basket teaching:

A. Only argon and N_2 B. CO_2, CH_4, N_2O, HFCs, PFCs, SF_6 (later NF_3) C. Only SO_2 and dust D. Only BOD and COD

Show answer

Ans: B. Kyoto six (+ NF_3 later). **Logic:** Basket contents.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Paddy fields — classic CH_4 source
2. Biogas — mainly CH_4 + CO_2
3. Biogas — CO_2-only always

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

Show answer

Ans: C. Only pair 3 is not correctly matched. **Logic:** Not CO_2-only.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): N_2O is not the same as N_2.

Reason (R): Molecular nitrogen is inert as a greenhouse gas.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R supports the separation. **A/R logic:** $N_2O \neq N_2$.

Q9. With reference to direct vs indirect GHGs, which of the following statements is/are correct?

1. Direct: CO_2, CH_4, N_2O, CFCs, SF_6, NF_3.
2. Indirect: NO_x, CO, NMVOCs, SO_2.
3. Argon is a direct Kyoto gas.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Argon not GHG.

Q10. Match List-I with List-II:

List-I	List-II
A. Tropospheric O_3	1. UV shield
B. Stratospheric O_3	2. GHG + pollutant
C. CO_2e	3. Convert GHGs via GWP

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** O_3 roles + CO_2e.

Q11. Arrange non-GHG set:

A. Only CO_2 B. Argon, N_2, O_2, helium, hydrogen, propane teaching C. Only CH_4 D. Only N_2O

💡 **Show answer**

Ans: B. Non-GHG list. **Logic:** Do not list argon as GHG.

Q12. With reference to methane, which of the following statements is/are correct?

1. Links to paddies, cattle, landfills, termites, wetlands neighbourhood.
2. Short-horizon GWP much higher than 100-year figure teaching.
3. Longer lifetime than CO_2 always.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** CH_4 lifetime shorter (~12 yr).

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): F-gases are small by mass but can be huge by GWP.

Reason (R): HFCs, PFCs, SF_6 sit in the F-gas set.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; membership list does not itself explain mass-vs-GWP asymmetry. **A/R logic:** Mass vs forcing.

Q14. Which one of the following correctly states carbon fertilization?

A. Only Dobson Unit rise B. Ocean acidification synonym C. Only methane hydrate melt D. Higher plant growth from elevated CO_2

💡 **Show answer**

Ans: D. Elevated-CO_2 growth effect. **Logic:** ≠ acidification.

Q15. How many of the following emitter facts are correct?

China largest CO_2; India ~3rd absolute; Bhutan often carbon-negative

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Emitter map.

Q16. With reference to cement / industry, which of the following statements is/are correct?

1. Cement process emissions are major industrial CO_2 beyond fuel combustion.
2. CO_2 useful in photosynthesis and harmful via enhanced GE.
3. CO_2 has GWP defined as 1.

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

💡 **Show answer**

Ans: B. All three. **Logic:** Industrial + dual role + GWP.

Q17. Match List-I with List-II:

List-I	List-II
A. Methane hydrates	1. Feedback amplifier
B. Water vapour	2. Arctic/seafloor CH_4 release risk
C. Rice	3. Emits CH_4 and links to N_2O fertiliser neighbourhood

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Hydrates / vapour / rice.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): GWP compares heat-trapping power to CO_2 over a stated horizon (usually 100 years).

Reason (R): CO_2e uses GWP to convert other gases into CO_2-equivalent.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; CO_2e definition does not explain what GWP itself measures. **A/R logic:** GWP vs CO_2e.

Q19. Which of the following pairs is/are correctly matched?

1. Fourier — greenhouse-effect framing neighbourhood
2. Tropospheric ozone — main GHG form of ozone
3. Stratospheric ozone — main GHG form of ozone

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Stratospheric O_3 = UV shield, not main GHG form.

Q20. Arrange CH_4 oxidation teaching:

A. CH_4 oxidises to CO_2 in about a decade or two teaching B. Never oxidises C. Instantly becomes argon D. Becomes only N_2

💡 **Show answer**

Ans: A. Oxidises to CO_2. **Logic:** Lifetime/two-decade idea.

Q21. With reference to N_2O, which of the following statements is/are correct?

1. Called laughing gas.
2. Strong fertiliser link.
3. GWP about equal to 1 like CO_2.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Very high GWP (~265).

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Water vapour rises as Earth warms and amplifies warming.

Reason (R): That feedback replaces fossil CO_2 as the primary driver.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Feedback does not replace fossil CO_2 driver.

Q23. How many of the following are greenhouse gases?

CO_2, CH_4, N_2O, argon

A. Only one B. Only two C. Only three D. All four

💡 **Show answer**

Ans: C. Only three. **Logic:** Argon out.

Q24. Which one of the following correctly states SF_6 / NF_3 neighbourhood?

A. Only BOD metrics B. Only natural water vapour C. Only argon isotopes D. High-GWP F-gases in Kyoto/expanded basket teaching

> 💡 **Show answer**
>
> **Ans: D.** F-gas basket members. **Logic:** Huge GWP.

Q25. With reference to relative shares, which of the following statements is/are correct?

1. CO_2 ~64%; CH_4 ~19%; N_2O ~6%; CFCs+others ~11% teaching.
2. Order CO_2 > CH_4 > CFCs > N_2O.
3. N_2O share exceeds CO_2.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1 and 2. **Logic:** CO_2 dominates share.

Q26. Match List-I with List-II:

List-I	List-II
A. Kigali	1. ODS + GHG
B. CFCs	2. HFC phase-down
C. Propane	3. Not a GHG in teaching list

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: B.** A-2, B-1, C-3. **Logic:** Kigali/CFC/propane.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Methane's shorter lifetime still matters because near-term warming impact is strong.

Reason (R): Short-horizon GWP for methane is much higher than the usual 100-year figure.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains why near-term impact matters. **A/R logic:** Horizon sensitivity.

Q28. Arrange CO_2 dual role:

A. Only harmful always with no photosynthesis role B. Only argon substitute C. Useful in photosynthesis; harmful via enhanced greenhouse effect D. Only ozone ODS

> 💡 **Show answer**
>
> **Ans: C.** Useful + harmful. **Logic:** Dual role.

Q29. With reference to GWP horizon, which of the following statements is/are correct?

1. Usually compared over 100 years.

2. Horizon choice changes methane's apparent power.
3. GWP of CO_2 is defined as 100.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** CO_2 GWP = 1.

Q30. Which one of the following correctly states laughing gas in this chapter?

A. Nitrous oxide (N_2O) B. Argon C. Molecular nitrogen only D. Helium

💡 **Show answer**

Ans: A. N_2O. **Logic:** Laughing gas = N_2O.

Common Traps

Trap	Correct
Argon / N_2 / O_2 are GHGs	**Not**
Hydrogen / propane are GHGs	**Not**
Paddy gas = CO_2 or CO	**Methane**
Laughing gas = ammonia / CFC	**N_2O**
Biogas = butane / CO	**CH_4 + CO_2**
Highest GWP = most total warming	**CO_2 dominates by volume**
Water vapour = primary human driver	**Feedback** / most abundant **natural** GHG
CFC is not a GHG	**It is** (and ODS)
SO_2 warms Earth directly	**Indirect** / acid-rain role
Fourier missing	**Joseph Fourier (1820s)**
HFC = ODS	**GHG only**; Kigali
Stratospheric O_3 = smog pollutant	**UV shield**; bad O_3 is tropospheric
CNG = butane	**Methane**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 38 — Pollution (Advanced)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Air Pollution (Sources | Air Quality | AQI | NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) | Smog | Photochemical Smog | Industrial | Vehicular | Indoor) | Water Pollution (Surface | Groundwater | Drinking | Standards) | Soil Pollution (Fertilizers | Pesticides | Heavy Metals) | Noise Pollution (Sources | Standards) | Marine Pollution (Coastal | Ocean) | Plastic Pollution (PWM)

Sources baked in: CPCB NAAQS/noise; NCAP 2019; SAFAR; IS 10500; CGWB; PWM Rules; MARPOL; UPPCS PYQs
Weight: ★★★ High — lichens; smog secondary; BOD; CGWB≠CPCB; Moradabad noise; fly ash **Last verified:** August 2026

Current Affairs (this topic)

- Advanced pollution CA mirrors Pollution & Waste — AQI / CAQM / waste-rule updates.
- Keep criteria-pollutant chemistry pairs consistent with the core pollution chapter.

💡 🎯 **Consolidated — 38 Must-Score Facts (Click to Expand)**

1. **Primary** air pollutants are emitted directly (SO_2, PM, CO, NOx, lead). **Secondary** pollutants form in air (O_3, PAN, photochemical smog, acid mist). 2. **Lichens** are pollution indicators and do **not** grow well in badly polluted air. 3. India's AQI uses **eight** pollutants (**PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb**) — **not CO_2/CH_4**. Bands: **Good (0–50)** to **Severe (401–500)**. NAQI launched **2014**. 4. **NCAP (2019)** targets non-attainment cities with about a **20–30%** particulate cut goal. 5. **London smog** is SO_2 plus smoke. **Photochemical smog** needs NOx, VOCs, and sunlight and yields **ozone + PAN**; **PAN** is a strong eye irritant. 6. **Fly ash** is a thermal-power primary pollutant usable in bricks/cement; oxides SiO_2/Al_2O_3/CaO **plus toxic metals**. **BS-VI** began **1 Apr 2020**. 7. **SAFAR** (सफर) is an IITM air-quality and weather forecast system. **WAYU** units were installed in **Delhi**. 8. **CO** forms carboxyhaemoglobin (**~200–300×** affinity vs O_2). **Radon** is the top indoor air-pollutant teaching fact. 9. **BOD** measures organic pollution through microbial oxygen demand. High BOD means worse water quality. 10. **COD** is usually greater than or equal to BOD. Clean water often shows DO above about **6 mg/L** and BOD below about **3 mg/L**. 11. **Eutrophication** (सुपोषण) is nutrient (N/P) overload → algal bloom → dissolved-oxygen crash — water, not air. 12. The **Central Ground Water Board (CGWB)** (केंद्रीय भूजल बोर्ड) issues the groundwater quality report. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) is not CGWB. 13. Classic arsenic hotspot teaching points to **Bangladesh** / Gangetic plains. **Knock-knee** = fluoride; **Black Foot** = arsenic; **Minamata** = Hg; **Itai-itai** = Cd. 14. **Damodar** (दामोदर) = biological desert. Ganga (गंगा) BOD peak classically **Kanpur (कानपुर)–Allahabad (इलाहाबाद)**. Leather industry = high chemical pollution. 15. Many pesticides are **not** readily biodegradable — a frequent true statement in papers. 16. Heavy-metal facts include **chromium** around Kanpur plus lead, cadmium, and mercury elsewhere. Mercury sources include **gold mining** and **coal TPPs**; **no known safe exposure level**. 17. Daytime noise standards (dB): Silence **50**, Residential **55**, Commercial **65**, Industrial **75**. **Green muffler** = tree belts for noise. 18. **Moradabad** was flagged among the world's noisiest cities in UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) Frontier reporting. 19. About **80%** of marine pollution comes from land. **MARPOL** governs ship-source pollution. **IMO HQ = London**. Ocean acidification (महासागरीय अम्लीकरण) harms calcareous life and corals.

20. Plastic Waste Management Rules date from **2016**. The SUP ban push is **2022**. EPR is the producer-responsibility tool. Microbeads harm marine ecosystems. 21. India ranks among the world's largest plastic-waste generators (often taught as about **third**). 22. Indoor air pollution is driven by solid cooking fuels, tobacco smoke, poor ventilation, and **radon**. 23. Surface-water pollution includes sewage, industrial effluent, and agricultural runoff into rivers and lakes. 24. Drinking-water quality in India is commonly referenced to **IS 10500** standards. 25. Vehicular emissions: **catalytic converters** (Pt/Pd/Rh); Euro-II diesel S **≤0.05%**; India leapfrogged to **BS-VI (2020)**. 26. Coastal pollution includes oil, plastics, sewage, and industrial discharge along the shoreline belt. 27. Do not call photochemical smog a simple SO_2–smoke fog. Ozone chemistry is the key. 28. SAMEER-type apps publish AQI for public warning; they do not replace NCAP city action plans. 29. **CCAC** focuses on **methane, black carbon, HFCs** — not a G20-only initiative. 30. **Asian Brown Cloud** is especially associated with **South Asia**. Bag filter ≠ cyclone divider. 31. **Bioremediation** cleans via microbes; **Cd/Pb** not readily treated; **Oilzapper** (TERI) for oily waste. 32. **Bhopal MIC (Dec 1984)**; **Chernobyl (1986)** radioactive disaster teaching pair. 33. **SWM Rules 2016** give elaborate landfill/site criteria. **R2 Code** = responsible electronics recycling. 34. **H-CNG** reduces CO/HC/CO_2 vs CNG but does **not eliminate** CO; up to ~**1/5** H_2 by volume; not cheaper than CNG. 35. Geoengineering (भू-इंजीनियरिंग) teaching: **cirrus thinning** and **sulphate aerosol** injection aim at cooling / offsetting warming — not cyclone control. 36. Steel industry emits **CO, CO_2, SOx, and NOx**. **Aflatoxins** = mould toxins on poorly stored grains. 37. Excess **nitrogen and phosphorus** fertilisers drive **eutrophication**, raise nitrate in groundwater, and link to **N_2O** climate forcing — Indo-Gangetic wheat–rice systems are high-input geography. 38. Fertiliser pollution is mostly **nutrient overload**, not heavy-metal tanning chemistry.

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Primary Air Pollutant	**Secondary Air Pollutant**	Emitted directly from identifiable sources (CO, SO_2, NO, PM) vs formed in atmosphere through photochemical reactions (Ozone, PAN, Smog, Acid Rain)	प्राथमिक प्रदूषक (सीधे उत्सर्जित) / द्वितीयक प्रदूषक (रासायनिक क्रिया)
Biological Oxygen Demand (BOD)	**Chemical Oxygen Demand (COD)**	Amount of dissolved O_2 required by aerobic bacteria to decompose organic matter vs amount of oxygen required to chemically oxidize all biodegradable and non-biodegradable matter	जैव रासायनिक ऑक्सीजन मांग (BOD) / रासायनिक ऑक्सीजन मांग (COD)
Classical (London) Smog	**Photochemical (Los Angeles) Smog**	Occurs in cool, humid winter mornings; sulfurous, reducing, coal-smoke based vs occurs in dry, sunny afternoons; oxidizing, auto-exhaust NOx + VOCs generating ground ozone and PAN	लंदन स्मॉग (सल्फ्यूरस, अपचायक) / प्रकाश रासायनिक स्मॉग (ऑक्सीकारक)
Central Pollution Control Board (CPCB)	**Central Ground Water Board (CGWB)**	Statutory body under Water Act 1974 for overall air/water quality monitoring vs apex subordinate office under Jal Shakti Ministry for groundwater management	सीपीसीबी (प्रदूषण नियंत्रण) / सीजीडब्ल्यूबी (भूजल बोर्ड)
SAFAR	**NCAP**	Ministry of Earth Sciences real-time metropolitan air quality forecasting system vs MoEFCC national targeted programme to cut PM2.5/PM10 by 20–30% across 131 non-attainment cities	सफर (सटीक पूर्वानुमान प्रणाली) / राष्ट्रीय स्वच्छ वायु कार्यक्रम

💡 📋 Must-score facts — AQI, smog, BOD, disease traps (Click to Expand)

Air / water

Item	Lock
Primary / Secondary	Direct emit / form in air
AQI 8	PM10 PM2.5 NO_2 CO O_3 SO_2 NH_3 Pb
NCAP / BS-VI	**2019** / **1 Apr 2020**
London / Photochemical	SO_2+smoke / NOx+VOC+sun → O_3+PAN
SAFAR	IITM AQ + weather forecast
BOD / COD / clean water	Microbial O_2 demand / COD≥BOD; DO>~6, BOD<~3 mg/L
Eutrophication	N/P overload → bloom → DO crash (**water**)
CGWB ≠ CPCB	GW quality report / ambient air–surface water
Disease locks	Fluoride knock-knee · As Black Foot · Hg Minamata · Cd Itai-itai
Damodar / Ganga BOD	Biological desert / Kanpur–Allahabad peak class

38.1 Sources of Air Pollution

Identity: Point / non-point | Sectoral | Primary pollutants

- **Point sources** are chimneys, stacks, and pipes; **non-point sources** include roads, farms, and open burning.
- Major sectors: power, industry, vehicles, construction dust, biomass/stubble burning, and households.
- Primary pollutants include **SO_2, NOx, CO, PM, and VOCs**.
- UP winter peaks link stubble smoke, vehicles, industry, and Indo-Gangetic inversion.

> **Logic:** Stubble smoke is a classic non-point seasonal source in north India.

38.2 Air Quality

Identity: NAAQS | Monitoring | Lichens | SAFAR

- **Air quality** is judged against CPCB **National Ambient Air Quality Standards (NAAQS)**.
- Key pollutants monitored: PM2.5, PM10, SO_2, NO_2, O_3, CO, NH_3, Pb, and related species.
- **Lichens** are sensitive bio-indicators of air pollution, especially SO_2 stress.
- They indicate pollution because they are **sensitive** — they do **not** grow well in polluted air.
- **SAFAR** (System of Air Quality and Weather Forecasting and Research) is run with **IITM** support for forecast products.

PYQ — UPPCS Prelims 2025, Q61

1. Lichens are very good indicators of pollution.
2. Lichens grow well in polluted areas.

Options: A. Only 2 B. Neither C. Both D. Only 1

> 💡 **Show answer**
>
> **Ans: D** — Stmt 1 true; stmt 2 false.

PYQ — UPPCS Prelims 2018, Q84

SAFAR inaugurated (July 2018) by:

> 💡 **Show answer**
>
> **Dr. Harsh Vardhan** (option C in that paper).

> **Logic:** Indicator ≠ thrives in dirty air.

38.3 Air Quality Index (AQI)

Identity: CPCB 0–500 | Eight pollutants | Health categories | Sub-indices

- India's **AQI** (CPCB) runs from **0 to 500**. **National Air Quality Index (NAQI)** launched **17 October 2014** under the "one number–one colour–one description" idea.
- Categories: **0–50 Good**, **51–100 Satisfactory**, **101–200 Moderate (नरम दल)**, **201–300 Poor**, **301–400 Very Poor**, **401–500 Severe**.
- Eight pollutants: **PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb**. **CO_2 and CH_4 are not AQI parameters** (they are climate/GHG metrics).

- CO and O_3 use **8-hour** averages; most others use **24-hour** averages. CO is reported in **mg/m³**; NO_2/O_3 typically in **µg/m³**.
- AQI uses sub-indices for pollutants and reports the worst as the day's AQI.
- PM2.5 and PM10 often drive winter AQI in UP NCR/plain cities.
- WHO guideline teaching (updated): annual PM2.5 **≤5 µg/m³**, 24-h PM2.5 **≤15 µg/m³**. **PM2.5** (not PM10) can enter the bloodstream; ozone can trigger asthma. Highest ozone is **not** typically in "inclement weather."

AQI	Category
0–50	Good
51–100	Satisfactory
101–200	Moderate
201–300	Poor
301–400	Very Poor
401–500	Severe

> **Logic:** Higher AQI = worse air. CO_2/CH_4 ≠ AQI.

38.4 National Clean Air Programme (NCAP)

Identity: 2019 | 122 cities | 20–30% PM reduction target

- **NCAP** (राष्ट्रीय स्वच्छ वायु कार्यक्रम) launched in **2019** to improve air quality in non-attainment cities.
- Covers **122 cities** with city action plans, monitoring, and source control.
- Target framework: about **20–30% reduction in PM concentrations** over the programme timeline (baseline-linked).
- UP NCAP cities include Lucknow (लखनऊ), Kanpur, Ghaziabad, Agra, Varanasi (वाराणसी), Meerut (मेरठ), Noida, Prayagraj (प्रयागराज), Moradabad, Bareilly, and others notified over time.
- Complements BS-VI fuel, dust control, and industrial CETP/stack measures.

> **Logic:** NCAP = city clean-air programme, not a groundwater body.

38.5 Smog

Identity: Smoke + fog origin | Secondary pollutant | London type

- **Smog** is a secondary air pollution mixture; classical **London smog** is linked to **SO_2 + smoke** in cold, humid air.
- Smog is a **secondary pollutant** (formed in the atmosphere), unlike CO or fly ash emitted directly.
- wording often links smog to **oxides of nitrogen and sulphur**.

PYQ — UPPCS Prelims 2018, Q121

Which is the secondary pollutant?

Options: A. Smog B. CO_2 C. CO D. Fly ash

> 💡 **Show answer**
>
> **Ans: A**

PYQ — UPPCS Prelims 2019, Q136

Smog is essentially caused by the atmospheric presence of

Options: A. Oxygen and ozone B. Ozone and nitrogen C. Oxygen and nitrogen D. Oxide of nitrogen and sulphur

💡 **Show answer**

Ans: D

> **Logic:** Smog = secondary. Fly ash = primary (and from thermal plants).

38.6 Photochemical Smog

Identity: NOx + VOC + sunlight → O_3 / PAN | LA type | Eye irritation

- **Photochemical smog** forms when **NOx and VOCs/hydrocarbons** react in strong sunlight.
- Engine path: $N_2 + O_2 \rightarrow NO$; then $NO \rightarrow NO_2$; NO_2 + light → NO + O; $O + O_2 \rightarrow$ **O_3**; hydrocarbons + NOx → **PAN**.
- It always involves **ozone** as a key secondary product; **PAN**, formaldehyde, and aldehydes also appear.
- **PAN (peroxyacetyl nitrate)** is a powerful **eye irritant** (lachrymator) and damages plant chloroplasts.
- Typical of traffic-rich sunny cities (Los Angeles type); also relevant to Indian summer/pre-monsoon urban (नगरीय) episodes.
- The brown-orange haze of photochemical smog is a **secondary** pollution state — SO_2 stack gas alone is **primary**.

PYQ — UPPCS Prelims 2022, Q46

Photochemical smog always contains—

Options: A. Phosphorus B. Ozone C. Methane D. Aluminium ion

💡 **Show answer**

Ans: B

> **Logic:** Photochemical smog ≠ classical London SO_2 fog. PAN ≠ primary stack emission.

38.7 Industrial Emissions

Identity: Stacks / SPM / SO_2 | Fly ash | CETP | Kanpur tanneries

- Industry emits PM, SO_2, NOx, VOCs, and process-specific toxins.
- **Fly ash** pollution is caused by **thermal power plants**.
- Fly ash is used for **bricks** and as a **Portland cement** replacement in concrete.
- Main chemical components: **SiO_2, Al_2O_3**, and sometimes **CaO**, plus toxic elements such as **lead, arsenic, and copper** — not "oxides only."
- Control tools: ESPs, **bag filters** (finer PM, often **<~50 µm**), **cyclone separators/collectors** (coarser, **>~50 µm**), FGD, wet scrubbers, and **CETPs** for clusters.
- **Cyclone divider** is **not** a standard instrumental device name for air-pollution control.
- Mercury teaching: **gold mining** and **coal TPPs** are major sources; **no known safe level** of mercury exposure.
- UP geography: Kanpur tanneries (chromium/organics), Singrauli thermal belt, brick kilns.

PYQ — UPPCS Prelims 2021, Q143

Flyash pollution is caused by

Options: A. Oil refining B. Fertilizer industry C. Thermal power plant D. Mining

💡 **Show answer**

Ans: C

> **Logic:** Fly ash = thermal power **primary** pollutant; smog = secondary. Statement "fly ash = SiO_2 and CaO only, no toxics" is false.

38.8 Vehicular Emissions

Identity: CO / NOx / PM / VOCs | BS-VI 2020 | Urban traffic

- Vehicles emit **CO, NOx, PM, VOCs**, and CO_2; incomplete combustion and cigarette smoke classically yield **CO**.
- Lead antiknock in petrol was a major source of airborne lead toxicity.
- India moved to **BS-VI** norms nationwide from **1 April 2020**, leapfrogging BS-V after **BS-IV** from **1 April 2017**.
- Euro norms limit CO, hydrocarbons, NOx, and PM. For **Euro-II**, ultra-low sulphur diesel needs sulphur **≤0.05%**.
- **Catalytic converters** use **platinum, palladium, and rhodium** to convert CO → CO_2, NOx → N_2, and HC → CO_2 + H_2O.
- Congestion, diesel fleets, and two-wheelers drive city NO_2/PM peaks.
- Complementary tools: EV push, PUC checks, odd-even-type episodic measures, CAFE efficiency norms.
- **H-CNG** (hydrogen-enriched CNG) can cut CO/HC/CO_2 versus CNG; hydrogen up to about **one-fifth** by volume can be blended; it does **not eliminate** CO and is **not** cheaper than CNG.
- **CCAC** (Climate and Clean Air Coalition (गठबंधन), **2012**, with UNEP) focuses on **methane, black carbon, and HFCs** — statement that it is a unique G20-only initiative is false.

> **Logic:** BS-VI cut sulphur in fuel and tightened NOx/PM limits versus BS-IV. CCAC ≠ G20 club.

38.9 Indoor Air Pollution

Identity: Biomass chulha | PM2.5 | Women/children exposure

- Indoor pollution comes from **biomass cookstoves**, kerosene, incense, and tobacco smoke.
- Cigarette smoke contains **CO, HCN, NOx, formaldehyde, acrolein, benzene**, and small amounts of **nicotine**.
- Benzene exposure sources include automobile exhaust, tobacco smoke, wood burning, varnished furniture, and polyurethane products.
- The most significant classic **indoor** pollutant gas in many stems is **radon** — colourless, odourless, radioactive, from soil/radium, linked to lung cancer.
- Rural indoor PM2.5 can exceed outdoor levels and heavily affects women and children.
- Clean cooking (LPG/PNG/electric) is the main mitigation path (Ujjwala-type coverage).
- **Asian Brown Cloud** / atmospheric brown cloud is especially noted over **South Asia** (often Jan–Mar), rich in aerosols including black carbon and sulphate.

> **Logic:** Indoor air is a household health issue, not only an outdoor AQI story. Radon ≠ SO_2 as "most important indoor."

38.10 Surface Water Pollution

Identity: BOD / COD / DO | Sewage / industry | Eutrophication

- Surface waters are polluted by sewage, industrial effluent, and agricultural runoff.
- **BOD** measures oxygen used by microbes decomposing organic matter — a standard **aquatic pollution assay**.
- Higher BOD means worse organic pollution; **COD ≥ BOD**.
- Clean-ish river fact: **DO > ~6 mg/L**, **BOD < ~3 mg/L**. Teaching banks often call DO **<4 mg/L** highly polluted.
- **Eutrophication:** excess **N and P** → algal bloom → DO crash → fish kill. Eutrophication is **not** an air-pollution process.
- Highest Ganga **BOD** stretch is classically **Kanpur–Allahabad**. **Damodar** is the classic **biological desert** river.
- Leather-industry effluent is a top chemical pollution source for water and soil.
- UP rivers: Ganga (Kanpur stretch), Gomti (Lucknow), Yamuna (Mathura (मथुरा)–Agra influence).

PYQ — UPPCS Prelims 2023, Q103

Biological Oxygen Demand (BOD) (जैव रासायनिक ऑक्सीजन मांग) is a standard criterion for—

Options: A. Pollution assay in aquatic ecosystems B. High-altitude O_2 C. Blood O_2 D. Forest O_2

💡 **Show answer**

Ans: A

> **Logic:** BOD = water organic pollution test, not blood oximetry.

38.11 Groundwater Pollution

Identity: Arsenic / fluoride / nitrate | CGWB reports

- Groundwater contaminants include **arsenic, fluoride, nitrate, iron, salinity**, uranium, and industrial leachates.
- **Arsenic** contamination is classically associated with the **Bangladesh** / lower Gangetic basin story; first Indian notice often dated to **West Bengal, 1978**.
- Affected Indian geography teaching: West Bengal, Jharkhand, Bihar, UP, Assam, Manipur, and parts of Chhattisgarh.
- Chronic arsenic exposure links to cancers and **Black Foot** disease teaching. WHO guideline often cited as **0.01 mg/L**.
- **Fluoride** excess causes fluorosis and **knock-knee** syndrome. **Nitrate** links to blue-baby risk.
- The **Annual Groundwater Quality Report** is prepared by the **Central Ground Water Board (CGWB)** — not CPCB.

PYQ — UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 prepared by:

1. CPCB 2. CWC (केंद्रीय जल आयोग) 3. CGWB

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: B

PYQ — UPPCS Prelims 2022, Q63

Which country has natural arsenic water contamination?

Options: A. Sri Lanka B. Bangladesh C. Bhutan D. Pakistan

💡 **Show answer**

Ans: B

> **Logic:** CGWB = groundwater. CPCB = pollution control board standards/monitoring.

38.12 Drinking Water Pollution

Identity: Pathogens | Coliform | Fluorosis / arsenicosis | IS 10500

- Unsafe drinking water carries pathogens (diarrhoea, cholera, typhoid risk) and chemical toxins.
- Microbial safety uses **coliform / E. coli** indicators.
- Chronic chemical risks: **fluorosis** (fluoride), **arsenicosis** (arsenic), nitrate (blue-baby risk in infants).
- Treatment chain: source protection → filtration/disinfection → safe storage.

> **Logic:** Microbial and chemical risks are different drinking-water failure modes.

38.13 Water Quality Standards

Identity: Water Act 1974 | BIS IS 10500 | CPCB river classes

- **Water (Prevention and Control of Pollution) Act, 1974** is the core water-pollution statute.
- **BIS IS 10500** sets drinking-water specification limits.
- CPCB classifies river stretches for designated best uses (A–E style classes).
- CPCB/SPCB enforce consent, standards, and monitoring for effluent and ambient water.

> **Logic:** Drinking standard (IS 10500) ≠ ambient river class alone.

38.14 Fertilizers and Environment (पर्यावरण)

Identity: N/P runoff | Eutrophication | Urea overuse

- Excess **nitrogen and phosphorus** fertilisers run off into water bodies.
- Result: eutrophication, nitrate groundwater rise, and N_2O climate link (Topic 37).
- Indo-Gangetic wheat–rice systems are high-input geography.

> **Logic:** Fertiliser pollution is mostly nutrient overload, not heavy-metal tanning.

38.15 Pesticides and Environment

Identity: Persistence | Biomagnification (जैव आवर्धन) | Not biodegradable

- Many pesticides are **persistent** and can **biomagnify** along food chains.
- UPPCS treats **pesticide** as **not a biodegradable pollutant** versus domestic waste/urine/faecal matter.

PYQ — UPPCS Prelims 2023, Q120

Which is not a biodegradable pollutant?

Options: A. Domestic waste B. Urine C. Faecal matter D. Pesticide

💡 **Show answer**

Ans: D

> **Logic:** Silent Spring-era logic: persistent pesticides harm ecosystems.

38.16 Heavy Metal Pollution

Identity: Pb / Cd / Cr / Hg | Bioaccumulation (जैव संचय) | Kanpur Cr

- Heavy metals do not degrade; they **bioaccumulate**.
- Key facts: **Pb** (batteries/paint/antiknock petrol legacy), **Cd** (batteries/industry — **Itai-itai**), **Cr** (tanneries — Kanpur), **Hg** (**Minamata**; gold mining + coal TPP sources; **no known safe exposure level**).
- Asbestos dust → lung disease / mesothelioma teaching. Soil particles → breathing ailments. NOx → cancer risk teaching in some match lists. CO displaces oxygen in blood.
- Soil and sediment act as long-term sinks near industrial clusters.

Metal / pollutant	Classic disease / effect
Cr	Kanpur tanneries
Hg	Minamata / neurological
Pb	Neurotoxin; IQ / CNS
Cd	Itai-itai / bone–kidney
As	Black Foot / cancers
F	Knock-knee / fluorosis
Asbestos	Lung scarring / cancer

> **Logic:** Heavy metals ≠ biodegradable organics. Fluoride ≠ "indigestion only."

38.17 Sources of Noise Pollution

Identity: Traffic / industry / construction | Moradabad UNEP fact

- Main sources: road traffic, industry, construction, railways, aircraft, loudspeakers, and generators.
- Jet flights are among the most intense common outdoor noise sources; sonic boom = Mach-scale teaching.
- **Green muffler** = multi-row roadside tree belts that absorb noise (and dust).
- **Moradabad** was listed as the world's **second most noise-polluted city** in UNEP **Frontier Report 2022**.
- **Ringelmann scale** (रिंगेलमैन पैमाना) measures smoke density, not noise.

PYQ — UPPCS Prelims 2022, Q81

UNEP Frontier Report 2022 — second most noise-polluted city:

Options: A. Kota B. Moradabad C. Indore D. Patna (पटना)

💡 **Show answer**

Ans: B

> **Logic:** Moradabad = noise PYQ city, not only brassware geography.

38.18 Noise Standards

Identity: CPCB zone limits | dB scale | Day/night

- Noise is measured in **decibels (dB)** on a logarithmic scale.
- CPCB ambient daytime limits (standard fact): **Silence 50**, **Residential 55**, **Commercial 65**, **Industrial 75** dB; night limits are lower.
- Silence zones include areas around hospitals, courts, and educational institutions.

PYQ — UPPCS Prelims 2022, Q115

Match area → permissible level (paper List-II used 50/55/65/70):

💡 **Show answer**

Intended Ans: C — Residential **55**, Silent **50**, Industrial **70** (as listed), Commercial **65**. Teach CPCB industrial day standard as **75 dB** in general notes.

> **Logic:** Silence < Residential < Commercial < Industrial.

38.19 Coastal Pollution

Identity: Land runoff | Ports / sewage | Mangrove stress

- Coastal pollution comes mainly from **land-based** sewage, industry, ports, aquaculture, and plastics.
- Damages mangroves, estuaries, turtle nesting beaches, and nearshore fisheries.
- India's first **National Centre for Marine Biodiversity (NCMB)** is at **Jamnagar (Gujarat)** — coastal biodiversity safeguarding, not a CPCB/NGT fact.

> **Logic:** Coastal pollution is mostly land-origin, not only ship dumps.

38.20 Ocean Pollution

Identity: MARPOL | Oil / plastic / nutrients | ~80% land-based

- About **80%** of marine pollution originates on **land**.
- **MARPOL** is the key international ship-pollution convention. **IMO** headquarters is in **London**.
- **Ocean acidification** (CO_2 → falling ocean pH) harms calcareous phytoplankton, coral reefs, and some animals with phytoplankton larvae — cloud seeding is **not** the standard concern list.
- Harmful algal blooms link to estuary (ज्वारनदमुख) nutrients, monsoon run-off, and upwelling.
- **SACEP** = South Asia Cooperative Environment Programme.
- Major issues: oil spills, nutrient dead zones, and plastic/microplastic accumulation in gyres.
- Coral stress combines warming and local pollution (cross-topic with climate/coral bleaching (प्रवाल विरंजन)).

> **Logic:** Ocean pollution ≠ only mid-ocean dumping. Acidification ≠ cloud seeding.

38.21 Plastic Waste Management

Identity: PWM Rules 2016 | SUP ban 2022 | EPR | Microplastics

- **Plastic Waste Management Rules, 2016** (amended later) govern collection, recycling, and producer duties.
- **Single-Use Plastic (SUP)** ban notification effective **July 2022** for identified items.
- **EPR** makes producers responsible for end-of-life plastic; EPR featured early in plastic/e-waste rules from **2011**.
- India is among the world's largest plastic-waste generators (~**3rd** in common framing).
- **Microplastics** are plastic particles **< 5 mm**. **Microbeads** in cosmetics are harmful to **marine ecosystems**.
- Polythene bags persist because they are **polymers** of ethylene, not because they are proteins.

> **Logic:** PWM/EPR = management law track; SUP ban = item-specific prohibition (प्रतिषेध) track.

Complete PYQ Bank (Topic 38)

Q1. UPPCS Prelims 2025, Q61

Consider the following statements:

1. Lichens are very good indicators of pollution.
2. Lichens grow well in polluted areas.

A. Only 2 B. Neither C. Both D. Only 1

💡 **Show answer**

Logic: Lichens are sensitive bio-indicators — they die back in polluted air; they do not thrive in it. **Ans: D (Only 1).** Statement 1 true; statement 2 false — lichens are **sensitive**, not pollution-tolerant. **Trap:** "Indicator" mistaken for "thrives in dirty air."

Q2. UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 was prepared by:

1. Central Pollution Control Board (CPCB)
2. Central Water Commission (CWC)
3. Central Ground Water Board (CGWB)

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Logic: Groundwater quality reporting is CGWB's mandate — not CPCB or CWC. **Ans: B (Only 3).** **CGWB** publishes the annual groundwater quality report. **Trap:** Swapping CPCB (air/surface water) with CGWB.

Q3. UPPCS Prelims 2023, Q103

Biological Oxygen Demand (BOD) is a standard criterion for—

A. Pollution assay in aquatic ecosystems B. High-altitude O_2 C. Blood O_2 D. Forest O_2

💡 **Show answer**

Logic: BOD measures microbial oxygen demand in water — an aquatic pollution assay, not blood oxygen. **Ans: A.** **BOD** assays organic pollution in rivers/lakes. **Trap:** "Oxygen" in the name → blood or altitude distractors.

Q4. UPPCS Prelims 2023, Q120

Which is not a biodegradable pollutant?

A. Domestic waste B. Urine C. Faecal matter D. Pesticide

> 💡 **Show answer**
>
> **Logic:** Pesticides persist and are not readily biodegradable unlike domestic organic waste. **Ans: D.** **Pesticides** are persistent; domestic waste, urine, and faecal matter biodegrade. **Trap:** Assuming all "natural" chemicals biodegrade quickly.

Q5. UPPCS Prelims 2022, Q46

Photochemical smog always contains—

A. Phosphorus B. Ozone C. Methane D. Aluminium ion

> 💡 **Show answer**
>
> **Logic:** Photochemical smog always contains ozone as a key secondary product. **Ans: B.** NOx + VOCs + sunlight → **O_3** (+ PAN). **Trap:** London SO_2 smog without ozone chemistry.

Q6. UPPCS Prelims 2022, Q81

According to UNEP Frontier Report 2022, the second most noise-polluted city was:

A. Kota B. Moradabad C. Indore D. Patna

> 💡 **Show answer**
>
> **Logic:** UNEP Frontier Report 2022 ranked Moradabad among the world's noisiest cities. **Ans: B.** **Moradabad** (UP brassware hub) was #2 in that report. **Trap:** Kota/Indore as generic UP/North India distractors.

Q7. UPPCS Prelims 2022, Q63

Which country has natural arsenic water contamination?

A. Sri Lanka B. Bangladesh C. Bhutan D. Pakistan

> 💡 **Show answer**
>
> **Logic:** Natural arsenic in groundwater is classically associated with Bangladesh / lower Gangetic basin. **Ans: B.** **Bangladesh** is the textbook arsenic hotspot (Gangetic plains geology). **Trap:** Assuming arsenic is India-exclusive only.

Q8. UPPCS Prelims 2022, Q115

Match the area with permissible noise level (daytime, as in paper):

Area	Level (dB)
Residential	55
Silence zone	50
Industrial	70
Commercial	65

Row order is not the answer code.

A. Residential–50; Silence–55; Industrial–65; Commercial–70 B. Residential–65; Silence–75; Industrial–50; Commercial–55 C. Residential–55; Silence–50; Industrial–70; Commercial–65 D. All zones at 75 dB

💡 **Show answer**

Logic: CPCB daytime noise limits — silence zone lowest, then residential, commercial, industrial highest. **Ans: C.** Residential **55**, Silence **50**, Industrial **70**, Commercial **65** — silence < residential < commercial < industrial (paper industrial **70**; teach CPCB day industrial **75 dB** in notes). **Trap:** Reversing silence vs residential or using one dB for all zones.

Q9. UPPCS Prelims 2021, Q143

Flyash pollution is caused by

A. Oil refining B. Fertilizer industry C. Thermal power plant D. Mining

💡 **Show answer**

Logic: Fly ash is a primary pollutant from coal thermal power plants — usable in bricks/cement. **Ans: C.** **Thermal power plants** emit fly ash (SiO_2/Al_2O_3/CaO plus toxic metals). **Trap:** Calling fly ash a secondary pollutant like smog.

Q10. UPPCS Prelims 2019, Q136 / 2018, Q121

With reference to smog, consider the following:

1. Smog is essentially caused by oxides of nitrogen and sulphur in the atmosphere.
2. Among CO, CO_2, and fly ash, smog is the secondary pollutant.

A. Only 1 B. Only 2 C. Both 1 and 2 D. Neither 1 nor 2

💡 **Show answer**

Logic: Smog is a secondary pollutant; classical smog links to oxides of nitrogen and sulphur. **Ans: C.** Statement 1 matches **2019 Q136** (oxides of N and S); statement 2 matches **2018 Q121** (smog secondary; fly ash/CO primary). **Trap:** Treating fly ash or stack SO_2 alone as “smog” without atmospheric formation.

Extra Drill (Ghatnachakra Pollution map)

Q31. Human-made pollutants / human-modified environment are called—

A. Natural only

B. Anthropogenic

C. Xenobiotic antibiotics only

D. Humalins

💡 **Show answer**

Logic: These stems recycle bank traps on anthropogenic pollution, AQI gases, control devices, diseases, and remediation. **Ans: B.** Anthropogenic means caused or influenced by humans.

Q32. Which is biodegradable among typical options?

A. Mercury

B. Plastic / asbestos

C. Sewage

D. Pesticide

💡 **Show answer**

Ans: C. Sewage/domestic organic waste biodegrades; pesticide/plastic/Hg/asbestos do not (or only very slowly).

Q33. India's AQI includes which set?

A. CO_2 and CH_4 only

B. PM10, PM2.5, NO_2, CO, O_3, SO_2, NH_3, Pb

C. Only methane and CFC

D. Only radon and neon

💡 **Show answer**

Ans: B. Eight criteria pollutants — not CO_2/CH_4.

Q34. Powerful eye irritant in photochemical smog—

A. Argon

B. Peroxyacetyl nitrate (PAN)

C. Pure nitrogen

D. Helium

💡 **Show answer**

Ans: B. PAN is a secondary lachrymator in photochemical smog.

Q35. Most important classic indoor air pollutant gas—

A. Outdoor ozone only

B. Radon

C. Stratospheric CFCs only

D. Pure oxygen

💡 **Show answer**

Ans: B. Radon from soil/radium accumulates indoors and raises lung-cancer risk.

Q36. Knock-knee syndrome is linked to—

A. Mercury Minamata only

B. Fluoride excess

C. Ozone hole only

D. Noise Rules 2000 only

💡 **Show answer**

Ans: B. Excess fluoride → fluorosis / knock-knee. Minamata = Hg; Itai-itai = Cd.

Q37. Which river is classically called a biological desert from pollution?

A. Only pristine Brahmaputra whole length

B. Damodar (Giridih–Durgapur industrial stretch)

C. Only Antarctic meltwater

D. Only pure glacial melt always

> 💡 **Show answer**
>
> **Ans: B.** Mining/industry load makes Damodar a biological desert teaching fact.

Q38. Not a standard air-pollution control device name—

A. Bag filter

B. Cyclone separator / collector

C. Cyclone divider

D. Wet scrubber

> 💡 **Show answer**
>
> **Ans: C.** Cyclone divider is the distractor; bag/cyclone separator/scrubber are real tools.

Q39. Bioremediation / Oilzapper facts — which is correct?

A. Cd and Pb are always fully cleaned by simple bioremediation

B. Oilzapper (TERI) remediates oily sludge/spills; heavy metals like Cd/Pb are not readily/completely treated by microbes

C. Bioremediation only means transplanting organs

D. Oilzapper is a maize biofuel variety

> 💡 **Show answer**
>
> **Ans: B.** Enhance natural biodegradation; Oilzapper for oil; metals remain hard.

Q40. CCAC focuses primarily on—

A. Only G20 membership rules for ozone hole

B. Methane, black carbon, and HFCs (short-lived climate pollutants)

C. Only tiger census

D. Only Ramsar wetland birds

> 💡 **Show answer**
>
> **Ans: B.** Founded 2012 with UNEP partners — not a G20-only club statement.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to air pollutants, which of the following statements is/are correct?

1. Primary — emitted directly (SO_2, PM, CO, NOx, lead).
2. primary — form in air (O_3, PAN, photochemical smog products).
3. CO_2 and CH_4 are India's AQI eight pollutants.

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

Show answer

Ans: C. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q2. Match List-I with List-II:

List-I	List-II
A. London smog	1. NOx + VOCs + sunlight → O_3 + PAN
B. Photochemical smog	2. SO_2 + smoke
C. PAN	3. Strong eye irritant

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

Show answer

Ans: C. A-2, B-1, C-3. **Logic:** Smog types + PAN.

Q3. Which of the following pairs is/are correctly matched?

1. India's AQI — eight pollutants including PM2.5, O_3, NH_3, Pb
2. AQI Good is 401–500 and Severe is 0–50.
3. Lichens grow best in badly polluted air

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

Show answer

Ans: D. Only 1 is correct. **Logic:** Statement 1 correctly ties India's AQI to eight pollutants. Statement 2 reverses Good/Severe bands. Statement 3 is false — lichens indicate cleaner air, not badly polluted air.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NCAP (2019) targets non-attainment cities with ~20–30% particulate cut goal.

Reason (R): SAMEER-type apps replace NCAP city action plans.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: C. A true, R false. **A/R logic:** Apps warn — do not replace NCAP plans.

Q5. With reference to water quality metrics, which of the following statements is/are correct?

1. High BOD = worse quality.
2. COD ≥ BOD usually.
3. Clean water often DO >~6 mg/L and BOD <~3 mg/L teaching.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

Show answer

Ans: A. All three. **Logic:** BOD/COD/DO.

Q6. Arrange eutrophication chain:

A. Air-only SO_2 fog B. Nutrient N/P overload → algal bloom → DO crash C. Only Dobson Unit fall D. Only catalytic converter failure

> 💡 **Show answer**
>
> **Ans: B.** N/P → bloom → DO crash. **Logic:** Water — not air.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Knock-knee — fluoride
2. Black Foot — arsenic
3. Minamata — fluoride

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only pair 3 is not correctly matched. **Logic:** Minamata = mercury.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): CGWB issues the groundwater quality report.

Reason (R): CPCB is identical to CGWB.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** CGWB ≠ CPCB.

Q9. With reference to plastics, which of the following statements is/are correct?

1. PWM Rules 2016; SUP ban push 2022.
2. EPR = producer responsibility.
3. ~80% of marine pollution from land; MARPOL for ships.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: D.** All three. **Logic:** Plastic + marine.

Q10. Match List-I with List-II:

List-I	List-II
A. CO poisoning	1. Top indoor pollutant teaching
B. Radon	2. Carboxyhaemoglobin (~200–300× vs O_2)
C. Fly ash	3. Thermal-power; bricks/cement use

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** CO / radon / fly ash.

Q11. Arrange daytime noise standards idea:

A. All zones 100 dB B. Silence 50; Residential 55; Commercial 65; Industrial 75 C. Only industrial 20 dB D. No standards taught

💡 **Show answer**

Ans: B. 50/55/65/75. **Logic:** Noise bands.

Q12. With reference to disasters/tools, which of the following statements is/are correct?

1. Bhopal MIC Dec 1984; Chernobyl 1986 pair.
2. Oilzapper (TERI) for oily waste bioremediation.
3. Bag filter identical to cyclone divider always.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Bag filter ≠ cyclone.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Photochemical smog is not a simple SO_2–smoke fog.

Reason (R): Ozone chemistry (NOx/VOCs/sunlight) is the key.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Photochemical ≠ London smog.

Q14. Which one of the following correctly states BS-VI start teaching?

A. Only Montreal ODS B. Only Euro-I forever C. Only 1950 coal norms D. India leapfrogged to BS-VI (2020 neighbourhood)

💡 **Show answer**

Ans: D. BS-VI ~2020. **Logic:** Fuel/vehicle norms.

Q15. How many of the following disease–toxin pairs are correct?

Arsenic–Gangetic/Bangladesh; Fluoride–knock-knee; Mercury–Minamata

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Classic toxin map.

Q16. With reference to Ganga/Damodar, which of the following statements is/are correct?

1. Damodar = biological desert teaching.
2. Ganga BOD peak classically Kanpur–Allahabad.
3. Leather industry = high chemical pollution.

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

💡 **Show answer**

Ans: B. All three. **Logic:** River pollution colour.

Q17. Match List-I with List-II:

List-I	List-II
A. SAFAR	1. Delhi units teaching
B. WAYU	2. IITM AQ + weather forecast
C. Green muffler	3. Tree belts for noise

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** SAFAR/WAYU/muffler.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): H-CNG reduces CO/HC/CO_2 vs CNG but does not eliminate CO.

Reason (R): Up to ~1/5 H_2 by volume teaching; not cheaper than CNG.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; blend share/cost does not itself explain residual CO. **A/R logic:** Benefit vs residual CO.

Q19. Which of the following pairs is/are correctly matched?

1. CCAC — methane, black carbon, HFCs focus
2. Asian Brown Cloud — South Asia association
3. CCAC — G20-only initiative

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not G20-only.

Q20. Arrange catalytic converter metals:

A. Pt/Pd/Rh neighbourhood B. Only Na/K C. Only Pb/Hg D. Only argon

💡 **Show answer**

Ans: A. Platinum group. **Logic:** Converter catalysts.

Q21. With reference to indoor air, which of the following statements is/are correct?

1. Solid cooking fuels, tobacco, poor ventilation, radon.
2. Radon top indoor pollutant teaching.
3. Indoor air pollution is only outdoor AQI copied indoors.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: B. Only 1 and 2. **Logic:** Distinct indoor drivers.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Many pesticides are not readily biodegradable.

Reason (R): That persistence is a frequent true statement in papers.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: D. Both true and R notes why the fact is tested. **A/R logic:** Persistence fact.

Q23. How many of the following SWM/electronics facts are correct?

SWM Rules 2016 landfill criteria; R2 Code responsible e-recycling

A. Only one B. None C. Both D. Three including Montreal

Show answer

Ans: C. Both. **Logic:** Waste rules pair.

Q24. Which one of the following correctly states Moradabad noise teaching?

A. Only LDN capital B. Silence zone only at 0 dB C. Only ozone hole capital D. Flagged among world's noisiest in UNEP Frontier reporting

Show answer

Ans: D. Noisiest-city flag. **Logic:** UP colour.

Q25. With reference to geoengineering teaching, which of the following statements is/are correct?

1. Cirrus thinning and sulphate aerosol injection aim at cooling/offsetting warming.
2. Not cyclone control tools.
3. Identical to BS-VI fuel norms.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: A. Only 1 and 2. **Logic:** Not fuel norms.

Q26. Match List-I with List-II:

List-I	List-II
A. IMO HQ	1. Mould toxins on grain
B. Aflatoxins	2. London
C. Steel industry	3. CO, CO_2, SOx, NOx

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** IMO / aflatoxin / steel.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Ocean acidification harms calcifiers.

Reason (R): About 80% of marine pollution comes from land.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; land-share statistic does not explain acidification chemistry. **A/R logic:** Two ocean facts.

Q28. Arrange Kanpur chromium neighbourhood:

A. Only Himalayan glaciers B. Only Dobson Units C. Heavy-metal teaching around Kanpur plus Pb/Cd/Hg elsewhere D. Only Earth Hour

💡 **Show answer**

Ans: C. Cr–Kanpur + other metals. **Logic:** Heavy-metal map.

Q29. With reference to bioremediation, which of the following statements is/are correct?

1. Cleans via microbes.
2. Cd/Pb not readily treated teaching.
3. Oilzapper for oily waste (TERI).

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Bioremediation set.

Q30. Which one of the following correctly states India's AQI pollutant set idea?

A. Eight pollutants — not CO_2/CH_4 B. Only CO_2 and CH_4 C. Only argon and helium D. Only BOD and COD

💡 **Show answer**

Ans: A. Eight criteria pollutants. **Logic:** AQI composition.

Common Traps

Trap	Correct
Lichens thrive in polluted air	**Sensitive; die back**
Fly ash = secondary pollutant	**Primary**; smog is secondary
Fly ash = SiO_2/CaO only, no toxics	Also **Pb/As/Cu**-type metals
BOD = blood oxygen test	**Aquatic pollution assay**
Groundwater report = CPCB	**CGWB**
Pesticide = biodegradable	**Not**

Trap	Correct
Photochemical smog without O_3	**Always contains ozone**
PAN = primary stack gas	**Secondary**; eye irritant
AQI includes CO_2/CH_4	**Eight** pollutants — **not** those GHGs
Most important indoor = SO_2	Classic answer often **radon**
Knock-knee = mercury	**Fluoride**; Minamata = **Hg**
Eutrophication = air pollution	**Water** nutrient overload
Cyclone divider = control device	**Not** a standard name
CCAC = unique G20 initiative	Voluntary partnership; focus **CH_4 / BC / HFCs**
Bioremediation cleans all Cd/Pb	**Not** readily/completely
Noisiest UP city distractors	**Moradabad** (Frontier 2022)
Arsenic only = India exclusive	Classic **Bangladesh** fact
NCAP = SAFAR	Programme vs forecast system
Industrial noise day = 55 dB	**75 dB** (CPCB day)
Green muffler = soil fix	**Noise** (tree belts)
Bhopal gas = CO/SO_2	**MIC**, Dec **1984**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 39 — Acid Rain (अम्ल वर्षा)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Acid Rain | Causes | Effects | Prevention

Sources baked in: NCERT; CPCB; CLRTAP/Gothenburg; Taj Trapezium; UPPCS PYQs **Weight:** ★★ Medium–High — pH traps; CO_2 vs SO_2/NOx; Taj gypsum; Montreal ≠ acid rain **Last verified:** August 2026

Current Affairs (this topic)

- Twin of Acid Rain core — chemistry fixed; rare CA only.
- Acid rain ≠ photochemical smog ≠ ozone depletion (ओजोन क्षरण).

💡 🎯 **Consolidated — 28 Must-Score Facts (Click to Expand)**

1. **Acid rain** means acidic deposition with **pH below 5.6**. 2. Normal rain (सामान्य वर्षा) is already slightly acidic near **pH 5.6** because CO_2 forms weak carbonic acid. It is **not** pH 7. 3. **Wet deposition** arrives with rain, snow, or fog. **Dry deposition** settles as acidic gases or particles without rain. 4. The main strong acid pathway is **$SO_2 \rightarrow H_2SO_4$**. **NOx → HNO_3** is the other major path. 5. **$CO_2 \rightarrow H_2CO_3$** sets the weak baseline. Carbon dioxide alone is **not** the main culprit of severe acid rain. 6. On the pH scale, acidic is below **7**, basic is above **7**, and neutral is **7**. 7. Reversed statements that call normal rain alkaline or acid rain alkaline are both false. 8. Lake waters below about **pH 5** can kill fish. Acid soils release toxic **Al^{3+}**. 9. Acid deposition harms conifers and weathers marble and limestone. The Taj marble → **gypsum** story is the classic Indian fact. 10. **Lichens** die in high SO_2 and act as pollution bio-indicators — they indicate pollution but do **not** thrive in polluted air. 11. Prevention tools include **FGD**, catalytic converters, cleaner fuel norms such as **BS-VI**, and regional protocols. 12. The **Gothenburg Protocol (1999)** under **CLRTAP (1979)** targets acidifying air pollutants. **Montreal** targets ozone ODS — different problem. 13. The **Taj Trapezium Zone** covers about **10,400 km^2** around the Taj Mahal (ताज महल), with Supreme Court-driven controls from **1996**. 14. **Mathura (मथुरा) refinery** is a classic SO_2 source in the Taj Trapezium narrative. 15. The **Singrauli** thermal belt and industrial NCR/Kanpur (कानपुर) airsheds are Indian acidifying-emission hotspots. 16. SO_2 is denser than air and can travel long distances before deposition. 17. Acid rain is a **secondary-pollutant** story built from primary SO_2 and NOx emissions. 18. Coal power and some industries are major SO_2 sources. Vehicles and combustion add NOx. 19. Liming lakes is an emergency response, not a substitute for cutting emissions at source. 20. Do not blame acid rain mainly on CO_2 just because normal rain is weakly acidic. 21. Dry deposition can damage materials even on rainless days. 22. Acid mist and acid fog are wet forms even when they are not classic rainfall. 23. Cultural-heritage damage and forest dieback are both valid effect facts alongside aquatic harm. 24. India's focus pairs chemistry (SO_2/NOx) with Taj protection geography. 25. Calling Montreal the acid-rain treaty is a standard distractor. 26. FGD is flue-gas desulphurisation at power-plant chimneys — mitigation hardware, not a treaty name. 27. Secondary pollutants H_2SO_4 and HNO_3 form in the atmosphere from primary SO_2 and NOx. 28. Acid rain is **not** "rain only"; dry deposition counts too.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Normal Rain** | **Acid Rain** | Slightly acidic with pH ~5.6 due to dissolved atmospheric CO_2 forming weak carbonic acid (H_2CO_3) vs precipitation with pH < 5.6 caused by anthropogenic SO_2 and NOx | सामान्य वर्षा (pH 5.6) / अम्ल वर्षा (pH < 5.6) | | **Sulfuric Acid (H_2SO_4)** | **Nitric Acid (HNO_3)** | Contributes ~60–70% of total acid precipitation (emitted mainly by coal thermal power plants) vs contributes ~30–40% (emitted by automobile engines and fertilizers) | सल्फ्यूरिक अम्ल (60-70%) / नाइट्रिक अम्ल (30-40%) | | **Wet Acid Deposition** | **Dry Acid Deposition** | Acidic compounds deposited on Earth via rain, fog, sleet, or snow vs acidic gases, sulfate, and nitrate particles settling directly on soil and buildings without precipitation | आर्द्र निक्षेपण (वर्षा/कोहरा) / शुष्क निक्षेपण (धूल/गैस कण) | | **Marble Cancer** | **Eutrophication** | Yellowing and pitting of calcium carbonate monuments (Taj Mahal) reacting with sulfuric acid to form soluble gypsum vs excess nutrient loading in water bodies causing algal blooms | संगमरमर कैंसर (ताजमहल क्षरण) / सुपोषण (शैवाल प्रस्फुटन) | | **Gothenburg Protocol (1999)** | **Montreal Protocol (1987)** | UNECE protocol to abate acidification, eutrophication, and ground-level ozone (SO_2, NOx, VOCs, NH_3) vs UNEP protocol phasing out ozone-depleting chlorofluorocarbons | गोथेनबर्ग प्रोटोकॉल (अम्लीकरण) / मॉन्ट्रियल प्रोटोकॉल (ओजोन) |

💡 📋 **Must-score facts — pH, Taj Trapezium, protocols (Click to Expand)**

Locks | Item | Lock | |---|---| | Acid rain | pH <**5.6** | | Normal rain | ~5.6 from CO_2 (≠ alkaline) | | Main acids | H_2SO_4 (SO_2) + HNO_3 (NOx) | | Taj marble | → gypsum ($CaSO_4$) under sulphuric attack | | Taj Trapezium | Pollution-control belt for Taj | | Gothenburg/CLRTAP | Acidifying air (≠ Montreal ozone) | | Prevention | FGD, converters, BS-VI, cleaner fuel |

39.1 Acid Rain

Identity: pH < 5.6 | H_2SO_4 + HNO_3 | Wet + dry | Secondary pollutant

- **Acid rain** means precipitation or atmospheric deposition with **pH below 5.6**, mainly from **sulphuric acid (H_2SO_4)** and **nitric acid (HNO_3)** (नाइट्रिक अम्ल).
- **Normal unpolluted rain** is already slightly acidic (~**pH 5.6**) because **CO_2** forms weak **carbonic acid (H_2CO_3)** — it is **not** pH 7.
- Anthropogenic episodes can fall to about **pH 4.0–4.5** (or lower in extreme cases).
- Includes **wet deposition** (rain, snow, fog) and **dry deposition** (acidic gases/particles settling without rain).
- Acids form in the air, so acid rain is a **secondary pollution** problem.
- On the pH scale: **acidic < 7**, **neutral = 7**, **basic > 7**. Each step down is **10×** more H^+.

PYQ — UPPCS Prelims 2025, Q103

1. Acidic solutions have pH more than 7.
2. Basic solutions have pH less than 7.

Options: A. Only 2 B. Neither 1 nor 2 C. Both D. Only 1

💡 **Show answer**

Ans: B — Acidic **< 7**; basic **> 7**.

PYQ — UPPCS Prelims 2024, Q108

(A) CO_2, SO_2, and NO_2 dissolve in rainwater and produce acid rain. (R) Air is polluted due to high CO_2, SO_2, and oxides of nitrogen.

Options: A. Both true; R explains A B. A false, R true C. Both true; R not explanation D. A true, R false

💡 **Show answer**

Ans: C — All three dissolve (A treated true). R is true about air pollution but does **not** explain why **CO_2 gives only weak acid** while **SO_2/NOx drive strong acid rain**.

pH	Meaning
< 5.6	Acid rain threshold
~5.6	Normal rain (CO_2)
7	Neutral (not natural rain)
> 7	Basic

> **Logic:** Normal rain ≠ neutral. CO_2 alone ≠ severe acid rain.

39.2 Causes

Identity: SO_2 → H_2SO_4 | NOx → HNO_3 | Coal TPPs + vehicles | CO_2 weak only

- **SO_2** from burning sulphur-rich **coal/oil** in thermal plants, refineries, smelters, brick kilns, and generators forms **H_2SO_4**.
- **NOx** from vehicles, power plants, and high-temperature combustion forms **HNO_3**.
- Natural sources (volcanoes for SO_2; lightning for NOx) exist but are minor versus anthropogenic loads in industrial regions.
- **CO_2** only sets the weak baseline (~5.6); it is **not** the main cause of damaging acid rain.
- **SO_2 is heavier than air** and can linger near ground-level sources.
- India/UP hotspots: Singrauli thermal belt, Mathura (मथुरा) refinery–Agra corridor, industrial NCR/Kanpur (कानपुर) airsheds.

PYQ — UPPCS Prelims 2025, Q43 (pattern)

Gases lighter than air among listed options — only **helium** fits; **SO_2 and HCl are heavier**.

> **Logic:** SO_2 sinks relative (सापेक्ष) to air — density trap.

39.3 Effects

Identity: Aquatic kill | Soil Al^{3+} | Conifers | Taj gypsum | Lichens

- Freshwater life fails when pH falls below about **5.0** — fish eggs and many aquatic species die; toxic **Al^{3+}** is mobilised.
- Soils lose **Ca^{2+}/Mg^{2+}** nutrients and gain toxic aluminium, damaging roots.
- **Conifers** (pine, spruce) are especially sensitive to acid fog and soil change.
- Marble/limestone monuments: **$CaCO_3$ + acid → gypsum ($CaSO_4 \cdot 2H_2O$) crust** — classic **Taj Mahal** (ताज महल) yellowing/damage.
- Taj damage is linked to **SO_2** from Mathura refinery and nearby industries, plus particulates.
- Human health: SO_2/NOx and acid aerosols irritate lungs.
- **Lichens** die in high-SO_2 air and act as pollution bio-indicators (Topic 38 crossover).

Receptor	Effect
Lakes/rivers	Fish kills below ~pH 5
Soil	Nutrient leach + Al^{3+}

Receptor	Effect
Forests	Conifer damage
Marble	Gypsum crust (Taj)
Lichens	Die in high SO_2

> **Logic:** Taj fact = **SO_2/acid**, not "only CO_2 yellowing."

39.4 Prevention

Identity: FGD | Catalytic converters | BS-VI | Gothenburg | Taj Trapezium

- **Flue Gas Desulphurisation (FGD)** scrubbers use lime/limestone to cut SO_2 from power-plant stacks.
- Vehicle **catalytic converters** and cleaner fuels cut NOx; **BS-VI** fuel has very low sulphur (~**10 ppm**).
- Switch from coal/diesel to **CNG/LPG/renewables** reduces acidifying emissions.
- CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) emission standards and consents limit SO_2/NOx from industry.
- International: **Gothenburg Protocol (1999)** (गोथेनबर्ग प्रोटोकॉल) under **CLRTAP** targets acidifying and related air pollutants — **not** the Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) (ozone).
- **Taj Trapezium Zone (~10,400 km^2)** — Supreme Court-driven controls (1996 onward) forced gas switch/relocation of polluting units around Agra.

Tool	Cuts
FGD	SO_2
Catalytic converter / BS-VI	NOx + fuel S
Fuel switch	Both
Taj Trapezium	Local heritage protection
Gothenburg/CLRTAP	Transboundary acidification

> **Logic:** Montreal = ozone. Acid rain treaty family = **CLRTAP / Gothenburg**.

Complete PYQ Bank (Topic 39)

Q1. UPPCS Prelims 2025, Q103

Consider the following statements:

1. Acidic solutions have pH more than 7.
2. Basic solutions have pH less than 7.

A. Only 2 B. Neither 1 nor 2 C. Both 1 and 2 D. Only 1

💡 **Show answer**

Logic: pH scale basics — acidic < 7, basic > 7; both statements in the stem are reversed. **Ans: B (Neither).** Acidic **< 7**; basic **> 7** — both statements are wrong. **Trap:** Normal rain ~5.6 confused with "neutral = 7."

Q2. UPPCS Prelims 2024, Q108

Assertion (A): CO_2, SO_2, and NO_2 dissolve in rainwater and produce acid rain.

Reason (R): Air is polluted due to high CO_2, SO_2, and oxides of nitrogen.

A. Both true; R explains A B. A false, R true C. Both true; R does not explain A D. A true, R false

> 💡 **Show answer**
>
> **A/R logic:** A tests whether CO_2/SO_2/NO_2 dissolve in rain; R tests air pollution — CO_2 gives only weak acid, so R does not explain strong acid rain. **Ans: C.** All three dissolve (A true); R true about pollution but **does not explain** why **CO_2 = weak H_2CO_3** while **SO_2/NOx → strong H_2SO_4/HNO_3**. **Trap:** Treating CO_2 as main severe acid-rain driver.

Q3. UPPCS Prelims 2025, Q43 (pattern)

Which of the following gases is lighter than air?

A. Sulphur dioxide B. Only helium C. Hydrogen chloride D. All of the above

> 💡 **Show answer**
>
> **Logic:** Gas density relative to air — SO_2 and HCl are heavier; helium is lighter. **Ans: B (Only helium).** **SO_2** and **HCl** are denser than air; helium rises. **Trap:** Assuming all pollutant gases sink — only SO_2 density is tested here.

Q4. UPPCS Prelims 2025, Q61

Consider the following statements:

1. Lichens are very good indicators of pollution.
2. Lichens grow well in polluted areas.

A. Only 2 B. Neither C. Both D. Only 1

> 💡 **Show answer**
>
> **Logic:** Lichens are SO_2-sensitive bio-indicators — they indicate pollution but do not thrive in it. **Ans: D (Only 1).** Lichens die in high-SO_2 air — indicator ≠ thrives in pollution. **Trap:** Same lichen stem as Topic 38; answer is **Only 1**.

Q5. UPPCS Prelims (pattern)

Acid rain damage to the Taj Mahal mainly involves conversion of marble to:

A. Gypsum ($CaSO_4 \cdot 2H_2O$) B. Pure gold plating C. Ozone layer depletion D. Biogas methane

> 💡 **Show answer**
>
> **Logic:** Taj marble damage — acid reacts with $CaCO_3$ to form gypsum crust; SO_2 from nearby industry is the classic Indian fact. **Ans: A.** **$CaCO_3$ + acid → gypsum crust** (yellowing/spalling); Mathura refinery SO_2 corridor. **Trap:** Blaming CO_2 alone or ozone hole for marble damage.

Q6. UPPCS Prelims (pattern)

Normal unpolluted rain has a pH of about:

A. 5.6 B. 7.0 exactly C. 14 D. 2.0

> 💡 **Show answer**
>
> **Logic:** Acid rain threshold pH < 5.6; normal rain ~5.6 from weak carbonic acid — not pH 7. **Ans: A.** **~5.6** from CO_2 → H_2CO_3; acid rain is **< 5.6**. **Trap:** "Natural = neutral 7."

Q7. UPPCS Prelims (pattern)

Which international agreement primarily targets acidifying air pollutants (SO_2, NOx)?

A. Montreal Protocol B. Gothenburg Protocol under CLRTAP C. CITES D. Minamata Convention

> 💡 **Show answer**
>
> **Logic:** Treaty pairing — Montreal = ozone ODS; Gothenburg/CLRTAP = acidifying pollutants. **Ans: B.** **Gothenburg/CLRTAP** cuts acidification; **Montreal** is ozone. **Trap:** Montreal as catch-all "environment" treaty.

Q8. UPPCS Prelims (pattern)

Flue Gas Desulphurisation (FGD) is mainly installed to reduce:

A. SO_2 from thermal power plants B. Noise in silence zones C. Microplastics in oceans D. Groundwater fluoride only

> 💡 **Show answer**
>
> **Logic:** FGD scrubbers remove SO_2 from power-plant stacks; catalytic converters cut vehicle NOx. **Ans: A.** **FGD** uses lime/limestone to trap **SO_2**. **Trap:** FGD for NOx only, or confusing with catalytic converters (mainly NOx/CO/HC).

Q9. UPPCS Prelims (pattern)

Dry deposition in acid rain refers to:

A. Acidic gases and particles settling without precipitation B. Only rainfall with pH below 5.6 C. Only ocean MARPOL oil spills D. Only indoor chulha smoke

> 💡 **Show answer**
>
> **Logic:** Acid deposition includes dry deposition — acidic gases/particles settling without rain. **Ans: A.** **Dry deposition** = acids settle without rain; **wet** = with rain/snow/fog. **Trap:** "Acid rain = only liquid rain."

Q10. UPPCS Prelims (pattern)

The main strong acids in damaging acid rain are formed in the atmosphere from:

A. SO_2 and NOx emissions B. Only argon and helium C. Only methane from paddies D. Only carbon dioxide alone

> 💡 **Show answer**
>
> **Logic:** H_2SO_4 and HNO_3 form in the atmosphere from primary SO_2 and NOx — a secondary-pollutant pathway. **Ans: A.** **SO_2 → H_2SO_4**; **NOx → HNO_3** in air; CO_2 gives only weak baseline. **Trap:** CO_2 as "main acid rain gas."

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to acid rain, which of the following statements is/are correct?

1. Acid rain is defined as deposition with pH above 7.0.
2. Normal rain ~pH 5.6 from CO_2 → weak H_2CO_3 — not pH 7.
3. Normal rain is alkaline (pH >7).

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

Show answer

Ans: B. Only 2 is correct. **Logic:** Statement 2 is the normal-rain ~pH 5.6 fact. Statement 1 is false — acid rain is below ~5.6, not above 7. Statement 3 is false — normal rain is slightly acidic, not alkaline.

Q2. Match List-I with List-II:

List-I	List-II
A. Wet deposition	1. Acidic gases/particles without rain
B. Dry deposition	2. Rain/snow/fog
C. Main strong path	3. $SO_2 \rightarrow H_2SO_4$ (and NOx → HNO_3)

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

Show answer

Ans: C. A-2, B-1, C-3. **Logic:** Wet/dry/chemistry.

Q3. Which of the following pairs is/are correctly matched?

1. $CO_2 \rightarrow H_2CO_3$ sets weak baseline
2. CO_2 alone is the main culprit of severe acid rain
3. Acid rain is secondary-pollutant story from secondary SO_2/NOx

A. Only 3 B. Only 1 C. 1 and 2 D. 2 and 3

Show answer

Ans: B. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Lake waters below about pH 5 can kill fish.

Reason (R): Acid soils can release toxic Al^{3+}.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: A. Both true; Al^{3+} release does not itself state the lake fish pH threshold. **A/R logic:** Aquatic vs soil effects.

Q5. With reference to Taj / materials, which of the following statements is/are correct?

1. Acid deposition weathers marble/limestone.
2. Taj marble → gypsum is the classic Indian fact.
3. Taj Trapezium is the protection belt.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

Show answer

Ans: A. All three. **Logic:** Taj set.

Q6. Arrange pH scale basics:

A. Acidic >7; basic <7 B. Acidic <7; basic >7; neutral 7 C. Neutral is 5.6 always D. No meaning to 7

💡 **Show answer**

Ans: B. Acidic <7 / basic >7. **Logic:** pH scale rule.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Gothenburg/CLRTAP — acidifying air pollutants
2. Montreal — ozone ODS
3. Montreal — acid-rain treaty

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Montreal ≠ acid rain.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Liming lakes is an emergency response.

Reason (R): It substitutes fully for cutting SO_2/NOx at source.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Liming ≠ source control substitute.

Q9. With reference to sources, which of the following statements is/are correct?

1. Coal power / some industries — major SO_2.
2. Vehicles/combustion — NOx.
3. SO_2 denser than air; can travel long distances.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Sources + transport.

Q10. Match List-I with List-II:

List-I	List-II
A. FGD	1. Cleaner fuel neighbourhood
B. BS-VI	2. Flue-gas desulphurisation
C. Catalytic converters	3. Cut NOx/others in vehicles

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Prevention tools.

Q11. Arrange dry deposition idea:

A. Only possible during rainfall B. Can damage materials even on rainless days C. Only ocean acidification D. Only Dobson Units

💡 **Show answer**

Ans: B. Rainless-day damage. **Logic:** Dry pathway.

Q12. With reference to effects, which of the following statements is/are correct?

1. Conifer harm / forest dieback.
2. Cultural-heritage damage.
3. Aquatic harm.
4. Makes normal rain alkaline.

A. 2, 3 and 4 only B. 1, 2 and 4 only C. 1, 2 and 3 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: C. 1, 2 and 3 only. **Logic:** Does not make rain alkaline.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Do not blame severe acid rain mainly on CO_2.

Reason (R): Normal rain's weak acidity comes from carbonic acid, but strong acids come from SO_2/NOx.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Baseline vs severe pathway.

Q14. Which one of the following correctly states acid mist/fog?

A. Only UV-C beams B. Only dry gypsum dust C. Only argon deposition D. Wet forms even when not classic rainfall

💡 **Show answer**

Ans: D. Wet deposition forms. **Logic:** Mist/fog.

Q15. How many of the following prevention tools are correct?

FGD; catalytic converters; BS-VI; regional acid protocols

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** Prevention set.

Q16. With reference to India's focus, which of the following statements is/are correct?

1. Chemistry SO_2/NOx.
2. Taj protection geography.
3. Montreal as the acid-rain statute.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Montreal distractor.

Q17. Match List-I with List-II:

List-I	List-II
A. H_2SO_4 path	1. NOx
B. HNO_3 path	2. SO_2
C. Weak baseline	3. $CO_2 \rightarrow H_2CO_3$

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Acid pathways.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Calling acid rain alkaline is false.

Reason (R): Acidic means pH below 7 on the scale; acid rain is below 5.6.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains why alkaline label fails. **A/R logic:** pH definition.

Q19. Which of the following pairs is/are correctly matched?

1. Secondary pollutant story — acid rain
2. Primary emissions — SO_2 and NOx
3. Only natural sea salt causes Taj gypsum

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Taj story = acid deposition neighbourhood.

Q20. Arrange Taj Trapezium:

A. Pollution-control belt to protect Taj Mahal B. Ozone Cell only C. Only Antarctic vortex D. Only UNCCD LDN

💡 **Show answer**

Ans: A. Taj protection belt. **Logic:** Geography tool.

Q21. With reference to reversed false statements, which of the following is/are false?

1. Normal rain is alkaline.
2. Acid rain is alkaline.
3. Acid rain has pH below 5.6.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. 1 and 2 are false; 3 is true. **Logic:** Reversed alkaline traps.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Acid deposition harms conifers.

Reason (R): Cultural-heritage weathering of marble is also a valid effect.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; marble weathering does not explain conifer harm. **A/R logic:** Two effect families.

Q23. How many of the following are true?

Wet with rain/snow/fog; dry without rain; $SO_2 \rightarrow H_2SO_4$ major path

A. Only one B. Only two C. All three D. None

💡 **Show answer**

Ans: C. All three. **Logic:** Deposition trio.

Q24. Which one of the following correctly states Gothenburg neighbourhood?

A. Only Chipko movement B. Ozone ODS phase-out only C. Desertification LDN only D. Acidifying air pollutants under CLRTAP family

💡 **Show answer**

Ans: D. Acidifying pollutants protocol. **Logic:** ≠ Montreal.

Q25. With reference to long-range transport, which of the following statements is/are correct?

1. SO_2 can travel long distances before deposition.
2. SO_2 is denser than air teaching.
3. Acid rain never crosses regions.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Regional problem.

Q26. Match List-I with List-II:

List-I	List-II
A. Acid rain pH	1. ~5.6
B. Normal rain pH	2. <5.6
C. Neutral	3. 7

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** <5.6 / ~5.6 / 7.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Vehicles and combustion add NOx to the acid-rain pathway.

Reason (R): NOx can form HNO_3.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the NOx path. **A/R logic:** NOx → nitric acid.

Q28. Arrange emergency vs source control:

A. Liming replaces emission cuts forever B. Only liming; never FGD C. Liming emergency; source cuts essential D. Only Montreal controls SO_2

💡 **Show answer**

Ans: C. Emergency vs prevention. **Logic:** Hierarchy.

Q29. With reference to secondary-pollutant framing, which of the following statements is/are correct?

1. Acid rain built from primary SO_2/NOx.
2. Forms strong acids in atmosphere/deposition pathway.
3. Identical to primary PM emission only.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Secondary story.

Q30. Which one of the following correctly separates Montreal from acid rain?

A. Montreal = ODS/ozone; acid rain = SO_2/NOx deposition chemistry B. Both identical treaties C. Both only desertification D. Both only noise standards

💡 **Show answer**

Ans: A. Different problems/treaties. **Logic:** Standard distractor.

Common Traps

Trap	Correct
Normal rain pH = 7	**~5.6**
Acidic pH > 7	**Acidic < 7**
CO_2 = main severe acid rain gas	**Weak only**; SO_2/NOx drive strong acids
Montreal = acid rain	**Ozone**; Gothenburg/CLRTAP for acidifying
Acid rain = only liquid rain	Includes **dry deposition**
SO_2 lighter than air	**Heavier**
Taj damage from CO_2 alone	**SO_2 + acid → gypsum**

Trap	Correct
Lichens love polluted SO_2 air	**Die / indicate pollution**
FGD removes only NOx	Mainly **SO_2**
Neutral rain = natural rain	Natural rain already **slightly acidic**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 40 — Desertification (मरुस्थलीकरण)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Desertification | Land Degradation (भू-क्षरण) | Drought (सूखा) | Soil Erosion

Sources baked in: NCERT; ISRO Desertification Atlas 2016; UNCCD; IMD; DPAP; UPPCS PYQs **Weight:** ★★ Medium — ASDM definition; sheet→ravine; UNCCD; LDN 2030; DPAP **Last verified:** August 2026

Current Affairs (this topic)

- Twin of Desertification & Land Degradation — UNCCD / LDN CA refresh.
- Keep desertification ≠ drought-only wording clear.

💡 🎯 **Consolidated — 22 Must-Score Facts (Click to Expand)**

1. **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** drylands. It is not merely “the desert expanding.” 2. Drivers combine climate variation with human pressures such as overgrazing, deforestation, and bad irrigation. 3. **Land degradation** is the broader productivity decline and can occur in any climate zone. 4. India’s degraded land share is often taught near **29–30%** (ISRO Atlas **2016**, about **96.4 million ha**). **Rajasthan** (थार) is the worst state hotspot. 5. Drought types run **meteorological → hydrological → agricultural**. Meteorological drought (मौसमी (मौसम विज्ञान) सूखा) is often linked to rainfall below about **75%** of long-period average. 6. **DPAP** began in **1973**. Drought and desertification reinforce each other in a feedback loop. 7. Bundelkhand’s **seven** Uttar Pradesh (उत्तर प्रदेश) districts are a classic UP drought–degradation hotspot. 8. Water erosion stages run **sheet → rill → gully → ravine**. The **Chambal** (चंबल) ravines are the textbook deep badlands. 9. Wind erosion works through **deflation** and **abrasion**, especially in the Thar (थार). 10. **Overgrazing** is the top dryland cause in many coaching facts. Afforestation helps prevent further degradation. 11. The **UNCCD** was adopted (अंगीकृत) in **1994**. **17 June** is Desertification and Drought Day. 12. **Land Degradation Neutrality (LDN)** (भूमि क्षरण तटस्थता) targets **no net loss** of healthy land by **2030** (SDG **15.3**). 13. UNCCD is the desertification convention. **UNFCCC** (यूएनएफसीसीसी) is the climate convention — different treaties. 14. Salinity from poor irrigation is a human-made degradation path even where rainfall is not zero. 15. Accelerated erosion from human misuse is the focus, not only slow geological erosion. 16. Desertification can reverse with sustainable land management; it is not an irreversible death sentence in every plot. 17. Do not define desertification as any bare land in a humid forest belt. 18. Sheet erosion (परत (चादर) अपरदन) removes a thin film; gullies cut channels; ravines are deep, dense badland networks. 19. IMD drought monitoring and DPAP geography often appear beside desertification stems. 20. SDG 15.3 LDN is the policy end-state; UNCCD is the global legal home. 21. Thar wind processes and Chambal water ravines are twin Indian landform facts. 22. Deforestation removes cover and raises runoff and erosion risk in drylands and beyond.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Desertification** | **Land Degradation** | Land degradation specifically occurring in arid, semi-arid, and dry sub-humid areas (drylands) from climatic variations and human activities vs any reduction

in productive potential of land everywhere | मरुस्थलीकरण (शुष्क क्षेत्र) / भूमि क्षरण (सामान्य भूमि) | | **Drought** | **Desertification** | Temporary natural period of deficient precipitation over months/years vs persistent, long-term degradation and loss of biological/economic productivity in dryland soils | सूखा (अस्थायी जल संकट) / मरुस्थलीकरण (स्थायी भूमि ह्रास) | | **Sheet Erosion** | **Gully Erosion** | Uniform, imperceptible removal of thin topsoil layer by surface runoff on gentle slopes vs formation of deep pronounced ravines and gorges rendering land unfit for cultivation (Chambal Badlands) | परत अपरदन (अदृश्य मिट्टी ह्रास) / अवनलिका अपरदन (बीहड़ निर्माण) | | **UNCCD (1994)** | **UNFCCC (1992)** | Only legally binding international agreement linking environment and development to sustainable land management in drylands vs framework treaty for combating greenhouse gases | मरुस्थलीकरण संधि (UNCCD) / जलवायु संधि (UNFCCC) | | **Land Degradation Neutrality (LDN)** | **Net Zero Emissions** | Target under SDG 15.3 ensuring no net loss of healthy land resources by 2030 vs balancing anthropogenic GHG emissions with equivalent carbon removals | भूमि क्षरण तटस्थता (LDN) / नेट जीरो उत्सर्जन |

💡 📋 **Must-score facts — ASDM, Chambal, UNCCD, LDN (Click to Expand)**

Locks | Item | Lock | |------|------| | Desertification | ASDM dryland degradation (≠ only desert expand) | | India share | ~29–30%; ~96.4 mha; Rajasthan worst hotspot | | Drought types | Met → hydro → agri; met often <75% normal rain (सामान्य वर्षा) | | DPAP | **1973** | | Bundelkhand UP | Seven districts drought–degradation hotspot | | Erosion sequence | Sheet → rill → gully → ravine (Chambal) | | UNCCD / Day | **1994** / **17 June** | | LDN | No net loss by **2030** (SDG 15.3) | | UNCCD ≠ UNFCCC | Desertification / climate |

40.1 Desertification

Identity: ASDM drylands | Productivity loss | Overgrazing | UNCCD 1994

- **Desertification** is land degradation in **arid, semi-arid, and dry sub-humid (ASDM)** areas from climate variation and human activities.
- It is **not** only the physical expansion of sand deserts like the Thar. Any qualifying dryland that loses fertility and vegetation can desertify.
- **Overgrazing** is the leading human cause in drylands; deforestation and unsustainable cropping follow.
- Improper irrigation can cause **salinization** and **waterlogging**.
- Drought and climate stress kill vegetation and speed erosion — a feedback loop.
- **Wind erosion** dominates true arid zones such as the **Thar** (थार).
- India's **ISRO Desertification Atlas (2016)** mapped about **96.4 M ha (~29.3%)** under desertification/land degradation; **Rajasthan** is the worst-affected state.
- **UNCCD** adopted **17 June 1994** (Paris); force **1996**.
- **17 June** = World Desertification and Drought Day.
- **SDG 15.3** targets **Land Degradation Neutrality (LDN) by 2030**.

PYQ — UPPCS Prelims 2019, Q42

India aims to achieve land degradation neutrality by the year

Options: A. 2025 B. 2030 C. 2035 D. 2040

💡 **Show answer**

Ans: B

> **Logic:** Bundelkhand (dry sub-humid) can desertify — not only sand deserts.

40.2 Land Degradation

Identity: Broader than desertification | Processes | Wasteland | Watershed fix

- **Land degradation** is any lasting decline in the land's productive capacity.
- **Desertification** is the ASDM subset; erosion, salinity, waterlogging, mining, and pollution are processes under the broader term.
- Continued degradation produces **wasteland** unsuitable for cultivation.
- Consequences: food insecurity, rural poverty, migration, reservoir silting, and higher flood risk on silted beds.
- Solutions: watershed management (वाटरशेड (जलग्रहण) प्रबंधन), afforestation, controlled grazing, soil-water conservation, and LDN restoration projects.
- **Afforestation reduces** degradation — it is not a cause.

Process	Meaning
Water/wind erosion	Topsoil loss
Salinization	Excess irrigation salts
Waterlogging	Poor drainage
Mining / pollution	Direct land damage

> **Logic:** UNCCD ≠ UNFCCC. LDN year = **2030**.

40.3 Drought

Identity: Water deficit types | IMD LPA | DPAP 1973 | Bundelkhand link

- **Drought** is a prolonged water deficit relative (सापेक्ष) to normal supply — not the same as desertification, though the two reinforce each other.
- **Meteorological drought:** rainfall well below normal (IMD often uses **< 75% of Long Period Average** as a severe marker).
- **Hydrological drought (जल-वैज्ञानिक सूखा):** rivers, reservoirs, and groundwater fall.
- **Agricultural drought (कृषि सूखा):** soil moisture fails crops even if some rain fell.
- Socio-economic drought (सामाजिक-आर्थिक सूखा) follows when livelihoods and prices collapse.
- Drought kills vegetation → bare soil → erosion → desertification risk rises.
- **Drought Prone Areas Programme (DPAP)** launched **1973–74** for dryland watershed-type works.
- Related lineage: Desert Development Programme (DDP), later watershed/IWMP-type convergence.
- UP hotspot: **Bundelkhand** seven districts — Jhansi (झांसी), Lalitpur, Jalaun, Hamirpur, Mahoba, Banda, Chitrakoot (चित्रकूट).

PYQ — UPPCS Prelims 2025, Q134

Arrange chronologically: 1. Food for Work 2. Community Development Programme 3. TRYSEM 4. Drought Prone Areas Programme

Options: A. 4, 2, 3, 1 B. 2, 4, 1, 3 C. 2, 4, 3, 1 D. 4, 2, 1, 3

💡 **Show answer**

Ans: B — CDP (सीडीपी) **1952** → DPAP **1973** → Food for Work **1977** → TRYSEM **1979**.

> **Logic:** Drought ≠ famine automatically; drought ≠ permanent desert.

40.4 Soil Erosion

Identity: Normal vs accelerated | Sheet→ravine | Wind | Conservation

- **Normal (geological) erosion** is slow and roughly balances soil formation.
- **Accelerated erosion** is human-driven and faster than formation — the focus.
- **Water erosion ladder:sheet → rill → gully → ravine**.
- **Chambal ravines** (चंबल) (MP–UP–Rajasthan, including Etawah (इटावा)–Agra belt) are the classic Indian ravine badlands.
- **Wind erosion:deflation** (lifting fine particles) and **abrasion** (sandblasting) — strong in the Thar.
- Conservation: contour bunding, terracing, shelter belts, check dams, gully plugging, cover crops, and afforestation.
- Healthy soil biology (e.g. earthworms) supports fertility and structure.

PYQ — UPPCS Prelims 2018, Q101

Thar / Rajasthan desert expanse is mainly of:

Options: A. Pliocene B. Paleocene C. Pleistocene and recent deposits D. Oligocene

💡 **Show answer**

Ans: C

Type	Feature
Sheet	Uniform thin topsoil loss
Rill	Small channels
Gully	Deeper channels
Ravine	Deep badlands (Chambal)
Wind	Deflation + abrasion

> **Logic:** Afforestation and shelter belts **prevent** erosion — they are not causes.

Complete PYQ Bank (Topic 40)

Q1. UPPCS Prelims 2019, Q42

India aims to achieve land degradation neutrality by the year

A. 2025 B. 2030 C. 2035 D. 2040

💡 **Show answer**

Logic: SDG 15.3 Land Degradation Neutrality target year for India is 2030. **Ans: B.** **LDN by 2030** under SDG 15.3 / UNCCD framing. **Trap:** 2025 or 2040 as round-number distractors.

Q2. UPPCS Prelims 2025, Q134

Arrange the following chronologically:

1. Food for Work Programme
2. Community Development Programme
3. Training of Rural Youth for Self-Employment (TRYSEM)

4. Drought Prone Areas Programme (DPAP)

A. 4, 2, 3, 1 B. 2, 4, 1, 3 C. 2, 4, 3, 1 D. 4, 2, 1, 3

💡 **Show answer**

Logic: Chronology of rural employment/welfare programmes — CDP 1952 → DPAP 1973 → Food for Work 1977 → TRYSEM 1979. **Ans: B.** Order: **CDP (1952) → DPAP (1973) → Food for Work (1977) → TRYSEM (1979)**. **Trap:** Placing DPAP before CDP or TRYSEM before Food for Work.

Q3. UPPCS Prelims 2018, Q101

Thar / Rajasthan desert expanse is mainly of:

A. Pliocene B. Paleocene C. Pleistocene and recent deposits D. Oligocene

💡 **Show answer**

Logic: Thar desert surficial deposits — Pleistocene and recent, not Paleocene/Oligocene alone. **Ans: C.** Thar cover is **Pleistocene + recent** aeolian/alluvial material. **Trap:** Picking older Tertiary epochs as "desert = ancient only."

Q4. UPPCS Prelims (pattern)

The UN Convention to Combat Desertification (UNCCD) was adopted in:

A. 1994 B. 1972 C. 1987 D. 2015

💡 **Show answer**

Logic: UNCCD adopted 1994; Desertification and Drought Day is 17 June. **Ans: A.** **UNCCD adopted 17 June 1994** (Paris); in force 1996. **Trap:** Confusing with Rio 1992 or UNFCCC alone.

Q5. UPPCS Prelims (pattern)

Desertification, under UN definition, means:

A. Land degradation in arid, semi-arid, and dry sub-humid areas B. Only physical expansion of the Thar sand sea eastward C. Any bare patch in a humid rainforest D. Permanent ocean acidification

💡 **Show answer**

Logic: Desertification is productivity loss in ASDM drylands — not merely sand-dune expansion. **Ans: A.** **ASDM dryland** productivity loss from climate + human pressure. **Trap:** "Desertification = dunes moving only."

Q6. UPPCS Prelims (pattern)

The correct sequence of water erosion stages is:

A. Sheet → rill → gully → ravine B. Ravine → sheet → gully → rill C. Gully → sheet only D. Wind abrasion → BOD assay

💡 **Show answer**

Logic: Water erosion stages — sheet → rill → gully → ravine; Chambal is the classic ravine belt. **Ans: A.** Thin **sheet** loss → **rill** channels → **gullies** → deep **ravines** (Chambal badlands). **Trap:** Reversing ravine and sheet.

Q7. UPPCS Prelims (pattern)

Which of the following helps **prevent** land degradation?

A. Afforestation and shelter belts B. Overgrazing C. Unchecked clear-felling D. Improper irrigation without drainage (salinization)

> 💡 **Show answer**
>
> **Logic:** Afforestation and shelter belts prevent erosion — they are mitigation, not causes of degradation. **Ans: A.** **Afforestation** stabilises soil; overgrazing and salinization **cause** degradation. **Trap:** Statement stems that list afforestation as a "cause."

Q8. UPPCS Prelims (pattern)

The leading human cause of desertification in drylands is often:

A. Overgrazing B. Contour bunding C. Check dams D. Watershed management

> 💡 **Show answer**
>
> **Logic:** Overgrazing is the leading human driver of dryland desertification in standard teaching. **Ans: A.** **Overgrazing** removes cover → erosion; conservation structures are responses. **Trap:** Picking a conservation measure as the "cause."

Q9. UPPCS Prelims (pattern)

Bundelkhand in Uttar Pradesh, often cited for drought stress, includes about how many UP districts in standard lists?

A. Seven B. Seventy C. Two D. Zero

> 💡 **Show answer**
>
> **Logic:** Bundelkhand spans seven UP districts — drought and degradation hotspot linked to desertification risk. **Ans: A.** **Seven districts** — Jhansi, Lalitpur, Jalaun, Hamirpur, Mahoba, Banda, Chitrakoot. **Trap:** Treating Bundelkhand as only Rajasthan.

Q10. UPPCS Prelims (pattern)

Meteorological drought is commonly indicated when rainfall falls to about:

A. Less than 75% of the long-period average B. More than 150% of LPA always C. Exactly 100% of LPA only D. Zero rainfall for one hour

> 💡 **Show answer**
>
> **Logic:** Meteorological drought — rainfall well below long-period average (IMD often uses <75% LPA as severe marker). **Ans: A.** **<75% LPA** is a standard severe meteorological drought marker. **Trap:** Confusing meteorological with agricultural soil-moisture drought only.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to desertification, which of the following statements is/are correct?

1. Desertification means land degradation only in humid equatorial forests.
2. Not merely "the desert expanding."
3. Identical to any bare land in a humid forest belt.

A. Only 3 B. Only 2 C. Only 1 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 2 is correct. **Logic:** Statement 2 is the teaching point (not merely desert expansion). Statement 1 is false — desertification is ASDM dryland degradation. Statement 3 wrongly equates it with any bare humid-forest land.

Q2. Match List-I with List-II:

List-I	List-II
A. UNCCD	1. 17 June
B. Desertification Day	2. Adopted 1994
C. LDN	3. No net loss healthy land by 2030 (SDG 15.3)

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-1, C-3. **Logic:** UNCCD / Day / LDN.

Q3. Which of the following pairs is/are correctly matched?

1. Desertification drivers exclude climate variation, overgrazing, deforestation and bad irrigation.
2. Land degradation — broader; any climate zone
3. UNCCD identical to UNFCCC

A. 1 and 2 B. Only 3 C. Only 2 D. Only 1

> 💡 **Show answer**
>
> **Ans: C.** Only 2 is correct. **Logic:** Statement 2 is true (land degradation is broader / any climate zone). Statement 1 denies the standard drivers. Statement 3 is false — UNCCD ≠ UNFCCC.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): India's degraded land share often taught ~29–30% (ISRO Atlas 2016 ~96.4 Mha).

Reason (R): Rajasthan is the worst state neighbourhood teaching.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; Rajasthan rank does not itself prove the national % figure. **A/R logic:** National % vs state worst.

Q5. With reference to drought ladder, which of the following statements is/are correct?

1. Meteorological → hydrological → agricultural teaching order variants exist; met often first.
2. DPAP began in 1973.
3. Drought and desertification reinforce each other.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** Drought–desertification link.

Q6. Arrange water erosion stages:

A. Ravine → sheet only B. Sheet → rill → gully → ravine C. Only wind abrasion D. Only LDN certificates

💡 **Show answer**

Ans: B. Sheet→rill→gully→ravine. **Logic:** Chambal ravines textbook.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Thar — wind deflation/abrasion
2. Chambal — water ravines / badlands
3. Chambal — only coastal mangroves

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Chambal = ravines.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Desertification can reverse with sustainable land management.

Reason (R): It is an irreversible death sentence in every plot.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Not irreversible always.

Q9. With reference to Bundelkhand, which of the following statements is/are correct?

1. Seven UP districts — classic drought–degradation hotspot.
2. Links meteorological and agricultural stress neighbourhood.
3. It is India's only coastal desertification capital.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not coastal capital.

Q10. Match List-I with List-II:

List-I	List-II
A. Overgrazing	1. Raises runoff/erosion
B. Afforestation	2. Top dryland cause in many facts
C. Deforestation	3. Helps prevent further degradation

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-3, C-1. **Logic:** Cause/prevention/deforestation.

Q11. Arrange salinity path:

A. Only natural rainforest process B. Poor irrigation → salinity — human-made degradation even if rain ≠ zero C. Only Antarctic ozone D. Only AQI Severe band

💡 **Show answer**

Ans: B. Irrigation salinity. **Logic:** Human-made path.

Q12. With reference to LDN / SDG, which of the following statements is/are correct?

1. LDN = no net loss of healthy land by 2030.
2. Linked to SDG 15.3.
3. UNCCD is the global legal home; UNFCCC is climate.

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: C. All three. **Logic:** LDN + treaty split.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Accelerated erosion from human misuse is the focus.

Reason (R): Only slow geological erosion matters in desertification teaching.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Human-accelerated focus.

Q14. Which one of the following correctly states sheet vs gully vs ravine?

A. All identical BOD metrics B. All identical wind dunes only C. All identical ozone holes D. Sheet thin film; gully channels; ravines deep dense badlands

💡 **Show answer**

Ans: D. Erosion stage meanings. **Logic:** Water-erosion vocabulary.

Q15. How many of the following twin landform facts are correct?

Thar wind processes; Chambal water ravines

A. Both B. Only one C. None D. Three including Sundarbans BR

💡 **Show answer**

Ans: A. Both. **Logic:** Indian twin facts.

Q16. With reference to IMD / DPAP, which of the following statements is/are correct?

1. IMD drought monitoring often appears beside desertification stems.
2. DPAP 1973 geography neighbourhood.

3. DPAP began in 1952 as CDP identical.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** DPAP 1973 — not 1952.

Q17. Match List-I with List-II:

List-I	List-II
A. Meteorological drought	1. Rivers/groundwater stress neighbourhood
B. Agricultural drought	2. Low rainfall
C. Hydrological drought	3. Soil moisture / crop stress

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-3, C-1. **Logic:** Drought types.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): UNCCD was adopted in 1994.

Reason (R): 17 June is Desertification and Drought Day.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; observance date does not explain adoption year. **A/R logic:** Year vs day.

Q19. Which of the following pairs is/are correctly matched?

1. Desertification — ASDM drylands
2. Land degradation — broader productivity decline
3. Desertification — only Yellowstone expansion

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Not Yellowstone.

Q20. Arrange feedback idea:

A. Drought and desertification reinforce each other B. Drought never links to desertification C. Only Montreal controls deserts D. Only AQI controls ravines

> 💡 **Show answer**
>
> **Ans: A.** Mutual reinforcement. **Logic:** Feedback loop.

Q21. With reference to overgrazing, which of the following statements is/are correct?

1. Top dryland cause in many coaching facts.

2. Afforestation helps prevent further degradation.
3. Overgrazing improves soil cover always.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** Statement 3 false.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): SDG 15.3 LDN is the policy end-state.

Reason (R): UNCCD is the global legal home for desertification.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; legal home does not itself define the 2030 LDN end-state wording. **A/R logic:** Policy vs convention.

Q23. How many of the following drivers are human pressures in teaching?

Overgrazing, deforestation, bad irrigation, only solar flares

A. Only one B. Only two C. Only three D. All four

> 💡 **Show answer**
>
> **Ans: C.** Only three. **Logic:** Solar flares not the dryland driver set.

Q24. Which one of the following correctly states wind erosion in Thar?

A. Only catalytic converters B. Only sheet→ravine water stages C. Only BOD crash D. Deflation and abrasion

> 💡 **Show answer**
>
> **Ans: D.** Deflation + abrasion. **Logic:** Thar wind.

Q25. With reference to reversibility, which of the following statements is/are correct?

1. Sustainable land management can reverse desertification in many plots.
2. Not every plot is an irreversible death sentence.
3. Definition requires humid Amazon only.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1 and 2. **Logic:** ASDM — not humid Amazon definition.

Q26. Match List-I with List-II:

List-I	List-II
A. Sheet erosion	1. Deep badland networks
B. Gully	2. Thin film removal
C. Ravine	3. Channel cutting

Code:

A. A-1, B-2, C-3 B. A-2, B-3, C-1 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-3, C-1. **Logic:** Sheet/gully/ravine.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Do not define desertification as any bare land in a humid forest belt.

Reason (R): Desertification is ASDM dryland degradation.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Definition boundary.

Q28. Arrange India's degraded land teaching figure neighbourhood:

A. 0% always B. 100% forests only C. ~29–30% / ~96.4 Mha (ISRO Atlas 2016 teaching) D. Only 2 February Wetlands

💡 **Show answer**

Ans: C. ~29–30%. **Logic:** ISRO Atlas fact.

Q29. With reference to met drought threshold teaching, which of the following statements is/are correct?

1. Often linked to rainfall below about 75% of normal neighbourhood.
2. Sits beside IMD monitoring stems.
3. Identical to World Ozone Day.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not Ozone Day.

Q30. Which one of the following correctly separates UNCCD from UNFCCC?

A. UNCCD desertification; UNFCCC climate — different treaties B. Both identical ozone protocols C. Both identical noise standards D. Both identical AQI eight pollutants

💡 **Show answer**

Ans: A. Desertification vs climate conventions. **Logic:** Treaty split.

Common Traps

Trap	Correct
Desertification = only desert expansion	**ASDM productivity loss**
UNCCD = climate treaty	**Desertification**; UNFCCC = climate
UNCCD signed at Rio 1992	**Adopted 1994**
Afforestation causes degradation	**Prevents** it

Trap	Correct
Gully = sheet	Different stages
Drought = desertification	Related, **not identical**
LDN year 2025/2040	**2030**
Chambal only in one state	**MP–UP–Rajasthan**
DPAP = 1994	**1973–74**
Normal erosion = focus	**Accelerated** erosion is

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 41 — Environmental Monitoring (पर्यावरणीय निगरानी)

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Environmental Indicators | Environmental Monitoring | Environmental Audit (पर्यावरण लेखापरीक्षा)

Sources baked in: CPCB; CGWB; ISO 14001; EPA Form V; SAFAR/IITM; UPPCS PYQs **Weight:** ★★ Medium — BOD; Ringelmann; CGWB≠CPCB; SAFAR; audit≠monitoring **Last verified:** August 2026

Current Affairs (this topic)

- CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) / SPCB monitoring network and satellite products are living CA.
- Keep ambient vs emission monitoring distinction clear.

💡 🎯 **Consolidated — 22 Must-Score Facts (Click to Expand)**

1. The **PSR** model runs **Pressure → State → Response** for environmental indicators. 2. Air indicators include **PM2.5, PM10, SO_2, NOx, and AQI**. Water indicators include **BOD, COD, DO, pH, and coliform**. 3. Noise is measured in **decibels**. Smoke opacity uses the **Ringelmann** scale **0–5**. 4. **Lichens** are classic bio-indicators for sulphur dioxide stress. 5. **Monitoring** means systematic repeated measurement against standards. It is not the same as an audit. 6. **CPCB** (केंद्रीय प्रदूषण नियंत्रण बोर्ड) runs **NAMP** (manual air), **CAAQMS** (real-time air), and **NWMP** (surface water). 7. **CGWB** monitors groundwater quality and levels and publishes the annual groundwater quality report. 8. **SAFAR** (सफर) is the IITM air-quality and weather forecast system associated with Harsh Vardhan's launch messaging in **July 2018**. 9. **NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम) (2019)** expands city monitoring and action. **SAMEER** is an AQI public app. 10. An **environmental audit** verifies compliance with law or clearance conditions — it is more than raw (रॉ) data logging. 11. **Form V** Environmental Statement under EPA Rule **14** goes to the SPCB yearly. 12. **ISO 14001** is the Environmental Management System standard. 13. Consent to Operate links industry permission to both monitoring data and audit compliance. 14. **BOD** is microbial oxygen demand. **COD** is chemical oxygen demand. COD is usually ≥ BOD. 15. NAMP shows long trend with manual stations. CAAQMS gives continuous real-time feeds. 16. CPCB is the air–surface-water regulator home. CGWB is the groundwater home. Do not swap them. 17. Green audits and EC compliance reports sit in the audit family, not in casual one-off sampling. 18. Indicators turn raw numbers into decision signals for policy and public warning. 19. Bio-indicators supplement instruments when living organisms reveal chronic pollution stress. 20. Ringelmann is about visible smoke shade, not a water-quality number. 21. Do not call an audit "just another AQI reading (रीडिंग)." Audit checks legal conformity. 22. Surface-water networks (NWMP) do not replace groundwater networks (CGWB).

💡 ⚡ **Confused Pairs (Click to Expand)**

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
Environmental Indicator	**Environmental Monitoring**	Specific quantitative metric reflecting state of an ecosystem (e.g., Lichens for SO_2, E. coli for fecal water contamination) vs continuous systematic collection and analysis of environmental data over time	पर्यावरणीय संकेतक / पर्यावरणीय निगरानी
Environmental Impact Assessment (EIA)	**Environmental Audit**	Prospective preventive study	

predicting ecological impacts of a project *before* environmental clearance vs retrospective verification of legal compliance and emissions *after* commissioning | पर्यावरण प्रभाव आकलन (EIA) / पर्यावरण लेखापरीक्षा | | **National Air Monitoring Programme (NAMP)** | **Continuous Ambient Air Quality (CAAQMS)** | Nationwide manual sampling network monitoring SO_2, NO_2, PM10, PM2.5 twice weekly vs automated sensors measuring 12 criteria pollutants with real-time online data | राष्ट्रीय वायु निगरानी (NAMP) / सतत परिवेशी वायु निगरानी (CAAQMS) | | **Bioindicator** | **Biomarker** | Organism whose presence/absence indicates environmental quality (Lichens for air, Diatoms/Mayfly for fresh water) vs cellular/biochemical alteration measured in an organism due to chemical exposure | जैव संकेतक (लाइकेन/मेफ्लाई) / जैव मार्कर | | **Ringelmann Scale** | **Decibel (dB)** | Visual chart scale (0 to 5) used to measure the apparent density/opacity of industrial smoke emissions vs logarithmic acoustic unit measuring environmental sound levels | रिंगेलमैन पैमाना (धुआं घनत्व) / डेसिबल (ध्वनि तीव्रता) |

💡 📋 **Must-score facts — PSR, NAMP, audit, BOD (Click to Expand)**

Tools | Item | Lock | |------|------| | PSR | Pressure → State → Response | | Noise / smoke | dB / Ringelmann **0–5** | | Lichens | SO_2 bio-indicators | | CPCB networks | NAMP · CAAQMS · NWMP | | CGWB | Groundwater quality + levels report | | SAFAR | IITM forecast (2018 launch messaging) | | SAMEER | AQI public app | | Form V | EPA Rule 14 Environmental Statement → SPCB yearly | | ISO 14001 | EMS standard | | Monitoring ≠ audit | Repeated measurement / compliance verification |

41.1 Environmental Indicators

Identity: PSR framework | BOD/COD/DO/AQI/dB | Bio-indicators | Ringelmann

- **Environmental indicators** turn raw data into decision signals for policy and public warning.
- **PSR model:Pressure** (emissions, resource use) → **State** (ambient quality) → **Response** (treatment capacity, action plans).
- Air indicators: **PM2.5, PM10, SO_2, NO_2, CO, O_3** (and AQI composite 0–500).
- Water indicators: **BOD, COD, DO, pH, coliform**. Higher BOD = more organic pollution; **COD ≥ BOD**; healthy streams need adequate **DO**.
- Noise indicator: **decibel (dB)** (डेसिबल) — logarithmic; CPCB zone limits apply.
- **Ringelmann scale (रिंगेलमैन पैमाना) (0–5)** measures **smoke plume density/opacity**, not water, fog, or noise.
- Bio-indicators: **lichens** (SO_2-sensitive air quality); aquatic macroinvertebrates for clean streams.

PYQ — UPPCS Prelims 2023, Q103

BOD is a standard criterion for—

Options: A. Pollution assay in aquatic ecosystems B. High-altitude O_2 C. Blood O_2 D. Forest O_2

💡 **Show answer**

Ans: A

PYQ — UPPCS Prelims 2021, Q34

Ringelmann scale is used to measure density of

Options: A. Smoke B. Polluted water C. Fog D. Noise

💡 **Show answer**

Ans: A

PYQ — UPPCS Prelims 2025, Q61

Lichens are good pollution indicators; they do **not** grow well in polluted areas → **Only statement 1**.

Indicator	Medium	High value means
PM2.5 / AQI	Air	Worse air
BOD	Water	More organic pollution
DO	Water	Low = stressed aquatic life
dB	Noise	Louder / possible breach
Ringelmann	Smoke	Denser plume

> **Logic:** BOD ≠ blood oximetry. Ringelmann ≠ fog meter.

41.2 Environmental Monitoring

Identity: Networks | CPCB vs CGWB | SAFAR | NCAP | SPCB consents

- **Environmental monitoring** is systematic, repeated measurement of air, water, soil, noise, and related parameters against standards.
- Cycle: sample → analyse → compare with norms → report → enforce/public alert.
- **CPCB** runs national **air** and **surface-water** programmes and sets ambient/effluent standards with SPCBs.
- **NAMP** = manual air stations for long-term trends; **CAAQMS** = continuous real-time ambient air (feeds AQI boards).
- **NWMP** = National Water Quality Monitoring Programme for rivers/lakes.
- **CGWB** monitors **groundwater** levels and quality; publishes the **Annual Groundwater Quality Report** — not CPCB.
- **SAFAR** (IITM) combines air-quality monitoring with weather forecasting; inaugurated by **Dr. Harsh Vardhan** (July 2018).
- **NCAP (2019)** expands monitoring and action in non-attainment cities (including multiple UP cities).
- **SAMEER** app disseminates CPCB AQI; **PARIVESH** tracks EC compliance reporting.
- **UPPCB** issues Consent to Establish/Operate and inspects industries in Uttar Pradesh (उत्तर प्रदेश).

PYQ — UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 prepared by:

1. CPCB 2. CWC (केंद्रीय जल आयोग) 3. CGWB

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Ans: B**

PYQ — UPPCS Prelims 2018, Q84

SAFAR inaugurated (July 2018) by:

> 💡 **Show answer**
>
> **Dr. Harsh Vardhan**

System	Nodal	Medium
NAMP / CAAQMS	CPCB–SPCB	Air

System	Nodal	Medium
NWMP	CPCB	Surface water
WQMN / GW Quality Report	CGWB	Groundwater
SAFAR	IITM	Air + forecast
Noise monitoring	CPCB–SPCB	Sound

> **Logic:** CPCB ≠ groundwater atlas. CGWB ≠ AQI boards.

41.3 Environmental Audit

Identity: Compliance verification | Form V | ISO 14001 | ≠ monitoring alone

- **Environmental audit** is a systematic, documented check of whether an organisation meets environmental laws, consent conditions, and internal EMS goals.
- Monitoring **collects data**; audit **verifies compliance** and management systems using that data (and inspections).
- Under **EPA Rules, Rule 14**, specified industries file an annual **Environmental Statement (Form V)** with the SPCB — India's statutory disclosure/audit-type tool.
- **ISO 14001** is the voluntary international standard for an **Environmental Management System (EMS)**.
- Post-EC projects submit periodic **compliance reports** (often six-monthly) — audit/compliance track linked to EIA (Topic 14).
- Green audits in institutions review energy, water, waste, and campus ecology (पारिस्थितिकी).
- Consent to Operate renewal typically depends on monitoring results plus inspection/audit findings.

Tool	Role
Form V statement	Annual statutory disclosure to SPCB
ISO 14001	Voluntary EMS certification
EC compliance report	Project clearance conditions
SPCB inspection	Enforcement audit on ground

> **Logic:** Passing an AQI reading ≠ completing an environmental audit.

Complete PYQ Bank (Topic 41)

Q1. UPPCS Prelims 2023, Q103

Biological Oxygen Demand (BOD) is a standard criterion for—

A. Pollution assay in aquatic ecosystems B. High-altitude O_2 C. Blood O_2 D. Forest O_2

💡 **Show answer**

Logic: BOD assays organic pollution in aquatic ecosystems — not blood or altitude oxygen. **Ans: A.**
BOD = microbial oxygen demand in water; higher BOD = dirtier water. **Trap:** "Oxygen" in name → medical/altitude options.

Q2. UPPCS Prelims 2021, Q34

Ringelmann scale is used to measure density of

A. Smoke B. Polluted water C. Fog D. Noise

> 💡 **Show answer**
>
> **Logic:** Ringelmann scale grades smoke plume opacity 0–5 — not water, fog, or noise. **Ans: A.** **Ringelmann 0–5** = visible **smoke** density from stacks. **Trap:** Fog meter or BOD/water quality distractors.

Q3. UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 was prepared by:

1. CPCB
2. CWC
3. CGWB

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Logic:** Annual Groundwater Quality Report is CGWB's product — CPCB handles air/surface water, not this report. **Ans: B (Only 3).** **CGWB** monitors groundwater quality/levels. **Trap:** CPCB name sounds like "all pollution."

Q4. UPPCS Prelims 2025, Q61

Consider the following statements:

1. Lichens are very good indicators of pollution.
2. Lichens grow well in polluted areas.

A. Only 2 B. Neither C. Both D. Only 1

> 💡 **Show answer**
>
> **Logic:** Lichens are sensitive SO_2 indicators — they do not grow well in polluted air. **Ans: D (Only 1).** Indicators **die back** in dirty air — statement 2 is false. **Trap:** "Good indicator" read as "likes pollution."

Q5. UPPCS Prelims 2018, Q84

SAFAR (System of Air Quality and Weather Forecasting and Research) was inaugurated in July 2018 by:

A. Dr. Harsh Vardhan B. Governor of RBI C. Director-General of WTO D. Secretary, UNFCCC alone

> 💡 **Show answer**
>
> **Logic:** SAFAR is IITM's air-quality and weather forecast system — inaugurated by Dr. Harsh Vardhan in July 2018. **Ans: A.** **Dr. Harsh Vardhan** launched SAFAR (IITM-backed). **Trap:** Attributing SAFAR to IMD alone or CPCB alone.

Q6. UPPCS Prelims (pattern)

National Clean Air Programme (NCAP) was launched in:

A. 2019 B. 1974 C. 1987 D. 1730

> 💡 **Show answer**
>
> **Logic:** NCAP (2019) expanded monitoring and city action plans for non-attainment cities — complements CPCB networks. **Ans: A.** **NCAP 2019** targets ~20–30% PM cut in 122 cities. **Trap:** Water Act 1974 or Montreal 1987 year swaps.

Q7. UPPCS Prelims (pattern)

Continuous Ambient Air Quality Monitoring Stations (CAAQMS) primarily provide:

A. Real-time ambient air quality data B. Only annual groundwater atlases C. Only Form V filing receipts D. Only noise silence-zone maps

> 💡 **Show answer**
>
> **Logic:** CAAQMS gives continuous real-time ambient air data; NAMP is mainly manual long-term trend stations. **Ans: A.** **CAAQMS** = continuous feeds for AQI boards; **NAMP** = manual trend network. **Trap:** Calling NAMP "real-time only."

Q8. UPPCS Prelims (pattern)

Under the Environment (Protection) Rules, the annual **Environmental Statement (Form V)** is filed with:

A. State Pollution Control Board (SPCB) B. Montreal Protocol Secretariat only C. CITES Management Authority only D. Ramsar Bureau only

> 💡 **Show answer**
>
> **Logic:** Form V Environmental Statement under EPA Rule 14 is the annual statutory disclosure to SPCB. **Ans: A.** **Form V** goes to **SPCB** yearly (Rule 14). **Trap:** Confusing with ISO 14001 certificate or EC portal alone.

Q9. UPPCS Prelims (pattern)

ISO 14001 standard relates to:

A. Environmental Management Systems B. Carbon credit issuance under Kyoto only C. CITES Appendix listing D. Ringelmann smoke photography only

> 💡 **Show answer**
>
> **Logic:** ISO 14001 is the voluntary Environmental Management System standard — not a treaty or AQI tool. **Ans: A.** **ISO 14001** = voluntary **EMS** certification. **Trap:** Equating ISO certificate with Form V statutory filing.

Q10. UPPCS Prelims (pattern)

Environmental **audit** differs from **monitoring** because audit:

A. Verifies compliance with laws, consents, and management systems B. Only publishes one daily AQI colour C. Only measures smoke once without standards D. Replaces all groundwater monitoring by CGWB

> 💡 **Show answer**
>
> **Logic:** Monitoring collects data; environmental audit verifies legal/EC compliance using that data. **Ans: A.** **Monitoring** = repeated measurement; **audit** = documented compliance check (Form V, EC reports, inspections). **Trap:** "We have AQI data, so audit is done."

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to PSR, which of the following statements is/are correct?

1. Pressure → State → Response.
2. Indicators are identical to raw instrument readings and never support decisions.
3. PSR means only Ringelmann smoke shade.

A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: D. Only 1 is correct. **Logic:** Statement 1 is the PSR chain. Statement 2 is false — indicators convert raw numbers into decision signals. Statement 3 confuses PSR with Ringelmann smoke shade.

Q2. Match List-I with List-II:

List-I	List-II
A. NAMP	1. Real-time air
B. CAAQMS	2. Manual air trends
C. NWMP	3. Surface water

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** CPCB network trio.

Q3. Which of the following pairs is/are correctly matched?

1. Standard air indicators exclude PM2.5, PM10, SO_2, NOx and AQI.
2. Water indicators — BOD, COD, DO, pH, coliform
3. CGWB — surface-water NWMP identical to CPCB

A. Only 1 B. 1 and 2 C. Only 3 D. Only 2

💡 **Show answer**

Ans: D. Only 2 is correct. **Logic:** Statement 2 is true (BOD/COD/DO/pH/coliform). Statement 1 wrongly excludes the usual air indicators. Statement 3 confuses CGWB groundwater with CPCB surface-water NWMP.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Monitoring is systematic repeated measurement against standards.

Reason (R): An audit is identical to an AQI reading.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Audit checks legal conformity — not just another AQI reading.

Q5. With reference to bio-indicators, which of the following statements is/are correct?

1. Lichens indicate SO_2 stress.
2. Living organisms can reveal chronic pollution stress.
3. Ringelmann is a water-quality number.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Ringelmann = smoke shade 0–5.

Q6. Arrange BOD vs COD:

A. BOD always > COD B. COD usually ≥ BOD; BOD = microbial O_2 demand C. Both identical to Noise dB D. Both identical to ISO 14001

💡 **Show answer**

Ans: B. COD ≥ BOD usually. **Logic:** Oxygen-demand pair.

Q7. Which of the following pairs is/are NOT correctly matched?

1. SAFAR — IITM AQ + weather forecast
2. SAMEER — AQI public app
3. SAFAR — CGWB groundwater report

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** SAFAR ≠ CGWB.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Form V Environmental Statement under EPA Rule 14 goes to SPCB yearly.

Reason (R): ISO 14001 is the Environmental Management System standard.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; ISO 14001 does not explain Form V filing. **A/R logic:** Statement vs EMS standard.

Q9. With reference to NCAP, which of the following statements is/are correct?

1. 2019 expands city monitoring and action.
2. SAMEER publishes AQI for public warning.
3. SAMEER replaces NCAP city action plans.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** App ≠ action-plan substitute.

Q10. Match List-I with List-II:

List-I	List-II
A. Noise	1. Visible smoke shade 0–5
B. Ringelmann	2. Decibels
C. Consent to Operate	3. Links permission to monitoring + audit

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Noise / Ringelmann / CTO.

Q11. Arrange CPCB vs CGWB homes:

A. Both identical always B. CPCB air–surface water; CGWB groundwater C. CGWB runs NAMP only D. CPCB publishes only ISFR

💡 **Show answer**

Ans: B. Do not swap regulator homes. **Logic:** Institution split.

Q12. With reference to environmental audit, which of the following statements is/are correct?

1. Verifies compliance with law or clearance conditions.
2. More than raw data logging.
3. Green audits / EC compliance sit in the audit family.

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: C. All three. **Logic:** Audit family.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): NWMP surface-water networks do not replace CGWB groundwater networks.

Reason (R): Different media need different monitoring homes.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Surface ≠ groundwater networks.

Q14. Which one of the following correctly states NAMP vs CAAQMS?

A. Both only ISO 14001 logos B. Both only groundwater C. Both only Form V to MoEFCC weekly D. NAMP manual long trends; CAAQMS continuous real-time

💡 **Show answer**

Ans: D. Manual trends vs real-time. **Logic:** Air network pair.

Q15. How many of the following are air indicators in teaching?

PM2.5, PM10, SO_2, NOx, AQI, BOD

A. Only five B. Only four C. Only three D. All six

💡 **Show answer**

Ans: A. Only five. **Logic:** BOD is water.

Q16. With reference to SAFAR launch neighbourhood, which of the following statements is/are correct?

1. IITM system.

2. Associated with July 2018 launch messaging teaching.
3. It is the annual CGWB groundwater quality report.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** Not CGWB report.

Q17. Match List-I with List-II:

List-I	List-II
A. BOD	1. EMS standard
B. COD	2. Microbial oxygen demand
C. ISO 14001	3. Chemical oxygen demand

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-3, C-1. **Logic:** BOD/COD/ISO.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Bio-indicators supplement instruments.

Reason (R): Lichens are classic SO_2 stress indicators.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; lichen example does not alone define the whole supplement role. **A/R logic:** Concept vs example.

Q19. Which of the following pairs is/are correctly matched?

1. Monitoring ≠ audit
2. Audit checks legal conformity
3. Monitoring = only one-off casual sampling identical to green audit

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Monitoring ≠ casual one-off audit.

Q20. Arrange Form V destination:

A. SPCB yearly under EPA Rule 14 B. UNEP Nairobi only C. Only IUCN Gland D. Only WWF Living Planet

> 💡 **Show answer**
>
> **Ans: A.** SPCB yearly. **Logic:** Environmental Statement.

Q21. With reference to Consent to Operate, which of the following statements is/are correct?

1. Links industry permission to monitoring data.
2. Links to audit compliance.
3. Replaces ISO 14001 entirely as a global EMS.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** CTO ≠ ISO substitute.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Ringelmann measures visible smoke shade on a 0–5 scale.

Reason (R): It is a standard water BOD substitute.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Smoke — not water BOD.

Q23. How many of the following CPCB products are correct?

NAMP, CAAQMS, NWMP

A. Only one B. Only two C. All three D. None

💡 **Show answer**

Ans: C. All three. **Logic:** CPCB trio.

Q24. Which one of the following correctly states CGWB's report role?

A. Only Form V to UNEP B. Only NAMP manual air C. Only Ringelmann charts D. Groundwater quality/levels; annual groundwater quality report

💡 **Show answer**

Ans: D. Groundwater home. **Logic:** CGWB identity.

Q25. With reference to water indicators, which of the following statements is/are correct?

1. Include BOD, COD, DO, pH, coliform.
2. COD usually ≥ BOD.
3. PM2.5 is a classic water coliform test.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** PM2.5 is air.

Q26. Match List-I with List-II:

List-I	List-II
A. NCAP	1. AQI app
B. SAMEER	2. 2019 city action/monitoring
C. Lichens	3. SO_2 bio-indicator

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: B.** A-2, B-1, C-3. **Logic:** NCAP/SAMEER/lichens.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Indicators are decision signals for policy and public warning.

Reason (R): Raw numbers alone without standards are already a full environmental audit.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** Indicators ≠ audit by themselves.

Q28. Arrange green audit family idea:

A. Only casual one-off sampling B. Only AQI colour bands C. Green audits and EC compliance reports sit in audit family D. Only Ringelmann 5 forever

> 💡 **Show answer**
>
> **Ans: C.** Audit family — not casual sampling. **Logic:** Compliance reports.

Q29. With reference to noise, which of the following statements is/are correct?

1. Measured in decibels.
2. Distinct from Ringelmann smoke scale.
3. Noise dB equals COD mg/L always.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Different metrics.

Q30. Which one of the following best separates monitoring from audit?

A. Monitoring = repeated measurement vs standards; audit = legal/clearance conformity check B. Both identical AQI apps only C. Both identical groundwater wells only D. Both identical ISO logos only

> 💡 **Show answer**
>
> **Ans: A.** Measurement vs conformity. **Logic:** Core distinction.

Common Traps

Trap	Correct
BOD = blood/altitude oxygen	**Aquatic pollution assay**
Ringelmann = fog/noise/water	**Smoke**
GW Quality Report = CPCB	**CGWB**
Monitoring = audit	Measure ≠ verify compliance

Trap	Correct
Lichens thrive in dirty air	**Sensitive indicators**
SAFAR = IMD alone / CPCB alone	**IITM** (+ MoEFCC support)
Form V = ISO certificate	Statutory **statement** to SPCB
COD < BOD always	**COD ≥ BOD**
NAMP = real-time only	Mostly **manual**; CAAQMS = continuous
High BOD = clean water	**Dirty** organics

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 42 — International Environmental Organizations

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) | UNDP (यूएनडीपी) | FAO | UNESCO | IUCN (आईयूसीएन) | WWF (डब्ल्यूडब्ल्यूएफ) | IPCC (आईपीसीसी) | WMO | GEF | Greenpeace

Sources baked in: IPCC AR6; UNEP; UNDP HDR; IUCN Red List; WWF LPR; FAO SOFO; UNESCO WH/MAB; UPPCS PYQs **Weight:** ★★★ High — org ↔ report; HQ traps; IUCN≠CITES; IPCC≠UNFCCC **Last verified:** August 2026

Current Affairs (this topic)

- UNEP / IUCN / IPCC / GEF role headlines refresh institutional CA.
- HQ and mandate traps stay fixed teaching.

💡 🎯 **Consolidated — 29 Must-Score Facts (Click to Expand)**

1. **UNEP** was created after **Stockholm 1972** (स्टॉकहोम सम्मेलन). Its HQ is **Nairobi**. Flagship products include the **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) and **GEO**. 2. **UNDP** sits in **New York** and publishes the **Human Development Report / HDI** (from **1990**). 3. **FAO** sits in **Rome** and publishes **SOFO**. It co-hosts Rotterdam work with UNEP. 4. **UNESCO** sits in **Paris** and runs World Heritage (विश्व धरोहर) and the **MAB** Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र) programme. 5. **IUCN** (आईयूसीएन) sits in **Gland** and publishes the **Red List** of extinction risk. 6. **WWF** also sits in **Gland** and publishes the **Living Planet Report / LPI**. 7. **IPCC** sits in **Geneva**. It was created by **WMO + UNEP** in **1988** and shared the **2007** Nobel Peace Prize. 8. Emissions Gap Report = **UNEP**, not IPCC or UNDP. HDR = **UNDP**, not UNEP. 9. IPCC does science assessment. **UNFCCC** (यूएनएफसीसीसी) is the climate treaty. Do not merge the two. 10. IUCN assesses extinction risk. **CITES** (साइट्स) regulates wildlife **trade** (पण्याध्यक्ष). Different jobs. 11. IUCN and WWF share a city but differ: assessment union versus conservation NGO campaigning and Living Planet metrics. 12. UNEP coordinates environment (पर्यावरण). UNDP coordinates development. HQ and report facts separate them. 13. UNESCO heritage–education–MAB is not the same as UNEP environment coordination. 14. The **GEF (1991)** is a multilateral funder. Donors are not "developed countries only" as a rigid fact. 15. IPCC Assessment Reports synthesize climate science. They do not set national NDCs. 16. FAO's food–forest–agriculture mandate explains why SOFO and pesticide PIC work sit there. 17. Red List categories are IUCN risk ranks, not CITES Appendix numbers. 18. Living Planet Index tracks vertebrate abundance trends for WWF messaging. 19. Nairobi versus Geneva versus New York versus Rome versus Paris versus Gland is a high-yield HQ ladder. 20. Do not credit IPCC with the Emissions Gap Report. 21. Do not credit UNEP with HDI. 22. Treaty secretariats hosted or supported by UNEP (Montreal, Basel, Stockholm POPs, Minamata) are still treaty bodies, not UNEP synonyms. 23. WWF is not the Red List author. IUCN is. 24. FAO HQ is Rome, not Paris or Geneva. 25. **WMO** headquarters is **Geneva**. It grew from the International Meteorological Organization (**1873**) and became WMO on **17 March 1951**. It co-created the **IPCC** with UNEP in **1988**. 26. **Greenpeace International** is an environmental NGO headquartered in **Amsterdam**, Netherlands. 27. The **US EPA** was established on **2 December 1970**. Do not confuse it with India's **Environment (Protection) Act, 1986**. 28. **OPCW** implements the **Chemical Weapons Convention** (force **29 April 1997**); HQ is **The Hague**. It won the Nobel Peace Prize in **2013**. It is **not** an EU, NATO, or WHO body. 29. **UN-Habitat** (UN Human Settlements

Programme; HQ **Nairobi**, from **1978**) promotes sustainable towns and shelter. Partners are **not** governments-only.

💡 ⚡ **Confused Pairs (Click to Expand)**

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
UNEP	**UNDP**	UN environmental coordinator established post-1972 Stockholm Conference (HQ: Nairobi, Kenya) vs UN development network producing the Human Development Index (HQ: New York)	यूएनईपी (नैरोबी, पर्यावरण) / यूएनडीपी (न्यूयॉर्क, विकास)
IPCC	**UNFCCC**	Scientific review panel established in 1988 by WMO/UNEP to assess peer-reviewed climate data (HQ: Geneva) vs legally binding treaty secretariat coordinating annual COPs (HQ: Bonn, Germany)	आईपीसीसी (वैज्ञानिक समीक्षा) / यूएनएफसीसीसी (वार्ता सचिवालय)
IUCN	**WWF**	Oldest global conservation union (1948, Gland) publishing the Red List of Threatened Species vs private international conservation NGO (1961, Gland) famous for Giant Panda logo and Earth Hour	आईयूसीएन (रेड लिस्ट) / डब्ल्यूडब्ल्यूएफ (पांडा प्रतीक)
CITES	**CMS (Bonn Convention)**	Regulates commercial cross-border trade in endangered species of wild flora and fauna (1973) vs promotes conservation and habitat protection for migratory wild animal species (1979)	साइट्स (वन्यजीव व्यापार) / बॉन कन्वेंशन (प्रवासी प्रजातियां)
Global Environment Facility (GEF)	**Green Climate Fund (GCF)**	Multilateral financial mechanism (World Bank 1991) serving CBD, UNCCD, Stockholm, Minamata, and UNFCCC vs dedicated operating financial entity of UNFCCC created at COP16 Cancun	जीईएफ (वैश्विक पर्यावरण कोष) / जीसीएफ (हरित जलवायु कोष)

💡 📋 **Must-score facts — UNEP UNDP IUCN IPCC HQs (Click to Expand)**

Org ↔ HQ / product | Item | Lock |
|---|---|
| UNEP | Nairobi; Emissions Gap + GEO; post-Stockholm 1972 |
| UNDP | New York; HDR/HDI |
| FAO | Rome; SOFO |
| UNESCO | Paris; WH + MAB |
| IUCN | Gland; Red List |
| WWF | Gland; Living Planet |
| IPCC | Geneva; WMO+UNEP 1988; Nobel 2007 |
| IUCN ≠ CITES | Risk assess / trade regulate |
| GEF | **1991** multilateral funder |

42.1 UNEP

Identity: Stockholm 1972 | Nairobi | Emissions Gap | GEO | Treaty host

- **UNEP (United Nations Environment Programme)** is the UN's principal environment coordination body, created after the **Stockholm Conference (1972)** (स्टॉकहोम सम्मेलन).
- **HQ: Nairobi, Kenya** — not Geneva or New York.
- Publishes the annual **Emissions Gap Report** (NDC shortfall vs 1.5–2°C pathways) and **GEO (Global Environment Outlook)**.
- Hosts/supports secretariats for treaties such as Montreal, Basel, Stockholm POPs, and Minamata.
- Co-created **IPCC** with WMO, but Emissions Gap is **UNEP's own** report.
- **GEF (Global Environment Facility, 1991)** finances projects under major environment conventions; UNEP is an implementing agency. Donors are **not** "developed countries only."
- GEF serves as a financial mechanism for five major conventions: **Minamata**, **Stockholm POPs**, **CBD**, **UNCCD**, and **UNFCCC**.
- Core implementing partners include **UNDP**, **UNEP**, and the **World Bank**.

PYQ — UPPCS Prelims 2024, Q6

Emissions Gap Report publisher among org–report match:

💡 **Show answer**

UNEP — Full key: HDR→UNDP, WEO→IMF, WIR→UNCTAD, Emissions Gap→**UNEP** → **A (3 1 4 2)**.

PYQ — UPPCS Prelims 2025, Q145

GEF: (1) established 1991 (2) only developed countries are donors

Options: A. Only 2 B. Neither C. Both D. Only 1

💡 **Show answer**

Ans: D — Year true; donor statement false.

> **Logic:** Emissions Gap ≠ IPCC Assessment Report.

42.2 UNDP

Identity: New York | HDR 1990 | HDI | Mahbub ul Haq + Amartya Sen | SDGs

- **UNDP (United Nations Development Programme)** is the UN's development network agency.
- **HQ: New York**.
- Publishes the **Human Development Report (HDR)** and **Human Development Index (HDI)** since **1990**.
- HDI was developed by **Mahbub ul Haq** with conceptual contribution from **Amartya Sen** — not a "Priesner-led" first team.
- Supports country-level **SDG** implementation; environment enters via sustainable development programming.
- Not the publisher of Emissions Gap, Red List, or Living Planet Report.

PYQ — UPPCS Prelims 2025, Q132

1. UNDP published first HDR in 1990 with HDI.
2. First HDI team led by Mahbub-ul-Haq and S. Priesner.

Options: A. Only 2 B. Neither C. Both D. Only 1

💡 **Show answer**

Ans: D — Stmt 1 true; stmt 2 false (Haq–Sen lineage, not Priesner).

PYQ — UPPCS Prelims 2018, Q35

HDI was first developed by:

Options: A. UNDP B. IMF C. UNICEF D. UNCTAD

💡 **Show answer**

Ans: A

> **Logic:** UNDP ≠ UNEP. HDR ≠ Emissions Gap.

42.3 FAO

Identity: Rome | Food–agriculture–forests | SOFO | Rotterdam with UNEP

- **FAO (Food and Agriculture Organization)** is the UN specialised agency for food and agriculture.
- **HQ: Rome, Italy**.
- Environment role: agriculture–forest–biodiversity–land-use links; pesticide and forestry governance.
- Co-publishes **State of the World's Forests (SOFO)** with UNEP.

- Jointly administers the **Rotterdam Convention** (PIC for hazardous chemicals in trade) with **UNEP**.
- Not the publisher of HDR, Emissions Gap, or Red List.

> **Logic:** FAO HQ = **Rome**, not Nairobi.

42.4 UNESCO

Identity: Paris | World Heritage | MAB Biosphere Reserves (जैवमंडल आरक्षित क्षेत्र) | ESD

- **UNESCO** leads education, science, and culture — environment enters via heritage and biosphere programmes.
- **HQ: Paris**.
- Designates **UNESCO World Heritage Sites** (natural/cultural/mixed) under the 1972 WH Convention.
- Runs the **Man and the Biosphere (MAB)** programme — World Network of Biosphere Reserves.
- Supports Education for Sustainable Development (ESD).
- **Limits to Growth** is **Club of Rome**, not UNESCO.

PYQ — UPPCS Prelims 2019, Q35

Jaipur listed as UNESCO World Heritage Site at the 43rd session (पंगु सत्र) held in:

Options: A. Baku B. Bishkek C. Istanbul D. Marrakesh

💡 **Show answer**

Ans: A (Baku)

PYQ — UPPCS Prelims 2019, Q146

"Limits to Growth" propounded by:

Options: A. Club of Rome B. UNESCO C. Brundtland D. Agenda 21

💡 **Show answer**

Ans: A

> **Logic:** UNESCO = WH/MAB. UNEP = environment programme.

42.5 IUCN

Identity: 1948 | Gland | Red List | Assess risk ≠ regulate trade

- **IUCN (International Union for Conservation of Nature)** is a global conservation union of governments and NGOs, founded **1948**.
- **HQ: Gland, Switzerland**.
- Publishes the **IUCN Red List** of extinction risk (LC, NT, VU, EN, CR, EW, EX).
- **Threatened** categories commonly = **VU + EN + CR**.
- Assesses scientific extinction risk; does **not** regulate wildlife trade — **CITES** does.
- Advises CBD/CITES scientifically but is a separate organisation.

> **Logic:** "IUCN bans trade" = false. **CITES** regulates trade.

42.6 WWF

Identity: Conservation NGO | Gland | Living Planet Report | Panda logo

- **WWF (World Wide Fund for Nature)** is a global conservation **NGO**, not a UN agency.
- Since **1961**, WWF's logo animal has been the **Giant Panda (*Ailuropoda melanoleuca*)**.
- It publishes the **Living Planet Report** and Living Planet Index.
- **HQ: Gland, Switzerland** — same city as IUCN, different organisation.
- Publishes the **Living Planet Report** and **Living Planet Index (LPI)** on vertebrate population (जनसंख्या) trends.
- Known for the **giant panda** logo and flagship-species campaigns.
- Campaigns and advocates; IUCN assesses via Red List.

> **Logic:** Living Planet = **WWF**. Red List = **IUCN**.

42.7 IPCC

Identity: 1988 WMO+UNEP | Geneva | Assessment Reports | Science ≠ policy

- **IPCC (Intergovernmental Panel on Climate Change (जलवायु परिवर्तन))** was established in **1988** by **WMO** and **UNEP**.
- **HQ: Geneva**.
- Publishes **Assessment Reports** (AR1→AR6; AR6 synthesis **2023**) plus special reports.
- Synthesises existing science; does **not** set NDCs or negotiate treaties.
- Shared the **Nobel Peace Prize 2007** with Al Gore.
- **UNFCCC/COP** is the policy–treaty track; IPCC informs it.

> **Logic:** IPCC ≠ UNFCCC. IPCC ≠ Emissions Gap Report.

42.8 WMO, Greenpeace & US EPA

Identity: WMO Geneva | Greenpeace Amsterdam | US EPA 1970

- The **World Meteorological Organization (WMO)** (डब्लूएमओ) provides the global framework for meteorology and operational hydrology.
- WMO originated from the **International Meteorological Organization (1873)** and became WMO on **17 March 1951**.
- **WMO headquarters is Geneva, Switzerland.** It co-created the **IPCC** with UNEP in **1988**.
- **Greenpeace International** is an environmental NGO headquartered in **Amsterdam, Netherlands**.
- The **US Environment Protection Agency (EPA)** was established on **2 December 1970** to write and enforce environmental health regulations in the United States.
- Do not confuse the **US EPA** with India's **Environment (Protection) Act, 1986**.

Body	HQ / year	Job
WMO	Geneva	Weather/climate cooperation; IPCC parent
Greenpeace	Amsterdam	Environmental NGO
US EPA	Est. 1970	US federal environment agency
UNEP	Nairobi	UN environment programme

Org ↔ Report ↔ HQ Card

Org	HQ	Flagship fact
UNEP	Nairobi	Emissions Gap / GEO
UNDP	New York	HDR / HDI
FAO	Rome	SOFO; Rotterdam (+UNEP)
UNESCO	Paris	World Heritage; MAB
IUCN	Gland	Red List
WWF	Gland	Living Planet
IPCC	Geneva	Assessment Reports
WMO	Geneva	Meteorology; IPCC co-parent
Greenpeace	Amsterdam	Environmental NGO

Complete PYQ Bank (Topic 42)

Q1. UPPCS Prelims 2024, Q6

Match List-I (Organisation) with List-II (Report):

List-I	List-II
1. UNDP	A. Emissions Gap Report
2. IMF	B. Human Development Report
3. UNEP	C. World Economic Outlook
4. UNCTAD	D. World Investment Report

Row order is not the answer code.

A. 3 1 4 2 B. 1 3 2 4 C. 2 4 1 3 D. 4 2 3 1

💡 **Show answer**

Logic: Org–report match — Emissions Gap Report is UNEP's; HDR is UNDP's; WEO is IMF's; WIR is UNCTAD's. **Ans: A (3 1 4 2).** HDR→**UNDP**; WEO→**IMF**; WIR→**UNCTAD**; Emissions Gap→**UNEP**. **Trap:** Crediting IPCC or UNDP with Emissions Gap Report.

Q2. UPPCS Prelims 2025, Q132

Consider the following statements:

1. UNDP published the first Human Development Report in 1990 with the HDI.
2. The first HDI team was led by Mahbub-ul-Haq and S. Priesner.

A. Only 2 B. Neither C. Both D. Only 1

💡 **Show answer**

Logic: HDR/HDI started 1990 under UNDP with Mahbub ul Haq and Amartya Sen — not "Priesner-led" team. **Ans: D (Only 1).** Stmt 1 true; stmt 2 false — **Haq–Sen** lineage, not Priesner. **Trap:** Accepting any plausible Western name on stmt 2.

Q3. UPPCS Prelims 2025, Q145

With reference to the Global Environment Facility (GEF):

1. It was established in 1991.
2. Only developed countries are donors to the GEF.

A. Only 2 B. Neither C. Both D. Only 1

> 💡 **Show answer**
>
> **Logic:** GEF established 1991; donors are not "developed countries only." **Ans: D (Only 1).** **1991** true; donor statement **false** — developing countries also contribute. **Trap:** Treating GEF as Annex-I-only club.

Q4. UPPCS Prelims 2018, Q35

HDI was first developed by:

A. UNDP B. IMF C. UNICEF D. UNCTAD

> 💡 **Show answer**
>
> **Logic:** HDI is UNDP's flagship index — not IMF, UNICEF, or UNCTAD. **Ans: A.** **UNDP** publishes HDR/HDI from **1990**. **Trap:** UNEP because "environment" sounds related.

Q5. UPPCS Prelims 2019, Q35

Jaipur was listed as a UNESCO World Heritage Site at the 43rd session held in:

A. Baku B. Bishkek C. Istanbul D. Marrakesh

> 💡 **Show answer**
>
> **Logic:** Jaipur inscribed as UNESCO World Heritage Site at the 43rd session in Baku (2019). **Ans: A (Baku).** **43rd WHC session — Baku, 2019.** **Trap:** Marrakesh/Istanbul as other recent host cities.

Q6. UPPCS Prelims 2019, Q146

"Limits to Growth" was propounded by:

A. Club of Rome B. UNESCO C. Brundtland Commission D. Agenda 21

> 💡 **Show answer**
>
> **Logic:** Limits to Growth (1972) was Club of Rome — not UNESCO, Brundtland, or Agenda 21. **Ans: A.** **Club of Rome** (1972); Brundtland = Our Common Future; Agenda 21 = Rio 1992. **Trap:** UNESCO because of "education/science" label.

Q7. UPPCS Prelims (pattern)

The Intergovernmental Panel on Climate Change (IPCC) was established in 1988 by:

A. World Meteorological Organization (WMO) and UNEP B. FAO and WHO alone C. IUCN and WWF alone D. UNDP and UNESCO alone

> 💡 **Show answer**
>
> **Logic:** IPCC created 1988 by WMO + UNEP; shared Nobel Peace Prize 2007 — does not set NDCs. **Ans: A.** **WMO + UNEP → IPCC (1988)**; Nobel **2007**. **Trap:** IPCC = UNFCCC treaty body (policy is UNFCCC/COP).

Q8. UPPCS Prelims (pattern)

Which statement is correct?

A. IUCN Red List assesses extinction risk; CITES regulates international trade in listed species B. IUCN bans all wildlife trade globally C. CITES publishes the Living Planet Report D. Red List Appendix I–III sets trade permits

> 💡 **Show answer**
>
> **Logic:** IUCN Red List assesses extinction risk; CITES regulates international wildlife trade — different jobs. **Ans: A.** **IUCN** = risk assessment; **CITES** = trade regulation (Appendices I–III). **Trap:** "IUCN bans trade" or WWF authorship of Red List.

Q9. UPPCS Prelims (pattern)

The **Living Planet Report** is published by:

A. WWF B. IUCN C. IPCC D. FAO

> 💡 **Show answer**
>
> **Logic:** WWF publishes Living Planet Report/LPI; IUCN publishes Red List — both HQ in Gland but different roles. **Ans: A.** **WWF** = Living Planet / LPI; **IUCN** = Red List. **Trap:** Same city (Gland) → same organisation.

Q10. UPPCS Prelims (pattern)

Which pairing of organisation and headquarters is **correct**?

A. UNEP — Nairobi; FAO — Rome; UNESCO — Paris B. UNEP — Geneva; FAO — Nairobi; UNESCO — New York C. UNEP — Rome; FAO — Paris; UNESCO — Nairobi D. UNEP — New York; FAO — Gland; UNESCO — Geneva

> 💡 **Show answer**
>
> **Logic:** HQ ladder — UNEP Nairobi; FAO Rome; UNESCO Paris; IPCC/WMO Geneva; UNDP New York. **Ans: A.** **UNEP Nairobi**, **FAO Rome**, **UNESCO Paris** (IPCC/WMO Geneva; UNDP New York). **Trap:** UNEP in Geneva because "environment = Geneva agencies."

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to UNEP, which of the following statements is/are correct?

1. Created after Stockholm 1972; HQ Nairobi.
2. UNEP's flagship products are the Human Development Report and World Development Report.
3. UNEP publishes HDI.

A. Only 1 B. 1 and 2 C. Only 3 D. 2 and 3

> 💡 **Show answer**
>
> **Ans: A.** Only 1 is correct. **Logic:** Statement 1 is true (Stockholm → UNEP Nairobi). Statement 2 confuses UNEP flagships (Emissions Gap, GEO) with UNDP/World Bank reports. Statement 3 is false — HDI is UNDP, not UNEP.

Q2. Match List-I with List-II:

List-I	List-II
A. UNDP	1. Rome; SOFO
B. FAO	2. New York; HDR/HDI
C. UNESCO	3. Paris; WH + MAB

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** UNDP/FAO/UNESCO.

Q3. Which of the following pairs is/are correctly matched?

1. IUCN — Gland; Red List
2. IUCN — Gland; Living Planet Report / LPI
3. WWF — Red List author

A. 2 and 3 B. Only 1 C. 1 and 2 D. Only 3

💡 **Show answer**

Ans: B. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Emissions Gap Report is by UNEP.

Reason (R): It is published by the IPCC.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Do not credit IPCC with Emissions Gap.

Q5. With reference to IPCC, which of the following statements is/are correct?

1. HQ Geneva; created by WMO + UNEP 1988.
2. Shared 2007 Nobel Peace Prize.
3. IPCC sets national NDCs.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Science assessment — not NDC setter.

Q6. Arrange HQ ladder teaching cities:

A. All in Nairobi only B. Nairobi UNEP; Geneva IPCC; New York UNDP; Rome FAO; Paris UNESCO; Gland IUCN/WWF C. All in Gland only D. All in Bonn only

💡 **Show answer**

Ans: B. Six-city HQ ladder. **Logic:** High-yield map.

Q7. Which of the following pairs is/are NOT correctly matched?

1. IUCN — extinction risk assessment
2. CITES — wildlife trade regulation
3. IUCN Red List categories = CITES Appendix numbers

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Risk ranks ≠ Appendices.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): IPCC does science assessment; UNFCCC is the climate treaty.

Reason (R): The two should not be merged.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R restates the separation. **A/R logic:** Science body ≠ treaty.

Q9. With reference to GEF, which of the following statements is/are correct?

1. Multilateral funder from 1991.
2. Donors are not “developed countries only” as a rigid fact.
3. GEF publishes the Living Planet Index.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** LPI = WWF.

Q10. Match List-I with List-II:

List-I	List-II
A. Emissions Gap	1. UNDP
B. HDR / HDI	2. UNEP
C. Red List	3. IUCN

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** Report–org map.

Q11. Arrange IUCN vs WWF same-city split:

A. Identical jobs always B. IUCN assessment union; WWF NGO campaigning + Living Planet metrics C. Both set NDCs D. Both publish SOFO only

💡 **Show answer**

Ans: B. Assessment vs campaigning NGO. **Logic:** Shared Gland — different jobs.

Q12. With reference to FAO, which of the following statements is/are correct?

1. HQ Rome — not Paris or Geneva.
2. Publishes SOFO.
3. Co-hosts Rotterdam work with UNEP neighbourhood.

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: C. All three. **Logic:** FAO mandate.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Do not credit UNEP with HDI.

Reason (R): HDR/HDI is UNDP's product from 1990 neighbourhood.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** UNEP ≠ HDI.

Q14. Which one of the following correctly states Living Planet Index?

A. FAO SOFO forest cover only B. IUCN Red List categories C. CITES Appendix I only D. Vertebrate abundance trends for WWF messaging

💡 **Show answer**

Ans: D. WWF LPI. **Logic:** Not Red List.

Q15. How many of the following HQ facts are correct?

UNEP Nairobi; IPCC Geneva; UNDP New York; FAO Rome

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** HQ ladder chunk.

Q16. With reference to treaty secretariats, which of the following statements is/are correct?

1. Montreal, Basel, Stockholm POPs, Minamata may be hosted/supported by UNEP.
2. They remain treaty bodies — not "UNEP itself."
3. They are identical to IPCC Assessment Reports.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Treaty bodies ≠ IPCC ARs.

Q17. Match List-I with List-II:

List-I	List-II
A. UNESCO	1. Environment coordination
B. UNEP	2. Heritage / education / MAB
C. UNDP	3. Development coordination

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Mandate split.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): IPCC Assessment Reports synthesize climate science.

Reason (R): They set each country's NDC targets.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Science ≠ NDC setter.

Q19. Which of the following pairs is/are correctly matched?

1. FAO food–forest–agriculture mandate explains SOFO + pesticide PIC neighbourhood
2. UNESCO ≠ UNEP environment coordination
3. FAO HQ is Paris

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** FAO = Rome.

Q20. Arrange IPCC creation:

A. 1988 by WMO + UNEP B. 1972 UNEP alone C. 1992 CBD alone D. 2015 Paris alone

💡 **Show answer**

Ans: A. 1988 WMO+UNEP. **Logic:** IPCC birth.

Q21. With reference to Gland twins, which of the following statements is/are correct?

1. IUCN and WWF both sit in Gland.
2. Different products: Red List vs Living Planet.
3. Both are identical to CITES Appendices.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Not CITES.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): IUCN assesses extinction risk; CITES regulates wildlife trade.

Reason (R): Different institutions, different jobs.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the separation. **A/R logic:** Risk vs trade.

Q23. How many of the following report–org pairs are correct?

Emissions Gap–UNEP; HDR–UNDP; Red List–IUCN; LPI–WWF

A. Only one B. Only two C. All four D. Only three

💡 **Show answer**

Ans: C. All four. **Logic:** Four classic pairs.

Q24. Which one of the following correctly states UNEP vs UNDP?

A. Both set CITES Appendices B. Both only publish Red List C. Both HQ only in Gland D. Environment coordination vs development coordination — different HQ/reports

💡 **Show answer**

Ans: D. Environment vs development. **Logic:** Separate lanes.

Q25. With reference to GEO, which of the following statements is/are correct?

1. UNEP flagship neighbourhood with Emissions Gap.
2. Not the UNDP Human Development Report.
3. Identical to IUCN Red List categories.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** GEO ≠ Red List.

Q26. Match List-I with List-II:

List-I	List-II
A. IUCN	1. Living Planet
B. WWF	2. Red List
C. IPCC	3. Geneva; science ARs

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** IUCN/WWF/IPCC.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Do not credit IPCC with the Emissions Gap Report.

Reason (R): Emissions Gap Report is UNEP's annual product.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains A. **A/R logic:** UNEP not IPCC.

Q28. Arrange UNESCO MAB neighbourhood:

A. Only Kyoto Annex-I B. Only CGWB groundwater C. Biosphere Reserve programme under UNESCO Paris D. Only BS-VI fuels

> 💡 **Show answer**
>
> **Ans: C.** MAB / Biosphere. **Logic:** UNESCO tool.

Q29. With reference to HDR start, which of the following statements is/are correct?

1. From 1990 neighbourhood.
2. UNDP New York.
3. Published by UNEP Nairobi as HDI.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Not UNEP.

Q30. Which one of the following correctly states FAO HQ?

A. Rome B. Paris C. Geneva D. Nairobi

> 💡 **Show answer**
>
> **Ans: A.** Rome. **Logic:** Not Paris/Geneva.

Common Traps

Trap	Correct
Emissions Gap = IPCC/UNDP	**UNEP**
HDR = UNEP	**UNDP**
UNEP HQ Geneva/NY	**Nairobi**
IPCC sets NDCs	**UNFCCC/COP**
IUCN regulates trade	**CITES**
WWF = IUCN	Same city, **different roles**
FAO HQ Nairobi	**Rome**
GEF donors = developed only	**False**
HDI team = Priesner	**Haq–Sen** lineage
Limits to Growth = UNESCO	**Club of Rome**

42.8 OPCW & UN-Habitat (bank add-ons)

Identity: Chemical Weapons Convention | The Hague | Habitat (वास स्थान) Nairobi

- **OPCW (Organisation for the Prohibition (प्रतिषेध) of Chemical Weapons)** implements the **Chemical Weapons Convention** (force **29 April 1997**); HQ **The Hague**, Netherlands. It verifies destruction/non-re-emergence of chemical weapons and assists states against chemical threats — **not** an EU/NATO/WHO body. Nobel Peace Prize **2013**.
- **UN-Habitat** (UN Human Settlements Programme; HQ **Nairobi**, from **1978**) is mandated to promote socially and environmentally sustainable towns/cities and adequate shelter; partners include governments, local authorities, and non-state actors; contributes to poverty reduction and safe water/sanitation access. Partners are **not** governments-only.

> **Logic:** OPCW ≠ EU organ. UN-Habitat partners ≠ governments only.

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 43 — International Environmental Agreements

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

UNFCCC (यूएनएफसीसीसी) | Kyoto (क्योटो) Protocol | Paris Agreement (पेरिस समझौता) | CBD | Ramsar (रामसर) | CITES (साइट्स) | Bonn Convention (CMS) | Basel | Rotterdam | Stockholm Convention | Minamata

Sources baked in: UNFCCC; CBD; Ramsar; CITES; UNEP; MoEFCC; UPPCS PYQs **Weight:** ★★★ High — Montreal≠Kyoto; treaty↔issue; Ramsar UP traps; Stockholm 1972≠POPs 2001 **Last verified:** August 2026

Current Affairs (this topic)

- Twin of Agreements & Conferences — COP / listing CA; treaty ladder identical.
- Ramsar India **~100** / UP **12** baseline unless a stem cites a later official total.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. The climate ladder is **UNFCCC 1992 → Kyoto 1997 → Paris 2015**. 2. **Kyoto** (क्योटो) set binding targets mainly for **Annex-I** parties. **Paris** uses universal **NDCs** for all parties. 3. **Montreal** protects the ozone layer. **Kyoto / Paris / UNFCCC** are climate instruments. Never swap Montreal for climate mitigation (जलवायु शमन). 4. **CBD (1992)** is the biodiversity convention. **Cartagena** is the biosafety protocol under CBD. 5. **Ramsar (1971)** covers wetlands. **CITES (1973)** (साइट्स) covers wildlife **trade** (पण्याध्यक्ष). **CMS / Bonn (1979)** covers **migratory** species. 6. **Basel (1989)** covers hazardous **waste**. **Rotterdam (1998)** covers hazardous **chemicals** through the PIC procedure (प्रक्रिया). 7. **Stockholm Convention (2001)** (स्टॉकहोम संधि) covers **POPs** such as DDT and PCBs. It is **not** the Stockholm Conference of **1972**. 8. **Minamata (2013)** covers **mercury**. 9. UNFCCC was opened at **Rio 1992**. Its secretariat sits in **Bonn**. Annual COP meetings drive negotiations. 10. CITES is not the IUCN (आईयूसीएन) Red List. CMS is not CITES. Trade rules and migration treaties are different tools. 11. Ramsar "wise use" allows sustainable human use of wetlands — it is not a total ban slogan. 12. Paris Agreement year fact is **2015**. Kyoto year fact is **1997**. Montreal year fact is **1987**. 13. A stem that says Kyoto is about "saving water" is false. Kyoto is climate. 14. Stockholm **1972** created the UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) moment. Stockholm **2001** is the POPs treaty. Keep the two Stockholms apart. 15. Basel moves on waste shipment and disposal. Rotterdam's PIC is prior informed consent for chemicals trade. 16. CBD's three objectives are conservation, sustainable use, and fair benefit-sharing from genetic resources. 17. CMS protects migratory animals across borders. CITES controls international commercial trade in listed species. 18. Minamata's mercury focus includes artisanal gold mining and product phase-downs in teaching notes. 19. UNFCCC is the parent framework; Kyoto and Paris are protocols/agreements under that climate family. 20. Do not put Ramsar, CITES, or CBD into the climate-mitigation answer slot when Kyoto is present. 21. UP Ramsar geography traps belong beside Ramsar treaty facts but do not change the 1971 identity. 22. Rotterdam is chemicals PIC, not mercury (Minamata) and not POPs listing (Stockholm 2001 (स्टॉकहोम संधि)). 23. Paris NDCs are nationally determined. Kyoto Annex-I targets were top-down binding for listed developed parties. 24. Bonn Convention means **CMS**, not the UNFCCC Bonn secretariat alone.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Kyoto Protocol (1997)** | **Montreal Protocol (1987)** | Legally binding treaty targeting reduction of 6 greenhouse gases causing climate change vs universally ratified protocol phasing out ozone-depleting chlorofluorocarbons | क्योटो प्रोटोकॉल (जलवायु परिवर्तन) / मॉन्ट्रियल (ओजोन क्षरण) | | **Kyoto Protocol (1997)** | **Paris Agreement (2015)** | Top-down legally binding reduction targets solely for developed Annex-I countries vs bottom-up voluntary Nationally Determined Contributions (NDCs) applicable to all countries | क्योटो (शीर्ष से नीचे बाध्यकारी) / पेरिस (स्वैच्छिक NDCs) | | **Basel Convention (1989)** | **Rotterdam Convention (1998)** | Regulates transboundary movements and disposal of hazardous wastes vs enforces Prior Informed Consent (PIC) procedure for hazardous international trade chemicals | बासेल (खतरनाक अपशिष्ट आवागमन) / रॉटरडैम (पूर्व सूचित सहमति) | | **Stockholm Convention (2001)** | **Minamata Convention (2013)** | Mandates global elimination and phase-out of Persistent Organic Pollutants (POPs - "Dirty Dozen") vs protects human health and environment from anthropogenic emissions of toxic mercury (Hg) | स्टॉकहोम (पीओपी कीटनाशक) / मिनामाता (पारा नियंत्रण) | | **Ramsar Convention (1971)** | **CITES (1973)** | Intergovernmental treaty dedicated specifically to the conservation and wise use of wetlands of international importance vs treaty controlling illegal commercial wildlife trade | रामसर कन्वेंशन (आर्द्रभूमि) / साइट्स (वन्यजीव व्यापार) | | **Stockholm Conference (1972)** | **Rio Earth Summit (1992)** | UN Conference on the Human Environment that led to creation of UNEP vs UN Conference on Environment & Development giving birth to Agenda 21, CBD, UNFCCC, and UNCCD | स्टॉकहोम सम्मेलन 1972 / रियो पृथ्वी शिखर सम्मेलन 1992 |

💡 📋 **Must-score facts — UNFCCC Kyoto Paris CBD Basel (Click to Expand)**

Ladders / tags | Item | Lock | |---|---| | Climate | UNFCCC **1992** → Kyoto **1997** → Paris **2015** | | Kyoto vs Paris | Annex-I binding / universal NDCs | | Montreal | Ozone (**1987**) — not climate | | CBD / Cartagena | **1992** / biosafety protocol | | Ramsar / CITES / CMS | **1971** / **1973** / Bonn **1979** | | Basel / Rotterdam / Stockholm POPs | **1989** / **1998** / **2001** | | Minamata | Mercury **2013** | | Two Stockholms | **1972** conference→UNEP / **2001** POPs treaty | | UNFCCC secretariat | **Bonn** |

43.1 UNFCCC

Identity: 1992 Rio framework | Stabilize GHGs | COP | Bonn secretariat

- The **UN Framework Convention on Climate Change (जलवायु परिवर्तन) (UNFCCC)** was opened at **Rio 1992**.
- Goal: stabilize greenhouse gas concentrations to prevent dangerous climate interference.
- Holds annual **COP** meetings; secretariat in **Bonn, Germany**.
- Parent framework for **Kyoto Protocol** and **Paris Agreement** (पेरिस समझौता).
- Does **not** itself set the Kyoto Annex targets or Paris NDCs — those are later instruments.

> **Logic:** UNFCCC = climate framework. IPCC (आईपीसीसी) = science body (Topic 42).

43.2 Kyoto Protocol

Identity: 1997 | Binding Annex-I targets | CDM / carbon markets | ≠ Montreal

- The **Kyoto Protocol (1997)** (क्योटो प्रोटोकॉल) set **legally binding emission reduction targets** for **Annex-I (developed) countries**.
- Flexible mechanisms: **CDM**, Joint Implementation, and Emissions Trading — origin of **carbon credits**.
- Climate mitigation treaty — **not** ozone (Montreal) and **not** "save water."

PYQ — UPPCS Prelims 2024, Q35

Climate change mitigation agreement:

Options: A. Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) B. Ramsar C. Kyoto Protocol D. CITES

> 💡 **Show answer**
>
> **Ans: C**

PYQ — UPPCS Prelims 2019, Q132

NOT correctly matched:

Options: A. Cartagena — Biosafety B. Stockholm — POPs C. Montreal — Ozone D. Kyoto — Save water

> 💡 **Show answer**
>
> **Ans: D** — Kyoto = climate/GHG.

PYQ — UPPCS Prelims 2021, Q125

Carbon credit (कार्बन क्रेडिट) concept originated from:

> 💡 **Show answer**
>
> **Kyoto Protocol**

> **Logic:** Montreal = ozone. Kyoto = climate.

43.3 Paris Agreement

Identity: 2015 COP21 | Universal NDCs | Well below 2°C / pursue 1.5°C | Force 2016

- The **Paris Agreement (2015)** (पेरिस समझौता) applies to **all parties** through **Nationally Determined Contributions (NDCs)**.
- Aims to hold warming **well below 2°C** and pursue **1.5°C**.
- Entered into force in **2016**.
- Differs from Kyoto's Annex-I-only binding model — bottom-up pledges with global stocktake.
- India's NDC/net-zero policy track sits under Paris (Topics 10/34).

> **Logic:** Paris ≠ Kyoto clone. NDCs are nationally determined.

43.4 Convention on Biological Diversity (CBD)

Identity: 1992 Rio | Three objectives | Nagoya ABS | Cartagena biosafety

- **CBD** opened at **Rio 1992** with three objectives: **conservation**, **sustainable use**, and **fair benefit-sharing** of genetic resources.
- **Cartagena Protocol** = biosafety / LMOs.
- **Nagoya Protocol (2010)** = Access and Benefit Sharing (ABS).
- India implements CBD via the **Biological Diversity Act 2002** (जैव विविधता अधिनियम 2002) (NBA/SBBs/BMCs).
- Biodiversity Day **22 May** commemorates CBD text adoption.

PYQ — UPPCS Prelims 2019, Q132

Cartagena Protocol — Biosafety is a **correct** match.

> **Logic:** CBD ≠ CITES. CBD = biodiversity framework; CITES = trade.

43.5 Ramsar Convention

Identity: 1971 | Wetlands | Wise Use | 2 February | UP geography traps

- The **Ramsar Convention (1971)** (रामसर कन्वेंशन) conserves wetlands of international importance under **"Wise Use."**
- **World Wetlands Day (विश्व आर्द्रभूमि दिवस) = 2 February** (Ramsar adoption date).
- India has about **100** Ramsar sites (**Tamil Nadu (नाडु)** leads state counts; **UP 12**, **Punjab 6** among high counts).
- **Montreux Record** = Ramsar sites with adverse / likely adverse ecological change from human interference.
- Negative geography: **Rudrasagar = Tripura**; **Sultanpur NP = Haryana** — not UP.

PYQ — UPPCS Prelims 2025, Q58

Consider the following Ramsar sites:

1. Sarsai Nawar
2. Samaspur
3. Rudrasagar Lake
4. Sultanpur National Park

Which of the above are **not** located in Uttar Pradesh (उत्तर प्रदेश)?

A. 1 and 2 B. 3 and 4 C. 1 and 3 D. 2 and 4

💡 **Show answer**

Logic: UP Ramsar geography — which listed sites are **outside** Uttar Pradesh. **Ans: B.** **Rudrasagar Lake** is in **Tripura**; **Sultanpur National Park** is in **Haryana**. **Sarsai Nawar** and **Samaspur** are Ramsar sites in **UP**. **Trap:** Treating every listed wetland as a UP site because India has many Ramsar wetlands.

> **Logic:** Ramsar = wetlands, not forests or ozone.

43.6 CITES

Identity: 1973 Washington | Wildlife trade | Appendices I–III | ≠ IUCN

- **CITES** regulates **international trade** in endangered wild fauna (प्राणीजात) and flora (वनस्पति).
- Uses **Appendices I, II, III** by trade-threat level.
- Adopted (अंगीकृत) **1973 (Washington)** — not a Rio 1992 product.
- **World Wildlife Day (विश्व वन्यजीव दिवस) = 3 March** (CITES adoption date).
- IUCN Red List assesses extinction risk; CITES regulates trade.

> **Logic:** CITES ≠ CMS. Trade ≠ migration focus.

43.7 Bonn Convention (CMS)

Identity: 1979 | Migratory species | Habitat (वास स्थान) corridors | Bonn

- The **Convention on the Conservation of Migratory Species of Wild Animals (CMS / Bonn Convention, 1979)** protects migratory species across range states.
- Focuses on **migration routes and habitats**, not commercial trade permits like CITES.
- Secretariat association with **Bonn, Germany** (name fact).

> **Logic:** CMS = migration. CITES = trade.

43.8 Basel Convention

Identity: 1989 | Transboundary hazardous waste | Prior notice / control

- The **Basel Convention (1989)** (बासेल कन्वेंशन) controls **transboundary movements of hazardous wastes** and their disposal.
- Aims to reduce hazardous waste generation and prevent dumping on weaker states.
- Part of the BRS chemicals/waste cluster with Rotterdam and Stockholm.

> **Logic:** Basel = hazardous **waste** shipment, not wildlife trade.

43.9 Rotterdam Convention

Identity: 1998 | PIC | Hazardous chemicals in trade | FAO + UNEP

- The **Rotterdam Convention (1998)** (रॉटरडैम कन्वेंशन) uses **Prior Informed Consent (PIC)** for certain hazardous chemicals and pesticides in international trade.
- Jointly administered by **FAO** and **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम).
- Does not ban all listed chemicals globally; it ensures importing countries are informed and consent.

> **Logic:** Rotterdam = PIC chemicals. Basel = waste. Stockholm = POPs.

43.10 Stockholm Convention

Identity: 2001 | POPs | DDT/PCBs etc. | ≠ Stockholm 1972 (स्टॉकहोम सम्मेलन) conference

- The **Stockholm Convention (2001)** targets **Persistent Organic Pollutants (POPs)** — long-lived toxic chemicals (e.g. certain pesticides, PCBs, dioxins).
- Seeks elimination or restriction of listed POPs.
- **Not** the same as the **Stockholm Conference 1972** that created UNEP.

PYQ — UPPCS Prelims 2019, Q132

Stockholm Convention — Persistent organic pollutant is a **correct** match.

> **Logic:** Stockholm POPs 2001 ≠ Stockholm Environment (पर्यावरण) Conference 1972.

43.11 Minamata Convention

Identity: 2013 | Mercury | Named after Japan disaster | India ratified

- The **Minamata Convention (2013)** (मिनामाता संधि) controls **mercury** use, emissions, and releases across the life cycle.

- Named after **Minamata disease** (mercury poisoning) in Japan.
- Covers products, processes, artisanal gold mining, and emissions to air/water/land.
- India ratified around **2018** (CA fact in many banks).

> **Logic:** Minamata = mercury. Basel = hazardous waste generally.

Treaty ↔ Issue Matrix

Treaty	Year	Issue
Ramsar	1971	Wetlands
CITES	1973	Wildlife trade
CMS (Bonn)	1979	Migratory species
Basel	1989	Hazardous waste
UNFCCC	1992	Climate framework
CBD	1992	Biodiversity
Kyoto	1997	Climate (Annex-I)
Rotterdam	1998	PIC chemicals
Stockholm	2001	POPs
Minamata	2013	Mercury
Paris	2015	Climate (NDCs)

Also remember (trap pair): Montreal 1987 = **ozone** (not on this syllabus list, but frequent distractor).

Complete PYQ Bank (Topic 43)

Q1. UPPCS Prelims 2024, Q35

Climate change mitigation agreement:

A. Montreal Protocol B. Ramsar Convention C. Kyoto Protocol D. CITES

💡 **Show answer**

Logic: Climate mitigation treaty among options — Kyoto/Paris/UNFCCC family, not Montreal (ozone) or Ramsar/CITES. **Ans: C.** **Kyoto Protocol (1997)** = binding Annex-I GHG cuts; Montreal = **ozone**. **Trap:** Montreal as generic "environment" answer.

Q2. UPPCS Prelims 2019, Q132

Which of the following is NOT correctly matched?

A. Cartagena Protocol — Biosafety B. Stockholm Convention — Persistent organic pollutants C. Montreal Protocol — Ozone D. Kyoto Protocol — Save water

💡 **Show answer**

Logic: Match-list trap — Kyoto ≠ "save water"; Cartagena=biosafety, Stockholm=POPs, Montreal=ozone are correct pairs. **Ans: D.** **Kyoto = climate/GHG mitigation**, not "save water." **Trap:** Accepting any Kyoto label that sounds like sustainable development.

Q3. UPPCS Prelims 2021, Q125

The concept of carbon credit originated from:

A. Ramsar Convention B. Kyoto Protocol C. CITES D. Basel Convention

> 💡 **Show answer**
>
> **Logic:** Carbon credits originated from Kyoto flexible mechanisms (CDM, JI, emissions trading). **Ans: B.** **Kyoto (1997)** created **CDM/JI/ET** credit architecture. **Trap:** Paris 2015 as "origin" — Paris builds on Kyoto market experience.

Q4. UPPCS Prelims 2025, Q58

Which of the following Ramsar sites are **NOT** located in Uttar Pradesh?

1. Sarsai Nawar
2. Samaspur Bird Sanctuary
3. Rudrasagar Lake
4. Sultanpur National Park

A. 1 and 2 B. 3 and 4 C. 1 and 4 D. 2 and 3

> 💡 **Show answer**
>
> **Logic:** UP Ramsar geography — Rudrasagar (Tripura) and Sultanpur NP (Haryana) are NOT in UP; Sarsai Nawar and Samaspur are in UP. **Ans: B (3 and 4).** **Rudrasagar = Tripura**; **Sultanpur = Haryana**; Sarsai Nawar and Samaspur **are in UP**. **Trap:** Assuming any famous wetland name is in UP.

Q5. UPPCS Prelims (pattern)

The Paris Agreement (2015) differs from Kyoto mainly because Paris relies on:

A. Nationally Determined Contributions (NDCs) by all parties B. Binding Annex-I-only targets for developed countries only C. Wetland wise-use principles D. PIC for hazardous chemicals in trade

> 💡 **Show answer**
>
> **Logic:** Paris Agreement 2015 uses universal NDCs; UNFCCC 1992 is the Rio parent framework. **Ans: A.** **Paris = universal NDCs**; Kyoto = **Annex-I binding** model. **Trap:** Paris as Kyoto clone with same Annex structure.

Q6. UPPCS Prelims (pattern)

The Convention on Biological Diversity (CBD) seeks:

A. Conservation, sustainable use, and fair benefit-sharing of genetic resources B. Only ozone layer protection C. Only transboundary hazardous waste shipment D. Only mercury phase-down

> 💡 **Show answer**
>
> **Logic:** CBD three objectives — conservation, sustainable use, fair benefit-sharing; Cartagena = biosafety; Nagoya = ABS. **Ans: A.** **CBD (1992 Rio)** triple objective; **Cartagena** = biosafety; **Nagoya** = ABS. **Trap:** CBD = CITES trade regulation.

Q7. UPPCS Prelims (pattern)

CITES (1973) primarily regulates:

A. International trade in endangered wild fauna and flora B. Migratory bird routes only without trade permits C. Wetland wise use only D. POPs elimination globally

💡 **Show answer**

Logic: CITES 1973 Washington regulates wildlife trade; CMS/Bonn 1979 protects migratory species — different tools. **Ans: A.** **CITES = trade** (Appendices I–III); **CMS = migration** routes/habitats. **Trap:** CMS and CITES swapped.

Q8. UPPCS Prelims (pattern)

The Stockholm Convention (2001) targets:

A. Persistent Organic Pollutants (POPs) B. Wetlands of international importance C. Annex-I CO_2 quotas only D. Prior Informed Consent for pesticide trade alone

💡 **Show answer**

Logic: Chemicals/waste cluster — Basel = hazardous waste; Rotterdam = PIC chemicals; Stockholm 2001 = POPs; Minamata = mercury. **Ans: A.** **Stockholm 2001 = POPs** (DDT, PCBs, etc.). **Trap:** **Stockholm Conference 1972** (UNEP birth) confused with **Stockholm Convention 2001**.

Q9. UPPCS Prelims (pattern)

Montreal Protocol is chiefly concerned with:

A. Substances that deplete the stratospheric ozone layer B. Universal NDC climate pledges C. Mercury emissions from gold mining D. Transboundary movement of hazardous wastes

💡 **Show answer**

Logic: Montreal Protocol = ozone/ODS — frequent distractor when stem asks for climate treaty. **Ans: A.** **Montreal 1987 = ODS/CFC phase-out**; climate = UNFCCC/Kyoto/Paris. **Trap:** "Montreal = climate mitigation."

Q10. UPPCS Prelims (pattern)

Which statement is **correct**?

A. The Stockholm Conference (1972) launched UNEP; the Stockholm Convention (2001) controls POPs — they are different instruments B. Both Stockholms are the same treaty on POPs C. Stockholm 1972 banned DDT globally immediately D. Stockholm 2001 created UNEP in Nairobi

💡 **Show answer**

Logic: Two Stockholms — 1972 conference created UNEP; 2001 convention controls POPs. **Ans: A.** **1972 conference ≠ 2001 POPs treaty** — highest-frequency treaty year trap. **Trap:** Single "Stockholm" answer for any chemical question.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to the climate ladder, which of the following statements is/are correct?

1. UNFCCC 1992 → Montreal 1997 → Paris 2015.
2. Kyoto mainly Annex-I binding; Paris universal NDCs.
3. Montreal is the main climate-mitigation treaty.

A. Only 2 B. Only 1 C. 1 and 2 D. Only 3

Show answer

Ans: A. Only 2 is correct. **Logic:** Only statement 2 remains true among the numbered claims. Statement 1 is false. Statement 2 is true. Statement 3 is false.

Q2. Match List-I with List-II:

List-I	List-II
A. Ramsar	1. 1973 wildlife trade
B. CITES	2. 1971 wetlands
C. CMS/Bonn	3. 1979 migratory species

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

Show answer

Ans: C. A-2, B-1, C-3. **Logic:** Wetlands/trade/migration.

Q3. Which of the following pairs is/are correctly matched?

1. Basel 1989 — hazardous waste
2. Rotterdam 2008 — PIC chemicals
3. Stockholm 2011 — POPs (≠ 1972 Conference)

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

Show answer

Ans: C. Only 1 is correct. **Logic:** Only statement 1 remains true among the numbered claims. Statement 1 is true. Statement 2 is false. Statement 3 is false.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Stockholm 1972 created the UNEP moment.

Reason (R): Stockholm 2001 is the same as the 1972 Conference.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: C. A true, R false. **A/R logic:** Keep two Stockholms apart.

Q5. With reference to CBD family, which of the following statements is/are correct?

1. CBD 1992 — conservation, sustainable use, fair benefit-sharing.
2. Cartagena — biosafety protocol under CBD.
3. CBD is identical to Kyoto Annex-I targets.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: A. Only 1 and 2. **Logic:** Biodiversity ≠ Kyoto climate.

Q6. Arrange Minamata focus:

A. Only wetlands wise use B. Mercury (2013); artisanal gold / product phase-downs neighbourhood C. Only ozone ODS D. Only forest SOFO

💡 **Show answer**

Ans: B. Mercury convention. **Logic:** Minamata.

Q7. Which of the following pairs is/are NOT correctly matched?

1. CITES — IUCN Red List
2. CMS — migratory species
3. CITES — international commercial trade in listed species

A. 1 and 3 only B. 2 and 3 only C. 1 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 1 is not correctly matched. **Logic:** CITES ≠ Red List.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Ramsar "wise use" allows sustainable human use of wetlands.

Reason (R): Ramsar is a total ban slogan with no human use.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Wise use ≠ total ban.

Q9. With reference to UNFCCC, which of the following statements is/are correct?

1. Opened at Rio 1992; secretariat Bonn.
2. Annual COP meetings drive negotiations.
3. Parent framework for Kyoto and Paris.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** UNFCCC identity.

Q10. Match List-I with List-II:

List-I	List-II
A. Montreal	1. 2015
B. Kyoto	2. 1987
C. Paris	3. 1997

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-3, C-1. **Logic:** 1987 / 1997 / 2015.

Q11. Arrange Bonn Convention meaning:

A. Only UNFCCC secretariat building B. CMS migratory species — not secretariat alone C. Only Minamata mercury D. Only Basel waste

💡 **Show answer**

Ans: B. Bonn = CMS. **Logic:** Name trap.

Q12. With reference to Rotterdam vs others, which of the following statements is/are correct?

1. Rotterdam = chemicals PIC.
2. Not mercury (Minamata).
3. Not POPs listing (Stockholm 2001).

A. 1 and 2 only B. 2 and 3 only C. 1, 2 and 3 D. 1 and 3 only

💡 **Show answer**

Ans: C. All three. **Logic:** PIC niche.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): A stem that says Kyoto is about "saving water" is false.

Reason (R): Kyoto is a climate instrument.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Kyoto ≠ water treaty.

Q14. Which one of the following correctly states Paris vs Kyoto design?

A. Both identical CITES Appendices B. Both identical Montreal ODS lists C. Both identical Ramsar wise use D. Paris = nationally determined NDCs; Kyoto = top-down Annex-I binding for listed parties

💡 **Show answer**

Ans: D. NDCs vs Annex-I. **Logic:** Design split.

Q15. How many of the following year facts are correct?

Montreal 1987; Kyoto 1997; Paris 2015; Minamata 2013

A. All four B. Only two C. Only three D. Only one

💡 **Show answer**

Ans: A. All four. **Logic:** Year ladder.

Q16. With reference to climate-slot traps, which of the following statements is/are correct?

1. Do not put Ramsar/CITES/CBD into climate-mitigation slot when Kyoto is present.
2. Montreal ≠ climate mitigation.
3. UNFCCC family = Kyoto/Paris neighbourhood.

A. 1 and 2 only B. 1, 2 and 3 C. 1 and 3 only D. 2 and 3 only

💡 **Show answer**

Ans: B. All three. **Logic:** Slot discipline.

Q17. Match List-I with List-II:

List-I	List-II
A. Basel	1. Mercury
B. Rotterdam	2. Hazardous waste
C. Minamata	3. PIC chemicals

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-3, C-1. **Logic:** Waste/PIC/mercury.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): CMS protects migratory animals across borders.

Reason (R): CITES controls international commercial trade in listed species.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; trade rules do not explain migration-treaty purpose. **A/R logic:** CMS vs CITES.

Q19. Which of the following pairs is/are correctly matched?

1. Cartagena — biosafety under CBD
2. Stockholm POPs — DDT/PCBs neighbourhood
3. Ramsar year — 1987 Montreal

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Ramsar = 1971.

Q20. Arrange UP Ramsar geography idea:

A. UP geography traps sit beside treaty facts but do not change 1971 identity B. Changes Ramsar 1971 identity C. Makes Ramsar a climate NDC D. Makes Ramsar = CITES

💡 **Show answer**

Ans: A. Treaty year stays 1971. **Logic:** Geography ≠ identity.

Q21. With reference to Basel vs Rotterdam, which of the following statements is/are correct?

1. Basel — waste shipment/disposal.
2. Rotterdam — prior informed consent for chemicals trade.
3. Both are identical mercury treaties.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Mercury = Minamata.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): UNFCCC is the parent framework.

Reason (R): Kyoto and Paris sit under that climate family.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the family tree. **A/R logic:** Parent vs protocols.

Q23. How many of the following are climate instruments?

UNFCCC, Kyoto, Paris, Montreal

A. Only one B. Only two C. Only three D. All four

💡 **Show answer**

Ans: C. Only three. **Logic:** Montreal = ozone.

Q24. Which one of the following correctly states CBD's three objectives?

A. Only ship MARPOL annexes B. Only Annex-I emission caps C. Only ozone ODS phase-out D. Conservation; sustainable use; fair benefit-sharing from genetic resources

💡 **Show answer**

Ans: D. Three CBD objectives. **Logic:** Biodiversity convention.

Q25. With reference to CITES vs CMS, which of the following statements is/are correct?

1. Trade rules ≠ migration treaties.
2. CMS is not CITES.
3. Both are identical to IUCN Red List.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** Neither is Red List.

Q26. Match List-I with List-II:

List-I	List-II
A. CBD	1. 1971
B. Ramsar	2. 1992
C. CITES	3. 1973

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** 1992 / 1971 / 1973.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Paris Agreement year fact is 2015.

Reason (R): Kyoto year fact is 1997.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; Kyoto year does not explain Paris year. **A/R logic:** Two year facts.

Q28. Arrange Montreal vs Kyoto one-liner:

A. Both identical climate NDCs B. Both identical Ramsar C. Montreal ozone; Kyoto/Paris/UNFCCC climate D. Both identical Minamata

💡 **Show answer**

Ans: C. Ozone vs climate. **Logic:** Never swap.

Q29. With reference to POPs, which of the following statements is/are correct?

1. Stockholm Convention 2001.
2. Covers POPs such as DDT and PCBs.
3. Identical to Stockholm Conference 1972 UNEP moment.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** 2001 ≠ 1972.

Q30. Which one of the following correctly states “Bonn” double meaning caution?

A. Bonn Convention = CMS; UNFCCC secretariat also sits in Bonn — do not collapse them B. Bonn means only Montreal C. Bonn means only Minamata D. Bonn means only SUP ban

💡 **Show answer**

Ans: A. CMS vs UNFCCC Bonn HQ. **Logic:** Name caution.

Common Traps

Trap	Correct
Montreal = climate mitigation	**Ozone**; Kyoto/Paris = climate
Kyoto = save water	**Climate / GHG**
Stockholm Convention = 1972 conference	**POPs 2001** vs UNEP conference 1972
CITES = IUCN Red List	Trade rules vs risk assessment
CMS = CITES	Migration vs trade
Ramsar = forests/ozone	**Wetlands**

Trap	Correct
CITES born at Rio 1992	**1973 Washington**
Paris = Kyoto Annex-only model	**Universal NDCs**
Basel = wildlife trade	**Hazardous waste**
Minamata = all hazardous waste	**Mercury**

✍️ S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)

Topic 44 — Current Environmental Issues

★ UPPCS Revision Sheet — Lucent-style (no repetition · Practice ≥25)

💡 **Covers syllabus (click to expand)**

Carbon Footprint | Ecological Footprint (पारिस्थितिक पदचिह्न) | Net Zero Emissions (नेट जीरो उत्सर्जन) | Green Economy (हरित अर्थव्यवस्था) | Blue Economy (नीली अर्थव्यवस्था) | Circular Economy (चक्रीय अर्थव्यवस्था) | Green Hydrogen | Microplastics | Climate Refugees | Nature-based Solutions | LiFE Mission

Sources baked in: IPCC; UNEP; Global Footprint Network; MoEFCC; MNRE; IUCN NbS; UPPCS PYQs 2018–2025
Weight: ★★★ High — LiFE COP trap; MISHTI year; India net zero 2070; microplastic <5 mm; EF (gha) ≠ carbon (tCO_2e) **Last verified:** August 2026

Current Affairs (this topic)

- Primary living CA chapter — NbS, plastics, heat, Himalaya (हिमालय) cryosphere, LiFE.
- Cross-link facts to Climate / Biodiversity / Pollution without relocating treaty years.

💡 🎯 **Consolidated — 24 Must-Score Facts (Click to Expand)**

1. A **carbon footprint** totals greenhouse gases in **tCO_2e**. Scope **1** is direct, Scope **2** is purchased energy, Scope **3** is the wider value chain. 2. India's **per capita** carbon footprint is about **2 tCO_2e** — low versus global and USA averages — even though absolute (निरपेक्ष) emissions are high. 3. An **ecological footprint** (पारिस्थितिक पदचिह्न) measures resource demand in **global hectares (gha)**. It is **not** the same unit as tCO_2e. The **Social Cost of Carbon** is the monetary long-term damage of **one tonne of CO_2**. 4. **Net zero** balances remaining emissions with removals. India's year fact is **2070**. It is not gross zero and not automatically **2050**. 5. NDC-style facts include about **45%** intensity cut, about **50%** non-fossil capacity, and a large forest sink goal. 6. The **Emissions Gap Report** (उत्सर्जन अंतराल रिपोर्ट) is by **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम). 7. A **green economy** (UNEP framing) is low-carbon, resource-efficient, and socially inclusive. 8. A **blue economy** is sustainable ocean use linked to **SDG 14**. **MISHTI** mangrove restoration is from the **2023** Budget, **not 2019**. 9. A **circular economy** closes material loops through 3R/9R and EPR. It sits inside the broader green-economy idea and links to **SDG 12**. 10. **Green hydrogen** uses electrolysis powered by renewables. India's mission (January **2023**) targets about **5 MMT** by **2030**. 11. **Grey hydrogen** reforms natural gas. **Blue hydrogen** adds CCS (सुरक्षा कैबिनेट समिति) to grey pathways. Do not call grey "green." 12. **Microplastics** are plastic pieces smaller than **5 mm**. They are primary (made small) or secondary (fragmented) and are not fast biodegradable. 13. The **SUP ban** push of **2022** targets single-use plastics that feed microplastic pathways. 14. **Climate refugees** are mostly **internally** displaced. They are **not** a separate legal category under the **1951** Refugee Convention. 15. **Nature-based Solutions** protect, manage, or restore ecosystems (IUCN (आईयूसीएन) framing). They are not geoengineering (भू-इंजीनियरिंग) (cirrus thinning / sulphate aerosols are geoengineering ideas for cooling, not NbS). 16. **LiFE** was an idea at **COP-26 (2021)** and launched in **June 2022** under MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) with a long action list. **COP-25 Madrid** is the false trap. 17. Carbon footprint ≠ ecological footprint. Units are **tCO_2e** versus **gha**. 18. Green economy ≠ circular economy as identical synonyms. Circular is the material-loop design inside a wider green agenda. 19. Net zero allows balancing removals. Gross zero means nearly eliminating emissions without relying on offsets. 20. MISHTI is a mangrove initiative. Pairing it with 2019 is wrong. 21. Microplastic size fact is

<5 mm, not 5 cm. 22. NbS examples include mangroves, wetlands, and urban (नगरीय) forests — ecosystem (पारिस्थितिकी तंत्र) actions, not aerosol injection. 23. LiFE is lifestyle and demand-side behaviour change, not a substitute for the entire NDC package. 24. Blue economy is oceans and coasts. Green hydrogen is energy. Do not merge the two labels.

💡 ⚡ **Confused Pairs (Click to Expand)**

| A | B | Distinguishing Fact / Correct Match | Hindi Terminology | |---|---|---|---| | **Carbon Footprint** | **Ecological Footprint** | Total amount of greenhouse gases (in tonnes of CO_2e) emitted directly/indirectly by an entity vs total biocapacity area (in global hectares, gha) required to sustain a lifestyle | कार्बन पदचिह्न (CO_2e उत्सर्जन) / पारिस्थितिक पदचिह्न (भूमि/संसाधन मांग) | | **Net Zero Emissions** | **Gross Zero Emissions** | Balancing residual carbon emissions with equivalent carbon removals via sinks and DACCS vs completely eliminating all greenhouse gas emissions across all sectors without offsets | नेट जीरो (उत्सर्जन-अवशोषण संतुलन) / सकल शून्य (पूर्ण उत्सर्जन शून्यता) | | **Green Economy** | **Circular Economy** | Broad macro-framework promoting low-carbon, resource-efficient, and socially inclusive growth vs industrial economic model eliminating waste through closed-loop recycling and repair | हरित अर्थव्यवस्था / चक्रीय अर्थव्यवस्था | | **Green Hydrogen** | **Grey Hydrogen / Blue Hydrogen** | Produced by electrolysis of water powered entirely by renewable electricity vs produced from natural gas without carbon capture (Grey) or with carbon capture & storage (Blue) | हरित हाइड्रोजन (नवीकरणीय) / धूसर व नीला हाइड्रोजन | | **Primary Microplastics** | **Secondary Microplastics** | Tiny plastic beads (<5mm) intentionally manufactured for cosmetics and industrial pellets vs microscopic plastic fragments resulting from physical weathering and breakdown of larger plastic debris | प्राथमिक माइक्रोप्लास्टिक (निर्मित मनके) / द्वितीयक माइक्रोप्लास्टिक (अपघटित टुकड़े) | | **Nature-based Solutions (NbS)** | **Geoengineering** | Ecosystem protection, restoration, and sustainable management to address societal challenges (e.g. mangrove restoration) vs large-scale technological manipulation of Earth's systems (e.g. solar radiation management) | प्रकृति-आधारित समाधान (NbS) / भू-इंजीनियरिंग | | **Mission LiFE Genesis** | **Mission LiFE Official Launch** | Conceptualized and introduced by PM Modi at COP26 Glasgow in **November 2021** vs officially launched jointly with UN Secretary-General António Guterres at Ekta Nagar (Kevadia), Gujarat in **October 2022** | मिशन लाइफ विचार (ग्लासगो 2021) / औपचारिक शुभारंभ (केवड़िया 2022) |

💡 📋 **Must-score facts — footprints, net zero, hydrogen, plastics (Click to Expand)**

Current locks | Item | Lock | |---|---| | Carbon footprint | tCO_2e; Scope 1/2/3 | | Ecological footprint | gha demand (≠ tCO_2e) | | India net zero | **2070** | | Emissions Gap | **UNEP** | | Green / Blue / Circular | Low-carbon inclusive / ocean SDG14 / 3R–EPR SDG12 | | MISHTI | Budget **2023** (≠ 2019) | | Green H_2 mission | Jan **2023**; ~5 MMT by 2030 | | Grey / Blue / Green H_2 | Gas reform / +CCS / RE electrolysis | | Microplastics | <5 mm; SUP ban push **2022** | | Climate refugees | Mostly IDPs; not 1951 Refugee Convention category |

44.1 Carbon Footprint

Identity: Total GHGs as tCO_2e | Scope 1/2/3 | India low per capita, high total

- A **carbon footprint** is the total greenhouse gases caused directly and indirectly by a person, organisation, product, or nation, expressed as **tonnes of CO_2 equivalent (tCO_2e)**.
- **Scope 1** covers **direct** emissions from owned sources (boilers, fleet fuel).
- **Scope 2** covers **purchased electricity, heat, or steam**.
- **Scope 3** covers other indirect emissions (supply chain, product use, travel, waste) — often the largest share and hardest to measure.
- India's **per capita** footprint is about **~2 tCO_2e**, below the global average (~4.7) and far below the USA (~14+), even though India's **absolute** emissions rank near the top globally.
- Energy/electricity is the largest Indian sector share; diet, transport mode, and grid mix drive lifestyle footprints.

- Carbon footprint is **not** the same as ecological footprint (land-area metric in **gha**).

Scope	Type	Example
Scope 1	Direct	Factory (फैक्टरी) boiler
Scope 2	Purchased energy	Grid electricity
Scope 3	Other indirect	Supplier emissions, flights

> **Logic:** "India has the highest per capita carbon footprint" = **FALSE**. Unit = **tCO_2e**, not gha.

44.2 Ecological Footprint

Identity: gha unit | Biocapacity | Global Footprint Network | Broader than carbon

- An **ecological footprint** measures how much biologically productive land and water a population (जनसंख्या) needs for resources and waste absorption, in **global hectares (gha)**.
- **Biocapacity** is nature's regenerative capacity; if footprint exceeds biocapacity, there is an **ecological deficit**.
- **Earth Overshoot Day** marks when annual demand exceeds that year's regeneration.
- Components include cropland, grazing land, forest, fishing grounds, built-up land, and carbon-absorption land.
- India per capita ecological footprint is roughly **~1.0–1.2 gha** — below the global average (~2.8 gha) but rising with urban consumption.
- Standard methodology is associated with the **Global Footprint Network**.
- shorthand: ecological footprint measures **human pressure on nature** — broader than GHG-only carbon accounting.

Metric	Unit	Rough India
Ecological footprint	gha	~1.0–1.2 / person
Carbon footprint	tCO_2e	~2 / person

> **Logic:** Same unit for both footprints = **FALSE**. EF = **gha**; carbon = **tCO_2e**.

Social Cost of Carbon

- The **Social Cost of Carbon (SCC)** is a **monetary measure** of the long-term damage done by emitting **one tonne of CO_2** in a given year.
- US EPA and other agencies use SCC estimates (प्राकलन समिति) to evaluate climate impacts of rule-making.
- SCC is **not** the same as a person's carbon footprint, fossil-fuel requirement of a country, or climate-refugee adaptation cost.

44.3 Net Zero Emissions

Identity: Emissions = removals | India 2070 COP26 | NDC 2030 | Gap Report = UNEP

- **Net zero** means anthropogenic emissions are **balanced by anthropogenic removals** over a period — residual emissions may remain if matched by sinks or removals.
- **Gross zero** means eliminating virtually all emissions — harder than net zero.
- India pledged **net zero by 2070** at **COP26 Glasgow (November 2021)**.
- Updated **NDC 2030:45%** emissions-intensity cut vs 2005; **50%** non-fossil electricity capacity; additional **2.5–3 $GtCO_2e$** forest/tree-cover sink.

- Country ladder often tested: EU/UK **2050**, China **2060**, India **2070**.
- **UNEP Emissions Gap Report** measures the gap between pledges and 1.5°C/2°C pathways.

Country/bloc	Net zero year
EU / UK	2050
China	2060
India	**2070**

PYQ — UPPCS Prelims 2021, Q70

With reference to 'NET-ZERO' goal for 2050: (1) It means emissions brought down to zero by 2050. (2) Emissions will be compensated by absorption and removal of GHGs.

💡 **Show answer**

Ans: B (Only 2) — Stmt 1 describes **gross zero**; net zero allows balance via removals.

PYQ — UPPCS Prelims 2024, Q6

Match: Emissions Gap Report → **UNEP**.

💡 **Show answer**

Ans: A (3 1 4 2) — HDR=UNDP (यूएनडीपी); WEO=IMF; WIR=UNCTAD; Emissions Gap=**UNEP**.

> **Logic:** India net zero = **2070**, not 2050. Gap Report ≠ IPCC (आईपीसीसी) Assessment Report.

44.4 Green Economy

Identity: UNEP 2011 | Low-carbon + inclusive | Decouple GDP from harm | Broader than circular

- UNEP's **green economy** (2011 framing) improves well-being and equity while cutting environmental risks — **low-carbon, resource-efficient, socially inclusive**.
- Goal is to **decouple GDP growth from environmental degradation (पर्यावरणीय क्षरण)**, not to stop growth.
- India levers include renewables (**500 GW non-fossil capacity by 2030** pledge track), EVs (FAME-II), efficiency (PAT/UJALA), and green hydrogen.
- **Green finance** (हरित वित्त) (sovereign green bonds, SEBI green-bond rules) funds this pathway — detailed market tools sit with Topic 24.
- Green economy is **broader** than circular economy; circular material loops are one component.

> **Logic:** Green economy ≠ circular economy as identical labels.

44.5 Blue Economy

Identity: SDG 14 | Ocean livelihoods + conservation | MISHTI 2023 | Sagarmala | Blue carbon (नीला कार्बन)

- **Blue economy** is sustainable use of ocean and coastal resources for growth and jobs while protecting marine ecosystems.
- **SDG 14** = **Life Below Water**.
- India has a large **EEZ (~2.37 million sq km)** — fisheries, ports, shipping, coastal tourism, offshore wind, marine biotech.

- **Blue carbon** is carbon stored in mangroves, seagrass, and salt (लवणाध्यक्ष) marshes.
- **MISHTI** (Mangrove Initiative for Shoreline Habitats and Tangible Income) was announced in **Union Budget 2023–24** to restore mangroves for shoreline protection and income.
- **Sagarmala** is port-led coastal development; **Deep Ocean Mission** covers deep-sea exploration (Samudrayaan).
- UP is **landlocked** — blue-economy PYQs are national schemes, not UP coastal projects.

PYQ — UPPCS Prelims 2025, Q93

MISHTI: (1) Initiated in 2019. (2) Aims to restore mangrove forests.

> 💡 **Show answer**
>
> **Ans: A (Only 2)** — Restoration true; year is **2023–24 Budget**, not 2019.

Sector	Fact
Mangroves	MISHTI 2023
Ports	Sagarmala
Deep sea	Deep Ocean Mission

> **Logic:** MISHTI = coastal mangroves, **not** Ganga-basin inland planting.

44.6 Circular Economy

Identity: Close loops | 3R/9R | EPR rules | SDG 12 | ⊂ green economy

- A **circular economy** designs out waste by keeping materials in use through reduce, reuse, repair, remanufacture, and recycle — opposite of linear **take-make-dispose**.
- Classic **3R** expands in some frameworks to a fuller **9R** ladder (refuse → recover).
- India operationalises circularity mainly through **Extended Producer Responsibility (EPR)** (विस्तारित उत्पादक दायित्व) — plastic, e-waste, batteries, tyres.
- **Plastic Waste Management Rules 2016** (amended 2022) and related EPR rules make producers responsible for end-of-life collection.
- Circular economy maps strongly to **SDG 12** (Responsible Consumption and Production).
- It is a **component** of green economy, not a synonym for the whole green-growth agenda.

Linear	Circular
Virgin in → dump out	Recycled/regenerative loops
Waste = disposal problem	Waste = design flaw

> **Logic:** "Circular = green economy identically" = **FALSE**.

44.7 Green Hydrogen

Identity: Renewable electrolysis | Mission Jan 2023 | 5 MMT by 2030 | Colour codes

- **Green hydrogen** is H_2 from **water electrolysis** powered by **renewable electricity**.
- Hydrogen is an **energy carrier**, not a primary source; at point of use it burns to water vapour.
- **Grey hydrogen** from natural-gas reforming dominates today's supply and is **not** climate-friendly.

- **Blue hydrogen** is grey production plus **CCS** — a debated transition label.
- **National Green Hydrogen Mission (January 2023)**, coordinated under **MNRE**, targets about **5 MMT/year by 2030** for refining, ammonia/fertiliser, steel, and heavy transport.
- Grey ≠ green — the colour word is an trap.

Colour	Method
Green	Electrolysis + renewables
Grey	Natural gas reforming
Blue	Grey + CCS
Black/brown	Coal gasification

> **Logic:** Mission = **2023**, not 2019. Target = **5 MMT by 2030**.

44.8 Microplastics

Identity: <5 mm | Primary vs secondary | Persistence | SUP ban 2022 | Ganga (गंगा) pathway

- **Microplastics** are plastic particles **smaller than 5 mm**.
- **Primary** types are manufactured small (microbeads, nurdles).
- **Secondary** types form when larger plastics fragment (bags, nets, tyre wear) — most environmental mass is secondary.
- Pathways run land → rivers → sea; Ganga/Yamuna (यमुना) studies show urban downstream contamination.
- Plastics photodegrade into **smaller** particles; they do **not** vanish into harmless minerals in weeks.
- India banned identified **single-use plastic items in July 2022** and uses EPR under plastic-waste rules.
- UNEA **2022** launched Global Plastics Treaty negotiations (INC process).

Type	Origin
Primary	Made small
Secondary	Fragmented from large items

> **Logic:** Threshold = **<5 mm**, not 5 cm. “Biodegrade in weeks” = **FALSE**.

44.9 Climate Refugees

Identity: Climate-linked mobility | Mostly internal | No 1951 legal category | Bundelkhand UP angle

- **Climate refugees / climate migrants** are people forced to move by climate impacts — cyclones, floods, drought (सूखा), sea-level rise, desertification (मरुस्थलीकरण).
- Preferred policy language often uses **climate migration**; **“climate refugee”** is **not** a formal category under the **1951 Refugee Convention** (persecution-based).
- Most climate-linked displacement is **internal** within countries, not cross-border asylum claims.
- Sudden-onset (floods, cyclones) differs from slow-onset (drought, sea-level rise).
- UP example: **Bundelkhand** drought stress drives outmigration toward cities.
- Mobility is **multi-causal** — climate interacts with poverty and livelihoods; not every migrant is “climate-only.”

Type	Example
Sudden-onset	Cyclone / flash flood displacement
Slow-onset	Drought / sea-level rise
Legal refugee (1951)	Persecution — not climate alone

> **Logic:** No dedicated Indian **Climate Refugee Act**. Most Indian cases = **IDPs / internal migrants**.

44.10 Nature-based Solutions (NbS) (प्रकृति-आधारित समाधान)

Identity: IUCN | Protect–manage–restore | Climate + society | MISHTI | ≠ geoengineering

- **Nature-based Solutions** protect, sustainably manage, or restore ecosystems to address societal challenges (climate, disasters, food, water) while aiding biodiversity and well-being.
- Mitigation NbS: forests, mangroves (blue carbon), soils, peatlands.
- Adaptation NbS: mangrove storm buffers, wetlands for flood peaks, urban greenery for heat.
- **MISHTI mangrove restoration** is a flagship Indian NbS example (Budget **2023–24**).
- Related India tracks: Green India Mission, Amrit Dharohar wetlands, Namami Gange (नमामि गंगे) riparian works.
- NbS ≠ **geoengineering**; planting trees is ecosystem management, not solar-radiation engineering.
- Geoengineering teaching contrast: **cirrus cloud thinning** and **stratospheric sulphate aerosol injection** are proposed to cool the planet / offset warming — they are **not** NbS and are not for cyclone control or solar-wind shielding.
- NbS may generate carbon credits but the concept is broader than offset markets alone.

> **Logic:** NbS = ecosystems for people + planet. Permanence risk (fire, clearing) still matters for carbon claims.

44.11 LiFE Mission

Identity: Lifestyle for Environment (जीवनशैली पर्यावरण) | MoEFCC | COP26 idea 2021 | Launch June 2022 | 75 actions

- **LiFE (Lifestyle for Environment)** nudges individual and community behaviour toward mindful consumption.
- The idea was proposed at **COP26 Glasgow, November 2021**.
- **Mission LiFE** was formally launched in **June 2022**; **MoEFCC** (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) is nodal.
- Catalogue of about **75** pro-planet actions covers energy, water, waste, food, and e-waste habits.
- It is a **behavioural nudge** programme — not a substitute for solar missions, carbon markets, or Green Credit rules.
- Supports demand-side cuts that help the **net zero 2070** pathway.

PYQ — UPPCS Prelims 2025, Q139

LiFE: (1) Launched in June 2022. (2) Idea proposed at COP-25 in Madrid.

💡 **Show answer**

Ans: D (Only 1) — Launch true; idea = **COP26 Glasgow 2021**, not COP-25 Madrid 2019.

Field	Detail
Idea	COP26 **2021**
Launch	**June 2022**

Field	Detail
Nodal	**MoEFCC**
Actions	**~75**

> **Logic:** COP-25 Madrid trap is the classic false statement.

Complete PYQ Bank (Topic 44)

Q1. UPPCS Prelims 2025, Q139

With reference to Mission LiFE, consider the following statements:

1. It was launched in June 2022.
2. The idea was proposed at COP-25 in Madrid.

A. Only 2 B. Neither C. Both D. Only 1

> 💡 **Show answer**
>
> **Logic:** LiFE — launch June 2022 true; idea was at COP26 Glasgow 2021, not COP-25 Madrid. **Ans: D (Only 1).** Launch **June 2022** true; idea = **COP26 Glasgow 2021**, not COP-25 Madrid. **Trap:** Madrid 2019 COP number swap.

Q2. UPPCS Prelims 2025, Q93

With reference to MISHTI, consider the following statements:

1. It was initiated in 2019.
2. It aims to restore mangrove forests.

A. Only 2 B. Neither C. Both D. Only 1

> 💡 **Show answer**
>
> **Logic:** MISHTI restores mangroves (true) but was announced in Union Budget 2023–24, not 2019. **Ans: A (Only 2).** Mangrove restoration aim true; year is **Budget 2023–24**, not 2019. **Trap:** Any coastal scheme dated to 2019 by habit.

Q3. UPPCS Prelims 2024, Q6

In the organisation–report match set, Emissions Gap Report corresponds to:

A. UNEP B. IPCC C. UNDP D. WWF

> 💡 **Show answer**
>
> **Logic:** Org–report match — Emissions Gap Report publisher is UNEP, not IPCC or UNDP. **Ans: A.** Full paper code **A (3 1 4 2)** — Gap→**UNEP**; HDR→UNDP; WEO→IMF; WIR→UNCTAD. **Trap:** IPCC because "emissions" sounds like climate science body.

Q4. UPPCS Prelims 2021, Q70

With reference to the 'NET-ZERO' goal for 2050:

1. It means emissions are brought down to zero by 2050.
2. Emissions will be compensated by absorption and removal of GHGs.

A. Only 1 B. Only 2 C. Both 1 and 2 D. Neither 1 nor 2

💡 **Show answer**

Logic: Net zero allows balance via removals — not the same as gross zero (all emissions to absolute zero by 2050). **Ans: B (Only 2).** Stmt 1 = **gross zero** wording; net zero **balances** with sinks/removals. **Trap:** Treating net zero as "no emissions at all."

Q5. UPPCS Prelims (pattern)

Ecological footprint is best described as measuring:

A. Human demand on biologically productive land and water in global hectares (gha) B. Total GHGs only in tCO_2e C. Noise in decibels around silence zones D. BOD in mg/L only

💡 **Show answer**

Logic: Ecological footprint measures human pressure on nature in global hectares (gha) — broader than GHG-only carbon accounting. **Ans: A.** **EF in gha** includes cropland, forest, carbon uptake land, etc. **Trap:** Same unit as carbon footprint (tCO_2e).

Q6. UPPCS Prelims (pattern)

Carbon footprint is expressed in:

A. Tonnes of CO_2 equivalent (tCO_2e) B. Global hectares only C. Decibels (dB) D. Dobson Units

💡 **Show answer**

Logic: Carbon footprint unit is tCO_2e; Scope 1 direct, Scope 2 purchased energy, Scope 3 value chain. **Ans: A.** **tCO_2e** for GHG totals; scopes 1/2/3 cover direct, purchased energy, and indirect chain. **Trap:** gha unit from ecological footprint.

Q7. UPPCS Prelims (pattern)

India's net-zero emissions pledge year is:

A. 2070 B. 2050 C. 2030 D. 2100

💡 **Show answer**

Logic: India pledged net zero by 2070 at COP26 Glasgow 2021 — not 2050 like EU/UK. **Ans: A.** **Net zero by 2070** (COP26, 2021); NDC 2030 has 45% intensity cut and 50% non-fossil capacity tracks. **Trap:** 2050 EU/UK year applied to India.

Q8. UPPCS Prelims (pattern)

Microplastics are plastic particles smaller than:

A. 5 mm B. 5 cm C. 5 m D. 50 km

💡 **Show answer**

Logic: Microplastics are plastic particles smaller than 5 mm — primary (microbeads) or secondary (fragmentation). **Ans: A.** Threshold **<5 mm**; India's **SUP ban July 2022** cuts primary sources. **Trap:** 5 cm or "biodegrade in weeks" distractors.

Q9. UPPCS Prelims (pattern)

Green hydrogen is produced mainly by:

A. Water electrolysis using renewable electricity B. Natural gas reforming without carbon capture (grey hydrogen) C. Coal gasification only D. Diesel generator electrolysis

💡 **Show answer**

Logic: Green hydrogen from renewable-powered electrolysis; National Green Hydrogen Mission January 2023 targets ~5 MMT/year by 2030. **Ans: A.** **Green H_2** = renewables + electrolysis; **Mission 2023 → ~5 MMT by 2030**. **Trap:** Grey hydrogen labelled "green."

Q10. UPPCS Prelims (pattern)

Which set is **correct**?

A. Blue economy → SDG 14; circular economy ⊂ green economy; NbS = protect/manage/restore ecosystems; "climate refugee" is not a formal 1951 Convention category B. Circular economy = identical to green economy always; microplastics >5 cm; India net zero 2050; LiFE idea at COP-25 Madrid C. Green hydrogen = grey natural-gas reforming; MISHTI started 2019; Emissions Gap Report by IPCC D. Ecological footprint in tCO_2e; carbon footprint in gha; NbS = sulphate aerosol injection

💡 **Show answer**

Logic: Blue economy = SDG 14 oceans; circular economy = material loops (SDG 12); NbS = ecosystem actions, not geoengineering; climate migrants lack 1951 Refugee Convention status. **Ans: A.** Consolidates syllabus traps: **SDG 14**, **circular ⊂ green**, **NbS ≠ geoengineering**, **no 1951 climate-refugee status**. **Trap:** Any option mixing footprint units or LiFE/MISHTI dates.

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to carbon footprint, which of the following statements is/are correct?

1. Carbon footprint is measured only in global hectares (gha), never in tCO_2e.
2. Scope 1 direct; Scope 2 purchased energy; Scope 3 value chain.
3. Identical unit to ecological footprint (gha).

A. Only 2 B. 1 and 3 C. Only 1 D. 2 and 3

💡 **Show answer**

Ans: A. Only 2 is correct. **Logic:** Statement 2 correctly states Scopes 1–3. Statement 1 is false — carbon footprint totals GHGs in tCO_2e. Statement 3 confuses carbon footprint with ecological footprint (gha).

Q2. Match List-I with List-II:

List-I	List-II
A. Grey hydrogen	1. RE electrolysis
B. Blue hydrogen	2. Natural gas reform
C. Green hydrogen	3. Grey + CCS

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-3, C-1. **Logic:** Grey/blue/green.

Q3. Which of the following pairs is/are correctly matched?

1. India's per capita ~2 tCO_2e is higher than USA averages in standard teaching.
2. Absolute emissions still high
3. Net zero India — 2050 identical to EU only

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

💡 **Show answer**

Ans: B. Only 2 is correct. **Logic:** Statement 2 is true (absolute emissions still high). Statement 1 is false — Indian per capita is low vs USA/global high emitters. Statement 3 is false — India net-zero teaching is 2070, not 2050 identical to EU.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Net zero balances remaining emissions with removals.

Reason (R): Net zero is identical to gross zero with no removals allowed.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Net zero ≠ gross zero.

Q5. With reference to NDC-style facts, which of the following statements is/are correct?

1. ~45% intensity cut teaching.
2. ~50% non-fossil capacity teaching.
3. Large forest sink goal neighbourhood.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** NDC cluster.

Q6. Arrange MISHTI year:

A. 2019 B. Budget 2023 mangrove restoration C. 1972 D. 1987

💡 **Show answer**

Ans: B. 2023 Budget — not 2019. **Logic:** MISHTI trap.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Green economy — low-carbon, resource-efficient, socially inclusive
2. Circular economy — material loops 3R/9R/EPR; SDG 12 neighbourhood
3. Green economy = circular economy as identical synonyms

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Circular ⊂ green agenda — not identical.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Blue economy is sustainable ocean use linked to SDG 14.

Reason (R): Blue economy is identical to green hydrogen energy mission.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: C. A true, R false. **A/R logic:** Oceans ≠ green hydrogen label.

Q9. With reference to microplastics, which of the following statements is/are correct?

1. <5 mm — not 5 cm.
2. Primary made small; secondary fragmented.
3. SUP ban push 2022 feeds pathway control neighbourhood.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: D. All three. **Logic:** Size + SUP.

Q10. Match List-I with List-II:

List-I	List-II
A. LiFE idea	1. June 2022
B. LiFE launch	2. COP26 2021
C. Green H_2 mission	3. Jan 2023; ~5 MMT by 2030

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: A. A-2, B-1, C-3. **Logic:** LiFE / GH2.

Q11. Arrange NbS vs geoengineering:

A. Identical aerosol injection B. NbS = protect/manage/restore ecosystems; not cirrus thinning / sulphate injection C. NbS only means BS-VI D. NbS only means CITES

💡 **Show answer**

Ans: B. Ecosystem actions ≠ geoengineering. **Logic:** IUCN NbS frame.

Q12. With reference to climate refugees, which of the following statements is/are correct?

1. Mostly internally displaced.
2. Not a separate legal category under 1951 Refugee Convention.
3. Automatically get permanent UNSC seats.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only 1 and 2. **Logic:** Statement 3 false.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): LiFE idea pitch is COP26 — not COP25 Madrid.

Reason (R): LiFE launched June 2022 under MoEFCC.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; launch date does not itself prove the COP26 pitch year. **A/R logic:** Pitch vs launch.

Q14. Which one of the following correctly states ecological footprint?

A. Only AQI Severe B. Only tCO_2e identical to carbon footprint C. Only Dobson Units D. Resource demand in global hectares (gha)

💡 **Show answer**

Ans: D. gha resource demand. **Logic:** ≠ carbon footprint unit.

Q15. How many of the following are true?

Emissions Gap = UNEP; India NZ 2070; do not call grey hydrogen green

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Trio.

Q16. With reference to green hydrogen mission, which of the following statements is/are correct?

1. January 2023 mission neighbourhood.
2. Targets about 5 MMT by 2030.
3. Grey hydrogen is the same as green.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Grey ≠ green.

Q17. Match List-I with List-II:

List-I	List-II
A. Carbon footprint	1. gha
B. Ecological footprint	2. tCO_2e
C. NbS examples	3. Mangroves, wetlands, urban forests

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** Units + NbS.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): LiFE is lifestyle / demand-side behaviour change.

Reason (R): It substitutes for the entire NDC package.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: C.** A true, R false. **A/R logic:** LiFE ≠ full NDC substitute.

Q19. Which of the following pairs is/are correctly matched?

1. Circular economy — closes loops via 3R/9R and EPR
2. Sits inside broader green-economy idea
3. Blue economy — only green hydrogen plants inland

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Blue = oceans/coasts.

Q20. Arrange gross zero vs net zero:

A. Gross zero ≈ eliminate emissions without relying on offsets; net zero allows balancing removals B. Identical always C. Both only mean 2019 MISHTI D. Both only mean argon

> 💡 **Show answer**
>
> **Ans: A.** Gross vs net. **Logic:** Offset reliance difference.

Q21. With reference to Social Cost of Carbon neighbourhood, which of the following statements is/are correct?

1. Distinct from ecological footprint gha teaching.
2. Carbon footprint uses tCO_2e.
3. Ecological footprint equals Scope 3 only always.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: B.** Only 1 and 2. **Logic:** gha ≠ Scope-3 identity.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Microplastics are not fast biodegradable in standard teaching.

Reason (R): Size fact is smaller than 5 mm.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: A.** Both true; size does not itself prove persistence. **A/R logic:** Size vs persistence.

Q23. How many of the following scheme-year facts are correct?

LiFE idea 2021; LiFE launch 2022; MISHTI 2023

A. Only one B. Only two C. All three D. None

Show answer

Ans: C. All three. **Logic:** Year trap trio.

Q24. Which one of the following correctly states NbS examples?

A. Only BS-VI sulphur caps B. Only sulphate aerosol injection C. Only cirrus thinning D. Mangroves, wetlands, urban forests — ecosystem actions

Show answer

Ans: D. Ecosystem NbS. **Logic:** Not geoengineering.

Q25. With reference to India's absolute vs per capita, which of the following statements is/are correct?

1. Per capita low (~2 tCO_2e teaching).
2. Absolute emissions still among the highest.
3. Per capita low means absolute emissions are zero.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

Show answer

Ans: A. Only 1 and 2. **Logic:** Low per capita ≠ zero absolute.

Q26. Match List-I with List-II:

List-I	List-II
A. SUP ban push	1. SDG 14
B. Blue economy	2. 2022
C. Circular economy	3. SDG 12 neighbourhood

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

Show answer

Ans: B. A-2, B-1, C-3. **Logic:** SUP/blue/circular.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Do not call grey hydrogen green.

Reason (R): Green hydrogen uses renewable-powered electrolysis.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

Show answer

Ans: D. Both true and R defines why grey fails the green test. **A/R logic:** Colour codes.

Q28. Arrange India net-zero year:

A. 2015 B. 1992 C. 2070 D. 1972

💡 **Show answer**

Ans: C. 2070. **Logic:** NZ year.

Q29. With reference to Emissions Gap Report, which of the following statements is/are correct?

1. UNEP product.
2. Cross-subject economy–environment fact neighbourhood.
3. Published by IUCN Red List office.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Not IUCN.

Q30. Which one of the following correctly separates carbon footprint from ecological footprint?

A. tCO_2e GHG total vs gha resource demand B. Both identical always C. Both only Dobson Units D. Both only Ringelmann shades

💡 **Show answer**

Ans: A. Unit and meaning split. **Logic:** Core confused pair.

Common Traps

Trap	Correct
LiFE idea at COP-25 Madrid	**COP26 Glasgow 2021**; launch **June 2022**
MISHTI initiated 2019	**Budget 2023–24**
India net zero 2050	**2070**
Net zero = emissions to absolute zero	Balance with **removals**
Emissions Gap = IPCC	**UNEP**
Ecological footprint in tCO_2e	Unit = **gha**
Microplastics <5 cm	**<5 mm**
Green H_2 = grey H_2	Grey = natural gas reforming
Green economy = circular identically	Circular = **material loops**; green = broader
Climate refugee = 1951 legal status	**Not** a formal Convention category
NbS = geoengineering	Ecosystem protect/manage/restore
MISHTI for Ganga basin inland	**Coastal mangroves**
Highest India per capita carbon footprint	High **total**, low **per capita**

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**

Topic 45 — Environment (पर्यावरण) PYQ Trend Analysis

★ Revision Meta-File — How UPPCS Actually Tests Environment

> **Purpose:** PYQ-derived intelligence — not a syllabus topic, but a **revision accelerator** after Topics 1–44 **Data source:** All files in pyq/ (UPPCS Prelims 2018–2025) — 121 Environment-tagged + ~20 cross-subject env-adjacent questions **Use when:** After completing topic notes; before mock tests; for last-week priority sorting **Last verified:** July 2026

💡 🎯 Consolidated — 18 Must-Score Facts (Click to Expand)

1. Recent UPPCS papers give roughly **15–20 tagged environment marks** most years. Planning for only 8–10 is outdated. 2. Tagged volume rose from about **12** questions (2018–20 average) to about **17.7** (2023–25 average). 3. **2023** was the peak tagged year (**25** questions). **2025** stayed strong at **19**. 4. **2020** and **2024** dips in tagged count still hide cross-subject environment leakage in Economy and Geography. 5. The hottest topic cluster by frequency is **SDGs and sustainable development**, then **pollution**, then **ecology (पारिस्थितिकी) and ecosystem (पारिस्थितिकी तंत्र)**. 6. Climate change (जलवायु परिवर्तन), wildlife conservation (वन्यजीव संरक्षण), biodiversity, biosphere reserves (जैवमंडल आरक्षित क्षेत्र), forests/ISFR, and Ramsar (रामसर)–UP traps also stay hot. 7. From **2023–2025**, rising lines include SDGs, wildlife, biodiversity, biosphere, forests, and current schemes such as LiFE and MISHTI. 8. The **2025** environment paper shifted toward **multi-statement, A/R, Match, and NOT-matched** formats — about two-thirds (ते-भागा) multi-format. 9. Practice must include two-statement and Assertion–Reason drills, not only one-line date or HQ recall. 10. 2025 blueprint clusters cover SDGs, current schemes, UP geography traps, ecology indicators, and international reports or funds. 11. Scheme year traps: **LiFE** idea **COP-26 2021** / launch **June 2022** — not COP-25. **MISHTI** = **2023** Budget — not 2019. 12. **DPAP = 1973**. India **net zero = 2070**. **GEF = 1991**. **Montreal = ozone**. **Kyoto (क्योटो)/Paris = climate**. 13. UP negative geography repeats: Rudrasagar = Tripura; Sultanpur = Haryana; Nokrek ≠ Manipur; many "UP BR" claims are false. 14. True environment ROI is higher than tagged count because reports and protected-area facts leak into other GS subjects. 15. Study ROI priority puts SDGs and UP protected-area traps near the top, then current issues, ecology, climate/pollution, and treaties. 16. Emissions Gap Report (उत्सर्जन अंतराल रिपोर्ट) = **UNEP** (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) remains a cross-subject economy–environment fact. 17. Format formula after 2025: prepare heavily for multi-statement and Match/NOT-matched, with a smaller share of pure direct MCQs. 18. Volume formula for the next attempt: budget about **15–20 tagged** plus a few cross-subject environment hits.

💡 ⚡ Confused Pairs (Click to Expand)

A	B	Distinguishing Fact / Correct Match	Hindi Terminology
UPPCS Direct Questions	**2024–2025 Multi-Statement Format**	Historical single-line factual recall questions vs modern complex Assertion-Reasoning, 4-statement matching, and chronological order puzzles	प्रत्यक्ष स्मृति प्रश्न / बहु-कथनीय कूट प्रारूप
Dedicated Environment Questions	**True Environmental Weightage**	Official subject classification (~10–12 questions) vs actual interdisciplinary paper presence (~18–22 questions when counting geography/economy overlaps)	प्रत्यक्ष पर्यावरण प्रश्न / वास्तविक विषय भारांक
MISHTI Scheme (2023)	**Mangrove Alliance for Climate (2022)**	Indian Union Budget 2023–24 programme for coastline mangrove plantation via MGNREGA/CAMPA vs international climate initiative launched by UAE and Indonesia at COP27 Sharm el-Sheikh	मिष्टी योजना (बजट 2023) / मैंग्रोव गठबंधन (MAC)
Nokrek Biosphere Reserve	**Keibul Lamjao National Park**	Biosphere Reserve located in Garo Hills,

Meghalaya (zero BRs in UP) vs floating National Park on Loktak Lake, **Manipur** | नोकरेक (मेघालय) / केइबुल लामजाओ (मणिपुर) | | **Kigali Amendment (2016)** | **Paris Agreement (2015)** | Binding phase-down of Hydrofluorocarbons (HFCs) under the Montreal Protocol vs global climate framework under UNFCCC capping warming at 1.5°C | किगाली संशोधन (एचएफसी कटौती) / पेरिस समझौता | | **Stockholm 1972** | **Stockholm 2001** | First UN global conference on the environment giving birth to UNEP vs binding international treaty banning Persistent Organic Pollutants (POPs) | स्टॉकहोम सम्मेलन (1972) / स्टॉकहोम संधि (2001) | | **National Green Tribunal (NGT)** | **Central Pollution Control Board (CPCB)** | Specialized judicial tribunal established under NGT Act 2010 for environmental disputes vs statutory executive regulatory board established under Water Act 1974 | राष्ट्रीय हरित अधिकरण (न्यायिक) / सीपीसीबी (कार्यकारी बोर्ड) |

💡 📋 **Must-score facts — volume, hot clusters, year traps (Click to Expand)**

Trend locks | Item | Lock | |------|------| | Typical tagged marks | ~**15–20**/year (not 8–10) | | Peak / recent strong | **2023** ~25; **2025** ~19 | | Hottest clusters | SDGs → pollution → ecology/ecosystem | | Also hot | Climate · wildlife · biodiversity · BR · ISFR · Ramsar–UP | | 2025 format shift | ~2/3 multi-statement / A/R / Match / NOT | | LiFE / MISHTI years | COP26'21→Jun'22 / Budget **2023** | | DPAP / Net zero / GEF | **1973** / **2070** / **1991** | | UP negatives | Rudrasagar Tripura · Sultanpur Haryana · no UP BR · Nokrek≠Manipur | | Montreal / Kyoto–Paris | Ozone / climate |

How to Use This File

When	Read what	Time
First pass	§45.1 Volume + §45.2 Topic Heat Map	10 min
Before mock test	§45.3 Format Shift + §45.5 2025 Blueprint	15 min
Last-week revision	Consolidated facts + Trap Checklist + Study ROI	10 min
After each PYQ year	§45.6 Year-wise snapshot	5 min

45.1 Volume Trend — Environment Weight Over Years

Year	Env-tagged Qs	~% of 150 Q paper	Notes
2018	14	~9%	Baseline
2019	14	~9%	Stable
2020	8	~5%	**Low year** — do not use as weight benchmark
2021	13	~9%	Recovery
2022	19	~13%	Upswing begins
2023	**25**	**~17%**	**Peak year** — maximum env tagging
2024	9	~6%	Dip (still high cross-subject leakage)
2025	**19**	**~13%**	Strong; format shift to multi-statement

Key Takeaways

- **Early period (2018–20):** average **12 questions** per paper.
- **Recent period (2023–25):** average **17.7 questions** per paper, **+47% increase**.
- ** planning rule: **budget** 15–20 environment marks** in every attempt. Treating env as a 8-question afterthought is outdated.

- **2020 and 2024 dips** are anomalies. Cross-subject env-adjacent questions (Economy reports, Geography protected areas) keep true environment ROI higher than tagged count alone.

Broad Environment Footprint (Tagged + Cross-Subject)

Year	Tagged Env	+ Env-adjacent in other subjects	Broad total
2018	14	12	~26
2019	14	10	~24
2020	8	8	~16
2021	13	4	~17
2022	19	11	~30
2023	25	3	~28
2024	9	6	~15
2025	19	2	~21

45.2 Topic Heat Map — PYQ Frequency → Your Notes

Tier 1 — Must Master (highest PYQ hits)

Rank	PYQ hits	Your topic file	What UPPCS keeps asking
1	16	**15 — SD & Governance**	SDG ↔ goal matching; carrying capacity; India's SDG role; Brundtland/Agenda 21 traps
2	12	**09/38 — Pollution**	Air/water/soil; AQI; NCAP (राष्ट्रीय स्वच्छ वायु कार्यक्रम); smog types; pollution sources
3	11	**02 — Ecology & Ecosystem**	Energy flow (10% law); lichens; niches; pyramids; ecological balance
4	9	**10/34 — Climate Change**	Net zero 2070; GHGs; impacts; mitigation vs adaptation
5	8	**07 — Wildlife Conservation**	Project Tiger (प्रोजेक्ट टाइगर); Operation Olivia; vulture crisis; WPA linkages

Tier 2 — Rising Fast (2023–25 surge)

Topic file	Δ vs 2018–20	pattern
15 — SDGs	+9	SDG 6/13/14/15 mapping; voluntary national review
07 — Wildlife	+3	Species-specific conservation ops
04 — Biodiversity	+2	IUCN (आईयूसीएन) categories; endemic; invasive alien
26 — Water	+2	CGWB groundwater report; river quality
33 — Biosphere Reserves	+2	"NOT in India" negative lists
08 — Forests	+2	ISFR 2023; forest cover (वन आवरण) change; carbon stock
06/32 — Ramsar & Protected	+1	**UP geography negative** — highest trap ROI
24/44 — Current Issues	New	LiFE, MISHTI, net zero, microplastics

Tier 3 — Moderate (नरम दल) but Trap-Heavy

Topic file	Trap type
11/36 — Ozone	Montreal ≠ Kyoto; CFC statements
18/42/43 — Intl orgs & treaties	GEF 1991; Emissions Gap = UNEP; treaty ↔ issue
16/17 — Org & Laws	NGT, MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय), EPA, WPA schedules
40 — Desertification (मरुस्थलीकरण)	DPAP 1973; UNCCD 1994; erosion sequence
29 — Movements	Chipko (चिपको), Appiko (अप्पिको), Narmada (नर्मदा)
31 — Environmental Days	Date ↔ event matching

Tier 4 — Revise Lightly (low standalone frequency)

Acid rain (अम्ल वर्षा) (39), EIA (14), renewable energy (नवीकरणीय ऊर्जा) detail (22/27), disaster-env (23), global env geography (25) — appear but rarely as primary Environment-tag questions. Cover via cross-links in Tier 1–3 files.

45.3 Question Format — Historical vs 2025 Shift

All-Time Format Distribution (121 env-tagged Qs)

Format	Count	Share
Direct MCQ	82	**68%**
Multi-statement (1-2-3)	15	12%
Assertion–Reason	9	7%
Match List-I / II	9	7%
NOT correctly matched	4	3%
Chronology	2	2%

2025 Environment Paper — Format Breakdown (19 Qs)

Format	2025 count	Share
Multi-statement	8	42%
Direct MCQ	4	21%
NOT-matched	3	16%
A/R	2	11%
Match	2	11%

> **Logic:** ** 2025 reversed the historical pattern — **68% multi-format** vs 32% historically. Your Practice Zones must mirror 2-statement + A/R, not only single MCQs.

UPPCS Favourite Question Types — Ranked

1. **2-statement correct?** — Scheme years, treaty purposes, pollution facts
2. **A/R** — SDGs, overgrazing→desertification, ecological concepts
3. **Match List-I / II** — SDG ↔ goal; org ↔ report; erosion ↔ landform
4. **NOT in UP / NOT in India** — Ramsar, sanctuaries, biosphere reserves
5. **NOT correctly matched** — Treaty ↔ wrong purpose; wrong HQ; wrong year

6. **Direct MCQ** — Movement identification; report publisher; single fact

---

45.4 Five Mega-Trends (2023–2025)

Trend 1 — SDG Explosion

SDG-related questions surged from ~3 (2018–20) to **12+ (2023–25)**. UPPCS now treats SDGs as a **standalone high-weight cluster**, not a footnote under sustainable development.

Must-know: SDG 6 (water), 13 (climate), 14 (oceans/blue economy (नीली अर्थव्यवस्था)), 15 (land/biodiversity); India's VNR; SDG ↔ goal number matching.

Files: Topic 15 §15.1–15.3; Topic 44 §44.4–44.6.

Trend 2 — Scheme Chronology Traps

Examiners systematically plant **wrong launch year** as statement 2:

Scheme/Concept	Correct year	Common trap
LiFE Mission	Idea COP26 **2021**; launch **June 2022**	COP-25 Madrid 2019
MISHTI	**2023–24 Union Budget**	Initiated 2019
DPAP	**1973**	1977, 1985, 1992
India Net Zero	**2070** (COP26 2021)	2050, 2030
GEF	**1991**	Only developed donors
National Green Hydrogen Mission	**2023**	2019, 2021
UNCCD	**1994**	Rio 1992 (same year trap)

Files: Topics 40, 44, 43.

Trend 3 — UP Negative Geography

"NOT in UP" / "NOT in India" is a **repeat high-ROI weapon**:

Year	Q#	Trap tested
2021	Q17	Ramsar site NOT in UP
2022	Q36	Wildlife Sanctuary of UP — district
2023	Q56	Chandraprabha WLS — UP district
2025	Q8	ISFR 2023 — UP forest cover districts
2025	Q58	Ramsar NOT in UP — **Rudrasagar (Tripura) + Sultanpur (Haryana)**
2025	Q90	NP/sanctuary matching
2025	Q144	Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र) NOT in India

Files: Topic 32 (critical); Topics 6, 33, 40.

Trend 4 — Cross-Subject Environment Leakage

Environment concepts appear outside `Subject: Environment` — especially in **Economy** (reports, green finance (हरित वित्त)), **Geography** (biosphere, climate zones), and **Current Affairs** (new schemes).

Year	Q#	Subject	Env concept
2024	Q6	Economy	Emissions Gap Report = **UNEP**

Year	Q#	Subject	Env concept
2024	Q35	Environment	Kyoto = climate mitigation (जलवायु शमन) (Montreal trap)
2021	Q125	Environment	Carbon credit (कार्बन क्रेडिट) → Kyoto Protocol

Rule: When revising Environment, also scan Economy/Geography PYQs for report ↔ org and treaty ↔ issue questions.

Files: Topics 42, 43, 44.

Trend 5 — Reports & Organisations Over Pure Theory

Rising share of questions on **who publishes what** — less "define ecosystem," more "match report to body."

Report / Data	Publisher	PYQ year
Emissions Gap Report	**UNEP**	2024 Q6
India State of Forest Report	**FSI** (MoEFCC)	2025 Q8
Annual Groundwater Quality Report	**CGWB**	2025 Q70
Human Development Report	**UNDP** (यूएनडीपी)	2024 Q6 (cross-subject)
Living Planet Report	**WWF** (डब्ल्यूडब्ल्यूएफ)	Pattern
IUCN Red List	**IUCN** (आईयूसीएन)	Pattern
IPCC Assessment Reports	**IPCC** (आईपीसीसी)	Pattern

Files: Topics 41, 42, 16.

45.5 — 2025 Environment Paper Blueprint (Nearest Model)

Full list of 19 Environment-tagged questions — use as mock-paper template:

Q#	Format	Topic cluster	Subtopic / concept
Q5	Multi-statement	Ecology	Energy flow — 10% law
Q8	Direct MCQ	Forestry	ISFR 2023 — UP districts
Q10	Multi-statement	Agriculture-Env	Bio-decomposer programme (stubble)
Q22	Multi-statement	Sustainable Development	Carrying capacity
Q45	Multi-statement	Atmosphere/Ozone	CFC + UV protection
Q51	A/R	Sustainable Development	SDGs / Agenda 21
Q58	NOT-matched	Protected Areas	Ramsar sites NOT in UP
Q61	Multi-statement	Ecology	Lichens — bio-indicators
Q62	NOT-matched	Important Days	World Wildlife Day (विश्व वन्यजीव दिवस) date
Q70	Direct MCQ	Environmental Reports	CGWB groundwater quality 2024
Q90	Match	Protected Areas	NP and wildlife sanctuaries
Q93	Multi-statement	Conservation / Current	MISHTI mangroves
Q120	A/R	SDGs	India's role in SDGs
Q122	Direct MCQ	Environmental Movements	Appiko Movement
Q139	Multi-statement	Current Issues	LiFE Mission — COP trap
Q142	Match	SDGs	SDG ↔ goal mapping

Q#	Format	Topic cluster	Subtopic / concept
Q143	Multi-statement	Wildlife Conservation	Operation Olivia
Q144	NOT-matched	Biosphere Reserves	BR NOT in India
Q145	Multi-statement	Intl Organizations	GEF 1991

2025 Cluster Weights

Cluster	Qs	Your files
SDGs & Sustainable Development	4	Topic 15
Protected Areas / UP geography	4	Topics 6, 32, 33
Current schemes (LiFE, MISHTI)	2	Topic 44
Ecology concepts	2	Topic 2
Reports & organisations	2	Topics 41, 42
Ozone / atmosphere	1	Topics 11, 36
Environmental movements	1	Topic 29
Environmental days	1	Topic 31
Agriculture-environment	1	Topic 2 / 40 crossover

45.6 Year-Wise Snapshot

2018–2019 (Baseline era)

- Heavy ecology, pollution, biodiversity MCQs
- ~14 env Qs, mostly direct format.
- International treaties appear as NOT-matched

2020 (Low year — 8 Qs)

- Do not benchmark weight from this paper
- Ramsar sites added to UP (Q48), early UP wetland CA.

2021–2022 (Upswing)

- Carbon credits originated under the Kyoto (Q125).
- Wildlife sanctuaries UP geography
- SDG questions begin increasing
- 2022: 19 env Qs, format diversifying.

2023 (Peak — 25 Qs)

- Maximum Environment & Ecology tagging
- SDG cluster expands heavily
- UP WLS district questions continue
- Environmental laws, population-environment appear

2024 (Dip — 9 tagged, but high leakage)

- Kyoto climate mitigation vs Montreal trap (Q35)
- Emissions Gap = UNEP (Q6, Economy subject)

- Treaty ↔ issue remains favourite

2025 (Model year — 19 Qs, format shift)

- Multi-statement dominance
- LiFE + MISHTI scheme year traps
- UP negative geography (Ramsar, ISFR, Biosphere)
- SDGs + current issues + ecology trifecta

45.7 Study ROI — Priority Order (Marks per Hour)

Priority	Topic #	Why (PYQ evidence)	Time alloc.
★★★	**15** SDGs & Governance	#1 frequency; 4 Qs in 2025 alone	3–4 hr
★★★	**32** UP Protected Areas	Every 2–3 years UP negative Q	2–3 hr
★★★	**44** Current Env Issues	LiFE, MISHTI, net zero — 2025 direct	2 hr
★★★	**02** Ecology & Ecosystem	11 all-time; energy flow, lichens	2 hr
★★☆	**42 + 43** Intl orgs & treaties	GEF, UNEP, Kyoto/Montreal traps	2 hr
★★☆	**10 + 38** Climate & Pollution	21 combined hits; ozone, AQI	2 hr
★★☆	**07** Wildlife Conservation	Operation Olivia, Project Tiger	1.5 hr
★★☆	**40** Desertification	DPAP 2025 Q134; erosion sequence	1.5 hr
★★☆	**33** Biosphere Reserves	2025 Q144 negative list	1 hr
★☆☆	**29 + 31** Movements + Days	1 Q each in 2025; quick raata	30 min
★☆☆	**39–41** Advanced block	Acid rain, monitoring — lower freq	As time permits

---

45.8 What's Cooling vs Heating

❄️ Cooling (deprioritise standalone prep)	🔥 Heating (prioritise now)
Pure acid rain theory alone	SDGs + goal mapping
Detailed food-web diagram drawing	Multi-statement scheme CA
Generic "define pollution"	UP-specific protected area traps
Kyoto CDM mechanism depth	LiFE, MISHTI, net zero 2070, NbS
Single-line ecology MCQs only	Report ↔ organisation matching
Standalone EIA process detail	Bio-decomposer, stubble burning link
Global env geography alone	ISFR, CGWB report-based questions

Master Trap Checklist — Highest PYQ Strike Rate

#	Trap	Correct	PYQ evidence
1	Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) = climate mitigation	Montreal = **ozone/ODS**; Kyoto/Paris = climate	2024 Q35, 2025 Q45
2	LiFE proposed at COP-25 Madrid	**COP26 Glasgow 2021**; launched June 2022	2025 Q139
3	MISHTI initiated 2019	**2023–24 Union Budget**	2025 Q93

#	Trap	Correct	PYQ evidence
4	India net zero by 2050	**2070** (COP26 2021)	2021 Q70 pattern
5	Emissions Gap Report by IPCC	**UNEP**	2024 Q6
6	Rudrasagar Ramsar site in UP	**Tripura**	2025 Q58
7	Sultanpur NP / Ramsar in UP	**Haryana** (Gurugram)	2025 Q58
8	Desertification = desert expanding only	**ASDM productivity loss**	Pattern
9	DPAP launched 1977/1985	**1973**	2025 Q134
10	Afforestation causes land degradation (भू-क्षरण)	**FALSE** — prevents degradation	UPSC (संघ लोक सेवा आयोग) pattern
11	GEF established 2001	**1991**	2025 Q145
12	World Wildlife Day = 22 March	**3 March** (WWD)	2025 Q62
13	Ecological footprint (पारिस्थितिक पदचिह्न) unit = tCO_2e	**global hectares (gha)**	2019 pattern
14	Green hydrogen from natural gas	**Grey** hydrogen; green = renewable electrolysis	Pattern
15	UNCCD signed at Rio 1992	**1994** (Paris) — 2 years after Rio	Topic 40
16	CITES (साइट्स) = IUCN Red List	CITES = **trade** (पण्याध्यक्ष); IUCN = **assessment**	Pattern

45.9 Cross-Reference — PYQ Cluster → Topic File

If PYQ asks...	Open this file	Section
SDG ↔ goal number	15_Sustainable_Development	§15.1–15.3
LiFE / MISHTI / net zero	44_Current_Environmental_Issues	§44.3, §44.5, §44.11
Ramsar / NP NOT in UP	32_National_Parks_Advanced	UP traps + Q58 answer
ISFR forest cover	08_Forests	+ 32 UP districts
Kyoto vs Montreal	43_International_Agreements	§43.2 + Montreal trap
GEF / UNEP / Emissions Gap	42_International_Organizations	§42.1 UNEP
DPAP / desertification	40_Desertification	§40.3
Energy flow / lichens	02_Ecology_Ecosystem	§2.3, §2.5
Operation Olivia / turtles	07_Wildlife_Conservation	§7.x marine
Appiko / Chipko	29_Indian_Environmental_Movements	Full file
Env days dates	31_Environmental_Days	Full file
Ozone / CFC statements	36_Ozone_Layer	§36.1–36.3
Bio-decomposer / stubble	02 + 40 crossover	Ecology + land degradation

45.10 Mock-Test Self-Audit Checklist

After every UPPCS mock, score yourself on these trend dimensions:

- [ ] Did I attempt all 15–20 env-linked Qs (including cross-subject)?
- [ ] Did I fall for any **scheme year trap** (LiFE, MISHTI, DPAP, GEF)?

- [ ] Did I confuse **Montreal vs Kyoto** for climate questions?
- [ ] Did I mark UP Ramsar/sanctuary/NP correctly on **negative** questions?
- [ ] Did I match **report ↔ organisation** correctly (UNEP, UNDP, FSI, CGWB)?
- [ ] Did I handle **2-statement** format by evaluating each stmt independently?
- [ ] Did I revise **SDG goal numbers** for matching questions?
- [ ] Did I know **2070** net zero and **1973** DPAP without hesitation?

Practice Zone — UPPCS Format Drill

> **30 questions** | Original drill | ≥60% multi-statement/application | A/R · Match · chronology · NOT-matched | Keys balanced | **Ans first, then Logic**

Q1. With reference to recent UPPCS environment weight, which of the following statements is/are correct?

1. Roughly 15–20 tagged marks most years.
2. Planning for only 8–10 tagged marks is still the current best estimate.
3. Tagged volume fell to zero after 2018.

A. 1 and 3 B. Only 2 C. 2 and 3 D. Only 1

> 💡 **Show answer**
>
> **Ans: D.** Only 1 is correct. **Logic:** Statement 1 is true (~15–20 tagged marks most years). Statement 2 is false — 8–10 planning is outdated. Statement 3 is false — tagged volume did not fall to zero after 2018.

Q2. Match List-I with List-II:

List-I	List-II
A. 2018–20 avg tagged	1. ~17.7
B. 2023–25 avg tagged	2. ~12
C. 2023 peak	3. 25 questions

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: C.** A-2, B-1, C-3. **Logic:** Volume ladder.

Q3. Which of the following pairs is/are correctly matched?

1. The hottest cluster is only Van Mahotsav ceremony dates.
2. Then pollution; then ecology/ecosystem
3. Hottest cluster — only Van Mahotsav dates

A. Only 1 B. 1 and 2 C. 2 and 3 D. Only 2

> 💡 **Show answer**
>
> **Ans: D.** Only 2 is correct. **Logic:** Statement 2 is true (pollution then ecology/ecosystem after the hottest SDG cluster). Statement 1 is false — SDGs/sustainable development is the hottest cluster. Statement 3 repeats the Van Mahotsav trap.

Q4. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): 2025 environment paper shifted toward multi-statement, A/R, Match, and NOT-matched.

Reason (R): About two-thirds multi-format teaching — practice must include two-statement and A/R drills.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains the practice implication. **A/R logic:** Format shift → drill mix.

Q5. With reference to scheme year traps, which of the following statements is/are correct?

1. LiFE idea COP26 2021 / launch June 2022 — not COP25.
2. MISHTI = 2023 Budget — not 2019.
3. DPAP = 1973; India net zero = 2070; GEF = 1991.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: A. All three. **Logic:** Year trap set.

Q6. Arrange Montreal vs Kyoto in trend checklist:

A. Montreal = climate; Kyoto = ozone B. Montreal = ozone; Kyoto/Paris = climate C. Both identical Ramsar D. Both identical noise dB

💡 **Show answer**

Ans: B. Ozone vs climate. **Logic:** Recurring trap.

Q7. Which of the following pairs is/are NOT correctly matched?

1. Rudrasagar — Tripura
2. Sultanpur — Haryana
3. Nokrek — Manipur

A. 1 and 2 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: C. Only pair 3 is not correctly matched. **Logic:** Nokrek = Meghalaya — not Manipur.

Q8. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): True environment ROI is higher than tagged count.

Reason (R): Reports and protected-area facts leak into other GS subjects.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains A. **A/R logic:** Cross-subject leakage.

Q9. With reference to 2020/2024 dips, which of the following statements is/are correct?

1. Tagged count dipped.

2. Cross-subject environment leakage in Economy and Geography still hides weight.
3. Dips mean environment can be skipped entirely.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

> 💡 **Show answer**
>
> **Ans: D.** Only 1 and 2. **Logic:** Do not skip.

Q10. Match List-I with List-II:

List-I	List-II
A. Emissions Gap	1. 2070
B. India net zero	2. UNEP
C. DPAP	3. 1973

Code:

A. A-2, B-1, C-3 B. A-1, B-2, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

> 💡 **Show answer**
>
> **Ans: A.** A-2, B-1, C-3. **Logic:** Report / NZ / DPAP.

Q11. Arrange study ROI priority teaching:

A. Only Tier-4 rare topics first B. SDGs + UP protected-area traps near top; then current issues, ecology, climate/pollution, treaties/orgs C. Only Earth Hour D. Only Ringelmann

> 💡 **Show answer**
>
> **Ans: B.** ROI order from heat map. **Logic:** Priority spine.

Q12. With reference to rising 2023–25 lines, which of the following statements is/are correct?

1. SDGs, wildlife, biodiversity, biosphere, forests.
2. Current schemes such as LiFE and MISHTI.
3. Only tubeless-tyre CFC stems.

A. 1 and 3 only B. 2 and 3 only C. 1 and 2 only D. 1, 2 and 3

> 💡 **Show answer**
>
> **Ans: C.** Only 1 and 2. **Logic:** Rising cluster.

Q13. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Many "UP BR" claims are false.

Reason (R): Uttar Pradesh has zero Biosphere Reserves in standard teaching.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

> 💡 **Show answer**
>
> **Ans: D.** Both true and R explains A. **A/R logic:** UP BR trap.

Q14. Which one of the following correctly states 2025 tagged count neighbourhood?

A. Only 2 B. Zero C. Exactly 8 always D. Stayed strong at 19

💡 **Show answer**

Ans: D. 19 in 2025. **Logic:** Strong year.

Q15. How many of the following UP negative geography facts are correct?

Rudrasagar≠UP; Sultanpur≠UP; Nokrek≠Manipur

A. All three B. Only two C. Only one D. None

💡 **Show answer**

Ans: A. All three. **Logic:** Negative geography set.

Q16. With reference to format formula after 2025, which of the following statements is/are correct?

1. Prepare heavily for multi-statement and Match/NOT-matched.
2. Smaller share of pure direct MCQ recall.
3. Only one-line HQ recall is enough.

A. 2 and 3 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: B. Only 1 and 2. **Logic:** Multi-format heavy.

Q17. Match List-I with List-II:

List-I	List-II
A. LiFE false date	1. 2019
B. MISHTI false date	2. COP25 Madrid
C. GEF year	3. 1991

Code:

A. A-2, B-3, C-1 B. A-1, B-2, C-3 C. A-2, B-1, C-3 D. A-3, B-1, C-2

💡 **Show answer**

Ans: C. A-2, B-1, C-3. **Logic:** False dates + GEF.

Q18. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): 2025 blueprint clusters cover SDGs, current schemes, UP geography traps, ecology indicators, and international reports/funds.

Reason (R): Practice should mirror those clusters — not only isolated date recall.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: D. Both true and R explains how to train. **A/R logic:** Blueprint → practice design.

Q19. Which of the following pairs is/are correctly matched?

1. Climate, wildlife, biodiversity, BRs, forests/ISFR, Ramsar–UP traps stay hot

2. Volume formula — budget ~15–20 tagged plus cross-subject hits
3. Planning only 8–10 tagged is still optimal

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** 8–10 planning outdated.

Q20. Arrange Emissions Gap in trend sheet:

A. UNEP — cross-subject economy–environment fact B. IUCN product C. Only CGWB D. Only Form V

💡 **Show answer**

Ans: A. UNEP Gap Report. **Logic:** Cross-subject.

Q21. With reference to mock self-audit ideas, which of the following statements is/are correct?

1. Check scheme year traps.
2. Check Montreal vs Kyoto.
3. Check UP negative geography.
4. Ignore 2-statement format entirely.

A. 1, 2 and 4 only B. 1, 2 and 3 only C. 2, 3 and 4 only D. 1, 2, 3 and 4

💡 **Show answer**

Ans: B. 1, 2 and 3 only. **Logic:** Must handle 2-statement.

Q22. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Tagged volume rose from ~12 (2018–20) to ~17.7 (2023–25).

Reason (R): 2023 was the peak tagged year at 25.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; 2023 peak does not itself compute the 2018–20 average. **A/R logic:** Average rise vs peak year.

Q23. How many of the following treaty splits are trend-critical?

Montreal≠climate; Kyoto/Paris=climate; GEF=1991

A. Only one B. Only two C. All three D. None

💡 **Show answer**

Ans: C. All three. **Logic:** Checklist facts.

Q24. Which one of the following correctly states why Practice Zones were reframed?

A. Only to delete A/R B. Only to add more Ans A keys C. Only to remove Match List D. Match 2024–25 multi-format paper — not only thin one-liner recall

💡 **Show answer**

Ans: D. Multi-format standard. **Logic:** Meta purpose of this rewrite.

Q25. With reference to hot vs cool, which of the following statements is/are correct?

1. SDGs heating.
2. Current schemes heating.
3. Ignoring UP PA traps is safe because tagged count dipped in some years.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

💡 **Show answer**

Ans: A. Only 1 and 2. **Logic:** UP traps stay high-ROI.

Q26. Match List-I with List-II:

List-I	List-II
A. 2025 tagged	1. 25
B. 2023 peak	2. 19
C. Outdated plan	3. Only 8–10

Code:

A. A-1, B-2, C-3 B. A-2, B-1, C-3 C. A-2, B-3, C-1 D. A-3, B-1, C-2

💡 **Show answer**

Ans: B. A-2, B-1, C-3. **Logic:** 19 / 25 / outdated 8–10.

Q27. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Report ↔ organisation matching remains high-yield.

Reason (R): Emissions Gap Report = UNEP is a recurring cross-subject fact.

A. Both (A) and (R) are true, but (R) is not the correct explanation of (A) B. (A) is false, but (R) is true C. (A) is true, but (R) is false D. Both (A) and (R) are true and (R) is the correct explanation of (A)

💡 **Show answer**

Ans: A. Both true; one example does not exhaust all report matches, but both stand. **A/R logic:** Reports over pure theory trend.

Q28. Arrange next-attempt volume budget:

A. Only 5 tagged B. Only 100 tagged guaranteed C. About 15–20 tagged plus a few cross-subject hits D. Zero environment

💡 **Show answer**

Ans: C. 15–20 + leakage. **Logic:** Volume formula.

Q29. With reference to ecology indicators in 2025 blueprint, which of the following statements is/are correct?

1. Sit among blueprint clusters with SDGs and UP traps.
2. International reports/funds also appear.
3. Blueprint means only Van Mahotsav July recall.

A. 1, 2 and 3 B. 2 and 3 only C. 1 and 3 only D. 1 and 2 only

💡 **Show answer**

Ans: D. Only 1 and 2. **Logic:** Broader than one festival.

Q30. Which one of the following best states the post-2025 practice rule from this sheet?

A. Heavy multi-statement / Match / NOT-matched / A/R — not only one-line recall B. Only HQ city lists forever C. Only single-date MCQs D. Skip SDGs entirely

💡 **Show answer**

Ans: A. Format formula. **Logic:** Closes the Environment Practice rewrite logic.

✍️ **S-PEN ACTIVE RECALL & MARGIN SCRIBBLES (CHAPTER SUMMARY)**